

HANDBOOK STRUCTURAL STEEL



YICK HOE

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All technical data, weights and measures shown in this handbook will be governed by standards of the respective specifications so offered. Care has been taken to ensure that all data and information herein is factual and the numerical values are accurate to the best of our knowledge. Yick Hoe Companies assumes no responsibility for errors in or misinterpretation of the information contained in this handbook or in its use.

COMPANY PROFILE



- **Yick Hoe Ferrous Steel Sdn. Bhd.** (37855-P)
- **Yick Hoe Steel Industries Sdn. Bhd.** (556274-P)
(Formerly known as Yick Hoe Special Steel & Engineering Sdn Bhd)
- **Sandakan Metal Sdn. Bhd.** (879779-K)
- **Lahad Datu Metal Sdn. Bhd.** (730343-A)

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Importer, Exporter & Distribution of Mild Steel, Stainless Steel materials like plates, structural steel, Long products, operating a steel coil center and manufacturing of GI Purlin products likes:

- GI COILS
- COLD ROLLED COILS
- EG COILS
- PURLIN
- PLATES
- BEAM
- PIPES
- DEFORMED BARS
- SECTIONS
- PIPE FITTINGS
- TUBES

COIL SERVICING CENTER



Slitting Machine

Capability : Thickness : 0.30 ~ 3.0mm
Width : 22 ~ 1250mm



Shearing Machine

Capability : Thickness : 0.30 ~ 3.0mm
Width : 300 ~ 1600mm
Length : 300 ~ 4000mm

GI PURLING MANUFACTURING



Thickness	Width	Grade	Strength
1.60mm	75mm	HT GI JIS G3302	Tensile : min 500N/MM2
2.00mm	100mm	SGC490M1	Yield : Min 500N/MM2
2.5mm	125mm		
	150mm		
	200mm		
	250mm		
	300mm		

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YICK HOE

EXPLANATORY NOTES

MATERIAL

The structural components referred to in this handbook are of steel to BS 4360, 'Weldable structural steels' and/or its related equivalents.

SECTIONS

The universal beams and columns and tees cut therefrom, the joists, channels, universal bearing piles and rolled tees are generally as listed in BS 4: Part 1: 1980. Some other joists, universal bearing piles and rolled tees in regular production are also included.

N.B. The universal beams listed in pages 48 and 49 all have 2° 52' tapered flanges.

Column core. This section, although listed separately, is rolled in the serial size 356 mm x 406 mm. It is the most appropriate one to be used as the core of a built-up stanchion if a heavier section than those rolled is required.

DIMENSIONAL UNITS

The dimensions of sections are given in millimetres (mm) and the calculated properties (centroidal distances, cross-sectional areas, radii of gyration, moments of inertia, elastic and plastic moduli) are given in centimetre (cm) units. Surface areas are in square metres (m²). For tolerances on dimensions and shape, reference should be made to BS 4848: Part 2.

OTHER UNITS

The units of force, mass and acceleration are those of the Systeme International (SI). They are the Newton (N), the kilogramme (kg) and the metre per second per second (m/s²) so that 1N = 1kg x 1m/s². The acceleration due to gravity varies slightly from place to place and for convenience a 'standard' value of 9.80665 m/s² has become generally accepted in structural engineering. With this convention, the force exerted by a mass of 1kg under the action of gravity is the 'technical unit' of 9.80665N. In the same way 9.80665 kilonewtons (kN) is the force exerted by a mass of 1 tonne (1000 kg) under gravity, and 1kN the force from a mass of 0.102 tonne.

EXPLANATORY NOTES

COMPARABLE SPECIFICATIONS AND CHEMICAL ANALYSIS

Specifications

The following specifications are normally readily available, but any offer will depend upon acceptance of full specification details and/or any other specifications not listed below.

Structural Sections

Quality	Grade	Tensile strength			Min. yield stress		
		lb/in ²	N/mm ²	kg/mm ²	lb/in ²	N/mm ²	kg/mm ²
BS4360	43A	62300/84000	430/580	44/59	38500	265	27
	50A	72500/93000	490/640	50/65	50000	345	36
ASTM	A36	58000/80000	400/550	41/56	36000	250	25
ASTM A572	50	65000 min	450 min	46 min	50000	345	36
Din 17100	RSt. 37.2	52000/74000	360/510	36/52	32500	225	23
	St 44.2	62300/83500	430/580	44/59	38500	265	27
	St 52.3	74000/98500	510/680	52/69	50000	345	36
JIS G 3101	SS 41	58000/74000	402/510	41/52	34000	235	24
	SS 50	72500/88500	490/608	50/62	40000	275	28
JIS G 3106	SM 41	58000/74000	402/510	41/52	34000	235	24
	SM 50	72500/88500	490/608	50/62	45000	314	32

Notes: Sections over 5/8 inch (16mm) up to and including 1 inch (25mm) thick
All figures are approximate and for guidance only
Other specifications are obtainable subject to acceptance of specification (eg. UNI 7070-72 and NFA 35-501)

Chemical analyses

Specification	Grade	Composition (Maxima unless stated)								
Quality		C	Si	Mn	P	S	Nb	V	N1	Others
BS4360 (1986)	43A	0.25	0.50	1.60	0.05	0.05	--	--	--	--
	50B	0.20	0.50	1.50	0.05	0.05	0.003/0.10	0.003/0.10	--	--
ASTM A36 (1984)		0.26	--	--	0.04	0.05	--	--	--	Cu 0.20 (min)
ASTM A572 (1984)	50	0.23	0.40	1.35	0.04	0.05	--	--	--	Cu 0.20 (min)
JIS 3101 (1976)	SS 41	--	--	--	0.05	0.05	--	--	--	--
	SS 50	--	--	--	0.05	0.05	--	--	--	--
JIS 3106 (1976)	SM 41 A	0.25	--	2.5 x C min	0.04	0.04	--	--	--	--
	SM 41 B	0.22	0.35	0.6 - 1.20	0.04	0.04	--	--	--	--
	SM 41 C	0.18	0.35	1.40	0.04	0.04	--	--	--	--
	SM 50 A	0.22	0.55	1.50	0.04	0.04	--	--	--	--
	SM 50 B	0.20	0.55	1.50	0.04	0.04	--	--	--	--
	SM 50 C	0.18	0.55	1.50	0.04	0.04	--	--	--	--

EXPLANATORY NOTES

AVAILABLE SPECIFICATIONS

Application		Standard			
		KS	JIS	ASTM	Others
Steel for General Structure		KS D3503 SS 34, 41, 50, 55	JIS G3101 SS 34, 41, 50, 55	A36, A113, A242 A233, A284, A441 A529, A573	BS 1449 Part 1 50/35 HR, HS DIN 17100 ST 33-1,2 ST 37-2,3 ST 52-3 RST 37-2 UST 37-2 IS 226 ST 42-S
Steel for Welded Structure		KS D3515 SWS 41A, B, C SWS 50A, B, C SWS 50YA, YB SWS 538, C	JIS G3106 SM 41A, B, C SM 50A, B, C SM 50YA, YB SM 53B, C	A36, A242, A283 A284, A440, A529 A573	BS 4360 43A, B IS 2062 ST 42-W AS 250
Hot Rolled Mild Steel Sheets & Strip		KS D3501 SHP 1, 2, 3	JIS G3131 SPHC, SPHD SPHE	A569 A570 A621	BS 1449 HR 15 HR 4 HR 3
Pipes & Tubes	Hot Rolled Carbon Steel Strip for Pipes & Tubes	KS D3555 HRS 1, 2, 3, 4 KS D3566 SPS 30, 41, 50, 51	JIS G3132 SPHT 1, 2, 3, 4 JIS G3444 STK 30, 41, 50, 51	A53 A252-2,3	
	Boiler & Heat Exchanger Tubes	KS D3563 STBH 33, 35, 42 KS D3571 STLT 39 KS D3572 STHA 12, 13, 20, 22	JIS G3461 STB 33, 35, 42 JIS G3462 STBA12, 13, 20, 22 JIS G3464 STBL 39		
	Carbon Steel Pipes for High Temperature Service	KS D3570 SPHT 38, 42	JIS G3456 STPT 38, 42		
	Steel Pipes for Low Temperature Service	KS D3571 SPLT 39	JIS G3460 STPL 39		
	Carbon Steel Tubes for Machine Structure	KS D3517 STKM 11 A STKM 12A, B, C STKM 13A, B, C STKM 14A, B, C STKM 15A, B, C STKM 18A, B, C	JIS G3445 STKM 11A STKM 12A, B, C STKM 13A, B, C STKM 14A, B, C STKM 15A, B, C STKM 18A, B, C		

EXPLANATORY NOTES

Specifications

Application		Standard			
		KS	JIS	ASTM	Others
Pipes & Tubes	Pipes for Oil Well Tubing & Line Pipes				API 5L(S)-A,B API 5LX(S)-X42, X46, X56, X60, X65, X70 API 5A- H40, J55, K55, N80
Ordinary Strength Hull Structural Steel					KR - RA, RB, RD AB - A LR - A,B NK- KA NV -A
Superior Atmospheric Corrosion Resisting Steel			JIS G3125 SPA-H	A441 A588	
Hot Rolled Sheets & Strips for Automobile Structure		KS D3519 SAPH 32, 38, 41, 45	JIS G3113 SAPH 32, 38, 41, 45		SAE 950B
Spring Steel		KS D3701 SPS 7	JIS G4801 SUP 9		
Chromium Molybdenum Steels		KS D3711 SCM 430	JIS G4106 SCM 430		
Manganese Steels			JIS G4106 SMn 420		
Carbon Steel for Machine Structure		KS D3752 SM10C ~ SM58C	JIS G4051 S10C ~ S58C		SAE 1006 ~ SAE 1050 AISI 1006 ~ AISI 1050
Low Alloy High Strength Steels for Structure				A607 - 45, 50, 55 A715 - 50TY7 A572 A242	
Steel for Boilers & Other Pressure Vessels		KS D3560 SBB 42, 46, 49	JIS G3103 SB 42, 46, 49	A285 A442 A455 A516	
Steel Sheets & Strip for Gas Cylinders		KS D3533 SG 26, 30, 33, 37	JIS G3116 SG 26, 30, 33, 37	A285-C A455	

EXPLANATORY NOTES

COMPARISON OF EN10025 & BS4360

EN 10025: 1993								BS 4360: 1986					
					Impact Energy (J °C) Nominal Thickness							Impact Energy (J °C) Nominal Thickness	
Grade	Former Grade	Tensile Strength	Min Yield Strength at 16mm N/mm ²	Max Thk for Specified Yield N/mm ²	Temp °C	<150mm -1	>150mm <250mm -1	Grade	Tensile Strength >3mm N/mm ²	Min Yield Strength at 16mm N/mm ²	Max thk Specified Yield N/mm ² (2)	Temp °C	<100mm -3
S185 (4)	Fe 310-0(4)	290/510	185	25	-	-	-	-	-	-	-	-	-
S235 (5)	Fe 360A (5)	340/470	235	250	-	-	-	40A	340/500	235	150	-	-
S235JR (4)	Fe 360B (4)	340/470	235	25	20	27	-	-	-	-	-	-	-
(4)	Fe 360(FU) (4)	340/470	235	25	+20 (6)	27	-	-	-	-	-	-	-
S235JRG2	Fe 360B (FN)	340/470	235	250	+20 (6)	27	23	40B	340/500	235	150	+20 (6)	27
S235JO	Fe 360C	340/470	235	250	0	27	23	40C	340/500	235	150	0	27
S235J2G3	Fe 360D1	340/470	235	250	-20	27	23	40D	340/500	235	150	-20	27
S235J2G4	Fe 360D2	340/470	235	250	-20	27	23	40D	340/500	235	150	-20	27
S275 (5)	Fe 430A (5)	410/560	275	250	-	-	-	43A	430/580	275	150	-	-
S275JR	Fe 430B	410/560	275	250	+20 (6)	27	23	43B	430/580	275	150	+20 (6)	27
S275JO	Fe 430C	410/560	275	250	0	27	23	43C	430/580	275	150	0	27
S275J2G3	Fe 430D1	410/560	275	250	-20	27	23	43D	430/580	275	150	-20	27
S275J2G4	Fe 430D2	410/560	275	250	-20	27	23	43D	430/580	275	150	-20	27
S355 (5)	Fe 510A	490/630	355	250	-	-	-	50A	490/640	355	150	-	-
S355JR	Fe 510B	490/630	355	250	+20 (6)	27	23	50B	490/640	355	150	+20 (6)	27
S355JO	Fe 510C	490/630	355	250	0	27	23	50C	490/640	355	150	0	27
S355J2G3	Fe 510D1	490/630	355	250	-20	27	23	50D	490/640	355	150	-20	27
S355J2G4	Fe 510D2	490/630	355	250	-20	27	23	50D	490/640	355	150	-20	27
S355K2G3	Fe 510DD1	490/630	355	250	-20	40	33	50DD	490/640	355	150	-30	27
S355K2G4	Fe 510DD2	490/630	355	250	-20	40	33	50DD	490/640	355	150	-30	27
E295	Fe 490-2	470/610	295	250	-	-	-	-	-	-	-	-	-
E335	Fe 590-2	570/710	335	250	-	-	-	-	-	-	-	-	-
E360	Fe 690-2	670/830	350	250	-	-	-	-	-	-	-	-	-

- (1) For sections up to and including 100mm only
- (2) For wide flats and Sections up to and including 63mm and 100mm respectively
- (3) For wide flats up to and including 50mm and sections no limit is stated
- (4) Only available up to and including 25mm thick
- (5) The Steel grades S235 (Fe 360A), S275 (Fe 430A), and S355 (Fe 510A) appear only in the English language version (BS EN 10025) as non conflicting additions and do not appear in European versions
- (6) Verification of the specified impact value is only carried out when agreed at time of enquiry and order

EXPLANATORY NOTES

Designation:		Equivalent Former Designations in													
EN 10025: 1993 Designation		According EN 10025: 1990	Germany Din 17100	France NFA 35501	UK BS 4360	Spain	Italy	Belgium	Sweden	Portugal	Austria	Norway NS12101	USA ASTM	USA API	Japan JIS G3101/6
Steel Name	Steel Number														
S185	1.0035	Fe 310-0	St 33	A33		A3 10-0	Fe 320	A 320	13 00-00	Fe 310-00	St 320				
S235JR	1.0037	Fe 360B	St 37-2	E24-2			Fe 360 B	AE 235-B	13 11-00	Fe 360-B		NS 12 120	A283 GrC		
S235JRG1	1.0036	Fe 360BFU	U St 37-2			AE 235 B-FU					B	NS 12 122	A283 GrD		SM41A
S235JRG2	1.0038	Fe 360BFN	R St 37-2		40B	AE 235 B-FN			13 12-00		R St 360 B	NS 12 123			SM41B
S235JO	1.0014	Fe 360c	St 37-3U	E 24-3	40C	AE 235 D	Fe 360 C	Ae 235-C		Fe 360-C	St 360 C	St 430 B			
											St 360 CE	NS 12 142			
S235J2G3	1.0116	S Fe 360 D1	St 37-3N	E 24-4	40D		Fe 360 D	AE 236-D		Fe 360-D	St 360 D	NS 12 124			
S234J2G4	1.0117	Fe 360 D2													
S275JR	1.0044	Fe 430B	ST 44-2	E 28-2	43B	AE 275 B	Fe 430 B	AE 255-B	14 12-00	Fe 430-B	St 430 B	NS 12-142	Gr65		SS400
S275JO	1.0143	Fe 430C	ST 44-3U	E 28-3	43C	AE 275 C	Fe 430 C	AE 255-C		Fe 430-C	St 430 C	NS 12-143			
											ST 430 CE				
S275J2G3	1.0144	Fe 430D1	ST 430 D1	E 28-4	43D	AE 275 D	Fe 430 D	AE 255-D	14 14-00	Fe 430-D	ST 430 D	NS 12-143	Gr42	2H GR42	
S275J2G4	1.0145	Fe 430D2							14 14-01						
S355JR	1.0045	Fe 510 B		E 36-2	50B	AE 355 B	Fe 310 B	AE 355B		Fe 510-B			A572 Gr50		SS490
S355JO	1.0553	Fe 510 C	ST 52-3U	E 36-3	50C	AE 355 C	Fe 310 C	AE 355C		Fe 510 C	St 510 C	Ns 12 153	A790 Gr50	2H Gr50	
S355J2G3	1.057	Fe 510 D1	ST 52-3N		50D	AE 355 D	Fe 310 D	AE 355D		Fe 510- D	St 510-D	NS 12 153	Gr50	2 W/ YGr50	
S355J2G4	1.0577	Fe 510 D2													
S355K2G3	1.0595	Fe 510 DD1		E36- 4	50DD			AE 355 DD		Fe 510DD			A533 GrD		
S355K2G4	1.0596	Fe 510 DD2			50DD										
E295	1.005	Fe 490-2	ST 50-2	A 50-2		A 490	Fe 480	A 490-2	15 50-00	FE 490-2	St 490				
									15 50-01						
E355	Fe 590-2	ST 590-2	St 60-2	A 60-2		A 590	Fe 580	A590-2	16 50-00	Fe 590-2	St 590				
									16 50-01						
E 360	1.007	Fe 690-2	St 70-2	A 70-2		A 690	Fe 650	A690-2	16 55-00	Fe 690-2	St 590				
									16 55-01						

EXPLANATORY NOTES

HIGH STRENGTH STEEL

OFFSHORE PLATFORM, F.P.S.O., BRIDGE GIRDERS, STRUCTURAL TUBULARS & FABRICATIONS. LOW TEMP CHARPY TESTED, + M = T.M.C.P QUALITY

National Standard		Grade	Min Yield N/mm²	UTS N/mm²	Charpy °C	Impacts (L) Joules	Former National Specifications					
							EU		GB		DE	
Year-ASTM + Shipbuilding = Latest							EN 10025: 1993		BS 43601: 1986		DIN 17100: 1983	
JP	JIS G3106 :2004	SM490B	325	490/610	0	27						
ANS	AS/NZS 3678:1996	350	340	480 min	N/R	N/R						
		350 L0	340	480 min	0	27						
USASTM	A572M Gr 50	345	450 min	N/R	N/R							
		A633M Gr C	345	450 min	0	26						
		A588 Gr B	345	485 min	N/R	N/R						
							Equivalent to	Min Yield	UTS	°C	J	
							COR-TEN A	355	470-630	-20	27	
							COR-TEN B	355	470-630	-20	27	
CN	GB/T 1591:1994	Q345 A	345	470/630	N/R	N/R						
		Q345 B	345	470/630	+20	27						
		Q345 C	345	470/630	0	27						
		Q345 D	345	470/630	-20	27						
IN	IS 2062:2006	E350 A	350	490 min	N/R	N/R	IS 8500: 1991					
		E350 BR	350	490 min	+20	27						
		E350 B0	350	490 min	0	27						
		E350 C	350	490 min	-20	27						
EUEN 10025-2:2004		S355 JR	355	470/630	+20	27	S355 JR	50B	St.52-3			
		S355 JO	355	470/630	0	27	S355 JO	50C	St.52-3			
		S355 J2+N	355	470/630	-20	27	S355 J2G3	50D	St.52-3 N			
		S355 J2	355	470/630	-20	27	S355 J2G4	50EE				
		S355 K2/ K2+N	355	470/630	-20	40	S355 K2G3	50DD				
EUEN 10025-3:2004		S355N	355	470/630	-20	40	EN 10113:	50				
		S355NL	355	470/630	-50	27	EN 10113:					
							1993 S355 NL	50EE				
EUEN 10025-4:2004		S355M	355	520/680	-20	40	EN 10113:					
		S355ML	355	520/680	-50	27	EN 10113:					
							1993 S355 ML					
EUEN 10225:2004		S355 G2/G3+N	355	470/630	-20/-40	50	API: 1993	BS 7191: 1989				
		S355 G5/G6+M	355	470/630	-20/-40	50		355D/E				
		S355 G7+M/N	355	470/630	-40	50		355D/E				
		S355 G8+M/N	355	470/630	-40	50	API - 2H Gr50	355EM				
		S355 G9+M/N	355	470/630	-40	50	API - 2W Gr50	355MZ				
		S355 G10+M/N	355	470/630	-40	50	API - 2W Gr50T					
JP	JIS G3106:2004	SM490YB	365	490/610	0	27						
		SM520B	365	520/640	0	27						
JP	JIS G3101:2004	SS540	390	540 min	N/R	N/R						

ROLLED TUBULARS

Availability: 20" O/dia + over

API-2B/ 2H: 1 or more "Cans" can be fabricated from above S355 plates to suit req'd OD x WT x Length

PRODUCTS STOCK SIZES

AVAILABLE		S355JR - S355G10+N/M	ASTM A572-50-A588 GrB
Plates		Wide: 1.25m - 2.0m, 2.5m, 3.05m & 3.5m	Wide: 48" - 120" (1219 - 3048mm)
		Long: 3.0m - 6.0m, 8.0m, 10.0m, 12.0m & 15.0m	Long: 3' - 40' > 60' (915mm - 12.2m > 18.3m)
		THK: 5mm - 350mm	THK: 3/16" - 12" (4.75mm - 305mm)
		S355G7 - G10+M = 100mm MAX S355G7 - G10+N = 150mm MAX	
Coils	S355JR S355J2	Wide: 1.0m, 1.25m, 1.5m - 2.0m THK: 1mm - 12mm	

PRODUCT LIST

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I-BEAMS

DESIGN OF BEAMS

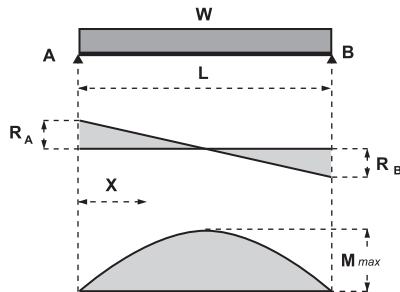
Design formulae for beams

	Notation
L	= length of span in millimetres
W	= total distributed or point load in Newtons
W_1 or W_2	= point load in Newtons
Σ	= resultant of point loads in Newtons
R_A, R_B, R_C etc.	= reaction at A, B, or C etc, in Newtons
F	= shearing force in Newtons
m	= applied moment in Newton millimetres
M_x	= bending moment in Newton millimetres at distance X from the left hand support A
δ_x	= deflection in millimetres
i_x	= slope in radians
M_A or M_B	= end fixing moment in Newton millimetres
M_{max}	= maximum bending moment in Newton millimetres
$M_{max\ max}$	= absolute maximum bending moment in Newton millimetres
M_{load}	= bending moment under the load in Newton millimetres
δ_{max}	= maximum deflection in millimetres
$\delta_{max\ max}$	= absolute maximum deflection in millimetres
$\delta_{negative}$	= negative (i.e. upward) deflection in millimetres
i_A or i_B	= slope at A or at B in radians
E	= modulus of elasticity, $2.1 \times 10^5 \text{ N/mm}^2$
I	= constant moment of inertia of uniform section beam in mm^4
Sign convention	
Loads	+ positive when acting downward
Support Reaction	+ positive when acting upward
Shearing Force	+ positive on a section where the upward left hand support reaction is greater than the algebraic sum of external loads located left of that section
Bending Moment	+ positive (shown above base line on diagrams) when causing convexity downward
Deflection	+ positive when downward
Slope	appropriate values in radians are given, but the signs depend upon which support or which section is being considered, and can be readily ascertained by inspection

Where space permits, general equations for M_x and i_x at any point of the beam, and also the equation to the elastic line (δ_x), have been included.

Values for Slope. These may be used in evaluating the angle of rotation for rubber bearings and similar constructional elements.

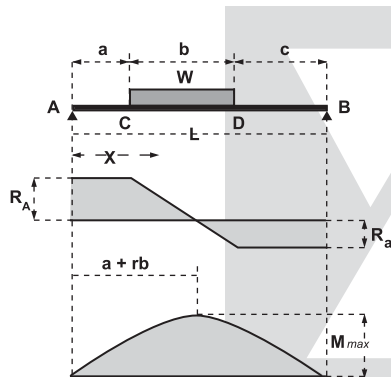
Simply supported beam



Uniform load on full span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total uniform load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL^2}{8} \\ \delta_{\max} &= \frac{5}{384} \cdot \frac{WL^3}{EI} \end{aligned} \right. \\
 i_A = i_B &= \frac{WL^2}{24EI} \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{2L}(L-X) \\ \delta_x &= \frac{WX}{24EI}(X^3-2X^2L+L^3) \\ i_x &= \frac{W}{24EI}(4X^3-6X^2L+L^3) \end{aligned} \right.
 \end{aligned}$$

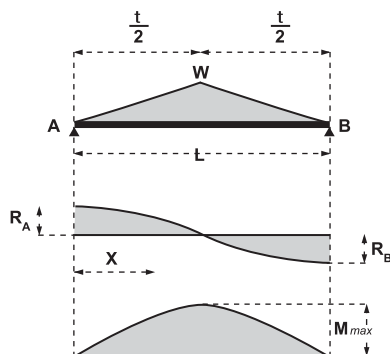
Simply supported beam



Uniform load on part of span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total uniform load} &= W \\
 \text{Let } r &= \frac{0.5b+c}{L} \\
 R_A = Wr; \quad R_B &= W(l-r) \\
 \text{at } X = a + rb, M_{\max} &= Wr(a+0.5rb) \\
 i_A = \frac{Wr(L^2-c^2-Lbr)}{6EI}; \quad i_B &= \frac{W(l-r)}{6EI}[L^2-a^2-Lb(l-r)] \\
 \text{Equation to elastic line between C and D, i.e. } a \leq X \leq a+b & \\
 \delta_x &= \frac{W}{24EIb}[X^4-4(a+rb)X^3+6a^2X^2+4\{rb(L^2-c^2-cb-\frac{b^2}{2})-a^2\}X+a^4]
 \end{aligned}$$

Simply supported beam

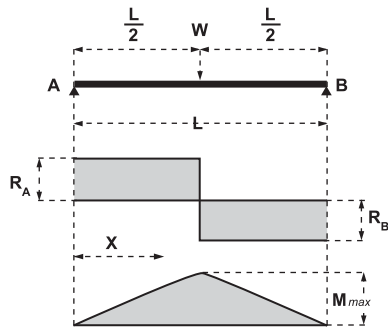


Triangular load on full span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL^2}{6} \\ \delta_{\max} &= \frac{WL^3}{60EI} \end{aligned} \right. \\
 i_A = i_B &= \frac{5WL^2}{96EI} \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{6L^2}(3L^2-4X^2) \\ \delta_x &= \frac{WX}{480EI}(16X^4-40X^2L^2+25L^4) \\ i_x &= \frac{W}{96EI}(16X^4-24X^2L^2+5L^4) \end{aligned} \right.
 \end{aligned}$$

I-BEAMS

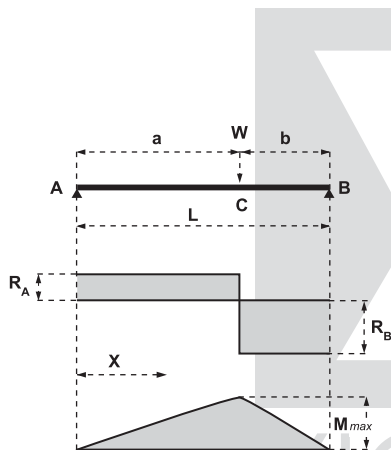
Simply supported beam



Point load at mid-span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL}{4} \\ \delta_{\max} &= \frac{1}{48} \cdot \frac{WL^3}{EI} \\ &= \frac{WL^2}{16EI} \end{aligned} \right. \\
 i_A = i_B & \\
 \text{at X from A} &\left\{ \begin{aligned} M_X &= \frac{WX}{2} \\ \delta_X &= \frac{WX}{48EI} (3L^2 - 4X^2) \\ i_X &= \frac{W}{16EI} (L^2 - 4X^2) \end{aligned} \right.
 \end{aligned}$$

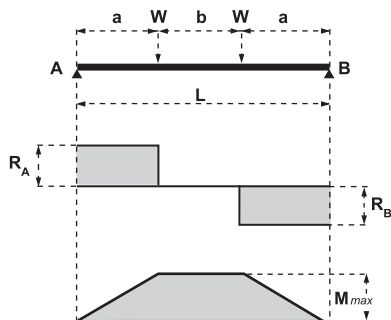
Simply supported beam



Point load at any position

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = \frac{Wb}{L} & \quad R_B = \frac{Wa}{L} \\
 \text{at C under load} &\left\{ \begin{aligned} M_{\max} &= \frac{Wab}{L} \\ \delta_c &= \frac{Wa^2b^2}{3EIL} \\ i_A &= \frac{Wab}{6EIL} (L+b); \quad i_B = \frac{Wab}{6EIL} (L+a) \end{aligned} \right. \\
 \text{When } a > b &\left\{ \begin{aligned} \delta_{\max} &= \frac{Wab(L+b)}{27EIL} \sqrt{3a(L+b)} \\ X &= \sqrt{\frac{a(L+b)}{3}} \end{aligned} \right.
 \end{aligned}$$

Simply supported beam

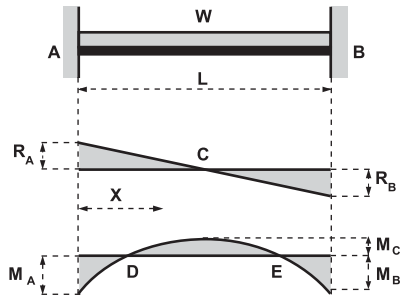


Two equal symmetrical point loads

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Two point loads, each} &= W \\
 R_A = R_B &= W \\
 M_{\max} \text{ over length } b &= Wa \\
 \delta_{\max} \text{ at mid-span} &= \frac{Wa}{24EI} (3L^2 - 4a^2) \\
 \delta \text{ under either load} &= \frac{Wa^2}{6EI} (3L - 4a) \\
 i_A = i_B &= \frac{Wa}{2EI} (L - a) \\
 \text{If } a=b=\frac{L}{3}, \delta_{\max} &= \frac{23}{648} \frac{WL^3}{EI}
 \end{aligned}$$

Beam fixed at both ends

Uniform load on full span

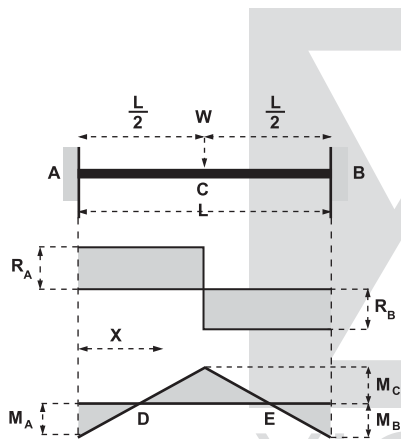


$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total Uniform load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 M_A = M_B &= -\frac{WL}{12} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_C &= \frac{WL}{24} \\ \delta_{\max} &= \frac{WL^3}{384EI} \end{aligned} \right. \\
 \text{at X from A} &\left\{ \begin{aligned} M_X &= \frac{W}{12L}(L^2 - 6LX + 6X^2) \\ \delta_X &= \frac{WX^2}{24EI}(L - X)^2 \\ i_X &= \frac{WX}{12EI}(L^2 - 3LX + 2X^2) \end{aligned} \right.
 \end{aligned}$$

at 0.211L from either end $M_D = M_E = 0$

Beam fixed at both ends

Point load at mid-span

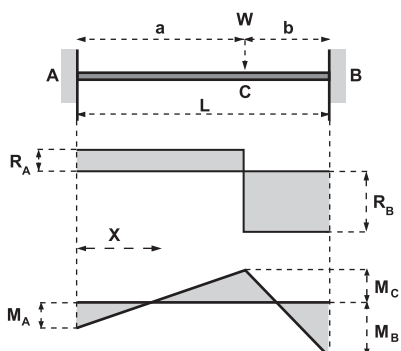


$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 M_A = M_B &= -\frac{WL}{8} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_C &= \frac{WL}{8} \\ \delta_{\max} &= \frac{WL^3}{192EI} \end{aligned} \right. \\
 \text{at X from A between A and C} &\left\{ \begin{aligned} M_X &= \frac{W}{8}(4X - L) \\ \delta_X &= \frac{WX^2}{48EI}(3L - 4X) \\ i_X &= \frac{WX}{8EI}(L - 2X) \end{aligned} \right.
 \end{aligned}$$

at 0.25L from either end $M_D = M_E = 0$

Beam fixed at both ends

Point load at any position



$$\begin{aligned}
 \text{Span} &= L \\
 \text{Point load} &= W \\
 R_A &= \frac{Wb^2(L+2a)}{L^3} & R_B &= \frac{Wa^2(L+2b)}{L^3} \\
 M_A &= -\frac{Wab^2}{L^2} & M_B &= -\frac{Wa^2b}{L^2} \\
 \text{at C, under load, } M_C &= \frac{2Wa^2b^2}{L^3} \\
 \text{at X from A between A and C} &\left\{ \begin{aligned} M_X &= \frac{Wab^2}{L^2} \frac{Wb^2(L+2a)X}{L^3} + \\ \delta_X &= \frac{Wb^2X^2[3La - (L+2a)X]}{6EIL^3} \\ i_X &= \frac{Wb^2X[2La - (L+2a)X]}{2EIL^3} \\ \delta_{\max} &= \frac{2Wa^3b^2}{3EI(L+2a)^2} \end{aligned} \right.
 \end{aligned}$$

when $a > b$ the maximum deflection is at $X = \frac{2La}{L+2a}$

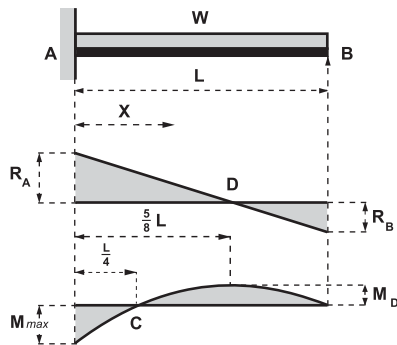
I-BEAMS

Cantilever	Uniform load on part of span	Special case: Uniform load on full span
	Span = L	Span = L = b
		a = c = 0
	Uniform load = W	Uniform load = W
	$R_A = W$	$R_A = W$
	$M_A = -W(a + \frac{b}{2})$	$M_A = -\frac{WL}{2}$
	$i_C = i_D = \frac{W}{6EI}(3a^2 + 3ab + b^2)$	$i_D = \frac{WL^2}{6EI}$
		$\delta_D = \frac{WL^3}{8EI}$
	$\delta_D = \frac{WL}{24EI}[8a^3 + 18a^2b + 12ab^2 + 3b^3 + 4c(3a^2 + 3ab + b^2)]$	

Cantilever	Triangular load on part of span	Special case: Triangular load on full span
	Span = L	Span = L = a
		b = 0
	Triangular load = W	Triangular load = W
	$R_A = W$	$R_A = W$
	$M_A = -\frac{Wa}{3}$	$M_A = -\frac{WL}{3}$
	$\delta_C = \frac{Wa^2}{15EI}(L + \frac{b}{4})$	$\delta_C = \frac{WL^3}{15EI}$
	$i_B = i_C = \frac{Wa^2}{12EI}$	$i_C = \frac{WL^2}{12EI}$

Cantilever	Point load at any position	Special case: Point load at free end
	Span = L	Span = L = a
		b = 0
	Point load = W	Point load = W
	$R_A = W$	$R_A = W$
	$M_A = -Wa$	$M_A = -WL$
	$\delta_C = \frac{Wa^2}{3EI}(L + \frac{b}{2})$	$\delta_C = \frac{WL^3}{3EI}$
	$i_B = i_C = \frac{Wa^2}{2EI}$	$i_C = \frac{WL^2}{2EI}$

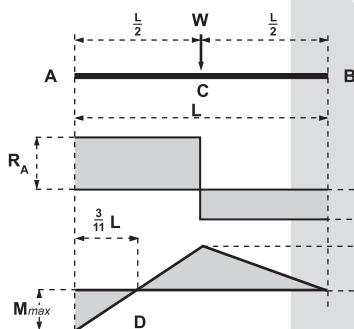
Propped cantilever



Uniform load on full span

Span	=	L
Total uniform load	=	W
$R_A = \frac{5}{8}W$	$R_B = \frac{3}{8}W$	
at A	$M_{\max} = -\frac{WL}{8}$	
at $\frac{5}{8}L$ from A	$M_D = \frac{9}{128}WL$	
at 0.5785L from A	$\delta_{\max} = \frac{WL^3}{185EI}$	
at B	$i_B = \frac{WL^2}{48EI}$	
at X	$M_X = \frac{W}{8L}(L^2 - 5LX + 4X^2)$	
from A	$\delta_X = \frac{WX^2}{48EI}(3L^2 - 5LX + 2X^2)$	
	$i_X = \frac{WX}{48EI}(6L^2 - 15LX + 8X^2)$	
at $\frac{1}{4}L$ from A, $M_C = 0$		

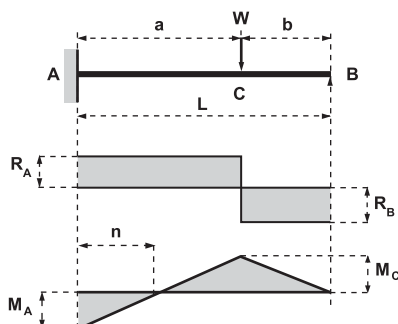
Propped cantilever



Point load at mid-span

Span	=	L
Point load	=	W
$R_A = \frac{11}{16}W$	$R_B = \frac{5}{16}W$	
at A	$M_{\max} = -\frac{3}{16}WL$	
at mid-span	$M_C = \frac{5}{32}WL$	
under load	$\delta_C = \frac{7WL^3}{768EI}$	
at 0.5528L from A	$\delta_{\max} = \frac{WL^3}{107EI}$	
at B	$i_B = \frac{WL^2}{32EI}$	
at $\frac{3}{11}L$ from A, $M = 0$		

Propped cantilever

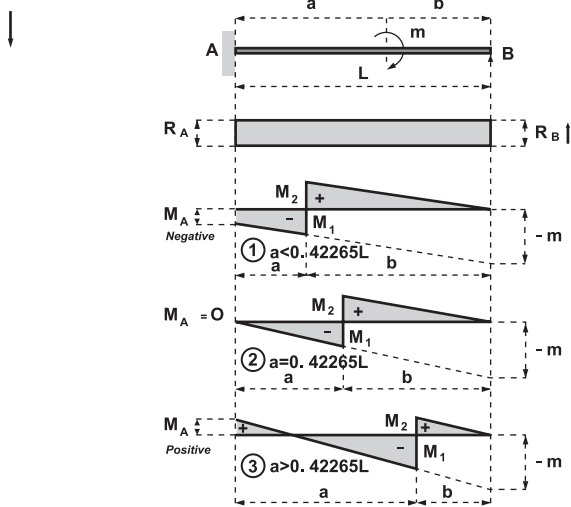


Point load at any position

Span	=	L
Point load	=	W
$R_A = \frac{Wb(3L^2 - b^2)}{2L^3}$	$R_B = \frac{Wa^2(2L+b)}{2L^3}$	
$M_A = -\frac{Wab(L+b)}{2L^2}$	$M_C = \frac{Wa^2b(2L+b)}{2L^3}$	
$i_B = \frac{Wa^2b}{4EI}$		
Absolute max deflection is under the load when $a=b\sqrt{2}=0.5858L$	$\delta_{\max \max} = \frac{WL^3}{102EI}$	
When $a > b\sqrt{2}$ max deflection is between A and C	$\delta_{\max} = \frac{Wa^3b}{3EI} \frac{(L+b)^3}{(3L^2 - b^2)^2}$	
When $a < b\sqrt{2}$ max deflection is between C and B	$\delta_{\max} = \frac{Wa^2b}{6EI} \sqrt{\frac{b}{2L+b}}$	
at $n=aL \frac{L+b}{3L^2 - b^2}$ from A, $M = 0$		

I-BEAMS

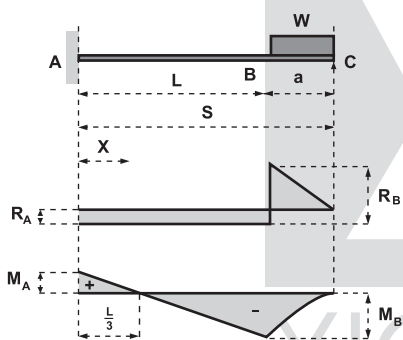
Propped cantilever



Moment applied at any point

$$\begin{aligned} \text{Span} &= L \\ \text{Applied moment} &= m \\ M_A &= \frac{L^2 - 3b^2}{2L^2} m & i_B &= \frac{ma}{4EI} (2b-a) \\ R_A = -R_B &= -\frac{3(L^2 - b^2)}{2L^3} m = -\frac{m + M_A}{L} \\ \begin{cases} M_1 &= -\frac{m}{L^3} (a^3 + \frac{3}{2}ab^2 + b^3) \\ M_2 &= \frac{3mab}{L^3} (b + \frac{a}{2}) = m + M_1 \end{cases} \\ \begin{cases} M_1 &= -0.42265m \\ M_2 &= 0.57735m \end{cases} \\ \begin{cases} M_1 &= \\ M_2 &= \end{cases} & \text{as for ①} \end{aligned}$$

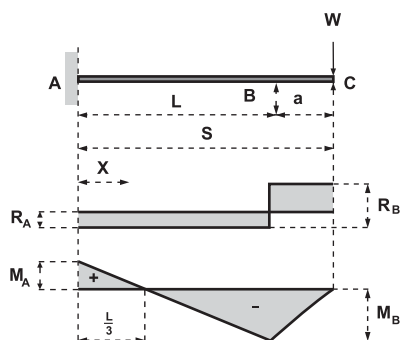
Propped cantilever



Uniform load on length beyond prop

$$\begin{aligned} \text{Span} &= L & \text{Full length} &= S \\ \text{Uniform load} &= W \\ R_A &= -\frac{3Wa}{4L} & R_B &= \frac{W}{L} (S - \frac{a}{4}) \\ M_A &= \frac{Wa}{4} & M_B &= -\frac{Wa}{2} \\ \text{Deflection at C} &= \delta_{\max} & &= \frac{Wa^2S}{8EI} \\ \text{Max. negative deflection} &= \delta_{\neg} & &= -\frac{WL^2a}{54EI} \\ \text{at } X = \frac{2}{3}L \\ \text{Slope at C} &= i_C & &= \frac{Wa}{8EI} (S + \frac{a}{3}) \\ \text{at } X = \frac{L}{3} \text{ from A, } M &= 0 \end{aligned}$$

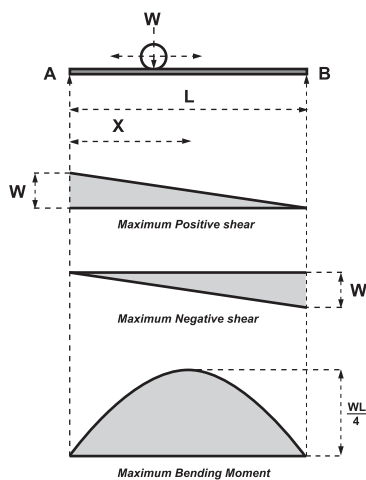
Propped cantilever



Point load at free end

$$\begin{aligned} \text{Span} &= L & \text{Full length} &= S \\ \text{Point load} &= W \\ R_A &= -\frac{3Wa}{2L} & R_B &= \frac{W}{L} (S + \frac{a}{2}) \\ M_A &= \frac{Wa}{2} & M_B &= -Wa \\ \text{Deflection at C} &= \delta_{\max} & &= \frac{Wa^2}{4EI} (S + \frac{a}{3}) \\ \text{Max. negative deflection} &= \delta_{\neg} & &= -\frac{WL^2a}{27EI} \\ \text{at } X = \frac{2}{3}L \\ \text{Slope at C} &= i_C & &= \frac{Wa}{4EI} (S + a) \\ \text{at } X = \frac{L}{3} \text{ from A, } M &= 0 \end{aligned}$$

Simply supported beam



Single concentrated moving load

Maximum Positive Shear at any section occurs when the load is immediately to the right of the section. Similarly, Maximum Negative Shear occurs when the load is to the left. For a section distance X from A:

$$\text{Positive } F_{x \max} = W \frac{L-X}{L} \quad \text{Negative } F_{x \max} = -W \frac{X}{L}$$

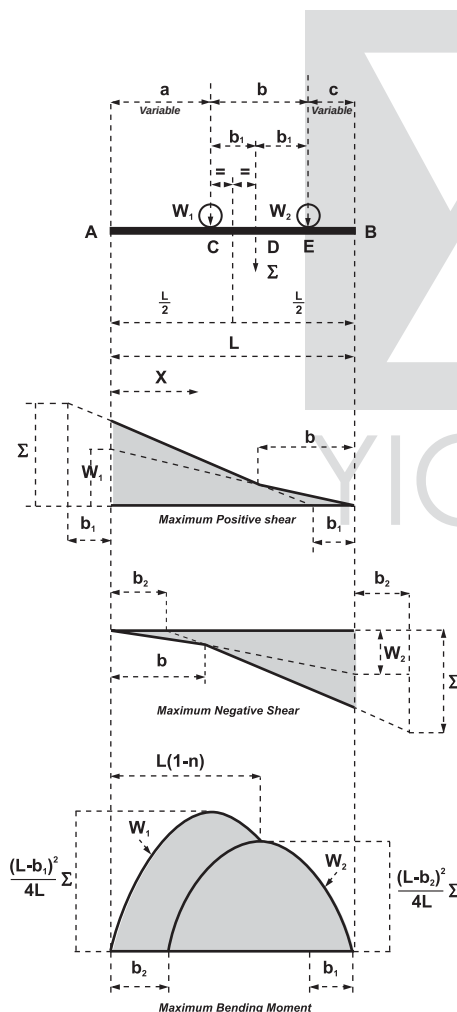
Maximum Bending Moment at any section occurs when the load is over the section. For a section distance X from A:

$$M_{x \max} = W \frac{X(L-X)}{L}$$

The Absolute Maximum Bending Moment and Deflection occur under the load at mid-span

$$\left. \begin{array}{l} M_{\max \max} = \frac{WL}{4} \\ \delta_{\max \max} = \frac{WL^3}{48EI} \end{array} \right\} \quad \left. \begin{array}{l} \text{Maximum end slope at A occurs} \\ \text{with the load at } X=0.42265L \\ \text{from A.} \end{array} \right\} i_{A \max} = 0.06415 \frac{WL^2}{EI}$$

Simply supported beam



Two concentrated moving load

Assume : $W_1 > W_2$ $W_1 + W_2 = \Sigma$ $W_2 = n\Sigma$
Fixed Distance $b = mL$

$$b_1 = \frac{W_2}{W_1 + W_2} b = nmL \quad b_2 = (m-n)L$$

Maximum Reaction at A and Absolute Maximum Positive Shear occur when W_1 is immediately to the right of A:

$$R_{A \max} = \text{Positive } F_{\max \max} = W_1 + W_2 \quad \frac{L-b}{L}$$

For a section distance X from A:

$$X \leq L-b \quad : \quad L-b \leq X$$

$$\text{Positive } F_{\max} = \frac{L-X}{L} mW_2 \quad : \quad \text{Positive } F_{\max} = \frac{L-X}{L} W_1$$

Note: 1. For $R_{B \max}$, interchange values of W_1 and W_2 in the formula for $R_{A \max}$

2. For Negative Shear, interchange W_1 and W_2 in formulae for Positive Shear, measuring X from B towards A

3. If $m > \frac{n}{1-n}$ calculate $R_{B \max}$ and Negative Shear values for W_1 only as single load.

Absolute Maximum Bending Moment occurs under W_1 when that load and the resultant of both loads are equidistant from mid-span (see loading diagram):

$$M_{\max \max} = \frac{(L-b_1)^2}{4L}$$

If $m < n$, the Maximum Bending Moment at any section occurs under one of the loads. For a section distance X from A:

$$X \leq L(1-n)$$

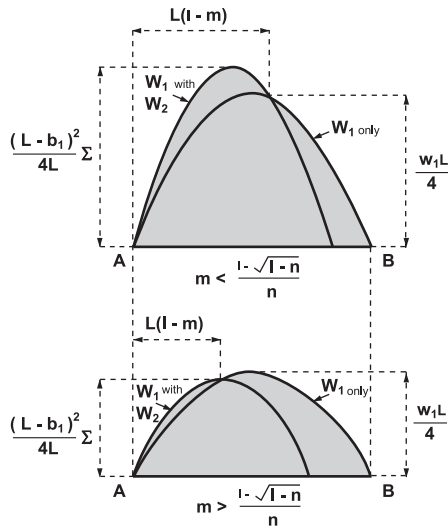
$$M_{\max} \text{ under } W_1 = \frac{(L-b_1-X)X}{L}$$

$$L(1-n) \leq X$$

$$M_{\max} \text{ under } W_2 = \frac{(X-b_2)(L-X)}{L}$$

I-BEAMS

Simply supported beam



Two concentrated moving loads (continued)

If $m > n$, the Maximum Bending Moment at any section always occurs under W_1 (the heavier load), whether W_2 is on or off the span.

For a section distance X from A:

$$X \leq L(1 - m) \quad : \quad L(1 - m) \leq X$$

$$M_{\max} = \frac{(L - b_1 - X)X}{L} \quad \Sigma \quad : \quad M_{\max} = \frac{(L - X)X}{L} W_1$$

If $n < m < \frac{L - \sqrt{L - n}}{n}$ the Absolute Maximum Bending

Moment occurs under W_1 with W_2 on the span.

If $n < m > \frac{L - \sqrt{L - n}}{n}$ the Absolute Maximum Bending

Moment occurs under W_1 at mid-span with W_2 off the span.

Note: When the two loads are equal ($W_1 = W_2$ and $n = 1/2$)

the critical value of $\frac{L - \sqrt{L - n}}{n} = 0.5858$.

Simply supported beams carrying several moving concentrated loads

The Maximum Reaction and the Maximum Shear due to several moving concentrated loads occur at one support with one of the loads at that support. The location producing the Absolute Maximum must be found by trial.

The Maximum Bending Moment due to several moving concentrated loads occurs under one of the loads when that load and the gravity centre of all loads are equidistant from mid-span. The Absolute Maximum must be determined by trial.

STRUCTURE PROPERTIES

FORMULAS FOR GEOMETRICAL PROPERTIES OF SECTIONS

Cross-section area

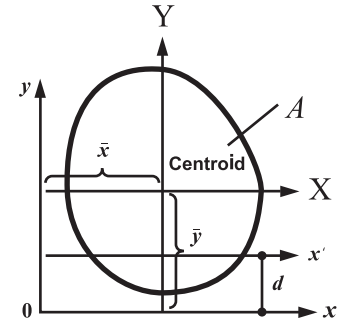
$$A = \int_A dA$$

First moment

$$S_x = \int_A y dA, S_y = \int_A x dA$$

Position of centroid

$$\bar{x} = \frac{S_y}{A}, \bar{y} = \frac{S_x}{A}$$



Moment of inertia

$$J_x = \int_A y^2 dA, J_y = \int_A x^2 dA$$

Product of inertia

$$J_{xy} = \int_A xy dA$$

Polar moment of inertia

$$J_p = \int_A r^2 dA = J_x + J_y$$

Radius of gyration

$$i_x = \sqrt{\frac{J_x}{A}}, i_y = \sqrt{\frac{J_y}{A}}$$

Polar radius of gyration

$$i_p = \sqrt{\frac{J_p}{A}}$$

Section modulus

$$Z_1 = \frac{J}{y_1}, Z_2 = \frac{J}{y_2}$$

Transition of axis

$$S_x' = S_x - Ad, S_y' = Ay - Ad$$

$$J_x' = J_x + Ay^2, J_y' = J_y + Ax^2, J_{xy}' = J_{xy} + xAy$$

Rotation of axis

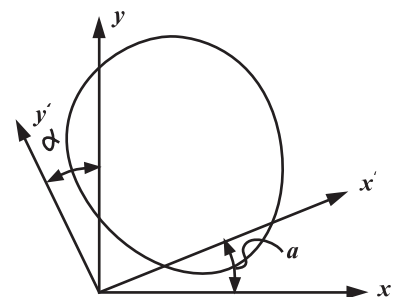
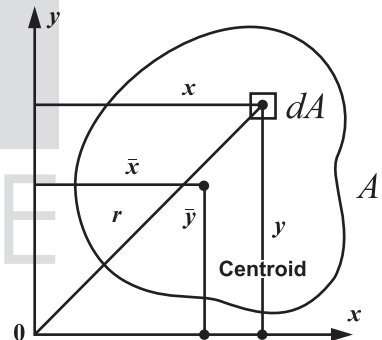
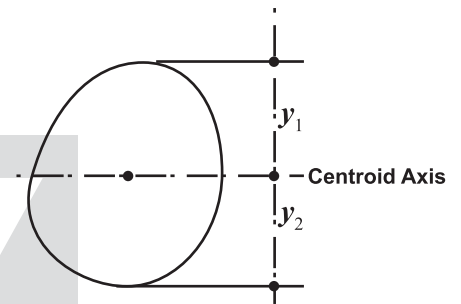
$$S_x' = S_x \cos \alpha - S_y \sin \alpha$$

$$J_x' = J_x \sin^2 \alpha + J_y \cos^2 \alpha + J_{xy} \sin 2\alpha$$

$$J_y' = J_x \cos^2 \alpha + J_y \sin^2 \alpha - J_{xy} \sin 2\alpha$$

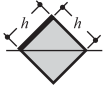
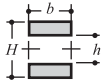
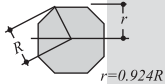
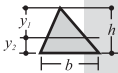
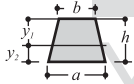
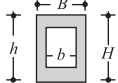
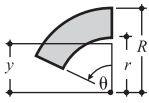
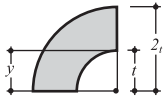
$$J_{xy}' = (J_y - J_x) \sin \alpha \cos \alpha + J_{xy} \cos 2\alpha$$

$$J_x' + J_y' = J_x + J_y = J_p$$

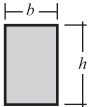
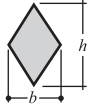
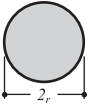
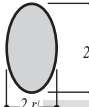
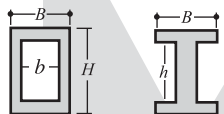
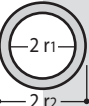


I-BEAMS

GEOMETRICAL PROPERTIES OF SECTIONS

Section	Centroid axis from edge: y cm	Moment of inertia: J cm ⁴	Section modulus: Z cm ³
	$\frac{h}{2}$	$\frac{bh^3}{12}$	$\frac{bh^2}{6}$
	$\frac{h}{\sqrt{2}}$	$\frac{h^4}{12}$	$\frac{\sqrt{2}h^3}{12}$
	$\frac{H}{2}$	$\frac{b}{12} (H^3 - h^3)$	$\frac{b}{6H} (H^3 - h^3)$
	$\frac{D}{2}$	$\frac{\pi D^4}{64}$	$\frac{\pi D^3}{32}$
	0.924 R	0.6381 R ⁴	0.6906 R ³
	$y_1 = \frac{2h}{3}$ $y_2 = \frac{h}{3}$	$\frac{bh^3}{36}$	$Z_1 = \frac{bh^2}{24}$ $Z_2 = \frac{bh^2}{12}$
	$y_1 = \frac{1}{3} \frac{2a+b}{a+b} h$	$\frac{a^2+4ab+b^2}{36(a+b)} h^3$	$Z_1 = \frac{a^2+4ab+b^2}{12(2a+b)} h^2$
	$\frac{H}{2}$	$\frac{BH^3 - bh^3}{12}$	$\frac{BH^3 - bh^3}{6H}$
	$R = \frac{D}{2}$	$\frac{\pi}{4} (R^4 - r^4)$	$\frac{\pi}{4} \cdot \frac{R^4 - r^4}{R}$ Around Upper Section
	$\frac{2(R^3 - r^3)}{3(R^2 - r^2)} \cdot \frac{\sin \theta}{\theta}$	$\frac{R^4 - r^4}{8} (\theta + \sin \theta \cdot \cos \theta)$ $\frac{-2(R^3 - r^3)^2}{9(R^2 - r^2)} \cdot \frac{\sin^2 \theta}{\theta}$	Around Upper Section $Z = \frac{J}{R - y}$ Around Lower Section $Z = \frac{J}{y - r \cos \theta}$
	0.9903 t	0.6364 t ⁴	Around Upper Section 0.6303 t ³

DISTRIBUTION DIAGRAM OF SHEAR STRESS IN BEAM SECTIONS

Section	Shear stress diagram	$T_{\max}$
		$\tau_{\max} = \frac{3Q}{2bh} = \frac{3}{2} \frac{Q}{A}$
		$\tau_{\max} = \frac{9}{4} \frac{Q}{bh} = \frac{9}{8} \frac{Q}{A}$
		$\tau_{\max} = \frac{4}{3} \frac{Q}{\pi r^2} = \frac{4}{3} \frac{Q}{A}$
		$\tau_{\max} = \frac{4}{3} \frac{Q}{\pi r r^1} = \frac{4}{3} \frac{Q}{A}$
		$\tau_{\max} = \frac{3}{2} \frac{(BH^2 - bh^2)(BH - bh)}{2(BH^3 - bh^3)(B - b)} \frac{Q}{A}$
		$\tau_{\max} = \frac{4(r_2^2 + r_2 r_1 + r_1^2)}{3(r_2^2 + r_1^2)} \frac{Q}{A}$

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STRESS AND DEFLECTION OF BEAMS

No.	Load condition	Bending moment	Reaction force and shear stress	Deflection
1		 $M_x = Px$ $M_{max} = Pa$	 $R_A = P$	 $dc = \frac{Pa^3}{3EI}$ $d_{max} = \frac{Pa^3}{3EI} (1 + \frac{3b}{2a})$
2		 $M_{max} = M_x = M_r$	$R_A = 0$	 $dc = \frac{Ma^2}{2EI}$ $d_{max} = \frac{Ma^2}{2EI} (1 + \frac{2b}{a})$
3		 $M_x = \frac{Wx^2}{2a}$ $M_{max} = \frac{Wa}{2}$	 $R_A = W$	 $dc = \frac{wa^3}{8EI}$ $d_{max} = \frac{wa^3}{8EI} (1 + \frac{4b}{3a})$
4		$M_{max} = W(a + \frac{b}{2})$	 $R_A = W$	 $d_{max} = \frac{w}{24EI} \times (8a^3 + 18a^2b + 12ab^2 + 3b^3 + 12a^2c + 12abc + 4b^3c)$
5		$M_x = \frac{Wx^2}{3a^2}$ $M_A = \frac{Wa}{3}$	 $R_A = W$	 $dc = \frac{Wa^3}{15EI}$ $d_{max} = \frac{Wa^3}{15EI} (1 + \frac{5b}{4a})$
6		$M_{max} = W(a + \frac{2b}{3})$	 $R_A = W$	 $d_{max} = \frac{W(20a^3 + 50a^2b + 40ab^2 + 11b^3)}{60EI}$
7		 $M_{max} = \frac{Pl}{4}$	 $R_A = R_B = \frac{P}{2}$	 $d_{max} = \frac{Pl^3}{48EI}$
8		 $M_{max} = \frac{Pab}{l}$	 $R_A = Pb/l$ $R_B = Pa/l$	 $d_{center} = \frac{Pl^3}{48EI} [\frac{3a}{l} - 4(\frac{a}{l})^3]$ d_{max} takes place in the range of 0.0774l from the span center.
9		$M_C = \frac{Pa(b+2c)}{l}$ $M_D = \frac{Pc(b+2a)}{l}$	 $R_A = \frac{P(b+2c)}{l}$ $R_B = \frac{P(b+2a)}{l}$	Deflection of span center is calculated by applying formula given in No.8 to each load.
10		$M_{max} = \frac{Pl}{3}$	 $R_A = R_B = P$	 $d_{max} = \frac{23Pl^3}{648EI}$

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, in-case of a beam subject to uniformly distributed load, w, W equals $w\ell$, where ℓ is span length.

No.	Load condition	Bending moment	Reaction force and shear stress	Deflection
11		$M_C = M_E = \frac{Pl}{4}$ $M_D = \frac{5Pl}{12}$	$R_A = R_B = \frac{3P}{2}$	$d_{max} = \frac{53Pl^3}{1296EI}$
12		$M_C = M_E = \frac{3Pl}{8}$ $M_D = \frac{Pl}{2}$	$R_A = R_B = \frac{3P}{2}$	$d_{max} = \frac{19Pl^3}{348EI}$
13		$M_x = \frac{Wx}{2} (l - \frac{x}{l})$ $M_{max} = \frac{Wl}{8}$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{5}{348} \cdot \frac{Wl^3}{EI}$
14		$M_{max} = \frac{W}{b} (\frac{x_1^2}{2} - a^2)$ where $x_1 = a + \frac{Rab}{W}$	$R_A = \frac{W}{l} (\frac{b}{2} + c)$ $R_B = \frac{W}{l} (\frac{b}{2} + a)$	$d_{max} = \frac{W}{384EI} (8l^3 + 4lb^2b^3)$ where a = c
15		$M_{max} = \frac{Wa}{4}$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{Wa(3l^3 - 2a^2)}{96EI}$
16		$M_x = \frac{Wx}{3} (l - \frac{x^2}{l^2})$ $M_{max} = 0.128Wl$ where $x_1 = 0.5774l$	$R_A = \frac{W}{3}$ $R_B = \frac{2W}{3}$	$d_{max} = \frac{0.01304 Wl^3}{EI}$ where $x = 0.5193l$
17		$M_x = Wx (\frac{l}{2} - \frac{2x^2}{3l})$ $M_{max} = Wl/6$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{Wl^3}{60EI}$
18		$M_{max} = \frac{W}{4} (l - \frac{b}{3})$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{W}{480EI} (8l^3 + 7al^2 + 4a^2l - 4a^3)$
19		$M_{max} = \frac{Wa}{6}$	$R_A = R_B = \frac{W}{2}$	$d_{max} = \frac{Wa}{240EI} (18a^3 + 20ab + 5b^3)$
20		$-M_A = -M_B = M_C = \frac{Pl}{8}$	$R_A = R_B = \frac{P}{2}$	$d_{max} = \frac{Pl^3}{192EI}$

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, incase of a beam subject to uniformly distributed load, w, W equals wℓ, where ℓ is span length.

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No.	Load condition	Bending moment	Reaction force and shear stress	Deflection
21		$M_A = -\frac{Pab^2}{l^2}$ $M_B = -\frac{Pa^2b}{l^2}$ $M_C = \frac{2Pa^2b^2}{l^3}$	$R_A = P \left(\frac{b}{l}\right)^2 \left(1+2\frac{a}{l}\right)$ $R_B = P \left(\frac{a}{l}\right)^2 \left(1+2\frac{b}{l}\right)$	$d_{max} = \frac{Pa^3b^3}{3EI l^3}$ $d_{max} = \frac{2Pa^2b^3}{3EI(3l-2a)} \times \frac{l^2}{3l-2a}$
22		$M_A = M_B = -2Pl/9$ $M_C = M_D = Pl/9$	$R_A = R_B = P$	$d_{max} = \frac{5Pl^3}{648EI}$
23		$M_A = M_B = -19Pl/72$ $M_D = 11Pl/72$	$R_A = R_B = 3P/2$	$d_{max} = \frac{41Pl^3}{5184EI}$
24		$M_A = M_B = -5Pl/16$ $M_D = 3Pl/16$	$R_A = R_B = 3P/2$	$d_{max} = \frac{Pl^3}{96EI}$
25		$M_A = M_B = -\frac{Wl^2}{12}$ $M_C = \frac{Wl^2}{24}$	$R_A = R_B = W/2$	$d_{max} = \frac{Wl^3}{384EI}$
26		$M_A = -\frac{W}{12Pb} [e^3(4l-3e)-c^3(4l-3c)]$ $M_B = -\frac{W}{12Pb} [d^3(4l-3d)-a^3(4l-3a)]$	$R_A = r_A + \frac{M_A - M_B}{l}$ $R_B = r_B + \frac{M_B - M_A}{l}$ <i>r indicates the reaction force of the single span.</i>	$d_{max} = \frac{W}{384EI} (l^3 + 2l^2a + 4la^2 - 8a^3)$ where $a = c$
27		$M_X = -\frac{Wl}{30} \left(\frac{10x^3}{b^3} - \frac{9x}{l} + 2 \right)$ where $+M_{max} = Wl/23.3$, $x = 0.55l$ $M_A = -Wl/15$, $M_B = -Wl/10$	$R_A = 0.3W$, $R_B = 0.7W$	$d_{max} = \frac{Wl^3}{382EI}$ where $x_1 = 0.525l$
28		$M_A = M_B = -\frac{5Wl}{48}$ $M_C = Wl/16$	$R_A = R_B = W/2$	$d_{max} = \frac{1.4Wl^3}{384EI}$
29		$M_A = M_B = -\frac{W}{48l} x$ ($5l^2 + 4al - 4a^2$)	$R_A = R_B = W/2$	$d_{max} = \frac{W}{1920EI} (7l^3 + 8al^2 + 4a^2l - 16a^3)$
30		$M_A = M_B = -\frac{Wa}{12l} (2l-a)$	$R_A = R_B = W/2$	$d_{max} = \frac{Wa^2}{480EI} (5l-4a)$

In case of beam subject to distributed loads, load symbol, W, indicates the total load of the beam. For example, in case of a beam subject to uniformly distributed load, w, W equals $w\ell$, where ℓ is span length.

WORKING STRESSES

The values given in the safe load tables are based on the allowable stresses in BS 449 as follows:

Thickness of material (mm)	Grade of steel	Allowable stresses (N/mm ²)		
		Bending	Shear*1	Bearing
Up to and including 40	43	165	100	190
Over 40		150	90	
Up to and including 65	50	230	140	260
Over 65		$Y_g/1.52^{*2}$		
Up to and including 40	55	280	170	320
Over 40		260	160	

Note: *1 On unstiffened web.

*2 Y_g = yield stress, to be agreed with the manufacturer and not to exceed 350N/mm². This applies only to universal column sections and plated compound sections and, in this handbook, affects certain universal column stanchions only for which a value of $Y_g = 325\text{N/mm}^2$ has been assumed.

Web buckling

The buckling values for the unstiffened webs have been calculated in accordance with the formula in Clause 28a(i) of BS 449, namely $W = p_c t B$ where

- W = allowable reaction or concentrated load in N, but tabulated in kN.
- p_c = allowable axial stress for struts as given in BS 449, Clause 30a for a slenderness ratio of $(d/3)/t$, in which
- d = clear depth of web between root fillets
- t = web thickness
- B = $\frac{1}{2}D + t_p + L_b$ for end bearings of simply supported beams
- = $D + 2t_p + L_b$ for intermediate bearings, in which
- D = overall depth of beam
- t_p = thickness of bearing and/or flange plate
- L_b = length of stiff portion of bearing which shall not be taken as greater than half depth of beam for simply supported beams and the full depth of beam for an intermediate bearing over which the beam is continuous, unless the web is stiffened.

Web crushing

To prevent local crushing of the web, the direction bearing stress at the root of the web is limited to the values given in BS 449, and included in the table above.

The length of web resisting crushing is determined on the assumption that the load is dispersed through the flange and the bearing and/or flange plate at an angle of 30° (BS 449, Clause 27e).

COMBINED STRESSES

Where bending and shear stresses, or bearing, bending and shear stresses, are co-existent, the beam should be checked in accordance with BS 449, Clauses 14.c and d.

DEFLECTION COEFFICIENTS

Clause 15 to BS 449 includes the requirement that the maximum deflection due to loads other than the weight of the structural floors or roof, steelwork and casing, if any, shall not exceed 1/360th of the span. For a simply supported beam, the uniformly distributed load W_D in kN to produce this deflection is:

$$W_D = \frac{384EI}{5 \times 360L^2} \times \frac{Cl}{1000}$$

where C = deflection coefficient tabulated for span L in m
 E = modulus of elasticity = 21000kN/cm² (=210000N/mm²)
 I = moment of inertia of beam, in cm⁴

The load W_D will be less than the tabular load if the span exceeds 12.17, 16.97 or 10.00 times the beam depth for grades 50, 43 or 55 steel respectively. For such cases, it may be necessary to confirm that the total load is within the capacity of the beam and that the loads to be considered for deflection purposes do not exceed W_D .

The table below gives limiting values of the span/depth ratio for uniformly loaded simply supported beams for different grades of steel for various ratios of W_d/W_t where W_d = load considered for deflection purposes and W_t = total load on beam. If the appropriate span/ depth ratio is exceeded, then the relevant deflection will exceed 1 /360th of the span unless the bending stress is reduced.

Material	Limiting values of span to depth ratio for W_d/W_t					
	1.0	0.9	0.8	0.7	0.6	0.5
Grade 50	12.17	13.53	15.22	17.39	20.29	24.35
Grade 43	16.97	18.86	21.21	24.24	28.28	33.94
Grade 55	10.00	11.11	12.50	14.29	16.67	20.00

SAFE LOADS (for as-rolled sections)

These are the safe uniformly distributed loads (including self weight) which can be carried by simply supported beams with adequate lateral restraint to the compression flange. They have therefore generally been calculated in accordance with the allowable bending stresses given in BS 449.

The loads are printed in three different type faces to draw attention to particular criteria which may affect the load-carrying capacity of a beam. Bold, italic and ordinary type faces have been used in that sequence for maximum clarity and the significance of each is explained below.

Bold face. Loads printed in this type are greater than the web buckling capacity of the beam (UB, joist or channel) alone. If sufficient additional capacity is not provided by the bearing, web stiffeners will be necessary to realise the full web capacity.

Italic face. Loads printed in this type are within the buckling capacity of the unstiffened web and produce a maximum deflection of less than 1/360th of the span.

Ordinary face. Loads printed in this type produce a maximum deflection exceeding 1/360th of the span. Such cases should therefore be examined as indicated above under 'Deflection coefficients'.

In addition to the criteria covered by the use of different type faces, two other matters require comment.

- Shear capacity. Where the shear capacity of the unstiffened web is less than the bending capacity of a beam, the safe load is calculated on the allowable average shear stress given in BS 449.
- Web crushing. Many of the loads tabulated, in the various type faces, exceed the beam component of the direct bearing capacity of the web at its junction with the flange. The designer should therefore ensure that the length of stiff bearing and thickness of flange plate or packing (if any) provide a sufficient additional length of web in bearing.

BEAMS WITHOUT LATERAL SUPPORT

The allowable compressive stress due to bending about the x-x axis for universal beams and columns, joists and channels is the lesser of the values of p_{bc} given in Tables 2 and 3 a, b or c (depending on the grade of steel) in BS 449. The values in Table 2 apply when the compression flange is so supported that lateral instability is obviated. When the compression flange is unsupported or insufficiently supported laterally, the appropriate Table 3 applies. The maximum effective length of compression flange which may be used without reduction of the allowable bending stresses in Table 2 is given as L_c with the safe load tables.

Example: Find the allowable stress, p_{bc} , and the safe uniformly distributed load, W , for a 533 = 210 UB 82 in grade 43 steel spanning 7m without lateral restraint and having the ends of the compression flange partially restrained.

Effective length = $0.85 \times 700 = 595$ cm (see BS 449, Clause 26)

$l/r_y 595/4.38 = 136$; $D/T = 40$.

From BS 449, Table 3a, $p_{bc} = 119 \text{ N/mm}^2$

The safe uniformly distributed load, W

$$\begin{aligned}
 &= 8fZ/L \\
 &= (8 \times 119 \times 1798 \times 10^3 / 7000) \text{ N} \\
 &= 245 \text{ kN}
 \end{aligned}$$

Lateral instability is not a criterion for these sections when bent about the y-y axis and the allowable stresses in Table 2 apply.



I-BEAMS

STRESS AND SAFE LOAD TABLES

Table 3a of BS 449. Allowable stress p_{bc} in bending (N/mm²) for beams of Grade 43 steel

l/r_y	D/T							
	10	15	20	25	30	35	40	50
90	165	165	165	165	165	165	165	165
95	165	165	165	163	163	163	163	163
100	165	165	165	157	157	157	157	157
105	165	165	160	152	152	152	152	152
110	165	165	156	147	147	147	147	147
115	165	165	152	141	141	141	141	141
120	165	162	148	136	136	136	136	136
130	165	155	139	126	126	126	126	126
140	165	149	130	115	115	115	115	115
150	165	143	122	104	104	104	104	104
160	163	136	113	95	94	94	94	94
170	159	130	104	91	85	82	82	82
180	155	124	96	87	80	76	72	71
190	151	118	93	83	77	72	68	62
200	147	111	89	80	73	68	64	59
210	143	105	87	77	70	65	61	55
220	139	99	84	74	67	62	58	52
230	134	95	81	71	64	59	55	49
240	130	92	78	69	61	56	52	47
250	126	90	76	66	59	54	50	44
260	122	88	74	64	57	52	48	42
270	118	86	72	62	55	50	46	40
280	114	84	70	60	53	48	44	39
290	110	82	68	58	51	46	42	37
300	106	80	66	56	49	44	41	36

Intermediate values may be obtained by linear interpolation.

Note : For materials over 40 mm thick the stress shall not exceed 150 N/mm².

AXIAL STRESSES

The safe loads on stanchions and struts have been calculated in accordance with the allowable axial stresses specified in BS 449, Clause 30, and shown in the tables that follow :

I-BEAMS

DESIGN OF STANCHIONS AND STRUTS

Table 17a of BS 449. Allowable stress p_c on gross section for axial compression for Grade 43 steel

l/r	p_c (N/mm ²) for Grade 43 steel									
	0	1	2	3	4	5	6	7	8	9
0	155	155	154	154	153	153	153	152	152	151
10	151	151	150	150	149	149	148	148	148	147
20	147	146	146	146	145	145	144	144	144	143
30	143	142	142	142	141	141	141	140	140	139
40	139	138	138	137	137	136	136	136	135	134
50	133	133	132	131	130	130	129	128	127	126
60	126	125	124	123	122	121	120	119	118	117
70	115	114	113	112	111	110	108	107	106	105
80	104	102	101	100	99	97	96	95	94	92
90	91	90	89	87	86	85	84	83	81	80
100	79	78	77	76	75	74	73	72	71	70
110	69	68	67	66	65	64	63	62	61	61
120	60	59	58	57	56	56	55	54	53	53
130	52	51	51	50	49	49	48	48	47	46
140	46	45	45	44	43	43	42	42	41	41
150	40	40	39	39	38	38	38	37	37	36
160	36	35	35	35	34	34	33	33	33	32
170	32	32	31	31	31	30	30	30	29	29
180	29	28	28	28	28	27	27	27	26	26
190	26	26	25	25	25	25	24	24	24	24
200	24	23	23	23	23	22	22	22	22	22
210	21	21	21	21	21	20	20	20	20	20
220	20	19	19	19	19	19	19	18	18	18
230	18	18	18	18	17	17	17	17	17	17
240	17	16	16	16	16	16	16	16	16	15
250	15									
300	11									
350	8									

Intermediate values may be obtained by linear interpolation.

Note : For material over 40 mm thick, other than rolled I-beams or channels, and for universal columns of thicknesses exceeding 40 mm, the limiting stress is 140 N/mm².

UNIVERSAL BEAMS AND COLUMNS

SAFE LOADS FOR GRADE 43 STEEL

Section size	Mass per metre	Safe distributed loads in kilonewtons for spans in metres and deflection coefficients													Critical span	Lc
mm	kg	2.00 112.0	2.50 71.68	3.00 49.78	3.50 36.57	4.00 28.00	4.50 22.12	5.00 17.92	5.50 14.81	6.00 12.44	7.00 9.143	8.00 7.000	9.00 5.531	10.00 4.480		m
203 x 133	25	153	122	102	87	77	68	61	56	51						2.80
	30	184	147	123	105	92	82	74	67	61	53					3.03
254 x 102	22	149	119	99	85	74	66	60	54	50	43	37				1.75
	25	175	140	117	100	88	78	70	64	58	50	37				1.87
	28	203	163	135	116	102	90	81	74	68	58	51				2.01
254 x 146	31	233	186	155	133	117	104	93	85	78	67	58				2.96
	37	286	229	191	164	143	127	115	104	95	82	72				3.22
	43	333	267	222	191	167	148	133	121	111	95	83				3.41
305 x 102	25	190	152	127	109	95	84	76	69	63	54	47	42	38		1.64
	28	232	185	154	132	116	103	93	84	77	66	58	51	46		1.79
	33	274	219	183	156	137	122	110	100	91	78	68	61	55		1.90
305 x 127	37	311	249	207	178	156	138	124	113	104	89	78	69	62		2.37
	42	351	280	234	200	175	156	140	127	117	100	88	78	70		2.45
	48	404	323	269	231	202	180	162	147	135	115	101	90	81		2.59
305 x 165	40	370	296	247	212	185	165	148	135	123	106	93	82	74		3.38
	46	412†	342	285	244	214	190	171	155	143	122	107	95	86		3.53
	54	479†	398	331	284	249	221	199	181	166	142	124	110	99		3.69
356 x 127	33	311	248	207	177	155	138	124	113	104	89	78	69	62		2.18
	39	377	302	252	216	189	168	151	137	126	108	94	84	75		2.33
356 x 171	45	453	363	302	259	227	201	181	165	151	130	113	101	91		3.23
	51	519†	420	350	300	263	234	210	191	175	150	131	117	105		3.38
	57	574†	473	394	338	296	263	237	215	197	169	148	131	118		3.50
	67	662†	567	472	405	354	315	283	258	236	202	177	157	142		3.72
406 x 140	39	414	331	276	236	207	184	165	150	138	118	103	92	83		2.41
	46	513	411	342	293	257	228	205	187	171	147	128	114	103		2.58
406 x 178	54	611	489	407	349	305	271	244	222	204	174	153	136	122		3.28
	60	634†	559	466	399	349	310	279	254	233	200	175	155	140		3.45
	67	721†	627	523	448	392	348	314	285	261	224	196	174	157		3.54
	74	801†	699	583	499	437	388	350	318	291	250	218	194	175		3.65
457 x 152	52	626	501	418	358	313	278	251	228	209	179	157	139	125		2.62
	60	728†	591	493	422	370	329	296	269	246	211	185	164	148		2.78
	67	825	660	550	471	413	367	330	300	275	236	206	183	165		2.81
	74	913†	742	619	530	464	412	371	337	309	265	282	206	186		2.92
	82	995†	822	685	587	514	457	411	374	343	294	257	228	206		3.03
457x191	67	771†	684	570	489	428	380	342	311	285	244	214	190	171		3.53
	74	832†	771	643	551	482	429	386	351	321	276	241	214	193		3.65
	82	911†	851	709	608	532	473	426	387	355	304	266	236	213		3.74
	89	983†	935	779	668	584	519	467	425	389	334	292	260	234		3.86
	98	1066†	1033	861	738	645	574	516	469	430	369	323	287	258		4.00

Loads printed in italic type do not cause overloading of the unstiffened web, and do not cause deflection exceeding span/360.

Loads printed in ordinary type should be checked for deflection

† Load is based on allowable shear of web and is less than allowable load in bending.

UNIVERSAL BEAMS AND COLUMNS

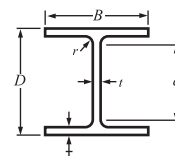
SAFE LOADS FOR GRADE 43 STEEL

Section size	Mass per metre	Safe distributed loads in kilonewtons for spans in metres and deflection coefficients													Critical span L _c m
		4.00 28.00	5.00 17.92	6.00 12.44	7.00 9.143	8.00 7.000	9.00 5.531	10.00 4.480	11.00 3.702	12.00 3.111	13.00 2.651	14.00 2.286	15.00 1.991	16.00 1.750	
533 x 210	82	593	475	396	339	297	264	237	216	198	183	170	158	148	3.70
	92	685	548	457	391	343	304	274	249	228	211	196	183	171	3.88
	101	758	607	506	433	379	337	303	276	253	233	217	202	190	3.99
	109	816	653	544	467	408	363	327	297	272	251	233	218	204	4.07
	122	924	739	616	528	462	411	369	336	308	284	264	246	231	4.24
610 x 229	101	830	664	553	474	415	369	332	302	277	255	237	221	207	4.01
	113	950	760	633	543	475	422	380	345	317	292	271	253	238	4.18
	125	1063	851	709	608	532	473	425	387	354	327	304	284	266	4.33
	140	1197	957	798	684	598	532	479	435	399	368	342	319	299	4.48
610 x 305	149	1350	1080	900	771	675	600	540	491	450	415	386	360	337	6.10
	179	1621	1297	1080	926	810	720	648	589	540	499	463	432	405	6.40
	238	2164	1732	1443	1237	1082	962	866	787	721	666	618	577	541	7.03
686 x 254	125	1149	919	766	656	574	511	459	418	383	353	328	306	287	4.41
	140	1316	1053	877	752	658	585	526	479	439	405	376	351	329	4.60
	152	1443	1154	962	824	721	641	577	525	481	444	412	385	361	4.73
	170	1621	1297	1080	926	810	720	648	589	540	499	463	432	405	4.88
762 x 267	147	1479	1184	986	845	740	658	592	538	493	455	423	395	370	4.52
	173	1777	1422	1185	1015	889	790	711	646	592	547	508	474	444	4.77
	197	2057	1646	1371	1176	1029	914	823	748	686	633	588	549	514	5.01
838 x 292	176	1945	1556	1297	1111	973	864	778	707	648	598	556	519	486	4.94
	194	2194	1755	1463	1254	1097	975	878	798	731	675	627	585	548	5.14
	226	2635	2108	1757	1506	1318	1171	1054	958	879	811	753	703	659	5.45
914 x 305	201	2379	1903	1586	1360	1190	1057	952	865	793	732	680	634	595	5.07
	224	2725	2180	1817	1557	1363	1211	1090	991	908	839	779	727	681	5.32
	253	3137	2510	2092	1793	1569	1394	1255	1141	1046	965	896	837	784	5.55
	289	3594	2875	2396	2054	1797	1597	1438	1307	1198	1106	1027	958	899	5.75
914 x 419	343		3536†	3019	2588	2264	2013	1811	1647	1509	1393	1294	1208	1132	8.39
	388		3958†	3436	2945	2577	2290	2061	1874	1718	1586	1472	1374	1288	8.73
686 x 254	125	1146	917	764	655	573	509	458	417	382	353	327	306	286	4.20
	140	1313	1050	875	750	657	584	525	477	438	404	375	350	328	4.43
	152	1440	1152	960	823	720	640	576	524	480	443	411	384	360	4.57
	170	1618	1294	1078	924	809	719	647	588	539	498	462	431	404	4.73
762 x 267	147	1475	1180	984	843	738	656	590	537	492	454	422	393	369	4.33
	173	1773	1419	1182	1013	887	788	709	645	591	546	507	473	443	4.61
	197	2054	1643	1369	1173	1027	913	821	747	685	632	587	548	513	4.86
838 x 292	176	1940	1552	1293	1109	970	862	776	705	647	597	554	517	485	4.72
	194	2189	1751	1459	1251	1094	973	876	796	730	674	625	584	547	4.94
	226	2630	2104	1754	1503	1315	1169	1052	957	877	809	752	701	658	5.29
914 x 305	201	2373	1899	1582	1356	1187	1055	949	863	791	730	678	633	593	4.86
	224	2720	2176	1813	1554	1360	1209	1088	989	907	837	777	725	680	5.14
	253	3132	2505	2088	1789	1566	1392	1253	1139	1044	964	895	835	783	5.39
	289	3588	2871	2392	2051	1794	1595	1435	1305	1196	1104	1025	957	897	5.60
914 x 419	343		3536†	3012	2582	2259	2008	1807	1643	1506	1390	1291	1205	1130	8.08
	388		3958†	3429	2939	2572	2286	2057	1870	1714	1583	1470	1372	1286	8.44

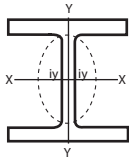
The loads listed are based on a bending stress of 165 N/mm² and assume adequate lateral support. Without such support the span must not exceed L_c unless the compressive stress is reduced in accordance with clause 19a of BS 449.

Loads printed in bold type may cause overloading of the unstiffened web, the capacity of which should be checked.

UNIVERSAL BEAMS AND COLUMNS



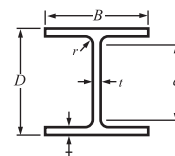
Section size	Unit weight M		Section area A		Section depth D		Flange width B		Flange thickness T		Web thickness t		Corner radius r	
in (mm)	lb/ft	kg/m	in²	cm²	in	mm	in	mm	in	mm	in	mm	in	mm
W4 4 x 4 (102 x 102)	13	19.35	3.83	24.71	4.16	105.7	4.06	103.1	0.345	8.76	0.28	7.11	0.25	6.35
W5 5 x 3 (127 x 76)	9	13.00	2.60	16.80	5.00	127.0	3.00	76.2	0.299	7.60	0.165	4.20	0.30	6.35
W5 5 x 5 (127 x 127)	16	23.81	4.71	30.39	5.01	127.3	5.00	127.0	0.36	9.14	0.24	6.10	0.30	7.62
	19	28.28	5.56	35.87	5.15	130.8	5.03	127.8	0.43	10.90	0.27	6.86	0.30	7.62
W6 6 x 3½ (152 x 89)	11	16.00	3.18	20.50	6.00	152.4	3.50	88.9	0.303	7.70	0.181	4.60	0.25	6.35
W6 6 x 4 (152 x 102)	9	13.39	2.68	17.29	5.90	149.9	3.94	100.1	0.215	5.46	0.17	4.32	0.25	6.35
	12	17.86	3.55	22.90	6.03	153.2	4.00	101.6	0.28	7.11	0.23	5.84	0.25	6.35
	16	23.81	4.74	30.58	6.28	159.5	4.03	102.4	0.405	10.20	0.26	6.60	0.25	6.35
W6 6 x 6 (152 x 152)	15	22.50	4.43	28.58	5.99	152.1	5.99	152.1	0.26	6.60	0.23	5.84	0.25	6.35
	15.7	23.00	4.62	29.80	6.00	152.4	6.00	152.4	0.269	6.80	0.24	6.10	0.25	6.35
	20	30.00	5.87	37.87	6.20	157.5	6.02	152.9	0.365	9.27	0.26	6.60	0.25	6.35
	25	37.20	7.34	47.35	6.38	162.1	6.08	154.4	0.455	11.50	0.32	8.13	0.25	6.35
W7 7 x 4 (178 x 102)	13	19.00	3.75	24.20	7.00	177.8	4.00	101.6	0.311	7.90	0.185	4.70	0.30	7.62
W8 8 x 4 (203 x 102)	10	14.88	2.96	19.10	7.89	200.4	3.94	100.1	0.205	5.21	0.17	4.32	0.30	7.62
	13	19.35	3.84	24.77	7.99	202.9	4.00	101.6	0.255	6.48	0.23	5.84	0.30	7.62
	15	22.32	4.44	28.65	8.11	206.0	4.015	102.0	0.315	8.00	0.245	6.22	0.30	7.62
	15.5	23.00	4.50	29.00	8.00	203.2	4.00	101.6	0.368	9.30	0.205	5.20	0.30	7.62
W8 8 x 5¼ (203 x 133)	17	25.30	5.01	32.30	8.00	203.2	5.25	133.4	0.308	7.80	0.23	5.84	0.30	7.62
	18	26.79	5.26	33.94	8.14	206.8	5.25	133.3	0.33	8.38	0.23	5.84	0.30	7.62
	20	30.00	5.89	38.00	8.14	206.8	5.268	133.8	0.378	9.60	0.248	6.30	0.30	7.62
	21	31.30	6.16	39.74	8.28	210.3	5.27	133.9	0.40	10.10	0.25	6.35	0.30	7.62
W8 8 x 6½ (203 x 165)	24	35.72	7.08	45.68	7.93	201.4	6.495	165.0	0.40	10.10	0.245	6.22	0.40	10.2
	28	41.67	8.25	53.23	8.06	204.7	6.535	166.0	0.465	11.80	0.285	7.24	0.40	10.2
W8 8 x 8 (203 x 203)	31	46.13	9.12	58.84	8.00	203.2	7.995	203.1	0.435	11.00	0.285	7.24	0.40	10.2
	35	52.09	10.3	66.45	8.12	206.2	8.02	203.7	0.495	12.50	0.31	7.87	0.40	10.2
	40	60.00	11.7	75.48	8.25	209.5	8.07	205.0	0.56	14.20	0.36	9.14	0.40	10.2
	48	71.43	14.1	90.97	8.50	215.9	8.11	206.0	0.685	17.40	0.40	10.16	0.40	10.2
	58	86.31	17.1	110.3	8.75	222.2	8.22	208.8	0.81	20.57	0.51	12.95	0.40	10.2
	67	100.0	19.7	127.1	9.00	228.6	8.28	210.3	0.935	23.75	0.57	14.48	0.40	10.2
W10 10 x 4 (254 x 102)	12	17.86	3.54	22.84	9.87	250.7	3.96	100.6	0.21	5.33	0.19	4.83	0.30	7.62
	15	22.32	4.41	28.45	9.99	253.7	4.00	101.6	0.27	6.86	0.23	5.84	0.30	7.62
	17	25.30	4.99	32.19	10.11	256.8	4.01	101.9	0.33	8.38	0.24	6.10	0.30	7.62
	19	28.28	5.62	36.26	10.24	260.1	4.02	102.1	0.395	10.00	0.25	6.35	0.30	7.62
W10 10 x 5¾ (254 x 146)	21	31.25	6.20	40.00	9.90	251.5	5.75	146.1	0.34	8.60	0.24	6.10	0.30	7.62
	22	32.74	6.49	41.87	10.17	258.3	5.75	146.0	0.36	9.14	0.24	6.10	0.30	7.62
	25	37.20	7.36	47.50	10.08	256.0	5.762	146.4	0.43	10.90	0.252	6.40	0.30	7.62
	26	38.69	7.61	49.10	10.33	262.4	5.77	146.6	0.44	11.10	0.26	6.60	0.30	7.62
	29	43.00	8.54	55.10	10.22	259.6	5.799	147.3	0.50	12.70	0.289	7.30	0.30	7.62
	30	44.80	8.85	57.10	10.47	265.9	5.819	147.8	0.51	12.90	0.30	7.62	0.30	7.62
W10 10 x 8 (254 x 203)	33	49.11	9.71	62.64	9.73	247.1	7.96	202.2	0.435	11.00	0.29	7.37	0.50	12.7
	39	58.04	11.5	74.19	9.92	252.0	7.985	202.8	0.53	13.40	0.315	8.00	0.50	12.7
	45	66.97	13.3	85.81	10.10	256.5	8.02	203.7	0.62	15.70	0.35	8.89	0.50	12.7



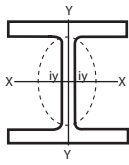
UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	I _x		I _y		i _x		i _y		Z _x		Z _y	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W4 4 x 4 (102 x 102)	11.3	470	3.86	161	1.72	4.37	1.00	2.54	5.46	89.5	1.90	31.1
W5 5 x 3 (127 x 76)	11.5	477	1.30	56	2.10	5.33	0.72	1.83	4.60	75	0.90	15.0
W5 5 x 5 (127 x 127)	21.4	891	7.51	313	2.13	5.41	1.26	3.20	8.55	140	3.00	49.2
	26.3	1,090	9.13	380	2.17	5.51	1.28	3.25	10.2	167	3.63	59.5
W6 6 x 3 1/2 (152 x 89)	20.1	838	2.20	90	2.52	6.40	0.83	2.10	6.70	110	1.20	20.0
W6 6 x 4 (152 x 102)	16.4	683	2.20	91.6	2.47	6.27	0.905	2.30	5.56	91.1	1.11	18.2
	22.1	920	2.99	124	2.49	6.32	0.918	2.33	7.31	120	1.50	24.6
	32.1	1,340	4.43	184	2.60	6.60	0.966	2.45	10.2	167	2.20	36.1
W6 6 x 6 (152 x 152)	29.1	1,210	9.32	388	2.56	6.50	1.45	3.68	9.72	159	3.11	51.0
	30.3	1,263	9.7	403	2.56	6.51	1.45	3.68	10.1	165.7	3.20	52.95
	41.4	1,720	13.3	554	2.66	6.76	1.50	3.81	13.4	220	4.41	72.3
	53.4	2,220	17.1	712	2.70	6.86	1.52	3.86	16.7	274	5.61	91.9
W7 7 x 4 (178 x 102)	32.6	1,357	3.30	138	2.95	7.49	0.940	2.39	9.30	153	1.60	27.0
W8 8 x 4 (203 x 102)	30.8	1,280	2.09	87	3.22	8.18	0.841	2.14	7.81	128	1.06	17.4
	39.6	1,650	2.73	114	3.21	8.15	0.843	2.14	9.91	162	1.37	22.4
	48.0	2,000	3.41	142	3.29	8.36	0.876	2.22	11.8	193	1.70	27.9
	50.2	2,091	3.90	163	3.34	8.49	0.93	2.37	12.6	206	2.00	32.0
W8 8 x 5 1/4 (203 x 133)	56.4	2,348	6.72	280	3.36	8.53	1.16	2.95	14.1	231	2.60	42.6
	61.9	2,580	7.97	332	3.43	8.71	1.23	3.12	15.2	249	3.04	49.8
	69.2	2,981	8.50	354	3.43	8.71	1.20	3.05	17.0	279	3.20	52.4
	75.3	3,130	9.77	407	3.49	8.86	1.26	3.20	18.2	298	3.71	60.8
W8 8 x 6 1/2 (203 x 165)	82.8	3,450	18.3	762	3.42	8.69	1.61	4.09	20.9	342	5.63	92.3
	98.0	4,080	21.7	903	3.45	8.76	1.62	4.11	24.3	398	6.63	109
W8 8 x 8 (203 x 203)	110	4,580	37.1	1,540	3.47	8.81	2.02	5.13	27.5	451	9.27	152
	127	5,290	42.6	1,770	3.51	8.92	2.03	5.16	31.2	511	10.6	174
	146	6,080	49.1	2,040	3.53	8.97	2.04	5.18	35.5	582	12.2	200
	184	7,660	60.9	2,530	3.61	9.17	2.08	5.28	43.3	710	15.0	246
	228	9,490	75.1	3,130	3.65	9.27	2.10	5.33	52.0	852	18.3	300
	272	11,300	88.6	3,690	3.72	9.45	2.12	5.38	60.4	990	21.4	351
W10 10 x 4 (254 x 102)	53.8	2,240	2.18	90.7	3.90	9.91	0.785	1.99	10.9	179	1.10	18.0
	68.9	2,870	2.89	120	3.95	10.0	0.81	2.06	13.8	226	1.45	23.8
	81.9	3,410	3.56	148	4.05	10.3	0.845	2.15	16.2	265	1.78	29.2
	96.3	4,010	4.29	179	4.14	10.5	0.874	2.22	18.8	308	2.14	35.1
W10 10 x 5 3/4 (254 x 146)	106	4,425	9.7	404	4.14	10.5	1.25	3.18	21.5	352	3.40	55.7
	118	4,910	11.4	474	4.27	10.8	1.33	3.38	23.2	380	3.97	65.1
	133	5,545	12.7	529	4.26	10.8	1.31	3.33	26.4	433	4.40	72.1
	144	5,990	14.1	587	4.35	11.0	1.36	3.45	27.9	457	4.89	80.1
	157	6,548	15.2	633	4.29	10.9	1.34	3.40	30.8	505	5.20	85.2
	170	7,080	16.8	699	4.38	11.1	1.38	3.51	32.5	533	5.76	94.4
W10 10 x 8 (254 x 203)	170	7,080	36.6	1,520	4.19	10.6	1.94	4.93	35.0	574	9.20	151
	209	8,700	45.0	1,870	4.27	10.8	1.98	5.03	42.1	690	11.3	185
	248	10,300	53.4	2,220	4.32	11.0	2.01	5.11	49.1	805	13.3	218

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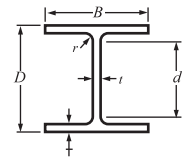
Section size	Unit weight M		Section area A		Section depth D		Flange width B		Flange thickness T		Web thickness t		Corner radius r	
in (mm)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W10 10 x 10 (254 x 254)	49	73.00	14.4	92.9	9.98	253.5	10.00	254.0	0.56	14.20	0.34	8.64	0.50	12.7
	54	80.36	15.8	101.9	10.09	256.3	10.03	254.8	0.615	15.60	0.37	9.40	0.50	12.7
	60	89.29	17.6	113.5	10.22	259.6	10.08	256.0	0.68	17.27	0.42	10.67	0.50	12.7
	68	101.2	20.0	129.0	10.40	264.2	10.13	257.3	0.77	19.56	0.47	11.94	0.50	12.7
	72	107.0	21.2	136.6	10.50	266.7	10.17	258.3	0.808	20.50	0.51	13.00	0.50	12.7
	77	115.0	22.6	145.8	10.60	269.2	10.19	258.8	0.87	22.10	0.53	13.46	0.50	12.7
	88	131.0	25.9	167.1	10.84	275.3	10.265	260.7	0.99	25.15	0.605	15.37	0.50	12.7
	89	132.0	26.2	168.9	10.88	276.4	10.275	261.0	0.998	25.30	0.615	15.60	0.50	12.7
	100	149.0	29.4	189.7	11.11	282.2	10.34	262.6	1.12	28.45	0.68	17.27	0.50	12.7
	112	167.0	32.9	212.3	11.36	288.5	10.415	264.5	1.25	31.75	0.755	19.18	0.50	12.7
W12 12 x 4 (305 x 102)	14	20.83	4.16	26.84	11.91	302.5	3.97	100.8	0.225	5.71	0.20	5.08	0.30	7.62
	16	23.81	4.71	30.39	11.99	304.5	3.99	101.3	0.265	6.73	0.22	5.59	0.30	7.62
	16.5	25.00	4.87	31.40	12.00	304.8	4.00	101.6	0.269	6.80	0.23	5.80	0.30	7.62
	19	28.28	5.57	35.94	12.16	308.9	4.005	101.7	0.35	8.89	0.235	5.97	0.30	7.62
	22	33.00	6.48	41.81	12.31	312.7	4.03	102.4	0.425	10.80	0.26	6.60	0.30	7.62
W12 12 x 5 (305 x 127)	25	37.00	7.36	47.50	11.96	303.8	4.864	123.5	0.421	10.70	0.284	7.20	0.30	7.62
	28	42.00	8.25	53.20	12.07	306.6	4.893	124.3	0.476	12.10	0.313	8.00	0.30	7.62
	32	48.00	9.42	60.80	12.22	310.4	4.93	125.2	0.551	14.00	0.350	8.90	0.30	7.62
W12 12 x 6 1/2 (305 x 165)	26	38.70	7.65	49.35	12.22	310.4	6.49	164.8	0.38	9.65	0.23	5.84	0.30	7.62
	27	40.00	7.98	51.50	11.96	303.8	6.50	165.1	0.40	10.20	0.24	6.10	0.30	7.62
	30	44.64	8.79	56.71	12.34	313.4	6.52	165.6	0.44	11.10	0.26	6.60	0.30	7.62
	31	46.00	9.13	58.90	12.09	307.1	6.525	165.7	0.465	11.80	0.265	6.70	0.30	7.62
	35	52.09	10.3	66.45	12.50	317.5	6.56	166.6	0.52	13.20	0.30	7.62	0.30	7.62
	36	54.00	10.6	68.40	12.24	310.9	6.565	166.8	0.54	13.70	0.305	7.70	0.30	7.62
W12 12 x 8 (305 x 203)	40	59.53	11.8	76.13	11.94	303.3	8.005	203.3	0.515	13.00	0.295	7.49	0.60	15.2
	45	66.97	13.2	85.16	12.06	306.3	8.045	204.3	0.575	14.60	0.335	8.51	0.60	15.2
	50	74.41	14.7	94.84	12.19	309.6	8.08	205.2	0.64	16.20	0.37	9.40	0.60	15.2
W12 12 x 10 (305 x 254)	53	78.87	15.6	100.6	12.06	306.3	9.995	253.9	0.575	14.60	0.345	8.76	0.60	15.2
	58	86.31	17.0	109.7	12.19	309.6	10.01	254.3	0.64	16.20	0.36	9.14	0.60	15.2
W12 12 x 12 (305 x 305)	65	97.0	19.1	123.2	12.12	307.8	12.00	304.8	0.605	15.30	0.39	9.91	0.60	15.2
	72	107.1	21.1	136.1	12.25	311.1	12.04	305.8	0.67	17.02	0.43	10.92	0.60	15.2
	79	118.0	23.2	149.7	12.38	314.5	12.08	306.8	0.735	18.67	0.47	11.94	0.60	15.2
	87	129.5	25.6	165.2	12.53	318.3	12.125	308.0	0.81	20.57	0.515	13.08	0.60	15.2
	92	137.0	27.1	174.6	12.62	320.4	12.155	308.7	0.856	21.70	0.545	13.80	0.60	15.2
	96	143.0	28.2	182.0	12.71	322.8	12.16	308.9	0.90	22.86	0.55	13.97	0.60	15.2
	106	158.0	31.2	201.3	12.89	327.4	12.22	310.4	0.99	25.15	0.61	15.49	0.60	15.2
	120	179.0	35.3	227.7	13.12	333.2	12.32	312.9	1.10	28.07	0.71	18.03	0.60	15.2
	133	198.0	39.1	252.3	13.38	339.9	12.365	314.1	1.236	31.40	0.755	19.20	0.60	15.2
	136	202.4	39.9	257.4	13.41	340.6	12.40	315.0	1.25	31.75	0.79	20.07	0.60	15.2
	152	226.2	44.7	288.4	13.71	348.2	12.48	317.0	1.40	35.56	0.87	22.10	0.60	15.2
	161	240.0	47.4	305.6	13.88	352.6	12.515	317.9	1.486	37.70	0.905	23.00	0.60	15.2
	170	253.0	50.0	322.6	14.03	356.4	12.57	319.3	1.56	39.62	0.96	24.38	0.60	15.2
	190	283.0	55.8	360.0	14.38	365.3	12.67	321.8	1.735	44.07	1.06	26.92	0.60	15.2
	210	312.5	61.8	398.7	14.71	373.6	12.79	324.9	1.90	48.26	1.18	29.97	0.60	15.2
	230	342.3	67.7	436.8	15.05	382.3	12.895	327.5	2.07	52.58	1.285	32.64	0.60	15.2
	252	375.0	74.0	477.4	15.41	391.4	13.005	330.3	2.25	57.15	1.395	35.43	0.60	15.2
	278	413.7	81.9	528.4	15.85	402.6	13.14	333.8	2.47	62.74	1.53	38.86	0.60	15.2
	305	453.9	89.6	578.1	16.32	414.5	13.235	336.2	2.705	68.71	1.625	41.27	0.60	15.2
	336	500.0	98.8	637.4	16.82	427.2	13.385	340.0	2.955	75.06	1.775	45.08	0.60	15.2
W14 14 x 5 (356 x 127)	22	33.0	6.49	41.87	13.74	349.0	5.00	127.0	0.335	8.51	0.23	5.84	0.40	10.2
	26	39.0	7.69	49.61	13.91	353.3	5.025	127.6	0.42	10.60	0.255	6.48	0.40	10.2



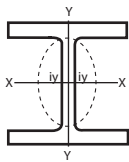
UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx	Zy		
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W10 10 x 10 (254 x 254)	272	11,300	93.4	3,890	4.35	11.0	2.54	6.45	54.6	895	18.7	306
	303	12,600	103	4,290	4.37	11.1	2.56	6.50	60.0	983	20.6	338
	341	14,200	116	4,830	4.39	11.2	2.57	6.53	66.7	1,090	23.0	377
	394	16,400	134	5,580	4.44	11.3	2.58	6.55	75.7	1,240	26.4	433
	421	17,510	142	5,903	4.46	11.3	2.59	6.58	80.1	1,312	27.9	457
	455	18,900	154	6,410	4.49	11.4	2.60	6.60	85.9	1,410	30.1	493
	534	22,200	179	7,450	4.54	11.5	2.63	6.68	98.5	1,610	34.8	570
	542	22,580	181	7,518	4.55	11.6	2.63	6.68	99.7	1,634	35.2	577
	624	26,000	207	8,620	4.61	11.7	2.65	6.73	112.0	1,840	40.0	655
	716	29,800	236	9,820	4.66	11.8	2.68	6.81	126.0	2,060	45.3	742
W12 12 x 4 (305 x 102)	88.6	3,690	2.36	98.2	4.62	11.7	0.753	1.91	14.9	244	1.19	19.5
	103	4,290	2.82	117	4.67	11.9	0.773	1.96	17.1	280	1.41	23.1
	105.4	4,387	2.9	120	4.65	11.8	0.770	1.96	17.6	288	1.40	23.6
	130	5,410	3.76	156	4.82	12.2	0.822	2.09	21.3	349	1.88	30.8
	156	6,490	4.66	194	4.91	12.5	0.848	2.15	25.4	416	2.31	37.9
W12 12 x 5 (305 x 127)	172	7,162	8.1	337	4.83	12.3	1.05	2.67	28.8	472	3.30	54.6
	195.6	8,143	9.3	388	4.87	12.4	1.06	2.70	32.4	531	3.80	62.5
	228.3	9,504	11.1	460	4.92	12.5	1.08	2.75	37.4	612	4.50	73.5
W12 12 x 6 1/2 (305 x 165)	204	8,490	17.3	720	5.17	13.1	1.51	3.84	33.4	547	5.34	87.5
	204.1	8,496	16.6	691	5.06	12.8	1.44	3.66	34.1	559	5.10	83.6
	238	9,910	20.3	845	5.21	13.2	1.52	3.86	38.6	633	6.24	102
	238.4	9,924	19.8	824	5.11	12.9	1.47	3.73	39.4	646	6.10	100
	280.8	11,690	23.7	986.6	5.15	13.1	1.50	3.81	45.9	752	7.20	118
	285	11,900	24.5	1,020	5.25	13.3	1.54	3.91	45.6	747	7.47	122
W12 12 x 8 (305 x 203)	310	12,900	44.1	1,840	5.13	13.0	1.94	4.93	51.9	850	11.0	180
	350	14,600	50.0	2,080	5.15	13.1	1.94	4.93	58.1	952	12.4	203
	394	16,400	56.3	2,340	5.18	13.2	1.96	4.98	64.7	1,060	13.9	228
W12 12 x 10 (305 x 254)	425	17,700	95.8	3,990	5.23	13.3	2.48	6.30	70.6	1,160	19.2	315
	475	19,800	107	4,450	5.28	13.4	2.51	6.38	78.0	1,280	21.4	351
W12 12 x 12 (305 x 305)	533	22,200	174	7,240	5.28	13.4	3.02	7.67	87.9	1,440	29.1	477
	597	24,800	195	8,120	5.31	13.5	3.04	7.72	97.4	1,600	32.4	531
	662	27,600	216	8,990	5.34	13.6	3.05	7.75	107	1,750	35.8	587
	740	30,800	241	10,000	5.38	13.7	3.07	7.80	118	1,930	39.7	651
	789	32,840	256	10,670	5.40	13.7	3.08	7.82	125	2,084	42.2	691
	833	34,700	270	11,200	5.44	13.8	3.09	7.85	131	2,150	44.4	728
	933	38,800	301	12,500	5.47	13.9	3.11	7.90	145	2,380	49.3	808
	1,070	44,500	345	14,400	5.51	14.0	3.13	7.95	163	2,670	56.0	918
	1,221	50,840	390	16,230	5.59	14.2	3.16	8.03	183	2,991	63.1	1,034
	1,240	51,600	398	16,600	5.58	14.2	3.16	8.03	186	3,050	64.2	1,050
	1,430	59,500	454	18,900	5.66	14.4	3.19	8.10	209	3,420	72.8	1,190
	1,542	64,180	486	20,240	5.70	14.5	3.20	8.13	222	3,641	77.1	1,273
	1,650	68,700	517	21,500	5.74	14.6	3.22	8.18	235	3,850	82.3	1,350
	1,890	78,700	589	24,500	5.82	14.8	3.25	8.25	263	4,310	93.0	1,520
	2,140	89,100	664	27,600	5.89	15.0	3.28	8.33	292	4,780	104	1,700
	2,420	101,000	742	30,900	5.97	15.2	3.31	8.41	321	5,260	115	1,880
	2,720	113,000	828	34,500	6.06	15.4	3.34	8.48	353	5,780	127	2,080
	3,110	129,000	937	39,000	6.16	15.6	3.38	8.59	393	6,440	143	2,340
	3,550	148,000	1,050	43,700	6.29	16.0	3.42	8.69	435	7,130	159	2,610
	4,060	169,000	1,190	49,500	6.41	16.3	3.47	8.81	483	7,910	177	2,900
W14 14 x 5 (356 x 127)	199	8,280	7.00	291	5.54	14.1	1.04	2.64	29.0	475	2.80	45.9
	245	10,200	8.91	371	5.65	14.4	1.08	2.74	35.3	578	3.55	58.2

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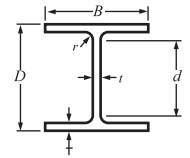
Section size	Unit weight M		Section area A		Section depth D		Flange width B		Flange thickness T		Web thickness t		Corner radius r	
in (mm)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W14 14 x 6 ³ / ₄ (356 x 171)34	30	45.0	8.85	57.10	13.84	351.5	6.73	170.9	0.385	9.78	0.27	6.86	0.40	10.2
	51.0	10.0	64.52	13.98	355.1	6.745	171.3	0.455	11.5	0.285	7.24	0.40	10.2	
	38	57.0	11.2	72.26	14.10	358.1	6.77	172.0	0.515	13.0	0.31	7.87	0.40	10.2
	45	67.0	13.2	85.40	14.33	364.0	6.82	173.2	0.618	15.7	0.357	9.10	0.40	10.2
W14 14 x 8 (356 x 203)48	43	63.99	12.6	81.29	13.66	347.0	7.995	203.1	0.53	13.4	0.305	7.75	0.60	15.2
	71.43	14.1	90.97	13.79	350.3	8.03	204.0	0.595	15.1	0.34	8.64	0.60	15.2	
	53	78.87	15.6	100.6	13.92	353.6	8.06	204.7	0.66	16.7	0.37	9.40	0.60	15.2
W14 14 x 10 (356 x 254)68	61	90.78	17.9	115.5	13.89	352.8	9.995	253.9	0.645	16.30	0.375	9.52	0.60	15.2
	101.2	20.0	129.0	14.04	356.6	10.035	254.9	0.723	18.36	0.415	10.54	0.60	15.2	
	74	110.1	21.8	140.6	14.17	359.9	10.07	255.8	0.785	19.94	0.45	11.43	0.60	15.2
	82	122.0	24.1	155.5	14.31	363.5	10.13	257.3	0.855	21.72	0.51	12.95	0.60	15.2
W14 14 x 14 ¹ / ₂ (356 x 368)90	87	129.0	25.6	164.9	14.00	355.6	14.50	368.3	0.688	17.50	0.42	10.70	0.60	15.2
	134.0	26.5	171.0	14.02	356.1	14.52	368.8	0.71	18.03	0.44	11.18	0.60	15.2	
	99	147.3	29.1	187.7	14.16	359.7	14.565	370.0	0.78	19.81	0.485	12.32	0.60	15.2
	103	153.0	30.3	195.2	14.25	362.0	14.575	370.2	0.813	20.70	0.495	12.60	0.60	15.2
	109	162.2	32.0	206.5	14.32	363.7	14.605	371.0	0.86	21.84	0.525	13.33	0.60	15.2
	119	177.0	35.0	225.7	14.50	368.3	14.65	372.1	0.938	23.80	0.57	14.50	0.60	15.2
	120	179.0	35.3	227.7	14.48	367.8	14.67	372.6	0.94	23.88	0.59	14.99	0.60	15.2
	132	196.4	38.8	250.3	14.66	372.4	14.725	374.0	1.03	26.16	0.645	16.38	0.60	15.2
	136	202.0	40.0	257.9	14.75	374.7	14.74	374.4	1.063	27.00	0.66	16.80	0.60	15.2
W14 14 x 16 (356 x 406)158	145	216.0	42.7	275.5	14.78	375.4	15.50	393.7	1.09	27.69	0.68	17.27	0.60	15.2
	235.0	46.5	299.8	15.00	381.0	15.55	395.0	1.188	30.20	0.73	18.50	0.60	15.2	
	159	237.0	46.7	301.3	14.98	380.5	15.565	395.4	1.19	30.23	0.745	18.92	0.60	15.2
	176	262.0	51.8	334.2	15.22	386.6	15.65	397.5	1.31	33.27	0.83	21.08	0.60	15.2
	193	287.2	56.8	366.4	15.48	393.2	15.71	399.0	1.44	36.58	0.89	22.61	0.60	15.2
	211	314.0	62.0	400.0	15.72	399.3	15.80	401.3	1.56	39.62	0.98	24.89	0.60	15.2
	228	340.0	67.1	432.7	16.00	406.4	15.865	403.0	1.688	42.90	1.045	26.50	0.60	15.2
	233	347.0	68.5	441.9	16.04	407.4	15.89	403.6	1.72	43.69	1.07	27.18	0.60	15.2
	257	382.5	75.6	487.7	16.38	416.1	15.995	406.3	1.89	48.01	1.175	29.84	0.60	15.2
	264	393.0	77.6	500.9	16.50	419.1	16.025	407.0	1.938	49.20	1.205	30.60	0.60	15.2
	283	421.1	83.3	537.4	16.74	425.2	16.11	409.2	2.07	52.58	1.29	32.77	0.60	15.2
	311	463.0	91.4	589.7	17.12	434.8	16.23	412.2	2.26	57.40	1.41	35.81	0.60	15.2
	314	467.0	92.3	595.5	17.19	436.6	16.235	412.4	2.283	58.00	1.415	35.90	0.60	15.2
	342	509.0	101	651.6	17.54	445.5	16.36	415.5	2.47	62.74	1.54	39.12	0.60	15.2
	370	551.0	109	703.2	17.92	455.2	16.475	418.5	2.66	67.56	1.655	42.04	0.60	15.2
	398	592.3	117	754.8	18.29	464.6	16.59	421.4	2.845	72.26	1.77	44.96	0.60	15.2
	426	634.0	125	806.4	18.67	474.2	16.695	424.1	3.035	77.09	1.875	47.62	0.60	15.2
	455	677.1	134	864.5	19.02	483.1	16.835	427.6	3.21	81.53	2.015	51.18	0.60	15.2
	500	744.1	147	948.4	19.60	497.8	17.01	432.1	3.50	88.90	2.19	55.63	0.60	15.2
	550	818.5	162	1,045	20.24	514.1	17.20	436.9	3.82	97.03	2.38	60.45	0.60	15.2
	605	900.3	178	1,148	20.92	531.4	17.415	442.3	4.16	105.6	2.595	65.91	0.60	15.2
	665	989.6	196	1,265	21.64	549.7	17.65	448.3	4.52	114.8	2.83	71.88	0.60	15.2
	730	1,086.0	215	1,387	22.42	569.5	17.89	454.4	4.91	124.7	3.07	77.98	0.60	15.2
W16 16 x 5 ¹ / ₂ (406 x 140)31	26	39.0	7.68	49.55	15.69	398.5	5.50	139.7	0.345	8.76	0.25	6.35	0.40	10.2
	46.13	9.12	58.84	15.88	403.4	5.525	140.3	0.44	11.1	0.275	6.98	0.40	10.2	



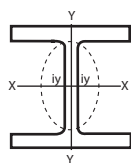
UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W14 14 x 6 ³ / ₄ (356 x 171)	291	12,100	19.6	816	5.73	14.6	1.49	3.78	42.0	688	5.82	95.4
	340	14,200	23.3	970	5.83	14.8	1.53	3.89	48.6	796	6.91	113
	385	16,000	26.7	1,110	5.88	14.9	1.55	3.94	54.6	895	7.88	129
	469	19,522	32.7	1,362	5.95	15.1	1.57	3.99	65.5	1,073	9.60	157
W14 14 x 8 (356 x 203)	428	17,800	45.2	1,880	5.82	14.8	1.89	4.80	62.7	1,030	11.3	185
	485	20,200	51.4	2,140	5.85	14.9	1.91	4.85	70.3	1,150	12.8	210
	542	22,600	57.7	2,400	5.89	15.0	1.92	4.88	77.8	1,270	14.3	234
W14 14 x 10 (356 x 254)	640	26,600	107	4,450	5.98	15.2	2.45	6.22	92.2	1,510	21.5	352
	725	30,200	122	5,080	6.01	15.3	2.47	6.27	103	1,690	24.3	398
	796	33,100	134	5,580	6.04	15.3	2.48	6.30	112	1,840	26.6	436
	882	36,700	148	6,160	6.05	15.4	2.48	6.30	123	2,020	29.3	480
W14 14 x 14 ¹ / ₂ (356 x 368)	966	40,250	350	14,560	6.15	15.6	3.70	9.40	138	2,263	48.2	790
	999	41,600	362	15,100	6.14	15.6	3.70	9.40	143	2,340	49.9	818
	1,110	46,200	402	16,700	6.17	15.7	3.71	9.42	157	2,570	55.2	905
	1,166	48,530	420	17,470	6.21	15.8	3.72	9.45	164	2,681	57.6	944
	1,240	51,600	447	18,600	6.22	15.8	3.73	9.47	173	2,830	61.2	1,000
	1,373	57,160	492	20,470	6.26	15.9	3.75	9.52	189	3,104	67.1	1,099
	1,380	57,400	495	20,600	6.24	15.8	3.74	9.50	190	3,110	67.5	1,110
	1,530	63,700	548	22,800	6.28	16.0	3.76	9.55	209	3,420	74.5	1,220
	1,593	66,310	568	23,630	6.31	16.0	3.77	9.58	216	3,540	77.0	1,262
W14 14 x 16 (356 x 406)	1,710	71,200	677	28,200	6.33	16.1	3.98	10.1	232	3,800	87.3	1,430
	1,900	79,100	745	31,010	6.40	16.2	4.00	10.2	253	4,153	95.8	1,570
	1,900	79,100	748	31,100	6.38	16.2	4.00	10.2	254	4,160	96.2	1,580
	2,140	89,100	838	34,900	6.43	16.3	4.02	10.2	281	4,600	107	1,750
	2,400	99,900	931	38,800	6.50	16.5	4.05	10.3	310	5,080	119	1,950
	2,660	111,000	1,030	42,900	6.55	16.6	4.07	10.3	338	5,540	130	2,130
	2,942	122,500	1,124	46,820	6.62	16.8	4.10	10.4	368	6,028	141	2,324
	3,010	125,000	1,150	47,900	6.63	16.8	4.10	10.4	375	6,150	145	2,380
	3,400	142,000	1,290	53,700	6.71	17.0	4.13	10.5	415	6,800	161	2,640
	3,526	146,800	1,331	55,420	6.74	17.1	4.14	10.5	427	7,005	166	2,772
	3,840	160,000	1,440	59,900	6.79	17.2	4.17	10.6	459	7,520	179	2,930
	4,330	180,000	1,610	67,000	6.88	17.5	4.20	10.7	506	8,290	199	3,260
	4,399	183,100	1,631	67,910	6.90	17.5	4.20	10.7	512	8,390	201	3,294
	4,900	204,000	1,810	75,300	6.98	17.7	4.24	10.8	559	9,160	221	3,620
	5,440	226,000	1,990	82,800	7.07	18.0	4.27	10.8	607	9,950	241	3,950
	6,000	250,000	2,170	90,300	7.16	18.2	4.31	10.9	656	10,700	262	4,290
	6,600	275,000	2,360	98,200	7.26	18.4	4.34	11.0	706	11,600	283	4,640
	7,190	299,000	2,560	107,000	7.33	18.6	4.38	11.1	756	12,400	304	4,980
	8,210	342,000	2,880	120,000	7.48	19.0	4.43	11.3	838	13,700	339	5,560
	9,430	392,000	3,250	135,000	7.63	19.4	4.49	11.4	931	15,300	378	6,190
	10,800	450,000	3,680	153,000	7.80	19.8	4.55	11.6	1,040	17,000	423	6,930
W16 16 x 5 ¹ / ₂ (406 x 140)	301	12,500	9.59	399	6.26	15.9	1.12	2.84	38.4	629	3.49	57.2
	375	15,600	12.4	516	6.41	16.3	1.17	2.97	47.2	773	4.49	73.6

UNIVERSAL BEAMS AND COLUMNS



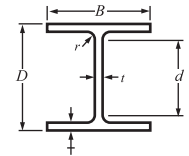
Section size	Unit weight M		Section area A		Section depth D		Flange width B		Flange thickness T		Web thickness t		Corner radius r	
in (mm)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W16 16 x 7 (406 x 178)	36	54.0	10.6	68.39	15.86	402.8	6.985	177.4	0.43	10.90	0.295	7.49	0.40	10.2
	40	60.0	11.8	76.13	16.01	406.7	6.995	177.7	0.505	12.80	0.305	7.75	0.40	10.2
	45	67.0	13.3	85.81	16.13	409.7	7.035	178.7	0.565	14.30	0.345	8.76	0.40	10.2
	50	74.41	14.7	94.84	16.26	413.0	7.07	179.6	0.63	16.00	0.38	9.65	0.40	10.2
	57	85.0	16.8	108.4	16.43	417.3	7.12	180.8	0.715	18.16	0.43	10.92	0.40	10.2
W16 16 x 10 ¹ / ₄ (406 x 260)77	67	99.71	19.7	127.1	16.33	414.8	10.235	260.0	0.665	16.89	0.395	10.03	0.40	10.2
	114.6	22.6	145.8	16.52	419.6	10.295	261.5	0.76	19.30	0.455	11.56	0.40	10.2	10.2
	89	132.4	26.2	169.0	16.75	425.4	10.365	263.3	0.875	22.22	0.525	13.33	0.40	10.2
	100	148.8	29.4	189.7	16.97	431.0	10.425	264.8	0.985	25.02	0.585	14.86	0.40	10.2
W18 18 x 6 (457 x 152)	35	52.09	10.3	66.45	17.70	449.6	6.00	152.4	0.425	10.80	0.30	7.62	0.40	10.2
	40	60.00	11.8	76.13	17.90	454.7	6.015	152.8	0.525	13.30	0.315	8.00	0.40	10.2
	45	67.00	13.2	85.4	18.00	457.2	5.982	151.9	0.589	15.00	0.36	9.10	0.40	10.2
	46	68.46	13.5	87.1	18.06	458.7	6.06	153.9	0.605	15.30	0.36	9.14	0.40	10.2
	50	74.00	14.7	95.0	18.16	461.3	6.011	152.7	0.669	17.00	0.389	9.90	0.40	10.2
	55	82.00	16.2	104.5	18.31	465.1	6.042	153.5	0.744	18.90	0.42	10.70	0.40	10.2
W18 18 x 7 ¹ / ₂ (457 x 191)50	45	67.0	13.2	85.4	17.86	453.6	7.476	189.9	0.50	12.70	0.334	8.50	0.40	10.2
	74.41	14.7	94.84	17.99	456.9	7.495	190.4	0.57	14.40	0.355	9.02	0.40	10.2	10.2
	55	82.0	16.2	104.5	18.11	460.0	7.53	191.3	0.63	16.00	0.39	9.91	0.40	10.2
	60	89.29	17.6	113.5	18.24	463.3	7.555	191.9	0.695	17.65	0.415	10.54	0.40	10.2
	65	97.0	19.1	123.2	18.35	466.1	7.59	192.8	0.75	19.05	0.45	11.43	0.40	10.2
	66	98.0	19.4	125.3	18.40	467.4	7.592	192.8	0.77	19.60	0.45	11.40	0.40	10.2
	71	106.0	20.8	134.2	18.47	469.1	7.635	193.9	0.81	20.57	0.495	12.57	0.40	10.2
W18 18 x 11 (457 x 279)	76	113.1	22.3	143.9	18.21	462.5	11.035	280.3	0.68	17.27	0.425	10.80	0.40	10.2
	86	128.0	25.3	163.2	18.39	467.1	11.09	281.7	0.77	19.56	0.48	12.19	0.40	10.2
	97	144.4	28.5	183.9	18.59	472.2	11.145	283.1	0.87	22.10	0.535	13.59	0.40	10.2
	106	157.7	31.1	200.6	18.73	475.7	11.20	284.5	0.94	23.88	0.59	14.99	0.40	10.2
	119	177.1	35.1	226.5	18.97	481.8	11.265	286.1	1.06	26.92	0.655	16.64	0.40	10.2
W21 21 x 6 ¹ / ₂ (533 x 165)50	44	66.0	13.0	83.87	20.66	524.8	6.50	165.1	0.45	11.40	0.35	8.89	0.50	12.7
	74.41	14.7	94.84	20.83	529.1	6.53	165.9	0.535	13.50	0.38	9.65	0.50	12.7	12.7
	57	85.0	16.7	107.7	21.06	534.9	6.555	166.5	0.65	16.51	0.405	10.29	0.50	12.7
W21 21 x 8 ¹ / ₄ (533 x 210)62	55	82.0	16.2	104.4	20.80	528.3	8.216	208.7	0.52	13.20	0.376	9.60	0.50	12.7
	92.27	18.3	118.1	20.99	533.1	8.24	209.3	0.615	15.62	0.40	10.16	0.50	12.7	12.7
	68	101.2	20.0	129.0	21.13	536.7	8.27	210.1	0.685	17.40	0.43	10.92	0.50	12.7
	73	109.0	21.5	138.7	21.24	539.5	8.295	210.7	0.74	18.80	0.455	11.56	0.50	12.7
	82	122.0	24.1	155.8	21.44	544.6	8.342	211.9	0.84	21.30	0.502	12.80	0.50	12.7
	83	123.5	24.3	156.8	21.43	544.3	8.355	212.2	0.835	21.21	0.515	13.08	0.50	12.7
	93	138.4	27.3	176.1	21.62	549.1	8.42	213.9	0.93	23.62	0.58	14.73	0.50	12.7
W21 21 x 12 ¹ / ₄ (533 x 311)111	101	150.3	29.8	192.3	21.36	542.5	12.29	312.2	0.80	20.32	0.50	12.70	0.50	12.7
	165.2	32.7	211.0	21.51	546.4	12.34	313.4	0.875	22.22	0.55	13.97	0.50	12.7	12.7
	122	181.6	35.9	231.6	21.68	550.7	12.39	314.7	0.96	24.38	0.60	15.24	0.50	12.7
	132	196.4	38.8	250.3	21.83	554.5	12.44	316.0	1.03	26.29	0.65	16.51	0.50	12.7
	147	218.8	43.2	278.7	22.06	560.3	12.51	317.8	1.15	29.21	0.72	18.29	0.50	12.7
W24 24 x 7 (610 x 178)	55	82	16.2	104.5	23.57	598.7	7.005	177.9	0.505	12.83	0.395	10.03	0.50	12.7
	61	91	17.9	116.5	23.72	602.5	7.023	178.4	0.591	15.00	0.419	10.60	0.50	12.7
	62	92.27	18.2	117.4	23.74	603.0	7.040	178.8	0.59	14.99	0.43	10.92	0.50	12.7



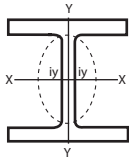
UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W16 16 x 7 (406 x 178)	448	18,600	24.5	1,020	6.51	16.5	1.52	3.86	56.5	926	7.00	115
	518	21,600	28.9	1,200	6.63	16.8	1.57	3.99	64.7	1,060	8.25	135
	586	24,400	32.8	1,370	6.65	16.9	1.57	3.99	72.7	1,190	9.34	153
	659	27,400	37.2	1,550	6.68	17.0	1.59	4.04	81.0	1,330	10.5	172
	758	31,600	43.1	1,790	6.72	17.1	1.60	4.06	92.2	1,510	12.1	198
W16 16 x 10 ¹ / ₄ (406 x 260)	954	39,700	119	4,950	6.96	17.7	2.46	6.25	117	1,920	23.2	380
	1,100	46,200	138	5,740	7.00	17.8	2.47	6.27	134	2,200	26.9	441
	1,300	54,100	163	6,780	7.05	17.9	2.49	6.32	155	2,540	31.4	515
	1,490	62,000	186	7,740	7.10	18.0	2.51	6.38	175	2,870	35.7	585
W18 18 x 6 (457 x 152)	510	21,200	15.3	637	7.04	17.9	1.22	3.10	57.6	944	5.12	83.9
	612	25,500	19.1	795	7.21	18.3	1.27	3.23	68.4	1,120	6.35	104
	686	28,577	21.1	878	7.20	18.3	1.26	3.21	76.3	1,250	7.00	116
	712	29,600	22.5	937	7.25	18.4	1.29	3.28	78.8	1,290	7.43	122
	779	32,435	24.3	1,012	7.28	18.5	1.28	3.26	85.8	1,406	8.10	133
	870	36,215	27.5	1,143	7.33	18.6	1.30	3.31	95.0	1,557	9.10	149
W18 18 x 7 ¹ / ₄ (457 x 191)	704	29,330	31.9	1,328	7.30	18.5	1.55	3.94	78.9	1,293	8.5	139
	800	33,300	40.1	1,670	7.38	18.7	1.65	4.19	88.9	1,460	10.7	175
	890	37,000	44.9	1,870	7.41	18.8	1.67	4.24	98.3	1,610	11.9	195
	984	41,000	50.1	2,090	7.47	19.0	1.68	4.27	108	1,770	13.3	218
	1,070	44,500	54.8	2,280	7.49	19.0	1.69	4.29	117	1,920	14.4	236
	1,098	45,717	56.3	2,343	7.52	19.1	1.70	4.33	119	1,956	14.8	243
	1,170	48,700	60.3	2,510	7.50	19.0	1.70	4.32	127	2,080	15.8	259
W18 18 x 11 (457 x 279)	1,330	55,400	152	6,330	7.73	19.6	2.61	6.63	146	2,390	27.6	452
	1,530	63,700	175	7,280	7.77	19.7	2.63	6.68	166	2,720	31.6	518
	1,750	72,800	201	8,370	7.82	19.9	2.65	6.73	188	3,080	36.1	592
	1,910	79,500	220	9,170	7.84	19.9	2.66	6.76	204	3,340	39.4	646
	2,190	91,200	253	10,500	7.90	20.1	2.69	6.83	231	3,790	44.9	736
W21 21 x 6 ¹ / ₂ (533 x 165)	843	35,100	20.7	862	8.06	20.5	1.26	3.20	81.6	1,340	6.37	104
	984	41,000	24.9	1,040	8.18	20.8	1.30	3.30	94.5	1,550	7.64	125
	1,170	48,700	30.6	1,270	8.36	21.2	1.35	3.43	111	1,820	9.35	153
W21 21 x 8 ¹ / ₄ (533 x 210)	1,140.7	47,490	44.0	1,832	8.40	21.3	1.65	4.19	109.7	1,798	10.7	175.4
	1,330	55,400	57.5	2,390	8.54	21.7	1.77	4.50	127	2,080	14.0	229
	1,480	61,600	64.7	2,690	8.60	21.8	1.80	4.57	140	2,290	15.7	257
	1,600	66,600	70.6	2,940	8.64	21.9	1.81	4.60	151	2,470	17.0	279
	1,830	76,200	81.4	3,390	8.67	22.0	1.83	4.65	170.8	2,799	19.5	320
	1,830.7	76,207	81.5	3,393	8.71	22.1	1.84	4.67	171	2,800	19.5	320.2
	2,070	86,200	92.9	3,870	8.70	22.1	1.84	4.67	192	3,150	22.1	362
W21 21 x 12 ¹ / ₄ (533 x 311)	2,420	101,000	248	10,300	9.02	22.9	2.89	7.34	227	3,720	40.3	660
	2,670	111,000	274	11,400	9.05	23.0	2.90	7.37	249	4,080	44.5	729
	2,960	123,000	305	12,700	9.09	23.1	2.92	7.42	273	4,470	49.2	806
	3,220	134,000	333	13,900	9.12	23.2	2.93	7.44	295	4,830	53.5	877
	3,630	151,000	376	15,600	9.17	23.3	2.95	7.49	329	5,390	60.1	985
W24 24 x 7 (610 x 178)	1,350	56,200	29.1	1,210	9.11	23.1	1.34	3.40	114	1,870	8.30	136
	1,540	64,085	34.3	1,432	9.25	23.5	1.38	3.51	130	2,124	9.76	159.5
	1,550	64,500	34.5	1,440	9.23	23.4	1.38	3.51	131	2,150	9.80	161

UNIVERSAL BEAMS AND COLUMNS



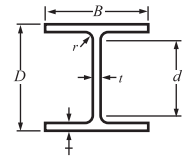
Section size	Unit weight M		Section area A		Section depth D		Flange width B		Flange thickness T		Web thickness t		Corner radius r	
in (mm)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W24 24 x 9 (610 x 229)	68	101.2	20.1	129.7	23.73	602.7	8.965	227.7	0.585	14.86	0.415	10.54	0.50	12.7
	76	113.1	22.4	144.5	23.92	607.6	8.99	228.3	0.680	17.27	0.44	11.18	0.50	12.7
	84	125	24.7	159.4	24.10	612.1	9.02	229.1	0.770	19.56	0.47	11.94	0.50	12.7
	94	140	27.7	178.7	24.31	617.5	9.065	230.2	0.875	22.22	0.515	13.08	0.50	12.7
W24 24 x 12 (610 x 305)	100	149	29.4	190.1	24.00	609.6	12.00	304.8	0.775	19.70	0.468	11.90	0.70	17.8
	120	179	35.3	227.9	24.31	617.5	12.088	307.0	0.93	23.60	0.556	14.10	0.70	17.8
	160	238	47.0	303.8	24.92	633.0	12.264	311.5	1.235	31.40	0.732	18.60	0.70	17.8
W24 24 x 12 ³ / ₄ (610 x 324)117	104	154.8	30.6	197.4	24.06	611.1	12.75	323.8	0.75	19.05	0.50	12.70	0.50	12.7
	174.1	34.4	221.9	24.26	616.2	12.80	325.1	0.85	21.59	0.55	13.97	0.50	12.7	
	131	194.9	38.5	248.4	24.48	621.8	12.855	326.5	0.96	24.38	0.605	15.37	0.50	12.7
	146	217.3	43.0	277.4	24.74	628.4	12.90	327.7	1.09	27.69	0.65	16.51	0.50	12.7
	162	241.1	47.7	307.7	25.00	635.0	12.955	329.1	1.22	30.99	0.705	17.91	0.50	12.7
W27 27 x 10 (686 x 254)	84	125	24.8	160.0	26.71	678.4	9.96	253.0	0.64	16.26	0.46	11.68	0.60	15.2
	94	140	27.7	178.7	26.92	683.8	9.99	253.7	0.745	18.92	0.49	12.45	0.60	15.2
	102	152	30.0	193.5	27.09	688.1	10.015	254.4	0.83	21.08	0.515	13.08	0.60	15.2
	114	170	33.5	216.1	27.29	693.2	10.07	255.8	0.93	23.62	0.57	14.48	0.60	15.2
W27 27 x 14 (686 x 356)	146	217.3	42.9	276.8	27.38	695.5	13.965	354.7	0.975	24.76	0.605	15.37	0.60	15.2
	161	239.6	47.4	305.8	27.59	700.8	14.02	356.1	1.08	27.43	0.66	16.76	0.60	15.2
	178	264.9	52.3	337.4	27.81	706.4	14.085	357.8	1.19	30.23	0.725	18.41	0.60	15.2
W30 30 x 10 ¹ / ₂ (762 x 267)108	99	147.3	29.1	187.7	29.65	753.1	10.45	265.4	0.67	17.02	0.52	13.21	0.65	16.5
	161	31.7	204.5	29.83	757.7	10.475	266.1	0.76	19.30	0.545	13.84	0.65	16.5	
	116	173	34.2	220.6	30.01	762.3	10.495	266.6	0.85	21.59	0.565	14.35	0.65	16.5
	124	185	36.5	235.5	30.17	766.3	10.515	267.1	0.93	23.62	0.585	14.86	0.65	16.5
	132	197	38.9	251.0	30.31	769.9	10.545	267.8	1.00	25.40	0.615	15.62	0.65	16.5
W30 30 x 15 (762 x 381)	173	257.5	50.8	327.7	30.44	773.2	14.985	380.6	1.06	27.05	0.655	16.64	0.65	16.5
	191	284.2	56.1	361.9	30.68	779.3	15.04	382.0	1.18	30.10	0.71	18.03	0.65	16.5
	211	314	62.0	400.0	30.94	785.9	15.105	383.7	1.31	33.40	0.775	19.68	0.65	16.5
W33 33 X 11 ¹ / ₂ 118 (838 x 292)130	176	34.7	223.9	32.86	834.6	11.48	291.6	0.74	18.80	0.55	13.97	0.70	17.8	
	194	38.3	247.1	33.09	840.5	11.51	292.4	0.855	21.72	0.58	14.73	0.70	17.8	
	141	210	41.6	268.4	33.30	845.8	11.535	293.0	0.96	24.38	0.605	15.37	0.70	17.8
	152	226.2	44.7	288.4	33.49	850.6	11.565	293.8	1.05	26.80	0.635	16.13	0.70	17.8
W33 33 x 15 ³ / ₄ (838 x 400)221	201	299.1	59.1	381.3	33.68	855.5	15.745	399.9	1.15	29.21	0.715	18.16	0.70	17.8
	328.9	65.0	419.4	33.93	861.8	15.805	401.4	1.27	32.38	0.775	19.68	0.70	17.8	
	241	358.6	70.9	457.4	34.18	868.2	15.86	402.8	1.40	35.56	0.83	21.08	0.70	17.8
W36 36 x 12 (914 x 305)150	135	201	39.7	256.1	35.55	903.0	11.95	303.5	0.79	20.07	0.60	15.24	0.75	19.0
	224	44.2	285.2	35.85	910.6	11.975	304.2	0.94	23.88	0.625	15.87	0.75	19.0	
	160	238.1	47.0	303.2	36.01	914.7	12.00	304.8	1.02	25.91	0.65	16.51	0.75	19.0
	170	253	50.0	322.6	36.17	918.7	12.03	305.6	1.10	27.94	0.68	17.27	0.75	19.0
	182	271	53.6	345.8	36.33	922.8	12.075	306.7	1.18	29.97	0.725	18.41	0.75	19.0
	194	289	57.0	367.7	36.49	926.8	12.115	307.7	1.26	32.00	0.765	19.43	0.75	19.0
	210	313	61.8	398.7	36.69	931.9	12.18	309.4	1.36	34.54	0.83	21.08	0.75	19.0
W36 36 x 16 ¹ / ₂ (914 x 419)245	230	343	67.6	436.1	35.90	911.9	16.47	418.3	1.26	32.00	0.76	19.30	0.95	24.1
	365	72.1	465.2	36.08	916.4	16.51	419.4	1.35	34.29	0.80	20.32	0.95	24.1	
	260	388	76.5	493.5	36.26	921.0	16.55	420.4	1.44	36.58	0.84	21.34	0.95	24.1
	280	417	82.4	531.6	36.52	927.6	16.595	421.5	1.57	39.88	0.885	22.48	0.95	24.1
	300	447	88.3	569.7	36.74	933.2	16.655	423.0	1.68	42.67	0.945	24.00	0.95	24.1



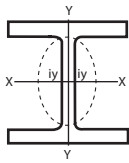
UNIVERSAL BEAMS AND COLUMNS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
in (mm)	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
W24 24 x 9 (610 x 229)	1,830	76,200	70.4	2,930	9.55	24.3	1.87	4.75	154	2,520	15.7	257
	2,100	87,400	82.5	3,430	9.69	24.6	1.92	4.88	176	2,880	18.4	302
	2,370	98,600	94.4	3,930	9.79	24.9	1.95	4.95	196	3,210	20.9	342
	2,700	112,000	109.0	4,540	9.87	25.1	1.98	5.03	222	3,640	24.0	393
W24 24 x 12 (610 x 305)	2,987.3	124,400	203.5	8,471	10.08	25.6	2.63	6.68	248.9	4,079.5	33.9	555.6
	3,635.3	151,300	254.0	10,570	10.15	25.78	2.68	6.81	299.1	4,902.2	42.0	688.4
	4,986.3	207,571	380.5	15,838	10.28	26.1	2.84	7.22	400.2	6,559	62.0	1,017
W24 24 x 12 ³ / ₄ (610 x 324)	3,100	129,000	259	10,800	10.1	25.7	2.91	7.39	258	4,230	40.7	667
	3,540	147,000	297	12,400	10.1	25.7	2.94	7.47	291	4,770	46.5	762
	4,020	167,000	340	14,200	10.2	25.9	2.97	7.54	329	5,390	53.0	869
	4,580	191,000	391	16,300	10.3	26.2	3.01	7.65	371	6,080	60.5	991
	5,170	215,000	443	18,400	10.4	26.4	3.05	7.75	414	6,780	68.4	1,120
W27 27 x 10 (686 x 254)	2,850	119,000	106	4,410	10.7	27.2	2.07	5.26	213	3,490	21.2	347
	3,270	136,000	124	5,160	10.9	27.7	2.12	5.38	243	3,980	24.8	406
	3,620	151,000	139	5,790	11.0	27.9	2.15	5.46	267	4,380	27.8	456
	4,090	170,000	159	6,620	11.0	27.9	2.18	5.54	299	4,900	31.5	516
W27 27 x 14 (686 x 356)	5,630	234,000	443	18,400	11.4	29.0	3.21	8.15	411	6,740	63.5	1,040
	6,280	261,000	497	20,700	11.5	29.2	3.24	8.23	455	7,460	70.9	1,160
	6,990	291,000	555	23,100	11.6	29.5	3.26	8.28	502	8,230	78.8	1,290
W30 30 x 10 ¹ / ₂ (762 x 267)	4,000	166,000	128	5,330	11.7	29.7	2.10	5.33	269	4,410	24.5	401
	4,470	186,000	146	6,080	11.9	30.2	2.15	5.46	299	4,900	27.9	457
	4,930	205,000	164	6,830	12.0	30.5	2.19	5.56	329	5,390	31.3	513
	5,360	223,000	181	7,530	12.1	30.7	2.23	5.66	355	5,820	34.4	564
	5,770	240,000	196	8,160	12.2	31.0	2.25	5.71	380	6,230	37.2	610
W30 30 x 15 (762 x 381)	8,200	341,000	598	24,900	12.7	32.3	3.43	8.71	539	8,830	79.8	1,310
	9,170	382,000	673	28,000	12.8	32.5	3.46	8.79	598	9,800	89.5	1,470
	10,300	429,000	757	31,500	12.9	32.8	3.49	8.86	663	10,900	100	1,640
W33 33 x 11 ¹ / ₂ (838 x 292)	5,900	246,000	187	7,780	13.0	33.0	2.32	5.89	359	5,880	32.6	534
	6,710	279,000	218	9,070	13.2	33.5	2.38	6.05	406	6,650	37.9	621
	7,450	310,000	246	10,200	13.4	34.0	2.43	6.17	448	7,340	42.7	700
	8,160	340,000	273	11,400	13.5	34.3	2.47	6.27	487	7,980	47.2	773
W33 33 x 15 ³ / ₄ (838 x 400)	11,500	479,000	749	31,200	14.0	35.6	3.56	9.04	684	11,200	95.2	1,560
	12,800	533,000	840	35,000	14.1	35.8	3.59	9.12	757	12,400	106	1,740
	14,200	591,000	932	38,800	14.1	35.8	3.63	9.22	829	13,600	118	1,930
W36 36 x 12 (914 x 305)	7,800	325,000	225	9,370	14.0	35.6	2.38	6.05	439	7,190	37.7	618
	9,040	376,000	270	11,200	14.3	36.3	2.47	6.27	504	8,260	45.1	739
	9,760	406,000	295	12,300	14.4	36.6	2.50	6.35	542	8,880	49.1	805
	10,500	437,000	320	13,300	14.5	36.8	2.53	6.43	580	9,500	53.2	872
	11,300	470,000	347	14,400	14.5	36.8	2.55	6.48	623	10,200	57.6	944
	12,100	504,000	375	15,600	14.6	37.1	2.56	6.50	664	10,900	61.9	1,010
	13,200	549,000	411	17,100	14.6	37.1	2.58	6.55	719	11,800	67.5	1,110
W36 36 x 16 ¹ / ₂ (914 x 419)	15,000	624,000	940	39,100	14.9	37.8	3.73	9.47	837	13,700	114	1,870
	16,100	670,000	1,010	42,000	15.0	38.1	3.75	9.52	895	14,700	123	2,020
	17,300	720,000	1,090	45,400	15.0	38.1	3.78	9.60	953	15,600	132	2,160
	18,900	787,000	1,200	49,900	15.1	38.4	3.81	9.68	1,030	16,900	144	2,360
	20,300	845,000	1,300	54,100	15.2	38.6	3.83	9.73	1,110	18,200	156	2,560

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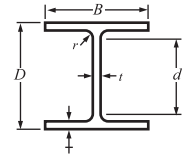
Section size	Unit weight M	Section depth D	Flange width B	Thickness		Corner radius r	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				t (web)	T (flange)			I _x	I _y	i _x	i _y	Z _x	Z _y
mm	kg/m	mm	mm	mm	mm	mm	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
100 x 50	9.3	100	50	5	7	8	11.85	187	14.8	3.98	1.12	37.5	5.91
100 x 100	17.2	100	100	6	8	10	21.90	333	134	4.18	2.47	76.5	26.7
125 x 60	13.2	125	60	6	8	9	16.84	413	29.2	4.95	1.32	66.1	9.73
125 x 125	23.8	125	125	6.5	9	10	30.31	847	293	5.29	3.11	136	47.0
150 x 75	14.0	150	75	5	7	8	17.85	666	49.5	6.11	1.66	88.8	13.2
150 x 100	21.1	148	100	6	9	11	26.84	1,020	151	6.17	2.37	138	30.1
150 x 150	31.5	150	150	7	10	11	40.14	1,640	563	6.39	3.75	219	75.1
175 x 90	18.1	175	90	5	8	9	23.04	1,210	97.5	7.26	2.06	139	21.7
175 x 125	23.3	169	125	5.5	8	12	29.65	1,530	261	7.18	2.97	181	41.8
175 x 175	40.2	175	175	7.5	11	12	51.21	2,880	984	7.50	4.38	330	112
200 x 100	18.2	198	99	4.5	7	11	23.18	1,580	114	8.26	2.21	160	23.0
	21.3	200	100	5.5	8	11	27.16	1,840	134	8.24	2.22	184	26.8
200 x 150	30.6	194	150	6	9	13	39.01	2,690	507	8.30	3.61	277	67.6
200 x 200	49.9	200	200	8	12	13	63.53	4,720	1,600	8.62	5.02	472	160
	56.2	200	204	12	12	13	71.53	4,980	1,700	8.35	4.88	498	167
	65.7	208	202	10	16	13	83.69	6,530	2,200	8.83	5.13	628	218
250 x 125	25.7	248	124	5	8	12	32.68	3,540	255	10.4	2.79	285	41.1
	29.6	250	125	6	9	12	37.66	4,050	294	10.4	2.79	324	47.0
250 x 175	44.1	244	175	7	11	16	56.24	6,120	984	10.4	4.18	502	113
250 x 250	64.4	244	252	11	11	16	82.06	8,780	2,940	10.3	5.98	720	233
	66.5	248	249	8	13	16	84.70	9,930	3,350	10.8	6.29	801	269
	72.4	250	250	9	14	16	92.18	10,800	3,650	10.8	6.29	867	292
	82.2	250	255	14	14	16	104.7	11,500	3,880	10.5	6.09	919	304
300 x 150	32.0	298	149	5.5	8	13	40.80	6,320	442	12.4	3.29	424	59.3
	36.7	300	150	6.5	9	13	46.78	7,210	508	12.4	3.29	481	67.7
300 x 200	56.8	294	200	8	12	18	72.38	11,300	1,600	12.5	4.71	771	160
	65.4	298	201	9	14	18	83.36	13,300	1,900	12.6	4.77	893	189
300 x 300	84.5	294	302	12	12	18	107.7	16,900	5,520	12.5	7.16	1,150	365
	87	298	299	9	14	18	110.8	18,800	6,240	13.0	7.51	1,270	417
	94	300	300	10	15	18	119.8	20,400	6,750	13.1	7.51	1,360	450
	106	300	305	15	15	18	134.8	21,500	7,100	12.6	7.26	1,440	466
	106	304	301	11	17	18	134.8	23,400	7,730	13.2	7.57	1,540	514
350 x 175	41.4	346	174	6	9	14	52.68	11,100	798	14.5	3.88	641	91
	49.6	350	175	7	11	14	63.14	13,600	984	14.7	3.95	775	112
350 x 250	69.2	336	249	8	12	20	88.15	18,500	3,090	14.5	5.92	1,100	248
	79.7	340	250	9	14	20	101.5	21,700	3,650	14.6	6.00	1,280	292



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Section size	Unit weight M	Section depth D	Flange width B	Thickness		Corner radius r	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				t (web)	T (flange)			Ix	Iy	ix	iy	Zx	Zy
mm	kg/m	mm	mm	mm	mm	mm	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
350 x 350	106	338	351	13	13	20	135.3	28,200	9,380	14.4	8.33	1,670	534
	115	344	348	10	16	20	146.0	33,300	11,200	15.1	8.78	1,940	646
	131	344	354	16	16	20	166.6	35,300	11,800	14.6	8.43	2,050	669
	137	350	350	12	19	20	173.9	40,300	13,600	15.2	8.84	2,300	776
	156	350	357	19	19	20	198.4	42,800	14,400	14.7	8.53	2,450	809
	159	356	352	14	22	20	202.0	47,600	16,000	15.3	8.90	2,670	909
400 x 200	56.6	396	199	7	11	16	72.16	20,000	1,450	16.7	4.48	1,010	145
	66.0	400	200	8	13	16	84.12	23,700	1,740	16.8	4.54	1,190	174
400 x 300	94.3	386	299	9	14	22	120.1	33,700	6,240	16.7	7.21	1,740	418
	107	390	300	10	16	22	136.0	38,700	7,210	16.9	7.28	1,980	481
400 x 400	140	388	402	15	15	22	178.5	49,000	16,300	16.6	9.54	2,250	809
	147	394	398	11	18	22	186.8	56,100	18,900	17.3	10.10	2,850	951
	168	394	405	18	18	22	214.4	49,700	20,000	16.7	9.65	3,030	985
	172	400	400	13	21	22	218.7	66,600	22,400	17.5	10.10	3,330	1,120
	197	400	408	21	21	22	250.7	70,900	23,800	16.8	9.75	3,540	1,170
	200	406	403	16	24	22	254.9	78,000	26,200	17.5	10.10	3,840	1,300
	232	414	405	18	28	22	295.5	92,800	31,000	17.7	10.20	4,480	1,530
	283	428	407	20	35	22	360.7	119,000	39,400	18.2	10.40	5,570	1,930
	415	458	417	30	50	22	528.6	187,000	60,500	18.8	10.70	8,170	2,900
	605	498	432	45	70	22	770.1	298,000	94,400	19.7	11.10	12,000	4,370
450 x 200	66.2	446	199	8	12	18	84.30	28,700	1,580	18.5	4.33	1,290	159
	76.0	450	200	9	14	18	96.76	33,500	1,870	18.6	4.40	1,490	187
450 x 300	106	434	299	10	15	24	135.0	46,800	6,690	18.6	7.04	2,160	448
	124	440	300	11	18	24	157.4	56,100	8,110	18.9	7.18	2,550	541
500 x 200	79.5	496	199	9	14	20	101.3	41,900	1,840	20.3	4.27	1,690	185
	89.6	500	200	10	16	20	114.2	47,800	2,140	20.5	4.33	1,910	214
	103	506	201	11	19	20	131.3	56,500	2,580	20.7	4.43	2,230	257
500 x 300	114	482	300	11	15	26	145.5	60,400	6,760	20.4	6.82	2,500	451
	128	488	300	11	18	26	163.5	71,000	8,110	20.8	7.04	2,910	541
600 x 200	94.6	596	199	10	15	22	120.5	68,700	1,980	23.9	4.05	2,310	199
	106	600	200	11	17	22	134.4	77,600	2,280	24.0	4.12	2,590	228
	120	606	201	12	20	22	152.5	90,400	2,720	24.3	4.22	2,980	271
	134	612	202	13	23	22	107.7	103,000	3,180	24.6	4.31	3,380	314
600 x 300	137	582	300	12	17	28	174.5	103,000	7,670	24.3	6.63	3,530	511
	151	588	300	12	20	28	192.5	118,000	9,020	24.8	6.85	4,020	601
	175	594	302	14	23	28	222.4	137,000	10,600	28.6	6.90	4,620	701
700 x 300	166	692	300	13	20	28	211.5	172,000	9,020	28.6	6.53	4,980	602
	185	700	300	13	24	28	235.5	201,000	10,800	29.3	6.78	5,760	722
	215	708	302	14	28	28	273.6	237,000	12,900	29.4	6.86	6,700	853
800 x 300	191	792	300	14	22	28	243.4	254,000	9,930	32.3	6.39	6,410	662
	210	800	300	14	26	28	267.4	292,000	11,700	33.0	6.62	7,290	782
	241	808	302	16	30	28	307.6	339,000	13,800	33.2	6.70	8,400	915
900 x 300	213	890	299	15	23	28	270.9	345,000	10,300	35.7	6.16	7,760	688
	243	900	300	16	28	28	309.8	411,000	12,600	36.4	6.39	9,140	843
	286	912	302	18	34	28	364.0	498,000	15,700	37.0	6.46	10,900	1,040

LIGHT BEAMS AND JOISTS

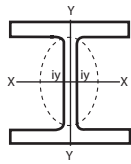


Section size	Unit weight		Section depth		Flange width		Web thickness		Flange thickness		Corner radius		Section area	
	lb/ft	kg/m	in	mm	in	mm	in	mm	in	mm	in	mm	in ²	cm ²
6 x 4 (J)	8.5	12.65	5.83	148	3.94	100	0.17	4.32	0.194	4.93	0.25	6.35	2.50	16.1
6 x 4 (L)	12	17.86	6.00	152	4.00	102	0.23	5.84	0.279	7.09	0.25	6.35	3.53	22.8
	16	23.81	6.25	159	4.03	102	0.26	6.60	0.404	10.26	0.25	6.35	4.72	30.5
8 x 4 (J)	10	14.88	7.90	201	3.94	100	0.17	4.32	0.204	5.18	0.30	7.62	2.95	19.0
8 x 4 (L)	13	19.35	8.00	203	4.00	102	0.23	5.84	0.254	6.45	0.30	7.62	3.83	24.7
	15	22.32	8.12	206	4.015	102	0.245	6.22	0.314	7.98	0.30	7.62	4.43	28.6
10 x 4 (J)	11.5	17.11	9.87	251	3.95	100	0.18	4.57	0.204	5.18	0.30	7.62	3.39	21.9
10 x 4 (L)	15	22.32	10.00	254	4.00	102	0.23	5.84	0.269	6.83	0.30	7.62	4.40	28.4
	17	25.30	10.12	257	4.01	102	0.24	6.10	0.329	8.36	0.30	7.62	4.98	32.1
	19	28.28	10.25	260	4.02	102	0.25	6.35	0.394	10.01	0.30	7.62	5.61	36.2
12 x 4 (J)	14	20.83	11.91	303	3.97	101	0.20	5.08	0.224	5.69	0.30	7.62	4.14	26.7
12 x 4 (L)	16.5	24.55	12.00	305	4.00	102	0.23	5.84	0.269	6.83	0.30	7.62	4.86	31.4
	19	28.28	12.16	309	4.01	102	0.24	6.10	0.349	8.86	0.30	7.62	5.62	36.3
	22	32.74	12.31	313	4.03	102	0.26	6.60	0.424	10.77	0.30	7.62	6.47	41.7
14 x 4 (L)	17.2*	25.60*	14.00*	356*	4.00*	102*	0.21*	5.33*	0.272	6.91*	0.43*	10.9*	5.05*	32.6*
14 x 5 (L)	22	32.74	13.72	348	5.00	127	0.23	5.84	0.335	8.51	0.43	10.9	6.47	41.7
	26	38.69	13.89	353	5.025	128	0.255	6.48	0.418	10.62	0.43	10.9	7.65	49.4
16 x 5 1/2 (L)	26	38.69	15.65	398	5.50	140	0.25	6.35	0.345	8.76	0.43	10.9	7.65	49.4
	31	46.13	15.84	402	5.525	140	0.275	6.99	0.442	11.23	0.43	10.9	9.12	58.8

L = Light beams, J = Joists

Note:

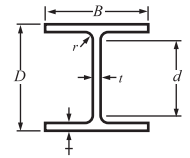
Sizes indicated by an asterisk (*) are not included in regular rolling schedules.



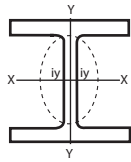
LIGHT BEAMS AND JOISTS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
	in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
6 x 4 (J)	14.8	616	1.89	78.66	2.43	6.17	0.87	2.21	5.07	83.1	0.96	15.73
6 x 4 (L)	21.7	903.3	2.89	120.3	2.48	6.30	0.90	2.29	7.24	118.7	1.44	23.60
	31.7	1,320	4.32	179.8	2.59	6.58	0.96	2.44	10.1	165.5	2.14	35.07
8 x 4 (J)	30.8	1,282	1.99	82.82	3.23	8.20	0.82	2.08	7.79	127.7	1.01	16.55
8 x 4 (L)	39.5	1,644	2.62	109.1	3.21	8.15	0.83	2.11	9.88	161.9	1.31	21.47
	48.0	1,994	3.30	137.4	3.29	8.36	0.86	2.18	11.8	193.4	1.65	27.04
10 x 4 (J)	51.9	2,160	2.01	83.66	3.92	9.96	0.77	1.96	10.5	172.1	1.02	16.72
10 x 4 (L)	68.8	2,864	2.79	116.1	3.95	10.03	0.80	2.03	13.8	226.2	1.39	22.78
	81.8	3,405	3.45	143.6	4.05	10.29	0.83	2.11	16.2	265.5	1.72	28.19
	96.2	4,005	4.19	174.4	4.14	10.52	0.86	2.18	18.8	308.1	2.08	34.09
12 x 4 (J)	88.2	3,671	2.25	93.65	4.61	11.71	0.74	1.88	14.8	242.5	1.13	18.52
12 x 4 (L)	105.3	4,383	2.79	116.1	4.65	11.81	0.76	1.93	17.5	286.8	1.39	22.78
	130.1	5,416	3.67	152.8	4.81	12.22	0.81	2.06	21.4	350.7	1.83	29.99
	155.7	6,482	4.55	189.4	4.91	12.47	0.84	2.13	25.3	414.7	2.26	37.04
14 x 4 (L)	147.3*	6,131*	2.65*	110.3 *	5.40*	13.72*	0.72*	1.83*	21.0 *	344.2*	1.32*	21.63*
14 x 5 (L)	197.4	8,212	6.40	266.4	5.52	14.02	0.99	2.51	28.8	472.0	2.56	41.96
	242.6	10,100	8.26	343.8	5.63	14.30	1.04	2.64	34.9	572.0	3.29	53.92
16 x 5 1/2 (L)	298.1	12,410	8.71	362.5	6.24	15.85	1.07	2.72	38.1	624.5	3.17	51.96
	372.5	15,500	11.57	481.5	6.39	16.23	1.13	2.87	47.0	770.3	4.19	68.67

LIGHT BEAMS AND JOISTS



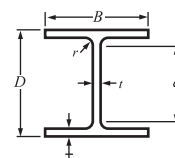
Section size	Unit weight M		Web thickness t		Flange thickness T		Radius Root r ₁		Toe r ₂		Section Area A	
	kg/m	lb/ft	mm	in	mm	in	mm	in	mm	in	cm ²	in ²
76 x 51 (3 x 2)	6.71	4.5	3.81	0.15	5.69	0.224	6.90	0.27	2.4	0.094	8.5	1.32
102 x 64 (4 x 2 1/2)	9.65	6.5	4.09	0.161	6.63	0.261	6.90	0.27	2.4	0.094	12.3	1.91
102 x 76 (4 x 3)	14.90	10	6.10	0.24	8.81	0.347	9.40	0.37	4.6	0.18	18.9	2.94
127 x 76 (5 x 3)	13.36	9	4.52	0.178	7.62	0.30	7.90	0.31	2.4	0.094	17.0	2.64
152 x 76 (6 x 3)	17.88	12	5.84	0.23	9.57	0.377	9.40	0.37	4.6	0.18	22.8	3.53
152 x 89 (6 x 3 1/2)	17.09	11.5	4.93	0.194	8.28	0.326	7.90	0.31	2.4	0.094	21.8	3.38
152 x 127 (6 x 5)	37.25	25	10.41	0.41	13.21	0.520	13.50	0.53	6.6	0.26	47.5	7.37
178 x 102 (7 x 4)	21.54	14.5	5.33	0.21	9.07	0.357	9.40	0.37	3.2	0.13	27.4	4.26
178 x 102 (7 x 4)	23.84	16	6.35	0.25	9.83	0.387	11.40	0.45	5.6	0.22	30.6	4.75
203 x 102 (8 x 4)	25.33	17	5.84	0.23	10.36	0.408	9.40	0.37	3.2	0.13	32.3	5.00
203 x 102 (8 x 4)	26.82	18	7.11	0.28	10.11	0.398	11.40	0.45	5.6	0.22	34.2	5.30
203 x 152 (8 x 6)	52.16	35	8.89	0.35	16.46	0.648	15.50	0.61	7.6	0.30	66.5	10.30
228 x 102 (9 x 4)	31.29	21	7.62	0.30	11.61	0.457	11.40	0.45	5.6	0.22	39.9	6.18
254 x 127 (10 x 5)44.71	30	9.14	0.36	14.02	0.552	13.50	0.53	6.6	0.26	57.1	8.85	
254 x 152 (10 x 6)59.61	40	9.14	0.36	18.01	0.709	15.50	0.61	7.6	0.30	75.9	11.77	
254 x 203 (10 x 8)81.97	55	10.16	0.40	19.89	0.783	19.60	0.77	9.7	0.38	104.4	16.18	
305 x 127 (12 x 5)47.69	32	8.89	0.35	11.61	0.550	13.50	0.53	6.6	0.26	60.9	9.45	
305 x 152 (12 x 6)65.57	44	10.16	0.40	18.21	0.717	15.50	0.61	7.6	0.30	83.9	13.00	



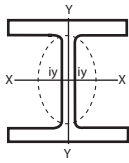
LIGHT BEAMS AND JOISTS

Section Size	Moment of inertia				Radius of gyration				Modulus of section			
	Ix		Iy		ix		iy		Zx		Zy	
mm (inch)	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm ³	in ³	cm ³	in ³
76 x 51 (3 x 2)	83	2.0	11.1	0.27	3.12	1.23	1.14	0.45	21.67	1.33	4.37	0.27
102 x 64 (4 x 2½)	218	5.2	25.3	0.61	4.21	1.66	1.43	0.57	42.84	2.62	7.97	0.49
102 x 76 (4 x 3)	325	7.8	55.4	1.33	4.14	1.63	1.70	0.67	63.7	3.89	14.4	0.88
127 x 76 (5 x 3)	476	11.5	50.2	1.21	5.29	2.08	1.72	0.68	74.9	4.58	13.17	0.81
152 x 76 (6 x 3)	874	21.5	60.8	1.46	6.20	2.44	1.63	0.64	114.7	7.00	15.9	0.97
152 x 89 (6 x 3½)	881	21.7	86.0	2.07	6.36	2.51	1.99	0.78	115.6	7.07	19.34	1.18
152 x 127 (6 x 5)	1,819	43.7	378.7	9.10	6.20	2.44	2.82	1.11	239.1	14.56	59.6	3.64
178 x 102 (7 X 4)	1,519	36.6	139.2	3.36	7.44	2.93	2.25	0.89	170.9	10.45	27.41	1.68
178 x 102 (7 x 4)	1,644	39.5	140.3	3.37	7.34	2.89	2.13	0.84	185.0	11.29	27.7	1.69
203 x 102 (8 x 4)	2,294	55.1	162.6	3.91	8.43	3.32	2.25	0.88	225.8	13.78	32.02	1.95
203 x 102 (8 x 4)	2,314	55.6	146.1	3.51	8.23	3.24	2.06	0.81	227.9	13.91	28.7	1.75
203 x 152 (8 x 6)	4,790	115.1	813.3	19.54	8.48	3.34	3.51	1.38	471.3	28.76	106.7	6.51
228 x 102 (9 x 4)	3,375	81.1	172.7	4.15	9.19	3.62	2.08	0.82	295.5	18.03	33.9	2.07
254 x 127 (10 x 5)	6,085	146.2	405.0	9.73	10.31	4.06	2.67	1.05	479.3	29.25	63.7	3.89
254 x 152 (10 x 6)	8,524	204.8	905.6	21.76	10.59	4.17	3.45	1.36	671.2	40.96	118.8	7.25
254 x 203 (10 x 8)	12,016	288.7	2,278.3	54.74	10.72	4.22	4.67	1.84	946.2	57.74	224.3	13.69
305 x 127 (12 x 5)	9,202	221.1	403.3	9.69	12.29	4.84	2.56	1.01	603.7	36.84	63.6	3.88
305 x 152 (12 x 6)	13,181	316.7	920.6	22.12	12.55	4.94	3.30	1.30	865.1	52.79	120.8	7.37

BEARING PILES



Section size	Unit weight M		Section depth D		Flange width B		Web & Flange thickness t T		Corner radius r		Section area A	
	lb/ft	kg/m	in	mm	in	mm	in	mm	in	mm	in ²	cm ²
8 x 8	30	45.0	7.88	200.2	8.085	205.4	0.373	9.5	0.4	10.2	8.83	57.0
	36	54.0	8.026	203.9	8.158	207.2	0.446	11.3	0.4	10.2	10.6	68.4
10 x 10	42	63.0	9.72	246.9	10.078	256.0	0.418	10.6	0.5	12.7	12.4	79.7
	48	71.0	9.838	249.9	10.137	257.5	0.477	12.1	0.5	12.7	14.1	91.0
	57	85.0	10.012	254.3	10.224	259.7	0.564	14.3	0.5	12.7	16.8	108.1
12 x 12	53	79.0	11.78	299.2	12.046	306.0	0.436	11.1	0.6	15.2	15.5	100.4
	59	88.0	11.878	301.7	12.095	307.2	0.485	12.3	0.6	15.2	17.36	112.0
	64	95.0	11.96	303.8	12.136	308.3	0.526	13.4	0.6	15.2	18.4	121.0
	74	110.0	12.122	307.9	12.217	310.3	0.607	15.4	0.6	15.2	21.8	140.4
	85	126.0	12.308	312.6	12.31	312.7	0.70	17.8	0.6	15.2	24.6	161.0
	100	149.0	12.528	318.2	12.42	315.5	0.81	20.6	0.6	15.2	29.4	190.0
	125	186.0	12.928	328.4	12.62	320.5	1.01	25.7	0.6	15.2	36.73	237.0
	150	223.0	13.308	338.0	12.81	325.4	1.20	30.5	0.6	15.2	44.17	285.0
14 x 14 1/2	73	109.0	13.636	346.4	14.586	370.5	0.506	12.9	0.6	15.2	21.4	138.4
	89	133.0	13.856	351.9	14.696	373.3	0.616	15.6	0.6	15.2	26.1	169.0
	102	152.0	14.032	356.4	14.784	375.5	0.704	17.9	0.6	15.2	30.0	193.6
	117	174.1	14.234	361.5	14.885	378.1	0.805	20.4	0.6	15.2	34.4	222.2



BEARING PILES

Section Size		Moment of inertia				Radius of gyration				Modulus of section			
		Ix		Iy		ix		iy		Zx		Zy	
in (mm)		in ⁴	cm ⁴	in ⁴	cm ⁴	in	cm	in	cm	in ³	cm ³	in ³	cm ³
8 x 8		98.0	4,079	37.0	1,539	3.33	8.46	1.93	4.90	24.9	408	8.1	133
		119.8	4,987	40.4	1,683	3.36	8.54	1.95	4.95	29.8	489	9.9	162
10 x 10	210.8	8,775	71.4	2,971	4.13	10.5	2.41	6.11	43.4	711	14.2	232	
		243.9	10,153	82.9	3,451	4.17	10.6	2.40	6.10	49.6	813	16.4	268
		294.6	12,264	100.6	4,188	4.21	10.7	2.45	6.22	58.9	965	19.7	323
12 x 12	394.0	16,400	127.1	5,292	5.04	12.8	2.86	7.26	66.9	1,096	21.1	346	
		442.1	18,402	143.2	5,959	5.04	12.8	2.87	7.30	74.4	1,220	23.7	388
		483.2	20,111	156.9	6,529	5.08	12.9	2.87	7.30	80.8	1,324	25.9	424
		566.5	23,580	184.7	7,689	5.12	13.0	2.91	7.40	93.5	1,532	30.3	496
		661.3	27,526	216.5	9,013	5.16	13.1	2.95	7.50	107.6	1,763	35.2	577
		793.8	33,040	261.1	10,869	5.20	13.2	2.99	7.60	126.6	2,075	42.0	689
		1,024.1	42,625	338.9	14,108	5.28	13.4	3.03	7.70	158.5	2,597	53.8	881
		1,268.9	52,817	422.1	17,570	5.35	13.6	3.07	7.80	190.8	3,126	65.9	1,080
14 x 14 ^{1/2}		733.1	30,515	261.9	10,901	5.83	14.8	3.49	8.87	107.5	1,762	35.9	588
		909.1	37,840	326.2	13,576	5.91	15.0	3.53	8.96	131.2	2,150	44.4	727
		1,055.1	43,916	379.6	15,799	5.94	15.1	3.56	9.03	150.4	2,464	51.3	841
		1,228.5	51,134	443.1	18,444	5.98	15.2	3.59	9.11	172.6	2,829	59.6	976



STEEL PILES

LARSEN AND FRODINGHAM SHEET PILES

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U-TYPE

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Z-TYPE / STRAIGHT WEB-TYPE

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YICK HOE

STEEL PILES

Larssen and Frodingham profiles of sheet piling are well known to civil engineers throughout the world.

Their usefulness and reliability in all types of earth and water retaining structures has been proven on countless occasions over many years. However, with the benefit of experience and research, the range and efficiency of the sections has continually been improved.

The following information contains full details of a major development - the new range of Larssen profiles recently developed.

The new range provides sections which are markedly superior to those previously available in terms of strength to weight ratios, but which have also been carefully proportioned to retain the traditionally highly valued driving characteristics.

More comprehensive information on the design and installation of all sections is available in the Piling Handbook, available from the publisher of this handbook - CHYE HIN HARDWARE PTE LTD.

STEEL QUALITIES

	Ultimate stress		Minimum yield stress				Minimum elongation on 200 mm	
			Up to and including 16 mm thick		Over 16 mm up to and including 40 mm thick		Up to and including 9 mm thick	Over 9mm thick
	kg/mm ²	N/mm ²	kg/mm ²	N/mm ²	kg/mm ²	N/mm ²	%	%
BS.4360:1986 (mild steel) Grade FE 430A	43.8/59.0	430/580	28.0	275	27.0	265	16	20
BS.4360:1986 (High yield steel) Grade FE 510A	49.9/65.2	490/640	36.2	355	35.2	345	15	18
Similar to ASTM-A328	49.2 min.	482.6 min.	27.0	265	27.0	265	--	17

Notes:

COPPER BEARING STEEL - all the grades of steel are available with 0.20% up to 0.35% or over 0.35% up to 0.50% Copper content.

1N/mm² = 0.101972 kg/mm²

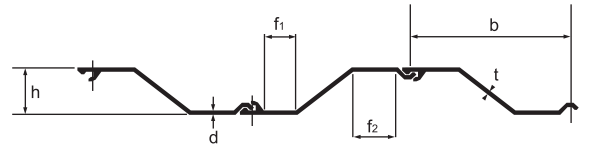
RECOMMENDED WORKING STRESSES FOR STEEL SHEET PILING

Class of work	BS.4360 Grade 43A ASTM A328 and CSA G40.7 (mild steel)		BS.4360 Grade 50A (high yield steel)	
	kg/mm ²	N/mm ²	kg/mm ²	N/mm ²
Permanent	14.2	140	18.3	180
Temporary	17.8	175	22.9	225

Note:

Stresses in temporary conditions occurring during construction may be varied at the discretion of the client's engineer.

Sheet



Frodingham Steel Sheet Piling - Dimensions and Properties

Section	b mm (nom.)	h mm (nom.)	d mm	t mm (nom.)	f ₁ mm (nom.)	f ₂ mm (nom.)	Section area sq. cm per metre of wall	Unit weight kg per linear metre		Moment of inertia cm ⁴ per metre	Modulus of section cm ³ per metre
1BXN	476	143	12.7	12.7	78	123	166.5	62.1	130.4	4,919	688
1N	483	170	9.0	9.0	105	137	126.0	47.8	99.1	6,048	713
2N	483	235	9.7	8.4	97	149	143.0	54.2	112.3	13,513	1,150
3N	483	283	11.7	8.9	89	145	175.0	66.2	137.1	23,885	1,688
3NA	483	305	9.7	9.5	96	146	165.0	62.6	129.8	25,687	1,690
4N	483	330	14.0	10.4	77	127	218.0	82.4	170.8	39,831	2,414
5	425	311	17.0	11.9	89	118	302.0	100.8	236.9	49,262	3,168

Rolling margin is within 4 per cent over and 2.5 per cent under theoretical mass; 75 mm over and 50 mm under on length.

*Please check on availability before specifying.

Special Sections

Sections may be "rolled up" (thickened) or "rolled down" (thinned) by special arrangement to increase or decrease the thickness of both webs and flanges by a maximum of 0.8 mm.

INTERLOCKING SECTIONS

Section (Tongue)	Interlocks with (Groove)	Section (Groove)	Interlocks with (Tongue)
1N	2N, 3NA	1N	NONE
1BXN	3N, 4N	1BXN	2N, 3NA
2N	1BXN, 3N, 3NA	2N	1N, 3NA
3N	4N	3N	1BXN, 2N, 3NA
3NA	3N, 1BXN, 2N	3NA	1N, 2N
4N	5	4N	1BXN, 3N
5	NONE	5	4N

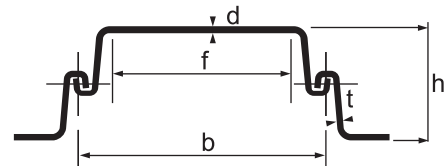
RECOMMENDED MAXIMUM LENGTHS FOR DRIVING

The maximum length for each piling section depends upon the type of strata encountered, penetration required, and the type of construction for which the piling is designed. The following table is provided as a guide only.

In hard driving conditions it may be necessary to move up a section size to achieve the required penetration. Alternatively Grade 50A may be used.

Section	Approx. max. length (metres)	Section	Approx. max. length (metres)	Section	Approx. max. length (metres)
6W	9	32W	26	2N	14
9W	14	3	18	3N	18
12W	17	4A	23	3NA	18
16W	20	6	30	4N	23
20W	23	1N	11	5	24
25W	23	1BXN	14		

STEEL PILES



LARSEN STEEL SHEET PILING - Dimensions and Properties

Section	b mm (nom.)	h mm (nom.)	d mm	t mm (nom.)	f Flat of pan mm	Section area sq. cm per metre of wall	Unit weight		Combined moment of inertia cm ⁴ per metre	Modulus of section cm ³ per metre
							kg per linear metre	kg per sq. metre of wall		
6W	525	212	7.8	6.4	331	108	44.7	85.1	6,371	601
9W	525	260	8.9	6.4	343	124	51.0	97.1	11,726	902
12W	525	306	9.0	8.5	343	147	60.4	115.1	18,345	1,199
16W	525	348	10.5	8.9	341	166	68.3	130.1	27,857	1,601
20W	525	400	11.3	9.4	333	188	77.3	147.2	40,180	2,009
25W	525	454	12.1	10.5	317	213	87.9	167.4	56,727	2,499
32W	525	454	17.0	10.5	317	252	103.6	197.4	73,003	3,216
3	400	250	14.1	8.5	248	198	62.2	155.5	16,980	1,360
4A	400	381	15.7	9.6	219	236	74.0	185.1	44,916	2,360
6	420	440	22.0	14.0	248	370	122.0	290.5	92,452	4,200
6	420	440	25.4	14.0	251	397	131.0	311.8	102,861	4,675
6	420	440	28.6	14.0	251	421	138.7	330.2	111,450	5,066
GSP3	400	250	13.0	8.6	271	191	60.0	150.0	16,800	1,340

Rolling margin is within 4 per cent over and 2.5 per cent under theoretical mass; 75mm over and 50 mm under on length.

	Section	Area	Dimension A	Moment of inertia about axis XX cm ⁴	Section modulus about axis XX cm ³
		cm ²	mm		
	6W	57	45	1,247	155
	9W	65	50	2,031	204
	12W	77	62	3,240	288
	16W	87	66	4,495	347
	20W	98	79	6,737	467
	25W	112	93	9,605	607
	32W	132	82	10,580	626
	3	79	49	2,227	227
	4A	94	76	5,650	412
	6 (22.0)	155	90	12,437	770
	6 (25.4)	167	85	12,882	777
	6 (28.6)	177	82	13,185	780
	GSP3	76.4	--	2,220	223

INTERLOCKING SECTIONS

Section	Interlocks with	Section	Interlocks with
6W	9W, 12W, 16W, 4A	25W	12W, 16W, 20W, 32W, 3, 4A, 6
9W	6W, 12W, 16W, 4A	32W	12W, 16W, 20W, 25W, 3, 4A, 6
12W	6W, 9W, 16W, 20W, 25W, 32W, 4A	3	20W, 25W, 32W, 4A
16W	6W, 9W, 12W, 20W, 25W, 32W, 4A	4A	6W, 9W, 12W, 16W, 20W, 25W, 32W, 3
20W	12W, 16W, 25W, 32W, 3, 4A, 6	6	20W, 25W, 32W

GENERAL INFORMATION

Steel sheet piling is used in all types of temporary works and permanent structures including cofferdams, retaining walls, river frontages, quays, wharves, dock and harbour works, permanent foundations, land reclamation and sea defence works.

The sections are designed to provide the maximum strength and durability at the lowest possible weight consistent with good driving qualities. The design of the section interlocks facilitates pitching and driving and results in a series of closely fitting joints, forming a continuous wall.

A comprehensive range of sections in widely differing sizes and weights is obtainable in various grades of steel which enables the most economical choice to be made to suit the nature and requirements of any given contract. This range can be extended if the tonnage required is economic.

Certain sections intended for use in permanent structures where stresses are not severe, such as cut-off-walls, are of uniform thickness throughout in order to provide the maximum effective useful life.

Where piles form temporary works, they can be extracted easily and without distortion for re-use.

Corner, junction and closure piles are available to cover all requirements.

Should any doubts or problems arise advice is available on application to the publisher of this handbook - CHYE HIN HARDWARE PTE LTD.

SPLICES

Steel sheet piling is generally available in lengths up to 30m. Where it is necessary to increase the pile length during driving, fishplated or site-welded joints can be used. Particulars of standard fish-plated joints or weld preparations are available on application.

Because of normal rolling tolerances there may be variations in the pile profile. Every effort should be made to match pile ends before welding.

CIRCULAR CONSTRUCTION

Steel sheet piling can be driven to form a complete circle without the water tightness of the locks being impaired.

The maximum angle of deviation in the locks is 9° for Larssen piles, and 3° for Frodingham piles. Deviation in the locks of Larssen Box Piles is limited to 3°.

The following table gives the approximate minimum diameters of circles which can be constructed using various sheet pile sections.

The diameters are only intended for guidance, as the possible deviation will vary according to length of piles, penetration required and section used.

Smaller diameters can be obtained if individual bent corners are introduced.

STEEL PILES

Type	Section	Minimum number of piles used	Approx. minimum diameter at internal face of wall (metres)	Approx. maximum driving length at minimum diameter (metres)
Larssen	6W	40	6.47	6
Larssen	9W	40	6.42	9
Larssen	12W	40	6.38	9
Larssen	16W	40	6.34	15
Larssen	20W	40	6.28	15
Larssen	25W	40	6.23	15
Larssen	32W	40	6.23	15
Larssen	3	40	4.85	9
Larssen	4A	40	4.71	15
Larssen	6	40	4.91	15
Frodingham	1N	120	18.36	6
Frodingham	1BXN	120	18.10	6
Frodingham	2N	120	18.32	9
Frodingham	3N, 3NA	120	18.30	15
Frodingham	4N	120	18.28	15
Frodingham	5	120	16.07	15

EFFECTIVE LIFE

The effective life of unpainted or otherwise unprotected steel piling to BS 4360 Grades 43A and 50A depends upon the combined effects of imposed stresses and corrosion.

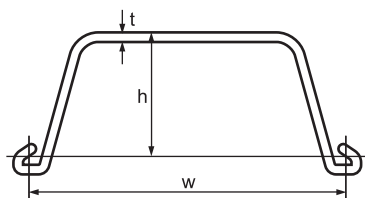
Performance is clearly optimised with combinations of high stress and low corrosion rate. Although other stress and corrosion combinations are less favourable, good design can ensure acceptable lives. The use of grade 50A steel at grade 43A stresses can also increase effective life. Copper-bearing steels are marginally advantageous but only in atmospheric exposures.

The table below indicates the minimum effective life for various environments assuming that maximum permissible design stress occurs in the zone under consideration. For regions of lower stress, the permissible loss in thickness may be increased and appropriate effective lives calculated from the corrosion data for each service environment.

Minimum Effective Life for Maximum Stress in Various Service Environments

Section type		Environments				
Frodingham	Larssen	Permissible loss in thickness (mm) at position of max stress in pile	Atmospheric corrosion	Splash and low water zones	Sea water immersion and tidal zone	Underground
			Corrosion rates, mm/year			
			0.05 (mean)	0.09 (mean)	0.05 (mean)	0.03 (max)
			Effective life in years			
	6W, 9W	3.5	70	*	*	117
	12W, 16W	4.5	90	50	90	120 +
	20W, 3, 4A	5.0	100	55	100	120 +
1N		5.2	104	58	104	120 +
2N		5.4	108	60	108	120 +
3NA		5.5	110	60	110	120 +
3N		5.9	118	65	118	120 +
	25W, 32W	6.0	120	67	120	120 +
	All No. 6	7.0	120 +	78	120 +	120 +
1BXN		7.2	120 +	80	120 +	120 +
4N		7.4	120 +	82	120 +	120 +
5		8.9	120 +	99	120 +	120 +

* Not suitable for Marine Environment.



U-TYPE

KSP DIMENSIONS & PROPERTIES

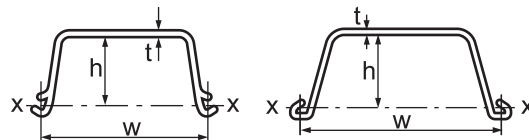
Section number	Dimensions			Section area Per pile	Unit weight		Moment of inertia		Modulus of section	
	w mm	h mm	t mm		Per pile kg/m	Per linear meter of wall kg/m ²	Per pile cm ⁴	Per linear meter of wall cm ⁴ /m	Per pile cm ³	Per linear meter of wall cm ³ /m
KSP-IA	400	85	8.0	45.2	35.5	88.8	598	4,500	88	529
KSP-II	400	100	10.5	61.2	48.0	120	1,190	8,700	146	870
KSP-IIA	400	120	9.2	55.0	43.2	108	1,450	10,600	162	880
KSP-III	400	130	13.0	76.4	60.0	150	2,320	17,400	232	1,340
KSP-IIIA	400	150	13.1	74.4	58.4	146	2,840	22,800	253	1,520
KSP-IV	400	160	16.0	96.9	76.1	190	4,110	34,400	334	2,150
KSP-IVA	400	180	16.2	94.2	74.0	185	4,910	40,600	374	2,250
KSP-VL	500	200	24.3	133.8	105.0	210	7,960	63,000	520	3,150
KSP-VIL	500	225	27.6	153.0	120.0	240	11,400	86,000	680	3,820

U-TYPE

Imperial units

Section number	Dimensions			Section area Per pile	Unit weight		Moment of inertia		Modulus of section	
	w in.	h in.	t in.		Per pile lb/ft	Per linear foot of wall lb/ft ²	Per pile in ⁴	Per linear foot of wall in ⁴ /ft	Per pile in ³	Per linear foot of wall in ³ /ft
KSP-IA	15.75	3.35	0.31	7.01	23.9	18.2	14	33	5.4	9.8
KSP-II	15.75	3.94	0.41	9.49	32.3	24.6	29	64	8.9	16.2
KSP-IIA	15.75	4.72	0.36	8.53	29.0	22.1	35	78	9.9	16.4
KSP-III	15.75	5.12	0.51	11.84	40.3	30.7	56	127	14.2	24.9
KSP-IIIA	15.75	5.91	0.52	11.53	39.2	29.9	68	167	15.4	28.3
KSP-IV	15.75	6.30	0.63	15.02	51.1	38.9	99	252	20.4	40.0
KSP-IVA	15.75	7.09	0.64	14.60	49.7	37.9	118	297	22.8	41.9
KSP-VL	19.68	7.87	0.96	20.74	70.6	43.0	191	461	31.7	58.6
KSP-VIL	19.68	8.86	1.09	23.72	80.6	49.2	274	630	41.5	71.1

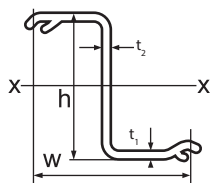
STEEL PILES



U-TYPE

Section	Dimensions			Section area Per pile	Unit weight		Moment of inertia		Modulus of section	
	w	h	t		Per pile	Per wall width	Per pile	Per wall width	Per pile	Per wall width
	mm in	mm in	mm in		kg/m lb/ft	kg/m ² lb/ft ²	cm ⁴ in ⁴	cm ⁴ /m in ⁴ /ft	cm ³ in ³	cm ³ /m in ³ /ft
YSP I	400 15.7	75 2.95	8.0 0.315	46.49 7.206	36.5 24.5	91.2 18.7	429 10.3	3,820 28.0	66.4 4.05	509 9.47
YSP U-5	400 15.7	80 3.15	7.6 0.299	45.21 7.008	35.5 23.9	88.8 18.2	454 10.9	4,220 30.9	64.7 3.95	527 9.80
FSP IA	400 15.7	85 3.35	8.0 0.315	45.21 7.008	35.5 23.9	88.8 18.2	598 14.4	4,500 33.0	88.0 5.37	529 9.84
YSP II	400 15.7	100 3.94	10.5 0.413	61.18 9.483	48.0 32.3	120 24.6	986 23.7	8,690 63.6	121 7.38	869 16.2
FSP II	400 15.7	100 3.94	10.5 0.413	61.18 9.483	48.0 32.3	120 24.6	1,240 29.8	8,740 64.0	152 9.28	874 16.3
YSP U-9	400 15.7	110 4.33	9.3 0.366	55.01 8.527	43.2 29.0	108 22.1	1,070 25.7	9,680 70.9	120 7.32	880 16.4
FSP IIA	400 15.7	120 4.72	9.2 0.362	55.01 8.527	43.2 29.0	108 22.1	1,460 35.1	10,600 77.6	160 9.76	880 16.4
YSP III	400 15.7	125 4.92	13.0 0.512	76.42 11.85	60.0 40.3	150 30.7	1,920 46.1	16,400 120	196 12.0	1,310 24.4
FSP III	400 15.7	125 4.92	13.0 0.512	76.42 11.85	60.0 40.3	150 30.7	2,220 53.3	16,800 123	223 13.6	1,340 24.9
YSP U-15	400 15.7	150 5.91	12.2 0.48	74.40 11.53	58.4 39.2	146 29.9	2,700 64.9	22,800 167	238 14.5	1,520 28.3
FSP IIIA	400 15.7	150 5.91	13.1 0.516	74.40 11.53	58.4 39.2	146 29.9	2,790 67.0	22,800 167	250 15.3	1,520 28.3
YSP IV	400 15.7	155 6.10	15.5 0.61	96.99 15.03	76.1 51.1	190 38.9	3,690 88.7	31,900 234	311 19.0	2,060 38.3
FSP IV	400 15.7	170 6.69	15.5 0.61	96.99 15.03	76.1 51.1	190 38.9	4,670 112	38,600 283	362 22.1	2,270 42.2
YSP U-23	400 15.7	175 6.89	14.7 0.579	94.21 14.60	74.0 49.7	185 37.9	4,380 105	39,400 289	330 20.1	2,250 41.9
FSP IVA	400 15.7	185 7.28	16.1 0.634	94.21 14.60	74.0 49.7	185 37.9	5,300 127	41,600 305	400 24.4	2,250 41.9
YSP V	420 16.5	175 6.89	22.0 0.866	134.0 20.77	105.0 70.6	250 51.2	5,950 143	55,200 404	433 26.4	3,150 58.6
FSP VL	500 19.7	200 7.87	24.3 0.957	133.8 20.74	105.0 70.6	210 43.0	7,960 191	63,000 461	520 31.7	3,150 58.6
FSP VIL	500 19.7	225 8.86	27.6 1.09	153.0 23.72	120.0 80.6	240 49.2	11,400 274	86,000 630	680 41.5	3,820 71.1

Note:
Weight per linear meter (foot) of wall is rounded off using the JIS Z 8401-specified formula: Weight per section x $\frac{1,000}{W \text{ (effective width)}}$

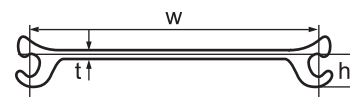


STEEL PILES

Z-TYPE

Section	Dimensions				Section area Per pile	Unit weight		Moment of inertia		Modulus of section	
	w	h	t1	t2		Per pile	Per wall width	Per pile	Per wall width	Per pile	Per wall width
	mm in	mm in	mm in	mm in	cm ² in ²	kg/m lb/ft	kg/m ² lb/ft ²	cm ⁴ in ⁴	cm ⁴ /m in ⁴ /ft	cm ³ in ³	cm ³ /m in ³ /ft
FSP Z-25	400	305	13.0	9.6	94.32	74.0	185	15,300	38,300	1,000	2,510
	15.7	12.0	0.512	0.378	14.62	49.7	37.9	368	280	61.0	46.7
FSP Z-32	400	344	14.2	10.4	107.7	84.5	211	22,000	55,000	1,280	3,200
	15.7	13.5	0.559	0.409	16.69	56.8	43.2	529	403	78.1	59.5
FSP Z-38	400	364	17.2	11.4	122.3	96.0	240	27,700	69,200	1,520	3,800
	15.7	14.3	0.677	0.449	18.96	64.5	49.2	665	507	92.8	70.7
FSP Z-45	400	367	21.9	13.2	148.2	116	290	33,400	83,500	1,820	4,550
	15.7	14.4	0.862	0.52	22.97	77.9	59.4	802	611	111	84.6

STRAIGHT WEB-TYPE



Section	Dimensions			Section area Per pile	Unit weight		Moment of inertia Per pile	Modulus of section Per pile	Interlock strength min.	
	w	h	t		Per pile	Per wall width				
	mm in	mm in	mm in	cm ² in ²	kg/m lb/ft	kg/m ² lb/ft ²	cm ⁴ in ⁴	cm ³ in ³	t/m lb/ft	
YSP F	400	44.5	9.5	69.07	54.2	136	190	47.8	400	
	15.7	1.75	0.374	10.71	36.4	27.9	4.56	2.92	268	780
YSP FA	400	44.5	12.7	77.50	60.8	152	196	48.3	400	
	15.7	1.75	0.5	12.01	40.9	31.1	4.71	2.95	268	780
YSP FX	400	47.0	12.7	85.66	67.2	168	224	56.1	600	
	15.7	1.85	0.5	13.28	45.2	34.4	5.38	3.42	403	170

Note:

1. YSP FA with minimum interlock strength of 500 t/m is subject to negotiation.
2. Normal interlock swing of YSP and YSP FA is at least 10° and that of YSP FX at least 8°.



API PIPES ERW & SEAMLESS PIPES

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API PIPES ERW & SEAMLESS PIPES

API MECHANICAL & CHEMICAL SPECIFICATION

TENSILE AND CHEMICAL REQUIREMENTS (EXCERPTS FROM API SPECIFICATIONS, 1983 EDITION)

Application	Grade	Tensile Requirements						Elongation	
		Yield Strength			Tensile Strength				% min
		psi min	MPa min	kg/mm ² min	psi min	MPa min	kg/mm ² min		
Line Pipe	API 5L Grade A	30,000	207	21.1	48,000	331	33.7	e = 625,000 x $\frac{A^{0.2}_{*2}}{U^{0.9}}$	
	API 5L Grade B	35,000	241	24.6	60,000	413	42.2		
	API 5L Grade X42	42,000	289	29.5	60,000	413	42.2		
	API 5L Grade X46	46,000	317	32.3	63,000	434	44.3		
	API 5L Grade X52	52,000	358	36.6	(1) 66,000* ¹ (2) 72,000	(1) 455 (2) 496	46.4 50.6		
	API 5L Grade X56* ⁶ 56,000	386	39.2	(1)	71,000* ¹ (2) 75,000	(1) 489 (2) 517	49.9 52.7		
	API 5L Grade X60* ³ 60,000 * ⁶	413	42.2	(1)	75,000* ¹ (2) 78,000	(1) 517 (2) 537	52.7 54.8		
	* ⁶ API 5L Grade X65* ⁸ 65,000 * ¹¹	448	45.7	(2)	(1) 77,000* ¹ 80,000	(1) 530 (2) 551	54.1 56.2		
	API 5L Grade X70* ⁶ 70,000 * ¹¹	482	49.2		82,000	565	57.6		
	API 5LU Grade X8080,000	551	56.2		95,000/ * ⁴ 125,000	655/ 862	66.8/ 87.8		
API 5LU Grade X100100,000	689	70.3		110,000/ * ⁴ 135,000	758/ 931	77.3/ 94.9			

- *1 (1) Wall thickness > 0.375 in., OD ≥ 20 in.
Any wall thickness, OD < 20 in.
(2) Wall thickness ≤ 0.375 in., OD ≥ 20 in.

- *2 The minimum elongation in 2 inches (50.8 mm) shall be determined by the following formula:

$$e = 625,000 \times \frac{A_{*2}^{0.2}}{U^{0.9}}$$

where:

e = minimum elongation in 2 inches (50.80 mm) in per cent rounded to the nearest 1/2 per cent.

A = cross sectional area of the tensile test specimen in square inches, based on specified outside diameter or nominal specimen width and specified wall thickness rounded to the nearest 0.01 sq. in., or 0.75 sq. in., whichever is smaller.

U = specified tensile strength, psi.

- *3 The minimum tensile strength for grade X60 electric-resistance welded pipe in all sizes and wall thicknesses shall be 75,000 psi (52.7 kg/mm²).

- *4 Tensile strength means min. value/max. value.

- *5 Figures in parentheses are values for seamless pipe. Upper line value means non-expanded. Lowerline value means cold-expanded. In grades X42 through X65, for each reduction of 0.01 per cent below the maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible, up to a maximum of 1.45 per cent for X52 and lower and up to a maximum of 1.60 per cent for grades higher than X52.

- *6 Other chemical analyses may be furnished by agreement between purchaser and manufacturer.

- *7 Either columbium, vanadium, titanium, or a combination thereof, shall be used at the discretion of the manufacturer.

API PIPES ERW & SEAMLESS PIPES

CHEMICAL REQUIREMENTS FOR LADLE ANALYSES, %

C ^{*5} max	Si max	Mn ^{*5} max	P max	S max	Cb min	V min	Ti min
0.21 (0.22)		0.90	0.04	0.05			
0.26 (0.27)		1.15	0.04	0.05			
0.28 (0.29) ^{*13}		1.25	0.04	0.05			
0.30 (0.31) ^{*13}		1.35	0.04	0.05			
0.28 (0.29) ^{*13}		1.25					
0.30 (0.31) ^{*13}		1.35	0.04	0.05			
0.28 (0.29) ^{*13}		1.25					
0.26		1.35	0.04	0.05	0.005 ^{*7}	0.02 ^{*7}	0.03 ^{*7}
0.26		1.35	0.04	0.05	0.005 ^{*7}	0.02 ^{*7}	0.03 ^{*7}
0.26		1.40	0.04	0.05	0.005 ^{*9}	0.02 ^{*9}	
0.23 ^{*10}		1.60 ^{*10}	0.04	0.05			
0.26 ^{*12}	0.35	1.40 ^{*12}	0.04	0.05			
0.26 ^{*12}	0.35	1.40 ^{*12}	0.04	0.05			

*8 For grades X65 in sizes 16 in. and larger with wall thickness 0.50 in. and less, the chemical composition shall be as shown or as agreed upon between the purchaser and manufacturer. For other sizes and wall thicknesses the chemical composition shall be as agreed upon between the purchaser and manufacturer. (Applicable to welded pipe only).

*9 Either columbium or vanadium or a combination of both shall be used at the discretion of the manufacturer.

*10 For each reduction of 0.01 per cent below the specified maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible.

*11 For seamless pipe of grades X65 and X70, the chemical composition shall be agreed upon between the purchaser and manufacturer.

*12 For each reduction of 0.01 per cent below the specified maximum carbon content, an increase of 0.05 per cent manganese above the specified maximum is permissible, up to a maximum of 1.50 per cent.

*13 For cold expanded seamless pipe in size 20 in. and larger, the maximum carbon content shall be 0.28 per cent.

API PIPES ERW & SEAMLESS PIPES

COMPARATIVE TABLES OF STEEL QUALITIES

MECHANICAL VALUES					CHEMICAL ANALYSIS							
Material	Tensile strength N/mm ²	Yield strength (min)	Elongation % min.	C %	Si %	Mn %	P % max.	S % max.	Mo %	Cr %	Ni %	Div %
A53 - Grade A	min. 330	205	variable	max. 0.25	--	max. 0.95	0.05	0.60				
A53 - Grade B	min. 415	240	variable	max. 0.30	--	max. 1.20	0.05	0.60				
A106 - Grade A	min. 330	205	35	max. 0.25	min 0.1	0.27 - 0.93	0.048	0.058				
A106 - Grade B	min. 415	240	30	max. 0.30	min 0.1	0.29 - 1.06	0.048	0.058				
A179	Hardness Rb. max. 72			0.06 - 0.18	--	0.27 - 0.63	0.048	0.058				
A333 - Grade 1	min. 379	207	35	max. 0.30	--	0.40 - 1.06	0.05	0.60				
A333 - Grade 6	min. 414	241	30	max. 0.30	min. 0.1	0.29 - 1.06	0.048	0.058				
A333 - Grade 3	min. 448	241	30	max. 0.19	0.18 - 0.37	0.31 - 0.64	0.05	0.05			3.18 - 3.82	
A335 - Grade P1	min. 379	207	30	0.1 - 0.2	0.1 - 0.5	0.3 - 0.8	0.045	0.045	0.44 - 0.65			
A335 - Grade P5	min. 414	207	30	max. 0.15	max. 0.5	0.3 - 0.6	0.03	0.03	0.44 - 0.65	4.0 - 6.0		
A335 - Grade P11	min. 414	207	30	max. 0.15	0.5 - 1.0	0.3 - 0.6	0.03	0.03	0.44 - 0.65	1.0 - 1.5		
A335 - Grade P12	min. 414	207	30	max. 0.15	max. 0.5	0.3 - 0.6	0.045	0.045	0.44 - 0.65	0.8 - 1.25		
A335 - Grade P22	min. 414	207	30	max. 0.15	max. 0.5	0.3 - 0.6	0.03	0.03	0.87 - 1.13	1.9 - 2.6		
API-5L - Grade A	min. 331	207	variable	max. 0.22	--	max. 0.9	0.04	0.05				
API-5L - Grade B	min. 413	241	variable	max. 0.27	--	max. 1.15	0.04	0.05				
API-5L - Grade X42	min. 413	289	variable	max. 0.29	--	max. 1.25	0.04	0.05				
API-5L - Grade X46	min. 434	317	variable	max. 0.29	--	max. 1.35	0.04	0.05				
API-5L - Grade X52	min. 455	358	variable	max. 0.31	--	max. 1.35	0.04	0.05				
API-5L - Grade X56	min. 489	386	variable	max. 0.26	--	max. 1.35	0.04	0.05				
API-5L - Grade X60	min. 517	413	variable	max. 0.26	--	max. 1.35	0.04	0.05				
BS3059/1 - 320	320-480	195	25	max. 0.16	--	min. 0.3	0.05	0.05				
BS3059/2 - 360	360-500	215	24	max. 0.17	max. 0.35	0.4 - 0.8	0.045	0.045				
BS3601 - ERW320	320-440	195	25	max. 0.16	--	0.3 - 0.7	0.05	0.05				
BS3601 - ERW360	360-480	215	24	max. 0.17	max. 0.35	0.4 - 0.8	0.045	0.045				
BS3602 - ERW/ SAW410	410-530	235	22	max. 0.21	max. 0.35	0.4 - 1.2	0.05	0.05				
BS3601-S360	360-480	215	24	max. 0.17	max. 0.35	0.4 - 0.8	0.05	0.05				
BS3601-S410	410-530	235	22	max. 0.21	max. 0.35	0.4 - 1.2	0.05	0.05				
BS3602/1-360	360-500	215	24	max. 0.17	max. 0.35	0.4 - 0.8	0.045	0.045				
BS3602/1-410	410-550	245	22	max. 0.21	max. 0.35	0.4 - 1.2	0.045	0.045				
BS3602/1-490Nb	490-630	340	20	max. 0.23	max. 0.35	0.8 - 1.5	0.045	0.045				Nb 0.01-0.1
BS3602/2-410	410-550	245	24	max. 0.2	max. 0.35	0.5 - 1.3	0.04	0.04				
BS3603-410	410-530	235	22	max. 0.2	max. 0.35	0.6 - 1.2	0.045	0.045				
BS3603-503	440-590	245	16	max. 0.15	0.15 - 0.35	0.3 - 0.8	0.025	0.02			3.25 - 3.75	
BS3604-620	440-590	290	22	0.1 - 0.18	0.10 - 0.35	0.4 - 0.7	0.05	0.05	0.45 - 0.65	0.7 - 1.1		
BS3604-622	440-640	275	20	0.08 - 0.15	max. 0.5	0.4 - 0.7	0.04	0.04	0.9 - 1.2	2.0 - 2.5		
BS3604-625	450-600	170	20	max. 0.15	max. 0.5	0.3 - 0.6	0.04	0.04	0.45 - 0.65	4.0 - 6.0		
BS4360-50D	490-640	355	20	max. 0.2	max. 0.4	max. 1.4	0.04	0.04				Nb 0.003-0.1 V 0.0003-0.1

API PIPES ERW & SEAMLESS PIPES

TECHNICAL STANDARDS			COMPARABLE QUALITIES					
Material	Specification standard	Dimensional standard	ASTM	API	BS	DIN	Euro Standard EU 25-72	DIV
A53	A53	ANSI-B 36.10		API-5L-Gr.A	3601-320/360	St. 37.0	FE 310	BS4360-43B/C
A53	A53	ANSI-B 36.10		API-5L-Gr.B	3601-410	St. 44.0	Fe 360B/430B	
A106/A530	A106/A530	ANSI-B 36.10		API-5L-Gr.A	3602-S360	St. 35.8		
A106/A530	A106/A530	ANSI-B 36.10		API-5L-Gr.B	3602-S410	St. 45.8		
A179/A450	A179/A450	ANSI-B 36.10			3059-320/360	St. 35.8		
A333/A530	A333/A530	ANSI-B 36.10			3601HFS27LT50	TT St. 35N		
A333/A530	A333/A530	ANSI-B 36.10			3603HFS503LT100	TT St. 35V		
A333/A530	A333/A530					10 Ni 14		
A335/A530	A335/A530	ANSI-B 36.10				15 Mo 3/16 Mo 5		
A335/A530	A335/A530	ANSI-B 36.10			3604-625	12 Cr Mo 19.5		
A335/A530	A335/A530	ANSI-B 36.10			3604-620-440-460	13 Cr Mo 44		
A335/A530	A335/A530	ANSI-B 36.10			3604-620-440-460	13 Cr Mo 44		
A335/A530	A335/A530	ANSI-B 36.10			3604-620-440	10 Cr Mo 910		
API-5L	API-5L	API-5L	A53 Gr.A		3601-320/360	St. 37.0	Fe 310	St.E 210.7
API-5L	API-5L	API-5L	A53 Gr.B		3601-410-4360-43B/C	St. 44.0	Fe 360B/430B	St.E 240.7
API-5L	API-5L	API-5L			4360-43 B/C	St. E 290.7	Fe 430B	St.E 355
API-5L	API-5L	API-5L				St. E 320.7		
API-5L	API-5L	API-5L			4360-50 B/C/D	St. E 360.7	Fe 510B/C/D	
API-5L	API-5L	API-5L				St. E 385.7		
API-5L	API-5L	API-5L				St. E 415.7		
BS3059 part1	BS3059 part1	BS3059 part1	A179	5L-Gr.A		St. 35.8/St. 37.8	Fe 310 Fe 360B Fe 430B/C Fe 360B	A672-B60 A672-B65/C65
BS3059 part2	BS3059 part2	BS3059 part2	A179			St. 35.8/St. 37.8		
BS3601	BS3601	BS1600/BS3600	A53 Gr.A			St. 33		
BS3601	BS3601	BS1600/BS3600	A53 Gr.B			St. 37.0/St. 37.2		
BS3601	BS3601	BS1600/BS3600	A53 Gr.B		4360-43 B/C	St. 44.0/St. 44.3		
BS3601	BS3601	BS1600/BS3600	A106 Gr.B			St. 37.0/St. 37.2		
BS3601	BS3601	BS1600/BS3600	A106 Gr.B			St. 44.0/St. 44.3		
BS3602 part1	BS3602 part1	BS1600/BS3600	A106 Gr.B			St. 35.8/St. 37.8		
BS3602 part1	BS3602 part1	BS1600/BS3600	A106 Gr.B			St. 45.8		
BS3602 part1	BS3602 part1	BS1600/BS3600			4360-50 B/C/D	St. 52/St 52.3	Fe 510B/C/D	
BS3602 part2	BS3602 part2	BS1600/BS3600	A672 B65/C65	5L-Gr.B	4360-43 B/C	St. E240.7	Fe 430B/C	
BS3603	BS3603	BS1600/BS3600	A333 Gr.6			TT St. 35V		
BS3603	BS3603	BS1600/BS3600	A333 Gr.3			10 Ni 14		
BS3604	BS3604	BS1600/BS3600	A335 P12			13 Cr Mo 44		
BS3604	BS3604	BS1600/BS3600	A335 P22			10 Cr Mo 910		
BS3604	BS3604	BS1600/BS3600	A335 P5			12 Cr Mo 195		
BS4360	BS4360	BS1600	A671 CC70	5L-X52		St. 52-3	Fe 510D	St. E 555

API PIPES ERW & SEAMLESS PIPES

TOLERANCES (EXCERPTS FROM API SPECIFICATIONS)

Pipe Specification		API 5L				API 5LU				
Outside Diameter	Pipe Body	1.90 in. & smaller	+0.016 in. -0.031 in.				Less than 20 in.	+0.75%	-0.75%	
		2-3/8 in. through 18 in.	+0.75% -0.75%							
		20 in. through 36 in.								
		Non-expanded	+ 1.00% -1.00%							
	Cold-expanded	+0.75% -0.25%								
	Larger than 36 in.									
Non-expanded	+1.00% -1.00%									
Cold -expanded*2	+1/4 in. -1/8 in.									
Pipe Ends *1	10-3/4 in. & smaller	+1/16 in. -1/64in. (measured with ring gage)				10-3/4 in & smaller	+1/16 in.	-1/64 in. (measured with ring gage)		
	12-3/4 in. through 20 in.	+3/32 in. -1/32 in. (measured with ring gage)				12-3/4 in. through 20in.	+3/32 in.	-1/32 in. (measured with ring gage)		
	Larger than 20 in. (22 in. & larger)	+3/32 in. -1/32 in. (measured with diameter tape)				Larger than 20 in.	+3/32 in.	-1/32 in. (measured with diameter tape)		
Wall Thickness		Welded Pipe:	Grade A, B X42 through X70				Welded Pipe:	+15.0%	-12.5%	
		2.875 in. OD & smaller	+20.0% -12.5% +15.0% -12.5%							18 in. OD & smaller
		3-1/2 in. OD	+18.0% -12.5% +15.0% -12.5%							
		4 in. through 18 in. OD	+15.0% -12.5% +15.0% -12.5%							
		20 in. OD & larger	+17.5% -10.0% +19.5% -8.0%				20 in. OD & larger	+19.5%	-8.0%	
		Seamless Pipe:	Grade A,B X42 through X70				Seamless Pipe:	+15.0%	-12.5%	
		2.875 in. OD & smaller	+20.0% -12.5% +15.0% -12.5%							18 in. OD & smaller
		3-1/2 in. OD	+18.0% -12.5% +15.0% -12.5%							
4 in. through 18 in. OD	+15.0% -12.5% +15.0% -12.5%									
20 in. OD & larger	+15.0% -12.5% +17.5% -10.0%				20 in. OD & larger	+17.5%	-10.0%			
Weight	Single Length	Standard weight					Regular weight	+10.0%	-3.5%	
		Regular weight	+10.0% -3.5%							
		Extra strong weight								
Double extra strong weight										
Special plain end weight	+10.0% -5.0%				Special light weight	+10.0%	-5.0%			
	Carload Lots	-1.75%				-1.75%				
Straightness		For pipe 4-1/2 in OD & larger max. 0.2%				max. 0.2%				
Out of Roundness		Max. OD - specified OD ≤ 1% x specified OD				Max. OD - specified OD ≤ 1% x specified_OD				
-pipe larger than 20in. OD		Specified OD - min. OD ≤ 1% x specified OD				Specified OD - min. OD ≤ 1% x specified_OD				
Pipe End Preparation		For pipe 2-3/8 in. OD & larger								
		Bevel angle 30°, +5° - 0°				Bevel angle 30°, +5° - 0°				
		Root face 1/16 in. ± 1/32 in.				Root face 1/16 in. ± 1/32 in.				
		Squareness for 10-3/4 in. OD & larger max. 1/16 in.				Squareness 10-3/4 in. OD & larger max. 1/16 in.				

Note:

*1 Pipe and tolerances are applicable 4 in. from either end.

*2 Not applicable to pipe hydrostatically tested to pressures in excess of standard test pressures.

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure											
Outside Diameter			Wall Thickness						API 5L				API 5LX							
									A		B									
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56	X60	X65	X70	
1/8	0.405	10.3	40 (std)	0.068	1.73	0.24	0.36	--	700	-	700	--	--	--	--	--	--	--	--	
			80 (XS)	0.095	2.41	0.31	0.46	--	850	--	850	--	--	--	--	--	--	--	--	
1/4	0.54	13.7	40 (Std)	0.088	2.24	0.42	0.63	--	700	--	700	--	--	--	--	--	--	--	--	
			80 (XS)	0.119	3.02	0.54	0.80	--	850	--	850	--	--	--	--	--	--	--	--	
3/8	0.675	17.1	40 (std)	0.091	2.31	0.57	0.85	--	700	--	700	--	--	--	--	--	--	--	--	
			80 (XS)	0.126	3.20	0.74	1.10	--	850	--	850	--	--	--	--	--	--	--	--	
1/2	0.84	21.3	40 (std)	0.109	2.77	0.85	1.27	--	700	--	700	--	--	--	--	--	--	--	--	
			80 (XS)	0.147	3.73	1.09	1.62	--	850	--	850	--	--	--	--	--	--	--	--	
			(XXS)	0.294	7.47	1.71	2.55	--	1000	--	1000	--	--	--	--	--	--	--	--	
3/4	1.05	26.7	40 (std)	0.113	2.87	1.13	1.68	--	700	--	700	--	--	--	--	--	--	--	--	
			80 (XS)	0.154	3.91	1.47	2.19	--	850	--	850	--	--	--	--	--	--	--	--	
			(XXS)	0.308	7.82	2.44	3.63	--	1000	--	1000	--	--	--	--	--	--	--	--	
1	1.315	33.4	40 (std)	0.133	3.38	1.68	2.50	0.76	700	--	700	--	--	--	--	--	--	--	--	
			80 (XS)	0.179	4.55	2.17	3.23	0.99	850	--	850	--	--	--	--	--	--	--	--	
			160	0.250	6.35	2.84	4.23	1.29	--	--	--	--	--	--	--	--	--	--	--	
			(XXS)	0.358	9.09	3.66	5.45	1.66	1000	--	1000	--	--	--	--	--	--	--	--	
1-1/4	1.66	42.2	40 (std)	0.140	3.56	2.27	3.38	1.03	1200	--	1300	--	--	--	--	--	--	--	--	
			80 (XS)	0.191	4.85	3.00	4.47	1.36	1800	--	1900	--	--	--	--	--	--	--	--	
			160	0.250	6.35	3.76	5.60	1.71	--	--	--	--	--	--	--	--	--	--	--	
			(XXS)	0.382	9.70	5.21	7.76	2.37	2200	--	2300	--	--	--	--	--	--	--	--	
1-1/2	1.90	48.3	40 (std)	0.145	3.68	2.72	4.05	1.23	1200	--	1300	--	--	--	--	--	--	--	--	
			80 (XS)	0.200	5.08	3.63	5.41	1.65	1800	--	1900	--	--	--	--	--	--	--	--	
			160	0.281	7.14	4.86	7.24	2.21	--	--	--	--	--	--	--	--	--	--	--	
			(XXS)	0.400	10.16	6.41	9.55	2.91	2200	--	2300	--	--	--	--	--	--	--	--	
2	2-3/8	60.3	40 (std)	0.083	2.11	2.03	3.02	0.92	1260	--	1470	--	1760	1930	2180	2350	2520	2730	2940	
				0.109	2.77	2.64	3.93	1.20	--	--	--	--	2310	2530	2860	3000	3000	3000	3000	
				0.125	3.18	3.00	4.47	1.36	--	--	--	--	2650	2910	3000	3000	3000	3000	3000	
				0.141	3.58	3.36	5.00	1.53	--	--	--	--	2990	3000	3000	3000	3000	3000	3000	
				0.154	3.91	3.65	5.44	1.66	2330	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.172	4.37	4.05	6.03	1.84	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.188	4.78	4.39	6.54	1.99	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.218	5.54	5.02	7.48	2.28	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
			80 (XS)	0.250	6.35	5.67	8.45	2.57	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
				0.281	7.14	6.28	9.35	2.85	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
				0.344	8.74	7.46	11.11	3.39	--	--	--	--	--	--	--	--	--	--	--	--
				(XXS)	0.436	11.07	9.03	13.45	4.10	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000
2-1/2	2-7/8	73.0	40 (std)	0.083	2.11	2.47	3.68	1.12	1040	--	1040	--	1460	1590	1800	1940	2080	2250	2430	
				0.109	2.77	3.22	4.80	1.46	--	--	--	--	1910	2090	2370	2550	2730	2960	3000	
				0.125	3.18	3.67	5.47	1.67	--	--	--	--	2190	2400	2710	2920	3000	3000	3000	
				0.141	3.58	4.12	6.14	1.87	--	--	--	--	2470	2710	3000	3000	3000	3000	3000	
				0.156	3.96	4.53	6.75	2.06	1950	--	2280	--	2730	3000	3000	3000	3000	3000	3000	
				0.172	4.37	4.97	7.40	2.26	2150	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.188	4.78	5.40	8.04	2.45	2350	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.203	5.16	5.79	8.62	2.63	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
			80 (XS)	0.216	5.49	6.13	9.13	2.78	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
				0.250	6.35	7.01	10.44	3.18	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
				0.276	7.01	7.66	11.41	3.48	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
				0.375	9.52	10.01	14.91	4.54	--	--	--	--	--	--	--	--	--	--	--	--
(XXS)	0.552	14.02	13.69	20.39	6.22	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000			

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure											
Outside Diameter			Wall Thickness						API 5L				API 5LX							
									A		B									
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56	X60	X65	X70	
3	3-1/2	88.9	40 (std)	0.083	2.11	3.03	4.51	1.38	850	--	1000	--	1200	1310	1480	1590	1710	1850	1990	
				0.109	2.77	3.95	5.88	1.79	--	--	--	--	1570	1720	1940	2090	2240	2430	2620	
				0.125	3.18	4.51	6.72	2.05	1290	--	1500	--	1800	1970	2230	2400	2570	2790	3000	
				0.141	3.58	5.06	7.54	2.30	--	--	--	--	2030	2220	2510	2710	2900	3000	3000	
				0.156	3.96	5.57	8.30	2.53	1600	--	1870	--	2250	2460	2780	3000	3000	3000	3000	
				0.172	4.37	6.11	9.10	2.77	1770	--	2060	--	2480	2710	3000	3000	3000	3000	3000	
				0.188	4.78	6.65	9.91	3.02	1930	--	2260	--	2710	2970	3000	3000	3000	3000	3000	
				0.216	5.49	7.58	11.29	3.44	2220	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.250	6.35	8.68	12.93	3.94	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
				0.281	7.14	9.66	14.39	4.39	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	
			80 (XS) 160 (XXS)	0.300	7.62	10.25	15.27	4.65	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
				0.438	11.13	14.32	21.33	6.50	--	--	--	--	--	--	--	--	--	--	--	--
				0.600	15.24	18.58	27.67	8.44	2500	--	2500	--	3000	3000	3000	3000	3000	3000	3000	3000
3-1/2	4	101.6	40 (std)	0.083	2.11	3.47	5.17	1.58	750	--	870	--	1050	1150	1290	1390	1490	1620	1740	
				0.109	2.77	4.53	6.75	2.06	980	--	1140	--	1370	1500	1700	1830	1960	2130	2290	
				0.125	3.18	5.17	7.70	2.35	1120	--	1310	--	1580	1720	1950	2100	2250	2440	2630	
				0.141	3.58	5.81	8.65	2.64	1270	--	1480	--	1780	1950	2200	2370	2540	2750	2960	
				0.156	3.96	6.40	9.53	2.91	1400	--	1640	--	1970	2150	2430	2620	2810	3000	3000	
				0.172	4.37	7.03	10.47	3.19	1550	--	1810	--	2170	2370	2680	2890	3000	3000	3000	
				0.188	4.78	7.65	11.39	3.47	1690	--	1970	--	2370	2590	2930	3000	3000	3000	3000	
				0.226	5.74	9.11	13.57	4.14	2030	--	2370	--	2850	3000	3000	3000	3000	3000	3300	
				0.250	6.35	10.01	14.91	4.54	2250	--	2620	--	3000	3000	3000	3000	3000	3000	3000	
				0.281	7.14	11.16	16.62	5.07	2530	--	2800	--	3000	3000	3000	3000	3000	3000	3000	
			80 (XS)	0.318	8.08	12.50	18.62	5.68	2800	--	2800	--	3000	3000	3000	3000	3000	3000	3000	
4	4-1/2	114.3	40 (std)	0.083	2.11	3.92	5.84	1.78	660	--	770	--	930	1020	1150	1240				
				0.109	2.77	5.11	7.61	2.32	870	--	1020	--	--	--	--	--				
				0.125	3.18	5.84	8.70	2.65	1000	--	1170	--	1400	1530	1730	1870				
				0.141	3.58	6.56	9.77	2.98	1130	--	1320	--	1580	1730	1960	2110				
				0.156	3.96	7.24	10.78	3.29	1250	--	1460	--	1750	1910	2160	2330				
				0.172	4.37	7.95	11.84	3.61	1380	--	1610	--	1930	2110	2390	2570				
				0.188	4.78	8.66	12.90	3.93	1500	--	1750	--	2110	2310	2610	2810				
				0.203	5.16	9.32	13.88	4.23	1620	--	1890	--	2270	2490	2810	3000				
				0.219	5.56	10.01	14.91	4.54	1750	--	2040	--	2450	2690	3000	3000				
				0.237	6.02	10.79	16.07	4.90	1900	--	2210	--	2650	2910	3000	3000				
			60	0.25	6.35	11.35	16.91	5.15	2000	--	2330	--	2800	3000	3000	3000				
				0.281	7.14	12.66	18.86	5.75	2250	--	2620	--	3000	3000	3000	3000				
				0.312	7.92	13.96	20.79	6.34	2500	--	2800	--	3000	3000	3000	3000				
				80 (XS)	0.337	8.56	14.98	22.31	6.80	2700	--	2800	--	3000	3000	3000	3000			
120	0.438	11.13	19.00	28.30	8.63	2800	--	2800	--	3000	3000	3000	3000							
	160	0.531	13.49	22.51	33.53	10.22	2800	--	2800	--	3000	3000	3000	3000						
	(XXS)	0.674	17.12	27.54	41.02	12.50	2800	--	2800	--	3000	3000	3000	3000						
5	5-9/16	141.3	40 (std)	0.083	2.11	4.86	7.24	2.21	540	--	630	--	--	--	--	--				
				0.125	3.18	7.26	10.81	3.30	810	--	940	--	--	--	--	--				
				0.156	3.96	9.01	13.42	4.09	1010	--	1180	--	--	--	--	--				
				0.188	4.78	10.79	16.07	4.90	1220	--	1420	--	--	--	--	--				
				0.219	5.56	12.50	18.62	5.68	1420	--	1650	--	--	--	--	--				
				0.258	6.55	14.62	21.76	6.63	1670	--	1950	--	--	--	--	--				
				0.281	7.14	15.85	23.61	7.20	1820	--	2120	--	--	--	--	--				
				0.312	7.92	17.50	26.05	7.94	2020	--	2360	--	--	--	--	--				
				0.344	8.74	19.17	28.55	8.70	2230	--	2600	--	--	--	--	--				
				80 (XS)	0.375	9.52	20.78	30.94	9.43	2430	--	2800	--	--	--	--	--			
			120		0.50	12.70	27.03	40.26	12.27	2800	--	2800	--	--	--	--	--			
			160		0.625	15.88	32.96	49.08	14.96	2800	--	2800	--	--	--	--	--			
			(XXS)		0.75	19.05	38.55	57.41	17.50	2800	--	2800	--	--	--	--	--			

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	A		B		X42	X46	X52	X56
									Std.	Alt.	Std.	Alt.				
6	6-5/8	168.3	30	0.083	2.11	5.80	8.64	2.63	450	560	530	660	790	860	980	1050
				0.109	2.77	7.59	11.31	3.45	590	740	690	860	1040	1140	1280	1380
				0.125	3.18	8.68	12.93	3.94	680	850	790	990	1190	1300	1470	1580
				0.141	3.58	9.76	14.54	4.43	770	960	890	1120	1340	1470	1660	1790
				0.156	3.96	10.78	16.06	4.89	850	1060	990	1240	1480	1620	1840	1980
				0.172	4.37	11.85	17.65	5.38	930	1170	1090	1360	1640	1790	2030	2180
				0.188	4.78	12.92	19.24	5.87	1020	1280	1190	1490	1790	1960	2210	2380
				0.203	5.16	13.92	20.73	6.32	1100	1380	1290	1610	1930	2110	2390	2570
				0.219	5.56	14.98	22.31	6.80	1190	1490	1390	1740	2080	2280	2580	2780
				0.250	6.35	17.02	25.35	7.73	1360	1700	1580	1980	2380	2600	2940	3000
			40 (std)	0.280	7.11	18.97	28.26	8.61	1520	1900	1780	2220	2660	2920	3000	3000
				0.312	7.92	21.04	31.34	9.55	1700	2120	1980	2470	2970	3000	3000	3000
				0.344	8.74	23.08	34.38	10.48	1870	2340	2180	2730	3000	3000	3000	3000
			80 (XS)	0.375	9.52	25.03	37.28	11.36	2040	2550	2380	2800	3000	3000	3000	3000
				0.432	10.97	28.57	42.56	12.97	2350	2800	2740	2800	3000	3000	3000	3000
				0.500	12.70	32.71	48.72	14.85	2720	2800	2800	2800	3000	3000	3000	3000
			120	0.562	14.27	36.39	54.20	16.52	2800	2800	2800	2800	3000	3000	3000	3000
				0.625	15.88	40.05	59.65	18.18	2800	2800	2800	2800	3000	3000	3000	3000
			160	0.719	18.26	45.35	67.55	20.59	2800	2800	2800	2800	3000	3000	3000	3000
				0.864	21.95	53.16	79.18	24.13	2800	2800	2800	2800	--	--	--	--
8	8-5/8	219.1	20	0.125	3.18	11.35	16.91	5.15	520	650	610	760	910	1000	1130	1220
				0.156	3.96	14.11	21.02	6.41	650	810	760	950	1140	1250	1410	1520
				0.188	4.78	16.94	25.23	7.69	780	980	920	1140	1370	1500	1700	1830
				0.203	5.16	18.26	27.20	8.29	--	--	--	--	1480	1620	1840	2000
				0.219	5.56	19.66	29.28	8.93	910	1140	1070	1330	1600	1750	1980	2130
				0.250	6.35	22.36	33.31	10.15	1040	1300	1220	1520	1830	2000	2260	2430
			30	0.277	7.04	24.70	36.79	11.21	1160	1450	1350	1690	2020	2220	2510	2700
				0.312	7.92	27.70	41.26	12.58	1300	1630	1520	1900	2280	2500	2820	3000
			40 (std)	0.322	8.18	28.55	42.53	12.96	1340	1680	1570	1960	2350	2580	2910	3000
				0.344	8.74	30.42	45.31	13.81	1440	1790	1680	2090	2510	2750	3000	3000
				0.375	9.52	33.04	49.21	15.00	1570	1960	1830	2280	2740	3000	3000	3000
			60	0.406	10.31	35.64	53.09	16.18	--	--	--	--	--	--	--	--
				0.438	11.13	38.30	57.05	17.39	1830	2290	2130	2670	3000	3000	3000	3000
			80 (XS)	0.500	12.70	43.39	64.63	19.70	2090	2610	2430	2800	3000	3000	3000	3000
				0.562	14.27	48.40	72.09	21.97	2350	2800	2740	2800	3000	3000	3000	3000
			100	0.594	15.09	50.95	75.89	23.13	--	--	--	--	--	--	--	--
				0.625	15.88	53.40	79.54	24.24	2610	2800	2800	2800	3000	3000	3000	3000
			120	0.719	18.26	60.71	90.43	27.56	2800	2800	2800	2800	3000	3000	3000	3000
			140	0.812	20.62	67.76	100.93	30.76	--	--	--	--	--	--	--	--
			(XXS)	0.875	22.22	72.42	107.87	32.88	2800	2800	2800	2800	--	--	--	--
				0.906	23.01	74.69	111.25	33.91	--	--	--	--	--	--	--	--
10	10-3/4	273.0	20	0.156	3.96	17.65	26.29	8.01	520	650	610	760	1040	1130	1280	1380
				0.188	4.78	21.21	31.59	9.63	630	790	730	920	1250	1370	1550	1660
				0.203	5.16	22.87	34.06	10.38	--	--	--	--	1350	1480	1670	1800
				0.219	5.56	24.63	36.69	11.18	730	920	860	1070	1450	1590	1800	1940
				0.250	6.35	28.04	41.77	12.73	840	1050	980	1220	1660	1820	2060	2210
				0.279	7.09	31.20	46.47	14.16	930	1170	1090	1360	1850	2030	2290	2470
			30	0.307	7.80	34.24	51.00	15.54	1030	1290	1200	1500	2040	2230	2520	2720
				0.344	8.74	38.23	56.94	17.36	1150	1440	1340	1680	2280	2500	2830	3000

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	A		B		X42	X46	X52	X56
									Std.	Alt.	Std.	Alt.				
10	10-3/4	273.0	40 (std)	0.365	9.27	40.48	60.29	18.38	1220	1530	1430	1780	2420	2660	3000	3000
				0.438	11.13	48.24	71.85	21.90	1470	1830	1710	2140	2910	3000	3000	3000
			60 (XS)	0.500	12.70	54.74	81.54	24.85	1670	2090	1950	2440	3000	3000	3000	3000
				0.562	14.27	61.15	91.08	27.76	1880	2350	2200	2740	3000	3000	3000	3000
			80	0.594	15.09	64.43	95.97	29.25	--	--	--	--	--	--	--	--
				0.625	15.88	67.58	100.66	30.68	2090	2620	2440	2800	3000	3000	3000	3000
			100	0.719	18.26	77.03	114.74	34.97	2410	2800	2800	2800	3000	3000	3000	3000
				0.812	20.62	86.18	128.37	39.13	2720	2800	2800	2800	--	--	--	--
			120	0.844	21.44	89.29	133.00	40.54	--	--	--	--	--	--	--	--
				1.000	25.40	104.13	155.10	47.28	--	--	--	--	--	--	--	--
12	12-3/4	323.8	20	0.172	4.37	23.11	34.42	10.49	490	610	570	710	960	1050	1190	1280
				0.188	4.78	25.22	37.57	11.45	530	660	620	770	1050	1150	1300	1400
				0.203	5.16	27.20	40.51	12.35	--	--	--	--	1140	1250	1410	1520
				0.219	5.56	29.31	43.66	13.31	620	770	720	900	1230	1340	1520	1640
				0.250	6.35	33.38	49.72	15.15	710	880	820	1030	1400	1530	1730	1870
				0.281	7.14	37.42	55.74	16.99	790	990	930	1160	1570	1720	1950	2100
			30	0.312	7.92	41.45	61.74	18.82	880	1100	1030	1280	1750	1910	2160	2330
				0.330	8.38	43.77	65.20	19.87	930	1160	1090	1360	1850	2020	2290	2460
				0.344	8.74	45.58	67.89	20.69	970	1210	1130	1420	1930	2110	2390	2570
				(std) 0.375	9.52	49.56	73.82	22.50	1060	1320	1240	1540	2100	2300	2600	2800
			40	0.406	10.31	53.52	79.72	24.30	--	--	--	--	2270	2490	2810	3000
				0.438	11.13	57.59	85.78	26.15	1240	1550	1440	1800	2450	2690	3000	3000
			(XS) 500	12.70	65.42	97.44	29.70	29.70	1410	1760	1650	2060	2800	3000	3000	3000
				0.562	14.27	73.15	108.96	33.21	1590	1980	1850	2310	3000	3000	3000	3000
			80	0.625	15.88	80.93	120.55	36.74	1760	2210	2060	2570	3000	3000	3000	3000
				0.688	17.48	88.63	132.01	40.24	1940	2430	2270	2800	3000	3000	3000	3000
				0.750	19.05	96.12	143.17	43.64	2120	2650	2470	2800	3000	3000	3000	3000
				0.812	20.62	103.53	154.21	47.00	2290	2800	2670	2800	3000	3000	3000	3000
			100	0.844	21.44	107.32	159.85	48.72	--	--	--	--	--	--	--	--
				0.875	22.22	110.97	165.29	50.38	2470	2800	2800	2800	3000	3000	3000	3000
			120	1.000	25.40	125.49	186.92	56.97	--	--	--	--	--	--	--	--
14	14	355.6	10	0.188	4.78	27.73	41.30	12.59	480	600	560	700	960	1050	1190	1280
				0.203	5.16	29.91	44.55	13.58	520	650	610	760	--	--	--	--
				0.210	5.33	30.93	46.07	14.04	--	--	--	--	1070	1170	1330	1430
				0.219	5.56	32.23	48.01	14.63	--	--	--	--	1120	1220	1380	1490
				0.250	6.35	36.71	54.68	16.67	640	800	750	940	1280	1400	1580	1700
				0.281	7.14	41.17	61.32	18.69	720	900	840	1050	1430	1570	1770	1910
			20	0.312	7.92	45.61	67.94	20.71	800	1000	940	1170	1590	1740	1970	2120
				0.344	8.74	50.17	74.73	22.78	880	1110	1030	1290	1750	1920	2170	2340
				(std) 0.375	9.52	54.57	81.28	24.77	960	1210	1120	1410	1910	2090	2370	2550
				0.406	10.31	58.94	87.79	26.76	--	--	--	--	2070	2270	2560	2760
			40	0.438	11.13	63.44	94.49	28.80	1130	1410	1310	1640	2230	2450	2770	2980
				0.469	11.91	67.78	100.96	30.77	--	--	--	--	2390	2620	2960	3000
			(XS) 500	12.70	72.09	107.38	32.73	32.73	1290	1610	1500	1880	2550	2790	3000	3000
				0.562	14.27	80.66	120.14	36.62	1450	1810	1690	2110	2870	3000	3000	3000
			60	0.594	15.09	85.05	126.68	38.61	--	--	--	--	--	--	--	--
				0.625	15.88	89.28	132.98	40.53	1610	2010	1880	2340	3000	3000	3000	3000
				0.688	17.48	97.81	145.69	44.41	1770	2210	2060	2580	3000	3000	3000	3000
				0.750	19.05	106.13	158.08	48.18	1930	2410	2250	2800	3000	3000	3000	3000
			80	0.812	20.62	114.37	170.35	51.92	2090	2610	2440	2800	3000	3000	3000	3000
				0.875	22.22	122.65	182.69	55.68	2250	2800	2620	2800	3000	3000	3000	3000
				0.938	23.83	130.85	194.90	59.41	2410	2800	2800	2800	3000	3000	3000	3000
				1.000	25.40	138.84	206.80	63.03	--	--	--	--	--	--	--	--

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure								
Outside Diameter			Wall Thickness						API 5L				API 5LX				
									A		B						
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56	
16	16	406.4	10	0.188	4.78	31.75	47.29	14.40	420	530	490	620	840	920	1040	1120	
				0.203	5.16	34.25	51.02	15.54	460	570	530	670	910	990	1120	1210	
				0.219	5.56	36.91	54.98	16.74	490	620	570	720	980	1070	1210	1300	
				0.250	6.35	42.05	62.63	19.07	560	700	660	820	1120	1220	1380	1490	
				0.281	7.14	47.17	70.26	21.40	630	790	740	920	1250	1370	1550	1670	
			20	0.312	7.92	52.27	77.86	23.71	700	880	820	1020	1390	1520	1720	1860	
				0.344	8.74	57.52	85.68	26.09	770	970	900	1130	1540	1680	1900	2050	
				0.375	9.52	62.58	93.21	28.39	840	1050	980	1230	1670	1830	2070	2230	
			30 (std)	0.406	10.31	67.62	100.72	30.67	--	--	--	--	1810	1980	2240	2420	
				0.438	11.13	72.80	108.44	33.02	990	1230	1150	1440	1950	2140	2420	2610	
				0.469	11.91	77.79	115.87	35.29	--	--	--	--	2090	2290	2590	2790	
				0.500	12.70	82.77	123.29	37.54	1120	1410	1310	1640	2230	2440	2760	2980	
			40 (XS)	0.562	14.27	92.66	138.02	42.03	1260	1580	1480	1840	2510	2750	3000	3000	
				0.625	15.88	102.63	152.87	46.55	1410	1760	1640	2050	2790	3000	3000	3000	
				60	0.656	16.66	107.50	160.12	48.80	--	--	--	--	--	--	--	--
					0.688	17.48	112.51	167.58	51.03	1550	1940	1810	2260	3000	3000	3000	3000
					0.750	19.05	122.15	181.94	55.41	1690	2110	1970	2460	3000	3000	3000	3000
				80	0.812	20.62	131.71	196.18	59.74	1830	2280	2130	2660	3000	3000	3000	3000
					0.844	21.44	136.61	203.48	62.02	--	--	--	--	--	--	--	--
					0.875	22.22	141.34	210.53	64.11	1970	2460	2300	2800	3000	3000	3000	3000
					0.938	23.83	150.89	224.75	68.44	2110	2640	2460	2800	3000	3000	3000	3000
					1.000	25.40	160.20	238.62	72.67	2250	2800	2620	2800	3000	3000	3000	3000
				1.062	26.97	169.43	252.37	76.85	2390	2800	2790	2800	3000	3000	3000	3000	
			1.125	28.58	178.72	266.20	81.07	2530	2800	2800	2800	3000	3000	3000	3000		
18	18	457.2	10	0.188	4.78	35.76	53.26	16.22	380	470	440	550	750	820	920	990	
				0.219	5.56	41.59	61.95	18.87	440	550	510	640	870	950	1080	1160	
				0.250	6.35	47.39	70.59	21.50	500	620	580	730	990	1090	1230	1320	
				0.281	7.14	53.18	79.21	24.12	560	700	660	820	1110	1220	1380	1490	
				0.312	7.92	58.94	87.79	26.74	620	780	730	910	1240	1360	1530	1650	
			20	0.344	8.74	64.87	96.62	29.43	690	860	800	1000	1360	1490	1690	1820	
				0.375	9.52	70.59	105.14	32.02	750	940	880	1090	1490	1630	1840	1980	
				0.406	10.31	76.29	113.63	34.61	--	--	--	--	1610	1760	1990	2150	
			30	0.438	11.13	82.15	122.36	37.26	880	1100	1020	1280	1740	1900	2150	2320	
				0.469	11.91	87.81	130.79	39.83	--	--	--	--	1860	2040	2300	2480	
				0.500	12.70	93.45	139.19	42.39	1000	1250	1170	1460	1980	2170	2460	2640	
			40	0.562	14.27	104.67	155.91	47.48	1120	1400	1310	1640	2230	2440	2760	2970	
				0.625	15.88	115.98	172.75	52.61	1250	1560	1460	1820	2480	2720	3000	3000	
				0.688	17.48	127.21	189.48	57.70	1380	1720	1610	2010	2730	2990	3000	3000	
				0.750	19.05	138.17	205.80	62.67	1500	1880	1750	2190	2980	3000	3000	3000	
			60	0.812	20.62	149.06	222.02	67.61	1620	2030	1890	2370	3000	3000	3000	3000	
				0.875	22.22	160.03	238.36	72.59	1750	2190	2040	2550	3000	3000	3000	3000	
				0.938	23.83	170.92	254.59	77.53	1880	2340	2190	2740	3000	3000	3000	3000	
				1.000	25.40	181.56	270.43	82.36	2000	2500	2330	2800	3000	3000	3000	3000	
				1.062	26.97	192.11	286.15	87.14	2120	2660	2480	2800	3000	3000	3000	3000	
				1.125	28.58	202.75	302.00	91.97	2250	2800	2620	2800	3000	3000	3000	3000	
				1.187	30.15	213.14	317.47	96.68	2370	2800	2770	2800	--	--	--	--	
				1.188	30.18	213.31	317.73	96.76	--	--	--	--	3000	3000	3000	3000	
				1.250	31.75	223.61	333.07	101.43	2500	2800	2800	2800	3000	3000	3000	3000	

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	A		B		X42	X46	X52	X56
									Std.	Alt.	Std.	Alt.				
20	20	508.0	10	0.219	5.56	46.27	68.92	20.99	390	490	460	570	830	910	1020	1100
				0.250	6.35	52.73	78.54	23.92	450	560	520	660	940	1040	1170	1260
				0.281	7.14	59.18	88.15	26.84	510	630	590	740	1060	1160	1320	1420
				0.312	7.92	65.60	97.71	29.76	560	700	660	820	1180	1290	1460	1570
			20	0.344	8.74	72.21	107.56	32.75	620	770	720	900	1300	1420	1610	1730
				0.375	9.52	78.60	117.07	35.65	680	840	790	980	1420	1550	1760	1890
				0.406	10.31	84.96	126.55	38.54	--	--	--	--	1530	1680	1900	2050
				0.438	11.13	91.51	136.30	41.51	790	990	920	1150	1660	1810	2050	2210
			30	0.469	11.91	97.83	145.72	44.38	--	--	--	--	1770	1940	2190	2360
				0.500	12.70	104.13	151.10	47.23	900	1120	1050	1310	1890	2070	2340	2520
				0.562	14.27	116.67	173.78	52.92	1010	1260	1180	1480	2120	2330	2630	2750
			60	0.625	15.88	129.33	192.64	58.66	1120	1410	1310	1640	2360	2590	2750	2750
				0.688	17.48	141.90	211.36	64.37	1240	1550	1440	1810	2600	2750	2750	2750
				0.750	19.05	154.19	229.67	69.94	1350	1690	1580	1970	2750	2750	2750	2750
				0.812	20.62	166.40	247.85	75.48	1460	1830	1710	2130	2750	2750	2750	2750
				0.875	22.22	178.72	266.20	81.07	1580	1970	1840	2300	2750	2750	2750	2750
				0.938	23.83	190.96	284.43	86.62	1690	2110	1970	2460	2750	2750	2750	2750
				1.000	25.40	202.92	302.25	92.04	1800	2250	2100	2620	2750	2750	2750	2750
				1.062	26.97	214.80	319.94	97.43	1910	2390	2230	2750	2750	2750	2750	2750
				1.125	28.58	226.78	337.79	102.87	2020	2530	2360	2750	2750	2750	2750	2750
				1.187	30.15	238.50	355.25	108.18	2140	2670	2490	2750	--	--	--	--
				1.188	30.15	238.68	--	--	--	--	--	--	2750	2750	2750	2750
				1.250	31.75	250.31	372.84	113.54	2250	2750	2620	2750	2750	2750	2750	2750
				1.312	33.32	261.86	390.04	118.78	2360	2750	2750	2750	2750	2750	2750	2750
				1.375	34.92	273.51	407.39	124.06	2480	2750	2750	2750	2750	2750	2750	2750
22	22	558.8	10	0.219	5.56	50.94	75.88	23.11	360	450	420	520	750	820	930	1000
				0.250	6.35	58.07	86.50	26.34	410	510	480	600	860	940	1060	1150
				0.281	7.14	65.18	97.09	29.57	460	570	540	670	970	1060	1200	1290
				0.312	7.92	72.27	107.65	32.78	510	640	600	740	1070	1170	1330	1430
			20	0.344	8.74	79.56	118.50	36.09	560	700	660	820	1180	1290	1460	1580
				0.375	9.52	86.61	129.01	39.29	610	770	720	890	1290	1410	1600	1720
				0.406	10.31	93.63	139.46	42.47	--	--	--	--	1400	1530	1730	1860
				0.438	11.13	100.86	150.23	45.75	720	900	840	1050	1510	1650	1860	2010
			30	0.469	11.91	107.85	160.64	48.92	--	--	--	--	1610	1770	2000	2150
				0.500	12.70	114.81	171.01	52.08	820	1020	950	1190	1720	1880	2130	2290
				0.562	14.27	128.67	191.65	58.36	920	1150	1070	1340	1930	2120	2390	2500
				0.625	15.88	142.68	212.52	64.72	1020	1280	1190	1490	2150	2350	2500	2500
			60	0.688	17.48	156.60	233.26	71.03	1130	1410	1310	1640	2360	2500	2500	2500
				0.750	19.05	170.21	253.53	77.21	1230	1530	1430	1790	2500	2500	2500	2500
				0.812	20.62	183.75	273.70	83.35	1330	1660	1550	1940	2500	2500	2500	2500
				0.875	22.22	197.41	294.04	89.55	1430	1790	1670	2090	2500	2500	2500	2500
				0.938	23.83	211.00	314.28	95.71	1530	1920	1790	2240	2500	2500	2500	2500
				1.000	25.40	224.28	334.07	101.73	1640	2050	1910	2390	2500	2500	2500	2500
				1.062	26.97	237.48	353.73	107.72	1740	2170	2030	2500	2500	2500	2500	2500
			80	1.125	28.58	250.81	373.58	113.77	1840	2300	2150	2500	2500	2500	2500	2500
				1.188	30.15	263.85	393.00	119.68	1940	2430	2270	2500	2500	2500	2500	2500
				1.250	31.75	277.01	412.61	125.65	2050	2500	2390	2500	2500	2500	2500	2500
				1.312	33.32	289.88	431.78	129.22	2150	2500	2500	2500	2500	2500	2500	2500
			100	1.375	34.92	302.88	451.14	137.39	2250	2500	2500	2500	2500	2500	2500	2500
				1.437	36.50	315.58	470.06	143.15	2350	2500	2500	2500	--	--	--	--
				1.438	36.53	--	--	--	--	--	--	--	2500	2500	2500	2500
				1.500	38.10	328.41	489.17	148.97	2450	2500	2500	2500	2500	2500	2500	2500

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	A		B		X42	X46	X52	X56
24	24	609.6	10	0.250	6.35	63.41	94.45	28.76	380	470	440	550	790	860	980	1050
				0.281	7.14	71.18	106.02	32.29	420	530	490	610	890	970	1100	1180
				0.312	7.92	78.93	117.57	35.80	470	580	550	680	980	1080	1220	1310
			20	0.344	8.74	86.91	129.45	39.42	520	640	600	750	1080	1190	1340	1440
				0.375	9.52	94.62	140.94	42.92	560	700	660	820	1180	1290	1460	1580
				0.406	10.31	102.31	152.39	46.41	--	--	--	--	1280	1400	1580	1710
				0.438	11.13	110.22	164.17	50.00	660	820	770	960	1380	1510	1710	1840
				0.469	11.91	117.86	175.55	53.46	--	--	--	--	1480	1620	1830	1970
			30	0.500	12.70	125.49	186.92	56.92	750	940	880	1090	1580	1720	1950	2100
				0.562	14.27	140.68	209.54	63.81	840	1050	980	1230	1770	1940	2190	2300
				0.625	15.88	156.03	232.41	70.78	940	1170	1090	1370	1970	2160	2300	2300
			40	0.688	17.48	171.29	255.14	77.70	1030	1290	1200	1500	2170	2300	2300	2300
				0.750	19.05	186.23	277.39	84.47	1120	1410	1310	1640	2300	2300	2300	2300
				0.812	20.62	201.09	299.52	91.21	1220	1520	1420	1780	2300	2300	2300	2300
				0.875	22.22	216.10	321.88	98.02	1310	1640	1530	1910	2300	2300	2300	2300
				0.938	23.83	231.03	344.12	104.80	1410	1760	1640	2050	2300	2300	2300	2300
				1.000	25.40	245.64	365.88	111.42	1500	1880	1750	2190	2300	2300	2300	2300
				1.062	26.97	260.17	387.52	118.01	1590	1990	1860	2300	2300	2300	2300	2300
				1.125	28.58	274.84	409.37	124.65	1690	2110	1970	2300	2300	2300	2300	2300
				1.187	30.15	289.20	430.76	131.18	1780	2230	2080	2300	--	--	--	--
				1.188	30.18	289.44	--	--	--	--	--	--	2300	2300	2300	2300
				1.250	31.75	303.71	452.38	137.76	1880	2300	2190	2300	2300	2300	2300	2300
				1.312	33.32	317.91	473.53	144.20	1970	2300	2300	2300	2300	2300	2300	2300
				1.375	34.92	332.25	494.89	150.71	2060	2300	2300	2300	2300	2300	2300	2300
				1.437	36.50	346.28	515.78	157.07	2160	2300	2300	2300	--	--	--	--
				1.438	36.53	346.50	--	--	--	--	--	--	2300	2300	2300	2300
				1.500	38.10	360.45	536.89	163.50	2250	2300	2300	2300	2300	2300	2300	2300
				1.562	39.67	374.31	557.53	169.79	2300	2300	2300	2300	2300	2300	2300	2300
26	26	660.4	10	0.250	6.35	68.75	102.40	31.19	350	430	400	500	730	800	900	970
				0.281	7.14	77.18	114.96	35.01	390	490	450	570	820	890	1010	1090
				0.312	7.92	85.60	127.50	38.83	430	540	500	630	910	990	1120	1210
				0.344	8.74	94.26	140.40	42.76	480	600	560	690	1000	1100	1240	1330
				0.375	9.52	102.63	152.87	46.55	520	650	610	760	1090	1190	1350	1450
				0.406	10.31	110.98	165.30	50.34	--	--	--	--	1180	1290	1460	1570
				0.438	11.13	119.57	178.10	54.24	610	760	710	880	1270	1390	1580	1700
			20	0.469	11.91	127.88	190.48	58.01	--	--	--	--	1360	1490	1690	1820
				0.500	12.70	136.17	202.83	61.77	690	870	810	1010	1450	1590	1800	1940
				0.562	14.27	152.68	227.42	69.26	780	970	910	1130	1630	1790	2000	2000
				0.625	15.88	169.38	252.29	76.83	870	1080	1010	1260	1820	1990	2000	2000
				0.688	17.48	185.99	277.03	84.37	950	1190	1110	1390	2000	2000	2000	2000
				0.750	19.05	202.25	301.25	91.74	1040	1300	1210	1510	2000	2000	2000	2000
				0.812	20.62	218.43	325.35	99.08	1120	1410	1310	1640	2000	2000	2000	2000
				0.875	22.22	234.79	349.72	106.50	1210	1510	1410	1770	2000	2000	2000	2000
				0.938	23.83	251.07	373.97	113.89	1300	1620	1520	1890	2000	2000	2000	2000
				1.000	25.40	267.00	397.70	121.11	1380	1730	1620	2000	2000	2000	2000	2000
28	28	711.2	10	0.250	6.35	74.09	110.36	33.61	320	400	370	470	680	740	840	900
				0.281	7.14	83.19	123.91	37.73	360	450	420	530	760	830	940	1010
				0.312	7.92	92.26	137.42	41.85	400	500	470	580	840	920	1040	1120
				0.344	8.74	101.61	151.35	46.09	--	--	--	--	930	1020	1150	1240
				0.375	9.52	110.64	164.80	50.19	480	600	560	700	1010	1110	1250	1350

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
									A		B					
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56
28	28	711.2	20	0.406	10.31	119.65	178.22	54.27	--	--	--	--	1100	1200	1360	1460
				0.438	11.13	128.93	192.04	58.48	560	700	660	820	1180	1300	1460	1580
				0.469	11.91	137.90	205.40	62.55	--	--	--	--	1270	1390	1570	1690
				0.500	12.70	146.85	218.73	66.61	640	800	750	940	1350	1480	1670	1800
				0.562	14.27	164.69	245.31	74.70	720	900	840	1050	1520	1660	1880	2020
				0.625	15.88	182.73	272.18	82.89	800	1000	940	1170	1690	1850	2090	2250
			30	0.688	17.48	200.68	298.91	91.03	880	1100	1030	1290	1860	2030	2300	2480
				0.750	19.05	218.27	325.11	99.01	960	1210	1120	1410	2020	2220	2510	2700
				0.812	20.62	235.78	351.19	106.95	1040	1300	1220	1520	2190	2400	2710	2920
				0.875	22.22	253.48	377.56	114.98	1120	1410	1310	1640	2360	2590	2920	3000
				0.938	23.83	271.10	403.80	122.97	1210	1510	1410	1760	2530	2770	3000	3000
				1.000	25.40	288.36	429.51	130.80	1290	1610	1500	1880	2700	2960	3000	3000
30	30	762.0	10	0.250	6.35	79.43	118.31	36.03	300	370	350	440	630	690	780	840
				0.281	7.14	89.19	132.85	40.46	340	420	390	490	710	780	880	940
				0.312	7.92	98.93	147.36	44.87	370	470	440	550	790	860	970	1050
				0.344	8.74	108.95	162.28	49.42	--	--	--	--	870	950	1070	1160
				0.375	9.52	118.65	176.73	53.82	450	560	520	660	940	1040	1170	1260
				0.406	10.31	128.32	191.13	58.21	--	--	--	--	1020	1120	1270	1360
			20	0.438	11.13	138.29	205.98	62.73	530	660	610	770	1100	1210	1370	1470
				0.469	11.91	147.92	220.33	67.10	--	--	--	--	1180	1290	1460	1580
				0.500	12.70	157.53	234.64	71.46	600	750	700	880	1260	1380	1560	1680
				0.562	14.27	176.69	263.18	80.15	670	840	790	980	1420	1550	1750	1890
				0.625	15.88	196.08	292.06	88.94	750	940	880	1090	1580	1720	1950	2100
				0.688	17.48	215.38	320.81	97.70	830	1030	960	1200	1730	1900	2150	2310
				0.750	19.05	234.29	348.97	106.27	900	1120	1050	1310	1890	2070	2340	2520
				0.812	20.62	253.12	377.02	114.82	970	1220	1140	1420	2050	2240	2530	2730
				0.875	22.22	272.17	405.40	123.46	1050	1310	1220	1530	2200	2420	2730	2940
				0.938	23.83	291.14	433.65	132.06	1130	1410	1310	1640	2360	2590	2930	3000
				1.000	25.40	309.72	461.33	140.49	1200	1500	1400	1750	2520	2760	3000	3000
				1.062	26.97	328.22	488.88	148.88	1270	1590	1490	1860	2680	2930	3000	3000
				1.125	28.58	346.93	516.75	157.37	1350	1690	1580	1970	2840	3000	3000	3000
				1.188	30.18	365.56	544.50	165.82	1430	1780	1660	2080	2990	3000	3000	3000
				1.250	31.75	383.81	571.68	174.10	1500	1880	1750	2190	3000	3000	3000	3000
32	32	812.8	10	0.250	6.35	84.77	126.26	38.45	280	350	330	410	590	650	730	790
				0.281	7.14	95.19	141.79	43.18	320	400	370	460	660	730	820	890
				0.312	7.92	105.59	157.28	47.90	350	440	410	510	740	810	910	980
				0.344	8.74	116.30	173.23	52.75	--	--	--	--	810	890	1010	1080
				0.375	9.52	126.66	188.66	57.45	420	530	490	620	890	970	1100	1180
				0.406	10.31	136.99	204.05	62.14	--	--	--	--	960	1050	1190	1280
			20	0.438	11.13	147.64	219.91	66.97	490	620	570	720	1030	1130	1280	1380
				0.469	11.91	157.94	235.25	71.64	--	--	--	--	1110	1210	1370	1480
				0.500	12.70	168.21	250.55	76.30	560	700	660	820	1180	1290	1460	1580
				0.562	14.27	188.70	281.07	85.59	630	790	740	920	1330	1450	1640	1770
				0.625	15.88	209.43	311.95	95.00	700	880	820	1030	1480	1620	1830	1970
				0.688	17.48	230.08	342.70	104.36	770	970	900	1130	1630	1780	2010	2170
			30	0.750	19.05	250.31	372.84	113.54	840	1050	980	1230	1770	1940	2190	2360
				0.812	20.62	270.47	402.87	122.69	910	1140	1070	1330	1920	2100	2380	2560
				0.875	22.22	290.86	433.24	131.93	980	1230	1150	1440	2070	2260	2560	2760
				0.938	23.83	311.17	463.49	141.15	1060	1320	1230	1540	2220	2430	2740	2950
				1.000	25.40	331.08	493.14	150.18	1120	1410	1310	1640	2360	2590	2920	3000
				1.062	26.97	350.90	522.74	159.17	1190	1490	1390	1740	2510	2750	3000	3000
				1.125	28.58	370.96	552.54	168.27	1270	1580	1480	1850	2660	2910	3000	3000
				1.188	30.18	390.94	582.31	177.33	1340	1670	1560	1950	2810	3000	3000	3000
			40	1.250	31.75	410.51	611.45	186.21	1410	1760	1640	2050	2950	3000	3000	3000

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
									A		B					
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56
34	34	863.6	10	0.250	6.35	90.11	134.22	40.87	260	330	310	390	560	610	690	740
				0.281	7.14	101.19	150.72	45.90	300	370	350	430	620	680	770	830
				0.312	7.92	112.25	167.20	50.92	330	410	390	480	690	760	860	920
				0.344	8.74	123.65	184.18	56.09	--	--	--	--	760	840	950	1020
				0.375	9.52	134.67	200.59	61.09	400	500	460	580	830	910	1030	1110
			0.406	10.31	145.67	216.98	66.08	--	--	--	--	900	990	1120	1200	
			0.438	11.13	157.00	233.85	71.22	460	580	540	680	970	1070	1210	1300	
			0.469	11.91	167.95	250.16	76.18	--	--	--	--	1040	1140	1290	1390	
			0.500	12.70	178.89	266.46	81.14	530	660	620	770	1110	1220	1380	1480	
			0.562	14.27	200.70	298.94	91.04	600	740	690	870	1250	1370	1550	1670	
			0.625	15.88	222.78	331.83	101.05	660	830	770	970	1390	1520	1720	1850	
			0.688	17.48	244.77	364.58	111.03	730	910	850	1060	1530	1680	1890	2040	
			0.750	19.05	266.33	396.70	120.81	790	990	930	1160	1670	1830	2060	2220	
			0.812	20.62	287.81	428.69	130.55	860	1070	1000	1250	1810	1980	2240	2410	
			0.875	22.22	309.55	461.07	140.41	930	1160	1080	1350	1950	2130	2410	2590	
			0.938	23.83	331.21	493.34	150.24	990	1240	1160	1450	2090	2280	2580	2780	
			1.000	25.40	352.44	524.96	162.13	1060	1320	1240	1540	2220	2440	2750	2960	
			1.062	26.97	373.59	556.46	169.46	1120	1410	1310	1640	2360	2590	2920	3000	
			1.125	28.58	394.99	588.34	179.17	1190	1490	1390	1740	2500	2740	3000	3000	
			1.188	30.18	416.31	620.09	188.84	1260	1570	1470	1830	2640	2890	3000	3000	
			1.250	31.75	437.21	651.22	198.32	1320	1650	1540	1930	2780	3000	3000	3000	
36	36	914.4	10	0.250	6.35	95.45	142.17	43.30	250	310	290	360	520	580	650	700
				0.281	7.14	107.20	159.67	48.63	280	350	330	410	590	650	730	790
				0.312	7.92	118.92	177.13	53.94	310	390	360	450	660	720	810	870
				0.344	8.74	131.00	195.12	59.42	--	--	--	--	720	790	890	960
				0.375	9.52	142.68	212.52	64.72	380	470	440	550	790	860	980	1050
			0.406	10.31	154.34	229.89	70.01	--	--	--	--	850	930	1060	1140	
			0.438	11.13	166.35	247.78	75.46	440	550	510	640	920	1010	1140	1230	
			0.469	11.91	177.97	265.09	80.73	--	--	--	--	980	1080	1220	1310	
			0.500	12.70	189.57	282.36	85.99	500	620	580	730	1050	1150	1300	1400	
			0.562	14.27	212.70	316.82	96.48	560	700	660	820	1180	1290	1460	1570	
			0.625	15.88	236.13	351.72	107.11	620	780	730	910	1310	1440	1620	1750	
			0.688	17.48	259.47	386.48	117.70	690	860	800	1000	1440	1580	1790	1930	
			0.750	19.05	282.35	420.56	128.07	750	940	880	1090	1580	1720	1950	2100	
			0.812	20.62	305.16	454.54	138.42	810	1020	950	1180	1710	1870	2110	2270	
			0.875	22.22	328.24	488.91	148.89	880	1090	1020	1280	1840	2010	2280	2450	
			0.938	23.83	351.25	523.19	159.33	940	1170	1090	1370	1970	2160	2440	2630	
			1.000	25.40	373.80	556.78	169.56	1000	1250	1170	1460	2100	2300	2600	2800	
			1.062	26.97	396.27	590.24	179.75	1060	1330	1240	1550	2230	2440	2760	2970	
			1.125	28.58	419.02	624.13	190.07	1130	1410	1310	1640	2360	2590	2930	3000	
			1.188	30.18	441.69	657.90	200.35	1190	1480	1390	1730	2490	2730	3000	3000	
			1.250	31.75	463.91	690.99	210.43	1250	1560	1460	1820	2630	2870	3000	3000	
38	38	965.2		0.312	7.92	125.58	187.05	56.96	300	370	340	430	620	680	770	830
				0.344	8.74	138.35	206.07	62.76	330	410	380	480	680	750	850	910
				0.375	9.52	150.69	224.45	68.35	360	440	410	520	750	820	920	990
				0.406	10.31	163.01	242.80	73.94	380	480	450	560	810	880	1000	1080
				0.438	11.13	175.71	261.72	79.70	410	520	480	610	870	950	1080	1160
				0.469	11.91	187.99	280.01	85.27	440	560	520	650	930	1020	1160	1240
				0.500	12.70	200.25	298.27	90.83	470	590	550	690	990	1090	1230	1330
				0.562	14.27	224.71	334.71	101.93	530	670	620	780	1120	1220	1380	1490
				0.625	15.88	249.48	371.60	113.16	590	740	690	860	1240	1360	1540	1560

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
									A		B					
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56
38	38	965.2		0.688	17.48	274.16	408.36	124.36	650	810	760	950	1370	1500	1690	1830
				0.750	19.05	298.37	444.42	135.34	710	890	830	1040	1490	1630	1850	1990
				0.812	20.62	322.50	480.36	146.29	770	960	900	1120	1620	1770	2000	2150
				0.875	22.22	346.93	516.75	157.37	830	1040	970	1210	1740	1910	2160	2320
				0.938	23.83	371.28	553.02	168.41	890	1110	1040	1300	1870	2040	2310	2490
				1.000	25.40	395.16	588.59	179.24	950	1180	1110	1380	1990	2180	2460	2650
				1.062	26.97	418.96	624.04	190.04	1010	1260	1170	1470	2110	2310	2620	2820
				1.125	28.58	443.05	659.92	200.97	1070	1330	1240	1550	2240	2450	2770	2980
				1.188	30.18	467.06	695.69	211.86	1130	1410	1310	1640	2360	2590	2930	3000
				1.250	31.75	490.61	730.76	222.54	1180	1480	1380	1730	2490	2720	3000	3000
40	40	1016.0		0.312	7.92	132.25	196.99	59.99	280	350	330	410	590	650	730	790
				0.344	8.74	145.69	217.01	66.08	310	390	360	450	650	710	800	870
				0.375	9.52	158.70	236.38	71.99	340	420	390	490	710	780	880	940
				0.406	10.31	171.68	255.72	77.87	370	460	430	530	770	840	950	1020
				0.438	11.13	185.06	275.65	83.94	390	490	460	570	830	910	1020	1100
				0.469	11.91	198.01	294.94	89.82	420	530	490	620	890	970	1100	1180
				0.500	12.70	210.93	314.18	95.68	450	560	520	660	940	1040	1170	1260
				0.562	14.27	236.71	352.58	107.37	510	630	590	740	1060	1160	1320	1420
				0.625	15.88	262.83	391.49	119.22	560	700	660	820	1180	1290	1460	1580
				0.688	17.48	288.86	430.26	131.03	620	770	720	900	1300	1420	1610	1730
				0.750	19.05	314.39	468.28	142.61	680	840	790	980	1420	1550	1760	1890
				0.812	20.62	339.84	506.19	154.15	730	910	850	1070	1530	1680	1900	2050
				0.875	22.22	365.62	544.59	165.85	790	980	920	1150	1650	1810	2050	2200
				0.938	23.83	391.32	582.87	177.50	840	1060	980	1230	1770	1940	2190	2360
				1.000	25.40	416.52	620.41	188.93	900	1120	1050	1310	1890	2070	2340	2520
				1.062	26.97	441.64	657.82	200.33	960	1190	1120	1390	2010	2200	2490	2680
				1.125	28.58	467.08	695.72	211.87	1010	1270	1180	1480	2130	2330	2630	2830
				1.188	30.18	492.44	733.49	223.37	1070	1340	1250	1560	2250	2460	2780	2990
				1.250	31.75	517.31	770.53	234.65	1130	1410	1310	1640	2360	2590	2930	3000
42	42	1066.8		0.344	8.74	153.04	227.95	69.42	290	370	340	430	620	680	770	830
				0.375	9.52	166.71	248.31	75.62	320	400	380	470	680	740	840	900
				0.406	10.31	180.35	268.63	81.81	350	430	410	510	730	800	900	970
				0.438	11.13	194.42	289.59	88.19	380	470	440	550	790	860	980	1050
				0.469	11.91	208.03	309.86	94.36	400	500	470	590	840	920	1050	1130
				0.500	12.70	221.61	330.09	100.52	430	540	500	620	900	990	1110	1200
				0.562	14.27	248.72	370.47	112.82	480	600	560	700	1010	1110	1250	1350
				0.625	15.88	276.18	411.37	125.28	540	670	620	780	1120	1230	1390	1500
				0.688	17.48	303.55	452.14	137.69	590	740	690	860	1240	1360	1530	1650
				0.750	19.05	330.41	492.15	149.87	640	800	750	940	1350	1480	1670	1800
				0.812	20.62	357.19	532.03	162.02	700	870	810	1020	1460	1600	1810	1950
				0.875	22.22	384.31	572.43	174.32	750	940	880	1090	1580	1720	1950	2100
				0.938	23.83	411.35	612.71	186.59	800	1000	940	1170	1690	1850	2090	2250
				1.000	25.40	437.88	652.22	198.62	860	1070	1000	1250	1800	1970	2230	2400
				1.062	26.97	464.32	691.60	210.62	910	1140	1060	1330	1910	2090	2370	2550
				1.125	28.58	491.11	731.51	222.77	960	1210	1130	1410	2030	2220	2510	2700
				1.188	30.18	517.82	771.29	234.88	1020	1270	1190	1480	2140	2340	2650	2850
				1.250	31.75	544.01	810.30	246.76	1070	1340	1250	1560	2250	2460	2790	3000

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
Nominal									A		B					
Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56
44	44	1117.6		0.344	8.74	160.39	238.90	72.75	280	350	330	410	590	650	730	790
				0.375	9.52	174.72	260.25	79.26	310	380	360	450	640	710	800	860
				0.406	10.31	189.03	281.56	85.74	330	420	390	480	700	760	860	930
				0.438	11.13	203.78	303.53	92.43	360	450	420	520	750	820	930	1000
				0.469	11.91	218.04	324.77	98.90	380	480	450	560	810	880	1000	1070
				0.500	12.70	232.29	346.00	105.37	410	510	480	600	860	940	1060	1150
				0.562	14.27	260.72	388.34	118.26	460	570	540	670	970	1060	1200	1290
				0.625	15.88	289.53	431.25	131.33	510	640	600	750	1070	1180	1330	1430
				0.688	17.48	318.25	474.03	144.36	560	700	660	820	1180	1290	1460	1580
				0.750	19.05	346.43	516.01	157.14	610	770	720	890	1290	1410	1600	1720
				0.812	20.62	374.53	557.86	169.89	660	830	780	970	1400	1530	1730	1860
				0.875	22.22	403.00	600.27	182.80	720	890	840	1040	1500	1650	1860	2000
				0.938	23.83	431.39	642.56	195.68	770	960	900	1120	1610	1770	2000	2150
				1.000	25.40	459.24	684.04	208.31	820	1020	950	1190	1720	1880	2130	2290
				1.062	26.97	487.01	725.40	220.91	870	1090	1010	1270	1820	2000	2260	2430
46	46	1168.4		1.125	28.58	515.14	767.30	233.67	920	1150	1070	1340	1930	2120	2390	2580
				1.188	30.18	543.19	809.08	246.39	970	1210	1130	1420	2040	2240	2530	2720
				1.250	31.75	570.71	850.07	258.87	1020	1280	1190	1490	2150	2350	2660	2860
				0.344	8.74	167.74	249.85	76.09	270	340	310	390	570	620	700	750
				0.375	9.52	182.73	272.18	82.89	290	370	340	430	620	680	760	820
				0.406	10.31	197.70	294.47	89.68	320	400	370	460	670	730	830	890
				0.438	11.13	213.13	317.46	96.68	340	430	400	500	720	790	890	960
				0.469	11.91	228.06	339.70	103.45	370	460	430	540	770	840	950	1030
				0.500	12.70	242.97	361.90	110.21	390	490	460	570	820	900	1020	1100
				0.562	14.27	272.73	406.23	123.71	440	550	510	640	920	1010	1140	1230
				0.625	15.88	302.88	451.14	137.39	490	610	570	710	1030	1120	1270	1370
				0.688	17.48	332.95	495.93	151.03	540	670	630	790	1130	1240	1400	1510
				0.750	19.05	362.45	539.87	164.41	590	730	680	860	1230	1350	1530	1640
				0.812	20.62	391.88	583.12	177.76	640	790	740	930	1330	1460	1650	1780
				0.875	22.22	421.69	628.11	191.28	680	860	800	1000	1440	1580	1780	1920
				0.938	23.83	451.42	672.39	204.76	730	920	860	1070	1540	1690	1910	2060
48	48	1219.2		1.000	25.40	480.60	715.85	218.00	780	980	910	1140	1640	1800	2030	2190
				1.062	26.97	509.69	759.18	231.20	830	1040	970	1210	1750	1910	2160	2330
				1.125	28.58	539.17	803.09	244.57	880	1100	1030	1280	1850	2020	2290	2470
				1.188	30.18	568.57	846.89	257.90	930	1160	1080	1360	1950	2140	2420	2600
				1.250	31.75	597.41	889.84	270.99	980	1220	1140	1430	2050	2250	2540	2740
				0.344	8.74	175.08	260.78	79.42	260	320	300	380	540	590	670	720
				0.375	9.52	190.74	284.11	86.52	280	350	330	410	590	650	730	790
				0.406	10.31	206.37	307.39	93.61	300	380	360	440	640	700	790	850
				0.438	11.13	222.49	331.40	100.92	330	410	380	480	690	760	850	920
				0.469	11.91	238.08	354.62	107.99	350	440	410	510	740	810	910	980
				0.500	12.70	253.65	377.81	115.06	380	470	440	550	790	860	980	1050
				0.562	14.27	284.73	424.11	129.15	420	530	490	610	890	970	1100	1180
				0.625	15.88	316.23	471.02	143.44	470	590	550	680	980	1080	1220	1310
				0.688	17.48	347.64	517.81	157.69	520	640	600	750	1080	1190	1340	1440
				0.750	19.05	378.47	563.73	171.67	560	700	660	820	1180	1290	1460	1580
				0.812	20.62	409.22	609.53	185.62	610	760	710	890	1280	1400	1580	1710
				0.875	22.22	440.38	655.95	199.76	660	820	770	960	1380	1510	1710	1840
				0.938	23.87	471.46	702.24	213.85	700	880	820	1030	1480	1620	1830	1970
				1.000	25.40	501.96	747.67	227.69	750	940	880	1090	1580	1720	1950	2100
				1.062	26.97	532.38	792.98	241.49	800	1000	930	1160	1670	1830	2070	2230
				1.125	28.58	563.20	838.89	255.47	840	1050	980	1230	1770	1940	2190	2360
				1.188	30.18	593.94	884.67	269.41	890	1110	1040	1300	1870	2050	2320	2490
				1.250	31.75	624.11	929.61	283.10	940	1170	1090	1370	1970	2160	2440	2620

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
									A		B					
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56
52	52	1320.8		0.375	9.52	206.76	307.97	93.79	--	--	--	--	--	--	--	--
				0.406	10.31	223.72	333.23	101.48	--	--	--	--	--	--	--	--
				0.438	11.13	241.20	359.27	109.41	300	380	350	440	640	700	790	850
				0.469	11.91	258.11	384.45	117.08	320	410	380	470	680	750	840	910
				0.500	12.70	275.01	409.63	124.74	350	430	400	500	730	800	900	970
				0.562	14.27	308.74	459.87	140.04	390	490	450	570	820	890	1010	1090
				0.625	15.88	342.93	510.79	155.55	430	540	500	630	910	1000	1130	1210
				0.688	17.48	377.03	561.59	171.02	480	600	560	690	1000	1100	1240	1330
				0.750	19.05	410.51	611.45	186.21	520	650	610	760	1090	1190	1350	1450
				0.812	20.62	443.91	661.20	201.36	560	700	660	820	1180	1290	1460	1570
				0.875	22.22	477.76	711.62	216.71	610	760	710	880	1270	1390	1580	1700
				0.938	23.83	511.53	761.92	232.03	650	810	760	950	1360	1490	1690	1820
				1.000	25.40	544.68	811.30	247.07	690	870	810	1010	1450	1590	1800	1940
				1.062	26.97	577.75	860.56	262.07	740	920	860	1070	1540	1690	1910	2060
				1.125	28.58	611.26	910.47	277.27	780	970	910	1140	1640	1790	2030	2180
				1.188	30.18	644.69	960.27	292.43	820	1030	960	1200	1730	1890	2140	2300
				1.250	31.75	677.51	1009.15	307.32	870	1080	1010	1260	1820	1990	2250	2420
56	56	1422.4		0.375	9.52	222.78	340.77	101.05	--	--	--	--	--	--	--	--
				0.406	10.31	241.06	359.06	109.34	--	--	--	--	--	--	--	--
				0.438	11.13	259.91	387.14	117.90	280	350	330	410	590	650	730	790
				0.469	11.91	278.15	414.30	126.17	300	380	350	440	630	690	780	840
				0.500	12.70	296.37	441.44	134.43	320	400	380	470	680	740	840	900
				0.562	14.27	332.75	495.63	150.94	360	450	420	530	760	830	940	1010
				0.625	15.88	369.63	550.56	167.66	400	500	470	590	840	920	1040	1120
				0.688	17.48	406.42	605.36	184.35	440	550	520	650	930	1020	1150	1240
				0.750	19.05	442.55	659.18	200.74	480	600	560	700	1010	1110	1250	1350
				0.812	20.62	478.60	712.87	217.09	520	650	610	760	1100	1200	1360	1460
				0.875	22.22	515.14	767.30	233.67	560	700	660	820	1180	1290	1460	1570
				0.938	23.83	551.60	821.61	250.21	600	750	700	880	1270	1390	1570	1690
				1.000	25.40	587.40	874.93	266.44	640	800	750	940	1350	1480	1670	1800
				1.062	26.97	623.12	928.14	282.65	680	850	800	1000	1430	1570	1780	1910
				1.125	28.58	659.32	982.06	299.07	720	900	840	1050	1520	1660	1880	2020
				1.188	30.18	695.45	1035.87	315.46	760	950	890	1110	1600	1760	1990	2140
				1.250	31.75	730.91	1088.69	331.54	800	1000	940	1170	1690	1850	2090	2250
60	60	1524.0		0.375	9.52	238.80	355.69	108.32	--	--	--	--	--	--	--	--
				0.406	10.31	258.40	384.89	117.21	--	--	--	--	--	--	--	--
				0.438	11.13	278.62	415.00	126.38	--	--	--	--	--	--	--	--
				0.469	11.91	298.19	444.15	135.26	--	--	--	--	--	--	--	--
				0.500	12.70	317.73	473.26	144.12	300	380	350	440	630	690	780	840
				0.562	14.27	356.76	531.39	161.83	340	420	390	490	710	780	880	940
				0.625	15.88	396.33	590.33	179.78	380	470	440	550	790	860	980	1050
				0.688	17.48	435.71	649.14	197.68	410	520	480	600	870	950	1070	1160
				0.750	19.05	474.59	706.90	215.27	450	560	530	660	950	1030	1170	1260
				0.812	20.62	513.29	764.55	232.83	490	610	570	710	1020	1120	1270	1360
				0.875	22.22	552.52	822.98	250.62	530	660	610	770	1100	1210	1370	1470
				0.938	23.83	591.67	881.29	268.38	560	700	660	820	1180	1290	1460	1580
				1.000	25.40	630.12	938.56	285.82	600	750	700	880	1260	1380	1560	1680
				1.062	26.97	668.48	995.70	303.22	640	800	740	930	1340	1470	1660	1780
				1.125	28.58	707.38	1053.64	320.87	680	840	790	980	1420	1550	1760	1890
				1.188	30.18	746.20	1111.46	338.48	710	890	830	1040	1500	1640	1850	2000
				1.250	31.75	784.31	1168.23	355.76	750	940	880	1090	1580	1720	1950	2100

API PIPES ERW & SEAMLESS PIPES

Size						Weight			Hydrostatic Test Pressure							
Outside Diameter			Wall Thickness						API 5L				API 5LX			
									A		B					
Nominal Size	in	mm	Sch No	in	mm	lb/ft	kg/m	kg/ft	Std.	Alt.	Std.	Alt.	X42	X46	X52	X56
64	64	1625.6		0.375	9.52	254.82	379.55	115.59	--	--	--	--	--	--	--	--
				0.406	10.31	275.75	410.73	125.08	--	--	--	--	--	--	--	--
				0.438	11.13	297.33	442.87	134.87	--	--	--	--	--	--	--	--
				0.469	11.91	318.22	473.99	144.34	--	--	--	--	--	--	--	--
				0.500	12.70	339.09	505.07	153.81	280	350	330	410	590	650	730	790
				0.562	14.27	380.76	567.14	172.71	320	400	370	460	660	730	820	890
				0.625	15.88	423.03	630.10	191.89	350	440	410	510	740	810	910	980
				0.688	17.48	465.21	692.93	211.02	390	480	450	560	810	890	1010	1080
				0.750	19.05	506.63	754.63	229.81	420	530	490	620	890	970	1100	1180
				0.812	20.62	547.98	816.22	248.56	460	570	530	670	960	1050	1190	1280
				0.875	22.22	589.90	878.66	267.58	490	620	570	720	1030	1130	1280	1380
				0.938	23.87	631.74	940.98	286.56	530	660	620	770	1110	1210	1370	1480
				1.000	25.40	672.84	1002.20	305.20	560	700	660	820	1180	1290	1460	1570
				1.062	26.97	713.85	1063.28	323.80	600	750	700	870	1250	1370	1550	1670
				1.125	28.58	755.44	1125.22	342.67	630	790	740	920	1330	1460	1650	1770
				1.188	30.18	796.95	1187.06	361.50	670	840	780	970	1400	1540	1740	1870
				1.250	31.75	837.71	1247.77	379.99	700	880	820	1030	1480	1620	1830	1970





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YICK HOE

COLD FORMED HOLLOW SECTIONS

STANDARD SPECIFICATIONS FOR WELDED CIRCULAR & NON-CIRCULAR STEEL TUBES

Standard Specification	Scope	Grade of Tubes	Chemical Composition Percentage (max. unless range is given)				Mechanical Properties			
			C	Mn	P	S	Condition	Tensile Strength	Yield Strength	Elongation
								(min.)	(min.)	(% min.)
1	2	3	4	5	6	7	8	9	10	11
ASTM-A53	E.R.W. Carbon Steel Pipes	Type E Grade A	0.25	0.95	0.05	0.06	As Rolled	ksi (MPa)	ksi (MPa)	(G.L. = 2" or 50mm)
								48 (331)	30 (207)	As per Table X7
ASTM-A120	E.R.W. Carbon Steel Pipes for ordinary use	Standard Pipe	--	--	--	--	--	--	--	--
ASTM-A135	Standard Specification for ERW Steel Pipe	Grade A	0.25	0.95	0.05	0.06	As Rolled	48 (331)	30 (207)	As per Table 1 Equation: (Elongation) = 56 x (Actual Thickness of Specimen, inch) + 17.50
ASTM-A500	E.R.W. Carbon Steel Structural Tubing in Round & Shapes	Round Grade A	0.26	--	0.04	0.05	As Rolled	45 (310)	33 (228)	25
		Grade B	0.26	--	0.04	0.05		58 (400)	42 (290)	23
		Shaped Grade A	0.26	--	0.04	0.05		45 (310)	39 (269)	25
		Grade B	0.26	--	0.04	0.05		58 (400)	46 (317)	23
ASTM-A513	E.R.W. Carbon Steel Mechanical Tubing	MT 1010	0.5-0.15	0.3-0.6	0.04	0.05	As Rolled	45 (310)	32 (221)	15
		MT 1015	0.1-0.2	0.3-0.6	0.04	0.05		48 (331)	35 (241)	15
		MT 1020	0.15-0.25	0.3-0.6	0.04	0.05		52 (359)	38 (262)	12
BS-31	E.R.W. Steel Conduits	Class A Plain End	--	--	--	--	As Rolled & Coated G.I. Strip	tonf/in ²	tonf/in ²	15 (G.L. = 8")
BS-980	E.R.W. Carbon Steel Pipes for Automobile Purposes	ERW 1	0.20	0.60	0.06	0.06	As Rolled	20	11	20
		ERW 2	0.30	0.60	0.06	0.06		25	15	15
		ERW 3	0.40	0.60	0.06	0.06		30	20	10 (G.L. = $4\sqrt{S_o}$)
BS-1139	E.R.W. Metal Scaffolding		--	--	0.06	0.06	As Rolled (Black Tubes)	20 - 30	13.5	$700/T.S.$ (ton/in ²) (G.L. = $4\sqrt{S_o}$)
BS-1387	E.R.W. Carbon Steel Tubes	Light	--	--	0.06	0.06	As Rolled	33 - 47	--	20 (G.L. = $5.65\sqrt{S_o}$)
		Medium	--	--	--	--				
		Heavy	--	--	--	--				
BS-1717	E.R.W. Carbon Steel Tubes for Bicycle and Motorcycle Purposes	ERW 101	0.10	0.60	0.06	0.06	As Rolled	20	11	--
		ERW 102	0.20	0.60	0.06	0.06		25	15	--
		ERW 103	0.30	0.60	0.06	0.06		30	20	--
BS-1775	E.R.W. Carbon Steel Tubes for Mechanical Structural and General Engineering Purposes	ERW 11	--	--	0.06	0.06	As Rolled	20	11	$700/T.S.$ (ton/in ²) (G.L. = $4\sqrt{S_o}$) or $600/T.S.$ (ton/in ²) (G.L. = $5.65\sqrt{S_o}$)
		ERW 16	--	--	0.06	0.06		25	16	
		ERW 20	--	--	0.06	0.06		30	20	
		ERW 23	--	--	0.06	0.06		32	23	

*1: G.L. = Gauge Length

*2: S_o = Original Cross-Section Area

*3: T.S. = Tensile Strength

*4: T.P. = Test Piece as per JIS Z2201

*5: H = Distance between Outside Surfaces

*6: D = Outside Diameter

*7: t = Wall Thickness of Tubes

*8: H' = Distance between Inside Surfaces

COLD FORMED HOLLOW SECTIONS

Other Tests			Dimensional Tolerances		
Flattening (min.)	Bend Test	Hydrostatic Test	On Diameter except for Non-Circular Tubes		On Wall Thickness
12	13	14	15		16
Apply for STD & XS of NB>2" Welded Portion: H= 2/3D The Other Side: H=1/3D	Apply fo NB ≤2" 12D at 90° When ordered for Close coiling: 8D at 180°	As per Table X2	1-1/2" & under 2" & over	+ 1/64" - 1/32" ± 1 % of OD	+ not limited -12.5%
--	--	1/2" to 1": P=700 psi 1-1/4" to 3": P= 1,000 psi 3-1/2" to 6": P=1,200 psi	1-1/2" & under 2" & over	+ 1/64" - 1/32" ± 1% of OD	+ not limited -12.5%
Welded Portion: H = 2/3D The Other Side: H = 1/3D	--	P=2St/D, Max. 2,500 psi S: Allowable Fiber Stress (16,000-18,000 psi) for Grade A. But in no case shall the stress produced exceed 80% of the specified Y.P	± 1% of OD		+ not limited -12.5%
Welded Portion: H = 2/3D The Other Side: H = 1/2D	--	--	Up to 1.90" 2" & over Under 2-1/2" 2-1/2" to 3-1/2" excluding 3-1/2" to 5-1/2" excluding 5-1/2" & over Square of Radii	± 0.5% ± 0.75% ± 0.02" ± 0.025" ± 0.03" ± 1% ± 2% max.	± 10%
Welded Portion: H = 2/3D The Other Side: H = 1/3D	--	P=2St/D S: Allowable Fiber Stress of 14,000 psi	As per Table 4 & 7		As per Table 6 & 8
--	6D at 90°	--	+ 0.001" - 0.005"		Nominal 0.04" - 0.072" Min. 0.036" - 0.068"
Max. 3t Max. 5t Max. 8t	--	--	1" & under Over 1" to 2" Over 2" to 2-1/2" Over 2-1/2" to 3" Over 3" to 3-1/2" Over 3-1/2" to 4" Over 4" to 4-1/2"	±0.004" ±0.005" ±0.007" ±0.008" ±0.010" ±0.012" ±0.015"	±8%
--	6D at 180°	--	Only 1.906" ±0.013" -0.017"		+ 20% - 10%
Apply for NB>2" Welded Portion: H=0.75D The Other Side: H=0.6D	Apply for NB≤2" Ungalvanized Tubes: 6D at 180°; Galvanized Tubes: 8D at 90°	P=700 lbf/in ² (50 kgf/cm ²)	Light Tubes: Medium, Heavy Tubes:	Table 1 Table 2	Light Tubes: -8%; + not limited Medium, Heavy Tubes: -10%; + not limited
Max. 3t Max. 5t Max. 8t	--	--	1-1/4" & under Over 1-1/4" to 1-5/8"	+ 0.001" -0.005" +0.002" - 0.006"	±8%
H'= 3t or 1/2D H'= 6t or 3/4D H'= 8t or 7/8D H'= 6t or 3/4D (whichever is the smaller)	--	--	1" & under over 1" to 2" over 2" to 2-1/2" over 2-1/2" to 3" over 3" to 3-1/2" over 3-1/2" to 4" over 4" to 4-1/2" over 4-1/2" to 5" over 5" to 5-1/2" over 5-1/2" to 6" over 6" to 6-5/8"	±0.006" ±0.007" ±0.009" ±0.010" ±0.012" ±0.014" ±0.016" ±0.018" ±0.020" ±0.022" ±0.025"	±8%

*9: P = Test Pressure

*10 t¹ = Thickness of Flat Portion

COLD FORMED HOLLOW SECTIONS

Standard Specification	Scope	Grade of Tubes	Chemical Composition Percentage (max. unless range is given)				Mechanical Properties				
			C	Mn	P	S	Condition (min.)	Tensile Strength (min.)	Yield Strength (% min.)	Elongation	
1	2	3	4	5	6	7	8	9	10	11	
JIS-G 3444	E.R.W. Carbon Steel Pipes for Civil Engineer Architecture, Steel Towers, Scaffolding, Pile Struts & other Structures.	STK - 30 STK - 41 STK - 51 STK - 50	-- 0.25 0.30 0.18	-- -- 0.3 - 1.0 1.50	0.05 0.04 0.04 0.04	0.05 0.04 0.04 0.04	As Rolled	kgf/mm²	kgf/mm²	longitudinal	transversal
										No. 11,12 T.P.	No. 5 T.P.
								30	—	30	25
								41	24	23	18
								51	36	15	10
			0.18	1.50	0.04	0.04		50	32	23	18
JIS-G 3445	E.R.W. Carbon Steel Pipes for Machine Structural Purposes	STKM-11A STKM-12A STKM-12B STKM-12C STKM-13A STKM-13B STKM-13C STKM-14A STKM-14B	0.12 0.20 0.25 0.30	0.25-0.6 0.25-0.6 0.30-0.9 0.30-1.0	0.04 0.04 0.04 0.04	0.04 0.04 0.04 0.04	As Rolled	30 35 40 48 38 45 52 42 51	-- 18 28 36 22 31 39 25 36	No. 4,11,12 T.P.	No. 4,5 T.P.
										35	30
										35	30
										25	20
										20	15
										30	25
										20	15
										15	10
										25	20
										15	10
JIS-G 3452	E.R.W. Carbon Steel Pipes for ordinary use	SGP	--	--	0.05	0.05	As Rolled (Black Tubes)	30	--	No. 11, 12 T.P.	No. 5 T.P.
										30	25
JIS-G 3466	E.R.W. Carbon Steel Square Tubes for General Structural Purposes	STKR-41 STKR-50	0.25 0.18	-- 1.50	0.04 0.04	0.04 0.04	As Rolled	41 50	25 33	No. 5 T.P.	--
										23	--
										23	--

*1: G.L. = Gauge Length
 *2: S_o = Original Cross-Section Area
 *3: T.S. = Tensile Strength

*4: T.P. = Test Piece as per JIS Z2201
 *5: H = Distance between Outside Surfaces

*6: D = Outside Diameter
 *7: t = Wall Thickness of Tubes
 *8: H' = Distance between Inside Surfaces

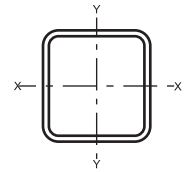
COLD FORMED HOLLOW SECTIONS

Other Tests			Dimensional Tolerances	
Flattening (min.)	Bend Test	Hydrostatic Test	On Diameter except for Non-Circular Tubes	On Wall Thickness
12	13	14	15	16
H = 2/3D H = 2/3D H = 7/8D H = 7/8D	6D at 90° 6D at 90° 8D at 90° 6D at 90°	-- -- --	Under 50 mm ± 0.25 mm 50 mm & over ± 0.50 %	Under 3mm ±0.30 mm 3mm - 12 mm ± 10% excluding
H = 1/2D H = 2/3D H = 2/3D -- H = 2/3D H = 3/4D -- H = 3/4D H = 7/8D	4D at 180° 6D at 90° 6D at 90° -- 6D at 90° 6D at 90° -- 6D at 90° 8D at 90°	--	As per Classification (No. 1, 2, 3) of Table 4	As per Classification (No. 1,2,3) of Table 5
H = 2/3D	6D at 90°	P=25 kgf/mm ²	10.5 to 48.6 mm ± 0.50mm 60.5 to 165. 2 mm ± 1%	+ not limited -12.5%
--	--	--	Up to 100 mm ± 1.5mm Over 100 mm ± 1.5% Squareness of side ± 1.5° Radii of corner Max. 3t ¹	Up to 3mm ±0.30 mm 3mm & over ± 10%

*9: P = Test Pressure

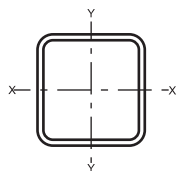
*10 t¹ = Thickness of Flat Portion

COLD FORMED HOLLOW SECTIONS



SQUARE

Section size	Wall thickness t	Unit weight M	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				I _x	I _y	i _x	i _y	Z _x	Z _y
mm	mm	kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
13 x 13	1.2	0.40	0.51	0.10	0.10	0.45	0.45	0.16	0.16
	1.6	0.50	0.64	0.12	0.12	0.43	0.43	0.19	0.19
16 x 16	1.2	0.52	0.67	0.23	0.23	0.59	0.59	0.29	0.29
	1.6	0.67	0.85	0.27	0.27	0.57	0.57	0.34	0.34
19 x 19	1.2	0.64	0.81	0.41	0.41	0.71	0.71	0.44	0.44
	1.6	0.82	1.04	0.50	0.50	0.69	0.69	0.53	0.53
25 x 25	1.2	0.86	1.10	1.02	1.02	0.96	0.96	0.81	0.81
	1.6	1.12	1.43	1.26	1.26	0.94	0.94	1.01	1.01
	2.3	1.53	1.95	1.59	1.59	0.90	0.90	1.27	1.27
	3.0	1.93	2.48	1.96	1.96	0.89	0.89	1.54	1.54
32 x 32	1.2	1.12	1.42	2.16	2.16	1.24	1.24	1.38	1.38
	1.6	1.46	1.87	2.75	2.75	1.22	1.22	1.74	1.74
	2.3	2.02	2.68	3.93	3.93	1.21	1.21	2.46	2.46
	3.0	2.53	3.26	4.29	4.29	1.15	1.15	2.70	2.70
38 x 38	1.2	1.35	1.72	3.85	3.85	1.49	1.49	2.02	2.02
	1.6	1.77	2.26	4.91	4.91	1.47	1.47	2.58	2.58
	2.3	2.47	3.14	6.51	6.51	1.43	1.43	3.42	3.42
	3.0	3.13	4.04	7.99	7.99	1.41	1.41	4.26	4.26
50 x 50	1.6	2.38	3.03	11.70	11.70	1.96	1.96	4.68	4.68
	2.3	3.34	4.25	15.90	15.90	1.93	1.93	6.34	6.34
	3.0	4.24	5.40	19.39	19.39	1.89	1.89	7.75	7.75
	3.2	4.50	5.73	20.40	20.40	1.89	1.89	8.16	8.16
	4.5	6.01	7.66	25.22	25.22	1.81	1.81	10.09	10.09
	6.0	7.56	9.63	28.76	28.76	1.72	1.72	11.50	11.50
60 x 60	1.6	2.88	3.67	20.70	20.70	2.37	2.37	6.89	6.89
	2.3	4.06	5.17	28.30	28.30	2.34	2.34	9.44	9.44
	3.0	5.18	6.60	35.04	35.04	2.30	2.30	11.68	11.68
	3.2	5.50	7.00	36.90	36.90	2.30	2.30	12.30	12.30
	4.5	7.43	9.46	46.88	46.88	2.22	2.22	15.62	15.62
	6.0	9.44	12.03	55.26	55.26	2.14	2.14	18.42	18.42
63.5 x 63.5	2.3	4.31	5.51	33.88	33.88	2.46	2.46	10.75	10.75
	3.0	5.52	7.16	42.46	42.46	2.44	2.44	13.44	13.44
	4.5	8.04	10.24	57.44	57.44	2.37	2.37	18.19	18.19
	6.0	10.57	13.48	70.34	70.34	2.28	2.28	22.12	22.12
75 x 75	2.3	5.14	6.55	57.10	57.10	2.95	2.95	15.20	15.20
	3.0	6.60	8.40	71.50	71.50	2.91	2.91	19.06	19.06
	3.2	7.01	8.92	75.50	75.50	2.91	2.91	20.10	20.10
	4.5	9.55	12.16	98.16	98.16	2.84	2.84	26.17	26.17
	6.0	12.27	15.63	119.2	119.2	2.76	2.76	31.78	31.78

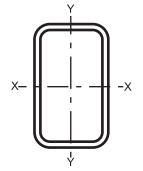


COLD FORMED HOLLOW SECTIONS

SQUARE

Section size	Wall thickness t	Unit weight M	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				I _x	I _y	i _x	i _y	Z _x	Z _y
mm	mm	kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
100 x 100	2.3	6.95	8.85	140.0	140	3.97	3.97	27.90	27.9
	3.0	8.95	11.40	176.9	176.9	3.93	3.93	35.38	35.38
	3.2	9.52	12.13	187.0	187	3.93	3.93	37.50	37.5
	4.5	13.1	16.67	249.0	249	3.87	3.87	49.90	49.9
	6.0	17.0	21.63	311.0	311	3.79	3.79	62.30	62.3
	9.0	24.1	30.67	408.0	408	3.65	3.65	81.60	81.6
112.5 x 112.5	2.3	7.85	10.00	200.97	200.97	4.48	4.48	35.72	35.72
	3.0	10.13	12.90	255.43	255.43	4.44	4.44	45.41	45.41
	4.5	14.85	18.91	362.01	362.01	4.37	4.37	64.35	64.35
	6.0	19.33	24.63	455.21	455.21	4.29	4.29	80.92	80.92
125 x 125	2.3	8.75	11.15	278.02	278.02	4.99	4.99	44.48	44.48
	3.0	11.31	14.40	354.32	354.32	4.95	4.95	56.69	56.69
	3.2	12.0	15.33	376	376	4.95	4.95	60.10	60.10
	4.5	16.61	21.17	506	506	4.89	4.89	80.90	80.90
	6.0	21.7	27.63	641	641	4.82	4.82	103	103
	9.0	31.1	39.67	865	865	4.67	4.67	138	138
150 x 150	4.5	20.1	25.67	896	896	5.91	5.91	120	120
	6.0	26.4	33.63	1,150	1,150	5.84	5.84	153	153
	9.0	38.2	48.67	1,580	1,580	5.69	5.69	210	210
175 x 175	6.0	31.1	39.63	1,860	1,860	6.86	6.86	213	213
	9.0	45.3	57.67	2,600	2,600	6.71	6.71	297	297
200 x 200	6.0	35.8	45.63	2,830	2,830	7.88	7.88	283	283
	8.0	46.9	59.79	3,620	3,620	7.78	7.78	362	362
	9.0	52.3	66.67	3,990	3,990	7.73	7.73	399	399
	12.0	67.9	86.53	4,980	4,980	7.59	7.59	498	498
250 x 250	6.0	45.2	57.63	5,670	5,670	9.92	9.92	454	454
	9.0	66.5	84.67	8,090	8,090	9.78	9.78	647	647
	12.0	86.8	110.5	10,300	10,300	9.63	9.63	820	820
300 x 300	6.0	54.7	69.63	9,960	9,960	12.0	12.0	664	664
	9.0	80.6	102.7	14,300	14,300	11.8	11.8	956	956
	12.0	106	134.5	18,300	18,300	11.7	11.7	1,220	1,220
	16.0	138	175.2	23,100	23,100	11.5	11.5	1,540	1,540
350 x 350	9.0	94.7	120.7	23,200	23,200	13.9	13.9	1,320	1,320
	12.0	124	158.5	29,800	29,800	13.7	13.7	1,700	1,700
	16.0	163	207.2	37,900	37,900	13.5	13.5	2,160	2,160
400 x 400	9.0	109	138.7	35,100	35,100	15.9	15.9	1,750	1,750
	12.0	143	182.5	45,300	45,300	15.8	15.8	2,270	2,270
	16.0	188	239.2	57,900	57,900	15.6	15.6	2,900	2,900

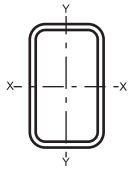
COLD FORMED HOLLOW SECTIONS



RECTANGULAR

Section size	Wall thickness t	Unit weight M	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				I _x	I _y	r _{ix}	r _{iy}	Z _x	Z _y
mm	mm	kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
19 x 9	1.2	0.421	0.0833	0.25	0.07	0.65	0.35	0.26	0.16
	1.6	0.54	0.107	0.30	0.86	0.63	0.33	0.32	0.19
25 x 12	1.2	0.67	0.85	0.62	0.21	0.87	0.50	0.51	0.33
	1.6	0.86	1.08	0.75	0.25	0.85	0.46	0.61	0.39
38 x 19	1.0	0.78	1.104	2.10	0.78	1.37	0.81	1.37	0.80
	1.2	1.02	1.30	2.41	0.81	1.36	0.79	1.27	0.85
	1.6	1.34	1.70	3.09	1.02	1.35	0.77	1.62	1.07
50 x 25	1.0	1.15	1.464	4.99	1.69	1.846	1.077	1.96	1.33
	1.2	1.36	1.86	5.99	2.04	1.84	1.07	2.36	1.59
	1.6	1.79	2.48	7.78	2.58	1.83	1.05	3.06	2.05
	2.3	2.48	3.19	9.86	3.23	1.76	1.01	3.94	2.59
	3.0	3.13	4.04	11.86	3.87	1.73	0.99	4.67	3.06
60 x 40	1.6	2.37	3.03	15.21	8.14	2.23	1.63	5.07	4.07
	2.3	3.33	4.25	20.61	10.96	2.20	1.60	6.87	5.48
	3.0	4.24	5.40	25.29	13.36	2.16	1.57	8.43	6.68
	4.5	6.01	7.66	33.01	17.17	2.07	1.49	11.00	8.58
75 x 38	1.6	2.70	3.30	24.97	8.69	2.72	1.60	6.55	4.57
	2.3	3.85	4.85	34.60	12.00	2.67	1.57	9.23	6.30
	3.0	4.92	6.35	45.20	15.15	2.66	1.54	11.86	7.95
75 x 50	1.9	3.60	4.606	36.62	20.22	2.84	2.09	9.61	7.96
	2.3	4.24	5.40	41.86	22.36	2.78	2.03	11.16	8.94
	3.0	5.42	6.90	52.06	27.67	2.74	2.00	13.88	11.06
	4.5	7.78	9.91	70.20	36.87	2.66	1.92	18.72	14.74
	6.0	10.56	13.47	91.98	47.86	2.62	1.88	24.09	18.85
100 x 50	1.9	4.36	5.484	73.67	25.47	3.65	2.15	14.50	10.02
	2.3	5.14	6.552	84.80	29.00	3.60	2.10	17.00	11.60
	3.0	6.60	8.40	106.33	35.95	3.55	2.06	21.26	14.38
	3.2	7.01	8.927	112.00	38.00	3.55	2.06	22.50	15.20
	4.5	9.55	12.16	146.18	48.51	3.46	1.99	29.23	19.40
	6.0	12.27	15.63	177.71	57.80	3.37	1.92	35.54	23.12
100 x 75	2.3	6.04	7.70	112.22	72.25	3.81	3.06	22.44	19.26
	3.0	7.77	9.90	141.61	90.94	3.78	3.02	28.32	24.25
	4.5	11.31	14.41	197.48	126.12	3.70	2.95	39.49	33.63
	6.0	14.72	18.63	243.98	154.91	3.61	2.88	48.79	41.31
125 x 50	2.3	6.04	7.70	148.17	35.44	4.38	2.14	23.70	14.17
	3.0	7.77	9.90	186.87	44.24	4.34	2.11	29.90	17.69
	4.5	11.31	14.41	260.18	60.16	4.24	2.04	41.62	24.06
	6.0	14.62	18.63	320.77	72.32	4.14	1.97	51.32	28.92

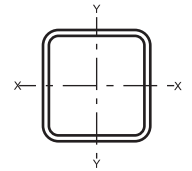
COLD FORMED HOLLOW SECTIONS



RECTANGULAR

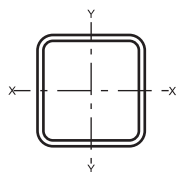
Section size	Wall thickness t	Unit weight M	Section area A	Moment of inertia		Radius of gyration		Modulus of section	
				I _x	I _y	i _x	i _y	Z _x	Z _y
mm	mm	kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³
125 x 75	2.3	6.95	8.852	192.00	87.50	4.65	3.14	30.60	23.30
	3.0	8.95	11.40	242.69	110.38	4.61	3.11	38.83	29.43
	3.2	9.52	12.13	257.00	117.00	4.60	3.10	41.10	31.10
	4.5	13.08	16.66	341.86	154.08	4.52	3.04	54.69	41.08
	6.0	16.98	21.63	426.98	190.62	4.44	2.96	68.31	50.83
150 x 75	3.0	10.13	12.90	379.41	129.82	5.42	3.17	50.58	34.62
	3.2	10.80	13.73	402.00	137.00	5.41	3.16	53.60	36.60
	4.5	14.85	18.91	538.32	182.03	5.33	3.10	71.77	48.54
	6.0	19.33	24.63	677.58	226.32	5.24	3.03	90.34	60.35
	9.0	27.60	35.17	905.00	297.00	5.07	2.91	121	79.20
150 x 100	3.0	11.31	14.40	460.45	247.47	5.65	4.14	61.39	49.49
	3.2	12.00	15.33	488.00	262.00	5.64	4.14	65.10	52.50
	4.5	16.60	21.17	658.00	352.00	5.58	4.08	87.70	70.40
	6.0	21.70	27.63	835.00	444.00	5.50	4.01	111.08	88.80
	9.0	31.10	39.67	1,430.00	595.00	5.33	3.87	151	119
200 x 100	4.5	20.1	25.67	1,330	455	7.20	4.21	133	90.9
	6.0	26.4	33.63	1,700	577	7.12	4.14	170	115
	9.0	38.2	48.67	2,350	782	6.94	4.01	235	156
	12.0	49.1	62.53	2,860	939	6.76	3.87	286	188
200 x 150	6.0	31.1	39.63	2,270	1,460	7.56	6.06	227	194
	9.0	45.3	57.67	3,170	2,020	7.41	5.93	317	270
	12.0	58.5	74.53	3,920	2,500	7.25	5.79	392	330
250 x 150	6.0	35.8	45.63	3,890	1,770	9.23	6.23	311	236
	9.0	52.3	66.67	5,480	2,470	9.06	6.09	438	330
	12.0	67.9	86.53	6,850	3,070	8.90	5.95	548	409
300 x 200	6.0	45.2	57.63	7,370	3,960	11.3	8.29	491	396
	9.0	66.5	84.67	10,500	5,630	11.2	8.16	702	563
	12.0	86.8	110.5	13,400	7,110	11.0	8.02	890	711
400 x 200	6.0	54.7	69.63	14,800	5,090	14.6	8.55	739	509
	9.0	80.6	102.7	21,300	7,270	14.4	8.42	1,070	727
	12.0	106.0	134.5	27,300	9,230	14.2	8.28	1,360	923

COLD FORMED HOLLOW SECTIONS



SQUARE

Section size		Wall thickness t		Unit weight M		Section area A	Moment of inertia I	Radius of gyration i	Modulus of section Z
in	mm	in	mm	lb/ft	kg/m	in ²	in ⁴	in	in ³
1/2 x 1/2	12.7 x 12.7	0.047	1.2	0.272	0.404	0.082	0.002	0.180	0.010
		0.063	1.6	0.364	0.541	0.102	0.003	0.173	0.012
5/8 x 5/8	15.9 x 15.9	0.047	1.2	0.353	0.525	0.106	0.005	0.231	0.018
		0.063	1.6	0.472	0.702	0.134	0.006	0.223	0.021
3/4 x 3/4	19.1 x 19.1	0.047	1.2	0.434	0.646	0.140	0.011	0.288	0.030
		0.063	1.6	0.580	0.863	0.186	0.014	0.282	0.037
		0.095	2.4	0.817	1.230	0.240	0.017	0.270	0.046
1 x 1	25.4 x 25.4	0.047	1.2	0.593	0.883	0.173	0.026	0.385	0.051
		0.063	1.6	0.770	1.150	0.226	0.032	0.376	0.064
		0.091	2.3	1.050	1.563	0.308	0.044	0.371	0.080
		0.095	2.4	1.100	1.640	0.321	0.045	0.358	0.082
		0.120	3.0	1.296	1.929	0.385	0.047	0.350	0.094
1-1/4 x 1-1/4	31.8 x 31.8	0.047	1.2	0.754	1.122	0.220	0.052	0.487	0.084
		0.063	1.6	0.983	1.463	0.290	0.066	0.479	0.106
		0.091	2.3	1.357	2.020	0.414	0.082	0.472	0.140
		0.095	2.4	1.420	2.113	0.416	0.089	0.461	0.142
		0.120	3.0	1.698	2.527	0.505	0.103	0.452	0.165
1-1/2 x 1-1/2	38.1 x 38.1	0.047	1.2	0.915	1.362	0.288	0.093	0.590	0.124
		0.063	1.6	1.200	1.785	0.384	0.120	0.581	0.159
		0.091	2.3	1.666	2.480	0.505	0.160	0.573	0.220
		0.095	2.4	1.740	2.592	0.526	0.163	0.564	0.220
		0.120	3.0	2.100	3.125	0.626	0.192	0.554	0.260
2 x 2	50.8 x 50.8	0.063	1.6	1.626	2.424	0.480	0.300	0.786	0.300
		0.091	2.3	2.282	3.396	0.689	0.410	0.776	0.416
		0.095	2.4	2.380	3.542	0.700	0.415	0.770	0.415
		0.120	3.0	2.943	4.380	0.870	0.494	0.756	0.494
		0.125	3.2	3.051	4.540	0.893	0.508	0.756	0.508
		0.156	4.0	3.70	5.510	1.088	0.589	0.740	0.589
		0.180	4.6	4.173	6.210	1.227	0.653	0.729	0.653
		0.1875	4.8	4.314	6.420	1.269	0.668	0.726	0.668
		0.250	6.4	5.403	8.041	1.589	0.766	0.694	0.766
2-1/2 x 2-1/2	63.5 x 63.5	0.091	2.3	2.898	4.313	0.86	0.82	0.980	0.656
		0.095	2.4	3.030	4.509	0.900	0.85	0.976	0.680
		0.120	3.0	3.760	5.590	1.110	1.02	0.961	0.820
		0.125	3.2	3.907	5.800	1.150	1.053	0.958	0.840
		0.156	4.0	4.760	7.090	1.400	1.241	0.942	0.993
		0.180	4.6	5.406	8.030	1.587	1.38	0.934	1.11
		0.1875	4.8	5.591	8.320	1.644	1.42	0.930	1.14
		0.250	6.4	7.122	10.600	2.089	1.69	0.899	1.35

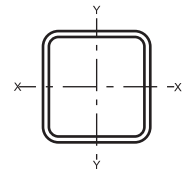


COLD FORMED HOLLOW SECTIONS

SQUARE

Section size		Wall thickness t		Unit weight M		Section area A	Moment of inertia I	Radius of gyration i	Modulus of section Z
in	mm	in	mm	lb/ft	kg/m	in ²	in ⁴	in	in ³
3 x 3	76.2 x 76.2	0.091	2.3	3.515	5.231	1.030	1.44	1.19	0.96
		0.095	2.4	3.670	5.480	1.080	1.503	1.18	1.002
		0.120	3.0	4.576	6.810	1.346	1.83	1.17	1.22
		0.125	3.2	4.751	7.070	1.400	1.89	1.163	1.26
		0.156	4.0	5.820	8.660	1.711	2.251	1.147	1.50
		0.180	4.6	6.620	9.850	1.947	2.52	1.14	1.68
		0.1875	4.8	6.860	10.200	2.019	2.60	1.13	1.73
		0.250	6.4	8.803	13.100	2.589	3.16	1.10	2.10
3-1/2 x 3-1/2	88.9 x 88.9	0.120	3.0	5.390	8.020	1.585	2.98	1.37	1.70
		0.125	3.2	5.600	8.330	1.651	3.10	1.367	1.76
		0.156	4.0	6.880	10.240	2.024	3.71	1.354	2.12
		0.177	4.5	7.739	11.518	2.27	4.10	1.34	2.34
		0.180	4.6	7.862	11.700	2.31	4.16	1.34	2.38
		0.1875	4.8	8.140	12.113	2.394	4.29	1.34	2.45
		0.236	6.0	10.006	14.891	2.99	5.25	1.33	3.00
		0.250	6.4	10.500	15.626	3.10	5.29	1.31	3.02
4 x 4	101.6 x 101.6	0.120	3.0	6.210	9.241	1.825	4.53	1.58	2.27
		0.125	3.2	6.450	9.681	1.902	4.70	1.57	2.34
		0.156	4.0	7.950	11.810	2.34	5.66	1.56	2.83
		0.177	4.5	8.945	13.312	2.63	6.28	1.55	3.14
		0.180	4.6	9.070	13.500	2.67	6.38	1.55	3.19
		0.1875	4.8	9.410	14.003	2.769	6.59	1.54	3.30
		0.236	6.0	11.614	17.285	3.39	7.82	1.52	3.91
		0.250	6.4	12.200	18.200	3.54	8.22	1.51	4.11
		0.3125	7.9	14.800	22.024	4.358	9.58	1.48	4.79
		0.375	9.5	17.300	25.745	5.075	10.7	1.45	5.35
		0.500	12.7	21.600	32.200	6.36	12.3	1.39	6.13
5 x 5	127 x 127	0.156	4.0	10.100	15.000	2.961	11.43	1.97	4.57
		0.177	4.5	11.356	16.901	3.33	12.7	1.97	5.08
		0.180	4.6	11.500	17.113	3.39	12.9	1.96	5.18
		0.1875	4.8	12.000	17.857	3.519	13.4	1.95	5.36
		0.236	6.0	14.830	22.070	4.34	16.0	1.92	6.42
		0.250	6.4	15.656	23.298	4.54	16.9	1.92	6.78
		0.3125	7.9	19.100	28.424	5.608	20.1	1.89	8.02
		0.375	9.5	22.400	33.334	6.575	22.8	1.86	9.11
		0.500	12.7	28.400	42.300	8.356	27.0	1.80	10.8
6 x 6	152.4 x 152.4	0.156	4.0	12.200	18.155	3.583	20.2	2.40	6.73
		0.177	4.5	13.768	20.490	4.04	22.5	2.36	7.50
		0.180	4.6	14.000	20.834	4.103	22.9	2.36	7.65

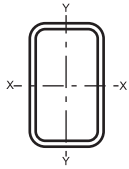
COLD FORMED HOLLOW SECTIONS



SQUARE

Section size		Wall thickness t		Unit weight M		Section area A	Moment of inertia I	Radius of gyration i	Modulus of section Z
in	mm	in	mm	lb/ft	kg/m	in ²	in ⁴	in	in ³
6 x 6	152.4 x 152.4	0.1875	4.8	14.5	21.60	4.269	23.8	2.36	7.93
		0.236	6.0	18.045	26.856	5.28	28.7	2.33	9.58
		0.250	6.4	19.016	28.30	5.535	30.3	2.33	10.1
		0.3125	7.9	23.317	34.70	6.858	36.3	2.30	12.1
		0.375	9.5	27.5	40.924	8.075	41.6	2.27	13.9
		0.500	12.7	35.2	52.40	10.360	50.5	2.21	16.8
7 x 7	177.8 x 177.8	0.180	4.6	16.4	24.405	4.827	37.1	2.77	10.6
		0.1875	4.8	17.1	25.447	5.019	38.5	2.77	11.0
		0.250	6.4	22.4	33.334	6.589	49.4	2.74	14.1
		0.3125	7.9	27.6	41.073	8.108	59.5	2.71	17.0
		0.375	9.5	32.0	48.00	9.575	68.7	2.68	19.6
		0.500	12.7	42.0	62.502	12.360	84.6	2.62	24.2
8 x 8	203.2 x 203.2	0.180	4.6	18.9	28.126	5.547	56.1	3.18	14.0
		0.1875	4.8	19.6	29.20	5.769	58.2	3.18	14.6
		0.250	6.4	25.8	38.40	7.589	75.1	3.15	18.8
		0.3125	7.9	31.8	47.323	9.358	90.9	3.12	22.7
		0.375	9.5	37.7	56.103	11.08	106	3.09	26.4
		0.500	12.7	48.8	72.622	14.36	131	3.03	32.9
10 x 10	254 x 254	0.250	6.4	32.5	48.512	9.589	151	3.96	30.1
		0.3125	7.9	40.3	60.00	11.86	183	3.93	36.7
		0.375	9.5	47.9	71.282	14.08	214	3.90	42.9
		0.500	12.7	62.4	92.90	18.36	271	3.84	54.2
		0.625	15.9	76.3	113.546	22.40	321	3.78	64.2
12 x 12	304.8 x 304.8	0.250	6.4	39.4	58.63	11.59	265	4.78	44.1
		0.3125	7.9	48.8	72.62	14.36	324	4.75	54.0
		0.375	9.5	58.0	86.40	17.08	380	4.72	63.4
		0.500	12.7	76.0	113.10	22.36	485	4.66	80.9
		0.625	15.9	93.3	139.00	27.43	580	4.60	96.7
14 x 14	355.6 x 355.6	0.375	9.5	68.2	102.00	20.08	615	5.54	87.9
		0.500	12.7	89.6	33.33	26.36	791	5.48	113
		0.625	15.9	110.0	164.00	32.43	952	5.42	136
16 x 16	406.4 x 406.4	0.500	12.7	103.0	154.00	30.36	1,200	6.29	150
		0.625	15.9	127.0	189.00	37.43	1,450	6.23	182

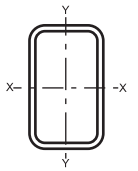
COLD FORMED HOLLOW SECTIONS



RECTANGULAR

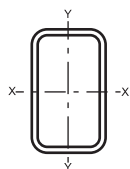
Section size		Wall thickness t		Unit weight M		Section area A	Moment of inertia Ix Iy		Radius of gyration ix iy		Modulus of section Zx Zy	
in	mm	in	mm	lb/ft	kg/m	in ²	in ⁴	in ⁴	in	in	in ³	in ³
1 x ½	25.4 x 12.7	0.039	1.0	0.373	0.555	0.110	0.013	0.004	0.355	0.204	0.027	0.018
		0.047	1.2	0.448	0.670	0.131	0.015	0.005	0.344	0.198	0.031	0.020
		0.063	1.6	0.580	0.863	0.167	0.018	0.006	0.335	0.191	0.037	0.024
1½ x 1	38.1 x 25.4	0.039	1.0	0.638	0.949	0.188	0.059	0.031	0.564	0.412	0.079	0.063
		0.047	1.2	0.767	1.141	0.229	0.070	0.037	0.555	0.405	0.094	0.075
		0.063	1.6	1.010	1.500	0.297	0.088	0.047	0.546	0.398	0.118	0.094
2 x 1	50.8 x 25.4	0.047	1.2	0.915	1.362	0.288	0.144	0.049	0.726	0.422	0.144	0.097
		0.0598	1.5	1.128	1.683	0.365	0.179	0.060	0.721	0.416	0.179	0.120
		0.063	1.6	1.200	1.785	0.384	0.187	0.062	0.720	0.415	0.187	0.125
		0.091	2.3	1.666	2.480	0.505	0.257	0.083	0.713	0.407	0.257	0.167
		0.095	2.4	1.740	2.592	0.526	0.260	0.085	0.702	0.401	0.260	0.170
		0.120	3.0	2.100	3.125	0.626	0.285	0.093	0.680	0.390	0.285	0.187
2½ x 1½	63.5 x 38.1	0.063	1.6	1.626	2.424	0.480	0.416	0.189	0.920	0.620	0.333	0.252
		0.091	2.3	2.282	3.396	0.692	0.55	0.248	0.912	0.611	0.439	0.331
		0.095	2.4	2.380	3.542	0.700	0.572	0.257	0.903	0.606	0.457	0.343
		0.120	3.0	2.904	4.322	0.870	0.685	0.306	0.889	0.595	0.548	0.409
3 x 1½	76.2 x 38.1	0.063	1.6	1.860	2.70	0.527	0.60	0.209	1.07	0.63	0.40	0.279
		0.075	1.9	2.166	3.224	0.629	0.72	0.246	1.06	0.623	0.48	0.328
		0.091	2.3	2.590	3.855	0.762	0.86	0.293	1.06	0.62	0.573	0.391
		0.095	2.4	2.700	4.018	0.795	0.901	0.304	1.06	0.619	0.601	0.406
		0.120	3.0	3.306	4.920	0.985	1.086	0.364	1.050	0.607	0.724	0.485
3 x 2	76.2 x 50.8	0.075	1.9	2.421	3.603	0.714	0.88	0.486	1.12	0.825	0.587	0.486
		0.091	2.3	2.898	4.313	0.863	1.06	0.578	1.112	0.818	0.707	0.578
		0.095	2.4	3.030	4.509	0.900	1.102	0.589	1.112	0.813	0.734	0.589
		0.120	3.0	3.760	5.560	1.110	1.33	0.707	1.097	0.800	0.886	0.707
		0.125	3.2	3.910	5.819	1.150	1.372	0.730	1.094	0.797	0.915	0.730
		0.156	4.0	4.760	7.090	1.400	1.62	0.860	1.08	0.782	1.080	0.860
		0.180	4.6	5.400	8.036	1.587	1.81	0.952	1.07	0.775	1.200	0.952
		0.1875	4.8	5.590	8.320	1.644	1.86	0.977	1.06	0.771	1.240	0.977
		0.250	6.4	7.100	10.600	2.089	2.21	1.150	1.03	0.742	1.47	1.150
4 x 2	101.6 x 50.8	0.075	1.9	2.930	4.361	0.85	1.77	0.612	1.44	0.848	0.885	0.612
		0.091	2.3	3.515	5.231	1.03	2.14	0.732	1.44	0.843	1.07	0.732
		0.095	2.4	3.670	5.480	1.080	2.233	0.761	1.44	0.840	1.116	0.761
		0.120	3.0	4.511	6.714	1.346	2.72	0.920	1.421	0.830	1.360	0.919
		0.125	3.2	4.790	7.129	1.400	2.81	0.950	1.42	0.824	1.405	0.950
		0.156	4.0	5.820	8.660	1.711	3.35	1.121	1.40	0.809	1.674	1.121
		0.180	4.6	6.620	9.851	1.947	3.75	1.25	1.39	0.802	1.88	1.25
		0.1875	4.8	6.860	10.209	2.019	3.87	1.29	1.38	0.798	1.93	1.29
		0.250	6.4	8.800	13.096	2.589	4.69	1.54	1.35	0.771	2.35	1.54
4 x 3	101.6 x 76.2	0.120	3.0	5.390	8.021	1.582	3.63	2.33	1.51	1.21	1.81	1.55
		0.125	3.2	5.647	8.404	1.651	3.75	2.41	1.51	1.21	1.88	1.61
		0.156	4.0	6.880	10.240	2.024	4.52	2.90	1.50	1.20	2.26	1.93

COLD FORMED HOLLOW SECTIONS



RECTANGULAR

Section size		Wall thickness t		Unit weight M		Section area A	Moment of inertia		Radius of gyration		Modulus of section	
							I _x	I _y	i _x	i _y	Z _x	Z _y
in	mm	in	mm	lb/ft	kg/m	in ²	in ⁴	in ⁴	in	in	in ³	in ³
4 x 3	101.6 x 76.2	0.177	4.5	7.739	11.517	2.27	4.99	3.19	1.48	1.19	2.50	2.13
		0.180	4.6	7.850	11.682	2.310	5.07	3.24	1.48	1.19	2.53	2.16
		0.1875	4.8	8.140	12.113	2.394	5.23	3.34	1.48	1.18	2.62	2.23
		0.236	6.0	10.006	14.891	2.99	6.43	4.07	1.46	1.16	3.21	2.71
		0.250	6.4	10.500	15.630	3.089	6.45	4.10	1.45	1.15	3.23	2.74
5 x 2	127 x 50.8	0.120	3.0	5.390	8.021	1.585	4.79	1.14	1.74	0.846	1.92	1.14
		0.125	3.2	5.647	8.404	1.647	4.96	1.17	1.73	0.844	1.98	1.17
		0.156	4.0	6.880	10.240	2.024	5.96	1.40	1.72	0.83	2.38	1.40
		0.177	4.5	7.739	11.518	2.27	6.57	1.52	1.70	0.82	2.63	1.52
		0.180	4.6	7.850	11.682	2.307	6.67	1.55	1.70	0.82	2.67	1.55
		0.1875	4.8	8.140	12.100	2.394	6.89	1.60	1.70	0.816	2.75	1.60
		0.236	6.0	10.006	14.891	2.99	8.50	1.91	1.68	0.799	3.39	1.91
		0.250	6.4	10.400	15.626	3.089	8.48	1.92	1.66	0.789	3.40	1.92
5 x 3	127 x 76.2	0.120	3.0	6.210	9.241	1.825	6.22	2.83	1.85	1.25	2.49	1.89
		0.125	3.2	6.505	9.781	1.902	6.44	2.93	1.84	1.24	2.58	1.95
		0.156	4.0	7.950	11.831	2.340	7.76	3.14	1.82	1.23	3.11	2.34
		0.177	4.5	8.945	13.312	2.63	8.63	3.90	1.81	1.22	3.45	2.60
		0.180	4.6	9.070	13.498	2.667	8.77	3.96	1.81	1.22	3.51	2.64
		0.1875	4.8	9.410	14.004	2.769	9.06	4.08	1.81	1.21	3.62	2.72
		0.236	6.0	11.614	17.284	3.47	11.2	4.98	1.79	1.19	4.48	3.32
		0.250	6.4	12.200	18.156	3.589	11.3	5.05	1.77	1.19	4.52	3.37
		0.3125	8.0	14.800	22.025	4.358	13.2	5.85	1.74	1.16	5.27	3.90
6 x 2	152.4 x 50.8	0.375	9.5	17.300	25.745	5.075	14.7	6.48	1.70	1.13	5.89	4.32
		0.120	3.0	6.210	9.241	1.825	7.64	1.35	2.05	0.859	2.55	1.35
		0.125	3.2	6.505	9.680	1.897	7.92	1.39	2.04	0.857	2.64	1.39
		0.156	4.0	7.950	11.831	2.340	9.56	1.66	2.02	0.843	3.19	1.66
		0.177	4.5	8.945	13.312	2.63	10.5	1.82	2.01	0.838	3.50	1.82
		0.180	4.6	9.070	13.500	2.667	10.7	1.85	2.01	0.833	3.58	1.85
		0.1875	4.8	9.410	14.003	2.769	11.1	1.90	2.00	0.829	3.70	1.90
		0.236	6.0	11.614	17.284	3.47	13.1	2.21	1.96	0.818	4.36	2.21
6 x 3	152.4 x 76.2	0.250	6.4	12.230	18.200	3.589	13.8	2.31	1.96	0.802	4.60	2.31
		0.125	3.2	7.320	10.893	2.130	10.06	3.44	2.17	1.27	3.35	2.30
		0.156	4.0	9.020	13.423	2.652	12.2	4.15	2.15	1.25	4.07	2.76
		0.177	4.5	10.150	15.105	2.98	13.6	4.60	2.13	1.24	4.53	3.07
		0.180	4.6	10.300	15.330	3.027	13.8	4.67	2.13	1.24	4.60	3.12
		0.1875	4.8	10.700	15.923	3.144	14.3	4.83	2.13	1.24	4.76	3.22
		0.236	6.0	13.222	19.676	3.87	17.0	5.72	2.10	1.22	5.67	3.81
		0.250	6.4	13.700	20.685	4.089	17.9	6.00	2.09	1.21	5.98	4.00
		0.3125	8.0	16.933	25.200	4.983	21.1	6.98	2.06	1.18	7.03	4.65
6 x 3	152.4 x 76.2	0.375	9.5	19.823	29.500	5.825	23.8	7.78	2.02	1.16	7.92	5.19



COLD FORMED HOLLOW SECTIONS

RECTANGULAR

Section size		Wall thickness t		Unit weight M		Section area A	Moment of inertia Ix Iy		Radius of gyration ix iy		Modulus of section Zx Zy	
in	mm	in	mm	lb/ft	kg/m	in ²	in ⁴	in ⁴	in	in	in ³	in ³
6 x 4	152.4 x 101.6	0.156	4.0	10.10	15.03	2.961	14.86	7.96	2.24	1.64	4.96	3.98
		0.177	4.5	11.356	16.90		16.5	8.87	2.23	1.63	5.50	4.44
		0.180	4.6	11.50	17.114	3.387	16.8	9.01	2.23	1.63	5.62	4.51
		0.1875	4.8	12.00	17.858	3.519	17.4	9.32	2.23	1.63	5.81	4.66
		0.236	6.0	14.83	22.07	4.34	21.0	11.1	2.20	1.60	7.00	5.55
		0.250	6.4	15.60	23.215	4.589	22.1	11.7	2.19	1.60	7.36	5.87
		0.3125	8.0	19.10	28.424	5.608	26.2	13.8	2.16	1.57	8.72	6.92
		0.375	9.5	22.40	33.335	6.575	29.7	15.6	2.13	1.54	9.9	7.82
		0.500	12.7	28.424	42.30	8.356	35.3	18.4	2.06	1.48	11.8	9.21
8 x 4	203.2 x 101.6	0.156	4.0	12.20	18.155	3.583	30.0	10.27	2.90	1.70	7.50	5.13
		0.177	4.5	13.768	20.49	3.98	32.0	10.9	2.89	1.69	8.12	5.55
		0.180	4.6	14.00	20.834	4.107	34.1	11.6	2.88	1.68	8.52	5.82
		0.1875	4.8	14.515	21.60	4.269	35.3	12.0	2.88	1.68	8.83	6.02
		0.236	6.0	18.045	26.854	5.21	40.8	13.9	2.84	1.66	10.4	7.02
		0.250	6.4	19.017	28.30	5.589	45.1	15.3	2.84	1.65	11.3	7.63
		0.3125	8.0	23.317	34.70	6.858	53.9	18.1	2.80	1.62	13.5	9.05
		0.375	9.5	27.50	40.924	8.075	61.9	20.6	2.77	1.60	15.5	10.3
		0.500	12.7	35.211	52.40	10.36	75.1	24.6	2.69	1.54	18.8	12.3
8 x 6	203.2 x 152.4	0.180	4.6	16.40	24.406	4.827	45.1	29.0	3.06	2.45	11.3	9.68
		0.1875	4.8	17.10	25.448	5.019	46.8	30.1	3.05	2.45	11.7	10.0
		0.250	6.4	22.40	33.335	6.589	60.1	38.6	3.02	2.42	15.0	12.9
		0.3125	8.0	27.60	41.073	8.108	72.4	46.4	2.99	2.39	18.1	15.5
		0.375	9.5	32.60	48.514	9.575	83.7	53.5	2.96	2.36	20.9	17.8
		0.500	12.7	42.00	62.503	12.36	103	65.7	2.89	2.31	25.8	21.9
10 x 6	254 x 152.4	0.180	4.6	18.90	28.126	5.547	76.9	35.1	3.72	2.52	15.4	11.7
		0.1875	4.8	19.621	29.20	5.769	79.8	36.5	3.72	2.51	16.0	12.2
		0.250	6.4	25.804	38.40	7.589	103	46.9	3.69	2.49	20.6	15.6
		0.3125	8.0	31.80	47.324	9.358	125	56.5	3.65	2.46	25.0	18.8
		0.375	9.5	37.70	56.104	11.08	145	65.4	3.62	2.43	29.0	21.8
		0.500	12.7	48.80	72.622	14.36	181	80.8	3.55	2.37	36.2	26.9
12 x 8	304.8 X 203.2	0.250	6.4	32.60	48.514	9.589	196	105	4.52	3.31	32.6	26.3
		0.3125	8.0	40.30	60.00	11.86	239	128	4.49	3.28	39.8	32.0
		0.375	9.5	47.90	71.283	14.08	279	149	4.45	3.26	46.5	37.3
		0.500	12.7	62.426	92.90	18.36	353	188	4.39	3.20	58.9	46.9
14 x 6	355.6 x 152.4	0.250	6.4	32.60	48.514	9.589	237	63.4	4.97	2.57	33.8	21.1
		0.3125	8.0	40.30	60.00	11.86	288	76.7	4.93	2.54	41.2	25.6
		0.375	9.5	47.90	71.283	14.08	337	89.1	4.89	2.52	48.1	29.7
		0.500	12.7	62.426	92.90	18.36	426	111	4.82	2.46	60.8	37.1
16 x 8	406.4 x 203.2	0.250	6.4	39.40	58.634	11.59	393	135	5.82	3.42	49.1	33.8
		0.3125	8.0	48.80	72.622	14.36	481	165	5.79	3.39	60.1	41.2
		0.375	9.5	58.058	86.40	17.08	565	193	5.75	3.36	70.6	48.2
		0.500	12.7	76.00	113.10	22.36	722	244	5.68	3.30	90.2	61.0

HOT FORMED HOLLOW SECTIONS

(excerpts from BRITISH STEEL TUBES)

The size range of Structural Hollow Sections manufactured by British Steel Tubes, is in accordance with BS 4848 Hot Rolled Structural Steel Sections: Part 2, Hollow Sections.

The Structural Hollow Section size range published in BS 4848: Part 2 relates generally to ISO 657 Hot Rolled Steel Sections: Part 14, Hot Finished Structural Hollow Sections, Dimensional and Sectional Properties.

STEEL

The steel is made by the basic oxygen or electric furnace process in accordance with BS 4360. For SHS there are three basic grades, 43, 50 and 55, each with sub-grades giving improved impact properties. For economic reasons, production has been standardised on grades 43C and 50C. Other grades and sub-grades can be supplied against mill orders and subject to minimum mill quantities.

CHEMICAL COMPOSITION AND MECHANICAL PROPERTIES

The ladle analyses and mechanical properties, in accordance with BS 4360 for hot finished sections are as follows:

	Grade 43C	Grade 50C		Grade 43C	Grade 50C
	%max	% max			
Carbon	0.21	0.20	Tensile Strength N/mm ²	430/580	490/640
Silicon	0.50	0.50	Yield Strength min. N/mm ²	275 [†]	355 [†]
Manganese	1.30	1.50	Min. Elongation on gauge		
Sulphur	0.05	0.045	length $5.65\sqrt{S_0}$	22%	20%
Phosphorus	0.05	0.045	Charpy V-Notch Impact Value		
Niobium	-	0.003/0.10	10 x 10mm specimen at		
Vanadium	-	0.003/0.15	0°C. Min. Ave Energy Joules	27*	27

[†] up to and including 16mm thick

* only by special arrangement

OVERSEAS SPECIFICATIONS

Other specifications, comparable to BS 4360 can be supplied; those most frequently used are as follows:

Ordinary Yield	High Yield
DIN 17100 St 44-2	DIN 17100 St 52-3
Euronorm 25 - 72 Fe 430C	Euronorm 25 - 72 Fe 510C
NBN A21-101 AE 255C	NBN A21-101 AE 355C
Onorm M3116 St 430C	Onorm M3116 St 510C
SS 14 14 12	SS 14 21 34
ASTM A36	UNI 7806 Fe 510C
UNI 7806 Fe 430C	

Other sub-grades are available

TENSILE TEST

The tensile strength, yield strength and elongation are determined from standard test pieces cut longitudinally from the sections, excluding the weld areas and away from the corners. The test pieces and testing comply with BS 18. Method for Tensile Testing of Metals.

HOT FORMED HOLLOW SECTIONS

IMPACT TEST

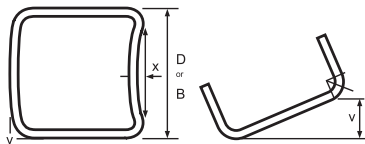
Standard impact specimens are cut longitudinally from the sections and machined on each face to 10mm x 10mm in cross section and a V notch, 2 mm deep, is cut into one face. The specimen is then tested at the appropriate temperature in accordance with BS 131: Part 2, The Charpy V-Notch Impact Test. Due to the machining, standard specimens can only be taken from material 11 mm thick, or thicker. For thinner material, tests are made on the next subsidiary standard test piece with average values in accordance with Table 15 of BS 4360.

Average value for standard test piece	Equivalent average value for subsidiary standard test piece	
10mm x 10mm	10mm x 7.5mm	10 mm x 5.0 mm
J	J	J
27	22	19

Unless agreed otherwise, impact tests are not carried out on Grade 43C material or where the section is less than 6 mm thick.

DIMENSIONAL TOLERANCES

The dimensional tolerances are in accordance with BS 4848, Part 2, as follows:



(a) Circular Hollow Sections Outside diameter	$\pm 0.5\text{mm}$ or $\pm 1\%$ whichever is the greater
(b) Rectangular Hollow Sections	
(i) Outside dimensions of sides	$\pm 0.5\text{mm}$ or $\pm 1\%$ whichever is the greater
(ii) Squareness of side	$90^\circ \pm 1^\circ$
(iii) Radii of corners	Outside - between the limits of $0.5t$ and $2.0t$ Inside - between the limits of $0.5t$ and $1.5t$ where t equals the specified thickness of section
(iv) Concavity/Convexity (x)	$\pm 1\%$ of the length of the side, (this tolerance measured independently of the tolerance on outside dimension.)
(v) Angular twist (v)	2 mm plus 0.5 mm per metre maximum. Twist is measured by laying the section, as produced, on a horizontal surface with the face at one end pressed flat. The difference in height above the surface between the two corners at the opposite end is measured.

MASS TOLERANCE

The rolling tolerance in accordance with BS 4848, Part 2 is $\pm 6\%$ on individual lengths; + 6% - 4% on lots of 10 tonnes and over.

STRAIGHTNESS TOLERANCE

Unless otherwise arranged, the hollow section shall not deviate from straightness by more than 0.2% of the total length as produced, measured at the centre of the length.

LENGTH TOLERANCE

Mill and Random lengths are supplied as in the following tables. Exacts may be supplied at our option, with a tolerance of +6 mm - 0. Lengths above 12 m in length are subject to an extra for transportation.

HOT FORMED HOLLOW SECTIONS

Rectangular Hollow Sections - Length Ranges & Tolerances

Size		Welded			Seamless	
Squares mm	Rectangles mm	Standard Mill Lengths m	Special Mill Lengths m	Tolerance mm	Standard Random Lengths m	Special Random Lengths m
20 x 20	--	6.4	5.4 - 7.5	+ 150 - 0		
25 x 25 & 30 x 30	--	6.4 & 7.5	5.4 - 7.5	+ 150 - 0		
--	50 x 25	7.5	5.4 - 7.5	+ 150 - 0		
40 x 40 uti	50 x 30 uti	7.5, 10 & 12	5.4 - 13.7	+ 150 - 0		
100 x 100 x 8	120 x 80 x 8					
100 x 100 x 10 uti	120 x 80 x 10 uti	7.5, 10 & 12	6.1 - 14.6	+ 150 - 0		
150 x 150 x 12.5	200 x 100 x 12.5					
150 x 150 x 16	200 x 100 x 16				10 - 11.2	5.6 - 11.2
180 x 180 to	250 x 150 to	10 & 12	9 - 14.8	+ 300 - 0		
400 x 400	500 x 300					
400 x 400 x 20	--	6 - 9 randoms	--	--		

Minimum quantities supplied from our works in one size, thickness, length and quality.

Welded - Standard Mill lengths uti 150 x 150 x 12.5 and 200 x 100 x 12.5, 2 bundles, above these sizes 10 tonnes or less at works discretion.

Special Mill lengths - 10 tonnes.

Seamless - Standard Random lengths, 2.5 tonnes.

Special Random lengths, 10 tonnes.

Circular Hollow Sections - Length Ranges & Tolerances

Outside Diameter mm	Welded				Seamless		
	Thickness mm	Standard Mill Lengths m	Special Mill Lengths m	Tolerance mm	Thickness mm	Standard Mill Lengths m	Tolerances mm
21.3 & 26.9	all	6.0 & 6.4	5.4 - 7.5	+ 150 - 0	--		
33.7 - 48.3	all	6.0, 6.4 & 7.5	5.4 - 7.5	+ 150 - 0	--		
60.3 - 114.3	all	6.0, 6.4, 7.5 & 10	5.4 - 12	+ 150 - 0	--		
139.7 - 168.3	all	7.5, 10 & 12	6.1 - 14.6	+ 150 - 0	--		
193.7	uti 12.5	7.5, 10 & 12	6.1 - 14.6	+ 150 - 0	16	8, 10 & 12	+ 300 - 0
219.1	uti 12.5	10 & 12	9 - 14.8	+ 300 - 0	16	8, 10 & 12	+ 300 - 0
					20	6, 8 & 10	+ 300 - 0
244.5	6.3 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	6, 8 & 10	+ 300 - 0
273.0	6.3 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	6, 8 & 10	+ 300 - 0
					25	4, 6 & 8	+ 300 - 0
323.9	6.3 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	6, 8 & 10	+ 300 - 0
					25	4, 6 & 8	+ 300 - 0
355.6	8.0 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	6, 8 & 10	+ 300 - 0
					25	4, 6 & 8	+ 300 - 0
406.4	10 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	8, 10 & 12	+ 300 - 0
					25	4, 6 & 8	+ 300 - 0
					32	2, 4 & 6	+ 300 - 0
457.0	10 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	8, 10 & 12	+ 300 - 0
					25	6, 8 & 10	+ 300 - 0
					32	4, 6 & 8	+ 300 - 0
					40	2, 4 & 6	+ 300 - 0
508.0	10 - 16	10 & 12	9 - 14.8	+ 300 - 0	20	6, 8 & 10	+ 300 - 0
					25	6, 8 & 10	+ 300 - 0
					32	4, 6 & 8	+ 300 - 0
					40	2, 4 & 6	+ 300 - 0
					50	3, 4 & 5	+ 300 - 0

Minimum quantities supplied from our works in one size, thickness, length and quality.

Welded - Standard Mill lengths uti 193.7 - 4/5 tonnes. Over 193.7 - 10 tonnes or less at works discretion.

Special Mill lengths - 10 tonnes.

Seamless - Standard Mill lengths - 10 tonnes, or less at works discretion.

HOT FORMED HOLLOW SECTIONS

COMPARABLE SPECIFICATIONS

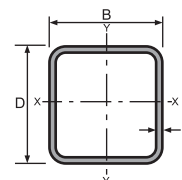
Compared by tensile and Charpy V-notch impacts

Specification	Grade	Tensile strength	Yield strength minimum	Charpy V-Notch Impact Values*	
		N/mm ²	N/mm ²	Joules	°C
BS 4360	43C	430 - 580	275	27	0
ISO 630	Fe430C	430 - 530	275	27	0
DIN 17100	St44 - 2	410 - 540	275	27	+ 20
DIN 17100	St44 - 3	410 - 540	275	27	0
EURONORM 25	Fe430C	430 - 510	275	28	0
SIS 14 14 12	--	430 - 540	270	--	--
UNI 7806	Fe430C	430 min.	275	28	0
NBN A21-101	AE255C	410 - 490	255	27	0
ONORM M3116	St430C	430 - 540	275	27	0
ASTM A36	--	400 - 550	250	--	--
ASTM A501	--	400 min.	250	--	--
NF A49-501	E-26/3	370 - 510	255	27	0
JIS G3466	STKR41	402 min.	245 min.	--	--
BS 4360	43D	430 - 580	275	27	-20
ISO 630	Fe430D	430 - 530	275	27	-20
DIN 17100	St44 - 3N	410 - 540	275	27	-20
EURONORM 25	Fe430D	430 - 510	275	28	-20
UNI 7806	Fe430D	430 min.	275	28	-20
NBN A21-101	AE255D	410 - 490	255	27	-20
ONORM M3116	St430D	430 - 540	275	27	-20
NF A49-501	E-26/4	370 - 510	255	27	-20
BS 4360	43EE	430 - 580	275	27	-50
BS 4360	50C	490 - 640	355	27	0
ISO 630	Fe 510C	490 - 630	355	27	0
DIN 17100	St52 - 3	490 - 630	355	27	0
EURONORM 25	Fe510C	510 - 610	355	28	0
SIS 14 21 32	--	510 min.	350	--	--
UNI 7806	Fe510C	510 min.	355	27	0
NBN A21-101	AE355C	510 - 610	355	27	0
ONORM M3116	St510C	510 - 630	355	27	0
NF A49-501	E-36/3	470 - 610	355	28	0
ASTM A618	--	450 min.	345	--	--
JIS G 3466	STKR50	490 min.	324 min.	--	--
BS 4360	50D	490 - 640	355	27	-20
ISO 630	Fe510D	490 - 630	355	27	-20
DIN 17100	St52 - 3N	490 - 630	355	27	-20
EURONORM 25	Fe510D	510 - 610	355	28	-20
SIS 14 21 34	--	470 - 630	350	27	-20
UNI 7806	Fe510D	510 min.	355	27	-20
ONORM M3116	St510D	510 - 630	355	27	-20
BS 4360	50EE	490 - 640	355	27	-50
NBN A21-101	AE355D	510 - 610	355	40	-20
NF A49-501	E-36/4	470 - 610	355	40	-20
NORSEC 360	--	490 - 630	360	100	-20
				50	-60
BS 4360	55C	550 - 700	450	27	0

Note:

* Average value on full size specimens, 10 mm x 10 mm

HOT FORMED HOLLOW SECTIONS



SQUARE

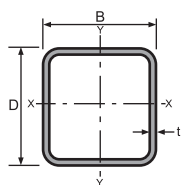
Section size D x B	Thickness t	Unit weight M	Section area A	Moment of inertia I	Radius of gyration r	Elastic modulus Z	Plastic modulus S	Torsional J	Constants C	Superficial area per m	Approx. length per tonne
mm	mm	kg/m	cm ²	cm ⁴	cm	cm ³	cm ³	cm ⁴	cm ³	m ²	m
20 x 20	2.0	1.12	1.42	0.76	0.73	0.76	0.95	1.22	1.07	0.076	893
	2.5*	1.35	1.72	0.87	0.71	0.88	1.12	1.41	1.21	0.075	741
25 x 25	2.0*	1.43	1.82	1.59	0.94	1.27	1.56	2.52	1.81	0.096	700
	2.5*	1.74	2.22	1.85	0.91	1.48	1.86	2.97	2.09	0.095	574
	3.0*	2.04	2.60	2.06	0.89	1.65	2.12	3.36	2.31	0.094	490
	3.2*	2.15	2.74	2.14	0.88	1.71	2.21	3.49	2.38	0.093	466
30 x 30	2.5*	2.14	2.72	3.40	1.12	2.27	2.79	5.40	3.22	0.115	469
	3.0*	2.51	3.20	3.84	1.10	2.56	3.21	6.17	3.61	0.114	399
	3.2	2.65	3.38	4.00	1.09	2.67	3.37	6.45	3.75	0.113	378
40 x 40	2.5*	2.92	3.72	8.67	1.53	4.33	5.21	13.6	6.23	0.155	343
	3.0*	3.45	4.40	9.96	1.51	4.98	6.07	15.7	7.11	0.154	290
	3.2	3.66	4.66	10.4	1.50	5.22	6.40	16.5	7.43	0.153	274
	4.0	4.46	5.68	12.1	1.46	6.07	7.61	19.5	8.56	0.151	225
	5.0*	5.40	6.88	13.8	1.42	6.92	8.92	22.6	9.65	0.149	185
50 x 50	2.5*	3.71	4.72	17.7	1.94	7.07	8.38	27.4	10.2	0.195	270
	3.0*	4.39	5.60	20.5	1.91	8.20	9.83	32.0	11.8	0.194	228
	3.2	4.66	5.94	21.6	1.91	8.62	10.4	33.8	12.4	0.193	215
	4.0	5.72	7.28	25.5	1.87	10.2	12.5	40.4	14.5	0.191	175
	5.0	6.97	8.88	29.6	1.83	11.9	14.9	47.6	16.7	0.189	144
	6.0	8.15	10.4	33.0	1.78	13.2	17.0	53.7	18.5	0.187	123
	6.3*	8.49	10.8	33.9	1.77	13.6	17.5	55.3	18.9	0.186	118
60 x 60	3.0*	5.34	6.80	36.6	2.32	12.2	14.5	56.9	17.7	0.234	188
	3.2	5.67	7.22	38.7	2.31	12.9	15.3	60.1	18.6	0.233	177
	4.0	6.97	8.88	46.1	2.28	15.4	18.6	72.4	22.1	0.231	144
	5.0	8.54	10.9	54.4	2.24	18.1	22.3	86.3	25.8	0.229	118
	6.0	10.0	12.8	61.5	2.19	20.5	25.7	98.6	28.9	0.227	100
	6.3*	10.5	13.3	63.4	2.18	21.1	26.6	102	29.7	0.226	95.3
	8.0*	12.8	16.3	72.4	2.11	24.1	31.4	119	33.5	0.223	78.1
70 x 70	3.0*	6.28	8.0	59.6	2.73	17.0	20.0	92.1	24.8	0.274	160
	3.6	7.46	9.5	69.5	2.70	19.9	23.6	108	28.7	0.272	135
	5.0	10.1	12.9	90.1	2.64	25.7	31.2	142	36.8	0.269	99.1
	6.0	11.9	15.2	103	2.60	29.4	36.2	163	41.7	0.267	84.0
	6.3*	12.5	15.9	106	2.59	30.4	37.6	169	43.0	0.266	80.0
	8.0*	15.3	19.5	123	2.51	35.3	45.0	200	49.4	0.263	65.4
75 x 75	3.0	6.75	8.6	74.0	2.93	19.7	23.2	114	28.8	0.294	148
	3.6	8.02	10.2	86.5	2.91	23.1	27.3	134	33.4	0.292	125
	5.0	10.9	13.9	113	2.85	30.0	36.3	177	43.1	0.289	91.7
	6.0	12.9	16.4	129	2.81	34.4	42.2	204	49.0	0.287	77.5
	6.3	13.4	17.1	134	2.79	35.6	43.8	212	50.6	0.286	74.6
	8.0	16.6	21.1	156	2.72	41.7	52.6	252	58.6	0.283	60.2
80 x 80	3.0*	7.22	9.2	90.6	3.14	22.7	26.5	139	33.1	0.314	139
	3.6	8.59	10.9	106	3.11	26.5	31.3	164	38.5	0.312	117
	5.0	11.7	14.9	139	3.05	34.7	41.7	217	49.8	0.309	85.5
	6.0	13.8	17.6	159	3.01	39.8	48.6	251	56.9	0.307	72.5
	6.3	14.4	18.4	165	3.00	41.3	50.5	261	58.8	0.306	69.5
	8.0*	17.8	22.7	194	2.92	48.6	60.9	312	68.5	0.303	56.2

Notes:
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

* Size not included in BS 4848 Part 2

† 43C only
 Length per tonne rounded up

HOT FORMED HOLLOW SECTIONS



SQUARE

Section size D x B	Thickness t	Unit weight M	Section area A	Moment of inertia I	Radius of gyration r	Elastic modulus Z	Plastic modulus S	Torsional J	Constants C	Superficial area per m	Approx. length per tonne
mm	mm	kg/m	cm ²	cm ⁴	cm	cm ³	cm ³	cm ⁴	cm ³	m ²	m
90 x 90	3.6	9.72	12.4	154	3.52	34.1	40.0	237	49.7	0.352	103
	5.0	13.3	16.9	202	3.46	45.0	53.6	315	64.9	0.349	75.2
	6.0	15.7	20.0	234	3.42	51.9	62.7	367	74.4	0.347	63.7
	6.3	16.4	20.9	242	3.41	53.9	65.3	381	77.1	0.346	61.0
	8.0*	20.4	25.9	288	3.33	64.0	79.2	459	90.7	0.343	49.0
100 x 100	4.0	12.0	15.3	234	3.91	46.8	54.9	361	68.2	0.391	83.4
	5.0	14.8	18.9	283	3.87	56.6	67.1	439	81.9	0.389	67.6
	6.0	17.6	22.4	328	3.83	65.6	78.6	512	94.4	0.387	56.8
	6.3	18.4	23.4	341	3.81	68.2	82.0	533	97.9	0.386	54.4
	8.0	22.9	29.1	408	3.74	81.5	99.9	646	116	0.383	43.7
	10.0	27.9	35.5	474	3.65	94.9	119	761	134	0.379	35.9
120 x 120	5.0	18.0	22.9	503	4.69	83.8	98.4	775	122	0.469	55.6
	6.0	21.3	27.2	586	4.64	97.7	116	910	141	0.467	46.9
	6.3	22.3	28.5	610	4.63	102	121	949	147	0.466	44.9
	8.0	27.9	35.5	738	4.56	123	149	1,159	176	0.463	35.9
	10.0	34.2	43.5	870	4.47	145	178	1,381	206	0.459	29.3
	12.5*†	41.6	53.0	1,009	4.36	168	212	1,624	237	0.453	24.0
140 x 140	5.0*	21.1	26.9	814	5.50	116	136	1,251	170	0.549	47.4
	5.6*	23.5	30.0	899	5.48	128	151	1,385	187	0.548	42.6
	6.0	25.1	32.0	954	5.46	136	160	1,473	198	0.547	39.8
	6.3*	26.3	33.5	994	5.45	142	168	1,538	206	0.546	38.0
	7.1*	29.4	37.5	1,099	5.41	157	186	1,706	227	0.545	34.0
	8.0*	32.9	41.9	1,212	5.38	173	207	1,889	249	0.543	30.4
	10.0*	40.4	51.5	1,441	5.29	206	250	2,269	294	0.539	24.8
	12.5*	49.5	63.0	1,691	5.18	242	299	2,695	342	0.533	20.2
150 x 150	5.0	22.7	28.9	1,009	5.91	135	157	1,548	197	0.589	44.1
	6.0	27.0	34.4	1,185	5.87	158	185	1,826	230	0.587	37.0
	6.3	28.3	36.0	1,236	5.86	165	194	1,907	240	0.586	35.4
	8.0	35.4	45.1	1,510	5.78	201	240	2,348	291	0.583	28.3
	10.0	43.6	55.5	1,803	5.70	240	290	2,829	345	0.579	23.0
	12.5	53.4	68.0	2,125	5.59	283	348	3,372	403	0.573	18.8
	16.0	66.4	84.5	2,500	5.44	333	421	4,029	468	0.566	15.1

Notes:

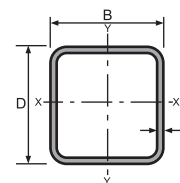
Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

* Size not included in BS 4848 Part 2

† 43C only

Length per tonne rounded up

HOT FORMED HOLLOW SECTIONS

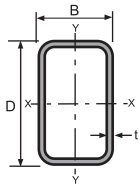


SQUARE

Section size D x B	Thickness t	Unit weight M	Section area A	Moment of inertia I	Radius of gyration r	Elastic modulus Z	Plastic modulus S	Torsional J	Constants C	Superficial area per m	Approx. length per tonne
mm	mm	kg/m	cm ²	cm ⁴	cm	cm ³	cm ³	cm ⁴	cm ³	m ²	m
180 x 180	6.0	32.6	41.6	2,093	7.09	233	271	3,211	340	0.707	30.7
	6.3	34.2	43.6	2,186	7.08	243	283	3,357	355	0.706	29.3
	8.0	43.0	54.7	2,689	7.01	299	352	4,156	434	0.703	23.3
	10.0	53.0	67.5	3,237	6.92	360	429	5,041	519	0.699	18.9
	12.5	65.2	83.0	3,856	6.82	428	519	6,062	613	0.693	15.4
	16.0	81.4	104	4,607	6.66	512	634	7,339	725	0.686	12.3
200 x 200	6.0	36.4	46.4	2,903	7.91	290	337	4,444	426	0.787	27.5
	6.3	38.2	48.6	3,033	7.90	303	353	4,647	444	0.786	26.2
	8.0	48.0	61.1	3,744	7.83	374	439	5,770	545	0.783	20.9
	10.0	59.3	75.5	4,525	7.74	452	536	7,020	655	0.779	16.9
	12.5	73.0	93.0	5,419	7.63	542	651	8,479	779	0.773	13.7
	16.0	91.5	117	6,524	7.48	652	799	10,330	929	0.766	11.0
250 x 250	6.0	45.8	58.4	5,784	9.95	463	534	8,816	681	0.987	21.8
	6.3	48.1	61.2	6,049	9.94	484	559	9,288	712	0.986	20.8
	8.0	60.5	77.1	7,510	9.87	601	699	11,511	880	0.983	16.6
	10.0	75.0	95.5	9,141	9.78	731	858	14,086	1,065	0.979	13.4
	12.5	92.6	118	11,050	9.68	884	1,048	17,139	1,279	0.973	10.8
	16.0	117	149	13,480	9.53	1,078	1,298	21,109	1,548	0.966	8.55
300 x 300	8.0*	73.1	93.1	13,209	11.90	881	1,018	20,171	1,294	1.18	13.7
	10.0	90.7	116	16,150	11.80	1,077	1,254	24,776	1,575	1.18	11.1
	12.5	112	143	19,630	11.70	1,309	1,538	30,290	1,905	1.17	8.93
	16.0	142	181	24,160	11.60	1,610	1,916	37,566	2,327	1.17	7.05
350 x 350	8.0*	85.7	109	21,240	14.00	1,214	1,398	32,352	1,789	1.38	11.7
	10.0	106	136	26,050	13.90	1,489	1,725	39,840	2,186	1.38	9.44
	12.5	132	168	31,810	13.80	1,817	2,122	48,869	2,655	1.37	7.58
	16.0	167	213	39,370	13.60	2,250	2,655	60,901	3,265	1.37	5.99
400 x 400	10.0	122	156	39,350	15.90	1,968	2,272	60,028	2,896	1.58	8.20
	12.5	152	193	48,190	15.80	2,409	2,800	73,815	3,530	1.57	6.58
	16.0	192	245	59,910	15.70	2,995	3,514	92,310	4,363	1.57	5.21
	20.0	237	302	72,396	15.50	3,620	4,292	112,324	5,240	1.56	4.22

Notes:
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.
 * Size not included in BS 4848 Part 2
 † 43C only
 Length per tonne rounded up

HOT FORMED HOLLOW SECTIONS



RECTANGULAR

Section size D x B	Thickness t	Unit weight M	Section area A	Moment of inertia		Radius of gyration r		Elastic modulus Z		Plastic modulus S		Torsional constants J C		Superficial area per m	Approx. length per tonne
				x-x	y-y	x-x	y-y	x-x	y-y	x-x	y-y	x-x	y-y		
mm	mm	kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³	cm ³	cm ³	cm ⁴	cm ³	m ²	m
50 x 25	2.5*	2.72	3.47	10.5	3.44	1.75	1.00	4.25	2.75	5.41	3.25	8.41	4.62	0.145	368
	3.0*	3.22	4.10	12.2	3.89	1.73	0.98	4.88	3.11	6.30	3.77	9.64	5.21	0.144	311
	3.2*	3.41	4.34	12.8	4.05	1.72	0.97	5.11	3.24	6.64	3.96	10.1	5.42	0.143	294
50 x 30	2.5*	2.92	3.72	12.0	5.30	1.80	1.19	4.81	3.53	6.01	4.16	11.7	5.74	0.155	343
	3.0*	3.45	4.40	13.9	6.04	1.78	1.17	5.54	4.03	7.01	4.83	13.5	6.52	0.154	290
	3.2	3.66	4.66	14.5	6.31	1.77	1.16	5.82	4.21	7.39	5.08	14.2	6.81	0.153	274
	4.0*	4.46	5.68	17.0	7.25	1.73	1.13	6.80	4.83	8.81	6.01	16.6	7.79	0.151	224
	5.0*	5.40	6.88	19.5	8.13	1.68	1.09	7.79	5.42	10.4	6.98	19.0	8.71	0.149	185
60 x 40	2.5*	3.71	4.72	23.1	12.2	2.21	1.61	7.71	6.10	9.43	7.09	25.0	9.74	0.195	270
	3.0*	4.39	5.60	26.9	14.1	2.19	1.59	8.96	7.04	11.1	8.29	29.2	11.2	0.194	228
	3.2	4.66	5.94	28.3	14.8	2.18	1.58	9.44	7.39	11.7	8.75	30.8	11.8	0.193	215
	4.0	5.72	7.28	33.6	17.3	2.15	1.54	11.2	8.67	14.1	10.5	36.6	13.7	0.191	175
	5.0*	6.97	8.88	39.2	20.0	2.10	1.50	13.1	10.0	16.8	12.4	43.0	15.8	0.189	143
	6.0*	8.15	10.4	43.9	22.1	2.06	1.46	14.6	11.0	19.2	14.1	48.3	17.3	0.187	129
	6.3*	8.49	10.8	45.1	22.6	2.04	1.45	15.0	11.3	19.9	14.6	49.7	17.7	0.186	118
80 x 40	3.0*	5.34	6.80	55.0	18.2	2.85	1.64	13.8	9.10	17.3	10.5	43.7	15.3	0.234	188
	3.2	5.67	7.22	58.1	19.1	2.84	1.63	14.5	9.56	18.3	11.1	46.1	16.1	0.233	177
	4.0	6.97	8.88	69.6	22.6	2.80	1.59	17.4	11.3	22.2	13.4	55.1	18.9	0.231	144
	5.0*	8.54	10.9	82.4	26.2	2.75	1.55	20.6	13.1	26.7	15.9	65.0	21.9	0.229	117
	6.0	10.0	12.8	93.5	29.1	2.70	1.51	23.4	14.5	30.8	18.2	73.5	24.3	0.227	100
	6.3*	10.5	13.3	96.5	29.8	2.69	1.50	24.1	14.9	31.9	18.8	75.8	24.9	0.226	95.2
	8.0*	12.8	16.3	111	33.1	2.61	1.42	27.7	16.6	37.8	21.8	86.3	27.6	0.223	78.1
90 x 50	3.0*	6.28	8.0	85.4	33.8	3.27	2.05	19.0	13.5	23.4	15.5	76.4	22.4	0.274	160
	3.6	7.46	9.5	99.8	39.1	3.24	2.03	22.2	15.6	27.6	18.1	89.3	25.9	0.272	135
	5.0	10.1	12.9	130	50.0	3.18	1.97	28.9	20.0	36.6	23.9	116	32.9	0.269	99.1
	6.0	11.9	15.2	149	56.4	3.13	1.93	33.1	22.6	42.5	27.5	133	37.0	0.267	84
	6.3*	12.5	15.9	154	58.1	3.12	1.91	34.2	23.3	44.2	28.5	138	38.2	0.266	80
	8.0*	15.3	19.5	180	66.3	3.04	1.84	40.0	26.5	53.0	33.7	161	43.4	0.263	65.4
100 x 50	3.0*	6.75	8.60	111	37.1	3.59	2.08	22.2	14.8	27.6	16.9	88.3	25.0	0.294	149
	3.2	7.18	9.14	117	39.1	3.58	2.07	23.5	15.6	29.2	17.9	93.3	26.4	0.293	140
	4.0	8.86	11.3	142	46.7	3.55	2.03	28.4	18.7	35.7	21.7	113	31.4	0.291	113
	5.0	10.9	13.9	170	55.1	3.50	1.99	34.0	22.0	43.3	26.1	135	37.0	0.289	91.8
	6.0	12.9	16.4	195	62.3	3.45	1.95	39.0	24.9	50.4	30.2	154	41.7	0.287	77.5
	6.3*	13.4	17.1	202	64.2	3.44	1.94	40.5	25.7	52.5	31.3	160	43.0	0.286	74.7
	8.0*	16.6	21.1	238	73.5	3.36	1.86	47.6	29.4	63.1	37.1	187	49.1	0.283	60.2
	8.0*	16.6	21.1	238	73.5	3.36	1.86	47.6	29.4	63.1	37.1	187	49.1	0.283	60.2
100 x 60	3.0*	7.22	9.2	125	56.2	3.69	2.47	25.0	18.7	30.5	21.3	121	30.7	0.314	139
	3.6*	8.59	10.9	147	65.4	3.66	2.45	29.3	21.8	36.0	25.1	142	35.6	0.312	117
	5.0	11.7	14.9	192	84.7	3.60	2.39	38.5	28.2	48.1	33.3	187	45.9	0.309	85.5
	6.0	13.8	17.6	222	96.6	3.55	2.34	44.4	32.2	56.1	38.6	216	52.2	0.307	72.5
	6.3	14.4	18.4	230	99.9	3.54	2.33	46.0	33.3	58.4	40.2	224	53.9	0.306	69.5
	8.0*	17.8	22.7	272	116	3.46	2.26	54.4	38.7	70.5	48.1	266	62.4	0.303	56.2
120 x 60	3.6	9.72	12.4	230	76.9	4.31	2.49	38.3	25.6	47.6	29.2	183	43.3	0.352	103
	5.0	13.3	16.9	304	99.9	4.24	2.43	50.7	33.3	63.9	38.8	242	56.0	0.349	75.2
	6.0	15.7	20.0	352	114	4.20	2.39	58.7	38.1	74.8	45.1	279	63.9	0.347	63.7
	6.3	16.4	20.9	366	118	4.18	2.38	61.0	39.4	78.0	46.9	290	66.0	0.346	61
	8.0*	20.4	25.9	437	138	4.10	2.31	72.8	45.9	94.8	56.4	344	76.8	0.343	49

Notes:

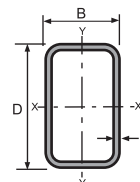
Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

* Size not included in BS 4848 Part 2

† 43C only

Length per tonne rounded up

HOT FORMED HOLLOW SECTIONS



RECTANGULAR

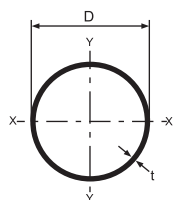
Section size	Thickness	Unit weight	Section area	Moment of inertia		Radius of gyration		Elastic modulus		Plastic modulus		Torsional constants		Superficial area per m	Approx. length per t
D x B	t	M	A	x-x	y-y	x-x	y-y	x-x	y-y	x-x	y-y	J	C	m ²	m
mm	mm	kg/m	cm ²	cm ⁴	cm ⁴	cm	cm	cm ³	cm ³	cm ³	cm ³	cm ⁴	cm ³		
120 x 80	5.0	14.8	18.9	370	195	4.43	3.21	61.7	48.8	75.4	56.7	401	77.9	0.389	67.6
	6.0	17.6	22.4	430	225	4.38	3.17	71.7	56.3	88.5	66.3	467	89.7	0.387	56.8
	6.3	18.4	23.4	447	234	4.37	3.16	74.6	58.4	92.3	69.1	486	93.0	0.386	54.4
	8.0	22.9	29.1	537	278	4.29	3.09	89.5	69.4	113	83.9	586	110	0.383	43.7
	10.0	27.9	35.5	628	320	4.20	3.00	105	80.0	134	99.4	688	126	0.379	35.9
150 x 100	5.0	18.7	23.9	747	396	5.59	4.07	99.5	79.1	121	90.8	806	127	0.489	53.5
	6.0	22.3	28.4	873	461	5.55	4.03	116	92.1	142	107	945	147	0.487	44.8
	6.3	23.3	29.7	910	479	5.53	4.02	121	95.9	148	111	985	153	0.486	43.0
	8.0	29.1	37.1	1,106	577	5.46	3.94	147	115	183	137	1,202	184	0.483	34.4
	10.0	35.7	45.5	1,312	678	5.37	3.86	175	136	220	164	1,431	215	0.479	28.1
	12.5*†	43.6	55.5	1,532	781	5.25	3.75	204	156	263	194	1,680	246	0.473	22.9
160 x 80	5.0	18.0	22.9	753	251	5.74	3.31	94.1	62.8	117	71.7	599	106	0.469	55.6
	6.0	21.3	27.2	880	291	5.69	3.27	110	72.8	138	84.1	700	123	0.467	46.9
	6.3	22.3	28.5	917	302	5.68	3.26	115	75.6	144	87.7	729	127	0.466	44.9
	8.0	27.9	35.5	1,113	361	5.60	3.19	139	90.2	177	107	882	151	0.463	35.9
	10.0	34.2	43.5	1,318	419	5.50	3.10	165	105	213	127	1,041	175	0.459	29.3
	12.5*†	41.6	53.0	1,536	476	5.38	3.00	192	119	254	150	1,206	199	0.453	24.0
200 x 100	5.0	22.7	28.9	1,509	509	7.23	4.20	151	102	186	115	1,202	172	0.589	44.1
	6.0	27.0	34.4	1,774	593	7.18	4.15	177	119	221	135	1,412	200	0.587	37.0
	6.3	28.3	36.0	1,851	618	7.17	4.14	185	124	231	141	1,473	208	0.586	35.4
	8.0	35.4	45.1	2,269	747	7.09	4.07	227	149	286	147	1,802	251	0.583	28.3
	10.0	43.6	55.5	2,718	881	7.00	3.98	272	176	346	209	2,154	296	0.579	23.0
	12.5	53.4	68.0	3,218	1,022	6.88	3.88	322	204	417	249	2,541	342	0.573	18.8
	16.0	66.4	84.5	3,808	1,175	6.71	3.73	381	235	505	297	2,988	393	0.566	15.1
250 x 150	6.0	36.4	46.4	3,997	1,807	9.28	6.24	320	241	387	272	3,873	396	0.787	27.5
	6.3	38.2	48.6	4,178	1,886	9.27	6.23	334	252	405	284	4,049	413	0.786	26.2
	8.0	48.0	61.1	5,167	2,317	9.19	6.16	413	309	505	353	5,014	506	0.783	20.9
	10.0	59.3	75.5	6,259	2,784	9.10	6.07	501	371	618	430	6,082	606	0.779	16.9
	12.5	73.0	93.0	7,518	3,310	8.99	5.97	601	441	751	520	7,317	717	0.773	13.7
	16.0	91.5	117	9,089	3,943	8.83	5.82	727	526	924	635	8,863	851	0.766	11.0
300 x 200	6.0	45.8	58.4	7,532	4,033	11.4	8.31	502	403	599	453	8,091	651	0.987	21.8
	6.3	48.1	61.2	7,880	4,216	11.3	8.30	525	422	627	475	8,468	681	0.986	20.8
	8.0	60.5	77.1	9,798	5,219	11.3	8.23	653	522	785	593	10,549	840	0.983	16.6
	10.0	75.0	95.5	11,940	6,331	11.2	8.14	796	633	964	726	12,890	1,016	0.979	13.4
	12.5	92.6	118	14,460	7,619	11.1	8.04	964	762	1,179	866	15,654	1,217	0.973	10.8
	16.0	117	149	17,700	9,239	10.9	7.89	1,180	924	1,462	1,094	19,227	1,469	0.966	8.55
400 x 200	10.0	90.7	116	24,140	8,138	14.5	8.39	1,207	814	1,492	916	19,236	1,377	1.18	11.1
	12.5	112	143	29,410	9,820	14.3	8.29	1,471	982	1,831	1,120	23,408	1,657	1.17	8.93
	16.0	142	181	36,300	11,950	14.2	8.14	1,815	1,195	2,285	1,388	28,835	2,011	1.17	7.05
450 x 250	10.0	106	136	37,180	14,900	16.6	10.5	1,653	1,192	2,013	1,338	33,247	1,986	1.38	9.44
	12.5	132	168	45,470	18,100	16.5	10.4	2,021	1,448	2,478	1,642	40,668	2,407	1.37	7.58
	16.0	167	213	56,420	22,250	16.3	10.2	2,508	1,780	3,103	2,047	50,478	2,948	1.37	5.99
500 x 300	10.0*	122	156	54,118	24,564	18.7	12.6	2,165	1,638	2,609	1,834	52,396	2,696	1.58	8.20
	12.5*	152	193	66,363	29,972	18.5	12.5	2,655	1,998	3,218	2,257	64,313	3,282	1.57	6.58
	16.0*	192	245	82,673	37,076	18.4	12.3	3,307	2,472	4,042	2,825	80,223	4,046	1.57	5.21

Notes:
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* Size not included in BS 4848 Part 2

† 43C only
 Length per tonne rounded up

HOT FORMED HOLLOW SECTIONS



CIRCULAR

Outside diameter	Thickness	Unit weight	Section area	Moment of inertia	Radius of gyration	Elastic modulus	Plastic modulus	Torsional	Constants	Superficial area per m	Approx. length per tonne
mm	t	kg/m	cm ²	cm ⁴	cm	cm ³	cm ³	cm ⁴	cm ³	m ²	m
21.3	3.2	1.43	1.82	0.77	0.65	0.72	1.06	1.54	1.44	0.067	700
26.9	3.2	1.87	2.38	1.70	0.846	1.27	1.81	3.41	2.53	0.085	535
33.7	2.6	1.99	2.54	3.09	1.10	1.84	2.52	6.19	3.67	0.106	503
	3.2	2.41	3.07	3.60	1.08	2.14	2.99	7.21	4.28	0.106	415
	4.0	2.93	3.73	4.19	1.06	2.49	3.55	8.38	4.97	0.106	342
42.4	2.6	2.55	3.25	6.46	1.41	3.05	4.12	12.9	6.10	0.133	393
	3.2	3.09	3.94	7.62	1.39	3.59	4.93	15.2	7.19	0.133	324
	4.0	3.79	4.83	8.99	1.36	4.24	5.92	18.0	8.48	0.133	264
48.3	3.2	3.56	4.53	11.6	1.60	4.80	6.52	23.2	9.59	0.152	281
	4.0	4.37	5.57	13.8	1.57	5.70	7.87	27.5	11.4	0.152	229
	5.0	5.34	6.80	16.2	1.54	6.69	9.42	32.3	13.4	0.152	188
60.3	3.2	4.51	5.74	23.5	2.02	7.78	10.4	46.9	15.6	0.189	222
	4.0	5.55	7.07	28.2	2.00	9.34	12.7	56.3	18.7	0.189	181
	5.0	6.82	8.69	33.5	1.96	11.1	15.3	67.0	22.2	0.189	147
76.1	3.2	5.75	7.33	48.8	2.58	12.8	17.0	97.6	25.6	0.239	174
	4.0	7.11	9.06	59.1	2.55	15.5	20.8	118	31.0	0.239	141
	5.0	8.77	11.2	70.9	2.52	18.6	25.3	142	37.3	0.239	115
88.9	3.2	6.76	8.62	79.2	3.03	17.8	23.5	158	35.6	0.279	148
	4.0	8.38	10.7	96.3	3.00	21.7	28.9	193	43.3	0.279	120
	5.0	10.3	13.2	116	2.97	26.2	35.2	233	52.4	0.279	97.1
114.3	3.6	9.83	12.5	192	3.92	33.6	44.1	384	67.2	0.359	102
	5.0	13.5	17.2	257	3.87	45.0	59.8	514	89.9	0.359	74.1
	6.3	16.8	21.4	313	3.82	54.7	73.6	625	109	0.359	59.6
139.7	5.0	16.6	21.2	481	4.77	68.8	90.8	961	138	0.439	60.3
	6.3	20.7	26.4	589	4.72	84.3	112	1,177	169	0.439	48.4
	8.0	26.0	33.1	720	4.66	103	139	1,441	206	0.439	38.5
	10.0	32.0	40.7	862	4.60	123	169	1,724	247	0.439	31.3
168.3	5.0	20.1	25.7	856	5.78	102	133	1,712	203	0.529	49.8
	6.3	25.2	32.1	1,053	5.73	125	165	2,107	250	0.529	39.7
	8.0	31.6	40.3	1,297	5.67	154	206	2,595	308	0.529	31.7
	10.0	39.0	49.7	1,564	5.61	186	251	3,128	372	0.529	25.7
193.7	5.0 *	23.3	29.6	1,320	6.67	136	178	2,640	273	0.609	43.0
	6.3	29.1	37.1	1,630	6.63	168	221	3,260	337	0.609	34.4
	8.0	36.6	46.7	2,016	6.57	208	276	4,031	416	0.609	27.4
	10.0	45.3	57.7	2,442	6.50	252	338	4,883	504	0.609	22.1
	12.5	55.9	71.2	2,934	6.42	303	411	5,869	606	0.609	17.9
	16.0 †	70.1	89.3	3,554	6.31	367	507	7,109	734	0.609	14.3
219.1	5.0 *	26.4	33.6	1,928	7.57	176	229	3,856	352	0.688	37.9
	6.3	33.1	42.1	2,386	7.53	218	285	4,772	436	0.688	30.3
	8.0	41.6	53.1	2,960	7.47	270	357	5,919	540	0.688	24.1
	10.0	51.6	65.7	3,598	7.40	328	438	7,197	657	0.688	19.4
	12.5	63.7	81.1	4,345	7.32	397	534	8,689	793	0.688	15.7
	16.0 †	80.1	102	5,297	7.20	483	661	10,590	967	0.688	12.5
	20.0 †	98.2	125	6,261	7.07	572	795	12,520	1143	0.688	10.2

Notes:

Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.

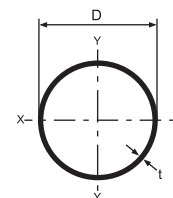
Sizes printed in italics may be supplied Hot Finished, Seamless at our option.

* Size not included in BS 4848 Part 2

† Grade 50C only

Details of larger sizes of CHS available on application. Length per tonne rounded up

HOT FORMED HOLLOW SECTIONS



CIRCULAR

Outside diameter	Thickness t	Unit weight MA	Section area I	Moment of inertia r	Radius of gyration Z	Elastic modulus S	Plastic modulus J	Torsional C	Constant per m	Superficial area per tonne	Approx. length
mm	mm	kg/m	cm ²	cm ⁴	cm	cm ³	cm ³	cm ⁴	cm ³	m ²	m
244.5	6.3	37.0	47.1	3,346	8.42	274	358	6,692	547	0.768	27.1
	8.0	46.7	59.4	4,160	8.37	340	448	8,321	681	0.768	21.5
	10.0	57.8	73.7	5,073	8.30	415	550	10,150	830	0.768	17.4
	12.5	71.5	91.1	6,147	8.21	503	673	12,290	1,006	0.768	14.0
	16.0	90.2	115	7,533	8.10	616	837	15,070	1,232	0.768	11.1
	20.0 †	111	141	8,957	7.97	733	1,011	17,910	1,465	0.768	9.01
273	6.3	41.4	52.8	4,696	9.43	344	448	9,392	688	0.858	24.2
	8.0	52.3	66.6	5,852	9.37	429	562	11,700	857	0.858	19.2
	10.0	64.9	82.6	7,154	9.31	524	692	14,310	1,048	0.858	15.5
	12.5	80.3	102	8,697	9.22	637	849	17,390	1,274	0.858	12.5
	16.0	101	129	10,710	9.10	784	1,058	21,410	1,569	0.858	9.91
	20.0 †	125	159	12,800	8.97	938	1,283	25,600	1,875	0.858	8.00
	25.0 †	153	195	15,130	8.81	1,108	1,543	30,250	2,216	0.858	6.54
323.9	6.3 *	49.3	62.9	7,929	11.2	490	636	15,850	980	1.02	20.3
	8.0	62.3	79.4	9,910	11.2	612	799	19,820	1,224	1.02	16.1
	10.0	77.4	98.6	12,160	11.1	751	986	24,320	1,501	1.02	13.0
	12.5	96.0	122	14,850	11.0	917	1,213	29,690	1,833	1.02	10.5
	16.0	121	155	18,390	10.9	1,136	1,518	36,780	2,271	1.02	8.27
	20.0 †	150	191	22,140	10.8	1,367	1,850	44,280	2,734	1.02	6.67
	25.0 †	184	235	26,400	10.6	1,630	2,239	52,800	3,260	1.02	5.44
355.6	8.0	68.6	87.4	13,200	12.3	742	967	26,400	1,485	1.12	14.6
	10.0	85.2	109	16,220	12.2	912	1,195	32,450	1,825	1.12	11.8
	12.5	106	135	19,850	12.1	1,117	1,472	39,700	2,233	1.12	9.44
	16.0	134	171	24,660	12.0	1,387	1,847	49,330	2,774	1.12	7.47
	20.0 †	166	211	29,790	11.9	1,676	2,255	59,580	3,351	1.12	6.03
	25.0 †	204	260	35,680	11.7	2,007	2,738	71,350	4,013	1.12	4.91
406.4	10.0	97.8	125	24,480	14.0	1,205	1,572	48,950	2,409	1.28	10.3
	12.5	121	155	30,030	13.9	1,478	1,940	60,060	2,956	1.28	8.27
	16.0	154	196	37,450	13.8	1,843	2,440	74,900	3,686	1.28	6.50
	20.0 †	191	243	45,430	13.7	2,236	2,989	90,860	4,472	1.28	5.24
	25.0 †	235	300	54,700	13.5	2,692	3,642	109,400	5,384	1.28	4.26
	32.0 †	295	376	66,430	13.3	3,269	4,497	132,900	6,539	1.28	3.39
457	10.0	110	140	35,090	15.8	1,536	1,998	70,180	3,071	1.44	9.10
	12.5	137	175	43,140	15.7	1,888	2,470	86,290	3,776	1.44	7.30
	16.0	174	222	53,960	15.6	2,361	3,113	107,900	4,723	1.44	5.75
	20.0 †	216	275	65,680	15.5	2,874	3,822	131,400	5,749	1.44	4.63
	25.0 †	266	339	79,420	15.3	3,476	4,671	158,800	6,951	1.44	3.76
	32.0 †	335	427	97,010	15.1	4,246	5,791	194,000	8,491	1.44	2.99
	40.0 †	411	524	114,900	14.8	5,031	6,977	229,900	10,060	1.44	2.44
508	10.0 *	123	156	48,520	17.6	1,910	2,480	97,040	3,821	1.60	8.14
	12.5 *	153	195	59,760	17.5	2,353	3,070	119,500	4,705	1.60	6.54
	16.0 *	194	247	74,910	17.4	2,949	3,874	149,800	5,898	1.60	5.16
	20.0 †	241	307	91,430	17.3	3,600	4,766	182,900	7,200	1.60	4.15
	25.0 †	298	379	110,900	17.1	4,367	5,837	221,800	8,734	1.60	3.36
	32.0 †	376	479	136,100	16.9	5,360	7,261	272,300	10,720	1.60	2.66
	40.0 †	462	588	162,200	16.6	6,385	8,782	324,400	12,770	1.60	2.17
	50.0 †	565	719	190,900	16.3	7,515	10,530	381,800	15,030	1.60	1.77

Notes:
 Sizes printed black on grey background are by the Continuous Weld Process. Sizes printed in bold type are by the Hot Finished Seamless Process.
 Sizes printed in italics may be supplied Hot Finished, Seamless at our option.
 * Size not included in BS 4848 Part 2
 † Grade 50C only
 Details of larger sizes of CHS available on application. Length per tonne rounded up

PIPES

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PIPES

TECHNICAL SPECIFICATION REFERENCE -STEEL PIPES

Excerpts from JIS G 3452 & ASTM A-53 Specifications

Classification	Specification	Grade	Chemical Composition					Tensile Test			Bend Test *1		Flattening Test H
			C max.	Mn max.	Si max.	P max.	S max.	Tensile Strength	Yield Strength	Elongation	Bending Angle	Bending Radius	
			%	%	%	%	%	N/mm ²	N/mm ²	%			
Carbon Steel Pipes for Ordinary Piping	JIS G 3452	SGP	--	--	--	0.04	0.04	290 min.	--	* Test Piece No. 11 & 12 * Test Piece No. 5	30 min. 25 min.	90° 6D	$\frac{2}{3} D$
ASTM Standard	ASTM A53	A	0.25	0.95	--	0.05	0.06	330 min.	205 min.	As specified in A-53 specification	90°	12D	$\frac{1}{3} D$
Welded Steel Pipes	ASTM A53	B	0.30	1.20	--	0.05	0.06	415 min.	240 min.				

Notes:

- * - When the tensile test is carried out on No. 12 or No. 5 test piece for the pipe under 8 mm in wall thickness, the minimum value of elongation shall be obtained by subtracting 1.5 % from the thickness values of elongation given in Table above for each 1 mm decrease in wall thickness, and rounding off to an integer in accordance with JIS Z 8401.
- * - The values of elongation given in Table above shall not applied to the pipe whose nominal size is 32 mm or smaller.
- *1 - Bend test in table above only applied to pipes of nominal size 2" (50 mm) or smaller.
- H - Distance between the plates
- D - Outside diameter of the pipe

YICK HOE

TECHNICAL SPECIFICATION REFERENCE - STEEL TUBES

Excerpts from JIS G 3444, JIS G 3445, BS 31 & BS 1139 Specifications

Classification	Specification	Grade	Chemical Composition					Tensile Test				Bend Test		Flattening Test H	
			C max. %	Mn max. %	Si max. %	P max. %	S max. %	Tensile Strength N/mm²	Yield Strength N/mm²	Elongation %		Bending Angle	Bending Radius		
Carbon Steel Tubes for General Structural Purposes	JIS G 3444	STK290	--	--	--	0.05	0.05	290 min.	--	* Test Piece No.11 & 12	30 min.	90°	6D	2/3D	
										* Test Piece No.5	25 min.				
	JIS G 3444	STK400	0.25	--	-	0.04	0.04	400 min.	235 min.	* Test Piece No. 11 &12	23 min.	90°	6D	2/3D	
										* Test Piece No. 5	18 min.				
	JIS G 3444	STK500	0.30	0.35	0.30- 1.00	0.04	0.04	500 min.	355 min.	* Test Piece No. 11 &12	15 min.	90°	8D	7/8D	
										* Test Piece No.5	10 min				
	JIS G 3444	STK540	0.23	0.55	1.50	0.04	0.04	540 min.	390 min.	* Test Piece No. 11 & 12	20 min.	90°	6D	7/8D	
										* Test Piece No.5	16 min.				
	Carbon Steel Tubes For Machine Structural Purposes	JIS G 3445	STKM 11A	0.12	0.35	0.60	0.04	0.04	290 min.	--	* Test Piece No. 11 & 12	35 min.	180°	4D	1/2D
											* Test Piece No.5	30 min.			
British Standard Galvanized Steel Conduit	BS31	Class B (screwed)	--	--	--	--	--	273 to 365	--	Gauge Length (8 in.)	15 min	90°	6D	--	
British Standard Steel Tubes for Scaffolding	BS1139		0.20	--	0.30	0.05	0.05	340 to 480	235 min.	Gauge Length Lo=5.65√So	24 min.	--	--	--	
Window Pipe	Manufacturer Standard	--	--	--	--	--	--	--	--	--	--	--	-	-	
Earth Pipe	Manufacturer Standard	--	--	--	--	--	--	--	--	--	--	--	--	--	

Notes:

- * - Elongation values in Table above only applied to tubes over 40 mm in outside diameter
- * - When the tensile test is carried out on No. 12 or No. 5 test piece for the tube under 8 mm in wall thickness the minimum elongation shall be determined by reducing 1.5% per 1 mm of decrease in wall thickness from the values given in the Table above and rounding off the value obtained to integer in accordance with JIS Z 8401.
- H - Distance between flat plates
- D - Outside diameter of the tube

PIPES

TECHNICAL SPECIFICATION REFERENCE -STEEL TUBES AND LIPPED CHANNEL

Classification	Standard Specification		Application	Chemical Requirements (%)					Physical Requirements (%)			
				C	Mn	Si	P	S	Others	Tensile Strength	Yield Strength	Elongation
				max. %	max. %	max. %	max. %	max. %	%	N/mm ²	N/mm ²	%
Circular Hollow Section	BS1387	L	Carbon Steel Tubes for Ordinary Uses, Water Pipes and Structural Purposes	0.20	1.20	--	0.045	0.045	--	320 - 460	195 min.	20 min.
		M										
		H										
	JIS G 3444 /Mfg Standard	STK 290	Carbon Steel Tubes for Structural Purposes (Scaffolding)	--	--	--	0.05	0.05	--	290 min.	--	25 min.
		STK 400		0.25	--	--	0.04	0.04	--	400 min.	235 min.	18 min.
		STK 500		0.30	0.30 - 1.00	0.35	0.04	0.04	--	500 min.	355 min.	10 min.
	JIS G 3445 /Mfg Standard	STKM11A	Carbon Steel Tubes for Machine Structural Purposes	0.12	0.60 max.	0.35	0.04	0.04	--	290 min.	--	30 min.
	JIS G 3452 /Mfg Standard	SGP	Carbon Steel Tubes for Ordinary Uses, Water Pipes and Structural Purposes	--	--	--	0.04	0.04	--	290 min.	--	25 min.
Square and Rectangular Hollow Section	ASTM A500/ Mfg Standard	A	Carbon Steel Tubes for Structural Purposes and Machine Structural Purposes	0.30	--	--	0.05	0.063	Cu (max) = 0.18	310 min.	269 min.	25 min.
		B		0.30	--	--	0.05	0.063	Cu (max) = 0.18	400 min.	317 min.	23 min.
Circular Hollow Section	Mfg Standard	AA	Carbon Steel Tubes for Ordinary Uses	0.20	1.20	--	0.045	0.045	--	250 min.	190 min.	20 min.
Light Gauge Lipped Channel	Mfg Standard		For Civil Engineering	0.25	--	--	0.05	0.05	--	295 min.	--	25 min.
Conduit Pipe	BS 31/ Mfg Standard		For Electrical Wiring	--	--	--	--	--	--	273 - 365	--	15 min.
Window Pipe	Mfg Standard		For Window Usage	--	--	--	--	--	--	--	--	--

BS 1387:1985 A1, LIGHT, MEDIUM AND HEAVY

Pertinent Excerpts from BS 1387:1985 Specification

Tolerances on Dimensions and Mass	Outside Diameter	:	As shown in Table on opposite page.
	Wall thickness	:	Light tubes -8%
		:	Medium and Heavy tubes -10%
	Mass	:	The mean consignment mass for quantities of 150 m and over of one size shall not deviate by more than $\pm 4\%$ from the mass of consignment calculated from the mass given in Table as appropriate. No single tube shall deviate by more than +10%, -8% from the mass given in Table as appropriate.

Bend Test Black tubes up to and including DN 50 shall be bent cold without any signs of fracture or failure, through 180°C round a former having a radius at the bottom of the groove equal to six times the outside diameter of the tube as given in Table. Hot-dip-zinc coated tubes shall be bent cold without cracking of the steel, through 90° round a former having a radius at the bottom of the groove equal to eight times the outside diameter of tube.

Leak Tightness Test	The test shall be either a hydraulic test at a pressure of 50 bar ($50 \times 10^5 \text{ N/M}^2$), or alternatively, an eddy current test.
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Rod Diameters	
Nominal Size (DN)	Diameter of rod (mm)
15.00	9.50
20.00	14.30
25.00	20.60

PIPES

BRITISH STANDARD WELDED STEEL PIPES

BS 1387:1985

Thickness Class	Nominal Size		Outside Diameter		Wall Thickness	Calculated Weight				No. of Threads per inch	Socket	
			Maximum	Minimum		Plain Ends		Threads and Coupling			Outer Diameter	Min. Length
	mm	in	mm	mm	mm	kg/m	kg/ft	kg/m	kg/ft	mm	mm	
A1	15	½	21.4	21.0	1.9	0.904	0.276	0.914	0.279	14	27.8	38.1
	20	¾	26.9	26.4	2.1	1.27	0.387	1.284	0.392	14	34.1	41.3
	25	1	33.8	33.2	2.3	1.77	0.54	1.787	0.545	11	42.1	47.6
	32	1¼	42.5	41.9	2.3	2.26	0.689	2.28	0.695	11	51.6	54.0
	40	1½	48.4	47.8	2.5	2.81	0.857	2.83	0.863	11	57.9	57.2
	50	2	60.2	59.6	2.6	3.67	1.119	3.693	1.126	11	70.6	63.5
	65	2½	76.0	75.2	2.9	5.20	1.585	5.228	1.594	11	87.3	69.9
	80	3	88.7	87.9	2.9	6.11	1.863	6.136	1.871	11	101.6	76.2
	100	4	113.9	113.0	3.2	8.70	2.652	8.736	2.663	11	128.6	88.9
Light	15	½	21.4	21.0	2.0	0.947	0.289	0.956	0.291	14	27.8	38.1
	20	¾	26.9	26.4	2.3	1.38	0.421	1.39	0.424	14	34.1	41.3
	25	1	33.8	33.2	2.6	1.98	0.604	2.00	0.61	11	42.1	47.6
	32	1¼	42.5	41.9	2.6	2.54	0.774	2.57	0.783	11	51.6	54.0
	40	1½	48.4	47.8	2.9	3.23	0.985	3.27	0.997	11	57.9	57.2
	50	2	60.2	59.6	2.9	4.08	1.24	4.15	1.26	11	70.6	63.5
	65	2½	76.0	75.2	3.2	5.71	1.74	5.83	1.78	11	87.3	69.9
	80	3	88.7	87.9	3.2	6.72	2.05	6.89	2.10	11	101.6	76.2
	100	4	113.9	113.0	3.6	9.75	2.97	10.00	3.05	11	128.6	88.9

Test Pressure: 50 Bar / 700Psi

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BRITISH STANDARD WELDED STEEL PIPES

BS 1387:1985

Thickness Class	Nominal Size		Outside Diameter		Wall Thickness	Calculated Weight				No. of Threads per inch	Socket	
			Maximum	Minimum		Plain Ends		Threads and Coupling			Outer Diameter	Min. Length
	mm	in	mm	mm	mm	kg/m	kg/ft	kg/m	kg/ft	mm	mm	
Medium	15	½	21.7	21.1	2.6	1.21	0.369	1.22	0.372	14	27.8	38.1
	20	¾	27.2	26.6	2.6	1.56	0.475	1.57	0.479	14	34.1	41.3
	25	1	34.2	33.4	3.2	2.41	0.735	2.43	0.741	11	42.1	47.6
	32	1¼	42.9	42.1	3.2	3.10	0.945	3.13	0.954	11	51.6	54.0
	40	1½	48.8	48.0	3.2	3.57	1.09	3.61	1.10	11	57.9	57.2
	50	2	60.8	59.8	3.6	5.03	1.53	5.10	1.55	11	70.6	63.5
	65	2½	76.6	75.4	3.6	6.43	1.96	6.55	2.00	11	87.3	69.9
	80	3	89.5	88.1	4.0	8.37	2.55	8.54	2.60	11	101.6	76.2
	100	4	114.9	113.3	4.5	12.2	3.72	12.5	3.81	11	128.6	88.9
	125	5	140.6	138.7	5.0	16.6	5.06	17.1	5.21	11	155.6	95.3
	150	6	166.1	164.1	5.0	19.7	6.00	20.3	6.19	11	184.2	95.3
Heavy	15	½	21.7	21.1	3.2	1.44	0.439	1.45	0.442	14	27.8	38.1
	20	¾	27.2	26.6	3.2	1.87	0.57	1.88	0.573	14	34.1	41.3
	25	1	34.2	33.4	4.0	2.94	0.896	2.96	0.902	11	42.1	47.0
	32	1¼	42.9	42.1	4.0	3.80	1.16	3.83	1.17	11	51.6	54.0
	40	1½	48.8	48.0	4.0	4.38	1.34	4.42	1.35	11	57.9	57.2
	50	2	60.8	59.8	4.5	6.19	1.89	6.26	1.91	11	70.6	63.5
	65	2½	76.6	75.4	4.5	7.93	2.42	8.05	2.45	11	87.3	69.9
	80	3	89.5	88.1	5.0	10.3	3.14	10.5	3.20	11	101.6	76.2
	100	4	114.9	113.3	5.4	14.5	4.42	14.8	4.51	11	128.6	88.9
	125	5	140.6	138.7	5.4	17.9	5.46	18.4	5.61	11	155.6	95.3
	150	6	166.1	164.1	5.4	21.3	6.49	21.9	6.68	11	184.2	95.3

Test Pressure: 50 Bar / 700Psi

PIPES

BRITISH STANDARD WELDED STEEL PIPES

BRITISH STANDARD STEEL TUBES FOR SCAFFOLDING (BS 1139: SECTION 1.1:1990)

Outside Diameter	Wall Thickness	Calculated Weight	Cross -Sectional Area	Second Moment of Inertia	Section Modulus	Radius of Gyration of Area	Plastic Modulus
mm	mm	kg/m	cm ²	cm ⁴	cm ³	cm	cm ³
48.30	4.00	4.37	5.57	13.80	5.70	1.57	7.87

Applicable Tolerances : Thickness $\pm 10\%$
Outside Diameter ± 0.50 mm

BRITISH STANDARD GALVANIZED STEEL CONDUIT (BS 31-1940 CLASS B SCREWED)

Nominal Size		Outside Diameter				Thickness				Calculated Weight with Coupler			No. of Threads	Length of Threads			
		Nominal		Minimum		Nominal		Minimum						Maximum		Minimum	
in	mm	in	mm	in	mm	in	mm	in	mm	kg/m	kg/ft	lb/ft	per inch	in	mm	in	mm
¾	19	0.75	19.05	0.7387	18.76	0.064	1.63	1.52	0.060	0.713	0.217	0.479	16	0.5625	14.3	0.500	12.7
1	25	1.00	25.40	0.9887	25.11	0.064	1.63	1.52	0.060	0.972	0.296	0.653	16	0.6875	17.5	0.625	15.9
1¼	32	1.25	31.75	1.2387	31.46	0.064	1.63	1.52	0.060	1.240	0.376	0.830	16	0.7500	19.1	0.6875	17.5
1½	38	1.50	38.10	1.4880	37.80	0.072	1.83	1.73	0.068	1.680	0.511	1.130	14	0.8125	20.6	0.750	19.1
2	50	2.00	50.80	1.9880	50.50	0.080	2.03	1.93	0.076	2.510	0.765	1.690	14	0.9375	23.8	0.875	22.2

Standard Length 3.81 m (12 ft. 6 in) without coupler

Features:

- * Made of hot-dip galvanized steel strip with extra-smooth surface and highly-adherent zinc coating by the unique tube-making process. The weld zone coating restored in-line
- * Inside weld bead controlled to a minimum for easier wire pulling
- * Screwed on both ends to BS 31 and fitted with a zinc-coated coupler on one end
- * Packed in bare bundles, but the unsocketed ends protected with plastic caps
- * Easier to cut, thread, bend and pull. Dimensionally accurate. Uniform quality in every respect.

EARTH PIPE - MANUFACTURER STANDARD

Nominal Size		Outside Diameter	Wall Thickness	Weight		
in	mm	mm	mm	kg/m	kg/ft	lb/ft
2	50	60.33	1.60	2.318	0.707	1.559

Standard Length: 4.78 m (16 ft.)

WINDOW PIPE - MANUFACTURER STANDARD

Nominal Size		Outside Diameter	Wall Thickness	Weight		
in	mm	mm	mm	kg/m	kg/ft	lb/ft
0.625	16	15.90	1.20	0.435	0.133	0.293

CARBON STEEL PIPES FOR GENERAL STRUCTURAL PURPOSES

LIGHT (AA) (MANUFACTURER STANDARD)

Nominal Size		Outside Diameter		Nominal Wall Thickness		Calculated Weight		
		Minimum	Maximum			Plain Ends		
in	mm	mm	mm	in	mm	kg/m	kg/ft	lb/ft
½	15	21.0	21.4	0.063	1.6	0.739	0.225	0.496
¾	20	26.4	26.9	0.063	1.6	0.982	0.229	0.659
1	25	33.2	33.8	0.069	1.8	1.364	0.416	0.916
1¼	32	41.9	42.3	0.069	1.8	1.739	0.530	1.167
1½	40	47.8	48.4	0.069	1.8	1.994	0.608	1.339
2	50	59.6	60.2	0.069	1.8	2.503	0.763	1.680
2½	65	75.2	76.0	0.072	1.9	3.368	1.026	2.261
3	80	87.9	88.7	0.080	2.1	4.330	1.320	2.907
3½	90	101.1	102.0	0.080	2.1	5.150	1.570	3.460
4	100	112.2	113.2	0.085	2.2	5.917	1.803	3.972
5	125	138.7	140.6	0.150	4.0	12.364	3.768	8.308
6	150	164.1	166.1	0.150	4.0	15.069	4.593	10.117

Wall Thickness Tolerance = -10%

CARBON STEEL CEMENT LINING PIPES (MANUFACTURER STANDARD)

Outside Diameter	Minimum Nominal Wall Thickness	Outside Diameter		Calculated Weight
		Maximum	Minimum	
mm	mm	mm	mm	kg/m
177.3	3.7	179.1	175.5	17.51
	4.1	179.1	175.5	19.22
232.2	4.1	234.5	229.9	25.32

PIPES

CARBON STEEL PIPES FOR SCAFFOLDING

JIS G 3444/88 (MANUFACTURER STANDARD))

Nominal Size	Standard Specification	Outside Diameter		Wall Thickness	Weight	Reference			
		Min	Max			Cross Sectional Area	Geometrical Moment of Inertia	Modulus of Section	Radius of Gyration of Area
		mm	mm			cm ²	cm ⁴	cm ³	cm
21.7 O/D	STK 290	21.45	21.95	1.20	0.607	0.773	0.407	0.375	0.726
				1.50	0.747	0.952	0.488	0.45	0.716
				1.60	0.793	1.010	0.513	0.473	0.713
				2.00	0.972	1.238	0.607	0.56	0.700
42.7 O/D	STK 400	42.45	42.95	2.30	2.29	2.919	5.97	2.80	1.43
				2.50	2.49	3.157	6.40	3.00	1.42
				2.80	2.76	3.510	7.02	3.29	1.41
48.6 O/D	STK 500	48.35	48.85	2.30	2.63	3.345	8.99	3.70	1.64
				2.50	2.84	3.621	9.65	3.97	1.63
				2.80	3.16	4.029	10.60	4.36	1.62
				3.20	3.58	4.564	11.80	4.86	1.61

Notes:

- 1) When Wall Thickness < 3.0 mm, tolerance = ± 0.3 mm
- 2) When $3.0 \text{ mm} \leq \text{wall thickness} < 12 \text{ mm}$, tolerance = $\pm 10\%$

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CARBON STEEL PIPES FOR ORDINARY PIPING

JIS G 3452-1968 (SGP)

Nominal Size		Outside Diameter		Thickness		Unit weight (plain end)		Test Pressure	
A	B	mm	in	mm	in	lb/ft	kg/m	kg/cm ²	Psi
6	1/8	10.5	0.413	2.0	0.079	0.282	0.419	25	360
8	¼	13.8	0.543	2.3	0.090	0.438	0.652	25	360
10	3/8	17.3	0.681	2.3	0.090	0.572	0.851	25	360
15	½	21.7	0.854	2.8	0.110	0.88	1.31	25	360
20	¾	27.2	1.071	2.8	0.110	1.13	1.68	25	360
25	1	34.0	1.338	3.2	0.126	1.63	2.43	25	360
32	1¼	42.7	1.681	3.5	0.138	2.27	3.38	25	360
40	1½	48.6	1.913	3.5	0.138	2.61	3.89	25	360
50	2	60.5	2.382	3.8	0.150	3.57	5.31	25	360
65	2½	76.3	3.004	4.2	0.165	5.02	7.47	25	360
80	3	89.1	3.508	4.2	0.165	5.91	8.79	25	360
90	3½	101.6	4.000	4.2	0.165	6.79	10.1	25	360
100	4	114.3	4.500	4.5	0.177	8.2	12.2	25	360
125	5	139.8	5.504	4.5	0.177	10.1	15.0	25	360
150	6	165.2	6.504	5.0	0.197	13.3	19.8	25	360
175	7	190.7	7.508	5.3	0.209	16.3	24.2	25	360
200	8	216.3	8.516	5.8	0.228	20.2	30.1	25	360
225	9	241.8	9.520	6.2	0.244	24.2	36.0	25	360
250	10	267.4	10.528	6.6	0.260	28.5	42.4	25	360
300	12	318.5	12.539	6.9	0.272	35.6	53.0	25	360
350	14	355.6	14.00	7.9	0.311	45.5	67.7	25	360
400	16	406.4	16.00	7.9	0.311	52.1	77.6	25	360
450	18	457.2	18.00	7.9	0.311	58.7	87.5	25	360
500	20	508.0	20.00	7.9	0.311	65.4	97.4	25	360

PIPES

CARBON STEEL PIPES FOR MACHINE STRUCTURAL PURPOSES

AVAILABLE IN COLD ROLLED, HOT ROLLED MATERIAL (JIS G 3445-1983-STKM 11A)

Nominal Size			Outside Diameter		Thickness												
					SWG	19				18				17			
					mm	1.0				1.2				1.4			
					in	0.039				0.047				0.065			
mm	in	mm		kg/m	kg/6m	kg/ft	lb/ft	kg/m	kg/6m	kg/ft	lb/ft	kg/m	kg/6m	kg/ft	lb/ft		
12	1/2	12.7		0.289	1.734	0.088	0.194	0.340	2.040	0.104	0.229	0.390	2.34	0.119	0.262		
16	5/8	15.9		0.368	2.208	0.112	0.247	0.435	2.610	0.133	0.293	0.501	3.006	0.153	0.337		
19	3/4	19.1		0.447	2.682	0.136	0.300	0.530	3.180	0.162	0.357	0.611	3.666	0.186	0.410		
22	7/8	22.2		0.523	3.138	0.159	0.351	0.622	3.732	0.189	0.417	0.718	4.308	0.219	0.483		
25	1	25.4		0.602	3.612	0.183	0.404	0.716	4.296	0.218	0.481	0.829	4.974	0.253	0.558		
28	1-1/8	28.6		0.681	4.086	0.208	0.459	0.811	4.866	0.247	0.545	0.939	5.634	0.286	0.631		
32	1-1/4	31.8		0.760	4.560	0.232	0.512	0.906	5.436	0.276	0.609	1.050	6.30	0.320	0.706		
35	1-3/8	34.9						1.000	6.000	0.304	0.670	1.160	6.96	0.354	0.781		
38	1-1/2	38.1						1.092	6.552	0.333	0.734	1.267	7.602	0.386	0.851		
41	1-5/8	41.3						1.187	7.122	0.362	0.798	1.378	8.268	0.420	0.926		
44	1-3/4	44.5						1.281	7.686	0.391	0.862	1.488	8.928	0.454	1.001		
47	1-7/8	47.6						1.373	8.238	0.419	0.924	1.595	9.570	0.486	1.072		
50	2	50.8						1.468	8.808	0.447	0.986	1.705	10.230	0.520	1.146		
54	2-1/8	54.0						1.563	9.378	0.476	1.049	1.816	10.896	0.554	1.221		
57	2-1/4	57.2										1.926	11.556	0.587	1.294		
60	2-3/8	60.3										2.033	12.198	0.620	1.367		
65	2-1/2	63.5										2.144	12.864	0.653	1.440		
80	3	76.2															

YICK HOE

CARBON STEEL PIPES FOR MACHINE STRUCTURAL PURPOSES

AVAILABLE IN COLD ROLLED, HOT ROLLED MATERIAL (JIS G 3445-1983-STKM 11A)

Nominal Size	Outside Diameter		Thickness												
			SWG	16				15				14			
			mm	1.6				1.8				2.0			
			in	0.063				0.071				0.079			
mm	in	mm		kg/m	kg/6m	kg/ft	lb/ft	kg/m	kg/6m	kg/ft	lb/ft	kg/m	kg/6m	kg/ft	lb/ft
12	1/2	12.7		0.438	2.628	0.134	0.296	0.484	2.904	0.148	0.326	0.528	3.168	0.161	0.355
16	5/8	15.9		0.564	3.384	0.172	0.379	0.626	3.756	0.191	0.421	0.686	4.116	0.209	0.461
19	3/4	19.1		0.691	4.146	0.211	0.465	0.768	4.608	0.234	0.516	0.844	5.064	0.257	0.567
22	7/8	22.2		0.813	4.878	0.248	0.547	0.906	5.436	0.276	0.609	0.996	5.976	0.304	0.670
25	1	25.4		0.939	5.634	0.286	0.631	1.048	6.288	0.319	0.703	1.154	6.924	0.352	0.776
28	1-1/8	28.6		1.066	6.396	0.325	0.717	1.190	7.140	0.363	0.80	1.312	7.872	0.400	0.882
32	1-1/4	31.8		1.192	7.152	0.363	0.800	1.332	7.992	0.406	0.895	1.470	8.820	0.448	0.988
35	1-3/8	34.9		1.318	7.908	0.402	0.886	1.469	8.814	0.448	0.988	1.623	9.738	0.495	1.091
38	1-1/2	38.1		1.440	8.640	0.439	0.968	1.612	9.672	0.491	1.083	1.781	10.686	0.543	1.197
41	1-5/8	41.3		1.567	9.402	0.478	1.054	1.754	10.524	0.534	1.177	1.938	11.628	0.591	1.303
44	1-3/4	44.5		1.693	10.158	0.516	1.138	1.896	11.376	0.578	1.274	2.096	12.576	0.639	1.409
47	1-7/8	47.6		1.815	10.890	0.553	1.219	2.033	12.198	0.620	1.367	2.249	13.494	0.686	1.512
50	2	50.8		1.942	11.652	0.592	1.305	2.175	13.050	0.663	1.462	2.407	14.442	0.734	1.618
54	2-1/8	54.0		2.068	12.408	0.630	1.389	2.317	13.902	0.706	1.557	2.565	15.390	0.782	1.724
57	2-1/4	57.2		2.194	13.164	0.669	1.475	2.459	14.754	0.750	1.654	2.723	16.338	0.830	1.830
60	2-3/8	60.3		2.316	13.896	0.706	1.557	2.597	15.582	0.792	1.746	2.876	17.256	0.877	1.934
65	2-1/2	63.5		2.443	14.658	0.745	1.642	2.739	16.434	0.835	1.841	3.033	18.198	0.925	2.039
80	3	76.2		2.944	17.644	0.897	1.978	3.303	19.818	1.007	2.220	3.660	21.960	1.116	2.460

The following size is also available upon request:

Nominal Size	Outside Diameter		Thickness				
			mm	1.60			
			in	0.063			
mm	in	mm		kg/m	kg/6m	kg/ft	lb/ft
36	1.417	36		1.357	8.144	0.414	0.912

PIPES

CARBON STEEL PIPES FOR GENERAL STRUCTURAL PURPOSES

JIS G 3444-1988-STK 290

JIS G 3444-1988-STK 500

JIS G 3444-1988-STK 400

JIS G 3444-1988-STK 540

Outside Diameter	Wall Thickness	Calculated Weight	Cross-Sectional Area	Geometrical Moment of Inertia	Modulus of Section	Radius of Gyration of Area
mm	mm	kg/mm	cm ²	cm ⁴	cm ³	cm
21.7	2.0	0.972	1.238	0.607	0.56	0.70
27.2	2.0	1.24	1.583	1.26	0.93	0.89
	2.3	1.41	1.799	1.41	1.03	0.88
34.0	2.3	1.80	2.291	2.89	1.70	1.12
42.7	2.3	2.29	2.919	5.97	2.80	1.43
	2.5	2.49	3.157	6.40	3.00	1.42
	2.8	2.76	3.510	7.02	3.29	1.41
48.6	2.3	2.63	3.345	8.99	3.70	1.64
	2.5	2.84	3.621	9.65	3.97	1.63
	2.8	3.16	4.029	10.6	4.36	1.62
	3.2	3.58	4.564	11.8	4.86	1.61
60.5	2.3	3.30	4.205	17.8	5.90	2.06
	3.2	4.52	5.760	23.7	7.84	2.03
	4.0	5.57	7.100	28.5	9.41	2.00
76.3	2.8	5.08	6.465	43.7	11.5	2.60
	3.2	5.77	7.349	49.2	12.9	2.59
	4.0	7.13	9.085	59.5	15.6	2.56
89.1	2.8	5.96	7.591	70.7	15.9	3.05
	3.2	6.78	8.636	79.8	17.9	3.04
	4.0	8.39	10.69	97.0	21.8	3.01
101.6	3.2	7.76	9.892	120	23.6	3.48
	4.0	9.63	12.26	146	28.8	3.45
	5.0	11.90	15.17	177	34.9	3.42
114.3	3.2	8.77	11.17	172	30.2	3.93
	3.6	9.83	12.52	192	33.6	3.92
	4.5	12.20	15.52	234	41.0	3.89
	5.6	15.00	19.12	283	49.6	3.85
139.8	3.6	12.10	15.40	357	51.1	4.82
	4.0	13.40	17.07	394	56.3	4.80
	4.5	15.00	19.13	438	62.7	4.79
	6.0	19.80	25.22	566	80.9	4.74
165.2	4.5	17.80	22.72	734	88.9	5.68
	5.0	19.80	25.16	808	97.8	5.67
	6.0	23.60	30.01	952	115	5.63
	7.0	27.30	34.79	1,090	132	5.60
190.7	4.5	20.70	26.32	1,140	120	6.59
	5.0	22.90	29.17	1,260	132	6.57
	6.0	27.30	34.82	1,490	156	6.53
	7.0	31.70	40.40	1,710	179	6.50
216.3	4.5	23.50	29.94	1,680	155	7.49
	6.0	31.10	39.64	2,190	203	7.44
	7.0	36.10	46.03	2,520	233	7.40

Applicable Tolerances:

Thickness	Under 3 mm ± 0.3mm
	3 mm and over to 8 mm ± 10%
Outside Diameter	Under 50 mm ± 0.25 mm
	50 mm or over ± 0.5%

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YICK HOE

CHANNELS

SAFE LOADS FOR GRADE 43 STEEL

Section size	Mass per metre	Safe distributed loads in kilonewtons for spans in metres and deflection coefficients													Critical span L_c
mm	kg	1.50 199.1	2.00 112.0	2.50 71.68	3.00 49.78	3.50 36.57	4.00 28.00	4.50 22.12	5.00 17.92	5.50 14.81	6.00 12.44	7.00 9.143	8.00 7.000	9.00 5.531	m
76 x 38	6.70	17	13	10											1.724
102 x 51	10.42	36	27	22	18										1.993
127 x 64	14.90	67	50	40	33	29	25								2.477
152 x 76	17.88	98	74	59	49	42	37	33	30						2.591
152 x 89	23.84	135	101	81	67	58	50	45	40						3.629
178 x 76	20.84	132	99	79	66	57	50	44	40	36	33				2.577
178 x 89	26.81	174	130	104	87	74	65	58	52	47	43				3.388
203 x 76	23.82	169	127	101	84	72	63	56	51	46	42				2.497
203 x 89	29.78	216	162	129	108	92	81	72	65	59	54				3.190
229 x 76	26.06	201	151	121	100	86	75	67	60	55	50	43			2.317
229 x 89	32.76	261	196	156	130	112	98	87	78	71	65	56			3.005
254 x 76	28.29	233	175	140	117	100	87	78	70	64	58	50	44		2.142
254 x 89	35.74	308	231	185	154	132	116	103	92	84	77	66	58		2.842
304 x 89	41.69	408	306	245	204	175	153	136	122	111	102	87	76	68	2.537
305 x 102	46.18	474	356	285	237	203	178	158	142	129	119	102	89	79	3.077
381 x 102	55.10	688	516	413	344	295	258	229	206	188	172	147	129	115	2.896
432 x 102	65.54	872	654	523	436	374	327	291	262	238	218	187	164	145	2.707

For explanation of table see page 25.

The loads listed are based on a bending stress of 165 N/mm², and assume adequate lateral support. Without such support the span must not exceed L_c unless the compressive stress is reduced in accordance with clause 19a (ii) of BS 449.

Loads printed in bold type may cause overloading of the unstiffened web, the capacity of which should be checked. See page 30.

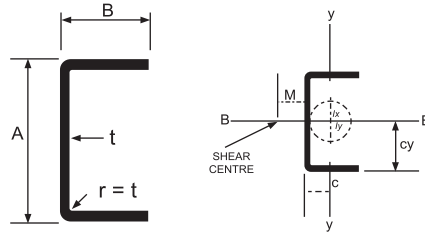
Loads printed in italic type do not cause overloading of the unstiffened web, and do not cause deflection exceeding span/360.

Loads printed in ordinary type should be checked for deflection. See pages 28 and 29.

† Load is based on allowable shear of web and is less than allowable load in bending.

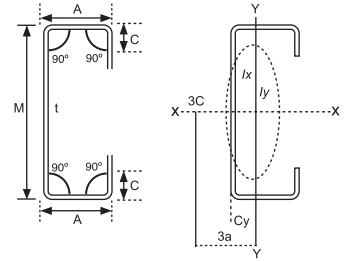
PLAIN CHANNELS

G 3350-1987



Section size A x B mm	Unit Weight M kg/m	Thick-ness t mm	Area A cm ²	Position of Centroid Cx cm Cy cm		Moment of Inertia Ix cm ⁴ Iy cm ⁴		Radius of gyration Rx cm Ry cm		Modulus of Section Zx cm ³ Zy cm ³		Moment Capacity Mx kg/m	Shear Centre M cm	Column Factor Q
60 x 30	1.44	1.6	1.83	0.82	3.00	10.33	1.63	2.37	0.94	3.44	0.75	46.08	1.12	0.77
	2.03	2.3	2.58	0.85	3.00	15.13	2.25	2.33	0.93	4.71	1.05	76.97	1.12	0.96
75 x 45	2.00	1.6	2.55	1.30	3.75	23.86	5.38	3.05	1.45	6.36	1.68	53.04	1.76	0.46
	2.84	2.3	3.62	1.33	3.75	33.08	7.51	3.02	1.44	8.82	2.37	114.32	1.76	0.76
	3.65	3.0	4.75	1.37	3.75	41.58	9.51	2.98	1.42	11.08	3.04	169.91	1.76	0.90
100 x 50	2.44	1.6	3.11	1.32	5.00	50.00	7.88	4.00	1.59	10.00	2.14	66.48	1.87	0.33
	3.47	2.3	4.42	1.35	5.00	69.83	11.05	3.97	1.58	13.96	3.03	165.23	1.87	0.66
	4.47	3.0	5.70	1.38	5.00	88.45	14.05	3.93	1.56	17.69	3.89	255.77	1.87	0.85
125 x 50	3.92	2.3	5.00	1.21	6.25	117.36	11.84	4.84	1.53	18.77	3.12	222.17	1.76	0.61
	5.06	3.0	6.45	1.24	6.25	149.25	15.07	4.80	1.52	23.88	5.01	345.27	1.75	0.81
	7.42	4.5	9.45	1.31	6.25	211.76	31.43	4.73	1.50	33.88	5.81	571.75	1.75	1.00
150 x 65	4.91	2.3	6.26	1.61	7.50	218.16	25.86	5.90	2.03	29.08	5.29	239.92	2.34	0.40
	6.36	3.0	8.10	1.64	7.50	278.84	33.12	5.86	2.02	37.17	6.82	441.26	2.34	0.63
	9.36	4.5	11.93	1.71	7.50	400.15	47.69	5.79	1.99	53.35	9.96	832.81	2.34	0.91
175 x 75	5.73	2.3	7.30	1.83	8.75	346.87	40.19	6.89	2.34	39.64	7.09	240.68	2.69	0.28
	7.42	3.0	9.45	1.86	8.75	444.65	51.59	6.85	2.33	50.81	9.15	515.38	2.69	0.51
	10.95	4.5	13.95	1.93	8.75	642.23	74.74	6.78	2.31	73.39	13.42	1061.21	2.69	0.82
200 x 75	6.18	2.3	7.87	1.70	10.00	473.31	41.76	7.75	2.30	47.33	7.21	287.35	2.48	0.26
	8.01	3.0	10.20	1.73	10.00	607.58	53.63	7.71	2.29	60.75	9.31	616.21	2.58	0.48
	11.84	4.5	15.08	1.80	10.00	880.30	77.78	7.63	2.27	88.03	13.65	1272.76	2.58	0.79
225 x 75	6.63	2.3	8.45	1.59	11.25	624.36	43.12	8.59	2.25	55.49	7.30	336.94	2.49	0.24
	8.59	3.0	10.95	1.63	11.25	802.40	55.40	8.55	2.24	71.32	9.43	723.37	2.48	0.45
	12.72	4.5	16.20	1.69	11.25	1145.51	80.39	8.47	2.22	103.60	13.85	1497.89	2.48	0.76
250 x 75	7.08	2.3	9.02	1.50	12.50	801.81	44.31	9.42	2.21	64.14	7.39	389.44	2.39	0.23
	9.18	3.0	11.70	1.53	12.50	1031.46	56.94	9.38	2.20	82.51	9.54	836.88	2.39	0.43
	13.60	4.5	17.33	1.60	12.50	1501.38	82.67	9.30	2.18	120.11	14.01	1736.58	2.39	0.72

CHANNELS



LIPPED CHANNELS (C-CHANNEL)

JIS G 3350-1987

Section size A x B x C mm	Unit Weight M kg/m	Thick-ness t mm	Area A cm ²	Position of Centroid Cx Cy cm cm		Moment of Inertia Ix Iy cm ⁴ cm ⁴		Radius of gyration Rx Ry cm cm		Modulus of Section Zx Zy cm ³ cm ³		Reduced Modulus Zx1 cm	Moment Capacity Mx kg/m	Shear Centre M cm	Column Factor Q
60 x 30 x 10	1.62	1.6	2.07	1.06	3.00	11.60	2.54	2.36	1.10	3.87	1.31	3.87	65.40	1.54	0.97
	2.25	2.3	2.87	1.05	3.00	15.50	3.28	2.32	1.07	5.18	1.69	5.13	87.50	1.53	1.00
75 x 45 x 15	2.31	1.6	2.95	1.72	3.75	27.10	8.69	3.03	1.71	7.23	3.13	7.23	122.00	2.43	0.94
	3.24	2.3	4.13	1.71	3.75	37.00	11.67	2.99	1.67	9.88	4.19	9.88	166.80	2.42	0.99
	4.12	3.0	5.25	1.71	3.75	45.90	14.16	2.95	1.64	12.24	5.07	12.24	206.50	2.42	1.00
100 x 50 x 20	2.88	1.6	3.67	1.86	5.00	58.30	13.97	3.98	1.95	11.67	4.46	11.66	196.80	2.73	0.85
	4.06	2.3	5.17	1.86	5.00	80.60	18.96	3.94	1.91	16.12	6.04	16.12	272.10	2.72	0.95
	5.18	3.0	6.60	1.85	5.00	100.90	23.30	3.90	1.87	20.18	7.42	20.18	340.60	2.71	0.99
125 x 50 x 20	4.51	2.3	5.74	1.68	6.25	136.40	20.54	4.87	1.89	21.82	6.20	21.82	368.20	2.55	0.89
	5.77	3.0	7.35	1.63	6.25	171.50	25.27	4.82	1.85	27.44	7.62	27.44	463.10	2.54	0.95
	8.31	4.5	10.59	1.67	6.25	237.10	33.21	4.73	1.77	37.94	10.00	37.94	640.20	2.51	1.00
150 x 65 x 20	5.50	2.3	7.01	2.11	7.50	247.70	40.99	5.94	2.41	33.03	9.35	33.03	557.40	3.16	0.84
	7.07	3.0	9.00	2.11	7.50	313.70	50.99	5.90	2.37	41.83	11.62	41.83	705.80	3.15	0.91
	10.25	4.5	13.06	2.10	7.50	440.70	68.79	5.80	2.29	58.77	15.64	58.77	991.70	3.12	0.99
175 x 75 x 20	6.31	2.3	8.04	2.34	8.75	389.30	60.97	6.95	2.75	44.49	11.83	44.31	747.70	3.52	0.80
	8.13	3.0	10.35	2.34	8.75	494.90	76.25	6.91	2.71	56.56	14.78	56.56	954.50	3.50	0.87
	11.84	4.5	15.09	2.33	8.75	701.60	104.12	6.81	2.62	80.18	20.14	80.18	1353.10	3.47	0.96
200 x 75 x 20	6.76	2.3	8.62	2.19	10.00	531.00	63.65	7.84	2.71	53.10	12.00	52.89	892.50	3.38	0.75
	8.71	3.0	11.10	2.19	10.00	676.10	79.62	7.80	2.67	67.61	15.00	67.61	1140.90	3.36	0.82
	12.73	4.5	16.12	2.18	10.00	962.00	108.77	7.70	2.58	96.20	20.46	96.20	1623.30	3.33	0.94
225 x 75 x 20	7.21	2.3	9.19	2.06	11.25	699.60	65.99	8.72	2.67	62.18	12.14	61.95	1045.40	3.26	0.71
	9.30	3.0	11.85	2.06	11.25	892.00	82.55	8.67	2.63	79.29	15.19	79.29	1338.00	3.24	0.78
	13.61	4.5	17.34	2.05	11.25	1273.10	112.81	8.56	2.55	113.16	20.73	113.16	1909.60	3.20	0.89
250 x 75 x 25	7.85	2.3	10.00	2.07	12.50	921.11	74.68	9.59	2.73	73.68	13.77	73.43	1239.14	3.34	0.68
	10.13	3.0	12.90	2.07	12.50	1176.57	93.68	9.54	2.69	94.12	17.27	94.12	1588.31	3.32	0.74
	14.85	4.5	18.91	2.07	12.50	1685.70	128.83	9.43	2.60	134.85	23.74	134.85	2275.61	3.27	0.86

DIN 1026 CHANNELS

Symbol U	Dimensions for										Cross Section F cm ²	Weight G kg/m	Surface Area U m ² /m	Relative to bending axis*2									Distance of axis y - y e _y cm X _M ¹⁵ cm		
	h	perm var.	b	perm var.	s	perm var.	t	perm var.*1	r ₁	r ₂				x - x			y - y			S _x ⁺³ cm ³	s _x ⁺⁴ cm				
														J _x cm ⁴	W _x cm ³	i _x cm	J _y cm ⁴	W _y cm ³	i _y cm						
30 x 15	30	±1,5	15	±1,5	4	±0,5	4,5	-0,5	4,5	2	2,21	1,74	0,103	2,53	1,69	1,07	0,38	0,39	0,42	--	--	0,52	0,74		
30	30		33		5		7		7	3,5	5,44	4,27	0,174	6,39	4,26	1,08	5,33	2,68	0,99	--	--	1,31	2,22		
40 x 20	40		20		5		5,5		5	2,5	3,66	2,87	0,142	7,58	3,79	1,44	1,14	0,86	0,56	--	--	0,67	1,01		
40	40		35		5		7		7	3,5	6,21	4,87	0,199	14,1	7,05	1,50	6,68	3,08	1,04	--	--	1,33	2,32		
50 x 25	50		25		5		6		6	3	4,92	3,86	0,181	16,8	6,73	1,85	2,49	1,48	0,71	--	--	0,81	1,34		
50	50		38		5		7		7	3,5	7,12	5,59	0,232	26,4	10,6	1,92	9,12	3,75	1,13	--	--	1,37	2,47		
60	60		30		6		6		6	3	6,46	5,07	0,215	31,6	10,5	2,21	4,51	2,16	0,84	--	--	0,91	1,50		
65	65		42		±1,5		5,5		7,5	7,5	4	9,03	7,09	0,273	57,5	17,7	2,52	14,1	5,07	1,25	--	--	1,42	2,60	
80	80	±2,0	45	±2,0	6	±0,5	8	-1,0	8	4	11,0	8,64	0,312	106	26,5	3,10	19,4	6,36	1,33	15,9	6,65	1,45	2,67		
100	100		50		6		8,5		8,5	4,5	13,5	10,6	0,372	206	41,2	3,91	29,3	8,49	1,47	24,5	8,42	1,55	2,93		
120	120		55		7		9		9	4,5	17,0	13,4	0,434	364	60,7	4,62	43,2	11,1	1,59	36,3	10,0	1,60	3,03		
140	140		60		7		10		10	5	20,4	16,0	0,489	605	86,4	5,45	62,7	14,8	1,75	51,4	11,8	1,75	3,37		
160	160		65		7,5		10,5		10,5	5,5	24,0	18,8	0,546	925	116	6,21	85,3	18,3	1,89	68,8	13,3	1,84	3,56		
180	180		70		8		11		11	5,5	28,0	22,0	0,611	1350	150	6,95	114	22,4	2,02	89,6	15,1	1,92	3,75		
200	200		75		8,5		11,5		11,5	6	32,2	25,3	0,661	1910	191	7,70	148	27,0	2,14	114	16,8	2,01	3,94		
220	220		80		9		12,5		-1,0	12,5	6,5	37,4	29,4	0,718	2690	245	8,48	197	33,6	2,30	146	18,5	2,14	4,20	
240	240	±3,0	85	±2,0	9,5	±0,7	13	-1,5	13	6,5	42,3	33,2	0,775	3600	300	9,22	248	39,6	2,42	179	20,1	2,23	4,39		
260	260		90		10		14		14	7	48,3	37,9	0,834	4820	371	9,99	317	47,7	2,56	221	21,8	2,36	4,66		
280	280		95		10		15		15	7,5	53,3	41,8	0,890	6280	448	10,9	399	57,2	2,74	266	23,6	2,53	5,02		
300	300		100		10		16		16	8	58,8	46,2	0,950	8030	535	11,7	495	67,8	2,90	316	25,4	2,70	5,41		
320	320		100		14		17,5		17,5	8,75	75,8	59,5	0,982	10870	679	12,1	597	80,6	2,81	413	26,3	2,60	4,82		
350	350		100		14		±0,7		16	-1,5	16	8	77,3	60,6	1,05	12840	734	12,9	570	75,0	2,72	459	28,6	2,40	4,45
380	380		102		±2,5		13,5		16	16	8	80,4	63,1	1,11	15760	829	14,0	615	78,7	2,77	507	31,1	2,38	4,58	
400	400		110		14		18		18	9	91,5	71,8	1,18	20350	1020	14,9	846	102	3,04	618	32,9	2,65	5,11		

*1 The permissible plus variation is limited by the permissible exceeding of weight.

*2 J = moment of inertia; W = section modulus; i = radius of gyration, referenced in each case to the bending axis concerned.

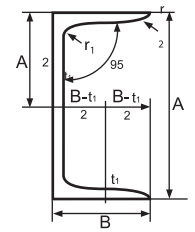
*3 S_x = static moment of half cross-section

*4 s_x = J_x : S_x = distance between compression and tension centres

*5 x_M = distance of shear centre M from the y - y axis

The cross-sections, weights, surface areas and static values have been calculated from the dimensions detailed in the Table.

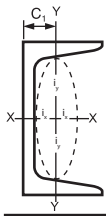
CHANNELS



U-CHANNELS (Inside slope 5°)

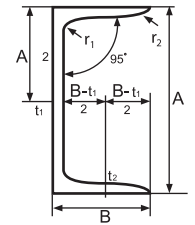
Section size A x B	Unit weight M		Section depth A		Flange width B		Flange thickness t1		Web thickness t2		Corner radius			
											r1		r2	
mm (in)	kg/m	lb/ft	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in
127 x 64 (5 x 2 1/2)	14.90	10.0	127.0	5.0	63.5	2.5	9.2	0.362	6.4	0.252	10.7	0.421	2.4	0.094
152 x 76 (6 x 3)	17.88	12.0	152.4	6.0	76.2	3.0	9.0	0.354	6.4	0.252	12.2	0.480	2.4	0.094
152 x 89 (6 x 3 1/2)	23.84	16.0	152.4	6.0	88.9	3.5	11.6	0.457	7.1	0.280	13.7	0.539	3.2	0.126
178 x 76 (7 x 3)	20.84	14.0	177.8	7.0	76.2	3.0	10.3	0.405	6.6	0.260	12.2	0.480	3.2	0.126
178 x 89 (7 x 3 1/2)	26.81	18.0	177.8	7.0	88.9	3.5	12.3	0.484	7.6	0.299	13.7	0.539	3.2	0.126
203 x 76 (8 x 3)	23.82	16.0	203.2	8.0	76.2	3.0	11.2	0.441	7.1	0.280	12.2	0.480	3.2	0.126
203 x 89 (8 x 3 1/2)	29.78	20.0	203.2	8.0	88.9	3.5	12.9	0.508	8.1	0.319	13.7	0.539	3.2	0.126
229 x 76 (9 x 3)	26.06	17.5	228.6	9.0	76.2	3.0	11.2	0.441	7.6	0.299	12.2	0.480	3.2	0.126
229 x 89 (9 x 3 1/2)	32.76	22.0	228.6	9.0	88.9	3.5	13.3	0.524	8.6	0.339	13.7	0.539	3.2	0.126
254 x 76 (10 x 3)	28.29	19.0	254.0	10.0	76.2	3.0	10.9	0.429	8.1	0.319	12.2	0.480	3.2	0.126
254 x 89 (10 x 3 1/2)	35.74	24.0	254.0	10.0	88.9	3.5	13.6	0.535	9.1	0.358	13.7	0.539	3.2	0.126
305 x 89 (12 x 3 1/2)	41.69	28.0	304.8	12.0	88.9	3.5	13.7	0.539	10.2	0.402	13.7	0.539	3.2	0.126
305 x 102 (12 x 4)	46.18	31.0	304.8	12.0	101.6	4.0	14.8	0.583	10.2	0.402	15.2	0.598	4.8	0.189
381 x 102 (15 x 4)	55.10	37.0	381.0	15.0	101.6	4.0	16.3	0.642	10.4	0.409	15.2	0.598	4.8	0.189
432 x 102 (17 x 4)	65.54	44.0	431.8	17.0	101.6	4.0	16.8	0.661	12.2	0.480	15.2	0.598	4.8	0.189

CHANNELS



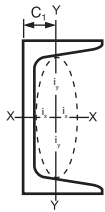
Section size	Section area		Centre of gravity		Moment of inertia				Radius of gyration				Modulus of section			
	A		y		Ix		Iy		ix		iy		Zx		Zy	
mm (in)	cm ²	in ²	cm	in	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm ³	in ³	cm ³	in ³
127 x 64 (5 x 2 1/2)	18.98	2.94	1.94	0.764	482.5	11.59	67.23	1.62	5.04	1.984	1.88	0.74	75.99	4.637	15.25	0.931
152 x 76 (6 x 3)	22.77	3.53	2.21	0.87	851.5	20.46	113.8	2.73	6.12	2.409	2.24	0.882	111.8	6.822	21.05	1.285
152 x 89 (6 x 3 1/2)	30.36	4.71	2.86	1.126	1166	28.01	215.1	5.17	6.20	2.441	2.66	1.047	153.0	9.337	35.70	2.178
178 x 76 (7 x 3)	26.54	4.11	2.20	0.866	1337	32.12	134.0	3.22	7.10	2.795	2.25	0.886	150.4	9.178	24.72	1.509
178 x 89 (7 x 3 1/2)	34.15	5.29	2.76	1.087	1753	42.12	241.0	5.79	7.16	2.819	2.66	1.047	197.2	12.034	39.29	2.398
203 x 76 (8 x 3)	30.34	4.70	2.13	0.839	1950	46.85	151.3	3.63	8.02	3.157	2.23	0.878	192.0	11.717	27.59	1.684
203 x 89 (8 x 3 1/2)	37.94	5.88	2.65	1.043	2491	59.85	264.4	6.35	8.10	3.189	2.64	1.039	245.2	14.963	42.34	2.584
229 x 76 (9 x 3)	33.20	5.15	2.00	0.787	2610	62.70	158.7	3.81	8.87	3.492	2.19	0.862	228.3	13.932	28.22	1.722
229 x 89 (9 x 3 1/2)	41.73	6.47	2.53	0.996	3387	81.37	285.0	6.85	9.01	3.547	2.61	1.028	296.4	18.087	44.82	2.735
254 x 76 (10 x 3)	36.03	5.58	1.86	0.732	3367	80.89	162.6	3.91	9.67	3.807	2.12	0.835	265.1	16.177	28.21	1.721
254 x 89 (10 x 3 1/2)	45.52	7.06	2.42	0.953	4448	106.86	302.4	7.27	9.88	3.89	2.58	1.016	350.2	21.370	46.70	2.850
305 x 89 (12 x 3 1/2)	53.11	8.23	2.18	0.858	7061	169.64	325.4	7.82	11.5	4.528	2.48	0.976	463.3	28.272	48.49	2.959
305 x 102 (12 x 4)	58.83	9.12	2.66	1.047	8214	197.34	499.5	12.00	11.8	4.646	2.91	1.146	539.0	32.892	66.59	4.064
381 x 102 (15 x 4)	70.19	10.88	2.52	0.992	14894	357.82	579.7	13.93	14.6	5.748	2.87	1.130	781.8	47.708	75.86	4.629
432 x 102 (17 x 4)	83.49	12.94	2.32	0.913	21399	514.11	628.6	15.10	16.0	6.299	2.74	1.079	991.1	60.480	80.14	4.890

CHANNELS



U-CHANNELS (Inside slope 5°)

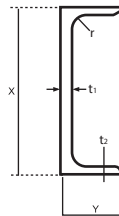
Section size A x B x t1	Unit weight M		Section depth A		Flange width B		Web thickness t1		Flange thickness t2		Corner radius			
	kg/m	lb/ft	mm	in	mm	in	mm	in	mm	in	r1		r2	
75 x 40 x 5	6.92	4.65	75	2.953	40	1.575	5.0	0.197	7.0	0.276	8	0.315	4.0	0.157
100 x 50 x 5	9.36	6.29	100	3.937	50	1.969	5.0	0.197	7.5	0.295	8	0.315	4.0	0.157
125 x 65 x 6	13.4	9.004	125	4.921	65	2.559	6.0	0.236	8.0	0.315	8	0.315	4.0	0.157
150 x 75 x 6.5 18.6	12.50	150	5.906	75	2.953	6.5	0.256	10.0	0.394	10	0.394	5.0	0.197	
150 x 75 x 9	24.0	16.13	150	5.906	75	2.953	9.0	0.354	12.5	0.492	15	0.591	7.5	0.295
180 x 75 x 7	21.4	14.35	180	7.087	75	2.953	7.0	0.276	10.5	0.413	11	0.433	5.5	0.217
180 x 90 x 7.5 27.1	18.21	180	7.087	90	3.543	7.5	0.295	12.5	0.492	13	0.512	6.5	0.256	
200 x 80 x 7.5 24.6	16.53	200	7.874	80	3.15	7.5	0.295	11.0	0.433	12	0.472	6.0	0.236	
200 x 90 x 8	30.3	20.40	200	7.874	90	3.543	8.0	0.315	13.5	0.531	14	0.551	7.0	0.276
230 x 80 x 8	28.4	19.08	230	9.055	80	3.15	8.0	0.315	12.0	0.472	13	0.512	6.5	0.256
230 x 90 x 8.5 33.1	22.24	230	9.055	90	3.543	8.5	0.335	13.5	0.531	15	0.591	7.5	0.295	
250 x 80 x 8	30.2	20.29	250	9.843	80	3.15	8.0	0.315	12.5	0.492	14	0.551	7.0	0.276
250 x 90 x 9	34.6	23.15	250	9.843	90	3.543	9.0	0.354	13.0	0.512	14	0.551	7.0	0.276
250 x 90 x 11 40.2	26.90	250	9.843	90	3.543	11.0	0.433	14.5	0.571	17	0.669	8.5	0.335	
280 x 100 x 9 38.8	26.07	280	11.024	100	3.937	9.0	0.354	13.0	0.512	14	0.551	7.0	0.276	
280 x 100 x 11.5	48.2	32.39	280	11.024	100	3.937	11.5	0.453	16.0	0.630	18	0.709	9.0	0.354
300 x 90 x 9	38.1	25.57	300	11.811	90	3.543	9.0	0.354	12.0	0.512	14	0.551	7.0	0.276
300 x 90 x 10 43.8	29.32	300	11.811	90	3.543	10.0	0.394	15.5	0.611	19	0.748	9.5	0.374	
380 x 100 x 10.5	54.5	36.60	380	14.961	100	3.937	10.5	0.413	16.0	0.630	18	0.709	9.0	0.354
380 x 100 x 13	62.0	41.66	380	14.961	100	3.937	13.0	0.512	16.5	0.650	18	0.709	9.0	0.354



CHANNELS

Section Size A x B x t1	Section area A		Centre of gravity Cy		Moment of inertia				Radius of gyration				Modulus of section			
	cm ²	in ²	cm	in	Ix		Iy		ix		iy		Zx		Zy	
	cm ²	in ²	cm	in	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm ³	in ³	cm ³	in ³
75 x 40 x 5	8.818	1.367	1.27	0.500	75.9	1.823	12.4	0.298	2.93	1.154	1.19	0.469	20.2	1.233	4.54	0.277
100 x 50 x 5	11.92	1.848	1.55	0.610	189	4.541	26.9	0.646	3.98	1.567	1.50	0.591	37.8	2.307	7.82	0.477
125 x 65 x 6	17.11	2.652	1.94	0.764	425	10.21	65.5	1.571	4.99	1.965	1.96	0.772	68.0	4.150	14.4	0.879
150 x 75 x 6.5	23.71	3.675	2.31	0.909	864	20.76	122	2.931	6.04	2.378	2.27	0.894	115	7.018	23.6	1.440
150 x 75 x 9	30.49	4.741	2.31	0.009	1060	25.47	151	3.628	5.87	2.311	2.22	0.874	141	8.604	29.1	1.776
180 x 75 x 7	27.20	4.316	2.15	0.846	1380	33.15	137	3.291	7.13	2.807	2.24	0.882	154	9.398	25.5	1.556
180 x 90 x 7.5	34.57	5.358	2.85	0.858	1840	44.21	258	6.198	7.29	2.870	2.73	1.075	204	12.45	42.0	2.563
200 x 80 x 7.5	31.33	4.856	2.24	0.882	1950	46.85	177	4.252	7.89	3.106	2.38	0.937	195	11.90	30.8	1.88
200 x 90 x 8	38.65	5.991	2.77	1.091	2490	59.82	286	6.871	8.03	3.161	2.72	1.071	249	15.19	45.9	2.801
230 x 80 x 8	36.12	5.499	2.15	0.846	2900	69.67	200	4.805	8.96	3.528	2.35	0.925	252	15.38	34.2	2.087
230 x 90 x 8.5	42.14	6.531	2.58	1.016	3490	83.85	303	7.280	9.10	3.583	2.68	1.055	304	18.55	47.3	2.886
250 x 80 x 8	38.51	5.969	2.11	0.831	3630	87.21	210	5.045	9.71	3.823	2.33	0.921	291	17.76	35.7	2.179
250 x 90 x 9	44.07	6.831	2.42	0.953	4180	100.4	306	7.352	9.74	3.835	2.64	1.039	335	20.44	46.5	2.838
250 x 90 x 11	51.17	7.931	2.39	0.941	4690	112.7	342	8.217	9.47	3.768	2.58	1.016	375	22.88	51.7	3.155
280 x 100 x 9	49.37	7.652	2.64	1.040	5930	142.5	428	10.28	11.0	4.331	2.95	1.161	423	25.81	58.2	3.552
280 x 100 x 11.5	61.37	9.512	2.68	1.055	7150	171.8	515	12.37	10.8	4.252	2.90	1.142	510	31.12	70.4	4.296
300 x 90 x 9	48.57	7.531	2.23	0.878	6440	154.7	325	7.808	11.5	4.528	2.59	1.020	429	26.18	48.0	2.929
300 x 90 x 10	55.74	8.64	2.33	0.917	7400	177.8	373	8.961	11.5	4.528	2.59	1.020	494	30.15	56.0	3.417
380 x 100 x 10.5	69.38	10.76	2.41	0.949	14500	348.4	557	13.38	14.5	5.709	2.83	1.114	762	46.50	73.3	4.473
380 x 100 x 13	78.96	12.54	2.29	0.902	15600	374.8	584	14.03	14.1	5.551	2.72	1.071	822	50.16	75.8	4.625

CHANNELS



INCH SERIES

Designation	Depth	Flange	Thickness		Radius of Fillet	Sectional Area	Unit Weight			Weight per		
			Web	Flange						20 ft	30 ft	40 ft
	H	B	t1	t2	r					kg	kg	kg
	in.	in.	in.	in.	in.	in ²	kg/m	kg/ft	lb/ft			
C3 x 4.1	3	1.41	0.17	0.273	0.27	1.21	6.10	1.86	4.1	37.2	55.8	74.4
C4 x 5.4	4	1.584	0.184	0.296	0.28	1.59	8.04	2.45	5.4	49.0	73.5	98
C5 x 6.7	5	1.75	0.19	0.32	0.29	1.97	9.97	3.04	6.7	60.8	91.2	122
C6 x 8.2	6	1.92	0.20	0.343	0.30	2.40	12.2	3.72	8.2	74.4	112	149
C8 x 11.5	8	2.26	0.22	0.39	0.32	3.30	17.1	5.22	11.5	104	157	209
C10 x 15.3	10	2.60	0.24	0.436	0.34	4.49	22.8	6.94	15.3	139	208	278
C12 x 20.7	12	2.942	0.282	0.501	0.38	6.09	30.8	9.39	20.7	188	282	376



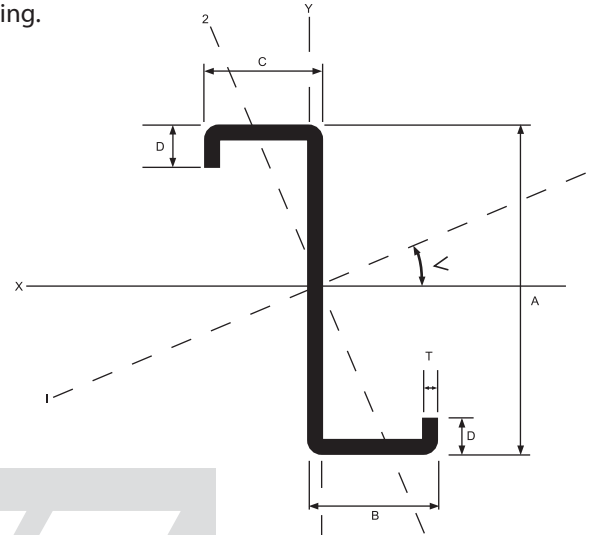
HIGH-TENSILE GALVANISED Z-PURLINS

High-Tensile Galvanised Z-Purlins are roll formed from high tensile Zinc coated steel with base steel Thickness of 1.60 mm, 2.00 mm and 2.50 mm.

They are suitable for roofing and wall cladding supports and for structural frames of buildings. Due to the light weight and high strength of the steel and the Zinc-coated surface, High-Tensile Galvanised Z-Purlins are versatile and economical and require minimal maintenance throughout the life span of the building.

MATERIAL SPECIFICATION

Base Steel Thickness	:	1.60 mm, 2.00 mm and 2.50 mm
Steel Grade	:	ASTM A446 Grade E Modified [AS 1397-G450]
Yield Stress	:	450 Mpa (minimum)
Zinc Coating	:	275g/m ² minimum coating mass
Mechanical Property	:	Y/P Min. 450 Mpa T/S Min. 510 Mpa E/LMin. 10%
Tolerances	:	Depth : 1 mm Flange width : 2 mm Length : 3 mm Hole Centres : 1.5mm



Dimensions											
Section Identification	Section Dimensions					Section Identification	Section Dimensions				
	A	B	C	D	t		A	B	C	D	t
	mm	mm	mm	mm	mm		mm	mm	mm	mm	mm
BSZ 1610	102	57	51	16	1.6	BSZ 1615	153	74	67.5	16	1.6
BSZ 2010	102	57	51	16	2.0	BSZ 2015	153	74	67.5	16	2.0
BSZ 2510	102	57	51	16	2.5	BSZ 2515	153	74	67.5	16	2.5
BSZ 1612	127	57	51	16	1.6	BSZ 1620	203	74	67.5	16	1.6
BSZ 2012	127	57	51	16	2.0	BSZ 2020	203	74	67.5	16	2.0
BSZ 2512	127	57	51	16	2.5	BSZ 2520	203	74	67.5	16	2.5

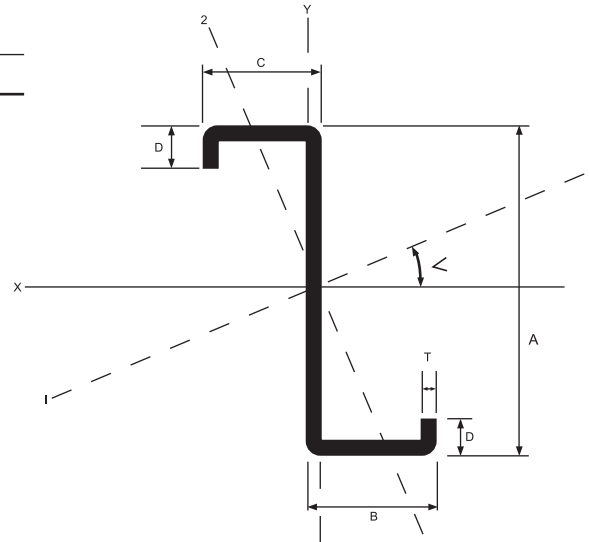
Properties

Section Identification	Mass	Area	XX-AXIS			YY-AXIS			11 -AXIS	22-AXIS	ANGLE
			IX	ZX	RX	IY	ZY	RY	I1	I2	
	kg/m	mm ²	10 ⁶ mm ⁴	10 ³ mm ³	mm	10 ⁶ mm ⁴	10 ³ mm ³	mm	10 ⁶ mm ⁴	10 ⁶ mm ⁴	
BSZ 1610	3.05	373	0.62	12.16	40.6	0.250	4.94	25.8	0.70	0.160	24.5
BSZ 2010	3.72	442	0.77	15.14	41.8	0.310	6.22	26.5	0.98	0.190	24.4
BSZ 2510	4.60	534	0.96	18.84	42.4	0.380	7.60	26.6	1.11	0.230	24.3
BSZ 1612	3.20	408	1.03	16.18	50.1	0.261	5.20	25.3	1.04	0.243	22.9
BSZ 2012	3.94	510	1.28	20.16	51.1	0.380	7.60	27.3	1.30	0.351	22.7
BSZ 2512	4.89	638	1.60	25.20	52.2	0.470	9.45	27.5	1.63	0.434	22.6
BSZ 1615	4.01	512	2.01	26.30	62.7	0.646	9.69	35.5	2.32	0.335	23.3
BSZ 2015	4.92	598	2.50	32.70	65.4	0.811	12.2	37.2	2.86	0.451	22.7
BSZ 2515	6.01	731	3.12	40.70	66.2	1.031	15.6	38.1	3.53	0.619	22.1
BSZ 1620	4.70	572	3.86	38.10	82.6	0.635	9.52	33.5	4.15	0.350	15.9
BSZ 2020	5.74	698	4.85	47.80	84.1	0.811	12.2	34.4	5.16	0.501	15.0
BSZ 2520	7.03	855	6.05	59.60	85.1	1.021	15.4	34.9	6.40	0.675	14.2

Z-PURLINS

HIGH-TENSILE GALVANISED Z-PURLINS (OTHER AVAILABLE SIZES)

Section Identification	Section Dimensions					Area mm ²	Mass Per Unit length kg/m
	A	B	C	D	t		
	mm	mm	mm	mm	mm		
SZ 100-16	102	53	49	14	1.6	344	2.78
SZ 100-20	102	53	48	14	2.0	430	3.45
SZ 150-16	152	65	61	16	1.6	472	3.84
SZ 150-20	152	65	61	16	2.0	590	4.77
SZ 150-25	152	66	60	16	2.5	738	5.93
SZ 200-16	203	79	74	20	1.6	592	4.82
SZ 200-20	203	79	74	20	2.0	750	6.00
SZ 200-25	203	79	73	20	2.5	938	7.46
SZ 250-20	250	78	72	20	2.0	840	6.76
SZ 250-25	250	78	72	20	2.5	1050	8.24



Section Identification	Section Dimensions					Area mm ²	Mass/ unit length kg/m	Second Moment of Area		Section Modulus		Radius of Gyration		Form Factor Q
	A	B	C	D	t			IX	IY	ZX	ZY	RX	RY	
	mm	mm	mm	mm	mm			10 ⁶ mm ⁴	10 ⁶ mm ⁴	10 ³ mm ³	10 ³ mm ³	mm	mm	
SZ 100-16	102	53	49	14	1.6	344	2.78	0.570	0.210	11.18	4.02	40.7	24.7	0.840
SZ 100-20	102	53	48	14	2.0	430	3.45	0.702	0.262	13.76	5.03	40.4	24.7	0.891
SZ 150-16	152	65	61	16	1.6	472	3.84	1.701	0.409	22.39	6.37	60.0	29.4	0.703
SZ 150-20	152	65	61	16	2.0	590	4.77	2.111	0.513	27.77	8.02	59.8	29.5	0.771
SZ 150-25	152	66	60	16	2.5	738	5.93	2.608	0.648	34.31	10.00	59.5	29.6	0.823
SZ 200-16	203	79	74	20	1.6	592	4.82	3.756	0.663	37.00	8.47	79.7	33.5	0.578
SZ 200-20	203	79	74	20	2.0	750	6.00	4.742	0.888	46.72	11.38	79.5	34.4	0.667
SZ 200-25	203	79	73	20	2.5	938	7.46	5.872	1.114	57.85	14.33	79.1	34.5	0.737
SZ 250-20	250	78	72	20	2.0	840	6.76	7.665	0.852	61.32	11.07	95.5	31.9	0.599
SZ 250-25	250	78	72	20	2.5	1050	8.24	9.522	1.075	76.18	14.01	95.2	32.0	0.658

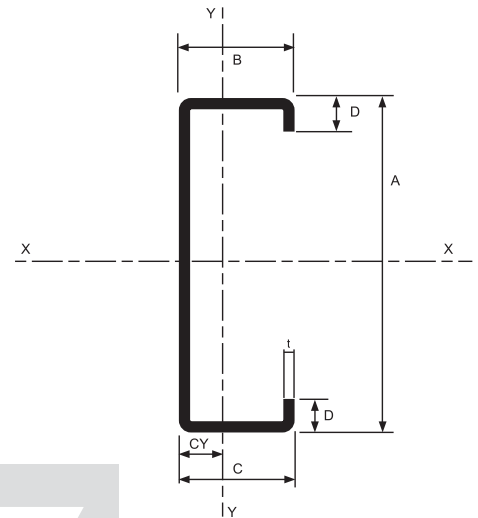
HIGH-TENSILE GALVANISED C-PURLINS

High-Tensile Galvanised C-Purlins (Sections) are roll formed from high-tensile Zinc coated steel with base steel Thickness of 1.60 mm, 2.00 mm and 2.50 mm.

They are suitable for roofing and wall cladding supports and for structural frames of buildings. Due to the light weight and high strength of the steel and the Zinc-coated surface. High-Tensile Galvanised C-Purlins are versatile and economical and require minimal maintenance throughout the life span of the building.

MATERIAL SPECIFICATION

Base Steel Thickness	:	1.60 mm, 2.00 mm and 2.50 mm
Steel Grade	:	ASTM A446 Grade E Modified [AS 1397-G450]
Yield Stress	:	450 Mpa (minimum)
Zinc Coating	:	275g/m ² minimum coating mass
Mechanical Property	:	Y/P Min. 450 Mpa T/S Min. 510 Mpa E/L Min. 10%
Tolerances	:	Depth : ± 1 mm Flange width : ± 2 mm Length : ± 3 mm Hole Centres : ± 1.5mm



Dimensions											
Section Identification	Section Dimensions					Section Identification	Section Dimensions				
	A	B	C	D	t		A	B	C	D	t
	mm	mm	mm	mm	mm		mm	mm	mm	mm	mm
BSC 1610	102	51	51	16	1.6	BSC 1615	153	71	71	16	1.6
BSC 2010	102	51	51	16	2.0	BSC 2015	153	71	71	16	2.0
BSC 2510	102	51	51	16	2.5	BSC 2515	153	71	71	16	2.5
BSC 1612	127	51	51	16	1.6	BSC 1620	203	71	71	16	1.6
BSC 2012	127	51	51	16	2.0	BSC 2020	203	71	71	16	2.0
BSC 2512	127	51	51	16	2.5	BSC 2520	203	71	71	16	2.5

Properties

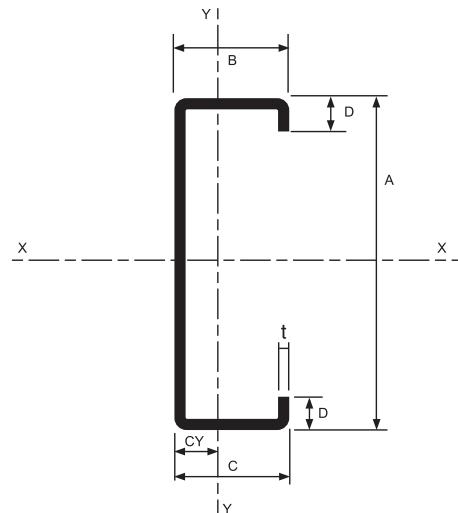
Section Identification	Mass	Area	XX-AXIS			YY-AXIS			CY
			IX	ZX	RX	IY	ZY	RY	
	kg/m	mm ²	10 ⁶ mm ⁴	10 ³ mm ³	mm	10 ⁶ mm ⁴	10 ³ mm ³	mm	
BSC 1610	3.05	373	0.60	11.77	40.6	0.14	4.11	19.5	18.24
BSC 2010	3.72	442	0.76	14.84	41.9	0.18	5.46	20.3	18.60
BSC 2510	4.60	534	0.95	18.56	42.7	0.22	7.01	20.9	19.05
BSC 1612	3.20	408	1.00	15.72	50.0	0.15	4.29	19.4	11.08
BSC 2012	3.94	510	1.25	19.67	50.1	0.19	5.49	19.5	16.42
BSC 2512	4.89	638	1.56	24.56	50.1	0.24	7.13	19.6	17.35
BSC 1615	4.01	512	1.99	26.30	62.3	0.34	6.90	25.8	21.69
BSC 2015	4.92	598	2.48	32.42	64.4	0.41	8.61	26.2	23.36
BSC 2515	6.01	731	3.11	40.65	65.2	0.50	10.24	26.3	24.11
BSC 1620	4.70	572	3.80	37.44	80.8	0.37	7.09	25.1	19.18
BSC 2020	5.74	698	4.75	46.80	82.5	0.45	9.39	25.4	20.16
BSC 2520	7.03	855	5.94	58.52	83.4	0.61	12.23	26.8	20.80

C-PURLINS

HIGH-TENSILE GALVANISED C-PURLINS (OTHER AVAILABLE SIZES)

The following purlin sizes provide an easy reference.

Section Identification	Dimensions					Mass kg/m
	A	B	C	D	t	
	mm	mm	mm	mm	mm	
SC100-16	100	50	50	14	1.6	2.78
SC100-20	100	50	50	14	2.0	3.45
SC150-16	150	65	65	16	1.6	3.84
SC150-20	150	65	65	16	2.0	4.77
SC150-25	150	65	65	16	2.5	5.93
SC200-16	200	75	75	20	1.6	4.82
SC200-20	200	75	75	20	2.0	6.00
SC200-25	200	75	75	20	2.5	7.46



DESIGN PRINCIPLE FOR PURLIN SELECTION TABLES

The purlin selection tables are derived basing on BS 5950 : Part 5 : 1987 "Code of Practice for design of cold formed sections".

- Purlin is simply supported with pinned joints over supports.
- Roof Pitch
The purlin selection tables are only applicable for roof pitch less than or equal to 30°.
- Loading
Besides dead load arising from roof self weight, the purlins are designed based on a live load of 0.25KN/m² (Malaysian Building Bylaw Clause 63) and wind load (uplift) of 0.50KN/m² respectively.
 - 1.4 D.L. + 1.6 L.L. (The most severe inward loading combination)
 - 1.0 D.L. + 1.4 W.L. (The most severe outward loading combination)
- Deflection
Under serviceability limit state the limiting deflection under total load is L/150 whereas under live load only is L/180 (where L is the purlin span).
- Design Strength P_y
The design strength P_y for purlins is to be taken as the minimum yield strength of 450N/mm²

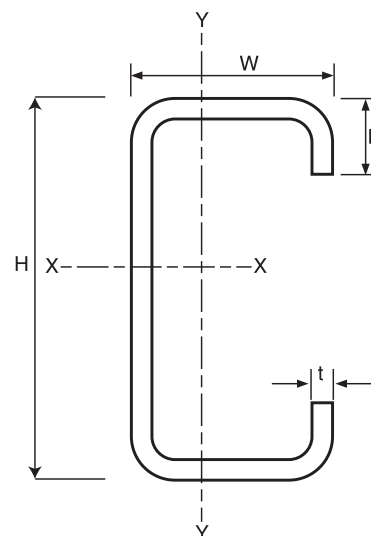


TABLE NO. 1 - PURLIN DIMENSIONS AND PROPERTIES

Section Identification	Dimensions				Mass per unit kg/m	Area mm ²	Effective section modulus		Full section modulus		Second moment of Area		Radius of Gyration	
	H	W	L	t			Zey	Zxx	Zyy	Ixx	lyy	rxx	ryy	
	mm	mm	mm	mm			10 ³ mm ³	10 ³ mm ³	10 ³ mm ³	10 ³ mm ³	10 ⁶ mm ⁴	10 ⁶ mm ⁴	mm	mm
SC100-16	100	50	14	1.6	2.78	355	11.36	3.60	11.60	3.82	0.579	0.125	40.4	18.8
SC100-20	100	50	14	2.0	3.45	440	14.22	4.52	14.30	4.65	0.713	0.153	40.3	18.6
SC150-16	150	65	16	1.6	3.84	489	21.78	4.51	23.60	6.31	1.770	0.282	60.2	24.0
SC150-20	150	65	16	2.0	4.77	608	28.44	5.67	29.20	7.74	2.187	0.346	60.0	23.9
SC150-25	150	65	16	2.5	5.93	755	35.80	7.08	35.90	9.44	2.696	0.422	59.8	23.6
SC200-16	200	75	20	1.6	4.82	614	34.11	6.28	38.50	8.94	3.848	0.472	79.2	27.7
SC200-20	200	75	20	2.0	6.00	764	44.89	7.92	47.70	11.00	4.768	0.580	79.0	27.6
SC200-25	200	75	20	2.5	7.46	950	58.07	9.93	58.90	13.50	5.893	0.710	78.8	27.3

Note : Effective section modulus are calculated based on the effective cross section which taken into consideration of local buckling effect (Section Four BS5950 Part 5 : 1987).

HIGH-TENSILE GALVANISED C-PURLINS

PURLIN SELECTION TABLES FOR ROOFING APPLICATION

The following purlin selection tables provide an easy reference for selection of appropriate purlin size in accordance with the bay width or column spacing of the building.

These tables can be generally used in Singapore and Malaysia. However, for optimum designs where specific roofing and insulation loads are to be used or for any other structure or for application outside Singapore and Malaysia, contact CHYE HIN HARDWARE PTE LTD for free technical advice.

TABLE NO. 2 - PURLIN SELECTION TABLE

Roof decking (0.61 mm TCT at 6.55kg/m² distributed load) on C-Purlin

Span mm	Purlin spacing 1200 mm c/c	Purlin spacing 1500 mm c/c	Purlin spacing 1800 mm c/c	Purlin spacing 2100 mm c/c
5000	SC100-16	SC100-20	SC150-16	SC150-16
6000	SC150-16	SC150-16	SC150-16	SC150-20
7000	SC150-20	SC200-16	SC200-16	SC200-16
7500	SC200-16	SC200-16	SC200-20	SC200-16
8000	SC200-16	SC200-16	SC200-16	SC200-16
9000	SC200-16	SC200-16	SC200-20	SC200-25
10000	SC200-20	SC200-25	--	--
11000	SC200-25	--	--	--

TABLE NO. 3 - PURLIN SELECTION TABLE

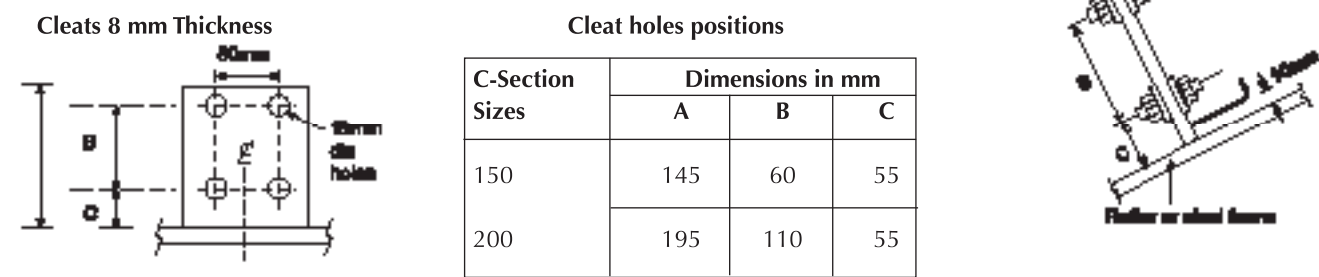
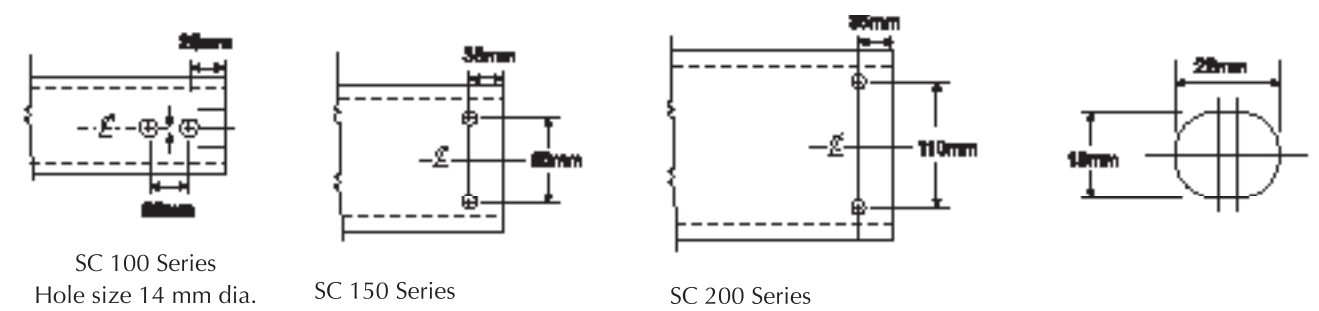
Roof decking (0.61 mm TCT at 6.55kg/m² distributed load) on
50mm thk insulation wool (32 kg/m³) on
1 layer Aluminium Foil on
1 layer Chicken mesh on C- Purlin

Span mm	Purlin spacing 1200 mm c/c	Purlin spacing 1500 mm c/c	Purlin spacing 1800 mm c/c	Purlin spacing 2100 mm c/c
5000	SC100-20	SC100-20	SC150-16	SC150-16
6000	SC150-16	SC150-16	SC150-16	SC150-20
7000	SC150-20	SC200-16	SC200-16	SC200-16
7500	SC200-16	SC200-16	SC200-20	SC200-16
8000	SC200-16	SC200-16	SC200-16	SC200-20
9000	SC200-16	SC200-20	SC200-25	SC200-25
10000	SC200-20	SC200-25	--	--

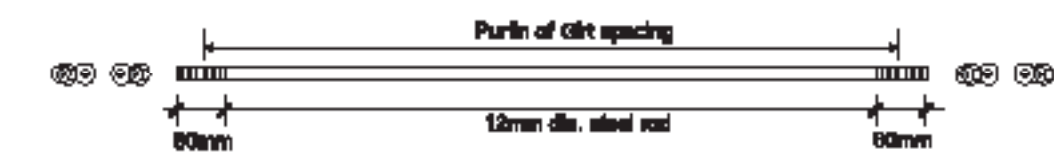
Note: All purlins shall have one bridging at midspan except for purlins in shaded area where 2 bridging are provided at L/3 and 2L/3 between supports.

C-PURLINS

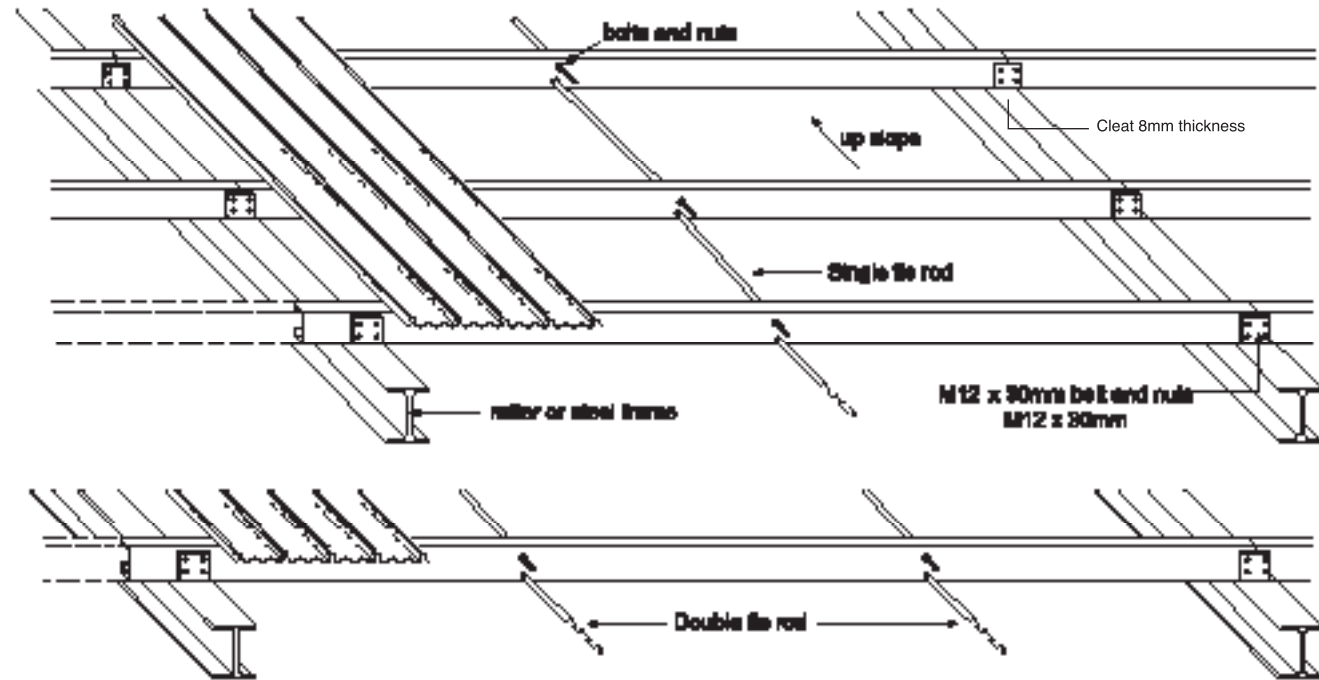
HOLING, CLEAT AND TIE ROD FOR C-PURLIN AND C-GIRTS



DETAIL OF STANDARD TIE ROD



TYPICAL ARRANGEMENT OF SIMPLE SPAN C-PURLINS



ANGLES

EQUAL ANGLES

142

UNEQUAL ANGLES

144

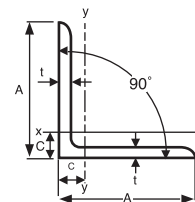
INVERTED ANGLES

150



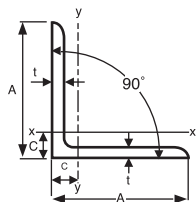
YICK HOE

ANGLES



EQUAL

Section size A x B mm	Thickness t mm	Unit weight M kg/m	Reduce of fillet r mm	Section area A cm ²	Centre of gravity Cx = Cy cm	Axis			
						Ix = Iy cm ⁴	ix = iy cm	iv cm	Zx = Zy cm ³
20 x 20	3.0	0.885	4.0	1.127	0.60	0.61	0.59	0.38	0.28
25 x 25	3.0	1.12	4.0	1.427	0.719	0.797	0.747	0.483	0.448
	5.0	1.76	4.0	2.246	0.79	1.18	0.72	0.48	0.69
30 x 30	3.0	1.36	4.0	1.727	0.844	1.42	0.908	0.585	0.661
	5.0	2.16	4.0	2.746	0.92	2.14	0.88	0.57	1.03
40 x 40	3.0	1.83	4.5	2.336	1.09	3.53	1.23	0.79	1.21
	4.0	2.42	6.0	3.079	1.12	4.48	1.21	0.78	1.56
	5.0	2.95	4.5	3.755	1.17	5.42	1.20	0.774	1.91
	6.0	3.52	6.0	4.48	1.20	6.33	1.19	0.77	2.26
45 x 45	4.0	2.74	6.5	3.492	1.24	6.50	1.36	0.880	2.00
	5.0	3.38	6.5	4.302	1.28	7.91	1.36	0.874	2.46
	6.0	3.96	6.5	5.044	1.32	9.00	1.34	0.86	2.83
50 x 50	3.0	2.33	7.0	2.96	1.31	6.86	1.52	0.99	1.86
	4.0	3.06	6.5	3.892	1.37	9.06	1.53	0.983	2.49
	5.0	3.77	6.5	4.802	1.41	11.1	1.52	0.976	3.08
	6.0	4.43	6.5	5.644	1.44	12.6	1.50	0.963	3.55
	8.0	5.78	6.5	7.364	1.52	16.1	1.48	0.96	4.62
60 x 60	5.0	4.55	6.5	5.802	1.66	19.6	1.84	1.18	4.52
	6.0	5.42	6.5	6.91	1.69	22.8	1.82	1.17	5.29
	7.0	6.21	6.5	7.914	1.73	25.9	1.81	1.16	6.06
65 x 65	5.0	5.00	8.5	6.367	1.77	25.3	1.99	1.28	5.35
	6.0	5.91	8.5	7.527	1.81	29.4	1.98	1.27	6.26
	8.0	7.66	8.5	9.761	1.88	36.8	1.94	1.25	7.96
	10.0	9.42	8.5	12.0	1.96	44.4	1.92	1.25	9.79
70 x 70	6.0	6.38	8.5	8.127	1.93	37.1	2.14	1.37	7.33
	7.0	7.38	8.5	9.40	1.97	42.4	2.12	1.37	8.43
	8.0	8.29	8.5	10.56	2.01	46.6	2.10	1.35	9.34
	10.0	10.20	8.5	13.0	2.09	56.4	2.08	1.35	11.5
75 x 75	6.0	6.85	8.5	8.727	2.06	46.1	2.30	1.48	8.47
	8.0	9.03	10.0	11.5	2.13	58.9	2.26	1.46	11.0
	9.0	9.96	8.5	12.69	2.17	64.4	2.25	1.45	12.1
	12.0	13.00	8.5	16.56	2.29	81.9	2.22	1.44	15.7
80 x 80	6.0	7.32	8.5	9.327	2.18	56.4	2.46	1.58	9.7
	7.0	8.49	10.0	10.8	2.21	64.2	2.44	1.57	11.1
	8.0	9.66	10.0	12.3	2.26	72.3	2.42	1.55	12.6
	9.0	10.7	10.0	13.59	2.30	79.2	2.41	1.55	13.9
	12.0	13.9	10.0	17.76	2.41	101.0	2.38	1.54	18.1
90 x 90	6.0	8.28	10.0	10.55	2.42	80.7	2.77	1.78	12.3
	7.0	9.59	10.0	12.22	2.46	93.0	2.76	1.77	14.2
	8.0	10.9	11.0	13.9	2.50	104.0	2.74	1.76	16.1
	9.0	12.2	11.0	15.5	2.54	116.0	2.74	1.76	18.0
	10.0	13.3	10.0	17.0	2.57	125.0	2.71	1.74	19.5
	12.0	15.9	11.0	20.3	2.70	148.0	2.70	1.75	23.3
	13.0	17.0	10.0	21.71	2.69	156.0	2.68	1.73	24.8

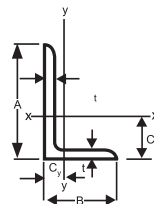


ANGLES

EQUAL

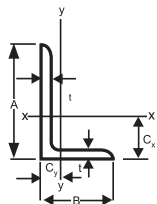
Section size A x B mm	Thickness t mm	Unit weight M kg/m	Reduce of fillet r mm	Section area A cm ²	Centre of gravity Cx = Cy cm	Axis			
						Ix = Iy cm ⁴	ix = iy cm	iv cm	Zx = Zy cm ³
100 x 100	6.0	9.2	8.5	11.67	2.67	111.3	3.09	1.95	15.2
	7.0	10.7	10.0	13.62	2.71	129.0	3.08	1.98	17.7
	8.0	12.2	12.0	15.5	2.74	145.0	3.06	1.96	19.9
	10.0	15.0	10.0	19.0	2.82	175.0	3.04	1.95	24.4
	12.0	17.8	12.0	22.7	3.02	207.0	3.10	1.94	29.1
	13.0	19.1	10.0	24.31	2.94	220.0	3.00	1.94	31.1
	15.0	21.9	12.0	27.9	2.98	249.0	3.05	1.93	35.6
120 x 120	8.0	14.7	13.0	18.7	3.69	255.0	3.96	2.37	29.1
	10.0	18.2	13.0	23.2	3.67	313.0	3.94	2.36	36.0
	12.0	21.6	13.0	27.5	3.65	368.0	3.91	2.35	42.7
	15.0	26.6	13.0	33.9	3.62	445.0	3.86	2.33	52.4
130 x 130	8.0	15.9	13.0	20.42	3.49	331.75	4.03	2.59	34.88
	9.0	17.9	12.0	22.74	3.53	366.0	4.01	2.57	38.7
	10.0	19.7	13.0	25.26	3.57	406.04	4.01	2.57	43.08
	12.0	23.5	12.0	29.76	3.64	467.0	3.96	2.54	49.9
	15.0	28.8	12.0	36.75	3.76	568.0	3.93	2.53	61.5
	16.0	30.7	13.0	39.38	3.81	669.86	3.90	2.51	66.31
150 x 150	10.0	23.0	16.0	29.3	4.62	624.0	4.76	2.97	56.90
	12.0	27.3	14.0	34.77	4.14	740.0	4.61	2.96	68.1
	15.0	33.6	14.0	42.74	4.24	888.0	4.56	2.92	82.6
	18.0	40.1	14.0	51.0	4.54	1,050.0	4.65	2.92	88.7
	19.0	41.9	14.0	53.38	4.40	1,090.0	4.52	2.91	103.0
175 x 175	12.0	31.8	15.0	40.52	4.73	1,170.0	5.37	3.44	91.6
	15.0	39.4	15.0	50.21	4.85	1,440.0	5.35	3.42	114.0
200 x 200	15.0	45.3	17.0	57.75	5.47	2,180.0	6.14	3.93	150.0
	16.0	48.5	18.0	61.8	6.16	2,340.0	6.30	3.94	162.0
	18.0	54.2	18.0	69.1	6.13	2,600.0	6.27	3.93	181.0
	20.0	59.9	17.0	76.3	5.67	2,820.0	6.09	3.90	197.0
	24.0	71.1	18.0	90.6	6.06	3,333.0	6.22	3.90	235.0
	25.0	73.6	17.0	93.75	5.87	3,420.0	6.04	3.88	242.0
250 x 250	25.0	93.7	24.0	119.4	7.10	6,950.0	7.63	4.89	388.0
	28.0	104.0	24.0	133.0	7.23	7,690.0	7.61	4.89	433.0
	32.0	118.0	24.0	150.0	7.38	8,600.0	7.57	4.87	488.0
	35.0	128.0	24.0	162.6	7.45	9,110.0	7.48	4.83	519.0

ANGLES



UNEQUAL

Section Size A x B x t	Thickness		Unit Weight		Leg Length				Corner radius				Section Area		Centre of Gravity			
	t		M		A		B		r1		r2		A		Cx		Cy	
	mm	in	kg/m	lb/ft	mm	in	mm	in	mm	in	mm	in	cm ²	in ²	cm	in	cm	in
40 x 20 x 3	3	0.118	1.37	0.921	40	1.575	20	0.787	5	0.197	2	0.079	1.747	0.271	1.41	0.555	0.44	0.173
40 x 20 x 5	5	0.197	2.16	1.451	40	1.575	20	0.787	5	0.197	3.5	0.138	2.751	0.426	1.50	0.591	0.51	0.201
50 x 35 x 4	4	0.157	2.58	1.734	50	1.969	35	1.378	6.5	0.256	3	0.118	3.292	0.510	1.58	0.622	0.85	0.335
50 x 35 x 6	6	0.236	3.72	2.500	50	1.969	35	1.378	6.5	0.256	4.5	0.177	4.744	0.735	1.66	0.654	0.92	0.362
60 x 50 x 5	5	0.197	4.16	2.795	60	2.362	50	1.969	6.5	0.256	3	0.118	5.302	0.822	1.79	0.705	1.30	0.512
60 x 50 x 7	7	0.276	5.66	3.803	60	2.362	50	1.969	6.5	0.256	4.5	0.177	7.214	1.118	1.87	0.736	1.37	0.539
65 x 50 x 5	5	0.197	4.36	2.930	65	2.559	50	1.969	6.5	0.256	3	0.118	5.552	0.861	1.99	0.783	1.25	0.492
65 x 50 x 7	7	0.276	5.94	3.992	65	2.559	50	1.969	6.5	0.256	4.5	0.177	7.564	1.172	2.07	0.815	1.33	0.524
65 x 50 x 9	9	0.354	7.49	5.033	65	2.559	50	1.969	6.5	0.256	4.5	0.177	9.544	1.479	2.14	0.843	1.40	0.551
70 x 60 x 6	6	0.236	5.91	3.971	70	2.756	60	2.362	8.5	0.335	4	0.157	7.527	1.167	2.07	0.815	1.58	0.622
70 x 60 x 8	8	0.315	7.66	5.147	70	2.756	60	2.362	8.5	0.335	6	0.236	9.761	1.513	2.14	0.843	1.65	0.661
70 x 60 x 10	10	0.394	9.42	6.33	70	2.756	60	2.362	8.5	0.335	6	0.236	12.00	1.86	2.24	0.874	1.73	0.681
75 x 50 x 6	6	0.236	5.67	3.810	75	2.953	50	1.969	8.5	0.335	4	0.157	7.227	1.120	2.43	0.957	1.20	0.472
75 x 50 x 8	8	0.315	7.35	4.939	75	2.953	50	1.969	8.5	0.335	6	0.236	9.261	1.451	2.50	0.984	1.27	0.500
75 x 50 x 10	10	0.394	9.03	6.068	75	2.953	50	1.969	8.5	0.335	6	0.236	11.50	1.782	2.58	1.016	1.35	0.531
75 x 65 x 6	6	0.236	6.38	4.287	75	2.953	65	2.559	8.5	0.335	4	0.157	8.127	1.260	2.19	0.862	1.70	0.669
75 x 65 x 8	8	0.315	8.29	5.571	75	2.953	65	2.559	8.5	0.335	6	0.236	10.56	1.637	2.26	0.890	1.77	0.697
75 x 65 x 10	10	0.394	10.2	6.854	75	2.953	65	2.559	8.5	0.335	6	0.236	13.00	2.015	2.34	0.921	1.85	0.728
80 x 60 x 6	6	0.236	6.38	4.287	80	3.15	60	2.362	8.5	0.335	4	0.157	8.127	1.260	2.46	0.969	1.48	0.583
80 x 60 x 8	8	0.315	8.29	5.571	80	3.15	60	2.362	8.5	0.335	6	0.236	10.56	1.637	2.54	1.000	1.55	0.610
80 x 60 x 10	10	0.394	10.2	6.854	80	3.15	60	2.362	8.5	0.335	6	0.236	13.00	2.015	2.62	1.031	1.63	0.642
80 x 70 x 6	6	0.236	6.85	4.603	80	3.15	70	2.756	8.5	0.335	4	0.157	8.727	1.353	2.32	0.913	1.83	0.72
80 x 70 x 9	9	0.354	9.96	6.693	80	3.15	70	2.756	8.5	0.335	6	0.236	12.69	1.967	2.43	0.957	1.93	0.76
80 x 70 x 12	12	0.472	13.0	8.736	80	3.15	70	2.756	8.5	0.335	6	0.236	16.56	2.567	2.54	1.00	2.05	0.807
90 x 60 x 6	6	0.236	6.85	4.603	90	3.543	60	2.362	8.5	0.335	4	0.157	8.727	1.353	2.88	1.134	1.40	0.551
90 x 60 x 9	9	0.354	9.96	6.693	90	3.543	60	2.362	8.5	0.335	6	0.236	12.69	1.967	2.99	1.177	1.51	0.594
90 x 60 x 12	12	0.472	13.0	8.736	90	3.543	60	2.362	8.5	0.335	6	0.236	16.56	2.567	3.11	1.244	1.63	0.642
90 x 75 x 6	6	0.236	7.56	5.08	90	3.543	75	2.953	8.5	0.335	4	0.157	9.626	1.492	2.64	1.039	1.90	0.748
90 x 75 x 9	9	0.354	11.0	7.392	90	3.543	75	2.953	8.5	0.335	6	0.236	14.04	2.176	2.75	1.083	2.01	0.791
90 x 75 x 12	12	0.472	14.4	9.676	90	3.543	75	2.953	8.5	0.335	6	0.236	18.36	2.846	2.87	1.13	2.12	0.835

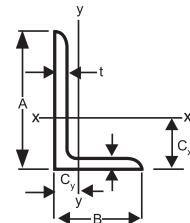


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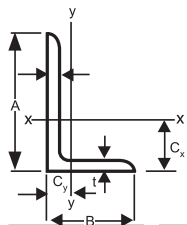
Section Size A x B x t	Moment of Inertia								Radius of Gyratation								tan	Modulus of Section			
	Ix		Iy		Max Iu		Min Iv		ix		iy		Max Iu		Min Iv		a	Zx		Zy	
	cm ⁴	in ⁴	cm ⁴	in ⁴	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm	in	cm	in		cm ³	in ³	cm ³	in ³
40 x 20 x 3	2.81	0.068	0.47	0.011	2.98	0.072	0.31	0.007	1.27	0.50	0.52	0.205	1.31	0.516	0.42	0.165	0.257	1.09	0.067	0.30	0.018
40 x 20 x 5	4.21	0.11	0.68	0.016	4.43	0.106	0.41	0.01	1.24	0.488	0.50	0.197	1.27	0.50	0.41	0.161	0.243	1.68	0.103	0.46	0.208
50 x 35 x 4	8.08	0.194	3.25	0.078	9.49	0.228	1.84	0.044	1.57	0.618	0.99	0.39	1.70	0.669	0.75	0.295	0.475	2.36	0.144	1.23	0.075
50 x 35 x 6	12.2	0.269	4.43	0.106	13.1	0.315	2.55	0.061	1.54	0.606	0.97	0.382	1.66	0.654	0.73	0.287	0.466	3.35	0.204	1.72	0.105
60 x 50 x 5	18.5	0.445	11.7	0.281	24.4	0.586	5.82	0.14	1.87	0.736	1.48	0.583	2.14	0.843	1.05	0.413	0.679	4.40	0.269	3.16	0.193
60 x 50 x 7	24.4	0.586	15.2	0.365	31.9	0.766	7.68	0.185	1.84	0.724	1.45	0.571	2.10	0.827	1.03	0.406	0.673	5.89	0.359	4.2	0.256
65 x 50 x 5	23.2	0.557	11.9	0.286	28.8	0.692	6.33	0.152	2.04	0.803	1.47	0.579	2.28	0.898	1.07	0.421	0.577	5.14	0.314	3.19	0.195
65 x 50 x 7	30.6	0.735	15.6	0.375	37.9	0.911	8.35	0.201	2.01	0.791	1.44	0.567	2.24	0.882	1.05	0.413	0.571	6.90	0.421	4.25	0.259
65 x 50 x 9	37.8	0.908	19.2	0.461	46.6	1.12	10.4	0.25	1.99	0.783	1.42	0.559	2.21	0.87	1.05	0.413	0.565	8.69	0.53	5.33	0.325
70 x 60 x 6	35.3	0.848	23.9	0.574	47.6	1.144	11.7	0.281	2.17	0.854	1.78	0.701	2.15	0.988	1.24	0.488	0.719	7.17	0.438	5.4	0.33
70 x 60 x 8	44.4	1.067	29.8	0.716	59.5	1.429	14.7	0.353	2.13	0.839	1.75	0.689	2.47	0.872	1.23	0.484	0.714	9.13	0.457	6.85	0.418
70 x 60 x 10	53.6	1.288	36.0	0.865	71.7	1.723	17.9	0.43	2.11	0.831	1.73	0.681	2.44	0.961	1.22	0.48	0.711	11.2	0.683	8.41	0.513
75 x 50 x 6	40.4	0.971	14.4	0.346	46.4	1.115	8.33	0.20	2.36	0.929	1.41	0.555	2.53	0.996	1.07	0.421	0.434	7.97	0.486	3.78	0.231
75 x 50 x 8	50.8	1.221	17.8	0.428	58.1	1.396	10.4	0.25	2.33	0.917	1.38	0.543	2.49	0.98	1.05	0.413	0.427	10.2	0.622	4.77	0.291
75 x 50 x 10	61.4	1.475	21.4	0.514	70.0	1.682	12.7	0.257	2.31	0.909	1.36	0.535	2.47	0.972	1.05	0.413	0.421	12.5	0.763	5.85	0.357
75 x 65 x 6	44.1	1.06	30.7	0.738	49.9	1.439	14.8	0.356	2.33	0.917	1.94	0.764	2.72	1.071	1.35	0.531	0.737	8.3	0.506	6.39	0.523
75 x 65 x 8	55.5	1.333	38.4	0.923	75.3	1.809	18.6	0.447	2.29	0.902	1.91	0.752	2.67	1.051	1.33	0.524	0.733	10.6	0.647	8.13	0.496
75 x 65 x 10	67.3	1.617	46.5	1.117	91.0	2.186	22.8	0.548	2.27	0.894	1.89	0.744	2.65	1.043	1.32	0.52	0.73	13.0	0.793	9.99	0.61
80 x 60 x 6	51.5	1.237	24.8	0.596	62.9	1.511	13.4	0.322	2.52	0.992	1.75	0.689	2.78	1.094	1.28	0.504	0.548	9.30	0.568	5.5	0.336
80 x 60 x 8	65.0	1.562	31.0	0.745	79.1	1.90	16.9	0.406	2.48	0.976	1.71	0.673	2.74	1.079	1.26	0.496	0.542	11.9	0.726	6.97	0.425
80 x 60 x 10	78.7	1.891	37.4	0.899	95.6	2.297	20.5	0.493	2.46	0.969	1.7	0.669	2.52	0.992	1.26	0.496	0.538	14.6	0.891	8.55	0.522
80 x 70 x 6	54.1	1.30	38.6	0.927	75.3	1.785	18.4	0.442	2.49	0.98	2.1	0.827	2.92	1.15	1.45	0.571	0.752	9.52	0.581	7.46	0.455
80 x 70 x 9	75.8	1.821	53.8	1.293	104	2.499	25.9	0.622	2.44	0.961	2.06	0.811	2.86	1.126	1.43	0.563	0.748	13.6	0.83	10.6	0.647
80 x 70 x 12	96.6	1.321	68.3	1.641	131	3.147	33.4	0.802	2.41	0.949	2.03	0.799	2.82	1.11	1.42	0.559	0.744	17.7	1.08	13.8	0.842
90 x 60 x 6	71.5	1.718	23.6	0.615	82.3	1.977	14.8	0.356	2.86	1.126	1.71	0.674	3.07	1.209	1.30	0.512	0.437	11.7	0.714	5.47	0.34
90 x 60 x 9	101	2.427	35.4	0.85	115	2.763	20.7	0.497	2.82	1.11	1.67	0.657	3.01	1.185	1.28	0.504	0.43	16.7	1.091	7.9	0.482
90 x 60 x 12	128	3.075	44.8	1.076	146	2.508	26.7	0.641	2.78	1.094	1.65	0.65	2.97	1.169	1.27	0.50	0.423	21.8	1.33	10.2	0.622
90 x 75 x 6	76.9	1.848	48.6	1.168	101	2.427	24.2	0.581	2.83	1.114	2.25	0.886	3.25	1.28	1.58	0.622	0.681	12.1	0.738	8.68	0.53
90 x 75 x 9	109	2.619	68.1	1.636	143	3.436	34.1	0.819	2.78	1.094	2.2	0.866	3.19	1.257	1.56	0.614	0.676	17.4	1.062	12.4	0.757
90 x 75 x 12	139	3.34	86.8	2.085	182	4.373	44.0	1.057	2.75	1.083	2.17	0.854	3.14	1.236	1.55	0.61	0.672	22.6	1.379	16.1	0.982

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Section Size A x B x t	Thickness		Unit Weight		Leg Length				Corner radius				Section Area		Centre of Gravity			
	t		M		A		B		r1		r2		A		Cx		Cy	
	mm	in	kg/m	lb/ft	mm	in	mm	in	mm	in	mm	in	cm ²	in ²	cm	in	cm	in
90 x 80 x 7	7	0.276	9.04	6.075	90	3.543	80	3.15	10	0.394	5	0.197	11.52	1.786	2.59	1.02	2.1	0.827
90 x 80 x 10	10	0.394	12.6	8.467	90	3.543	80	3.15	10	0.394	7	0.276	16.0	2.48	2.71	1.067	2.21	0.87
90 x 80 x 13	13	0.512	16.0	10.751	90	3.543	80	3.15	10	0.394	7	0.276	20.41	3.164	2.82	1.11	2.33	0.917
100 x 65 x 7	7	0.276	8.77	5.893	100	3.937	65	2.559	10	0.394	5	0.197	11.17	1.731	3.23	1.272	1.52	0.598
100 x 65 x 8	8	0.315	9.94	6.679	100	3.937	65	2.559	10	0.394	4.8	0.189	12.7	1.984	3.27	1.287	1.55	0.61
100 x 65 x 9	9	0.354	11.0	7.392	100	3.937	65	2.559	10	0.394	7	0.276	14.04	2.176	3.31	1.303	1.59	0.626
100 x 65 x 10	10	0.394	12.3	8.265	100	3.937	65	2.559	10	0.394	4.8	0.189	15.6	2.418	3.36	1.323	1.63	0.642
100 x 65 x 12	12	0.472	14.4	9.676	100	3.937	65	2.559	10	0.394	7	0.276	18.36	2.846	3.43	1.35	1.7	0.669
100 x 75 x 7	7	0.276	9.32	6.263	100	3.937	75	2.953	10	0.394	5	0.197	11.87	1.84	3.06	1.205	1.84	0.724
100 x 75 x 8	8	0.315	10.6	7.123	100	3.937	75	2.953	10	0.394	4.8	0.189	13.5	2.093	3.1	1.22	1.87	0.736
100 x 75 x 10	10	0.394	13.0	8.736	100	3.937	75	2.953	10	0.394	7	0.276	16.5	2.557	3.18	1.252	1.95	0.764
100 x 75 x 12	12	0.472	15.4	10.348	100	3.937	75	2.953	10	0.394	4.8	0.189	19.7	3.054	3.27	1.287	2.03	0.799
100 x 75 x 13	13	0.512	16.5	11.087	100	3.937	75	2.953	10	0.394	7	0.276	21.06	3.264	3.3	1.3	2.06	0.811
100 x 80 x 7	7	0.276	9.59	6.444	100	3.937	80	3.15	10	0.394	5	0.196	12.22	1.894	2.98	1.173	2.00	0.787
100 x 80 x 10	10	0.394	13.3	8.937	100	3.937	80	3.15	10	0.394	7	0.276	17.0	2.635	3.1	1.22	2.11	0.831
100 x 80 x 13	13	0.512	17.0	11.423	100	3.937	80	3.15	10	0.394	7	0.276	21.71	3.365	3.22	1.268	2.23	0.878
100 x 90 x 7	7	0.276	10.1	6.787	100	3.937	90	3.543	10	0.394	5	0.197	12.92	2.003	2.84	1.118	2.35	0.925
100 x 90 x 10	10	0.394	14.1	9.475	100	3.937	90	3.543	10	0.394	7	0.276	18.00	2.79	2.96	1.165	2.46	0.969
100 x 90 x 13	13	0.512	18.1	12.162	100	3.937	90	3.543	10	0.394	7	0.276	23.01	3.567	3.07	1.209	2.58	1.016
125 x 75 x 6.5	6.5	0.256	9.98	6.706	125	4.921	75	2.953	11	0.433	4.8	0.189	12.7	1.968	4.06	1.599	1.61	0.634
125 x 75 x 7	7	0.276	10.7	7.19	125	4.921	75	2.953	10	0.394	5	0.197	13.62	2.111	4.1	1.615	1.64	0.646
125 x 75 x 8	8	0.315	12.2	8.198	125	4.921	75	2.953	11	0.433	4.8	0.189	15.5	2.403	4.14	1.63	1.68	0.661
125 x 75 x 10	10	0.394	15.0	10.012	125	4.921	75	2.953	10	0.394	7	0.276	19.0	2.945	4.23	1.665	1.75	0.689
125 x 75 x 12	12	0.472	17.8	11.961	125	4.921	75	2.953	11	0.433	4.8	0.189	22.7	3.519	4.31	1.697	1.84	0.724
125 x 75 x 13	13	0.512	19.1	12.835	125	4.921	75	2.953	10	0.394	7	0.276	24.31	3.768	4.35	1.713	1.87	0.736
125 x 90 x 7	7	0.276	11.5	7.728	125	4.921	90	3.543	10	0.394	5	0.197	14.67	2.274	3.84	1.512	2.11	0.831
125 x 90 x 10	10	0.394	16.1	10.819	125	4.921	90	3.543	10	0.394	7	0.276	20.5	3.177	3.95	1.555	2.22	0.874
125 x 90 x 13	13	0.512	20.6	13.843	125	4.921	90	3.543	10	0.394	7	0.276	26.26	4.07	4.08	1.606	2.34	0.921
137 x 102 x 6.4	6.4	0.252	11.7	7.862	137	5.394	102	4.016	11	0.433	4.8	0.189	14.8	2.294	4.09	1.61	2.36	0.929
137 x 102 x 7.9	7.9	0.311	14.5	9.744	137	5.394	102	4.016	11	0.433	4.8	0.189	18.4	2.852	4.17	1.642	2.44	0.961
137 x 102 x 9.5	9.5	0.374	17.3	11.625	137	5.394	102	4.016	11	0.433	4.8	0.189	21.9	3.395	4.22	1.661	2.49	0.98

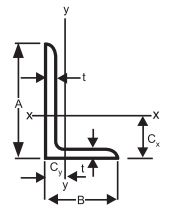


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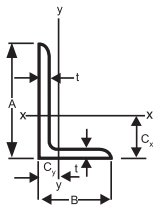
Section Size A x B x t	Moment of Inertia				Radius of Gyratation								tan				Modulus of Section				
	Ix		Iy		Max Iu		Min Iv		ix		iy		Max Iu		Min Iv		a	Zx		Zy	
	cm ⁴	in ⁴	cm ⁴	in ⁴	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm	in	cm	in		cm ³	in ³	cm ³	in ³
90 x 80 x 7	89.6	2.153	66.5	1.598	125	3.003	31.3	0.752	2.79	1.098	2.4	0.945	3.29	1.295	1.65	0.65	0.777	14.0	0.854	11.3	0.69
90 x 80 x 10	121	2.907	89.1	2.141	167	4.018	42.1	1.011	2.74	1.078	2.36	0.929	3.24	1.276	1.62	0.638	0.774	19.1	1.166	15.4	0.94
90 x 80 x 13	151	3.628	111	2.668	208	4.997	53.3	1.281	2.72	1.071	2.33	0.917	3.19	1.256	1.62	0.638	0.77	24.4	1.489	19.6	1.196
100 x 65 x 7	112	2.091	37.6	0.903	128	3.075	22.0	0.529	3.17	1.248	1.84	0.724	3.39	1.335	1.4	0.551	0.415	16.6	1.013	7.55	0.461
100 x 65 x 8	127	3.051	42.2	1.014	144	3.46	24.8	0.596	3.16	1.244	1.83	0.72	3.37	1.327	1.4	0.551	0.414	18.9	1.153	8.54	0.521
100 x 65 x 9	138	3.315	45.6	1.096	157	3.772	26.8	0.644	3.14	1.236	1.8	0.709	3.34	1.335	1.38	0.543	0.411	20.6	1.257	9.28	0.566
100 x 65 x 10	154	3.7	51.0	1.225	175	4.204	30.1	0.723	3.14	1.236	1.81	0.713	3.35	1.319	1.39	0.547	0.41	23.2	1.416	10.5	0.641
100 x 65 x 12	177	4.252	58.1	1.396	200	4.882	34.7	0.834	3.11	1.224	1.78	0.701	3.3	1.3	1.37	0.539	0.405	27.0	1.648	12.1	0.738
100 x 75 x 7	118	2.835	57.0	1.369	144	3.46	30.7	0.738	3.15	1.24	2.19	0.862	3.49	1.374	1.61	0.634	0.548	17.0	1.037	10.1	0.616
100 x 75 x 8	133	3.195	64.1	1.54	163	3.916	34.6	0.831	3.14	1.236	2.18	0.858	3.47	1.366	1.6	0.63	0.547	19.3	1.178	11.4	0.696
100 x 75 x 10	159	3.82	76.1	1.828	194	4.661	41.3	0.992	3.11	1.224	2.15	0.846	3.43	1.35	1.58	0.622	0.543	23.3	1.422	13.7	0.836
100 x 75 x 12	189	4.541	90.2	2.167	230	5.526	49.5	1.189	3.1	1.22	2.14	0.843	3.42	1.346	1.59	0.626	0.54	28.0	1.709	16.5	1.007
100 x 75 x 13	199	4.781	94.8	2.278	242	5.814	52.2	1.254	3.08	1.213	2.12	0.835	3.39	1.335	1.57	0.618	0.538	29.7	1.812	17.4	1.062
100 x 80 x 7	120	2.883	68.6	1.643	154	3.7	35.3	0.848	3.14	1.236	2.37	0.933	3.55	1.398	1.7	0.669	0.625	17.2	1.05	11.4	0.696
100 x 80 x 10	163	3.916	91.9	2.208	207	4.973	47.6	1.144	3.09	1.217	2.33	0.917	3.49	1.374	1.67	0.659	0.621	23.6	1.44	15.6	0.952
100 x 80 x 13	204	4.901	115	2.763	258	6.198	60.1	1.444	3.06	1.205	2.3	0.906	3.45	1.358	1.66	0.654	0.616	30.1	1.837	19.9	1.215
100 x 90 x 7	125	3.003	95.9	2.304	177	4.252	44.5	1.069	3.11	1.224	2.73	1.075	3.7	1.457	1.85	0.728	0.799	17.5	1.068	14.5	0.879
100 x 90 x 10	169	4.06	129	3.099	238	5.718	60.1	1.444	3.07	1.209	2.68	1.055	3.64	1.433	1.83	0.72	0.797	24.0	1.465	19.8	1.208
100 x 90 x 13	212	5.093	162	3.892	298	7.159	76.0	1.826	3.06	1.205	2.65	1.043	3.6	1.417	1.82	0.717	0.794	30.6	1.867	25.2	1.538
125 x 75 x 6.5	204	4.901	56.1	1.348	228	4.821	34.3	0.825	4.01	1.579	2.1	0.827	4.23	1.665	1.64	0.646	0.36	24.2	1.477	9.5	0.58
125 x 75 x 7	219	5.261	60.4	1.451	243	5.138	36.4	0.875	4.01	1.579	2.11	0.831	4.23	1.665	1.63	0.642	0.362	26.1	1.593	10.3	0.629
125 x 75 x 8	247	5.934	67.6	1.624	274	6.583	40.9	0.983	4.0	1.575	2.09	0.823	4.2	1.654	1.63	0.642	0.359	29.6	1.806	11.6	0.708
125 x 75 x 10	298	7.159	80.9	1.944	330	7.928	49.0	1.177	3.86	1.52	2.06	0.811	4.17	1.642	1.61	0.634	0.357	36.1	2.203	14.1	0.86
125 x 75 x 12	354	8.505	95.5	2.294	391	9.394	58.5	1.405	3.95	1.555	2.05	0.807	4.15	1.634	1.61	0.634	0.353	43.2	2.636	16.9	1.031
125 x 75 x 13	376	9.033	101	2.427	414	9.946	61.9	1.487	3.92	1.543	2.04	0.803	4.13	1.626	1.6	0.63	0.352	46.1	2.813	17.9	1.092
125 x 90 x 7	233	5.598	102	2.451	279	6.703	56.2	1.35	3.99	1.571	2.64	1.039	4.36	1.717	1.96	0.772	0.51	26.9	1.642	14.8	0.903
125 x 90 x 10	318	7.64	138	3.315	380	9.129	76.1	1.828	3.94	1.551	2.59	1.02	4.3	1.693	1.93	0.76	0.506	37.2	2.27	20.4	1.245
125 x 90 x 13	401	9.634	165	3.964	479	11.51	87.2	2.095	3.91	1.539	2.51	0.988	4.27	1.681	1.82	0.717	0.499	47.5	2.899	24.8	1.513
137 x 102 x 6.4	287	6.895	137	3.291	352	8.457	74.7	1.795	4.37	1.72	3.02	1.189	4.84	1.91	2.23	0.878	0.544	29.9	1.825	17.6	1.074
137 x 102 x 7.9	353	8.481	169	4.06	428	10.283	90.4	2.172	4.34	1.709	3.0	1.181	4.82	1.9	2.22	0.874	0.544	37.3	2.276	21.8	1.33
137 x 102 x 9.5	417	10.018	194	4.661	506	12.157	107	2.571	4.34	1.709	3.00	1.181	4.80	1.89	2.2	0.866	0.543	43.8	2.673	25.8	1.574

ANGLES



UNEQUAL

Section Size A x B x t	Thickness t		Unit Weight M		Leg Length				Corner radius				Section Area A		Centre of Gravity			
					A		B		r1		r2				Cx		Cy	
	mm	in	kg/m	lb/ft	mm	in	mm	in	mm	in	mm	in	cm ²	in ²	cm	in	cm	in
150 x 75 x 7	7	0.276	12.1	8.131	150	5.906	75	2.953	12	0.472	5	0.197	15.46	2.396	5.17	2.035	1.5	0.591
150 x 75 x 9	9	0.354	15.3	10.281	150	5.906	75	2.953	12	0.472	8.5	0.335	19.44	3.013	5.25	2.067	1.56	0.614
150 x 75 x 10	10	0.394	17.0	11.423	150	5.906	75	2.953	11	0.433	4.8	0.189	21.6	3.348	5.32	2.094	1.61	0.634
150 x 75 x 12	12	0.472	20.1	13.574	150	5.906	75	2.953	12	0.472	8.5	0.335	25.56	3.962	5.39	2.122	1.68	0.661
150 x 75 x 15	15	0.591	24.8	16.665	150	5.906	75	2.953	11	0.433	4.8	0.189	31.6	4.898	5.53	2.177	1.81	0.713
150 x 90 x 9	9	0.354	16.4	11.02	150	5.906	90	3.543	12	0.472	6	0.236	20.94	3.246	4.96	1.953	2.0	0.787
150 x 90 x 10	10	0.394	18.2	12.23	150	5.906	90	3.543	12	0.472	4.8	0.189	23.2	3.596	5.0	1.969	2.04	0.803
150 x 90 x 12	12	0.472	21.6	14.515	150	5.906	90	3.543	12	0.472	8.5	0.335	27.36	4.241	5.07	1.996	2.1	0.828
150 x 90 x 15	15	0.591	26.6	17.874	150	5.906	90	3.543	12	0.472	8.5	0.335	33.75	5.231	5.2	2.047	2.22	0.874
150 x 100 x 9	9	0.354	17.1	11.49	150	5.906	100	3.937	12	0.472	6	0.236	21.84	3.385	4.77	1.878	2.32	0.913
150 x 100 x 12	12	0.472	22.4	15.052	150	5.906	100	3.937	12	0.472	8.5	0.335	28.56	4.427	4.88	1.921	2.41	0.949
150 x 100 x 15	15	0.591	27.7	18.614	150	5.906	100	3.937	12	0.472	8.5	0.335	35.25	5.464	5.1	1.972	2.53	0.996
175 x 90 x 9	9	0.354	18.2	12.23	175	6.89	90	3.543	12	0.472	6	0.236	23.19	3.594	6.04	2.373	1.85	0.728
175 x 90 x 12	12	0.472	23.8	15.993	175	6.89	90	3.543	12	0.472	8.5	0.335	30.36	4.706	6.16	2.425	1.96	0.772
175 x 90 x 15	15	0.591	29.4	19.756	175	6.89	90	3.543	12	0.472	8.5	0.335	37.5	5.812	6.29	2.476	2.08	0.819
200 x 100 x 10	10	0.394	23.0	15.455	200	7.874	100	3.937	15	0.591	4.8	0.189	29.2	4.526	6.93	2.728	2.01	0.791
200 x 100 x 12	12	0.472	27.3	18.345	200	7.874	100	3.937	15	0.591	4.8	0.189	34.8	5.394	7.03	2.768	2.1	0.827
200 x 100 x 15	15	0.591	33.7	22.645	200	7.874	100	3.937	15	0.591	4.8	0.189	43.0	6.665	7.16	2.819	2.22	0.874
200 x 150 x 12	12	0.472	32.0	21.503	200	7.874	150	5.906	15	0.591	4.8	0.189	40.8	6.324	6.08	2.394	3.61	1.421
200 x 150 x 15	15	0.591	39.6	26.61	200	7.874	150	5.906	15	0.591	4.8	0.189	50.5	7.828	6.21	2.445	3.73	1.469
200 x 150 x 18	18	0.709	47.1	31.65	200	7.874	150	5.906	15	0.591	4.8	0.189	60.0	9.3	6.33	2.492	3.85	1.516



UNEQUAL

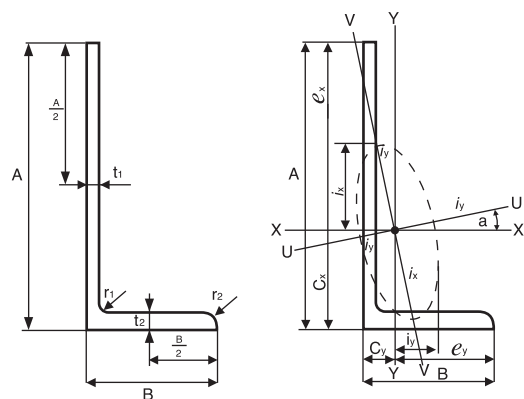
Section Size A x B x t	Moment of Inertia				Radius of								tan				Modulus of				
	Ix		Iy		Max Iu		Min Iv		Gyration				Max Iu		Min Iv		a	Section			
									ix		iy							Zx		Zy	
	cm ⁴	in ⁴	cm ⁴	in ⁴	cm ⁴	in ⁴	cm ⁴	in ⁴	cm	in	cm	in	cm	in	cm	in		cm ³	in ³	cm ³	in ³
150 x 75 x 7	364	8.745	63.1	1.516	386	9.274	40.5	0.973	4.85	1.909	2.02	0.795	5.0	1.969	1.62	0.638	0.265	37.0	2.258	10.5	0.641
150 x 75 x 9	446	7.151	75.6	1.816	473	11.36	48.5	1.17	4.79	1.886	1.97	0.776	4.93	1.941	1.58	0.622	0.26	45.8	2.795	12.7	0.775
150 x 75 x 10	501	12.036	85.8	2.061	532	12.78	55.3	1.329	4.81	1.894	1.99	0.783	4.96	1.953	1.6	0.63	0.261	51.8	3.161	14.6	0.891
150 x 75 x 12	580	13.935	97.4	2.34	614	14.75	63.3	1.521	4.76	1.874	1.95	0.768	4.9	1.929	1.57	0.618	0.257	60.3	3.68	16.7	1.019
150 x 75 x 15	713	17.13	120	2.883	754	18.11	78.8	1.893	4.75	1.87	1.94	0.764	4.88	1.921	1.58	0.622	0.254	75.3	4.595	21.0	1.281
150 x 90 x 9	484	16.63	133	3.195	537	12.9	80.2	1.927	4.81	1.894	2.52	0.992	5.06	1.992	1.96	0.772	0.362	48.2	2.941	19.0	1.159
150 x 90 x 10	533	12.8	146	3.508	591	14.2	88.3	2.121	4.8	1.89	2.51	0.988	5.05	1.988	1.95	0.768	0.36	53.3	3.253	21.0	1.281
150 x 90 x 12	619	14.87	168	4.036	684	16.43	102	2.451	4.75	1.87	2.47	0.972	5.0	1.969	1.93	0.76	0.357	62.3	3.802	24.3	1.483
150 x 90 x 15	753	18.09	202	4.853	831	19.96	124	2.979	4.72	1.858	2.45	0.965	4.99	1.965	1.92	0.756	0.353	76.8	4.687	29.9	1.825
150 x 100 x 9	502	12.06	179	4.3	580	13.93	101	2.427	4.79	1.886	2.86	1.126	5.15	2.028	2.15	0.846	0.441	49.0	2.99	23.3	1.422
150 x 100 x 12	642	15.42	229	5.502	738	17.73	133	3.195	4.74	1.866	2.83	1.114	5.08	2.0	2.15	0.846	0.435	63.4	3.869	30.2	1.843
150 x 100 x 15	781	18.76	276	6.631	896	21.55	161	3.868	4.71	1.854	2.8	1.102	5.04	1.984	2.14	0.843	0.432	78.2	4.772	37.0	2.258
175 x 90 x 9	741	17.8	138	3.315	790	18.93	78.4	1.884	5.65	2.224	2.44	0.961	5.84	2.299	1.84	0.724	0.277	64.6	3.942	19.3	1.178
175 x 90 x 12	950	22.82	174	4.18	1010	24.37	111	2.667	5.6	2.205	2.39	0.941	5.78	2.276	1.91	0.752	0.274	83.8	5.114	24.7	1.507
175 x 90 x 15	1160	27.87	210	5.045	1230	29.55	136	3.267	5.56	2.189	2.37	0.933	5.74	2.26	1.9	0.748	0.27	103	6.285	30.4	1.855
200 x 100 x 10	1220	29.31	210	5.045	1290	30.99	135	3.243	6.46	2.543	2.68	1.055	6.65	2.618	2.15	0.846	0.263	93.2	5.687	26.3	1.605
200 x 100 x 12	1440	34.6	247	5.934	1530	36.76	159	3.82	6.43	2.531	2.67	1.051	6.63	2.61	2.14	0.843	0.262	111	6.774	31.3	1.91
200 x 100 x 15	1758	42.24	299	7.183	1863	44.76	194	4.661	6.4	2.52	2.64	1.039	6.58	2.591	2.13	0.839	0.259	137	8.36	38.4	2.343
200 x 150 x 12	1652	39.69	803	19.29	2024	48.63	431	10.35	6.36	2.504	4.44	1.748	7.04	2.772	3.25	1.28	0.552	119	7.262	70.5	4.302
200 x 150 x 15	2022	48.58	979	23.52	2475	59.46	527	12.66	6.33	2.492	4.4	1.732	7.0	2.756	3.23	1.272	0.55	147	8.97	86.9	5.303
200 x 150 x 18	2376	57.08	1146	27.53	2902	69.72	618	14.85	6.29	2.476	4.37	1.72	6.95	2.736	3.21	1.264	0.548	174	10.62	103	6.285

ANGLES

INVERTED ANGLES

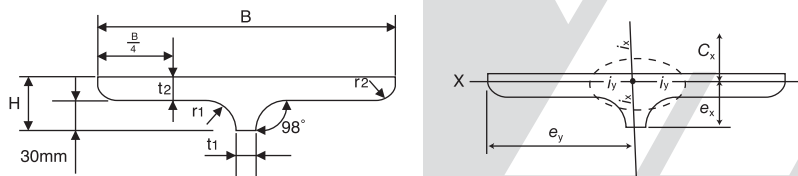
(ANGLES WITH UNEQUAL LEGS AND THICKNESS)

Geometrical moment of inertia : $I = ai^2$
 Radius of gyration of area : $i = \sqrt{I/a}$
 Modulus of section : $Z = I/e$
 (a - sectional area)



Standard Sectional Dimensions of Unequal Leg and Unequal Thickness Angles and Their Sectional Area, Unit Mass and Sectional Characteristics

Standard sectional dimension					Sec-tional area	Unit mass	Informative reference												
A x B	t1	t2	r1	r2			Position of centre of gravity (cm)		Geometrical moment of inertia (cm ⁴)				Radius of gyration of area (cm)				tan a	Modulus of section (cm ³)	
mm	mm	mm	mm	mm	cm ²	kg/m	Cx	Cy	Ix	Iy	Iu max	Iv min	ix	iy	Iu max	iv min		Zx	Zy
200 x 90	9	14	14	7	29.66	23.3	6.36	2.15	1210	200	1290	125	6.39	2.60	6.58	2.05	0.263	88.7	29.2
250 x 90	10	15	17	8.5	37.47	29.4	8.61	1.92	2440	223	2520	147	8.08	2.44	8.20	1.98	0.182	149	31.5
	12	16	17	8.5	42.95	33.7	8.99	1.89	2790	238	2870	160	8.07	2.35	8.18	1.93	0.173	174	33.5
300 x 90	11	16	19	9.5	46.22	36.3	11.0	1.76	4370	245	4440	168	9.72	2.30	9.80	1.90	0.136	229	33.8
	13	17	19	9.5	52.67	41.3	11.3	1.75	4940	259	5020	181	9.68	2.22	9.76	1.85	0.128	265	35.8
350 x 100	12	17	22	11	57.74	45.3	13.0	1.87	7440	362	7550	251	11.3	2.50	11.4	2.08	0.124	338	44.5
400 x 100	13	18	24	12	68.59	53.8	15.4	1.77	11500	388	11600	277	12.9	2.38	13.0	2.01	0.0996	467	47.1



Standard Sectional Dimensions of T Section and Their Sectional Area, Unit Mass and Sectional Characteristics

Standard sectional dimension							Unit mass	Informative reference							
Nominal section	B	H	t1	t2	r1	r2		Position of centre of gravity (cm)		Geometrical moment of inertia (cm ⁴)		Radius of gyration of area (cm)		Modulus of section (cm ³)	
B x t2	mm	mm	mm	mm	mm	mm	kg/m	Cx	Cy	Ix	Iy	ix	iy	Zx	Zy
150 x 9	150	39	12	9	8	3	18.52	0.934	0	16.5	254	0.942	3.70	5.55	33.8
150 x 12	150	42	12	12	8	3	23.02	1.02	0	20.7	338	0.949	3.83	6.52	45.1
150 x 15	150	45	12	15	8	3	27.52	1.13	0	25.9	423	0.971	3.92	7.70	56.4
200 x 12	200	42	12	12	8	3	29.02	0.935	0	22.3	799	0.877	5.25	6.83	79.9
200 x 16	200	46	12	16	8	3	37.02	1.09	0	30.5	1070	0.907	5.37	8.68	107
200 x 19	200	49	12	19	8	3	43.02	1.22	0	38.5	1270	0.946	5.43	10.4	127
200 x 22	200	52	12	22	8	3	49.02	1.35	0	48.3	1470	0.993	5.47	12.6	147
250 x 16	250	46	12	16	20	3	46.05	1.06	0	33.6	2080	0.854	6.72	9.49	167
250 x 19	250	49	12	19	20	3	53.55	1.19	0	43.1	2470	0.897	6.80	11.6	198
250 x 22	250	52	12	22	20	3	61.05	1.33	0	55.0	2870	0.949	6.85	14.2	229
250 x 25	250	55	12	25	20	3	68.55	1.46	0	69.6	3260	1.01	6.90	17.2	261

MILD STEEL TEE BARS - WEIGHT TABLE

Size (mm)	30 x 30 x 4	35 x 35 x 4.5	40 x 40 x 4	40 x 40 x 5	45 x 45 x 5.5	50 x 50 x 6
kg/m	1.77	2.33	2.38	2.96	3.67	4.44
6m	10.62	13.98	14.28	17.76	22.02	26.64

Size (mm)	60 x 60 x 7	70 x 70 x 8	80 x 80 x 9	100 x 100 x 11	120 x 120 x 13
kg/m	6.23	8.32	10.68	16.40	28.20
6m	37.38	49.92	64.08	98.40	169.20

BARS	
FLAT BARS	152
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YICK HOE

BARS

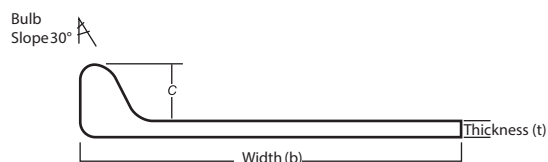
FLAT

Section size		Unit weight M kg/m	Section area A cm ²	Section size		Unit weight M kg/m	Section area A cm ²
thickness	width			thickness	width		
mm	mm			mm	mm		
3.0	12.0	0.283	0.36	9.0	19.0	1.34	1.71
3.0	16.0	0.377	0.48	9.0	25.0	1.77	2.25
3.0	19.0	0.447	0.57	9.0	32.0	2.26	2.88
3.0	25.0	0.589	0.75	9.0	35.0	2.47	3.15
3.0	32.0	0.754	0.96	9.0	38.0	2.68	3.42
3.0	38.0	0.895	0.14	9.0	44.0	3.11	3.96
3.0	50.0	1.18	1.50	9.0	50.0	3.53	4.50
4.5	12.0	0.424	0.54	9.0	65.0	4.59	5.85
4.5	16.0	0.565	0.72	9.0	75.0	5.30	6.75
4.5	19.0	0.67	0.85	9.0	90.0	6.36	8.10
4.5	25.0	0.88	1.125	9.0	100.0	7.06	9.999
4.5	32.0	1.13	1.44	9.0	125.0	8.83	11.25
4.5	38.0	1.34	1.71	9.0	150.0	10.6	13.50
4.5	44.0	1.55	1.98	9.0	180.0	12.7	16.20
4.5	50.0	1.77	2.25	9.0	200.0	14.1	18.00
4.5	65.0	2.30	2.93	9.0	230.0	16.2	20.70
4.5	75.0	2.60	3.37	9.0	250.0	17.7	22.50
4.5	100.0	3.55	4.50	9.0	300.0	21.2	27.00
5.0	25.0	0.98	1.25	12.0	25.0	2.36	3.00
5.0	50.0	1.96	2.50	12.0	32.0	3.01	3.84
6.0	12.0	0.57	0.72	12.0	35.0	3.31	4.20
6.0	16.0	0.754	0.96	12.0	38.0	3.58	4.56
6.0	19.0	0.895	1.14	12.0	44.0	4.14	5.28
6.0	25.0	1.13	1.50	12.0	50.0	4.71	6.00
6.0	32.0	1.51	1.92	12.0	65.0	6.12	7.80
6.0	38.0	1.79	2.28	12.0	75.0	7.06	9.00
6.0	44.0	2.07	2.64	12.0	90.0	8.48	10.80
6.0	50.0	2.36	3.00	12.0	100.0	9.42	12.00
6.0	65.0	3.06	3.90	12.0	125.0	11.8	15.00
6.0	75.0	3.53	4.50	12.0	150.0	14.1	18.00
6.0	90.0	4.24	5.40	12.0	180.0	17.0	21.60
6.0	100.0	4.71	6.00	12.0	200.0	18.8	24.00
6.0	125.0	5.89	7.50	12.0	230.0	21.7	27.60
6.0	150.0	7.05	9.00	12.0	250.0	23.6	30.00
6.0	200.0	9.42	12.00	12.0	280.0	26.4	33.60
6.0	250.0	11.78	15.00	12.0	300.0	28.3	36.00
6.0	300.0	14.13	18.00	16.0	25.0	3.14	4.00
8.0	25.0	1.57	2.00	16.0	32.0	4.02	5.12
8.0	32.0	2.01	2.56	16.0	38.0	4.77	6.08
8.0	38.0	2.39	3.04	16.0	44.0	5.53	7.04
8.0	44.0	2.76	3.52	16.0	50.0	6.28	8.00
8.0	50.0	3.14	4.00	16.0	65.0	8.16	10.40
8.0	65.0	5.65	5.20	16.0	75.0	9.42	12.00
8.0	75.0	4.71	6.00	16.0	90.0	11.3	14.40
8.0	90.0	5.65	7.20	16.0	100.0	12.6	16.00
8.0	100.0	6.28	8.00	16.0	125.0	15.7	20.00
8.0	125.0	7.85	10.00	16.0	150.0	18.8	24.00

FLAT

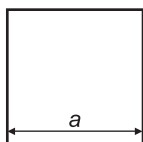
Section size		Unit weight M kg/m	Section area A cm ²	Section size		Unit weight M kg/m	Section area A cm ²
thickness	width			thickness	width		
mm	mm			mm	mm		
16.0	180.0	22.6	28.0	25.0	90.0	17.7	22.5
16.0	200.0	25.1	32.0	25.0	100.0	19.6	25.0
16.0	230.0	28.9	36.8	25.0	125.0	24.5	31.25
16.0	250.0	31.4	50.0	25.0	150.0	29.4	37.5
16.0	280.0	35.2	44.8	25.0	180.0	35.3	45.0
16.0	300.0	37.7	48.0	25.0	200.0	39.2	50.0
19.0	38.0	5.67	7.22	25.0	230.0	45.1	57.5
19.0	44.0	6.56	8.36	25.0	250.0	49.1	62.5
19.0	50.0	7.46	9.50	25.0	280.0	55.0	70.0
19.0	65.0	9.69	12.35	25.0	300.0	58.9	75.0
19.0	75.0	11.2	14.25	28.0	100.0	22.0	28.0
19.0	90.0	13.4	17.1	28.0	125.0	27.5	35.0
19.0	100.0	14.9	19.0	28.0	150.0	33.0	42.0
19.0	125.0	18.6	23.75	28.0	180.0	39.6	50.4
19.0	150.0	22.4	28.5	28.0	200.0	44.0	56.0
19.0	180.0	26.8	34.2	28.0	230.0	50.6	64.4
19.0	200.0	29.8	38.0	28.0	250.0	55.0	70.0
19.0	230.0	34.3	43.7	28.0	280.0	61.5	78.4
19.0	250.0	37.3	47.5	28.0	300.0	65.9	84.0
19.0	280.0	41.8	53.2	32.0	100.0	25.1	32.0
19.0	300.0	44.7	57.0	32.0	125.0	31.4	40.0
22.0	50.0	8.64	11.0	32.0	150.0	37.7	48.0
22.0	65.0	11.2	14.3	32.0	180.0	45.2	57.6
22.0	75.0	13.0	16.5	32.0	200.0	50.2	64.0
22.0	90.0	15.5	19.8	32.0	230.0	57.8	73.6
22.0	100.0	17.3	22.0	32.0	250.0	62.8	80.0
22.0	125.0	21.6	27.5	32.0	280.0	70.3	89.6
22.0	150.0	25.9	33.0	32.0	300.0	75.4	96.0
22.0	180.0	31.1	39.6	36.0	100.0	28.3	36.0
22.0	200.0	34.5	44.0	36.0	125.0	35.3	45.0
22.0	230.0	39.7	50.6	36.0	150.0	42.4	54.0
22.0	250.0	43.2	55.0	36.0	180.0	50.9	64.8
22.0	280.0	48.4	61.0	36.0	200.0	56.5	72.0
22.0	300.0	51.8	66.0	36.0	230.0	65.0	82.8
25.0	50.0	9.81	12.5	36.0	250.0	70.6	90.0
25.0	65.0	12.8	16.25	36.0	280.0	79.1	100.8
25.0	75.0	14.7	18.75	36.0	300.0	84.8	108.0

BARS



BULB FLATS

Width b mm	Thickness t mm	Bulb Height C mm	Area of Section A mm ²	Mass Per Metre kg/m
120	6.0	17.0	9.31	7.31
	7.0	17.0	10.5	8.25
	8.0	17.0	11.7	9.19
140	6.5	19.0	11.7	9.21
	7.0	19.0	12.4	9.74
	8.0	19.0	13.8	10.83
	10.0	19.0	16.6	13.03
160	7.0	22.0	14.6	11.4
	8.0	22.0	16.2	12.7
	9.0	22.0	17.8	14.0
	11.5	22.0	21.8	17.3
180	8.0	25.0	18.9	14.8
	9.0	25.0	20.7	16.2
	10.0	25.0	22.5	17.6
	11.5	25.0	25.2	19.7
200	8.5	28.0	22.6	17.8
	9.0	28.0	23.6	18.5
	10.0	28.0	25.6	20.1
	11.0	28.0	27.6	21.7
	12.0	28.0	29.6	23.2
220	9.0	31.0	26.8	21.0
	10.0	31.0	29.0	22.8
	11.0	31.0	31.2	24.5
	12.0	31.0	33.4	26.2
240	9.5	34.0	31.2	24.4
	10.0	34.0	32.4	25.4
	11.0	34.0	34.9	27.4
	12.0	34.0	37.3	29.3
260	10.0	37.0	36.1	28.3
	11.0	37.0	38.7	30.3
	12.0	37.0	41.3	32.4
280	10.5	40.0	41.2	32.4
	11.0	40.0	42.6	33.5
	12.0	40.0	45.5	35.7
	13.0	40.0	48.4	37.9
300	11.0	43.0	46.7	36.7
	12.0	43.0	49.7	39.0
	13.0	43.0	52.8	41.5
320	11.5	46.0	52.6	41.2
	12.0	46.0	54.2	42.5
	13.0	46.0	57.4	45.0
	14.0	46.0	60.6	47.5
340	12.0	49.0	58.8	46.1
	13.0	49.0	62.2	48.8
	14.0	49.0	65.5	51.5
	15.0	49.0	69.0	54.2
370	12.5	53.5	67.8	53.1
	13.0	53.5	69.6	54.6
	14.0	53.5	73.3	57.5
	15.0	53.5	77.0	60.5
	16.0	53.5	80.7	63.5
400	13.0	58.0	77.4	60.8
	14.0	58.0	81.4	63.9
	15.0	58.0	85.4	67.0
	16.0	58.0	89.4	70.2
430	14.0	62.5	89.7	70.6
	15.0	62.5	94.1	73.9
	17.0	62.5	103.0	80.6
	20.0	62.5	115.0	90.8



SQUARE

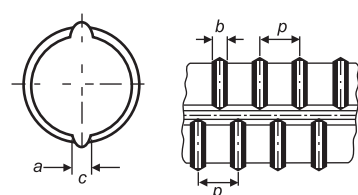
Section size	Unit weight M		Side length a		Section Area A		Moment of inertia	Radius of gyration	Modulus of section
	kg/m	lb/ft	mm	in	cm ²	in ²			
9	0.64	0.43	9	0.354	0.81	0.126	--	--	--
10	0.79	0.531	10	0.393	1.0	0.1549	--	--	--
12	1.13	0.759	12	0.47	1.44	0.2231	--	--	--
16	2.01	1.351	16	0.63	2.56	0.3968	0.0132	0.181	0.0415
18	2.54	1.707	18	0.709	3.24	0.5022	0.0209	0.205	0.0592
19	2.83	1.902	19	0.748	3.61	0.5596	0.0262	0.217	0.0696
20	3.14	2.11	20	0.787	4.0	0.612	--	--	--
22	3.80	2.553	22	0.866	4.84	0.7502	0.0468	0.252	0.108
25	4.91	3.299	25	0.984	6.25	0.9688	0.0783	0.283	0.159
28	6.15	4.133	28	1.102	7.84	1.215	0.123	0.319	0.223
30	7.07	4.751	30	1.181	9.00	1.395	0.162	0.343	0.275
32	8.04	5.403	32	1.260	10.24	1.587	0.210	0.362	0.333
35	9.62	6.464	35	1.378	12.25	1.899	0.300	0.398	0.436
36	10.2	6.854	36	1.417	12.96	2.009	0.336	0.409	0.475
38	11.3	7.593	38	1.496	14.44	2.238	0.418	0.433	0.558
44	15.2	10.21	44	1.732	19.36	3.001	--	--	--
50	19.6	13.17	50	1.969	25.00	3.875	1.252	0.567	1.269
55	23.7	15.93	55	2.165	30.25	4.689	1.833	0.626	1.69
60	28.3	19.02	60	2.362	36.00	5.580	2.595	0.681	2.197
65	33.2	22.31	65	2.559	42.25	6.549	3.580	0.740	2.795
70	38.5	25.87	70	2.756	49.00	7.595	4.805	0.795	3.491
75	44.2	29.70	75	2.953	56.25	8.719	6.342	0.854	4.29
80	50.2	33.73	80	3.150	64.00	9.92	8.192	0.909	5.205
85	56.7	38.10	85	3.346	72.25	11.20	10.450	0.965	6.224
90	63.6	42.74	90	3.543	81.00	12.56	13.141	1.024	7.384
95	70.8	47.58	95	3.740	90.25	13.99	16.312	1.079	8.726
100	78.5	52.75	100	3.937	100.0	15.50	20.012	1.138	10.191
110	95.0	63.84	110	4.331	121.0	18.76	29.309	1.252	13.547
120	113.0	75.93	120	4.724	144.0	22.32	41.562	1.362	17.575
130	133.0	89.37	130	5.118	169.0	26.20	57.177	1.476	22.334
140	154.0	103.5	140	5.512	196.0	30.38	76.877	1.591	27.888
150	177.0	118.9	150	5.906	225.0	34.88	101.381	1.705	34.295
160	201.0	135.1	160	6.299	256.0	39.68	131.195	1.819	41.679

BARS

DEFORMED AND ROUND

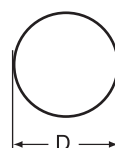
DEFORMED

Section size	Unit weight M	Cross section area A
mm	kg/m	cm ²
6.0	0.222	0.283
10.0	0.617	0.785
12.0	0.888	1.131
13.0	1.04	1.327
14.0	1.21	1.539
16.0	1.58	2.011
18.0	2.00	2.545
19.0	2.23	2.835
20.0	2.47	3.142
22.0	2.98	3.801
24.0	3.55	4.524
25.0	3.85	4.909
26.0	4.17	5.309
28.0	4.83	6.158
29.0	5.19	6.605
30.0	5.55	7.069
32.0	6.31	8.042
35.0	7.55	9.621
38.0	8.90	11.34



ROUND

Section size	Unit weight M	Cross section area A
mm	kg/m	cm ²
6.0	0.222	0.283
10.0	0.617	0.785
12.0	0.888	1.131
13.0	1.04	1.327
14.0	1.21	1.539
16.0	1.58	2.011
18.0	2.00	2.545
19.0	2.23	2.835
20.0	2.47	3.142
22.0	2.98	3.801
24.0	3.55	4.524
25.0	3.85	4.909
26.0	4.17	5.309
28.0	4.83	6.158
29.0	5.19	6.605
30.0	5.55	7.069
32.0	6.31	8.042
35.0	7.55	9.621
38.0	8.90	11.34
44.0	11.9	15.21
50.0	15.4	19.63
65.0	26.0	33.18
75.0	34.7	44.18
90.0	49.9	63.62
100.0	61.7	78.54
125.00	96.3	122.7
150.00	139.0	176.7
200.00	247.0	314.2



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PLATES

PLATES-SPECIFICATIONS

Application	Standard					
	KS	JIS	ASTM	DIN	BS	Others
Structural Steel Plates	KS D3503 SS34,41,50 KS D3514 SWS 41A,B,C SWS 50A,B,C SWS 50YA, YB SWS 53B,C SWS 58	JIS G3101 SS 34,41,50 JIS G3106 SM 41A,B,C SM 50A,B,C SM 50YA,B SM 58 JIS G3128 SHY 70N, NS	A283 Gr A,B,C,D A36 A284 Gr A,B,C,D A572 Gr 42,50,60,65 A633 Gr A,C,D,E A678 Gr B,C A656 A514 B,F	DIN 17100 U, RST 34-1,2 ST 37-1,2,3 U, RST 42-1 RST 42-2,3 RST 46-2,3 ST 50-1,2 ST 52-3 ST 60-1,2 ST 70-2	BS 4360 Gr 40A,B,C,D,E Gr 43A,B,C,D,E Gr 50A,B,C,D,E Gr 55C, E, F	ISO R630 Fe 33, 37 CAS G4021 IS 226 ST 42 (S) ST 42 (S,C) IS 2062 ST 42 (W) ST 42 (W,C) ISO R630 Fe 42,44,52
Weathering & Corrosion Resistant Steel Plates		JIS G3114 SMA 41A,B,C SMA 50A,B,C SMA 58	A441 A588 Gr A,B,C,D,E,F			
Steel Plates for Shipbuilding	KR RA,RB,RD,RE RA32,RD32,RE32 RA36,RD36,RE36	NK KA,KB,KD,KE KA32, KD32 KA36, KD36	A131 A,B,D,E ABS A,B,D,E AH32, DH32, EH32	GL A32, D32, E32 A36, D36, E36	LR AH32, DH32, EH32 AHS 34, DHS 34, EHS 34 AH36, DH36, EH36	NV, CR, BV, RINA A,B,C,D NV, RINA A27,D27,E27 A32,D32,E32 CR, BV A32,D32,E32 A36,D36,E36
Carbon Steel Plates for Boilers & Pressure Vessels	KS D3560 SBB 35,42, 46,49 KS D3540 SG 11,42,46,49 KS D3521 SPPV 24,32,36, 46,50	JIS G3103 SB 35,42,49 JIS G3118 SGV 42,46,49 JIS G3115 SPV 24,32,46 50	A285 Gr A,B,C A515 Gr 55,60,65,70 A516 Gr 55,60,65,70 A442 Gr 55, 60 A537 Class 1,2 A612	DIN 17155 HII, HII HIII, HIV 17Mn4 19Mn5 VDEhw680 TTST 41 TTST45	BS 1501-151 Gr 360A,B Gr 400A,B Gr 430A,B BS 1501-221 Gr 460A,B BS 1501-224 Gr 400A,B Gr 430A,B Gr 460A Gr 490A,B	

PLATES-SPECIFICATIONS

Application	Standard					
	KS	JIS	ASTM	DIN	BS	Others
Low Alloy Steel Plates for Boilers & Pressure Vessels	KS D3560 SBB46M,49M	JIS G3103 SB46M,49M JIS G3119 SBV 1A, 1B, 2,3 JIS G3120 SQV 1A,B, SQV 3A,B	A202 Gr A,B A204 Gr A,B,C A533 Type A Class 1,2,3 A533 Type C Class 1,2,3	WSN 5423 16Mo5 DIN 17155 15Mo3 16Mo5	BS 1501-240 1/4Mo BS 1501-261 1/4Mo	
Steel Plates for Low Temperature		JIS G3126 SLA 24 SLA 33A,B SLA 37 JIS G3127 SL3N26 SL3N28 SL3N45	A662 Gr A,B,C A203 Gr A	WDEhw680 TTST 41 TTST 45	BS 1501-224 Gr 430 A,B Gr 460 A,B	
Carbon Steel Pipes for Pressure Service	KS D3562 SPPS 38,42	JIS G3454 STPG 38,42				
Steel Plates for Line Pipes			A53 API 5L- A,B API 5LX - X42, X46,X52,X56, X60,X65,X70			
Automobile & Machine Steel Plates	KS D3519 SAPH 32 SAPH 38 SAPH 41 SAPH 45	JIS G3113 SAPH 32 SAPH 38 SAPH 41 SAPH 45				
Machine Structural Steel Plates	KS D3752 SM10C~ SM58C	JIS G4051 S10C~ S58C	SAE 1010 ~ SAE 1060			

PLATES

SPECIFICATIONS

(1) ABSTRACT FROM MAIN SPECIFICATION (Classified by Standards Association)

1. JIS Specifications (Japanese Industrial Standards) <table> <tr><td>G3101</td><td>Steel plates for general structures</td></tr> <tr><td>G3103</td><td>Steel plates for boilers</td></tr> <tr><td>G3106</td><td>Steel plates for welded structures</td></tr> <tr><td>G3114</td><td>Hot-rolled atmospheric corrosion resisting steel plates for welded structures</td></tr> <tr><td>G3115</td><td>Steel plates for pressure vessels</td></tr> </table>	G3101	Steel plates for general structures	G3103	Steel plates for boilers	G3106	Steel plates for welded structures	G3114	Hot-rolled atmospheric corrosion resisting steel plates for welded structures	G3115	Steel plates for pressure vessels	6. BS Specifications (British Standards Institution) <table> <tr><td>BS15</td><td>Steel plates for general structures</td></tr> <tr><td>BS1501-151</td><td>Carbon steel plates for pressure vessels</td></tr> <tr><td>BS2762</td><td>Notch ductile steel plates for bridges and general building construction</td></tr> <tr><td>BS4360</td><td>Weldable structural steel plates</td></tr> </table>	BS15	Steel plates for general structures	BS1501-151	Carbon steel plates for pressure vessels	BS2762	Notch ductile steel plates for bridges and general building construction	BS4360	Weldable structural steel plates								
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2. Kawasaki Specifications <table> <tr><td>VP</td><td>Steel plates for gas vessels</td></tr> <tr><td>AG</td><td>Steel plates for galvanizing pots</td></tr> <tr><td>PV</td><td>Steel plates for pressure vessels</td></tr> <tr><td>RIVER ACE</td><td>Quenched and tempered high tensile strength steel plates</td></tr> <tr><td>KH, HTP</td><td>High tensile strength steel plates</td></tr> <tr><td>RIVER TEN</td><td>Atmospheric corrosion resisting steel plates</td></tr> <tr><td>RIVER TEN</td><td>Atmospheric corrosion resisting steel plates for welded structures</td></tr> <tr><td>RIVER TEN-R</td><td>Atmospheric corrosion resisting steel plates for rolling stock</td></tr> <tr><td>RIVER TEN</td><td>Sulfuric-acid resisting steel plates</td></tr> <tr><td>MARINER</td><td>Anti-marine steel plates</td></tr> <tr><td>KL, RIVER ACE</td><td>Steel plates for low temperature service</td></tr> </table>	VP	Steel plates for gas vessels	AG	Steel plates for galvanizing pots	PV	Steel plates for pressure vessels	RIVER ACE	Quenched and tempered high tensile strength steel plates	KH, HTP	High tensile strength steel plates	RIVER TEN	Atmospheric corrosion resisting steel plates	RIVER TEN	Atmospheric corrosion resisting steel plates for welded structures	RIVER TEN-R	Atmospheric corrosion resisting steel plates for rolling stock	RIVER TEN	Sulfuric-acid resisting steel plates	MARINER	Anti-marine steel plates	KL, RIVER ACE	Steel plates for low temperature service	7. BV Specifications (Bureau Veritas) <table> <tr><td>BV</td><td>Steel for construction of hulls and engine parts</td></tr> </table> 8. CR Specifications (China Corporation Register of Shipping) <table> <tr><td>CR</td><td>Quality and testing of materials of hull, equipment, boiler and machinery</td></tr> </table>	BV	Steel for construction of hulls and engine parts	CR	Quality and testing of materials of hull, equipment, boiler and machinery
VP	Steel plates for gas vessels																										
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3. AB Specifications (American Bureau of Shipping Rules) <table> <tr><td>AB</td><td>Ordinary strength hull structural steel plates</td></tr> <tr><td>AB</td><td>High strength hull structural steel plates</td></tr> </table>	AB	Ordinary strength hull structural steel plates	AB	High strength hull structural steel plates	9. DIN Specifications (Deutscher Normenausschuss) <table> <tr><td>DIN 17100</td><td>Steel plates for general structural purposes</td></tr> </table>	DIN 17100	Steel plates for general structural purposes																				
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DIN 17100	Steel plates for general structural purposes																										
4. AAR Specifications (Association of American Railroads) <table> <tr><td>M128</td><td>High tensile strength plates for tank car tanks and other pressure vessels</td></tr> </table>	M128	High tensile strength plates for tank car tanks and other pressure vessels	10. CSA Specifications (Canadian Standards Association) <table> <tr><td>G40.8</td><td>Structural steel plates with improved resistance to brittle fracture</td></tr> </table>	G40.8	Structural steel plates with improved resistance to brittle fracture																						
M128	High tensile strength plates for tank car tanks and other pressure vessels																										
G40.8	Structural steel plates with improved resistance to brittle fracture																										
5. ASTM Specifications (American Society for Testing of Materials) <table> <tr><td>A113</td><td>Structural steel plates for locomotives and cars</td></tr> <tr><td>A131</td><td>Structural carbon steel plates for ships</td></tr> <tr><td>A242</td><td>High strength low alloy structural steel plates</td></tr> <tr><td>A283</td><td>Low and intermediate tensile strength carbon-steel plates of structural quality</td></tr> <tr><td>A387</td><td>Chromium-molybdenum alloy steel plates for pressure vessels</td></tr> <tr><td>A455</td><td>Carbon-manganese steel plates for pressure vessels</td></tr> <tr><td>A515</td><td>Carbon steel plates for pressure vessels for intermediate and higher temperature service</td></tr> <tr><td>A516</td><td>Carbon steel plates for pressure vessels for moderate and lower temperature service</td></tr> </table>	A113	Structural steel plates for locomotives and cars	A131	Structural carbon steel plates for ships	A242	High strength low alloy structural steel plates	A283	Low and intermediate tensile strength carbon-steel plates of structural quality	A387	Chromium-molybdenum alloy steel plates for pressure vessels	A455	Carbon-manganese steel plates for pressure vessels	A515	Carbon steel plates for pressure vessels for intermediate and higher temperature service	A516	Carbon steel plates for pressure vessels for moderate and lower temperature service	11. LR Specifications (Lloyd's Register of Shipping Rules) <table> <tr><td>LR</td><td>Mild steel plates for hull structures</td></tr> <tr><td>LR</td><td>Higher tensile steel plates for hull structures</td></tr> </table>	LR	Mild steel plates for hull structures	LR	Higher tensile steel plates for hull structures						
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	12. NK Specifications (Nippon Kaiji Kyokai) <table> <tr><td>NK</td><td>Ordinary steel plates for hull construction</td></tr> <tr><td>NK</td><td>High strength steel plates for hull construction</td></tr> </table>	NK	Ordinary steel plates for hull construction	NK	High strength steel plates for hull construction																						
NK	Ordinary steel plates for hull construction																										
NK	High strength steel plates for hull construction																										
	13. NV Specifications (Norske Veritas) <table> <tr><td>NV</td><td>Hull structural steel plates</td></tr> <tr><td>NV</td><td>Shipbuilding steel plates with controlled yield strength</td></tr> </table>	NV	Hull structural steel plates	NV	Shipbuilding steel plates with controlled yield strength																						
NV	Hull structural steel plates																										
NV	Shipbuilding steel plates with controlled yield strength																										
	14. GL Specifications (Germanischer Lloyd) <table> <tr><td>GL</td><td>Hull structural steel plates</td></tr> <tr><td>GL</td><td>High tensile hull structural steel plates</td></tr> </table>	GL	Hull structural steel plates	GL	High tensile hull structural steel plates																						
GL	Hull structural steel plates																										
GL	High tensile hull structural steel plates																										

(2) LIST OF SPECIFICATION (Classified by Tensile Strength)

Tensile strength class	Uses	Specifications	Tensile strength kg/mm ²	Yield point kg/mm ²
41	General structures	JIS G3101 SS41	41	t≤16 25 16<t≤40 24 40<t 22
	Welded structures	JIS G3106 SM41 A,B,C	41	t≤16 25 16<t≤40 24 40<t 22
		BS4360-40A.B.C.D.E	41	t≤16 23 16<t≤38 22 38<t≤64 22 38<t≤ 102 21
	Structures	ASTM A36	41	25
	Bridges, general structures	BS 2762 ND IA, IIA, IIIA	41	10<t≤19 24 19≤t 22
	Atmospheric corrosion resist steel	RIVER TEN 41	41	t≤16 25 16<t≤40 24 40<t 22
	Atmospheric corrosion resist steel, welded structures	RIVER TEN 41S	41	t≤16 25
	Boilers	JIS G31.14 SMA41 A,B,C	41	t≤16 19 16<t 22 40<t 24
		RIVER TEN 41 A,B,C	41	t≤16 25 16≤t 40 24 40<t 22
		NV2-0, 1,2, 3	41	--
		BV BP41	41	--
		CR Grade CRI	41	--
		DIN 17155 HII	41	t≤16 26 16<t≤ 40 25 40<t≤ 60 24
	Pressure vessels	JIS G3115 SPV24	41	t≤40 24 40<t 22
		BS 1501-151-26 A,B	41	t≤16 23 16<t≤ 32 23 32<t≤ 64 22
		BS 1501-211-26A, 221-26A.B	41	t≤16 24.1 16<t≤ 32 23.5 32<t≤ 64 22.7
		BS 1501-224-26	41	t≤16 26.8 16<t≤ 32 25.2 32<t≤ 64 23.6
	Gas vessels	RIVER TEN VP41	41	28
	Low temperature service	NK KT35	41	--
		RIVER TEN KL33A,B	41	33
	Automobiles	RIVER TEN KFR41	41	t≤6 26 t=8 24
	Wheels	RIVER TEN WH41	41	--
	Hull structures	NK Grade A, B, C, D, E	41	--
		AB Grade A, R, B, C, CS, D, E	41	--
		LR Grade A, B, C, D, E	41	--
		NV Grade A, W, C, D, E	41	--
		BV Grade A, B, C, D	41	--
		CR Grade A, B, C, D, E	41	--
		NDS XXG3101	41	23
		GL	41	--
		ASTM- A131A	41	22
42	General structures	DIN 17100 Ust 42-1, 2 RSt	42	t≤16 26 16<t ≤ 40 25 40<t≤ 100 24
		42-1, 2 St 42-3		
	Locomotives, cars	ASTM A113A	42	23
	Structures	ASTM A283D	42	23
	Machine structures	ASTM A284C	42	21
		ASTM A284D	42	23
	Structures	ASTM A441 (101.6 < t ≤ 203)	42	28
	Structures	IS 226, IS 2062	42	t<6-6<t≤ 20 26 20<t ≤40 24 40<t 23
	Structures	LR LM42	42	t≤16 23.5 16<t≤40 22.5 40t≤63 21.5

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SPECIFICATION

Tensile strength class	Uses	Specifications	Tensile strength kg/mm ²	Yield point kg/mm ²
42	Boilers	JIS G3103-SB42	42	23
		LR LP421, 422	42	--
		NK KP42A, B	42	0.5T
		AB APE, APL	42	22.5
	Pressure vessels	ASTM A387C	42	25
		ASTM A387D	42	21
	Low temperature vessels	ASTM A410LT5	42	21
	Pressure vessels	ASTM A442-60	42	22
	Pressure vessels, intermediate and higher temperature	ASTM A515-60	42	22
	Pressure vessels, moderate and lower temperature	ASTM A516-60	42	22
	Pressure vessels	RIVER TEN PV42A	42	TSX0.5
44	Structures	ASTM A242 Type	44	29.5
	Structures	ASTM A440 (38.1<t≤101.6)	44	30
		ASTM A441 (38.1<t≤101.6)	44	30
	General structures	BS 15-2	44	9.53<t≤19.05 25 19.05<t≤38.1 24
		BS 15-3	44	38.1≤t 23
		BS 1449 En2D/A, 2D/B, 2D	44	25
	Railroads	BS 24-621, 631	44	--
	Bridges, general structures	BS 2762 NDIB, IIB, IIIB, IVB	44	10<t≤19 25 19<t 24
	Welded structures	BS4360-43A.B.C.D.E	44	t≤16 25 16<t≤38 24
			44	38<t≤64 23 64<t≤102 22
	Structures	DIN 17100 RSt46-2, St46-3	44	t≤16 29 16<t≤40 28
			44	40≤t≤100 27
	Structures	IRS-M5	44	--
	Boilers	NV NV3-0, 1,2	44	t≤16 28 16<t≤40 26
			44	46<t≤100 25
		NV NV7-1	44	t≤16 28 16<t≤40 27
	Pressure vessels	NV NV7-2	44	46<t≤100 27
		BS 1501-151-28A	44	t≤16 31 16<t≤40 30
			44	40<t≤100 30
		BS 1501-151-28B	44	t≤16 25 16<t≤32 24
			44	32<t≤64 23
		BS 1501-161-28A	44	t≤16 25 16<t≤32 25
			44	32<t≤64 24
		BS 1501-161-28B	44	t≤16 25.2 16<t≤32 23.6
			44	32<t≤64 23.3
		BS 1501-211-28A, 221-28A	44	t≤16 25.2 16<t≤32 25.2
			44	32<t≤64 23.9
		BS 1501-211-28B, 221-28B	44	t≤16 26 16<t≤32 25.2
			44	32<t≤64 24.4
	Boilers	BS 1501-224-28	44	t≤16 26 16<t≤32 26
			44	32<t≤64 24.7
		DIN 17155 HIII	44	t≤16 29.1 16<t≤32 27.6
			44	32<t≤64 26
		DIN 17155 15Mo3	44	t≤16 28 16<t≤40 27

SPECIFICATIONS

Tensile strength class	Uses	Specifications	Tensile strength kg/mm ²	Yield point kg/mm ²
44	Boilers	DIN 17155 13CrMo44	44	t≤16 31 16<t≤40 30 40<t≤60 30
45	Gas vessels	RIVER TEN VP45	45	30
	Wheels	RIVER TEN WH45	45	31
	Hull structures	GL-F32, G32	45	t≤35 32 35<t≤50 31
	Hull structures	NDS XXG3102B NS30C	45	30
	Boilers	NV7-3	45	29
	Low temperature service	RIVER TEN KL36B	45	36
	Structures	CSA G40.8	45	t≤15.9 28 15.9<t≤25.4 26 25.4<t≤38.1 25
	Vessels, intermediate and higher temperature	ASTM A515-65	45	24
46	Hull structures	NDS XXG3102B-NS30B	46	31
	Boilers	JIS G3103 SB46	46	25
	Boilers	JIS G3103 SB46M	46	25
		NK KP46	46	0.5T
		AB APF, APM	46	24.5
		AB APH	46	26
	Boilers, pressure vessels	ASTM A203A, A203D	46	26
		ASTM A204A	46	26
	Pressure vessels	ASTM A387B	46	28
	Boilers, pressure vessels, intermediate and higher temperature	ASTM A515-65	46	25
	Pressure vessels, moderate and lower temperature	ASTM A516-65	46	25
	Low temperature service	NK KT50N, 50Q	46	--
	Low temperature pressure vessels	ASTM A203ALT, DLT	46	26
	Rolling stock (Atmospheric corrosion resisting)	RIVER TEN-R(C)	46	32
47	Structures	ASTM A440 (19<t≤38)	47	32
		ASTM A441 (19<t≤38)	47	32
	Pressure vessels	BS 1501-211-30A, 221-30A	47	t≤16 27.9 16<t≤32 26.9 32<t≤64 26.5
		BS 1501-211-30B, 221-30B	47	t≤16 27.9 16<t≤32 27.9 32<t≤26.5
		BS 1501-213-30	47	t≤16 32.3 16<t≤32 31.5 32<t≤64 30.7
		BS 1501-224-30	47	t≤16 31.5 16<t≤32 29.9 32<t≤64 28.3
	Boilers	LP LP471,472	47	--
		NV Grade NV5-0, 1,2	47	t≤16 29 16<t≤40 28 40<t≤100 26
		DIN 17155 HIV	47	t≤16 29 16<t≤40 28 40<t≤6 27
		DIN 17155 17Mn4	47	t≤16 29 16<t≤40 28 40<t≤60 28

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SPECIFICATIONS

Tensile strength class	Uses	Specifications	Tensile strength kg/mm ²	Yield point kg/mm ²
49	Structures	ASTM A242 Type 1	49	35
	Structures	ASTM A440 (t≤19)	49	35
		ASTM A441 (t≤19)	49	35
		Boilers	JIS G3103 SB49, 49M	49
	Boilers, pressure vessels	NK KP49	49	27
		ASTM A203B, A203E	49	28
		ASTM A204B	49	28
		ASTM A225A	49	28
		AB APG, API	49.5	27
	Boilers, pressure vessels, intermediate and higher temperature	ASTM A515-70	49	26
Boilers, pressure vessels, moderate and lower temperature	ASTM A516-70FR, F	49	27	
Low temperature pressure vessels	ASTM A203ELT	49	28	
Rolling stock (Atmospheric corrosion resisting)	RIVER TEN-R(H,P)	49	35	
50	General structures	JIS G3101 SS50	50	t≤16 29 16<t≤40 28 40<t 26
	Welded structures	JIS G3106 SM50A,B,C	50	t≤16 33 16<t≤40 32 40<t 30
		JIS G3106 SM50YA, YB	50	t≤16 37 16<t≤40 36 40<t 34
		BS4360-50A.B.C.D.E	50	t≤16 36 16<t≤38 35
			39<t≤64 34 64<t≤102 33	
	Gas vessels	RIVER TEN VP50	50	36
	High yield point structures	BS 968	50	t≤9.5 36 9.5<t≤16 36
				16<t≤31.7 35 31.7<t≤50.8 36
		50.8 ≤ t-		
	Structures	DIN 17100 St50-1, 50-2	50	t≤16 30 16<t≤ 40 29
			40<t≤100 28	
	Hull structures	NK Kind of No. 1 Grade A,D,E	50	32
		AB Grade AH,BH,CH,DHS,EH	50	32
		GL-F36, G36	50	t≤35 36 35<t≤50 35
	High tensile steel	RIVER TEN KH36A,B	50	t≤16 37 16<t≤40 36 40<t 34
	Atmospheric corrosion resist, steel	RIVER TEN 50M	50	t≤16 33 16<t≤40 32 40<t 30
	Atmospheric corrosion resist, welded structures	JIS G3114 SMA50A,B,C	50	t≤16 19 16<t 22 40<t 24
		RIVER TEN 50A,B,C	50	t≤16 37 16<t≤40 36 40<t 34
	Sulfuric-acid corrosion resisting steel	RIVER TEN 50K	50	33
	Anti-marine steel	RIVER TEN MARINER	50	36
	Boilers, pressure vessels	JIS G3115 SPV32	50	t≤40 32 40<t 30
		NV Grade NV4-0,1,2,3,4	50	t≤16 34 16<t≤40 33
			40<t≤100 32	
	Pressure vessels	NV Grade NV25-1,2,3,4	50	22
		BS 1501-211-32A, 221-32A	50	t≤16 29.6 16<t≤32 29
			32<t≤64 27.9	
	BS 1501-211-32B, 221-32B	50	t≤16 29.6 16<t≤32 29.6	
			32<t≤64 28.8	

SPECIFICATIONS

Tensile strength class	Uses	Specifications	Tensile strength kg/mm ²	Yield point kg/mm ²
50	Pressure vessels	BS 1501-213-32	50	t≤16 34.6 16<t≤32 33.9 32<t≤64 33.1
		BS 1501-224-32	50	t≤16 33.1 16<t≤32 31.5 32<t≤64 29.9
	Low temperature service Structures	NK Kind of No.2 K5T-50Q	50	--
		DIN 17100 St50, 50-2	50	t≤16 30 16<t≤40 29 40<t≤100 28
52	Structures	DIN 17100 St52-3	52	t<16 36 16<t≤30 35 30<t≤50 34
	Boilers	LR LP521,522	52	--
		IS 2002-2B	52	T.S. x 0.50
		NV Grade NV6	52	t≤16 33 16<t≤40 32 40<t≤100 31
		DIN 17155 19Mn5	52	t≤16 33 16<t≤40 32 46<t≤60 32
	Pressure vessels	ASTM A302A	52	31
	Pressure vessels	ASTM A302A	52	31
53	Welded structures	JIS G3106 SM53B,C	53	t≤16 37 16<t≤40 36 40<t 34
	High tensile steel, welded structures	WES (HPI) Kind of No.1 HW36	53	36
	Atmospheric corrosion resist, welded structures	RIVER TEN 53B,C	53	t≤16 37 16<t≤40 36 40<t 34
	Boilers, pressure vessels	AB APJ	53	30.5
		ASTM A204C	53	30
		ASTM A299	53	30
		ASTM A302A	53	32
		JIS G3115 SPV36	53	t≤40 36 40<t 34
	Pressure vessels	ASTM A387D, A387E	53	32
	Unfired pressure vessel	ASTM A455A	53	26
55	General structures	JIS G3101 SS55	55	t<16 41 16<t≤40 40 40<t -
	Structures	IS 961 St58-HT, HTC	55	6<t≤28 36 28<t≤45 35 45<t≤63 52 63<t 50
	Welded structures	BS4360-55C.E	55	t<16 45 16<t≤38 44 38<t≤64 42
	Railroads	BS24-632A, B	55	t≤25.4 33 t>25.4 31
	Hull structures	GL-G42	55	t≤35 42 35<t≤50 41
	Gas vessels	RIVER TEN VP55	55	38
	High tensile steel	RIVER TEN KH40A, B	55	t≤16 41 16<t≤40 40 40<t 38
		RIVER TEN HP55W,WA,WB	55	36
	Abrasion resist steel	RIVER TEN CrMn55	55	34
	Abrasion resist steel	RIVER TEN CrMn55	55	34
56	Boilers	JIS G3103 SB56M	56	35
	Boilers, pressure vessels	ASTM A302B,C,D	56	35
57	Tank car tanks, pressure vessels	AAR M128	57	35
	High tensile steel, welded structures	RIVER TEN HTP57W,WA,WB	57	40
		WES (HPI) Kind of No.2 HW40	57	40
58	Welded structures	JIS G3106 SM58	58	t≤16 47 16<t≤40 46 40<t 44
	Bridges, general structures	BS548	58	6.4<t≤31.8 36 31.8<t≤44.5 35 44.5<t≤57.1 33 57.1<t≤69.9 32 69.9≤t 30

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SPECIFICATIONS

Tensile strength class	Uses	Specifications	Tensile strength kg/mm ²	Yield point kg/mm ²
58	Structures	IS 961 St55-HTW. HTWC	58	6<t≤16 36 16<t≤32 35 32<t≤63 34 63<t 29
	Hull structures	NDS XXG3103 NS46B, 46C	58	46
	Pressure vessels	IIS G3115 SPV 46	58	t≤40 46 40<t 46
	Atmospheric corrosion resist, welded structures	JIS G3114 SMA58 RIVER TEN 58	58 58	t≤16 19 16<t 26 40 < t 20 t≤16 47 16<t≤40 46 40<t 44
60	Structures	DIN 17100 St60-1, 2	60	t≤16 34 16<t≤40 33 40<t≤100 32
	Hull structures	NK Kind of No.2 Grade A,D,E	60	46
	Quenched and tempered high tensile steel	RIVER ACE 60	60	50
	Non-tempered high tensile steel	RIVER TEN HTP60W	60	45
	High tensile steel, welded structures	WES (HPI) Kind of No.3 HW45	60	45
	Abrasion resist steel	RIVER TEN Cr Mn60	60	40
62	Quenched and tempered high tensile steel	RIVER ACE 60H	62	50
	Atmospheric corrosion resist. high tensile, welded structures	WES (HPI) Kind of No.4 HW50	62	50
	Atmospheric corrosion resist, welded structures	RIVER TEN 62	62	t≤16 50 16<t≤40 50 40<t 50
	Low temperature pressure vessels	JIS G3115 SPV50	62	t≤40 50 40<t -
	Low temperature service	RIVER ACE 60L	62	50
68	High tensile steel, welded structures	WES (HPI) Kind of No.5 HW56	68	56
70	Structures	DIN 17100 St70-2	70	t≤16 37 16<t≤40 36 40<t≤100 35
	Quenched and tempered high tensile steel	RIVER ACE 70	70	60
	Low temperature pressure vessels	ASTM A353LT4	70	53
74	Quenched and tempered high tensile steel	RIVER ACE 70H	74	63
	High tensile steel, welded structures	WES (HPI) Kind of No.6 HW63	74	63
	Low temperature service	RIVER ACE 70L	74	63
80	Quenched and tempered high tensile steel	RIVER ACE K-O	80	70
	High tensile steel, welded structures	WES (HPI) Kind of No.7 HW70	80	70
88	High tensile steel, welded structures	WES (HPI) Kind of No. 8 HW80	88	80
97	High tensile steel, welded structures	WES (HPI) Kind of No.9 HW90	97	90

Imperial Units JIS (18.524 kg/in ft²)

Thickness		Unit weight	Width x Length, ft							
t		M	4 x 8	5 x 10	5 x 20	5 x 30	6 x 20	6 x 24	6 x 30	8 x 30
in	mm	kg/ft ²	32	50	100	150	120	144	180	240
1/16	1.59	1.158	37.1	57.9	116	174	139	167	208	278
1/8	3.18	2.316	74.1	116	232	347	278	334	417	556
3/16	4.76	3.473	111	174	347	521	417	500	625	834
1/4	6.35	4.631	148	232	463	695	556	667	834	1,111
5/16	7.94	5.789	185	289	579	868	695	834	1,042	1,389
3/8	9.53	6.946	222	347	695	1,042	834	1,000	1,250	1,667
7/16	11.11	8.104	259	405	810	1,256	972	1,167	1,459	1,945
1/2	12.70	9.262	296	463	926	1,389	1,111	1,334	1,667	2,223
9/16	14.24	10.42	333	521	1,042	1,563	1,250	1,500	1,876	2,501
5/8	15.88	11.58	371	579	1,158	1,737	1,390	1,668	2,084	2,779
11/16	17.46	12.74	408	637	1,274	1,911	1,529	1,835	2,293	3,058
3/4	19.05	13.89	444	694	1,389	2,084	1,667	2,000	2,500	3,334
13/16	20.64	15.05	482	752	1,505	2,254	1,806	2,167	2,709	3,612
7/8	22.23	16.21	519	810	1,621	2,432	1,945	2,334	2,918	3,890
15/16	23.81	17.37	556	868	1,737	2,606	2,084	2,501	3,127	4,169
1	25.40	18.52	593	926	1,852	2,778	2,222	2,667	3,334	4,445
1 1/8	28.58	20.84	667	1,042	2,084	3,126	2,501	3,001	3,751	5,002
1 1/4	31.75	23.16	741	1,158	2,316	3,474	2,779	3,335	4,169	5,558
1 3/8	34.93	25.47	815	1,274	2,547	3,820	3,056	3,668	4,585	6,113
1 1/2	38.10	27.79	889	1,390	2,779	4,168	3,335	4,002	5,002	6,670
1 5/8	41.28	30.10	963	1,505	3,010	4,515	3,612	4,334	5,418	7,224
1 3/4	44.45	32.42	1,037	1,621	3,242	4,863	3,890	4,668	5,836	7,781
1 7/8	47.63	34.73	1,111	1,736	3,473	5,210	4,168	5,001	6,251	8,335
2	50.80	37.05	1,186	1,852	3,705	5,558	4,446	5,335	6,669	8,892
2 1/2	63.50	46.31	1,482	2,316	4,631	6,946	5,557	6,669	8,336	11,114
3	76.20	55.57	1,778	2,778	5,557	8,336	6,668	8,002	10,003	13,337
4	101.60	74.10	2,371	3,705	7,410	11,115	8,892	10,670	13,338	17,784
5	127.00	92.62	2,964	4,631	9,262	13,893	11,114	13,337	16,672	22,229
6	152.40	111.1	3,557	5,558	11,114	16,672	13,338	16,005	20,006	26,675

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Metric Units (7.85 kg/mm m²-0.7293 kg/mm ft²)

Thickness	Unit weight	Width x Length, ft										
t	M	3 x 6	4 x 8	4 x 10	4 x 16	4 x 20	5 x 10	5 x 20	5 x 30	5 x 40	6 x 30	6 x 40
mm	kg	18	32	40	64	80	50	100	150	200	180	240
1.20.8752	15.8	28.0	35.0	56.0	70.0	43.8	87.5	131	175	158	210	
1.61.167	21.0	37.3	46.7	74.7	93.4	58.4	117	175	233	210	280	
1.91.386	24.9	44.4	55.4	88.7	111	69.3	139	208	277	249	333	
2.31.677	30.2	53.7	67.1	107	134	83.9	168	252	335	302	402	
2.61.896	34.1	60.7	75.8	121	152	94.8	190	284	379	341	455	
3.02.188	39.4	70.0	87.5	140	175	109	219	328	438	394	525	
3.22.334	42.0	74.7	93.4	148	187	117	233	350	467	420	560	
4.53.282	59.1	105	131	210	263	164	328	492	656	591	788	
5.03.646	65.6	117	146	233	292	182	365	547	729	656	875	
6	4.376	78.8	140	175	280	350	219	438	656	875	788	1,050
7	5.105	91.9	163	204	327	408	255	510	766	1,021	919	1,225
8	5.834	105	187	233	373	467	292	583	875	1,167	1,050	1,400
9	6.564	118	210	263	420	525	328	656	985	1,313	1,182	1,575
10	7.293	131	233	292	467	583	365	729	1,094	1,459	1,313	1,750
11	8.022	144	257	321	513	642	401	802	1,203	1,604	1,444	1,925
12	8.752	158	280	350	560	700	438	875	1,313	1,750	1,575	2,100
12.7	9.262	167	296	370	593	741	463	926	1,389	1,852	1,667	2,223
13	9.481	171	303	379	607	758	474	948	1,422	1,896	1,707	2,275
14	10.21	184	327	408	653	817	510	1,021	1,532	2,042	1,838	2,450
15	10.94	197	350	438	700	875	547	1,094	1,641	2,188	1,969	2,626
16	11.67	210	373	467	747	934	584	1,167	1,750	2,334	2,101	2,801
17	12.40	223	397	496	794	992	620	1,240	1,860	2,480	2,232	2,976
18	13.13	236	420	525	840	1,050	656	1,313	1,970	2,625	2,363	3,151
19	13.86	249	444	554	887	1,109	693	1,386	2,079	2,772	2,495	3,326
20	14.59	263	467	584	934	1,167	730	1,459	2,188	2,918	2,626	3,502
21	15.32	276	490	613	980	1,226	766	1,532	2,298	3,064	2,758	3,677
22	16.04	289	513	642	1,027	1,283	802	1,604	2,406	3,208	2,887	3,850
23	16.77	302	537	671	1,073	1,342	838	1,677	2,516	3,354	3,019	4,025
24	17.50	315	560	700	1,120	1,400	875	1,750	2,625	3,500	3,150	4,200
25	18.23	328	583	729	1,167	1,458	912	1,823	2,734	3,646	3,281	4,375
25.4	18.52	333	593	741	1,185	1,482	926	1,852	2,778	3,704	3,334	4,445
26	18.96	341	607	758	1,213	1,517	948	1,896	2,844	3,792	3,413	4,550
27	19.69	354	630	788	1,260	1,575	984	1,969	2,954	3,938	3,544	4,726
28	20.42	368	653	817	1,307	1,634	1,021	2,042	3,063	4,084	3,676	4,901
29	21.15	381	677	846	1,354	1,692	1,058	2,115	3,172	4,230	3,807	5,076
30	21.88	394	700	875	1,400	1,750	1,094	2,188	3,282	4,376	3,938	5,251
32	23.34	420	747	934	1,494	1,867	1,167	2,334	3,501	4,668	4,201	5,602
34	24.80	446	794	992	1,587	1,984	1,240	2,480	3,720	4,960	4,464	5,952
36	26.25	472	840	1,050	1,680	2,100	1,312	2,625	3,938	5,250	4,725	6,300
38	27.71	499	887	1,108	1,773	2,217	1,386	2,771	4,156	5,542	4,988	6,650
40	29.17	525	933	1,167	1,867	2,334	1,458	2,917	4,376	5,834	5,251	7,001
45	32.82	591	1,050	1,313	2,100	2,626	1,641	3,282	4,923	6,564	5,908	7,877
50	36.46	656	1,167	1,458	2,334	2,917	1,823	3,646	5,469	7,292	6,563	8,750
55	40.11	722	1,284	1,604	2,567	3,209	2,006	4,011	6,016	8,022	7,220	9,626
60	43.76	788	1,400	1,750	2,801	3,501	2,188	4,376	6,564	8,752	7,877	10,502
65	47.40	853	1,517	1,896	3,034	3,792	2,370	4,740	7,110	9,480	8,532	11,376
70	51.05	919	1,634	2,042	3,267	4,084	2,552	5,105	7,658	10,210	9,189	12,252
80	58.34	1,050	1,867	2,334	3,734	4,667	2,917	5,834	8,751	11,670	10,500	14,002
90	65.64	1,182	2,100	2,626	4,201	5,251	3,282	6,564	9,846	13,128	11,815	15,754
100	72.93	1,313	2,334	2,917	4,668	5,834	3,647	7,293	10,940	14,586	13,127	17,503
110	80.22	1,444	2,567	3,209	5,134	6,418	4,011	8,022	12,033	16,044	14,440	19,253
120	87.52	1,575	2,801	3,501	5,601	7,002	4,376	8,752	13,128	17,504	15,754	21,005
130	94.81	1,707	3,034	3,792	6,080	7,585	4,741	9,481	14,222	18,962	17,066	22,754
140	102.11,838	3,267	4,084	6,534	8,168	5,105	10,210	15,315	20,420	18,378	24,504	
150	109.4	1,969	3,501	4,376	7,002	8,752	5,470	10,940	16,410	21,880	19,692	26,256

TECHNICAL SPECIFICATION REFERENCE - STEEL PLATES, SHEETS

Excerpts from JIS G 3131, JIS G 3101, JIS G 3302 & JIS G 3141 Specifications

Classification	Specification	Grade	Chemical Composition					Tensile Test					Bend Test ^{*2}	
			C max. %	Mn max. %	Si max. %	P max. %	S max. %	Tensile Strength N/mm ²	Yield Strength N/mm ²	Elongation			Bending Angle	Bending Radius
										Test PC	Thickness	%		
Hot Rolled Steel Plates and Sheets & Chequered Plates	JIS G 3131	SPHC	0.15	0.60	--	0.05	0.05	270 min.	--	No.5	1.2mm & over, up to 1.6 mm	27 min.	180°	Flat on itself 0.5t
											1.6mm & over, up to 3.2 mm	29 min.		
											3.2mm & over	31 min.		
	JIS G 3101	SS330	--	--	--	0.05	0.05	330 - 430	205 min.	No.5	5 mm & under	26 min.	180°	0.5t
										No.1A	over 5mm- 16mm	21 min.		
	JIS G 3101	SS400	--	--	--	0.05	0.05	400 - 510	245 min.	No.5	5 mm & under	21 min.	180°	1.5t
										No.1A	over 5mm- 16mm	17 min.		
Galvanised Steel Sheets	JIS G 3302	SGCC	--	--	--	--	--	270 min.	205 min.	--	Up to 1.6 mm	--	180°	t ^{*1} t ^{*1} 2t ^{*1}
											1.6 mm & over, up to 3.0 mm			
											3.0 mm & over			
	JIS G 3302	SGHC	--	--	--	--	--	270 min.	205 min.	--	Up to 1.6 mm	--	180°	t ^{*1} t ^{*1} 2t ^{*1}
											1.6 mm & over, Up to 3.0 mm			
											3.0 mm & over			
Cold Rolled Steel Sheets	JIS G 3141	SPCC	0.12	0.50	--	0.04	0.045	270 min.	--	No. 5	0.6 mm & over, up to 1.0 mm	36 min.	180°	Flat on itself
											1.0 mm & over, up to 1.6 mm	37 min.		
											1.6mm & over, up to 2.5 mm	38 min.		
											2.5 mm & over	39 min.		

Notes:

*1 - Up to grade Z27 maximum

*2 - Bend Test test piece No. 3

t - Thickness

PLATES

CHEQUERED (FLOOR) PLATES

WEIGHT TABLE

Thickness	Unit weight	Width x Length							
		(mm)							
mm	kg/m ²	914 x 1,829	914 x 3,658	1,219 x 2,438	1,219 x 3,048	1,219 x 4,877	1,219 x 6,096	1,524 x 3,048	1,524 x 6,096
2.3	19.73	33.0	66.0	58.6	73.3	117	147	91.6	183
3.0	25.26	42.2	84.5	75.1	93.9	150.2	187.7	117.3	235
3.2	26.79	44.8	89.6	79.6	99.6	159	199	124	249
4.5	36.99	61.8	124	110	137	220	275	172	344
6.0	48.77	81.5	163	145	181	290	362	227	453
8.0	64.47	108	216	192	240	383	479	299	599
9.0	72.32	121	242	215	269	430	537	336	672
10	80.17	134	268	238	298	477	596	372	745
12	95.87	167	320	285	356	570	712	445	891

WEIGHT TABLE

Thickness	Unit weight	Width x Length							
		(inch)							
in	lb/ft ²	3 x 6	3 x 12	4 x 8	4 x 10	4 x 16	4 x 20	5 x 10	5 x 20
0.091	4.04	72.75	145.5	129.2	161.6	257.9	324.1	201.9	403.4
0.118	5.17	93.09	186.2	165.5	206.9	331.0	413.7	258.6	517.2
0.126	5.49	98.77	197.5	175.5	219.6	350.5	438.7	273.4	549
0.177	7.58	136.2	273.4	242.5	302	485	606.3	379.2	758.4
0.236	9.99	179.7	359.4	319.7	399	639.3	798.1	500.4	998.7
0.315	13.2	238.1	476.2	423.3	529.1	844.4	1,056	659.2	1,321
0.354	14.8	266.8	533.5	474	593.9	948	1,184	740.8	1,482
0.394	16.4	295.4	590.8	524.7	657	1,052	1,314	820.1	1,642
0.472	19.6	368.2	705.5	628.3	748.8	1,257	1,570	981.1	1,964

COLD ROLLED COILS AND SHEETS - SPECIFICATIONS

Application		Standard						
		KS	JIS	ASTM	BS	DIN		POSCO
						1623	1624	
Commercial Quality		SCP 1	SPCC	A366	CR 4 CS 4	St 12	St 0 St 1	CSP 1 CSP 1D
Drawing Quality		SCP 2	SPCD	A619	CR 3 CS 3	St 13	St 2	CSP 2
Deep Drawing Quality		SCP 3 SCP 3N	SPCE SPCEN	A620	CR 2 CS 2	St 14	St 3 St 4	CSP 3 CSP 3N
Extra Deep Drawing Quality		--	--	--	CR 1 CS 1	--	--	CSP 3E
Structural Quality		--	--	A611	CR 34/20 CR 37/23	--	--	--
Porcelain Enameling	Commercial	--	SPP	A424	--	--	--	CESP 1
	Drawing							CESP 2
Superior Atmospheric Corrosion Resisting		--	SPA-C	--	--	--	--	CASP
High Strength Steel		--	--	--	--	--	--	CHSP

GENERAL COLD ROLLED STEELS

Classification	Standard			Characteristics	End use
	KS	JIS	POSCO		
Class 1	SCP 1	SPCC	CSP 1 CSP 1D	General steel used for flatting, bending process (CSP 1) and light drawing (CSP 1D)	Refrigerator, Laundry machine, Other electrical appliances, Drum, etc.
Class 2	SCP 2	SPCD	CSP 2	This quality has a better characteristics of processing for drawing quality	Drawing parts of automobile door, etc.
Class 3	SCP 3	SPCE	CSP 3	This quality improved characteristics of processing of deep drawing material through metallurgical adjustment of metal grain boundary	Automobile fender, Electrical cable case
	SCP 3N	SPCEN	CSP 3N	(Ç: Guarantee non-aging characteristics when designated by the buyer)	
	--	--	CSP 3E	Extra deep drawing steel which has superior characteristics of .drawability than SCP 3N	Automobile rear floor panel, etc.

PLATES

COLD ROLLED STEEL SHEETS

(JIS G3141)

DIMENSIONS AND WEIGHTS

For Reference only		Thickness	Sizes	lb/pce	kg/pce	Pcs/M. Ton
Gauge No.	mm	mm	ft			
30 SWG	0.315	0.3	3 x 6	8.6821	3.9382	253.9231
30 BG	0.3124		3 x 8	11.5762	5.2509	190.4435
30 USG	0.318		4 x 8	15.4349	7.0012	142.8327
30 SWG	0.300					
28 BWG	0.360	0.35	3 x 6	10.1292	4.5946	217.6468
29 SWG	0.3454		3 x 8	13.5056	6.1261	163.2360
29 USG	0.357		4 x 8	18.0075	8.1682	122.4260
29 BG	0.3531					
26 BWG	0.460	0.45	3 x 6	13.0233	5.9074	169.2792
26 SWG	0.457		3 x 8	17.3644	7.8765	126.9600
27 BG	0.4432		4 x 8	23.1525	10.5019	95.2209
25 BWG	0.510	0.5	3 x 6	14.4703	6.5637	152.3531
25 SWG	0.508		3 x 8	19.2938	8.7517	114.2636
26 BG	0.4981		4 x 8	25.7250	11.6688	85.6986
24 BWG	0.560	0.55	3 x 6	15.9174	7.2201	138.5023
24 SWG	0.559		3 x 8	21.2231	9.6268	103.8767
25 BG	0.5598		4 x 8	28.2975	12.8357	77.9078
23 SWG	0.610	0.6	3 x 6	17.3644	7.8765	126.9600
23 BWG	0.640		3 x 8	23.1525	10.5019	95.2209
24 USG	0.635		4 x 8	30.8700	14.0026	72.5624
24 BG	0.6298					
22 SWG	0.711	0.7	3 x 6	20.2584	9.1892	108.8234
22 BWG	0.710		3 x 8	27.0112	12.2522	81.6180
23 USG	0.714		4 x 8	36.0150	16.3363	61.2134
23 BG	0.7066					
21 BWG	0.810	0.8	3 x 6	23.1525	10.5019	95.2209
21 SWG	0.813		3 x 8	30.8700	14.0026	71.4153
22 USG	0.794		4 x 8	41.1600	18.6701	53.5616
20 BWG	0.890	0.89	3 x 6	25.7572	11.6834	85.5916
21 BG	0.8865		3 x 8	34.3429	15.5779	64.1935
			4 x 8	45.7905	20.7705	48.1452
			4 x 10	57.2400	25.9639	38.5046
20 SWG	0.914	0.914	3 x 6	26.4517	11.9985	83.3438
			3 x 8	35.2690	15.9980	62.5079
			4 x 8	47.0253	21.3306	46.8810
19 BWG	1.070	1.0	3 x 6	28.9406	13.1274	76.1765
19 SWG	1.016		3 x 8	38.5875	17.5032	57.1325
20 USG	0.952		4 x 8	51.4500	23.3376	42.8493
20 BG	0.9957					
18 SWG	1.219	1.2	3 x 6	34.7287	15.7529	63.4804
18 BG	1.257		3 x 8	46.3050	21.0039	47.6102
18 BWG	1.240		4 x 8	61.7399	28.0051	35.7078
18 USG	1.270					
16 USG	1.588	1.5	3 x 6	43.4109	19.6911	50.7844
17 BWG	1.470		3 x 8	57.8812	26.2548	38.0883
			4 x 8	77.1749	35.0063	28.5663
14 SWG	2.032	2.0	3 x 6	57.8812	26.2548	38.0883
14 USG	1.984		3 x 8	77.1749	35.0063	28.5663
14 BG	1.994		4 x 8	102.8999	46.6751	21.4247
12 SWG	2.642	2.6	3 x 6	75.2454	34.1311	29.2908
10 BG	2.590		3 x 8	100.3272	45.5081	21.9681
			4 x 8	133.7696	60.6775	16.4761

ELECTROLYTIC GALVANISED COILS AND SHEETS - SPECIFICATIONS

Application	Standard										
	Pure Zn				Fe-Zn	Ni-Zn	Pb-Sn	Base Metal			
	KS	JIS	ASTM	POSCO	POSCO Designation			KS	JIS	ASTM	POSCO
Commercial	SECC		A591-CQ	EGSC	EFSC	ENSC	ETSC	--	SPCC	A366	CSP 1
Drawing	SECD		A591-DQ	EGSD	EFSD	ENSD	ETSD	SCP 2	SPCD	A619	CSP 2
Deep Drawing	SECE		A591-DQ SK	EGSE	EFSP	ENSP	ETSP	SCP 3	SPCE	A620	CSP 3
Non-Aging Deep Drawing	SECEN			EGSN	EFSN	ENSN	--	SCP 3N	SPCEN	A620	CSP 3N
Extra Deep Drawing	--		--	EGSE	EFSE	ENSE	--	--	--	--	CSP 3E
High Strength Steel	--		--	EGCHSP 35/60	EFCHSP 35/60	ENCHSP 35/60	--	--	--	--	CHSP 35/60

WEIGHT TABLE (Standard Width & Length - 4' X 8')

Thickness	Weight
mm	kg/sht
0.36	8.46
0.40	9.44
0.45	10.62
0.50	11.77
0.55	12.94
0.60	14.11
0.70	16.44
0.80	18.77
0.85	19.94
0.90	21.11
1.00	23.44
1.20	28.11
1.40	32.79

Thickness	Weight
mm	kg/sht
1.50	36.127
1.55	36.22
1.60	37.44
1.80	42.12
1.90	44.43
2.00	46.77
2.30	53.77
2.50	58.43
2.60	60.75
2.80	65.43
2.90	67.72
3.00	70.12
3.20	74.72

PLATES

HOT DIP GALVANISED COILS AND SHEETS - SPECIFICATIONS

Application	Standard					
	KS	JIS	ASTM	BS	DIN 17162	
	D3506	G3302	A525	2989	Part 1	Part 2
Commercial Quality	SBHG 1	SPGC	A526	Z1	St 01Z	--
Lock Forming Quality	SBHG 2	--	A527	Z1	St 02Z	--
Drawing Quality	SBHG 3	SPGD	A587	Z3	St 03Z	--
Deep Drawing Quality	--	SPGDD	--	Z4	St 04Z	--
Deep Drawing Special Killed	--	--	A642	Z5	St 05Z	--
Extra Deep Drawing Quality	--	--	--	--	--	--
Roofing & Siding Quality	--	SPGR SPGA	A361	--	--	--
Structural Quality	SBHG4	SPGS	A446	Z25, Z28 Z32, Z55	--	St E250 St E280 St E320 St E350
Corrugating General Quality	SBHG 1-W	SPGH	A361	BS 3083	DIN 59231	--
Corrugating Building Quality	--	SPGW	--	--	--	--
High Strength Steel	--	--	--	--	--	--

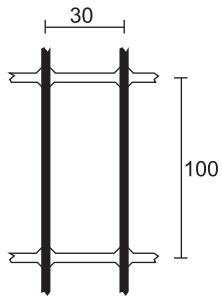
GALVANISED STEEL SHEETS

JIS G3302 - 1987 SGCC JIS G3302 - 1987 SGHC

Thickness	Width x Length								
	4'x 8'								
	1219 mm x 2438 mm								
	Z 18			Z 22 / Z 25			Z 27		
Calculated Weight									
mm	kg/pc	lb/pc	pcs/mt	kg/pc	lb/pc	pcs/mt	kg/pc	lb/pc	pcs/mt
0.30	7.724	17.028	129.5	7.905	17.428	126.5	8.131	17.926	123.0
0.35	8.891	19.600	112.5	9.072	20.000	110.2	9.298	20.498	107.6
0.36	9.124	20.114	109.6	9.305	20.514	107.5	9.531	21.012	104.9
0.40	10.057	22.172	99.4	10.238	22.571	97.7	10.464	23.069	95.6
0.45	11.223	24.743	89.1	11.405	25.143	87.7	11.631	25.641	86.0
0.50	12.390	27.315	80.7	12.571	27.715	79.5	12.797	28.212	78.1
0.55	13.556	29.886	73.8	13.738	30.286	72.8	13.964	30.784	71.6
0.60	14.723	32.458	67.9	14.904	32.858	67.1	15.130	33.356	66.1
0.65	15.889	35.030	62.9	16.071	35.429	62.2	16.297	35.927	61.4
0.70	17.056	37.601	58.6	17.237	38.001	58.0	17.463	38.499	57.3
0.75	18.222	40.173	54.9	18.404	40.573	54.3	18.629	41.071	53.7
0.80	19.389	42.745	51.6	19.570	43.144	51.1	19.796	43.642	50.5
0.85	20.555	45.316	48.6	20.737	45.716	48.2	20.962	46.214	47.7
0.90	21.722	47.888	46.0	21.903	48.287	45.7	22.129	48.785	45.2
1.00	24.055	53.031	41.6	24.236	53.431	41.3	24.462	53.929	40.9
1.10	26.388	58.174	37.9	26.569	58.574	37.6	26.795	59.072	37.3
1.20	28.721	63.318	34.8	28.902	63.717	34.6	29.128	64.215	34.3
1.40	33.387	73.604	30.0	33.568	74.004	29.8	33.794	74.502	29.6
1.45	34.553	76.176	28.9	34.734	76.575	28.8	34.960	77.073	28.6
1.50	35.720	78.747	28.0	35.901	79.147	27.9	36.127	79.645	27.7
1.55	36.886	81.319	27.1	37.067	81.719	27.0	37.293	82.217	26.8
1.60	38.052	83.891	26.3	38.234	84.290	26.2	38.460	84.788	26.0
1.80	42.718	94.177	23.4	42.900	94.577	23.3	43.126	95.075	23.2
1.90	45.051	99.320	22.2	45.233	99.720	22.1	45.459	100.218	22.0
2.00	47.384	104.463	21.1	47.566	104.863	21.0	47.791	105.361	20.9
2.20	52.050	114.750	19.2	52.232	115.150	19.1	52.457	115.648	19.1
2.30	54.383	119.893	18.4	54.564	120.293	18.3	54.790	120.791	18.3
2.50	59.049	130.180	16.9	59.230	130.579	16.9	59.456	131.077	16.8
2.60	61.382	135.323	16.3	61.563	135.723	16.2	61.789	136.221	16.2
2.80	66.048	145.609	15.1	66.229	146.009	15.1	66.455	146.507	15.0
2.90	68.381	150.753	14.6	68.562	151.152	14.6	68.788	151.650	14.5
3.00	70.714	155.896	14.1	70.895	156.296	14.1	71.121	156.794	14.1

GALVANISED SERRATED GRATINGS

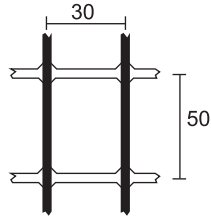
STANDARD TYPES



Series 3/1

Load factor 1.26

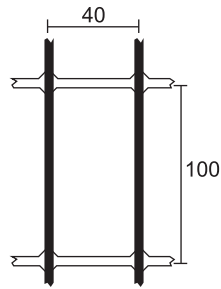
B/bars/M 34



Series 3/5

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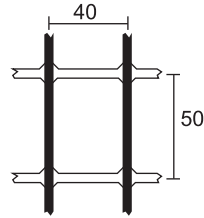
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Series 4/1

Load factor 1.0

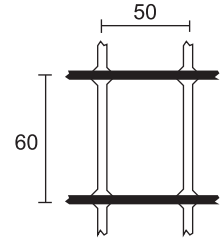
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Series 4/5

Load factor 1.0

B/bars/M 26



Series 6/5

Load factor 0.63

B/bars/M 17

GRATINGS LOADING TABLE "4/1" SERIES - based on type "4/1" plan (factor 1)

B/bar	B/bar type	Serrate factor "Z"	Banded mass/kg/m ²	Load	Span in mm											Max. recomm
							Heavy	Medium	Light						Class	
					500	750	900	1000	1250	1500	1800	2100	2400	2750		
25 x 3	1	0.84	20	U	4224	1770	1024	746	382	221					Light	1250
				d	1.76	3.75	4.5	5	6.25	7.5					Medium	1100
				C	1055	703	575	466	296	207					Heavy	1000
				d	1.42	3.19	4.5	5	6.25	7.5						
25 x 4.5	2	0.84	28	U	6336	2656	1536	1119	572	332					Light	1500
				d	1.76	3.75	4.5	5	6.25	7.5					Medium	1250
				C	1583	1055	863	699	447	310					Heavy	1100
				d	1.42	3.19	4.5	5	6.25	7.5						
30 x 3	4	0.87	23.1	U	6082	2703	1751	1276	653	378	218	131			Light	1500
				d	1.49	3.35	4.5	5	6.25	7.5	9	10			Medium	1300
				C	1520	1014	844	760	510	354	245	172			Heavy	1200
				d	1.19	2.68	3.86	4.77	6.25	7.5	9	10				
30 x 4.5	5	0.87	32.7	U	9124	4055	2627	1915	980	567	327	197			Light	1800
				d	1.49	3.35	4.5	5	6.25	7.5	9	10			Medium	1500
				C	2281	1521	1267	1140	765	531	368	258			Heavy	1300
				d	1.19	2.68	3.86	4.77	6.25	7.5	9	10				
40 x 3	7	0.9	29.5	U	10813	4806	3337	2703	1544	893	517	310	181	113	Light	2100
				d	1.12	2.52	3.63	4.48	6.25	7.5	9	10	10	10	Medium	1800
				C	2703	1802	1502	1351	1081	837	581	407	272	191	Heavy	1500
				d	0.9	2.02	2.91	3.59	5.61	7.5	9	10	10	10		
40 x 4.5	8	0.9	42.2	U	16222	7209	5006	4055	2316	1340	775	464	272	170	Light	2350
				d	1.12	2.52	3.63	4.48	6.25	7.5	9	10	10	10	Medium	2000
				C	4055	2703	2253	2027	1633	1255	872	610	408	286	Heavy	1800
				d	0.9	2.02	2.91	3.59	5.61	7.5	9	10	10	10		
50 x 5	11	0.92	57.2	U	28162	12516	8692	7040	4506	2925	1691	1014	594	370	Light	2750
				d	0.89	2	2.89	3.56	5.57	7.5	9	10	10	10	Medium	2400
				C	7040	4694	3911	3520	2816	2347	1903	1332	892	626	Heavy	2200
				d	0.71	1.61	2.31	2.85	4.46	6.42	9	10	10	10		

U=kg/m² UDL C=kg/m width line load d= Deflection at centre

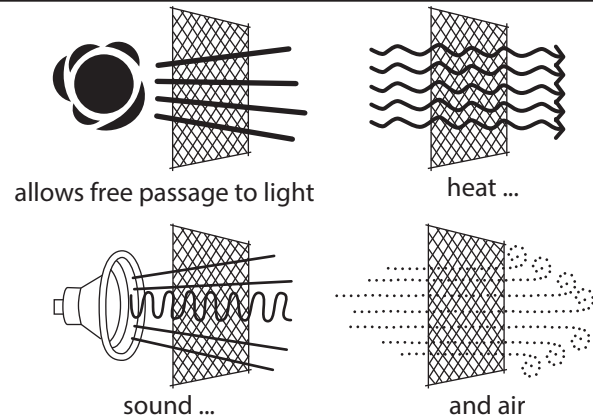
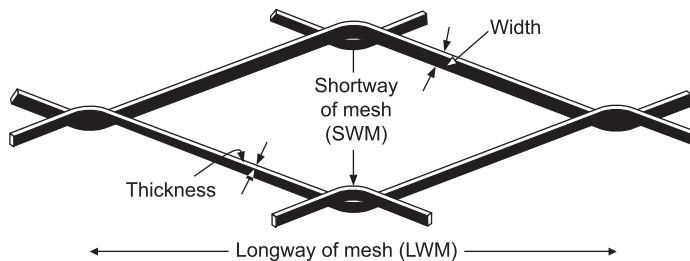
Note: Galvanizing will increase masses/m² by 9% on average

Deflection criteria : 1/200 of span or 10mm whichever is the lesser

Bending stress criteria : 165 MPa

EXPANDED METAL

PROPERTIES

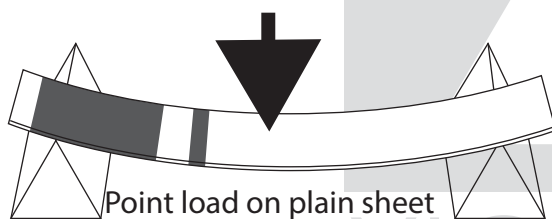


The main feature of Expanded Metal is that the mesh is formed from a single piece of metal. A metal sheet, plate or foil is passed through an expanding machine having a serrated knife which shears and stretches the metal into strands but leaves them joined at the knuckles.

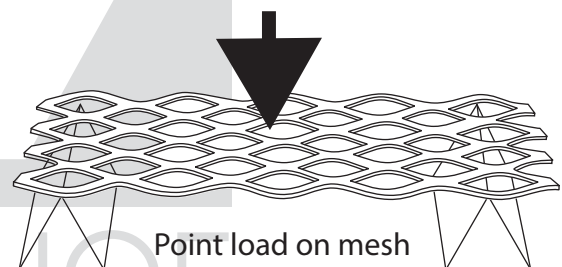
This high strength-to-weight ratio material is technically functional in many protective and structural applications. The jointless expanded metal ensures that every strand will perform as a structural member, distributing the load over a wide area.

Advantages

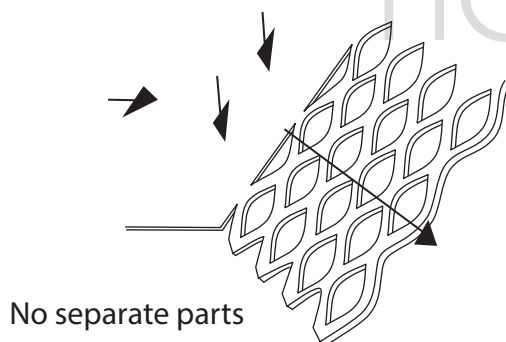
1. No separate parts. Nothing to work loose, no fretting strands. Stays together under stress, even when cut into irregular shapes.
2. The strands are angled to the plane of the complete sheet of mesh in a truss-like formation which ensures excellent load bearing properties.
3. Protection. The range includes meshes small enough to exclude even the smallest fingers where moving parts are within finger length of the guards.
4. Screening effect - especially in anti-dazzle fences.
5. Reinforcement. The meshes anchor immovably and the angle strand formation further strengthens the reinforced material.
6. Provides anti-slip surface e.g. catwalk, stairtreads and ramps.



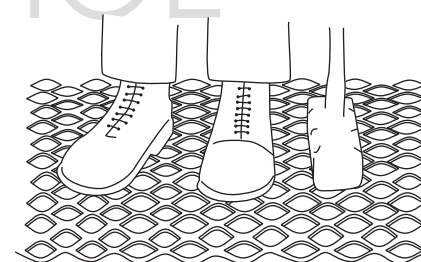
Point load on plain sheet



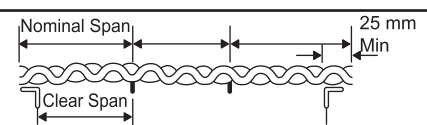
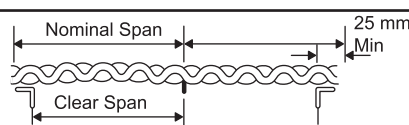
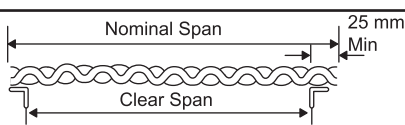
Point load on mesh



No separate parts



Anti-slip



QUICK SELECTION TABLE FOR WALKWAYS

LOAD	NOMINAL SPAN OF MESH - SIMPLY SUPPORTED		
	600 mm	750 mm	900 mm
Distributed - 2.5 kPa	1000 - G1201 - 1502	1000 - G 1201	1000 (3.8) - 1501
Distributed - 5.0 kPa	1000 - G 1201 - 1502	1000 (3.8)	-
Concentrated - 1.1 kN @ 1.2 m CTRS	1000, 1502, 1000 (3.8)	1000 - G 1201 - 1501	1000 (3.8) - 1501

* These meshes whilst safe for the spans shown, may be springy, underfoot.

EXPANDED METAL

SPECIFICATIONS

B.S No.	Code No.	Material	Nominal Size of Mesh		Thickness	Strand Width	Approx. Nominal Wt. kg/m ² ± 10%
			SWM	LWM			
			mm	mm		mm	mm
XM 21	3001	STEEL	76	200	6.0	12.5	16.5
XM 22	3002	STEEL			6.0	11.0	14.5
XM 23	3003	STEEL			4.5	12.0	12.4
XM 24	3004	STEEL			6.0	7.5	10.4
XM 25	3005	STEEL			4.5	9.0	9.3
XM 26	3006	STEEL			6.0	6.0	8.3
XM 27	3007	STEEL			4.5	7.0	7.0
XM 28	3008	STEEL			4.5	6.0	6.2
XM 29	3009	STEEL			3.0	7.5	5.1
XM 30	3010	STEEL			4.5	4.5	4.6
XM 31	3011	STEEL			3.0	6.0	4.1
XM 33	3013	STEEL			3.0	3.0	2.0
XM 41	1501	STEEL	38	114	4.5	9.5	18.0
XM 42	1502	STEEL			4.5	7.5	15.5
XM 43	1503	STEEL			4.5	6.0	12.4
XM 44	1504	STEEL			3.0	6.0	8.3
XM 45	1505	STEEL			3.0	4.5	6.2
XM 46	1506	STEEL			3.0	3.0	4.1
XM 47	1507	STEEL			1.5	6.0	4.1
XM 48	1508	STEEL			1.5	4.5	3.1
XM 49	1509	STEEL			1.5	3.0	2.0
XM 61	1001	STEEL	25	76	3.0	4.5	8.1
XM 63	1003	STEEL			3.0	3.0	5.4
XM 65	1005	STEEL			1.5	4.5	4.0
XM 66	1006	STEEL			1.5	3.0	2.7
XM 71	7502	STEEL	19	50	3.0	4.5	10.4
XM 72	7503	STEEL			3.0	3.0	7.1
XM 73	7504	STEEL			1.5	6.0	6.1
XM 75	7506	STEEL			1.5	3.0	3.5
XM 76	7507	STEEL			1.5	2.3	2.7

Code	Nominal size of mesh (mm)		Dimensions of strands (mm)		Approx. nominal weight kg/m ² ± 1 0%
	S.W.M.	L.W.M.	Thickness	Width	
G 1020	16	38	1.2	2.0	1.9
G 1032	25	40	3.0	3.0	5.4
G 1201	32	114	4.5	7.5	13.1
G 1232	33	102	3.0	3.0	4.5
G 2263	65	185	3.0	6.0	3.2
G 3092	38	76	3.0	4.5	4.5
G 5011	45	135	4.5	11.0	15.0
1000	30	76	4.5	6.0	15.7
1000 (3.8)	30	76	4.5	10.0	22.1
½ X18G	13	28	1.2	1.5	2.3

Tolerances :

Reference to sheet size, strand width, thickness and weights are approximate only. Whilst every effort is made for these figures to be accurate, it must be understood that the specification of goods manufactured and supplied by the company is subject to normal trade tolerances.

The company policy is one of continuous development : we therefore reserve the right to alter specifications, etc without notice.

WROUGHT STEEL BUTT WELDING FITTINGS	180
FLANGES	
JIS FLANGES (JIS B2220)	187
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SLIP-ON FLANGES (BS 4504)	193
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SLIP-ON FLANGES (BS 10)	200



YICK HOE

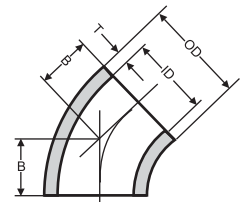
WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

DIMENSIONS OF FITTING ENDS

Nominal Pipe Size	Outside Diameter OD	Inside Diameter ID				Wall Thickness T			
		STD	S-40	XS	S-80	STD	S-40	XS	S-80
½	21.3	15.7	15.7	13.9	13.9	2.8	2.8	3.7	3.7
¾	26.7	20.9	20.9	18.9	18.9	2.9	2.9	3.9	3.9
1	33.4	26.6	26.6	24.4	24.4	3.4	3.4	4.5	4.5
1¼	42.2	35.0	35.0	32.4	32.4	3.6	3.6	4.9	4.9
1½	48.3	40.9	40.9	38.1	38.1	3.7	3.7	5.1	5.1
2	60.3	52.5	52.5	49.3	49.3	3.9	3.9	5.5	5.5
2½	73.0	62.6	62.6	59.0	59.0	5.2	5.2	7.0	7.0
3	88.9	77.9	77.9	73.7	73.7	5.5	5.5	7.6	7.6
3½	101.6	90.2	90.2	85.4	85.4	5.7	5.7	8.1	8.1
4	114.3	102.3	102.3	97.1	97.1	6.0	6.0	8.6	8.6
5	141.3	128.1	128.1	122.3	122.3	6.6	6.6	9.5	9.5
6	168.3	154.1	154.1	146.3	146.3	7.1	7.1	11.0	11.0
8	219.1	202.7	202.7	193.7	193.7	8.2	8.2	12.7	12.7
10	273.1	254.5	254.5	247.7	242.9	9.3	9.3	12.7	15.1
12	323.9	304.9	303.3	298.5	289.1	9.5	10.3	12.7	17.4
14	355.6	336.6	333.4	330.2	317.6	9.5	11.1	12.7	19.0
16	406.4	387.4	381.0	381.0	363.6	9.5	12.7	12.7	21.4
18	457.2	438.2	428.6	431.8	409.6	9.5	14.3	12.7	23.8

Note: Dimensions of outside diameter and wall thickness conform to ANSI B 36.10



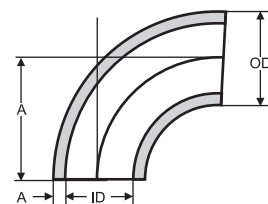
45° ELBOWS

ANSI B 16.9

Nominal Pipe Size	Outside Diameter OD	STD		XS		Centre to Face B	Approx. Weight kg	
		ID	T	ID	T		STD	XS
½	21.3	15.7	2.8	13.9	3.7	15.9	0.0379	0.0485
¾	26.7	20.9	2.9	18.9	3.9	11.1	0.0378	0.0494
1	33.4	26.6	3.4	24.4	4.5	22.2	0.0748	0.0968
1¼	42.2	35.0	3.6	32.4	4.9	25.4	0.127	0.167
1½	48.3	40.9	3.7	38.1	5.1	28.6	0.182	0.243
2	60.3	52.5	3.9	49.3	5.5	34.9	0.326	0.448
2½	73.0	62.6	5.2	59.0	7.0	44.5	0.646	0.854
3	88.9	77.9	5.5	73.7	7.6	50.8	1.01	1.37
3½	101.6	90.2	5.7	85.4	8.1	57.2	1.42	1.95
4	114.3	102.3	6.0	97.1	8.6	63.5	1.92	2.67
5	141.3	128.1	6.6	122.3	9.5	79.4	3.26	4.63
6	168.3	154.1	7.1	146.3	11.0	95.3	5.08	7.64
8	219.1	202.7	8.2	193.7	12.7	127.0	10.2	15.5
10	273.1	254.5	9.3	247.7	12.7	158.8	18.1	24.4
12	323.9	304.9	9.5	298.5	12.7	190.5	26.5	35.0
14	355.6	336.6	9.5	330.2	12.7	222.3	34.1	45.0
16	406.4	387.4	9.5	381.0	12.7	254.0	44.6	59.0
18	457.2	438.2	9.5	431.8	12.7	285.8	56.6	74.9

WROUGHT STEEL BUTT-WELDING FITTINGS

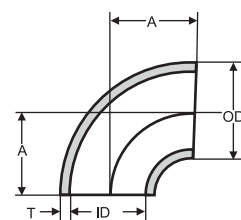
Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



90° ELBOWS Long Radius

ANSI B 16.9

Nominal Pipe Size	Outside Diameter OD	STD		XS		Centre to Face A	Approx. Weight kg	
		ID	T	ID	T		STD	XS
1/2	21.3	15.7	2.8	13.9	3.7	38.1	0.0759	0.0971
3/4	26.7	20.9	2.9	18.9	3.9	28.6	0.0756	0.0987
1	33.4	26.6	3.4	24.4	4.5	38.1	0.15	0.194
1 1/4	42.2	35.0	3.6	32.4	4.9	47.6	0.253	0.334
1 1/2	48.3	40.9	3.7	38.1	5.1	57.2	0.364	0.486
2	60.3	52.5	3.9	49.3	5.5	76.2	0.652	0.895
2 1/2	73.0	62.6	5.2	59.0	7.0	95.3	1.29	1.71
3	88.9	77.9	5.5	73.7	7.6	114.3	2.03	2.74
3 1/2	101.6	90.2	5.7	85.4	8.1	133.4	2.84	3.90
4	114.3	102.3	6.0	97.1	8.6	152.4	3.85	5.34
5	141.3	128.1	6.6	122.3	9.5	190.5	6.52	9.26
6	168.3	154.1	7.1	146.3	11.0	228.6	10.2	15.3
8	219.1	202.7	8.2	193.7	12.7	304.8	20.4	30.9
10	273.1	254.5	9.3	247.7	12.7	381.0	36.1	48.8
12	323.9	304.9	9.5	298.5	12.7	457.2	53.0	69.9
14	355.6	336.6	9.5	330.2	12.7	533.4	68.1	89.9
16	406.4	387.4	9.5	381.0	12.7	609.6	89.3	118.0
18	457.2	438.2	9.5	431.8	12.7	685.8	113.0	150.0



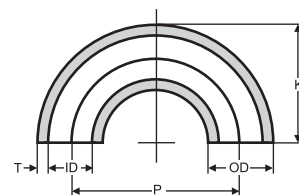
90° ELBOWS Short Radius

ANSI B 16.9

Nominal Pipe Size	Outside Diameter OD	STD		XS		Centre to Face A	Approx. Weight	
		ID	T	ID	T		STD	XS
1	33.4	26.6	3.4	24.4	4.5	25.4	0.0998	0.129
1 1/4	42.2	35.0	3.6	32.4	4.9	31.8	0.169	0.223
1 1/2	48.3	40.9	3.7	38.1	5.1	38.1	0.243	0.324
2	60.3	52.5	3.9	49.3	5.5	50.8	0.435	0.597
2 1/2	73.0	62.6	5.2	59.0	7.0	63.5	0.861	1.14
3	88.9	77.9	5.5	73.7	7.6	76.2	1.35	1.83
3 1/2	101.6	90.2	5.7	85.4	8.1	88.9	1.89	2.60
4	114.3	102.3	6.0	97.1	8.6	101.6	2.56	3.56
5	141.3	128.1	6.6	122.3	9.5	127.0	4.35	6.17
6	168.3	154.1	7.1	146.3	11.0	152.4	6.77	10.2
8	219.1	202.7	8.2	193.7	12.7	203.2	13.6	20.6
10	273.1	254.5	9.3	247.7	12.7	254.0	24.1	32.5
12	323.9	304.9	9.5	298.5	12.7	304.8	35.3	46.6
14	355.6	336.6	9.5	330.2	12.7	355.6	45.4	60.0
16	406.4	387.4	9.5	381.0	12.7	406.4	59.5	78.7
18	457.2	438.2	9.5	431.8	12.7	457.2	75.5	99.9

WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

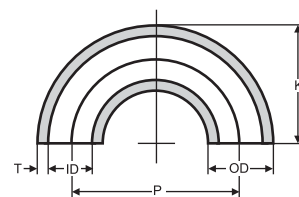


180° RETURNS Long Radius

ANSI B 16.9

Nominal Pipe Size	Outside Diameter OD	STD		XS		Centre to centre P	Centre to Face K	Approx. Weight kg	
		ID	T	ID	T			STD	XS
½	21.3	15.7	2.8	--	--	76.2	47.6	0.152	--
¾	26.7	20.9	2.9	18.9	3.9	57.2	42.9	0.151	0.197
1	33.4	26.6	3.4	24.4	4.5	76.2	55.6	0.299	0.387
1¼	42.2	35.0	3.6	32.4	4.9	95.3	69.9	0.506	0.668
1½	48.3	40.9	3.7	38.1	5.1	114.3	82.6	0.728	0.972
2	60.3	52.5	3.9	49.3	5.5	152.4	106.4	1.30	1.79
2½	73.0	62.6	5.2	59.0	7.0	190.5	131.8	2.58	3.42
3	88.9	77.9	5.5	73.7	7.6	228.6	158.8	4.05	5.48
3½	101.6	90.2	5.7	85.4	8.1	266.7	184.2	5.68	7.81
4	114.3	102.3	6.0	97.1	8.6	304.8	209.6	7.69	10.7
5	141.3	128.1	6.6	122.3	9.5	381.0	261.9	13.0	18.5
6	168.3	154.1	7.1	146.3	11.0	457.2	312.7	20.3	30.6
8	219.1	202.7	8.2	193.7	12.7	609.6	414.3	40.7	61.9
10	273.1	254.5	9.3	247.7	12.7	762.0	517.5	72.2	97.5
12	323.9	304.9	9.5	298.5	12.7	914.4	619.1	106.0	140.0

YICK HOE



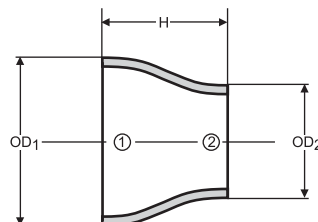
180° RETURNS Short Radius

ANSI B 16.9

Nominal Pipe Size	Outside Diameter OD	STD		XS		Centre to centre P	Centre to Face K	Approx. Weight kg	
		ID	T	ID	T			STD	XS
1	33.4	26.6	3.4	24.4	4.5	50.8	41.3	0.200	0.258
1¼	42.2	35.0	3.6	32.4	4.9	63.5	52.4	0.338	0.446
1½	48.3	40.9	3.7	38.1	5.1	76.2	61.9	0.485	0.647
2	60.3	52.5	3.9	49.3	5.5	101.6	81.0	0.87	1.19
2½	73.0	62.6	5.2	59.0	7.0	127.0	100.0	1.72	2.28
3	88.9	77.9	5.5	73.7	7.6	152.4	120.7	2.70	3.65
3½	101.6	90.2	5.7	85.4	8.1	177.8	139.7	3.79	5.20
4	114.3	102.3	6.0	97.1	8.6	203.2	158.8	5.13	7.12
5	141.3	128.1	6.6	122.3	9.5	254.0	196.9	8.69	12.3
6	168.3	154.1	7.1	146.3	11.0	304.8	236.5	13.5	20.4
8	219.1	202.7	8.2	193.7	12.7	406.4	312.7	27.1	41.2
10	273.1	254.5	9.3	247.7	12.7	508.0	390.5	48.1	65.0
12	323.9	304.9	9.5	298.5	12.7	609.6	466.7	70.7	93.3

WROUGHT STEEL BUTT-WELDING FITTINGS

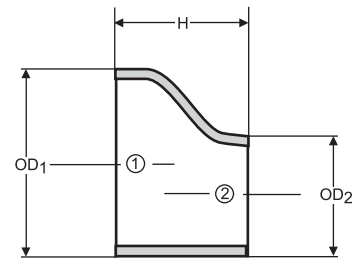
Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



REDUCERS		ANSI B 16.9			
Nominal Pipe Size ① x ②	Large End	Small End	Length	Approx. Weight	
	OD ₁	OD ₂	H	kg	
				STD	XS
1/4 x 1/2	26.7	21.3	38.1	0.0571	0.0741
1 x 3/4	33.4	26.7	50.8	0.113	0.145
1 x 1/2		21.3		0.102	0.131
1 1/4 x 1		33.4		0.153	0.201
1 1/4 x 3/4	42.2	26.7	50.8	0.139	0.182
1 1/4 x 1/2		21.3		0.128	0.167
1 1/2 x 1 1/4		42.2		0.240	0.319
1 1/2 x 1	48.3	33.4	63.5	0.216	0.286
1 1/2 x 3/4		26.7		0.198	0.261
1 1/2 x 1/2		21.3		0.184	0.241
2 x 1 1/2		48.3		0.372	0.509
2 x 1 1/4	60.3	42.2	76.2	0.351	0.479
2 x 1		33.4		0.321	0.437
2 x 3/4		26.7		0.298	0.404
2 1/2 x 2		60.3		0.697	0.919
2 1/2 x 1 1/2	73.0	48.3	88.9	0.633	0.832
2 1/2 x 1 1/4		42.2		0.602	0.788
2 1/2 x 1		33.4		0.556	0.727
3 x 2 1/2		73.0		0.911	1.23
3 x 2	88.9	60.3	88.9	0.842	1.13
3 x 1 1/2		48.3		0.778	1.04
3 x 1 1/4		42.2		0.747	0.999
3 1/2 x 3		88.9		1.29	1.77
3 1/2 x 2 1/2		73.0		1.18	1.62
3 1/2 x 2	101.6	60.3	101.6	1.10	1.50
3 1/2 x 1 1/2		48.3		1.03	1.40
3 1/2 x 1 1/4		42.2		0.984	1.34
4 x 3 1/2		101.6		1.54	2.13
4 x 3		88.9		1.45	2.01
4 x 2 1/2	114.3	73.0	101.6	1.35	1.86
4 x 2		60.3		1.27	1.75
4 x 1 1/2		48.3		1.19	1.64

WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



REDUCERS

ANSI B 16.9

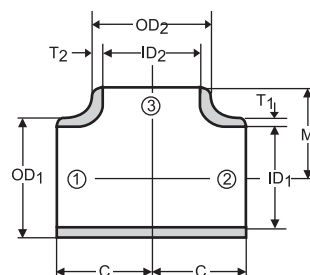
Nominal Pipe Size ①x ②	Large End	Small End	Length	Approx. Weight kg	
	OD ₁	OD ₂	H	STD	XS
5 x 4	141.3	114.3	127.0	2.50	3.55
5 x 3½		101.6		2.39	3.38
5 x 3		88.9		2.28	3.21
5 x 2½		73.0		2.14	3.02
5 x 2		60.3		2.03	2.86
6 x 5	168.3	141.3	139.7	3.64	5.46
6 x 4		114.3		3.35	5.02
6 x 3½		101.6		3.22	4.82
6 x 3		88.3		3.09	4.62
6 x 2½		73.0		2.94	4.38
8 x 6	219.1	168.3	152.4	5.78	8.75
8 x 5		141.3		5.45	8.24
8 x 4		114.3		5.15	7.77
8 x 3½		101.6		5.01	7.55
10 x 8		219.1		9.74	13.1
10 x 6	273.1	168.3	177.8	8.96	12.1
10 x 5		141.3		8.58	11.3
10 x 4		114.3		8.21	11.0
12 x 10		273.1		13.9	18.3
12 x 8	323.9	219.1	203.2	12.9	17.0
12 x 6		168.3		12.1	15.9
12 x 5		141.3		11.6	15.3
14 x 12		323.9		25.5	33.8
14 x 10	355.6	273.1	330.2	23.7	31.4
14 x 8		219.1		21.9	27.7
16 x 14		355.6		31.0	41.1
16 x 12	406.4	323.9	355.6	29.8	39.5
16 x 10		273.1		28.0	37.0
18 x 16		406.4		37.7	50.1
18 x 14	457.2	355.6	381.0	35.7	47.4
18 x 12		323.9		34.5	45.7

Note: (1) Wall thickness and inside diameter conform to ANSI B 36.10.

(2) Shows available sizes of eccentric reducer.

WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)



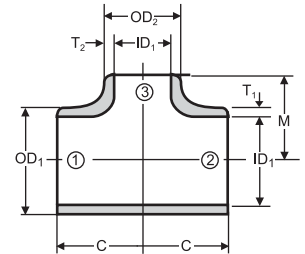
TEES

ANSI B 16.9

Nominal Pipe Size ①x ②x ③○	Outside Diameter		Center to End		STD				XS				Approx. Weight (kg)	
	OD ₁	OD ₂	C	M	ID ₁	T ₁	ID ₂	T ₂	ID ₁	T ₁	ID ₂	T ₂	STD	XS
½ x ½ x ½	21.3	21.3	25.4	25.4	15.7	2.8	15.7	2.8	13.9	3.7	13.9	3.7	0.11	0.16
¾ x ¾ x ¾	26.7	26.7	28.6	28.6	20.9	2.9	20.9	2.9	18.9	3.9	18.9	3.9	0.17	0.21
¾ x ½	26.7	21.3	28.6	28.6	20.9	2.9	15.7	2.8	18.9	3.9	13.9	3.7	0.15	0.19
1x1x1	33.4	33.4	38.1	38.1	26.6	3.4	26.6	3.4	24.4	4.5	24.4	4.5	0.35	0.42
1x ¾	33.4	26.7	38.1	38.1	26.6	3.4	20.9	2.9	24.4	4.5	18.9	3.9	0.30	0.36
1x ½	33.4	21.3	38.1	38.1	26.6	3.4	15.7	2.8	24.4	4.5	13.9	3.7	0.29	0.36
1¼ x 1¼ x 1¼	42.2	42.2	47.6	47.6	35.0	3.6	35.0	3.6	32.4	4.9	32.4	4.9	0.61	0.75
1¼ x 1	42.2	33.4	47.6	47.6	35.0	3.6	26.6	3.4	32.4	4.9	24.4	4.5	0.53	0.66
1¼ x ¾	42.2	26.7	47.6	47.6	35.0	3.6	20.9	2.9	32.4	4.9	18.9	3.9	0.50	0.63
1¼ x ½	42.2	21.3	47.6	47.6	35.0	3.6	15.7	2.8	32.4	4.9	13.9	3.7	0.49	0.61
1½ x 1½ x 1½	48.3	48.3	57.2	57.2	40.9	3.7	40.9	3.7	38.1	5.1	38.1	5.1	0.92	1.13
1½ x 1¼	48.3	42.2	57.2	57.2	40.9	3.7	35.0	3.6	38.1	5.1	32.4	4.9	0.80	1.00
1½ x 1	48.3	33.4	57.2	57.2	40.9	3.7	26.6	3.4	38.1	5.1	24.4	4.5	0.76	0.95
1½ x ¾	48.3	26.7	57.2	57.2	40.9	3.7	20.9	2.9	38.1	5.1	18.9	3.9	0.73	0.93
1½ x ½	48.3	21.3	57.2	57.2	40.9	3.7	15.7	2.8	38.1	5.1	13.9	3.7	0.72	0.90
2 x 2 x 2	60.3	60.3	63.5	63.5	52.5	3.9	52.5	3.9	49.3	5.5	49.3	5.5	1.34	1.70
2 x 1½	60.3	48.3	63.5	60.3	52.5	3.9	40.9	3.7	49.3	5.5	38.1	5.1	1.21	1.54
2 x 1¼	60.3	42.2	63.5	57.2	52.5	3.9	35.0	3.6	49.3	5.5	32.4	4.9	1.12	1.48
2 x 1	60.3	33.4	63.5	50.8	52.5	3.9	26.6	3.4	49.3	5.5	24.4	4.5	1.11	1.44
2 x ¾	60.3	26.7	63.5	44.5	52.5	3.9	20.9	2.9	49.3	5.5	18.9	3.9	1.10	1.12
2½ x 2½ x 2½	73.0	73.0	76.2	76.2	62.6	5.2	62.6	5.2	59.0	7.0	59.0	7.0	2.37	2.93
2½ x 2	73.0	60.3	76.2	69.9	62.6	5.2	52.5	3.9	59.0	7.0	49.3	5.5	2.08	2.62
2½ x 1½	73.0	48.3	76.2	66.7	62.6	5.2	40.9	3.7	59.0	7.0	38.1	5.1	2.05	2.57
2½ x 1¼	73.0	42.2	76.2	63.5	62.6	5.2	35.0	3.6	59.0	7.0	32.4	4.9	2.00	2.52
3 x 3 x 3	88.9	88.9	85.7	85.7	77.9	5.5	77.9	5.5	73.7	7.6	73.7	7.6	3.38	4.29
3 x 2½	88.9	73.0	85.7	82.6	77.9	5.5	62.6	5.2	73.7	7.6	59.0	7.0	3.09	3.91
3 x 2	88.9	60.3	85.7	76.2	77.9	5.5	52.5	3.9	73.7	7.6	49.3	5.5	2.96	3.79
3 x 1½	88.9	48.3	85.7	73.0	77.9	5.5	40.9	3.7	73.7	7.6	38.1	5.1	2.93	3.77
3 x 1¼	88.9	42.2	85.7	69.9	77.9	5.5	35.0	3.6	73.7	7.6	32.4	4.9	2.88	3.71
3½ x 3½ x 3½	101.6	101.6	95.3	95.3	90.2	5.7	90.2	5.7	85.4	8.1	85.4	8.1	4.48	5.73
3½ x 3	101.6	88.9	95.3	92.1	90.2	5.7	77.9	5.5	85.4	8.1	73.7	7.6	4.10	5.32
3½ x 2½	101.6	73.0	95.3	88.9	90.2	5.7	62.6	5.2	85.4	8.1	59.0	7.0	4.00	5.23
3½ x 2	101.6	60.3	95.3	82.6	90.2	5.7	52.5	3.9	85.4	8.1	49.3	5.5	3.86	5.10
3½ x 1½	101.6	48.3	95.3	79.4	90.2	5.7	40.9	3.7	85.4	8.1	38.1	5.1	3.86	5.04

WROUGHT STEEL BUTT-WELDING FITTINGS

Standard Sizes of Butt-Welding Fittings (Metric Unit: mm)

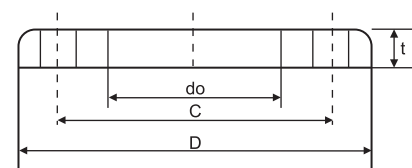


REDUCING

ANSI B 16.9

Nominal Pipe Size ①x ②x ③	Outside		Center		STD				XS				Approx.	
	Diameter		to End		Run		Branch		Run		Branch		Weight (kg)	
	OD ₁	OD ₂	C	M	ID ₁	T ₁	ID ₂	T ₂	ID ₁	T ₁	ID ₂	T ₂	STD	XS
4 x 4 x 4	114.3	114.3	104.8	104.8	102.3	6.0	102.3	6.0	97.1	8.6	97.1	8.6	5.73	7.47
x 3½	114.3	101.6	104.8	101.6	102.3	6.0	90.2	5.7	97.1	8.6	85.4	8.1	5.28	6.96
x 3	114.3	88.9	104.8	98.4	102.3	6.0	77.9	5.5	97.1	8.6	73.7	7.6	5.18	6.82
x 2½	114.3	73.0	104.8	95.3	102.3	6.0	62.6	5.2	97.1	8.6	59.0	7.0	5.13	6.68
x 2	114.3	60.3	104.8	88.9	102.3	6.0	52.5	3.9	97.1	8.6	49.3	5.5	5.00	6.59
x 1½	114.3	48.3	104.8	85.7	102.3	6.0	40.9	3.7	97.1	8.6	38.1	5.1	4.96	6.54
5 x 5 x 5	141.3	141.3	123.8	123.8	128.1	6.6	128.1	6.6	122.3	9.5	122.3	9.5	9.00	12.0
x 4	141.3	114.3	123.8	117.5	128.1	6.6	102.3	6.0	122.3	9.5	97.1	8.6	8.23	11.0
x 3½	141.3	101.6	123.8	114.3	128.1	6.6	90.2	5.7	122.3	9.5	85.4	8.1	8.10	10.9
x 3	141.3	88.9	123.8	111.1	128.1	6.6	77.9	5.5	122.3	9.5	73.7	7.6	8.00	10.8
x 2½	141.3	73.0	123.8	108.0	128.1	6.6	62.6	5.2	122.3	9.5	59.0	7.0	7.90	10.7
x 2	141.3	60.3	123.8	104.8	128.1	6.6	52.5	3.9	122.3	9.5	49.3	5.5	7.80	10.6
6 x 6 x 6	168.3	168.3	142.9	142.9	154.1	7.1	154.1	7.1	146.3	11.0	146.3	11.0	13.3	19.1
x 5	168.3	141.3	142.9	136.5	154.1	7.1	128.1	6.6	146.3	11.0	122.3	9.5	12.1	17.2
x 4	168.3	114.3	142.9	130.2	154.1	7.1	102.3	6.0	146.3	11.0	97.1	8.6	11.9	16.9
x 3½	168.3	101.6	142.9	127.0	154.1	7.1	90.2	5.7	146.3	11.0	85.4	8.1	11.8	16.9
x 3	168.3	88.9	142.9	123.0	154.1	7.1	77.9	5.5	146.3	11.0	73.7	7.6	11.8	16.8
x 2½	168.3	73.0	142.9	120.7	154.1	7.1	62.6	5.2	146.3	11.0	59.0	7.0	11.7	16.8
8 x 8 x 8	219.1	219.1	177.8	177.8	202.7	8.2	202.7	8.2	193.7	12.7	193.7	12.7	24.4	34.8
x 6	219.1	168.3	177.8	168.3	202.7	8.2	154.1	7.1	193.7	12.7	122.3	9.5	22.1	32.1
x 5	219.1	141.3	177.8	161.9	202.7	8.2	128.1	6.6	193.7	12.7	97.1	8.6	21.8	31.6
x 4	219.1	114.3	177.8	155.6	202.7	8.2	102.3	6.0	193.7	12.7	85.4	8.1	21.8	31.5
x 3½	219.1	101.6	177.8	152.4	202.7	8.2	90.2	5.7	193.7	12.7	73.7	7.6	21.7	31.4
10 x 10 x 10	273.1	273.1	215.9	215.9	254.5	9.3	254.5	9.3	247.7	12.7	247.7	12.7	41.4	55.9
x 8	273.1	219.1	215.9	203.2	254.5	9.3	202.7	8.2	247.7	12.7	193.7	12.7	37.7	52.9
x 6	273.1	168.3	215.9	193.7	254.5	9.3	154.1	7.1	247.7	12.7	122.3	9.5	37.1	50.4
x 5	273.1	141.3	215.9	190.5	254.5	9.3	128.1	6.6	247.7	12.7	97.1	8.6	37.0	49.3
12 x 12 x 12	323.9	323.9	254.0	254.0	304.9	9.5	304.9	9.5	298.5	12.7	298.5	12.7	60.0	78.7
x 10	323.9	273.1	254.0	241.3	304.9	9.5	254.5	9.3	298.5	12.7	247.7	12.7	55.0	75.2

JIS FLANGES (JIS B2220)



JIS FLANGES (5K)

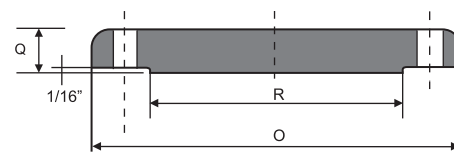
Nominal Dia. of Flange		Outer Dia. of Pipe	Inner Dia. of Flange do	Outer Dia. of Flange D	Thickness of Flange t	Hole of Bolt			Size of Bolt	Weight kg
						Distance of Hole	No. of Holes	Bore C		
in	mm									
1/2	15	21.7	22.2	80	9	60	4	12	M10	0.30
3/4	20	27.2	27.7	85	10	65	4	12	M10	0.37
1	25	34.0	34.5	95	10	75	4	12	M10	0.45
1 1/4	32	42.7	43.2	115	12	90	4	15	M12	0.78
1 1/2	40	48.6	49.1	120	12	95	4	15	M12	0.83
2	50	60.5	61.1	130	14	105	4	15	M12	1.07
2 1/2	65	76.3	77.1	155	14	130	4	15	M12	1.49
3	80	89.1	90.0	180	14	145	4	19	M16	1.99
3 1/2	90	101.6	102.6	190	14	155	4	19	M16	2.08
4	100	114.3	115.4	200	16	165	8	19	M16	2.39
5	125	139.8	141.2	235	16	200	8	19	M16	3.23
6	150	165.2	166.6	265	18	230	8	19	M16	4.41
8	200	216.3	218.0	320	20	280	8	23	M20	6.33
10	250	267.4	269.5	385	22	345	12	23	M20	9.45
12	300	318.5	321.0	430	22	390	12	23	M20	10.30
14	350	355.6	358.1	480	24	435	12	25	M22	14.00
16	400	406.4	409.0	540	24	495	16	25	M22	16.90
18	450	457.2	460.0	605	24	555	16	25	M22	21.36
20	500	508.0	511.0	665	24	605	20	25	M22	23.50
22	550	558.8	562.0	720	26	665	20	27	M24	33.40
24	600	609.6	613.0	770	26	715	20	27	M24	36.90

JIS FLANGES (10K)

Nominal Dia. of Flange		Outer Dia. of Pipe	Inner Dia. of Flange do	Outer Dia. of Flange D	Thickness of Flange t	Hole of Bolt			Size of Bolt	Weight kg
						Distance of Hole	No. of Holes	Bore C		
in	mm									
1/2	15	21.7	22.2	95	12	70	4	15	M12	0.57
3/4	20	27.2	27.7	100	14	75	4	15	M12	0.73
1	25	34.0	34.5	125	14	90	4	19	M16	1.13
1 1/4	32	42.7	43.2	135	16	100	4	19	M16	1.48
1 1/2	40	48.6	49.1	140	16	105	4	19	M16	1.56
2	50	60.5	61.1	155	16	120	4	19	M16	1.88
2 1/2	65	76.3	77.1	175	18	140	4	19	M16	2.60
3	80	89.1	90.0	185	18	150	8	19	M16	2.61
3 1/2	90	101.6	102.6	195	18	160	8	19	M16	2.76
4	100	114.3	115.4	210	18	175	8	19	M16	3.14
5	125	139.8	141.2	250	20	210	8	23	M20	4.77
6	150	165.2	166.6	280	22	240	8	23	M20	6.34
8	200	216.3	218.0	330	22	290	12	23	M20	7.53
10	250	267.4	269.5	400	24	355	12	25	M22	11.80
12	300	318.5	321.0	445	26	400	16	25	M22	13.60
14	350	355.6	358.1	490	26	445	16	25	M22	17.20
16	400	406.4	409.0	560	28	510	16	27	M24	24.00
18	450	457.2	460.0	620	30	565	20	27	M24	32.00
20	500	508.0	511.0	675	30	620	20	27	M24	35.00
22	550	558.8	562.0	745	32	680	20	33	M30	47.00
24	600	609.6	613.0	795	32	730	24	33	M30	49.00

FLANGES

ANSI FLANGES (ANSI B16.5)

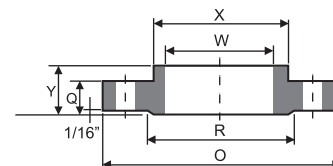


150LB. BLIND FLANGES

Nominal Pipe Size	Outside Dia. of Flange	Thickness of Flange	Dia. of Raised Face	Drilling Template			Approx. Weight
				No. of Holes	Dia. of Bolts	Dia. of Bolt Circle	
	O	Q	R				kg
1/2	3 1/2	7/16	1 3/8	4	1/2	2 3/8	0.5
3/4	3 7/8	1/2	1 11/16	4	1/2	2 3/4	0.9
1	4 1/4	9/16	2	4	1/2	3 1/8	0.9
1 1/4	4 5/8	5/8	2 1/2	4	1/2	3 1/2	1.4
1 1/2	5	11/16	2 7/8	4	1/2	3 7/8	1.8
2	6	3/4	3 5/8	4	5/8	4 3/4	2.3
2 1/2	7	7/8	4 1/8	4	5/8	5 1/2	3.2
3	7 1/2	15/16	5	4	5/8	6	4.1
3 1/2	8 1/2	15/16	5 1/2	8	5/8	7	5.9
4	9	15/16	6 3/16	8	5/8	7 1/2	7.7
5	10	15/16	7 5/16	8	3/4	8 1/2	9.1
6	11	1	8 1/2	8	3/4	9 1/2	11.8
8	13 1/2	1 1/8	10 5/8	8	3/4	11 3/4	20.4
10	16	1 3/16	12 3/4	12	7/8	14 1/4	31.8
12	19	1 1/4	15	12	7/8	17	49.9
14	21	1 3/8	16 1/4	12	1	18 3/4	63.5
16	23 1/2	1 7/16	18 1/2	16	1	21 1/4	81.6
18	25	1 9/16	21	16	1 1/8	22 3/4	99.8
20	27 1/2	1 11/16	23	20	1 1/8	25	129.3
24	32	1 7/8	27 1/2	20	1 1/4	29 1/2	195.0

300LB. BLIND FLANGES

Nominal Pipe Size	Outside Dia. of Flange	Thickness of Flange	Dia. of Raised Face	Drilling Template			Approx. Weight
				No. of Holes	Dia. of Bolts	Dia. of Bolt Circle	
	O	Q	R				kg
1/2	3 3/4	9/16	1 3/8	4	1/2	2 5/8	0.9
3/4	4 5/8	5/8	1 11/16	4	5/8	3 1/4	1.4
1	4 7/8	11/16	2	4	5/8	3 1/2	1.4
1 1/4	5 1/4	3/4	2 1/2	4	5/8	3 7/8	2.0
1 1/2	6 1/8	13/16	2 7/8	4	3/4	4 1/2	2.7
2	6 1/2	7/8	3 5/8	8	5/8	5	3.6
2 1/2	7 1/2	1	4 1/8	8	3/4	5 7/8	5.4
3	8 1/4	1 1/8	5	8	3/4	6 5/8	7.3
3 1/2	9	1 3/16	5 1/2	8	3/4	7 1/4	10.0
4	10	1 1/4	6 3/16	8	3/4	7 7/8	12.3
5	11	1 3/8	7 5/16	8	3/4	9 1/4	17.1
6	12 1/2	1 7/16	8 1/2	12	3/4	10 5/8	22.2
8	15	1 5/8	10 5/8	12	7/8	13	36.7
10	17 1/2	1 7/8	12 3/4	16	1	15 1/4	56.7
12	20 1/2	2	15	16	1 1/8	17 3/4	83.9
14	23	2 1/8	16 1/4	20	1 1/8	20 1/4	113.4
16	25 1/2	2 1/4	18 1/2	20	1 1/4	22 1/2	133.8
18	28	2 3/8	21	24	1 1/4	24 3/4	179.2
20	30 1/2	2 1/2	23	24	1 1/4	27	229.1
24	36	2 3/4	27 1/4	24	1 1/2	32	358.3



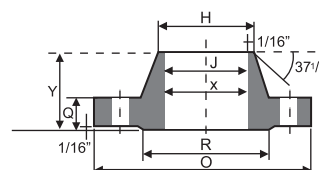
150LB. SLIP-ON FLANGES

Nominal Pipe Size	Outside Dia. of Flange	Thickness of Flange	Dia. of Raised Face	Dia. of Hub at Base	Length thru Hub	Dia. of Bore	Drilling Template			Approx. Weight
							No. of Holes	Dia. of Bolts	Dia. of Bolt Circle	
	O	Q	R	X	Y	W				kg
1/2	3 1/2	7/16	1 3/8	1 3/16	5/8	0.88	4	1/2	2 3/8	0.4
3/4	3 7/8	1/2	1 11/16	1 1/2	5/8	1.09	4	1/2	2 3/4	0.6
1	4 1/4	9/16	2	1 15/16	1 1/16	1.36	4	1/2	3 1/8	0.8
1 1/4	4 5/8	5/8	2 1/2	2 5/16	1 3/16	1.70	4	1/2	3 1/2	1.1
1 1/2	5	11/16	2 7/8	2 9/16	7/8	1.95	4	1/2	3 7/8	1.4
2	6	3/4	3 5/8	3 1/16	1	2.44	4	5/8	4 3/4	2.3
2 1/2	7	7/8	4 1/8	3 9/16	1 1/8	2.94	4	5/8	5 1/2	3.5
3	7 1/2	15/16	5	4 1/4	1 3/16	3.57	4	5/8	6	4.0
3 1/2	8 1/2	15/16	5 1/2	4 13/16	1 1/4	4.07	8	5/8	7	4.9
4	9	15/16	6 3/16	5 5/16	1 5/16	4.57	8	5/8	7 1/2	5.6
5	10	15/16	7 5/16	6 7/16	1 7/16	5.66	8	3/4	8 1/2	6.8
6	11	1	8 1/2	7 9/16	1 9/16	6.72	8	3/4	9 1/2	8.5
8	13 1/2	1 1/8	10 5/8	9 11/16	1 3/4	8.72	8	3/4	11 3/4	13.4
10	16	1 3/16	12 3/4	12	1 15/16	10.88	12	7/8	14 1/4	19.5
12	19	1 1/4	15	14 3/8	2 3/16	12.88	12	7/8	17	29.0
14	21	1 3/8	16 1/4	15 3/4	2 1/4	14.14	12	1	18 3/4	38.6
16	23 1/2	1 7/16	18 1/2	18	2 1/2	16.16	16	1	21 1/4	45.0
18	25	1 9/16	21	19 7/8	2 11/16	18.18	16	1 1/8	22 3/4	54.4
20	27 1/2	1 11/16	23	22	2 7/8	20.20	20	1 1/8	25	70.3
24	32	1 7/8	27 1/4	26 1/8	3 1/4	24.25	20	1 1/4	29 1/2	90.0

300LB. SLIP-ON FLANGES

Nominal Pipe Size	Outside Dia. of Flange	Thickness of Flange	Dia. of Raised Face	Dia. of Hub at Base	Length thru Hub	Dia. of Bore	Drilling Template			Approx. Weight
							No. of Holes	Dia. of Bolts	Dia. of Bolt Circle	
	O	Q	R	X	Y	W				kg
1/2	3 3/4	9/16	1 3/8	1 1/2	7/8	0.88	4	1/2	2 5/8	0.7
3/4	4 5/8	5/8	1 11/16	1 7/8	1	1.09	4	5/8	3 1/4	1.2
1	4 7/8	11/16	2	2 1/8	1 1/16	1.36	4	5/8	3 1/2	1.5
1 1/4	5 1/4	3/4	2 1/2	2 1/2	1 1/16	1.70	4	5/8	3 7/8	1.8
1 1/2	6 1/8	13/16	2 7/8	2 3/4	1 3/16	1.95	4	3/4	4 1/2	2.7
2	6 1/2	7/8	3 5/8	3 5/16	1 5/16	2.44	8	5/8	5	3.1
2 1/2	7 1/2	1	4 1/8	3 15/16	1 1/2	2.94	8	3/4	5 7/8	4.5
3	8 1/4	1 1/8	5	4 5/8	1 11/16	3.57	8	3/4	6 5/8	6.2
3 1/2	9	1 3/16	5 1/2	5 1/4	1 3/4	4.07	8	3/4	7 1/4	7.6
4	10	1 1/4	6 3/16	5 3/4	1 7/8	4.57	8	3/4	7 7/8	10.0
5	11	1 3/8	7 5/16	7	2	5.66	8	3/4	9 1/4	12.7
6	12 1/2	1 7/16	8 1/2	8 1/8	2 1/16	6.72	12	3/4	10 5/8	16.6
8	15	1 5/8	10 5/8	10 1/4	2 7/16	8.72	12	7/8	13	26.5
10	17 1/2	1 7/8	12 3/4	12 5/8	2 5/8	10.88	16	1	15 1/4	36.7
12	20 1/2	2	15	14 3/4	2 7/8	12.88	16	1 1/8	17 3/4	52.2
14	23	2 1/8	16 1/4	16 3/4	3	14.14	20	1 1/8	20 1/4	74.4
16	25 1/2	2 1/4	18 1/2	19	3 1/4	16.16	20	1 1/4	22 1/2	99.8
18	28	2 3/8	21	21	3 1/2	18.18	24	1 1/4	24 3/4	127.0
20	30 1/2	2 1/2	23	23 1/8	3 3/4	20.20	24	1 1/4	27	147.0
24	36	2 3/4	27 1/4	27 5/8	4 3/16	24.25	24	1 1/2	32	222.0

FLANGES



150LB. WELDING NECK FLANGES

Nominal Pipe Size	Outside Dia. of Flange	Thickness of Flange	Dia. of Raised Face	Dia. of Hub at Base	Min. Dia. of Hub at Pt. of Welding	Length thru Hub	Dia. of Bore	Drilling Template		
								No. of Holes	Dia. of Bolts	Dia. of Bolt Circle
	O	Q	R	X	H	Y	J			
1/2	3 1/2	7/16	1 3/8	1 3/16	0.84	1 7/8	0.62	4	1/2	2 3/8
3/4	3 7/8	1/2	1 11/16	1 1/2	1.05	2 1/16	0.82	4	1/2	2 3/4
1	4 1/4	9/16	2	1 15/16	1.32	2 3/16	1.05	4	1/2	3 1/8
1 1/4	4 5/8	5/8	2 1/2	2 15/16	1.66	2 1/4	1.38	4	1/2	3 1/2
1 1/2	5	11/16	2 7/8	2 9/16	1.90	2 7/16	1.61	4	1/2	3 7/8
2	6	3/4	3 5/8	3 1/16	2.38	2 1/2	2.07	4	5/8	4 3/4
2 1/2	7	7/8	4 1/8	3 9/16	2.88	2 3/4	2.47	4	5/8	5 1/2
3	7 1/2	15/16	5	4 1/4	3.50	2 3/4	3.07	4	5/8	6
3 1/2	8 1/2	15/16	5 1/2	4 13/16	4.00	2 13/16	3.55	8	5/8	7
4	9	15/16	6 3/16	5 5/16	4.50	3	4.03	8	5/8	7 1/2
5	10	15/16	7 9/16	6 7/16	5.56	3 1/2	5.05	8	3/4	8 1/2
6	11	1	8 1/2	7 5/16	6.63	3 1/2	6.07	8	3/4	9 1/2
8	13 1/2	1 1/8	10 5/8	9 11/16	8.63	4	7.98	8	3/4	11 3/4
10	16	1 3/16	12 3/4	12	10.75	4	10.02	12	7/8	14 1/4
12	19	1 1/4	15	14 3/8	12.75	4 1/2	12.00	12	7/8	17
14	21	1 3/8	16 1/4	15 3/4	14.00	5	13.25	12	1	18 3/4
16	23 1/2	1 7/16	18 1/2	18	16.00	5	15.25	16	1	21 1/4
18	25	1 9/16	21	19 7/8	18.00	5 1/2	17.25	16	1 1/8	22 3/4
20	27 1/2	1 11/16	23	22	20.00	5 11/16	19.25	20	1 1/8	25
24	32	1 7/8	27 1/4	26 1/8	24.00	6	23.25	20	1 1/4	29 1/2

300LB. WELDING NECK FLANGES

Nominal Pipe Size	Outside Dia. of Flange	Thickness of Flange	Dia. of Raised Face	Dia. of Hub at Base	Min. Dia. of Hub at Pt. of Welding	Length thru Hub	Dia. of Bore	Drilling Template			Approx. Weight
								No. of Holes	Dia. of Bolts	Dia. of Bolt Circle	
	O	Q	R	X	H	Y	J				kg
1/2	3 3/4	9/16	1 3/8	1 1/2	0.84	2 1/16	0.62	4	1/2	2 5/8	0.9
3/4	4 5/8	5/8	1 11/16	1 7/8	1.05	2 1/4	0.82	4	5/8	3 1/4	1.5
1	4 7/8	11/16	2	2 1/8	1.32	2 7/16	1.05	4	5/8	3 1/2	1.7
1 1/4	5 1/4	3/4	2 1/2	2 1/2	1.66	2 9/16	1.38	4	5/8	3 7/8	2.1
1 1/2	6 1/8	13/16	2 7/8	2 3/4	1.90	2 11/16	1.61	4	3/4	4 1/2	3.2
2	6 1/2	7/8	3 5/8	3 5/16	2.38	2 3/4	2.07	8	5/8	5	3.6
2 1/2	7 1/2	1	4 1/8	3 15/16	2.88	3	2.47	8	3/4	5 7/8	5.4
3	8 1/4	1 1/8	5	4 5/8	3.50	3 1/8	3.07	8	3/4	6 5/8	7.5
3 1/2	9	1 3/16	5 1/2	5 1/4	4.00	3 3/16	3.55	8	3/4	7 1/4	9.2
4	10	1 1/4	6 3/16	5 3/4	4.50	3 3/8	4.03	8	3/4	7 7/8	12.7
5	11	1 3/8	7 5/16	7	5.56	3 7/8	5.05	8	3/4	9 1/4	16.0
6	12 1/2	1 7/16	8 1/2	8 1/8	6.63	3 7/8	6.07	12	3/4	10 5/8	21.0
8	15	1 5/8	10 5/8	10 1/4	8.63	4 3/8	7.98	12	7/8	13	33.0
10	17 1/2	1 7/8	12 3/4	12 5/8	10.75	4 5/8	10.02	16	1	15 1/4	46.6
12	20 1/2	2	15	14 3/4	12.75	5 1/8	12.00	16	1 1/8	17 3/4	69.0
14	23	2 1/8	16 1/4	16 3/4	14.00	5 5/8	13.25	20	1 1/8	20 1/4	93.4
16	25 1/2	2 1/4	18 1/2	19	16.00	5 3/4	15.25	20	1 1/4	22 1/2	119.6
18	28	2 3/8	21	21	18.00	6 1/4	17.25	24	1 1/4	24 3/4	150.5
20	30 1/2	2 1/2	23	23 1/8	20.00	6 3/8	19.25	24	1 1/4	27	184.4
24	36	2 3/4	27 1/4	27 5/8	24.00	6 5/8	23.25	24	1 1/2	32	274.7

CLASS 600

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	3 ³ / ₄ (95)	2 ⁵ / ₈ (67)	4	1/2 (13)	5/8 (16)	1 ³ / ₈ (35)	1/4 (6)	-	-	9/16 (14)
3/4	4 ⁵ / ₈ (117)	3 ¹ / ₄ (83)	4	5/8 (16)	3/4 (19)	1 ¹¹ / ₁₆ (43)	1/4 (6)	-	-	5/8 (16)
1	4 ⁷ / ₈ (124)	3 ¹ / ₂ (89)	4	5/8 (16)	3/4 (19)	2 (51)	1/4 (6)	-	-	1 ¹¹ / ₁₆ (17)
1 ¹ / ₄	5 ¹ / ₄ (133)	3 ⁷ / ₈ (98)	4	5/8 (16)	3/4 (19)	2 ¹ / ₂ (64)	1/4 (6)	-	-	1 ³ / ₁₆ (21)
1 ¹ / ₂	6 ¹ / ₈ (156)	4 ¹ / ₂ (114)	4	3/4 (19)	7/8 (22)	2 ⁷ / ₈ (73)	1/4 (6)	-	-	7/8 (22)
2	6 ¹ / ₂ (165)	5 (127)	8	5/8 (16)	3/4 (19)	3 ⁵ / ₈ (92)	1/4 (6)	-	-	1 (25)
2 ¹ / ₂	7 ¹ / ₂ (191)	5 ⁷ / ₈ (149)	8	3/4 (19)	7/8 (22)	4 ¹ / ₈ (105)	1/4 (6)	-	-	1 ¹ / ₈ (29)
3	8 ¹ / ₄ (210)	6 ⁵ / ₈ (168)	8	3/4 (19)	7/8 (22)	5 (127)	1/4 (6)	-	-	1 ¹ / ₄ (32)
4	10 ³ / ₄ (273)	8 ¹ / ₂ (216)	8	7/8 (22)	1 (25)	6 ³ / ₁₆ (157)	1/4 (6)	-	-	1 ¹ / ₂ (38)
5	13 (330)	10 ¹ / ₂ (267)	8	1 (25)	1 ¹ / ₈ (29)	7 ⁵ / ₁₆ (186)	1/4 (6)	-	-	1 ³ / ₄ (44)
6	14 (356)	11 ¹ / ₂ (292)	12	1 (25)	1 ¹ / ₈ (29)	8 ¹ / ₂ (216)	1/4 (6)	-	-	1 ⁷ / ₈ (48)
8	16 ¹ / ₂ (419)	13 ³ / ₄ (349)	12	1 ¹ / ₈ (29)	1 ¹ / ₄ (32)	10 ⁵ / ₈ (270)	1/4 (6)	-	-	2 ³ / ₁₆ (56)
10	20 (508)	17 (432)	16	1 ¹ / ₄ (32)	1 ³ / ₈ (35)	12 ³ / ₄ (324)	1/4 (6)	-	-	2 ¹ / ₂ (64)
12	22 (559)	19 ¹ / ₄ (489)	20	1 ¹ / ₄ (32)	1 ³ / ₈ (35)	15 (381)	1/4 (6)	-	-	2 ⁵ / ₈ (67)
14	23 ³ / ₄ (603)	20 ³ / ₄ (527)	20	1 ³ / ₈ (35)	1 ¹ / ₂ (38)	16 ¹ / ₄ (413)	1/4 (6)	-	-	2 ³ / ₄ (70)
16	27 (686)	23 ³ / ₄ (603)	20	1 ¹ / ₂ (38)	1 ⁵ / ₈ (41)	18 ¹ / ₂ (470)	1/4 (6)	-	-	3 (76)
18	29 ¹ / ₄ (743)	25 ³ / ₄ (654)	20	1 ⁵ / ₈ (41)	1 ³ / ₄ (44)	21 (533)	1/4 (6)	-	-	3 ¹ / ₄ (83)
20	32 (813)	28 ¹ / ₂ (724)	24	1 ⁵ / ₈ (41)	1 ³ / ₄ (44)	23 (584)	1/4 (6)	-	-	3 ¹ / ₂ (89)
24	37 (940)	33 (838)	24	1 ⁷ / ₈ (48)	2 (51)	27 ¹ / ₄ (692)	1/4 (6)	-	-	4 (102)

CLASS 900

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	4 ³ / ₄ (121)	3 ¹ / ₄ (83)	4	3/4 (19)	7/8 (22)	1 ³ / ₈ (35)	1/4 (6)	-	-	7/8 (22)
3/4	5 ¹ / ₈ (130)	3 ¹ / ₂ (89)	4	3/4 (19)	7/8 (22)	1 ¹¹ / ₁₆ (43)	1/4 (6)	-	-	1 (25)
1	5 ⁷ / ₈ (149)	4 (102)	4	7/8 (22)	1 (25)	2 (51)	1/4 (6)	-	-	1 ¹ / ₈ (29)
1 ¹ / ₄	6 ¹ / ₄ (159)	4 ³ / ₈ (111)	4	7/8 (22)	1 (25)	2 ¹ / ₂ (64)	1/4 (6)	-	-	1 ¹ / ₈ (29)
1 ¹ / ₂	7 (178)	4 ⁷ / ₈ (124)	4	1 (25)	1 ¹ / ₈ (29)	2 ⁷ / ₈ (73)	1/4 (6)	-	-	1 ¹ / ₄ (32)
2	8 ¹ / ₂ (216)	6 ¹ / ₂ (165)	8	7/8 (22)	1 (25)	3 ⁵ / ₈ (92)	1/4 (6)	-	-	1 ¹ / ₂ (38)
2 ¹ / ₂	9 ⁵ / ₈ (244)	7 ¹ / ₂ (191)	8	1 (25)	1 ¹ / ₈ (29)	4 ¹ / ₈ (105)	1/4 (6)	-	-	1 ⁵ / ₈ (41)
3	9 ¹ / ₂ (241)	7 ¹ / ₂ (192)	8	7/8 (22)	1 (25)	5 (127)	1/4 (6)	-	-	1 ¹ / ₂ (38)
4	11 ¹ / ₂ (292)	9 ¹ / ₄ (235)	8	1 ¹ / ₈ (29)	1 ¹ / ₄ (32)	6 ³ / ₁₆ (157)	1/4 (6)	-	-	1 ³ / ₄ (44)
5	13 ³ / ₄ (349)	11 (279)	8	1 ¹ / ₄ (32)	1 ³ / ₈ (35)	7 ⁵ / ₁₆ (186)	1/4 (6)	-	-	2 (51)
6	15 (381)	12 ¹ / ₂ (318)	12	1 ¹ / ₈ (29)	1 ¹ / ₄ (32)	8 ¹ / ₂ (216)	1/4 (6)	-	-	2 ³ / ₁₆ (56)
8	18 ¹ / ₂ (470)	15 ¹ / ₂ (394)	12	1 ³ / ₈ (35)	1 ¹ / ₂ (38)	10 ⁵ / ₈ (270)	1/4 (6)	-	-	2 ¹ / ₂ (64)
10	21 ¹ / ₂ (546)	18 ¹ / ₂ (470)	16	1 ³ / ₈ (35)	1 ¹ / ₂ (38)	12 ³ / ₄ (324)	1/4 (6)	-	-	2 ³ / ₄ (70)
12	24 (610)	21 (533)	20	1 ³ / ₈ (35)	1 ¹ / ₂ (38)	15 (381)	1/4 (6)	-	-	3 ¹ / ₈ (80)
14	25 ¹ / ₄ (641)	22 (559)	20	1 ¹ / ₂ (38)	1 ⁵ / ₈ (41)	16 ¹ / ₄ (413)	1/4 (6)	-	-	3 ³ / ₈ (86)
16	27 ³ / ₄ (705)	24 ¹ / ₄ (616)	20	1 ⁵ / ₈ (41)	1 ³ / ₄ (44)	18 ¹ / ₂ (470)	1/4 (6)	-	-	3 ¹ / ₂ (89)
18	31 (787)	27 (686)	20	1 ⁷ / ₈ (48)	2 (51)	21 (533)	1/4 (6)	-	-	4 (102)
20	33 ³ / ₄ (857)	29 ¹ / ₂ (749)	20	2 (51)	2 ¹ / ₈ (54)	23 (584)	1/4 (6)	-	-	4 ¹ / ₄ (108)
24	41 (1041)	35 ¹ / ₂ (902)	20	2 ¹ / ₂ (64)	2 ⁵ / ₈ (67)	27 ¹ / ₄ (692)	1/4 (6)	-	-	5 ¹ / ₂ (140)

FLANGES

CLASS 1500

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	4 3/4(121)	3 1/4(83)	4	3/4(19)	7/8(22)	1 3/8(35)	1/4(6)	-	-	7/8(22)
3/4	5 1/8(130)	3 1/2(89)	4	3/4(19)	7/8(22)	1 11/16(43)	1/4(6)	-	-	1(25)
1	5 7/8(149)	4(102)	4	7/8(22)	1(25)	2(51)	1/4(6)	-	-	1 1/8(29)
1 1/4	6 1/4(159)	4 3/8(111)	4	7/8(22)	1(25)	2 1/2(64)	1/4(6)	-	-	1 1/8(29)
1 1/2	7(178)	4 7/8(124)	4	1(25)	1 1/8(29)	2 7/8(73)	1/4(6)	-	-	1 1/4(32)
2	8 1/2(216)	6 1/2(165)	8	7/8(22)	1(25)	3 5/8(92)	1/4(6)	-	-	1 1/2(38)
2 1/2	9 5/8(244)	7 1/2(191)	8	1(25)	1 1/8(29)	4 1/8(105)	1/4(6)	-	-	1 5/8(41)
3	10 1/2(267)	8(203)	8	1 1/8(29)	1 1/4(32)	5(127)	1/4(6)	-	-	1 7/8(48)
4	12 1/4(311)	9 1/2(241)	8	1 1/4(32)	1 3/8(35)	6 3/16(157)	1/4(6)	-	-	2 1/8(54)
5	14 3/4(375)	11 1/2(292)	8	1 1/2(38)	1 5/8(41)	7 5/16(186)	1/4(6)	-	-	2 7/8(73)
6	15 1/2(394)	12 1/2(318)	12	1 3/8(35)	1 1/2(38)	8 1/2(216)	1/4(6)	-	-	3 1/4(83)
8	19(483)	15 1/2(394)	12	1 5/8(41)	1 3/4(44)	10 5/8(270)	1/4(6)	-	-	3 5/8(92)
10	23(584)	19(483)	12	1 7/8(48)	2(51)	12 3/4(324)	1/4(6)	-	-	4 1/4(108)
12	26 1/2(673)	22 1/2(571)	16	2(51)	2 1/8(54)	15(381)	1/4(6)	-	-	4 7/8(124)
14	29 1/2(749)	25(635)	16	2 1/4(57)	2 3/8(60)	16 1/4(413)	1/4(6)	-	-	5 1/4(133)
16	32 1/2(826)	27 3/4(705)	16	2 1/2(64)	2 5/8(67)	18 1/2(470)	1/4(6)	-	-	5 3/4(146)
18	36(914)	30 1/2(775)	16	2 3/4(70)	2 7/8(73)	21(533)	1/4(6)	-	-	6 3/8(162)
20	38 3/4(984)	32 3/4(832)	16	3(76)	3 1/8(79)	23(584)	1/4(6)	-	-	7(178)
24	46(1168)	39(991)	16	3 1/2(89)	3 5/8(92)	27 1/4(692)	1/4(6)	-	-	8(203)

YICK HOE

SLIP-ON FLANGES(BS 4504)

PN 6

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	80	55	4	M10	11	40	2	12	-	-
3/4	90	65	4	M10	11	50	2	14	-	-
1	100	75	4	M10	11	60	2	14	-	-
1 1/4	120	90	4	M12	14	70	2	16	-	-
1 1/2	130	100	4	M12	14	80	3	16	-	-
2	140	110	4	M12	14	90	3	16	-	-
2 1/2	160	130	4	M12	14	110	3	16	-	-
3	190	150	4	M16	18	128	3	18	-	-
4	210	170	4	M16	18	148	3	18	-	-
5	240	200	8	M16	18	178	3	20	-	-
6	265	225	8	M16	18	202	3	20	-	-
8	320	280	8	M16	18	258	3	22	-	-
10	375	335	12	M16	18	312	3	24	-	-
12	440	395	12	M20	22	365	4	24	-	-
14	490	445	12	M20	22	415	4	26	-	-
16	540	495	16	M20	22	465	4	28	-	-
18	595	550	16	M20	22	520	4	30	-	-
20	645	600	20	M20	22	570	4	30	-	-
24	755	705	20	M24	26	670	5	30	-	-

PN 10

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	95	65	4	M12	14	45	2	14	-	-
3/4	105	75	4	M12	14	58	2	16	-	-
1	115	85	4	M12	14	68	2	16	-	-
1 1/4	140	100	4	M16	18	78	2	18	-	-
1 1/2	150	110	4	M16	18	88	3	18	-	-
2	165	125	4	M16	18	102	3	20	-	-
2 1/2	185	145	4	M16	18	122	3	20	-	-
3	200	160	8	M16	18	138	3	22	-	-
4	220	180	8	M16	18	158	3	24	-	-
5	250	210	8	M16	18	188	3	26	-	-
6	285	240	8	M20	22	212	3	26	-	-
8	340	295	8	M20	22	268	3	26	-	-
10	395	350	12	M20	22	320	3	28	-	-
12	445	400	12	M20	22	370	4	28	-	-
14	505	460	16	M20	22	430	4	30	-	-
16	565	515	16	M24	23	482	4	32	-	-
18	615	565	20	M24	26	532	4	32	-	-
20	670	620	20	M24	26	585	4	34	-	-
24	780	725	20	M27	30	685	5	36	-	-

FLANGES

PN 16

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	95	65	4	M12	14	45	2	14	6	16
3/4	105	75	4	M12	14	58	2	16	6	18
1	115	85	4	M12	14	68	2	16	8	18
1 1/4	140	100	4	M16	18	78	2	18	8	18
1 1/2	150	110	4	M16	18	88	3	18	9	18
2	165	125	4	M16	18	102	3	20	11	20
2 1/2	185	145	4	M16	18	122	3	20	13	18
3	200	160	8	M16	18	138	3	22	13	20
4	220	180	8	M16	18	158	3	24	16	20
5	250	210	8	M16	18	188	3	26	-	22
6	285	240	8	M20	22	212	3	26	-	22
8	340	295	12	M20	22	268	3	30	-	24
10	405	355	12	M24	26	320	3	32	-	26
12	460	410	12	M24	26	378	4	32	-	28
14	520	470	16	M24	26	438	4	36	-	30
16	580	525	16	M27	30	490	4	38	-	32
18	640	585	20	M27	30	550	4	40	-	-
20	715	650	20	M30	33	610	4	42	-	36
24	840	770	20	M33	36	725	5	48	-	40

PN 25

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	95	65	4	M12	14	45	2	16	8	16
3/4	105	75	4	M12	14	58	2	18	8	18
1	115	85	4	M12	14	68	2	18	9	18
1 1/4	140	100	4	M16	18	78	2	20	9	18
1 1/2	150	110	4	M16	18	88	3	20	11	18
2	165	125	4	M16	18	102	3	22	11	20
2 1/2	185	145	8	M16	18	122	3	24	13	22
3	200	160	8	M16	18	138	3	26	14	24
4	235	190	8	M20	22	162	3	28	17	24
5	270	220	8	M24	26	188	3	30	-	26
6	300	250	8	M24	26	218	3	34	-	28
8	360	310	12	M24	26	278	3	34	-	30
10	425	370	12	M27	30	335	3	36	-	32
12	485	430	16	M27	30	395	4	40	-	34
14	555	490	16	M30	33	450	4	44	-	38
16	620	550	16	M33	36	505	4	48	-	40
20	730	660	20	M33	36	615	4	52	-	44
24	845	770	20	M36	39	720	5	-	-	46

PN 40

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	95	65	4	M12	14	45	2	-	9	16
3/4	105	75	4	M12	14	58	2	-	9	18
1	115	85	4	M12	14	68	2	-	11	18
1 1/4	140	100	4	M16	18	78	2	-	11	18
1 1/2	150	110	4	M16	18	88	3	-	13	18
2	165	125	4	M16	18	102	3	-	13	20
2 1/2	185	145	8	M16	18	122	3	-	14	22
3	200	160	8	M16	18	138	3	-	16	24
4	235	190	8	M20	22	162	3	-	19	24
5	270	220	8	M24	26	188	3	-	-	26
6	300	250	8	M24	26	218	3	-	-	28
8	375	320	12	M27	30	285	3	-	-	34
10	450	385	12	M30	33	345	3	-	-	38
12	515	450	16	M30	33	410	4	-	-	42
14	580	510	16	M33	36	465	4	-	-	46
16	660	585	16	M36	39	535	4	-	-	50
20	755	670	20	M39	42	615	4	-	-	52
24	890	795	20	M45	48	735	5	-	-	60

PN 64

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	105	75	4	M12	14	45	2	-	-	20
3/4	130	90	4	M16	18	58	2	-	-	22
1	140	100	4	M16	18	68	2	-	-	24
1 1/4	155	110	4	M20	22	78	2	-	-	26
1 1/2	170	125	4	M20	22	88	3	-	-	28
2	180	135	4	M20	22	102	3	-	-	26
2 1/2	205	160	8	M20	22	122	3	-	-	26
3	215	170	8	M20	22	138	3	-	-	28
4	250	200	8	M24	26	162	3	-	-	30
5	295	240	8	M27	30	188	3	-	-	34
6	345	280	8	M30	33	218	3	-	-	36
8	415	345	12	M33	36	285	3	-	-	42
10	470	400	12	M33	36	345	3	-	-	46
12	530	460	16	M33	36	410	4	-	-	52
14	600	525	16	M36	39	465	4	-	-	56
16	670	585	16	M39	42	535	4	-	-	60
20	800	705	20	M45	48	615	4	-	-	68
24	930	820	20	M52	56	735	4	-	-	76

FLANGES

PN 100

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	105	75	4	M12	14	45	2	-	-	20
3/4	130	90	4	M16	18	58	2	-	-	22
1	140	100	4	M16	18	68	2	-	-	24
1 1/4	155	110	4	M20	22	78	2	-	-	26
1 1/2	170	125	4	M20	22	88	3	-	-	28
2	195	145	4	M24	26	102	3	-	-	30
2 1/2	220	170	8	M24	26	122	3	-	-	34
3	230	180	8	M24	26	138	3	-	-	36
4	265	210	8	M27	30	162	3	-	-	40
5	315	250	8	M30	33	188	3	-	-	40
6	355	290	12	M30	33	218	3	-	-	44
8	430	360	12	M33	36	285	3	-	-	52
10	505	430	12	M36	39	345	3	-	-	60
12	585	500	16	M39	42	410	4	-	-	68
14	655	560	16	M45	48	465	4	-	-	74
16	715	620	16	M45	48	535	4	-	-	78
20	870	760	20	M52	56	615	4	-	-	94
24	990	875	20	M56	62	735	4	-	-	104

PN 160

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	105	75	4	M12	14	45	2	-	-	20
3/4	130	90	4	M16	18	58	2	-	-	22
1	140	100	4	M16	18	68	2	-	-	24
1 1/4	155	110	4	M20	22	78	2	-	-	26
1 1/2	170	125	4	M20	22	88	3	-	-	28
2	195	145	4	M24	26	102	3	-	-	30
2 1/2	220	170	8	M24	26	122	3	-	-	34
3	230	180	8	M24	26	138	3	-	-	36
4	265	210	8	M27	30	162	3	-	-	40
5	315	250	8	M30	33	188	3	-	-	44
6	355	290	12	M30	33	218	3	-	-	50
8	430	360	12	M33	36	285	3	-	-	60
10	515	430	12	M39	42	345	3	-	-	68
12	585	500	16	M39	42	410	4	-	-	78

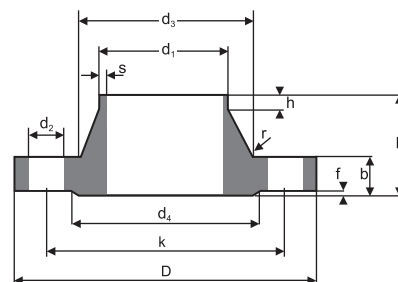
PN 250

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	130	90	4	M16	18	45	2	-	-	26
3/4	135	95	4	M16	18	58	2	-	-	28
1	150	105	4	M20	22	68	2	-	-	28
1 1/4	165	120	4	M20	22	78	2	-	-	32
1 1/2	185	135	4	M24	26	88	3	-	-	34
2	200	150	8	M24	26	102	3	-	-	38
2 1/2	230	180	8	M24	26	122	3	-	-	42
3	255	200	8	M27	30	138	3	-	-	46
4	300	230	8	M30	33	162	3	-	-	54
5	340	275	12	M30	33	188	3	-	-	60
6	390	320	12	M33	36	218	3	-	-	68
8	485	400	12	M39	42	285	3	-	-	82
10	585	490	12	M45	48	345	3	-	-	100
12	690	590	16	M48	52	410	4	-	-	120

YICK HOE

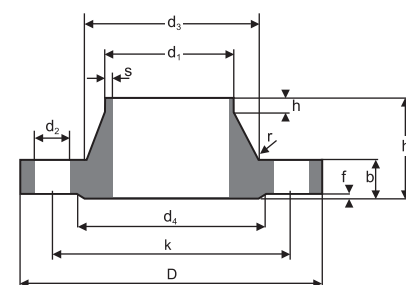
FLANGES

WELDING NECK FLANGES



DIN 2633 PN16 Raised face to DIN 2526, form C

Tube DN	Flange				Hub					Raised face		Drilling Template			
	d ₁ mm	D mm	b mm	k mm	h ₁ mm	d ₃ mm	s mm	r mm	h ₂ mm ≈	d ₄ mm	f mm	Number	Thread	d ₂ mm	Weight kg
15	20 21.3	95	14	65	35	30 32	2	4	6	45	2	4	M 12	14	0.648
20	25 26.9	105	16	75	38	38 40	2.3	4	6	58	2	4	M 12	14	0.952
25	30 33.7	115	16	85	38 45	42	2.6	4	6	68	2	4	M 12	14	1.140
32	38 42.4	140	16	100	40 56	52	2.6	6	6	78	2	4	M 16	18	1.690
40	44.5 48.3	150	16	110	42 64	60	2.6	6	7	88	3	4	M 16	18	1.860
50	57 60.3	165	18	125	45 75	72	2.9	6	8	102	3	4	M 16	18	2.530
65	76.1	185	18	145	45	90	2.9	6	10	122	3	4	M 16	18	3.060
80	88.9	200	20	160	50	105	3.2	8	10	138	3	8	M 16	18	3.700
100	108 114.3	220	20	180	52 131	125	3.6	8	12	158	3	8	M 16	18	4.620
125	133 139.7	250	22	210	55 156	150	4	8	12	188	3	8	M 16	18	6.300
150	159 168.3	285	22	240	55 184	175	4.5	10	12	212	3	8	M 20	22	7.750
200	219.1	340	24	295	62	235	5.9	10	16	268	3	12	M 20	22	11.000



DIN 2635 PN 40 Raised face to DIN 2526, form C

Tube DN	Flange				Hub					Raised face		Drilling Template			
	d ₁ mm	D mm	b mm	k mm	h ₁ mm	d ₃ mm	s mm	r mm	h ₂ mm	d ₄ mm ≈	f mm	Number	Thread	d ₂ mm	Weight kg
10	14 17.2	90	16	60	35	25 28	1.8	4	6	40	2	4	M 12	14	0.661
15	20 21.3	95	16	65	38	30 32	2	4	6	45	2	4	M 12	14	0.746
20	25 26.9	105	18	75	40	38 40	2.3	4	6	58	2	4	M 12	14	1.060
25	30 33.7	115	18	85	40	42 46	2.6	4	6	68	2	4	M 12	14	1.290
32	38 42.4	140	18	100	42	52 56	2.6	6	6	78	2	4	M 16	18	1.880
40	44.5 48.3	150	18	110	45	60 64	2.6	6	7	88	3	4	M 16	18	2.330
50	57 60.3	165	20	125	48	72 75	2.9	6	8	102	3	4	M 16	18	2.820
65	76.1	185	22	145	52	90	2.9	6	10	122	3	8	M 16	18	3.740
80	88.9	200	24	160	58	105	3.2	8	12	138	3	8	M 16	18	4.750
100	108 114.3	235	24	190	65	128 134	3.6	8	12	162	3	8	M 20	22	6.520
125	133 139.7	270	26	220	68	155 162	4	8	12	188	3	8	M 24	26	9.070
150	159 168.3	300	28	250	75	182 192	4.5	10	12	218	3	8	M 24	26	11.800
200	219.1	375	34	320	88	244	6.3	10	16	285	3	12	M 27	30	21.500

FLANGES

SLIP-ON FLANGES (BS10)

TABLE A

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	3 ³ / ₄ (95)	2 ⁵ / ₈ (67)	4	1/2(13)	9/16(14)	-	-	1/2(13)	1/4(6)	-
3/4	4(102)	2 ⁷ / ₈ (73)	4	1/2(13)	9/16(14)	-	-	1/2(13)	1/4(6)	-
1	4 ¹ / ₂ (114)	3 ¹ / ₄ (83)	4	1/2(13)	9/16(14)	-	-	1/2(13)	5/16(8)	-
1 ¹ / ₄	4 ³ / ₄ (121)	3 ⁷ / ₁₆ (87)	4	1/2(13)	9/16(14)	-	-	5/8(16)	5/16(8)	-
1 ¹ / ₂	5 ¹ / ₄ (133)	3 ⁷ / ₈ (98)	4	1/2(13)	9/16(14)	-	-	5/8(16)	3/8(10)	-
2	6(152)	4 ¹ / ₂ (114)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	5/8(16)	3/8(10)	-
2 ¹ / ₂	6 ¹ / ₂ (165)	5(127)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ¹ / ₁₆ (17)	7/16(11)	-
3	7 ¹ / ₄ (184)	5 ³ / ₄ (146)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ¹ / ₁₆ (17)	1/2(13)	-
4	8 ¹ / ₂ (216)	7(178)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	3/4(19)	5/8(16)	-
5	10(254)	8 ¹ / ₄ (210)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	3/4(19)	1 ¹ / ₁₆ (17)	-
6	11(279)	9 ¹ / ₄ (235)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ³ / ₁₆ (21)	1 ¹ / ₁₆ (17)	-
8	13 ¹ / ₄ (337)	11 ¹ / ₂ (292)	8	5/8(16)	1 ¹ / ₁₆ (17)	-	-	7/8(22)	3/4(19)	-
10	16(406)	14(356)	8	3/4(19)	7/8(22)	-	-	1 ⁵ / ₁₆ (24)	3/4(19)	-
12	18(457)	16(406)	8	3/4(19)	7/8(22)	-	-	1 ⁵ / ₁₆ (24)	7/8(22)	-
14	20 ³ / ₄ (527)	18 ¹ / ₂ (470)	8	7/8(22)	1(25)	-	-	1(25)	1(25)	-
16	22 ³ / ₄ (578)	20 ¹ / ₂ (521)	12	7/8(22)	1(25)	-	-	1 ¹ / ₁₆ (27)	1(25)	-
18	25 ¹ / ₄ (641)	23(584)	12	7/8(22)	1(25)	-	-	1 ¹ / ₁₆ (27)	1 ¹ / ₁₆ (27)	-
20	27 ³ / ₄ (705)	25 ¹ / ₄ (641)	12	7/8(22)	1(25)	-	-	1 ¹ / ₈ (29)	1 ¹ / ₈ (29)	-
24	32 ¹ / ₂ (826)	29 ³ / ₄ (756)	12	1(25)	1 ¹ / ₈ (29)	-	-	1 ³ / ₁₆ (30)	1 ³ / ₁₆ (30)	-

TABLE D

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	3 ³ / ₄ (95)	2 ⁵ / ₈ (67)	4	1/2(13)	9/16(14)	-	-	1/2(13)	1/4(6)	3/8(10)
3/4	4(102)	2 ⁷ / ₈ (73)	4	1/2(13)	9/16(14)	-	-	1/2(13)	1/4(6)	3/8(10)
1	4 ¹ / ₂ (114)	3 ¹ / ₄ (83)	4	1/2(13)	9/16(14)	-	-	1/2(13)	5/16(8)	3/8(10)
1 ¹ / ₄	4 ³ / ₄ (121)	3 ⁷ / ₁₆ (87)	4	1/2(13)	9/16(14)	-	-	5/8(16)	5/16(8)	1/2(13)
1 ¹ / ₂	5 ¹ / ₄ (133)	3 ⁷ / ₈ (98)	4	1/2(13)	9/16(14)	-	-	5/8(16)	3/8(10)	1/2(13)
2	6(152)	4 ¹ / ₂ (114)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ¹ / ₁₆ (17)	3/8(10)	9/16(14)
2 ¹ / ₂	6 ¹ / ₂ (165)	5(127)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ¹ / ₁₆ (17)	7/16(11)	9/16(14)
3	7 ¹ / ₄ (184)	5 ³ / ₄ (146)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	3/4(19)	1/2(13)	9/16(14)
4	8 ¹ / ₂ (216)	7(178)	4	5/8(16)	1 ¹ / ₁₆ (17)	-	-	3/4(19)	5/8(16)	1 ¹ / ₁₆ (17)
5	10(254)	8 ¹ / ₄ (210)	8	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ³ / ₁₆ (21)	1 ¹ / ₁₆ (17)	1 ¹ / ₁₆ (17)
6	11(279)	9 ¹ / ₄ (235)	8	5/8(16)	1 ¹ / ₁₆ (17)	-	-	1 ³ / ₁₆ (21)	1 ¹ / ₁₆ (17)	1 ¹ / ₁₆ (17)
8	13 ¹ / ₄ (337)	11 ¹ / ₂ (292)	8	5/8(16)	1 ¹ / ₁₆ (17)	-	-	7/8(22)	3/4(19)	3/4(19)
10	16(406)	14(356)	8	3/4(19)	7/8(22)	-	-	1(25)	3/4(19)	3/4(19)
12	18(457)	16(406)	12	3/4(19)	7/8(22)	-	-	1(25)	7/8(22)	7/8(22)
14	20 ³ / ₄ (527)	18 ¹ / ₂ (470)	12	7/8(22)	1(25)	-	-	1 ¹ / ₈ (29)	1(25)	1(25)
16	22 ³ / ₄ (578)	20 ¹ / ₂ (521)	12	7/8(22)	1(25)	-	-	1 ¹ / ₈ (29)	1(25)	1(25)
18	25 ¹ / ₄ (641)	23(584)	12	7/8(22)	1(25)	-	-	1 ¹ / ₄ (32)	1 ¹ / ₈ (29)	1 ¹ / ₈ (29)
20	27 ³ / ₄ (705)	25 ¹ / ₄ (641)	16	7/8(22)	1(25)	-	-	1 ¹ / ₄ (32)	1 ¹ / ₄ (32)	1 ¹ / ₄ (32)
24	32 ¹ / ₂ (826)	29 ³ / ₄ (756)	16	1(25)	1 ¹ / ₈ (29)	-	-	1 ³ / ₈ (35)	1 ³ / ₈ (35)	1 ³ / ₈ (35)

TABLE E

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	3 3/4(95)	2 5/8(67)	4	1/2(13)	9/16(14)	-	-	1/2(13)	1/4(6)	3/8(10)
3/4	4(102)	2 7/8(73)	4	1/2(13)	9/16(14)	-	-	1/2(13)	1/4(6)	3/8(10)
1	4 1/2(114)	3 1/4(83)	4	1/2(13)	9/16(14)	-	-	1/2(13)	5/16(8)	3/8(10)
1 1/4	4 3/4(121)	3 7/16(87)	4	1/2(13)	9/16(14)	-	-	5/8(16)	5/16(8)	1/2(13)
1 1/2	5 1/4(133)	3 7/8(98)	4	1/2(13)	9/16(14)	-	-	5/8(16)	3/8(10)	1/2(13)
2	6(152)	4 1/2(114)	4	5/8(16)	1 1/16(17)	-	-	3/4(19)	3/8(10)	9/16(14)
2 1/2	6 1/2(165)	5(127)	4	5/8(16)	1 1/16(17)	-	-	3/4(19)	7/16(11)	9/16(14)
3	7 1/4(184)	5 3/4(146)	4	5/8(16)	1 1/16(17)	-	-	3/4(19)	1/2(13)	9/16(14)
4	8 1/2(216)	7(178)	8	5/8(16)	1 1/16(17)	-	-	7/8(22)	5/8(16)	1 1/16(17)
5	10(254)	8 1/4(210)	8	5/8(16)	1 1/16(17)	-	-	7/8(22)	1 1/16(17)	1 1/16(17)
6	11(279)	9 1/4(235)	8	3/4(19)	7/8(22)	-	-	7/8(22)	1 1/16(17)	1 1/16(17)
8	13 1/4(337)	11 1/2(292)	8	3/4(19)	7/8(22)	-	-	1(25)	3/4(19)	3/4(19)
10	16(406)	14(356)	12	3/4(19)	7/8(22)	-	-	1(25)	7/8(22)	7/8(22)
12	18(457)	16(406)	12	7/8(22)	1(25)	-	-	1 1/8(29)	1(25)	1(25)
14	20 3/4(527)	18 1/2(470)	12	7/8(22)	1(25)	-	-	1 1/4(32)	1(25)	1(25)
16	22 3/4(578)	20 1/2(521)	12	7/8(22)	1(25)	-	-	1 1/4(32)	1(25)	1(25)
18	25 1/4(641)	23(584)	16	7/8(22)	1(25)	-	-	1 3/8(35)	1 1/8(29)	1 1/8(29)
20	27 3/4(705)	25 1/4(641)	16	7/8(22)	1(25)	-	-	1 1/2(38)	1 1/4(32)	1 1/4(32)
24	32 1/2(826)	29 3/4(756)	16	1 1/8(29)	1 1/4(32)	-	-	1 5/8(41)	1 1/2(38)	1 1/2(38)

TABLE F

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	3 3/4(95)	2 5/8(67)	4	1/2(13)	9/16(14)	-	-	1/2(13)	5/16(8)	3/8(10)
3/4	4(102)	2 7/8(73)	4	1/2(13)	9/16(14)	-	-	1/2(13)	5/16(8)	3/8(10)
1	4 3/4(121)	3 7/16(87)	4	5/8(16)	1 1/16(17)	-	-	1/2(13)	3/8(10)	3/8(10)
1 1/4	5 1/4(133)	3 7/8(98)	4	5/8(16)	1 1/16(17)	-	-	5/8(16)	3/8(10)	1/2(13)
1 1/2	5 1/2(140)	4 1/8(105)	4	5/8(16)	1 1/16(17)	-	-	5/8(16)	7/16(11)	1/2(13)
2	6 1/2(165)	5(127)	4	5/8(16)	1 1/16(17)	-	-	3/4(19)	7/16(11)	5/8(16)
2 1/2	7 1/4(184)	5 3/4(146)	8	5/8(16)	1 1/16(17)	-	-	3/4(19)	1/2(13)	5/8(16)
3	8(203)	6 1/2(165)	8	5/8(16)	1 1/16(17)	-	-	3/4(19)	9/16(14)	5/8(16)
4	9(229)	7 1/2(191)	8	5/8(16)	1 1/16(17)	-	-	7/8(22)	1 1/16(17)	3/4(19)
5	11(279)	9 1/4(235)	8	3/4(19)	7/8(22)	-	-	1(25)	3/4(19)	7/8(22)
6	12(305)	10 1/4(260)	12	3/4(19)	7/8(22)	-	-	1(25)	7/8(22)	7/8(22)
8	14 1/2(368)	12 3/4(324)	12	3/4(19)	7/8(22)	-	-	1 1/8(29)	1(25)	1(25)
10	17(432)	15(381)	12	7/8(22)	1(25)	-	-	1 1/8(29)	1(25)	1(25)
12	19 1/4(489)	17 1/4(438)	16	7/8(22)	1(25)	-	-	1 1/4(32)	1 1/8(29)	1 1/8(29)
14	21 3/4(552)	19 1/2(495)	16	1(25)	1 1/8(29)	-	-	1 3/8(35)	1 1/4(32)	1 1/4(32)
16	24(610)	21 3/4(552)	20	1(25)	1 1/8(29)	-	-	1 3/8(35)	1 1/4(32)	1 1/4(32)
18	26 1/2(673)	24(610)	20	1 1/8(29)	1 1/4(32)	-	-	1 1/2(38)	1 3/8(35)	1 3/8(35)
20	29(737)	26 1/2(673)	24	1 1/8(29)	1 1/4(32)	-	-	1 5/8(41)	1 1/2(38)	1 1/2(38)
24	33 1/2(851)	30 3/4(781)	24	1 1/4(32)	1 3/8(35)	-	-	1 3/4(44)	1 5/8(41)	1 5/8(41)

FLANGES

TABLE H

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	4 1/2(114)	3 1/4(83)	4	5/8(16)	11/16(17)	2 1/4(57)	1/16(2)	5/8(16)	3/8(10)	1/2(13)
3/4	4 1/2(114)	3 1/4(83)	4	5/8(16)	11/16(17)	2 1/4(57)	1/16(2)	5/8(16)	3/8(10)	1/2(13)
1	4 3/4(121)	3 7/16(87)	4	5/8(16)	11/16(17)	2 1/2(64)	1/16(2)	3/4(19)	7/16(11)	9/16(14)
1 1/4	5 1/4(133)	3 7/8(98)	4	5/8(16)	11/16(17)	3(76)	1/16(2)	7/8(22)	7/16(11)	11/16(17)
1 1/2	5 1/2(140)	4 1/8(105)	4	5/8(16)	11/16(17)	3 1/4(83)	1/16(2)	7/8(22)	1/2(13)	11/16(17)
2	6 1/2(165)	5(127)	4	5/8(16)	11/16(17)	4(102)	1/16(2)	1(25)	1/2(13)	3/4(19)
2 1/2	7 1/4(184)	5 3/4(146)	8	5/8(16)	11/16(17)	4 1/2(114)	1/16(2)	1(25)	9/16(14)	3/4(19)
3	8(203)	6 1/2(165)	8	5/8(16)	11/16(17)	5(127)	1/16(2)	1 1/8(29)	5/8(16)	7/8(22)
4	9(229)	7 1/2(191)	8	5/8(16)	11/16(17)	6(152)	1/16(2)	1 1/4(32)	3/4(19)	1(25)
5	11(279)	9 1/4(235)	8	3/4(19)	7/8(22)	7(178)	1/16(2)	1 3/8(35)	7/8(22)	1 1/8(29)
6	12(305)	10 1/4(260)	12	3/4(19)	7/8(22)	8 1/4(210)	1/16(2)	1 3/8(35)	1(25)	1 1/8(29)
8	14 1/2(368)	12 3/4(324)	12	3/4(19)	7/8(22)	10 1/4(260)	1/16(2)	1 1/2(38)	1 1/4(32)	1 1/4(32)
10	17(432)	15(381)	12	7/8(22)	1(25)	12 1/4(311)	1/16(2)	1 5/8(41)	1 3/8(35)	1 3/8(35)
12	19 1/4(489)	17 1/4(438)	16	7/8(22)	1(25)	14 1/4(362)	1/16(2)	1 3/4(44)	1 1/2(38)	1 1/2(38)
14	21 3/4(552)	19 1/2(495)	16	1(25)	1 1/8(29)	16 1/2(419)	1/16(2)	1 7/8(48)	1 5/8(41)	1 5/8(41)
16	24(610)	21 3/4(552)	20	1(25)	1 1/8(29)	19(483)	1/16(2)	2(51)	1 3/4(44)	1 3/4(44)
18	26 1/2(673)	24(610)	20	1 1/8(29)	1 1/4(32)	21(533)	1/16(2)	2 1/8(54)	1 7/8(48)	1 7/8(48)
20	29(737)	26 1/2(673)	24	1 1/8(29)	1 1/4(32)	23 1/2(597)	1/16(2)	2 1/4(57)	2(51)	2(51)
24	33 1/2(851)	30 3/4(781)	24	1 1/4(32)	1 3/8(35)	27 1/2(699)	1/16(2)	2 1/2(64)	2 1/4(57)	2 1/4(57)

TABLE J

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	4 1/2(114)	3 1/4(83)	4	5/8(16)	11/16(17)	2 1/4(57)	1/16(2)	-	5/8(16)	5/8(16)
3/4	4 1/2(114)	3 1/4(83)	4	5/8(16)	11/16(17)	2 1/4(57)	1/16(2)	-	5/8(16)	5/8(16)
1	4 3/4(121)	3 7/16(87)	4	5/8(16)	11/16(17)	2 1/2(64)	1/16(2)	-	3/4(19)	3/4(19)
1 1/4	5 1/4(133)	3 7/8(98)	4	5/8(16)	11/16(17)	3(76)	1/16(2)	-	3/4(19)	3/4(19)
1 1/2	5 1/2(140)	4 1/8(105)	4	5/8(16)	11/16(17)	3 1/4(83)	1/16(2)	-	7/8(22)	7/8(22)
2	6 1/2(165)	5(127)	4	3/4(19)	7/8(22)	4(102)	1/16(2)	-	1(25)	1(25)
2 1/2	7 1/4(184)	5 3/4(146)	8	3/4(19)	7/8(22)	4 1/2(114)	1/16(2)	-	1(25)	1(25)
3	8(203)	6 1/2(165)	8	3/4(19)	7/8(22)	5(127)	1/16(2)	-	1 1/4(32)	1 1/4(32)
4	9(229)	7 1/2(191)	8	3/4(19)	7/8(22)	6(152)	1/16(2)	-	1 3/8(35)	1 3/8(35)
5	11(279)	9 1/4(235)	8	7/8(22)	1(25)	7(178)	1/16(2)	-	1 1/2(38)	1 1/2(38)
6	12(305)	10 1/4(260)	12	7/8(22)	1(25)	8 1/4(210)	1/16(2)	-	1 1/2(38)	1 1/2(38)
8	14 1/2(368)	12 3/4(324)	12	7/8(22)	1(25)	10 1/4(260)	1/16(2)	-	1 5/8(41)	1 5/8(41)
10	17(432)	15(381)	12	1(25)	1 1/8(29)	12 1/4(311)	1/16(2)	-	-	1 7/8(48)
12	19 1/4(489)	17 1/4(438)	16	1(25)	1 1/8(29)	14 1/4(362)	1/16(2)	-	-	2(51)
14	21 3/4(552)	19 1/2(495)	16	1 1/8(29)	1 1/4(32)	16 1/2(419)	1/16(2)	-	-	2 1/8(54)
16	24(610)	21 3/4(552)	20	1 1/8(29)	1 1/4(32)	19(483)	1/16(2)	-	-	2 1/4(57)
18	26 1/2(673)	24(610)	20	1 1/4(32)	1 3/8(35)	21(533)	1/16(2)	-	-	2 3/8(60)
20	29(737)	26 1/2(673)	24	1 1/4(32)	1 3/8(35)	23 1/2(597)	1/16(2)	-	-	2 1/2(64)
24	33 1/2(851)	30 3/4(781)	24	1 3/8(35)	1 1/2(38)	27 1/2(699)	1/16(2)	-	-	2 3/4(70)

TABLE K

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	4 1/2(114)	3 1/4(83)	4	5/8(16)	1 1/16(17)	2 1/2(64)	1/16(2)	-	3/4(19)	3/4(19)
3/4	4 1/2(114)	3 1/4(83)	4	5/8(16)	1 1/16(17)	2 1/2(64)	1/16(2)	-	3/4(19)	3/4(19)
1	5(127)	3 3/4(95)	4	5/8(16)	1 1/16(17)	3(76)	1/16(2)	-	7/8(22)	7/8(22)
1 1/4	5 1/4(133)	3 7/8(98)	4	5/8(16)	1 1/16(17)	3(76)	1/16(2)	-	7/8(22)	7/8(22)
1 1/2	6(152)	4 1/2(114)	4	3/4(19)	7/8(22)	3 1/2(89)	1/16(2)	-	1(25)	1(25)
2	6 1/2(165)	5(127)	8	5/8(16)	1 1/16(17)	4(102)	1/16(2)	-	1(25)	1(25)
2 1/2	7 1/4(184)	5 3/4(146)	8	3/4(19)	7/8(22)	4 1/2(114)	1/16(2)	-	1 1/8(29)	1 1/8(29)
3	8(203)	6 1/2(165)	8	3/4(19)	7/8(22)	5(127)	1/16(2)	-	1 1/4(32)	1 1/4(32)
4	9 1/2(241)	7 3/4(197)	8	7/8(22)	1(25)	6(152)	1/16(2)	-	1 3/8(35)	1 3/8(35)
5	11(279)	9 1/4(235)	12	7/8(22)	1(25)	7(178)	1/16(2)	-	1 5/8(41)	1 5/8(41)
6	12(305)	10 1/4(260)	12	7/8(22)	1(25)	8 1/4(210)	1/16(2)	-	1 5/8(41)	1 5/8(41)
8	14 1/2(368)	12 1/2(318)	12	1(25)	1 1/8(29)	10 1/4(260)	1/16(2)	-	1 7/8(48)	1 7/8(48)
10	17(432)	15(381)	16	1(25)	1 1/8(29)	12 1/4(311)	1/16(2)	-	-	2(51)
12	19 1/4(489)	17(432)	16	1 1/8(29)	1 1/4(32)	14 1/4(362)	1/16(2)	-	-	2 1/4(57)
14	22 1/2(572)	20(508)	16	1 1/4(32)	1 3/8(35)	16 1/2(419)	1/16(2)	-	-	2 3/8(60)
16	24 3/4(629)	22 1/4(565)	20	1 1/4(32)	1 3/8(35)	19(483)	1/16(2)	-	-	2 5/8(67)
18	28 1/4(718)	25 3/4(654)	20	1 3/8(35)	1 1/2(38)	22 1/2(572)	1/16(2)	-	-	3(76)
20	31(787)	28(711)	20	1 1/2(38)	1 5/8(41)	24 1/2(622)	1/16(2)	-	-	3 1/4(83)

TABLE R

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	4 1/2(114)	3 1/4(83)	4	5/8(16)	1 1/16(17)	2 1/2(64)	1/16(2)	-	-	3/4(19)
3/4	4 1/2(114)	3 1/4(83)	4	5/8(16)	1 1/16(17)	2 1/2(64)	1/16(2)	-	-	3/4(19)
1	5(127)	3 3/4(95)	4	5/8(16)	1 1/16(17)	3(76)	1/16(2)	-	-	7/8(22)
1 1/4	5 1/4(133)	3 7/8(98)	4	5/8(16)	1 1/16(17)	3(76)	1/16(2)	-	-	7/8(22)
1 1/2	6(152)	4 1/2(114)	4	3/4(19)	7/8(22)	3 1/2(89)	1/16(2)	-	-	1(25)
2	6 1/2(165)	5(127)	8	5/8(16)	1 1/16(17)	4(102)	1/16(2)	-	-	1(25)
2 1/2	7 1/4(184)	5 3/4(146)	8	3/4(19)	7/8(22)	4 1/2(114)	1/16(2)	-	-	1 1/8(29)
3	8(203)	6 1/2(165)	8	3/4(19)	7/8(22)	5(127)	1/16(2)	-	-	1 1/4(32)
4	9 1/2(241)	7 3/4(197)	8	7/8(22)	1(25)	6(152)	1/16(2)	-	-	1 3/8(35)
5	11(279)	9 1/4(235)	12	7/8(22)	1(25)	7(178)	1/16(2)	-	-	1 5/8(41)
6	12(305)	10 1/4(260)	12	7/8(22)	1(25)	8 1/4(210)	1/16(2)	-	-	1 5/8(41)
8	14 1/2(368)	12 3/4(324)	12	1(25)	1 1/8(29)	10 1/4(260)	1/16(2)	-	-	2(51)
10	17(432)	15 1/4(387)	16	1(25)	1 1/8(29)	12 1/4(311)	1/16(2)	-	-	2 1/4(57)
12	20(508)	18(457)	16	1 1/8(29)	1 1/4(32)	14 1/4(362)	1/16(2)	-	-	2 1/2(64)
14	23(584)	20 3/4(527)	16	1 1/4(32)	1 3/8(35)	16 1/2(419)	1/16(2)	-	-	2 3/4(70)
16	25 1/4(641)	23(584)	20	1 1/4(32)	1 3/8(35)	19(483)	1/16(2)	-	-	3(76)
18	29(737)	26 1/2(673)	20	1 3/8(35)	1 1/2(38)	22 1/2(572)	1/16(2)	-	-	3 1/4(83)
20	31 3/4(806)	28 3/4(730)	20	1 1/2(38)	1 5/8(41)	24 1/2(622)	1/16(2)	-	-	3 1/2(89)

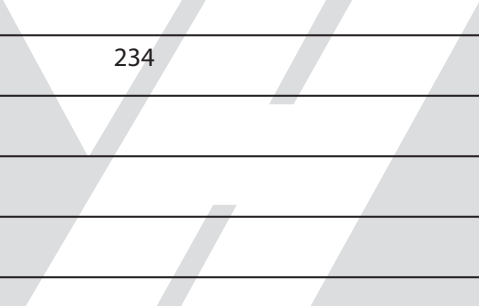
FLANGES

TABLE S

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	5(127)	3 1/2(89)	4	3/4(19)	7/8(22)	2 1/2(64)	1/16(2)	-	-	7/8(22)
3/4	5(127)	3 1/2(89)	4	3/4(19)	7/8(22)	2 1/2(64)	1/16(2)	-	-	7/8(22)
1	5 1/2(140)	4(102)	4	3/4(19)	7/8(22)	3(76)	1/16(2)	-	-	1(25)
1 1/4	5 3/4(146)	4 1/4(114)	4	3/4(19)	7/8(22)	3 1/4(83)	1/16(2)	-	-	1 1/8(29)
1 1/2	6 1/4(159)	4 3/4(121)	4	3/4(19)	7/8(22)	3 1/2(89)	1/16(2)	-	-	1 1/8(29)
2	6 3/4(171)	5 1/4(133)	8	3/4(19)	7/8(22)	3 1/2(89)	1/16(2)	-	-	1 1/4(32)
2 1/2	7 1/4(184)	5 3/4(146)	8	3/4(19)	7/8(22)	4(102)	1/16(2)	-	-	1 1/4(32)
3	8(203)	6 1/2(165)	8	7/8(22)	1(25)	4 1/2(114)	1/16(2)	-	-	1 3/8(35)
4	9 3/4(248)	8(203)	8	1(25)	1 1/8(29)	6 1/4(159)	1/16(2)	-	-	1 5/8(41)
5	11 1/4(286)	9 1/4(235)	12	7/8(22)	1(25)	7 1/2(191)	1/16(2)	-	-	1 3/4(44)
6	12 3/4(324)	10 3/4(273)	12	1(25)	1 1/8(29)	8 1/4(210)	1/16(2)	-	-	2(51)
8	16 1/4(413)	14(356)	12	1 1/4(32)	1 3/8(35)	10 3/4(273)	1/16(2)	-	-	2 1/2(64)
10	19(483)	16 3/4(425)	16	1 1/4(32)	1 3/8(35)	13(330)	1/16(2)	-	-	2 7/8(73)
12	22 3/4(578)	20(508)	16	1 1/2(38)	1 5/8(41)	15(381)	1/16(2)	-	-	3 1/4(83)
14	25 1/2(648)	22 3/4(578)	20	1 1/2(38)	1 5/8(41)	17 1/4(438)	1/16(2)	-	-	3 3/4(95)
16	29 1/4(743)	26(660)	20	1 3/4(44)	1 7/8(48)	19 1/2(495)	1/16(2)	-	-	4 1/4(108)

TABLE T

Nominal Size in	Dia. of Flange	Bolt Circle Dia.	No. of Bolts	Dia. of Bolts	Dia. of Holes	Dia. of Raised Face	Height of Raised Face	Thickness of Flange		
								Grey Cast Iron	Copper Alloy	Cast and Forged Steel
1/2	5 1/2(140)	4(102)	4	3/4(19)	7/8(22)	2 1/2(64)	1/16(2)	-	-	1(25)
3/4	5 1/2(140)	4(102)	4	3/4(19)	7/8(22)	2 1/2(64)	1/16(2)	-	-	1(25)
1	5 3/4(146)	4 1/4(114)	4	3/4(19)	7/8(22)	3(76)	1/16(2)	-	-	1 1/8(29)
1 1/4	6 1/4(159)	4 3/4(121)	4	7/8(22)	1(25)	3 1/4(83)	1/16(2)	-	-	1 1/4(32)
1 1/2	6 3/4(171)	5 1/4(133)	8	3/4(19)	7/8(22)	3 1/2(89)	1/16(2)	-	-	1 3/8(35)
2	7 1/4(184)	5 3/4(146)	8	3/4(19)	7/8(22)	4(102)	1/16(2)	-	-	1 3/8(35)
2 1/2	8(203)	6 1/2(165)	8	7/8(22)	1(25)	4 1/2(114)	1/16(2)	-	-	1 5/8(41)
3	9 1/4(235)	7 1/2(191)	8	1(25)	1 1/8(29)	5(127)	1/16(2)	-	-	1 7/8(48)
4	11 1/4(286)	9 1/4(235)	8	1 1/8(29)	1 1/4(32)	6 1/4(159)	1/16(2)	-	-	2 1/4(57)
5	12 3/4(324)	10 3/4(273)	12	1 1/8(29)	1 1/4(32)	8 1/4(210)	1/16(2)	-	-	2 5/8(67)
6	14 3/4(375)	12 1/2(318)	12	1 1/4(32)	1 3/8(35)	9(229)	1/16(2)	-	-	2 7/8(73)
8	18 3/4(476)	16(406)	12	1 1/2(38)	1 5/8(41)	11 3/4(298)	1/16(2)	-	-	3 1/2(89)
10	22(559)	19 1/4(489)	16	1 1/2(38)	1 5/8(41)	14(356)	1/16(2)	-	-	4 1/4(108)
12	25 3/4(654)	22 1/2(572)	16	1 3/4(44)	1 7/8(48)	16 1/4(413)	1/16(2)	-	-	4 3/4(121)



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STAINLESS STEEL PRODUCTS

GENERAL INFORMATION OF STAINLESS STEELS

RESISTANT TO CORROSION

Addition of the element chromium to steel enhances its resistance to corrosion and staining. Steels that contain a comparatively large amount of chromium are therefore called stainless steels.

Stainless steels have many areas of application, including household articles, building materials, rolling stock as well as chemical and petrochemical equipment. They are produced in a wide variety of types to suit specific end-uses.

ATTRACTIVENESS EXPANDS APPLICABILITY

Needless to say, the bulk of stainless steels are used where corrosion resistance is a basic requirement. Recently, however, their unique, attractive appearance has come to be widely appreciated, causing a sharp increase in their architectural applications.

HEAT AND LOW-TEMPERATURE RESISTANCE

Stainless steels retain high strength in elevated temperature service; the austenitic grades, in particular, have superior ductility at low temperatures.

SHEETS IN GREATEST DEMAND

While stainless steels are produced in various product forms such as sheets, shapes, rods, bars and tubes, those in sheet form presently register by far the fastest gain in market demand. This is because they have come to be used in large quantities as exterior and interior building materials, thus expanding a new market.

THREE BASIC CLASSES

There are many types of stainless steel. Although different in both mechanical properties and chemical composition, they may be generally grouped into three basic classes.

13 Cr CLASS (MARTENSITIC)

Generally, a group of steels with the carbon content of up to 0.2% and the chromium content ranging between 11.5% and 15.0%. These steels become martensitic in structure when heat treated. Since they are hardenable by heat treatment, steels of this class can be produced in many different degrees of hardness by varying heating conditions. Martensitic stainless steels are hardly corroded in air and are outstanding in hardness, although in weldability and workability they are inferior to the other two classes. Steels of this class are primarily used for the manufacture of cutlery.

18 Cr CLASS(FERRITIC)

Generally, steels having carbon up to 0.2% and chromium between 16.0% and 20.0%. Not hardenable by heat treatment, they are basically used in the annealed condition. These steels are more workable, more weldable, and more corrosion-resistant than martensitic steels. Ferritic steels have good mechanical and physical properties and effectively resist both atmospheric corrosion and strong oxidizing solutions. They are widely used in nitric acid-base chemical processing equipment as well as in buildings, kitchen sinks, and appliances.

18 Cr-8 Ni CLASS (AUSTENITIC)

Generally, steels containing a maximum of 0.15% carbon, a minimum of 16.0% chromium and a minimum of 6.0% nickel. In plain carbon steel, an austenitic structure is formed only at high temperatures. However, stainless steels of this class, because they retain a stable austenite form even at room temperature, are called austenitic stainless steels.

Of the three classes, this has the highest corrosion resistance. Steels of this class combine low yield point with high tensile strength and good elongation, thus offering the best cold-working properties. While it is not possible to change their mechanical properties by heat treatment, their tensile strength and hardness can be greatly enhanced by work hardening. Because of these characteristics, these steels are most extensively used in chemical plants and equipment

STAINLESS STEEL PRODUCTS

In using these data it should be understood that they are results of laboratory tests and are indicative only of the conditions under which they are run, and may be considered as a basis for recommendation, but not for guarantee. As actual operating conditions can rarely be duplicated in the laboratory it is desirable to test further under actual service. The letters A, B, C, etc., represent the approximate corrosion penetration per month as follows:

CORROSION RESISTANCE DATA

substance and condition	temp F	304	316	substance and condition	temp F	304	316	substance and condition	temp F	304	316
Acetic acid				beer		A	A	5% agitated	70°	A	A
5% agitated	70°	A	A	benzene	70°	A	A	5% aerated	70°	A	A
5% aerated	70°	A	A	benzol	hot	A	A	50% aqueous solution	hot	A	A
10% agitated	70°	A	A	blood (meat juices)	cold	A1	A	copper sulphate			
10% aerated	70°	A	A	boric acid				5% agitated still	70°	A	A
20% agitated	70°	A	A	5% solution	70°	A	A	5% aerated	70°	A	A
20% aerated	70°	A	A	5% solution hot				saturated solution	boiling	A	A
50%	70°	A	A	5% solution boiling			A1	creosote (coal tar)	hot	A	A
50%	boiling	C	B	saturated solution	70°	A1	A1	creosote oil	hot	A	A
80%	70°	A	A	boiling			A1	dyewood liquor	70°	A3	A
80%	boiling	D	B	buttermilk	70°	A	A	ether	70°	A	A
100%	70°	A	A	Calcium carbonate	70°	A	A	ethyl acetate			
100%	boiling	C	B	calcium chlorate				concentrated solution	70°	A	
100%-150 lb. pressure	400°	E	C	dilute solution	70°	A	A	ethyl chloride	70°	A	A
acetone	boiling	A	A	dilute solution	hot	A	A	ethylene chloride	70°	A	A
	70°	A	A	calcium chloride				ethylene glycol			
acetylene	70°			dilute solution	70°	B2	A1	concentrated	70°	A	
concentrated	70°	A		concentrated solution	70°	B2	A1	Ferric chloride			
commercially pure	70°			calcium hydroxide				1% solution still	70°	B41	A1
alcohol, ethyl	70°	A	A	10%	boiling	A	A	boiling	D41	C1	
	boiling	A	A	20%	boiling	A	A	5% solution still	70°	D431	C1
alcohol, methyl	70°	A	A	50%	boiling	C	B	5% agitated	70°	C431	C1
	150°	C1	B	calcium sulphate				5% aerated	70°	C431	C1
alum (chrome) 5%	70°	A	A	saturated	70°	A	A	ferric hydroxide			
aluminium chloride	70°	D	C	carbonated water		A	A	(hydrated iron oxide)	70°	A	A
aluminium	molten	E	E	carbonic acid				ferric nitrate			
aluminium sulphate				saturated solution	70°	A		1% still	70°	A	A
10%	70°	A	A	carbon monoxide gas	870°C.	A	A	5% still	70°	A	A
	boiling	B	A		760°C.	A	A	1% agitated	70°	A	A
saturated	70°	A	A	carbon tetrachloride				5% agitated	70°	A	A
	boiling	B	A	pure	70°	A	A	1% aerated	70°	A	A
ammonia				aqueous solution 5-10%	70°	C1		5% aerated	70°	A	A
all concentrations	70°	A	A	chloracetic acid	70°	D	C	ferric sulphate			
gas	hot	D		chloric acid	70°	E	D	1% and 5% still	70°	A1	A
ammonia liquor	70°	A	A	chlorinated water				aerated or agitated	70°	A1	A
	boiling	A	A	saturated	70°	C1	B1	ferrous chloride			
ammonium chloride				chlorine gas - dry	70°	C	B	saturated solution	70°		A
1% still	70°	A	A	gas - moist	70°	D	C	ferrous sulphate			
aerated	70°	A	A	gas	100°C.	E	D	dilute solution	70°	A	A
agitated	70°	A	A	chloroform	70°	A	A	fluorine	70°	E	E
10% solution	boiling	A1	A1	chromic acid				formaldehyde			
28% solution	boiling	B1	A1	5%	70°	A	A	40% solution		A1	A1
50% solution	boiling	B1	A1	10% CP	boiling	C	B	formic acid			
ammonium nitrate				50% commercial				5% still	70°	B	A
all concentrations				(cont. SO2)	boiling	D1	C	fruit juices	150°	B	A
agitated	70°	A	A	cider	70°	A	A	fuel oil	70°	A	A
aerated	70°	A	A	citric acid				containing sulphuric	hot	A	A
saturated	boiling	A	A	5% still	70°	A	A	gasoline		C	B
ammonium phosphate 5%	70°	A	A	5% still	150°	A	A	gelatine	70°	A	A
ammonium sulphate				15%	70°	A	A	gule - dry	70°	A	A
1% & 5% agitated	70°	A	A	concentrated	boiling	B	A	solution - acid	70°	A	A
aerated	70°	A	A	coca cola syrup (pure)	boiling	C	B		140°	B1	A
10%	boiling	B1	A1	coffee	70°	A	A	glycerine	70°	A	A
saturated	boiling	B1	A1	copper carbonate	boiling	A	A	Hydrachloric acid			
ammonium sulphite	cold	A	A	sat. sol. in 50% NH ₃ OH		A	A	all concentrations	70°	E	E
	boiling	A	A	copper chloride				hydrocyanic acid		A	A
aniline 3%	70°	A	A	1% agitated	70°	B1	A1	hydrofluosilicic acid	70°	E	D
concentrated crude	70°	A	A	1% aerated	70°	B1	A1	hydrogen sulphide			
barium chloride 5%	70°	A	A	5% agitated	70°	C1	B1	dry		A	A
saturated	70°	A	A	5% aerated	70°	E1	D1	wet		B3	A3
aqueous solution	hot	B1	A1	copper cyanide				Ink		B3	A
barium nitrate				saturated solution	boiling	A	A	iodine		E	D
aqueous solution	hot	A	A	copper nitrate				iodoform		A	A
barium sulphate				1% still	70°	A	A	Kerosene	70°	A	A
barytes-blanc fixe	70°	A	A	1% agitated	70°	A	A	ketchup	70°	A1	A
barium sulphide				1% aerated	70°	A	A				
saturated solution	70°	A		5% still	70°	A	A				
solution	70°		A								

- 1 Subject to pitting at air line or when allowed to dry.
- 2 Keep solutions alkaline.

STAINLESS STEEL PRODUCTS

fully resistant : A less than .00035 inches
satisfactorily resistant : B .00035-.0035 inches
fairly resistant : C .0035-.010 inches
slightly resistant : D .010-.035 inches
not resistant : E over .035 inches

substance and condition	temp F	304	316	substance and condition	temp F	304	316	substance and condition	temp F	304	316
Lactic acid	70°	A	A	5% still	70°	A1	A1	sodium sulphate	70°	A	A
5%	150°	B	A	5% agitated	70°	A	A	5% still	70°	A	A
10%	boiling	D	B	5% aerated	70°	A	A	all concentrations	70°	A	A
10%	150°	C	B		boiling	A	A	sodium sulphide		B1	A
lard	70°	A	A	potassium ferricyanide	70°	A	A	saturated			
lead	molten	B	B	5%				sodium sulphite	70°	A	A
linseed oil	70°	A	A	potassium ferrocyanide	70°	A	A	5%	150°	A	A
Magnesium chloride				5%				10%			
1% & 5% still	70°	A1	A	potassium hydroxide	70°	A	A	sodium thiosulphate	70°	A	A3
	hot	C1	B1	5% still	70°	A	A	saturated solution	70°		A
magnesium sulphate	cold, hot	A	A	5% agitated	70°	A	A	acid fixing bath (hypo)	70°		A3
malic acid	cold, hot	B	A	5% aerated	70°	A	A	25% solution	70°		A3
mayonnaise	70°	A1	A	27%	boiling	A	A		boiling		
mercuric chloride				50%	boiling	B	A	stearic acid		A	A
dilute solutions		E1	D1	potassium nitrate				sugar juice		A	A
mercury		A	A	1% still	70°	A	A	sulphur chloride		E	D
methanol				1% agitated	70°	A	A	sulphur dioxide			
(methyl alcohol)		A	A	1% aerated	70°	A	A	gas-moist	70°	B	A
milk, fresh or sour	hot, cold	A	A	5% aerated	70°	A	A	gas	575°	A	A
mixed acids				5% still & agitated	70°	A	A	sulphur-dry	molten	A	A
53% H ₂ SO ₄	cold	A	A	potassium nitrate	hot	A	A	wet		B1	A1
45% HNO ₃	cold	A	A	potassium oxalate		A	A	sulphuric acid			
mustard	70°	A1	A1	potassium permanganate				5%	70°	C	B
Naphtha	70°	A	A	5%	70°	A	A	5%	boiling	E	C
naphtha - crude	70°	A	A	potassium sulphate	70°	A	A	10%	70°	C	B
nickel chloride				1% still	70°	A	A	10%	boiling	E	D
solution	70°	A1	A1	1% agitated	70°	A	A	50%	70°	D	C
nickel sulphate	hot, cold	A	A	1% aerated	70°	A	A	50%	boiling	E	D
nitric acid				5% still	70°	A	A	concentrated	70°	A	A
5% solution	70°	A	A	5% agitated	70°	A	A	concentrated	boiling	D	D
20% solution	70°	A	A	5% aerated	70°	A	A	concentrated	300°	E	E
50% solution	70°	A	A		hot	A	A	fuming	70°	C	B
50% solution	boiling	A	A	potassium sulphide				sulphurous acid			
65% solution	boiling	B	B	(salt)		A	A	saturated	70°	C	B
concentrated	70°	A	A	pyrogalllic acid		A	A	saturated 60 lb.			
concentrated	boiling	D	D	Rosin		A	A	pressure	250°	C	B
Oils, crude	hot, cold	A3	A3	Sea water	molten	A1	A1	saturated 70/125 lb.			
oils, vegetable, mineral	hot, cold	A3	A	silver bromide		B1	A1	pressure	310°	C	B
oleic acid	70°	A1	A	silver chloride		E	E	150 lb. pressure	375°	C	B
oxalic acid				silver nitrate		A	A	sulphurous spray	70°	D1	D1
5%	hot, cold	A	A	soap	70°	A	A	Tannic acid	70°	A	A
10%	70°	A	A	sodium acetate-moist		A1	A		150°	B	A
10%	boiling	D	C	sodium bicarbonate				tin	molten	C	C
Paraffin	hot, cold	A	A	all concentrations	70°	A	A	trichloroacetic acid	70°	E	E
petroleum ether		A	A	5% still	150°	A	A	trichlorethylene	70°	A1	
phenol		A	A	sodium bisulphate				Varnish	70°	A	A
phosphoric acid				solution	70°	A	A		hot	A	A
1%	70°	A4	A4	saturated solution	70°	E		Vegetable juices		A	A
5% still	70°	A	A	2g + 1gH ₂ SO ₄				vinegar fumes		B	A
5% agitated	70°	A	A	per liter	68°		A	vinegar-still	70°	A	A
5% aerated	70°	A	A		212°		B	agitated		A	A
10% still	70°	C	A	sodium carbonate				aerated	70°	A	A
10% agitated	70°	C	B	5%	70°	A	A	Zinc	molten	E	E
10% aerated	70°	C	B	5%	150°	A	A	zinc chloride			
picric acid	70°	A	A	sodium chloride				5% still	70°	A1	A1
potassium bichromate	70°	A	A	5% still	70°	A1	A		boiling	B1	B1
potassium bromide	70°	A	A	5% still	150°	A1	A	zinc sulphate			
potassium carbonate	70°	B1	A1	20% aerated	70°	A1	A	5%	70°	A	A
1% still	70°	A	A	saturated	70°	A1	A	saturated	70°	A	A
1% agitated	70°	A	A	saturated	boiling	B1	A	25%	boiling	A	A
1% aerated	70°	A	A	sodium fluoride							
potassium carbonate	hot	A	A	5% solution		B1	A1				
potassium chlorate		A	A	sodium hyroxide		A	A				
potassium chloride				sodium hypochlorite							
1% still	70°	A1	A1	5% still		B1	A1				
1% agitated	70°	A	A	sodium hyposulphite	70°	A3	A				
1% aerated	70°	A	A	sodium nitrate	fused	C	B				

3. May attack when sulphuric acid is present.
4. May attack when hydrochloric acid is present.

STAINLESS STEEL COILS & SHEETS

SPECIFICATIONS

Application		Standard			
		AISI	JIS	BS	DIN
Stainless Steel	Austenitic	301,302 304, 304L 316, 316L 321, 347 385	SUS 301, 302 SUS 304, 304L SUS 316, 316L SUS 321, 347 SUS 385	301S21, 302S17 302S25 304S15, 304S12 304S16 316S16, 316S12 316S49, 316S37 321S12, 347S17 321S49, 347S49	X12CrNi177 X12CrNi188 X5CrNi1810 X2CrNi1911 X5CrNiMo17122 X2CrNiMo17132 X6CrNiMoTi1810 X6CrNi1
	Ferritic	405 409 430	SUS 405 SUS 409 SUS 430	405S17 409S17 430S15	X6CrAl13 X6CrTi12 X6Cr17
	Martensitic	403,410 420, 420J2 431,501 502	SUS 403, 410 SUS 420J1, 420J2 SUS 431	430S17, 410S21 420S29, 420S45 431S29	X15Cr13, X6Cr13 X20Cr13, X46Cr13 X22CrNi117
Heat Resisting Steel		314,446 H-46, 661	SUH 446 SUH 600, 661		X15CrNiSi2520

YICK HOE

STAINLESS STEEL SHEETS & PLATES

SPECIFICATIONS: CHEMICAL COMPOSITION OF STAINLESS STEEL SHEETS

Type of Structure	Type of Composition	Corresponding Designation			Chemical Composition, %								Other Elements
		JIS	AISI	DIN	C	Si max.	Mn	P max.	S max.	Ni	Cr	Mo	
Austenite	17Cr-5Ni-7Mn	SUS 201	201	--	0.15 max.	1.00	5.5-7.5	0.06	0.03	3.5 - 5.5	16.0 - 18.0	--	N: 0.25 max.
	18Cr-6Ni-10Mn	SUS 202	202	--	0.15 max.	1.00	7.5 - 10.0	0.06	0.03	4.0-5.5	16.0 - 18.0	--	N: 0.25 max.
	17Cr-7Ni	SUS 301	301	4310	0.15 max.	1.00	2.00	0.04	0.03	6.0-8.0	16.0 - 18.0	--	--
	18Cr-8Ni-high C	SUS 302	302	4300	0.15 max.	1.00	2.00	0.04	0.03	8.0 - 10.0	17.0 - 19.0	--	--
	18Cr-8Ni	SUS 304	304	4301	0.08 max.	1.00	2.00	0.04	0.03	8.0 - 10.5	18.0 - 20.0	--	--
	18Cr-8Ni -extra-low C	SUS 304L	304L	4306	0.03 max.	1.00	2.00	0.04	0.03	9.0 - 13.0	18.0 - 20.0	--	--
	18Cr-12Ni	SUS 305	305	3955	0.12 max.	1.00	2.00	0.04	0.03	10.5 - 13.0	17.0 - 19.0	--	--
	23Cr-12Ni	SUS 309S	309S	4845	0.03 max.	1.00	2.00	0.04	0.03	12.0-15.0	22.0 - 24.0	--	--
	25Cr-20Ni	SUS 310S	310S	--	0.08 max.	1.50	2.00	0.04	0.03	19.0-22.0	24.0 - 26.0	--	--
	18Cr-12Ni-2.5Mo	SUS 316	316	4401	0.08 max.	1.00	2.00	0.04	0.03	10.0-14.0	16.0 - 18.0	2.0-3.0	--
	18Cr-12Ni-7.5 Mo-extra-low C	SUS 316L	316L	4404	0.03 max.	1.00	2.00	0.04	0.03	12.0 - 15.0	16.0 - 18.0	2.0-3.0	--
	18Cr-12Ni-2Mo -2Cu	SUS 316J1		4505	0.03 max.	1.00	2.00	0.04	0.03	10.0 - 14.0	17.0 - 19.0	1.2 - 2.75	Cu: 1.0-2.5
	18Cr-13Ni-3.5Mo	SUS 317	317	4402	0.03 max.	1.00	2.00	0.04	0.03	18.0-15.0	19.0 - 20.0	3.0-4.0	--
	18Cr-13Ni-3.5 Mo-extra-low C	SUS 317L	317L	--	0.03 max.	1.00	2.00	0.04	0.03	11.0 - 15.0	18.0 - 20.0	3.0 - 4.0	--
	18Cr-8Ni-Ti	SUS 321	321	4541	0.03 max.	1.00	2.00	0.04	0.03	9.0 - 13.0	17.0 - 19.0	--	Ti: 5 x C% min
	18Cr-9Ni-Nb	SUS 347	347	4550	0.03 max.	1.00	2.00	0.04	0.03	9.0-13.0	17.0 - 19.0	--	Nb+Ta: 10xC% min
Ferrite	13Cr-Al	SUS 405	405	4002	0.03 max.	1.00	1.00	0.04	0.03	0.60 max.	11.5 - 16.0	--	Al: 0.1-0.3
	16Cr	SUS 429	429	4009	0.12 max.	1.00	1.00	0.04	0.03	--	14.0 - 16.0	--	--
	18Cr	SUS 430	430	4016	0.12 max.	0.75	1.00	0.04	0.03	0.60 max.	16.0 - 18.0	--	--
	18Cr-Mo	SUS 434	434	4113	0.12 max.	1.00	1.00	0.04	0.03	--	16.0 - 18.0	0.75 - 1.25	--
Martensite	13Cr-low Si	SUS 403	403	4024	0.15 max.	0.50	1.00	0.04	0.03	0.60 max.	11.5 - 13.0	--	--
	13Cr	SUS 410	410	4000	0.15 max.	1.00	1.00	0.04	0.03	0.60 max.	11.5 - 13.5	--	--
	13Cr-high C	SUS 420J2	420	4021	0.26-0.40	1.00	1.00	0.04	0.03	--	12.0 - 14.0	--	--
	18Cr-high C	SUS 440A	440A	--	0.6 - 0.75	1.00	1.00	0.04	0.03	--	16.0 - 18.0	0.75 max.	--
Precipitation Hardened	17Cr-7Ni-1Al	SUS 631	631	--	0.09 max.	1.00	1.00	0.04	0.03	6.5-7.75	16.0-18.0	--	Al: 0.75-1.5

STAINLESS STEEL SHEETS & PLATES

SPECIFICATIONS: MECHANICAL PROPERTIES OF STAINLESS STEEL SHEETS

Type of Structure	Type of Composition	Corresponding Designation			Mechanical Properties							
					Tensile Test					Hardness Test		Bend Test 180° Radius (t, min)
		Tensile Strength min.		Yield Strength min.		Elongation %, min.						
		JIS	AISI	DIN	kg/mm²	psi	kg/mm²	psi	JIS No.13 Specimen	Rockwell B Scale max.	Vickers Hardness max.	
Austenite	17Cr-5Ni-7Mn	SUS 201	201	--	65	93,000	25	35,800	40	100	253	--
	18Cr-6Ni-10Mn	SUS 202	202	--	60	85,500	25	35,800	40	95	218	--
	17Cr-7Ni	SUS 301	301	4310	53	75,800	21	30,000	40	90	200	--
	18Cr-8Ni-high C	SUS 302	302	4300	53	75,800	21	30,000	40	90	200	--
	18Cr-8Ni	SUS 304	304	4301	53	75,800	21	30,000	40	90	200	--
	18Cr-8Ni -extra -low C	SUS 304L	304L	4306	49	69,500	18	25,500	40	90	200	--
	18Cr-12Ni	SUS 305	305	3955	49	69,500	18	25,500	40	90	200	--
	23Cr-12Ni	SUS 309S	309S	4845	53	75,800	21	30,000	40	90	200	--
	25Cr-20Ni	SUS 310S	310S	--	53	75,800	21	30,000	40	90	200	--
	18Cr-12Ni-2.5Mo	SUS 316	316	4401	53	75,800	21	30,000	40	90	200	--
	18Cr-12Ni-7.5 Mo-extra-low C	SUS 316L	316L	4404	49	69,500	18	25,500	40	90	200	--
	18Cr-12Ni-2Mo -2Cu	SUS 316J1		4505	53	75,800	21	30,000	40	90	200	--
	18Cr-13Ni-3.5Mo	SUS 317	317	4402	53	75,800	21	30,000	40	90	200	--
	18Cr-13Ni-3.5 Mo-extra- low C	SUS 317L	317L	--	49	69,500	18	25,800	40	90	200	--
	18Cr-8Ni-Ti	SUS 321	321	4541	53	75,800	21	30,000	40	90	200	--
	18Cr-9Ni-Nb	SUS 347	347	4550	53	75,800	21	30,000	40	90	200	--
Ferrite	13Cr-Al	SUS 405	405	4002	42	60,000	18	25,500	20	88	200	t < 8.05 t
	16Cr	SUS 429	429	4009	46	65,800	21	30,000	22	88	200	1.0 t
	18Cr	SUS 430	430	4016	46	65,800	21	30,000	22	88	200	1.0 t
	18Cr-Mo	SUS 434	434	4113	46	65,800	21	30,000	22	88	200	1.0 t
Martensite	13Cr-low Si	SUS 403	403	4024	45	64,000	21	30,000	20	88	200	1.0 t
	13Cr	SUS 410	410	4000	45	64,000	21	30,000	20	88	200	1.0 t
	13Cr-high C	SUS 420J2	420	4021	55	78,500	23	32,750	18	93	210	--
	18Cr-high C	SUS 440A	440A	--	60	85,500	25	35,800	15	97	230	--
Precipitation Hardened	17Cr-7Ni-1Al	SUS 631	631	--	105	148,000	39	55,500	20	92	200	--

STAINLESS STEEL SHEETS & PLATES

MAIN USES

JIS Designation	AISI Designation	Characteristics and Main Uses
SUS 410	410	The most common grade of economical 13 Cr steels. In the annealed condition, this grade may be drawn or formed. Used for machine structures, home appliances, etc.
SUS 403	403	Suited for steam turbine blades, jet engines, compressor blades and other applications involving high stresses.
SUS 405	405	Possesses corrosion resistance equivalent to that of 13 Cr steel. Used in applications involving welding. Machine structures, food-processing machinery, chemical equipment, etc.
SUS 420 J2	420	High carbon content results in high abrasion resistance. Greater hardness may be obtained by heat treatment. Used for cutlery, machine parts, etc.
SUS 434	434	Designed to resist atmospheric corrosion especially for use as automotive trim. Used for wheel covers, hub caps, grilles, mouldings, etc.
SUS 430	430	The most common and widely used of the ferritic grades. This steel may be drawn or formed and has high corrosion resistance. Used for office and kitchen equipment, buildings, automobiles, etc.
SUS 301	301	This grade can be cold worked to very high strength levels. Corrosion resistance is somewhat inferior to SUS 304. Used for rolling stock, automobiles, home appliances, etc.
SUS 302	302	More resistant to corrosion than SUS 301. Suited for oil, chemical, paper and food-processing machinery, dairy installations, buildings, kitchen equipment and utensils, etc.
SUS 304	304	Slightly more resistant to general corrosion than SUS 302. Suited for the same range of applications as SUS 302.
SUS 304L	304L	A special SUS 304 grade for use where carbide precipitation after welding is undesirable. Used for chemical and petrochemical equipment, nuclear reactor equipment, oil-refining equipment, etc.
SUS 321	321	Suited for use in a 500° to 850°C temperature range in which carbides tend to precipitate and in applications where stress relieving (or annealing) is not feasible. Used in the chemical industry, etc.
SUS 316	316	High resistant to such reducing acids as sulphuric, sulphurous, acetic and hydrochloric acids. Also, high creep strength at elevated temperatures. Used in applications in the paper, textile, and chemical industries.
SUS 316L	316L	A special SUS 316 grade intended for use where carbide precipitation after welding is undesirable.
SUS 347	347	Suited for the same applications as SUS 321. Widely used for the construction of manufacturing equipment involving welding.

STAINLESS STEEL SHEETS & PLATES

FINISHES

Finish Designation	Finishing Method and Degree of Finish
No. 1	Hot-rolled, annealed and descaled. Generally used in industrial applications, where heat or corrosion resistance is required but surface smoothness is not particularly important.
No. 2D	Dull, smooth finish produced by cold rolling, annealing and pickling. Steel is soft and has a silver white surface, suited for deep drawing as well as general uses.
No. 2B	Bright, smooth cold-rolled finish obtained by skinpass rolling of No. 2D. Specified for general use.
No. 3	Intermediate polished finish obtained with rather coarse abrasives (#100 to #120 mesh). A higher luster can be attained by further polishing after fabrication.
No. 4	A standard polished finish produced with finer abrasives (#150 to #180 mesh). Widely specified for restaurant and kitchen equipment and dairy processing equipment.
#240	Polish finish of about #240. Used when finer finish than No. 4 finish is desired such as equipment for restaurants and household kitchens.
#320	Polishing finish of about #320.
#400	High luster finish produced by polishing No. 2B first with #400 buffing iron and then with buffing cloth. Suited for general use.
H.L.	Special polished finish having continuous lines produced with abrasives of an appropriate grain size. Mainly used in architectural applications.
BA	Highly reflective finish produced by cold rolling, bright annealing and-temper rolling. Used where a lustrous surface is specially required.

Note:

Upon request, strip surface protection is provided such as bonding of vinyl sheets or vinyl coating.

YICK HOE

STAINLESS STEEL SHEETS & PLATES

Sizes and Unit Weights

Size	1219 x 2438 (4'x 8')				1524 x 3048 (5'x 10')			
AISI	Type 304, 304L		Type 316, 316L		Type 304, 304L		Type 316, 316L	
Th / Wgt.	kg/pc	pcs/mt	kg/pc	pcs/mt	kg/pc	pcs/mt	kg/pc	pcs/mt
0.3	7.07	141.4	7.11	140.70	11.06	90.42	11.13	89.85
0.4	9.42	106.2	9.48	105.50	14.75	67.80	14.84	67.39
0.45	10.60	94.34	10.67	93.72	16.59	60.28	16.70	59.88
0.5	11.78	84.89	11.85	84.39	18.44	54.23	18.56	53.88
0.55	12.95	77.22	13.04	76.69	20.28	49.31	20.41	49.00
0.6	14.13	70.77	14.22	70.32	22.12	45.21	22.27	44.90
0.7	16.49	60.64	16.59	60.28	25.81	38.74	25.98	38.49
0.8	18.84	53.08	18.96	52.74	29.50	33.90	29.69	33.68
0.9	21.20	47.17	21.33	46.88	33.18	30.14	33.40	29.94
1	23.55	42.46	23.70	42.19	36.87	27.12	37.11	26.95
1.2	28.26	35.39	28.44	35.16	44.24	22.60	44.53	22.46
1.5	35.33	28.30	35.55	28.13	55.31	18.08	55.67	17.96
2	47.10	21.23	47.40	21.10	73.74	13.56	74.22	13.47
2.3	54.17	18.46	54.51	18.35	84.80	11.79	85.35	11.72
2.5	58.88	16.98	59.25	16.88	92.18	10.85	92.78	10.78
3	70.65	14.15	71.10	14.06	110.6	9.04	111.33	8.98
4	94.20	10.62	94.80	10.55	147.5	6.78	148.4	6.74
4.5	106.0	9.44	106.7	9.38	165.9	6.03	167.0	5.99
5	117.8	8.49	118.5	8.44	184.4	5.42	185.6	5.39
6	141.3	7.08	142.2	7.03	221.2	4.52	222.7	4.49
7.5	176.6	5.66	177.8	5.63	276.5	3.62	278.3	3.59
8	188.4	5.31	189.6	5.27	295.0	3.39	296.9	3.37
9	212.0	4.72	213.3	4.69	331.9	3.01	334.0	2.99
10	235.5	4.25	237.0	4.22	368.7	2.71	371.1	2.69
12	282.6	3.54	284.4	3.52	442.4	2.26	445.3	2.25
15	353.3	2.83	355.5	2.81	553.1	1.81	556.7	1.80
18	423.9	2.36	426.6	2.34	663.7	1.51	668.0	1.50
20	471.0	2.12	474.0	2.11	737.4	1.36	742.2	1.35
22	518.1	1.93	521.4	1.92	811.2	1.23	816.4	1.22
25	588.8	1.70	592.5	1.69	921.8	1.08	927.8	1.08
30	706.5	1.42	711.0	1.41	1,106.0	0.90	1,113.0	0.90
32	753.6	1.33	758.4	1.32	1,180.0	0.85	1,188.0	0.84
38	894.9	1.12	900.6	1.11	1,401.0	0.71	1,410.0	0.71
40	942.0	1.06	948.0	1.05	1,475.0	0.68	1,484.0	0.67
44	1,036.0	0.97	1,043.0	0.96	1,622.0	0.62	1,633.0	0.61
50	1,178.0	0.85	1,185.0	0.84	1,844.0	0.54	1,856.0	0.54
65	1,531.0	0.65	1,541.0	0.65	2,397.0	0.42	2,412.0	0.41
75	1,766.0	0.57	1,778.0	0.56	2,765.0	0.36	2,783.0	0.36

STAINLESS STEEL ANGLES & FLATS

Stainless Steel Angles and Flats are produced by hot-rolling.

After fully annealed, they are subjected to special chemical treatment for descaling, which gives the products the following advantages:

1. Clean and silvery coloured surface.
2. Smooth and tight skinned surface by this descaling process, which enables to achieve fine surface finish by polishing.
3. Excellent corrosion resistance and bending quality by full annealing.

Hot rolled stainless steel angles and flats are superior to cold formed products with regard to minimum internal stress, better corrosion resistance and lower cost.

For applications where the corrosion resistance after welding is essential, Extra Low Carbon Stainless Steel Angles and Flats (Ex. AISI 304 L, 316 L) is particularly recommended.

CHEMICAL COMPOSITION AISI, ASTM A276-78

Type	C max.	Mn max.	P max.	S max.	Si max.	Cr	Ni	Mo	Ti
302	0.15	2.0	0.045	0.03	1.0	17.0 - 19.0	8.0 - 10.0	--	
304	0.08	2.0	0.045	0.03	1.0	18.0 - 20.0	8.0 - 10.5	--	
304L	0.03	2.0	0.045	0.03	1.0	18.0 - 20.0	8.0 - 12.0	--	
316	0.08	2.0	0.045	0.03	1.0	16.0 - 18.0	10.0 - 14.0	2.0 - 3.0	
316L	0.03	2.0	0.045	0.03	1.0	16.0 - 18.0	10.0 - 14.0	2.0 - 3.0	
321	0.08	2.0	0.045	0.03	1.0	17.0 - 19.0	9.0 - 12.0	--	5 x C min.
430	0.12	1.0	0.040	0.03	1.0	16.0 - 18.0	--	--	

STAINLESS STEEL ANGLES & FLATS

TYPICAL CHARACTERISTICS FOR CHEMICAL COMPOSITION

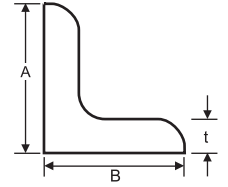
AISI	Typical Characteristics
302	Basic, general purpose type of non-magnetic, austenite stainless steel with good corrosion resistance and mechanical properties. Retains ductility and impact strength at low temperature. Good cold workability.
304	Low carbon variation of TYPE 302. Superior corrosion resistance and weldability to TYPE 302. Widely used for general purposes as well as parts fabricated by welding.
304L	Extra low carbon eliminates the harmful carbide precipitation due to welding. Suitable for applications where post-weld annealing is unpracticable.
316	Additional of Molybdenum strengthens corrosion resistance to reducing acids such as sulphuric, acetic, phosphoric and chloric acid. Improved pitting resistance and superior creep strength at high temperature to TYPE 302, 304. Used for various chemical industries such as pulp-, photograph-, dye-, textile-, fertilizer industry.
316L	Extra low carbon eliminates the harmful carbide precipitation due to welding. Suitable for applications where post-weld annealing is unpracticable and which are used in severely corrosive atmosphere.
321	Addition of Titanium prevents intergranular corrosion due to welding. Strong creep strength at high temperature. Can be used at high temperature (800 - 1500°F).
430	Typical ferritic stainless steel, magnetic. Superior corrosion resistance to TYPE 410. Suitable for applications used in weak corrosive atmosphere, kitchen equipment and utensils.

MECHANICAL PROPERTIES ASTM A276-78

Type	Condition	Size	Tension Test			
			Y.S (0.2%) p.s.i.	T.S p.s.i.	EL. (2 in.) %	R.A %
			min.	min.	min.	min.
302	annealed	all	30,000	75,000	40	50
304	annealed	all	30,000	75,000	40	40
304L	annealed	all	25,000	70,000	40	50
316	annealed	all	30,000	75,000	40	50
316L	annealed	all	25,000	70,000	40	50
321	annealed	all	30,000	75,000	40	50
430	annealed	all	40,000	70,000	20	45

STAINLESS STEEL ANGLES & FLATS

ANGLE (Sizes and Unit Weights)



IMPERIAL UNITS: APPROX. WT. LBS/FT

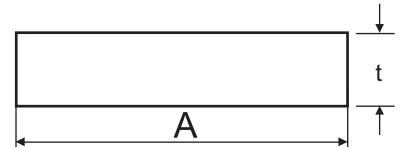
Legs (A x B) in.	Thickness (t) in.					
	1/8	3/16	1/4	5/16	3/8	1/2
3/4 x 3/4	0.599	0.856				
1 x 1	0.814	1.18	1.52			
1 1/4 x 1 1/4	1.03	1.51	1.96			
1 1/2 x 1 1/2	1.25	1.83	2.39			
2 x 2	1.68	2.49	3.26	4.00	4.73	
2 1/2 x 2 1/2		3.13	4.14	5.08	6.02	
3 x 3			5.01	6.19	7.34	9.57
3 1/2 x 3 1/2			5.88	7.27	8.63	11.30
4 x 4			6.75	8.35	9.95	13.00

METRIC UNITS: APPROX. WT. KG/M

Legs (A x B) mm	Thickness (t) mm										
	3	4	5	6	7	8	9	10	11	12	13
20 x 20	0.908	1.16	1.41								
25 x 25	1.15	1.48	1.81	2.12							
30 x 30	1.39	1.80	2.21	2.58							
35 x 35	1.64	2.13	2.62	3.00							
40 x 40	1.88	2.45	3.03	3.56							
50 x 50	2.41	3.14	3.87	4.55			6.60	7.26			
60 x 60			4.68	5.53	6.39						
65 x 65			5.13	6.07	6.98	7.87	8.78	9.67			
70 x 70				6.55	7.54	8.51	9.50	10.5			
75 x 75				7.03	8.11	9.16	10.2	11.3	12.3	13.3	14.3
80 x 80				7.52	8.67	9.80	11.0	12.0	13.2	14.3	15.4
90 x 90				8.53	9.85	11.1	12.5	13.7	15.0	16.2	17.5
100 x 100				9.50	11.0	12.4	13.9	15.3	16.7	18.2	19.5

STAINLESS STEEL ANGLES & FLATS

FLAT (Sizes and Unit Weights)



IMPERIAL UNIT: APPROX. WT. LBS/FT

Width (A) in.	Thickness (t) in.								
	1/8	3/16	1/4	5/16	3/8	1/2	5/8	3/4	1
1/2	0.217	0.326							
5/8	0.272	0.408	0.543						
3/4	0.326	0.489	0.652	0.816	0.979	1.30			
1	0.435	0.652	0.87	1.09	1.30	1.74	2.18	2.61	
1 1/4	0.544	0.816	1.09	1.36	1.63	2.18	2.72	3.26	
1 1/2	0.653	0.979	1.30	1.63	1.96	2.61	3.26	3.92	5.22
1 3/4	0.761	1.14	1.52	1.90	2.28	3.04			6.09
2	0.87	1.30	1.74	2.18	2.61	3.48	4.35	5.22	6.96
2 1/2	1.09	1.63	2.18	2.72	3.26	4.35	5.44	6.52	8.70
3		1.96	2.61	3.26	3.91	5.22	6.52	7.83	10.40
3 1/2			3.04	3.81	4.57	6.09	7.61	9.14	12.20
4			3.48	4.35	5.22	6.96	8.70	10.40	13.90

METRIC UNIT: APPROX. WT. KG/M

Width (A) mm	Thickness (t) mm											
	3	4	5	6	8	9	10	12	13	15	20	25
12	0.29			0.58								
13	0.314	0.419										
14	0.339	0.451	0.564	0.677								
15	0.363	0.484	0.604									
16	0.387	0.516	0.645	0.774								
20	0.484	0.645	0.806	0.967	1.29	1.45	1.61	1.93				
25	0.604	0.806	1.01	1.21	1.61	1.81	2.02	2.42	2.62	3.02	4.03	
30	0.725	0.967	1.21	1.45	1.93	2.18	2.42	2.90	3.14	3.63	4.84	
35	0.846	1.13	1.41	1.69	2.26	2.54	2.82	3.39	3.67	4.23	5.64	
40	0.967	1.29	1.61	1.93	2.58	2.90	3.22	3.87	4.19	4.84	6.45	8.06
45	1.09	1.45	1.81	2.18	2.90	3.26	3.63	4.35	4.72			9.07
50	1.21	1.61	2.02	2.42	3.22	3.63	4.03	4.84	5.24	6.04	8.06	10.1
60			2.42	2.90	3.87	4.35	4.84	5.80	6.29	7.25	9.67	12.1
65			2.62	3.14	4.19	4.72	5.24	6.29	6.81	7.86	10.5	13.1
70			2.82	3.39	4.51	5.08	5.64	6.77	7.33	8.46	11.3	14.1
75			3.02	3.63	4.84	5.44	6.04	7.25	7.86	9.07	12.1	15.1
80				3.87	5.16	5.80	6.45	7.74	8.38	9.67	12.9	16.1
90				4.35	5.80	6.38	7.25	8.70	9.43	10.9	14.5	18.1
100				4.84	6.45	7.25	8.06	9.67	10.5	12.1	16.1	20.2

STAINLESS STEEL BARS

ROUND BARS

Type Number	UNS Number	Equivalent to			Chemical Composition, %								Other Elements
		JIS	BS	DIN	C	Mn	P	S	Si	Cr	Ni	Mo	
201	(S20100)	SUS201	--	--	0.15	5.50 ~ 7.50	0.06	0.03	1.00	16.00 ~ 18.00	3.50 ~ 5.50	--	N 0.25
202	(S20200)	SUS202	284S16	--	0.15	7.50 ~ 10.00	0.06	0.03	1.00	17.00 ~ 19.00	4.00 ~ 6.00	--	N 0.25
205	(S20500)	--	--	--	0.12 ~ 0.25	14.00 ~ 15.50	0.06	0.03	1.00	16.50 ~ 18.00	1.00 ~ 1.75	--	N 0.32 / 0.40
301	(S30100)	SUS301	301S21	4310	0.15	2.00	0.045	0.03	1.00	16.00 ~ 18.00	6.00 ~ 8.00	--	
302	(S30200)	SUS302	302S25	4300	0.15	2.00	0.045	0.03	1.00	17.00 ~ 19.00	8.00 ~ 10.00	--	
302B	(S30215)	SUS302B	--	--	0.15	2.00	0.045	0.03	2.00 ~ 3.00*	17.00 ~ 19.00	8.00 ~ 10.00	--	
303	(S30300)	SUS303	303S21	4305	0.15	2.00	0.20	0.15	1.00	17.00 ~ 19.00	8.00 ~ 10.00	0.60	
303Se	(S30323)	SUS303Se	303S41	--	0.15	2.00	0.20	0.06	1.00	17.00 ~ 19.00	8.00 ~ 10.00	--	Se 0.15 min.
304	(S30400)	SUS304	304S15	4301	0.08	2.00	0.045	0.03	1.00	18.00 ~ 20.00	8.00 ~ 10.50	--	
304L	(S30403)	SUS304L	304S12	4306	0.03	2.00	0.045	0.03	1.00	18.00 ~ 20.00	8.00 ~ 12.00	--	
	(S30430)	--	--	--	0.08	2.00	0.045	0.03	1.00	17.00 ~ 19.00	8.00 ~ 10.00	--	Cu 3.00/4.00
304N	(S30451)	SUS304N1	--	--	0.08	2.00	0.045	0.03	1.00	18.00 ~ 20.00	8.00 ~ 10.00	--	N 0.10/0.16
305	(S30500)	SUS305	305S19	--	0.12	2.00	0.045	0.03	1.00	17.00 ~ 19.00	10.50 ~ 13.00	--	
308	(S30800)	SUS308	305S19	--	0.08	2.00	0.045	0.03	1.00	19.00 ~ 21.00	10.00 ~ 12.00	--	
309	(S30900)	SUH309	309S24	--	0.20	2.00	0.045	0.03	1.00	22.00 ~ 24.00	12.00 ~ 15.00	--	
309S	(S30908)	SUS309S	309S24	--	0.08	2.00	0.045	0.03	1.00	22.00 ~ 24.00	12.00 ~ 15.00	--	
310	(S31000)	SUH310	310S24	--	0.25	2.00	0.045	0.03	1.50	24.00 ~ 26.00	19.00 ~ 22.00	--	
310S	(S31008)	SUS310S	310S24	4845	0.08	2.00	0.045	0.03	1.50	24.00 ~ 26.00	19.00 ~ 22.00	--	
314	(S31400)	--	--	--	0.25	2.00	0.045	0.03	1.50 ~ 3.00	23.00 ~ 26.00	19.00 ~ 22.00	--	
316	(S31600)	SUS316	316S16	4401	0.08	2.00	0.045	0.03	1.00	16.00 ~ 18.00	10.00 ~ 14.00	2.00 ~ 3.00	
316F	(S31620)	--	--	--	0.08	2.00	0.20	0.10	1.00	16.00 ~ 18.00	10.00 ~ 14.00	1.75 ~ 2.50	
316L	(S31603)	SUS316L	316S16	4402	0.03	2.00	0.045	0.03	1.00	16.00 ~ 18.00	10.00 ~ 14.00	2.00 ~ 3.00	

Note:

- * May be added at the manufacturer's discretion.
- Products excluded from the above table are also available upon request.

STAINLESS STEEL BARS

ROUND BARS

Type Number	UNS Number	Equivalent to			Chemical Composition, %								Other Elements
		JIS	BS	DIN	C	Mn	P	S	Si	Cr	Ni	Mo	
316N	(S31651)	SUS316N	--	--	0.08	2.00	0.045	0.03	1.00	16.00 ~ 18.00	10.00 ~ 14.00	2.30 ~ 3.00	N 0.10/0.16
317	(S31700)	SUS317	316S12	--	0.08	2.00	0.045	0.03	1.00	18.00 ~ 20.00	11.00 ~ 15.00	3.00 ~ 4.00	
317L	(S31703)	SUS317L	--	4438	0.03	2.00	0.045	0.03	1.00	18.00 ~ 20.00	11.00 ~ 15.00	3.00 ~ 4.00	
321	(S32100)	SUS321	317S16	4541	0.08	2.00	0.045	0.03	1.00	17.00 ~ 19.00	9.00 ~ 12.00	--	Ti 5 x C min.
329	(S32900)	SUS327J1	317S12	--	0.10	2.00	0.04	0.03	1.00	25.00 ~ 30.00	3.00 ~ 6.00	1.00 ~ 2.00	
330	(N08330)	SUH330	321S12 321S20	--	0.08	2.00	0.04	0.03	0.75 ~ 1.50	17.00 ~ 20.00	34.00 ~ 37.00	--	
347	(S34700)	SUS347	347S17	4450	0.08	2.00	0.045	0.03	1.00	17.00 ~ 19.00	9.00 ~ 13.00	--	Cb + Ta 10 x C min.
348	(S34800)	--	--	--	0.08	2.00	0.045	0.03	1.00	17.00 ~ 19.00	9.00 ~ 13.00	--	Cb + Ta 10 x C min. Ta 0.10 max. Co 0.20 max.
384	(S38400)	SU384	--	--	0.08	2.00	0.045	0.03	1.00	11.00 ~ 17.00	17.00 ~ 19.00	--	
403	(S40300)	SUS403	403S17	--	0.15	1.00	0.04	0.03	0.50	11.50 ~ 13.00			
405	(S40500)	SUS405	405S17	4002	0.08	1.00	0.04	0.03	1.00	11.50 ~ 14.50	--	--	Al 0.10/0.30
409	(S40900)	SUH409	--	--	0.08	1.00	0.045	0.045	1.00	10.50 ~ 11.75			Ti 6 x C min / 0.75 max
410	(S41000)	SUS410	410S21	4006	0.15	1.00	0.04	0.03	1.00	11.50 ~ 13.50			
414	(S41400)	--	--	--	0.15	1.00	0.04	0.03	1.00	11.50 ~ 13.50	1.25 ~ 2.50		
416	(S41600)	SUS416	416S21	--	0.15	1.25	0.06	0.15	1.00	12.00 ~ 14.00	--	0.60*	
416Se	(S41623)	--	--	--	0.15	1.25	0.06	0.06	1.00	12.00 ~ 14.00	--	--	Se 0.15 min.
420	(S42000)	SUS420J1 SUS420J2	420S37 420S45	4021	0.15	1.00	0.04	0.03	1.00	12.00 ~ 14.00	--	--	
420F	(S42020)	SUS420F	--	--	0.15	1.25	0.06	0.15	1.00	12.00 ~ 14.00	--	0.60*	
422	(S42200)	SUH616	--	--	0.20 ~ 0.25	1.00	0.025	0.025	0.75	11.00 ~ 13.00	0.50 ~ 1.00	0.75 ~ 1.25	V 0.15/0.30 W 0.75/1.25
429	(S42900)	SUS429	--	--	0.12	1.00	0.04	0.03	1.00	14.00 ~ 16.00			
430	(S43000)	SUS430	430S15	4016	0.12	1.00	0.04	0.03	1.00	16.00 ~ 18.00			

Note:

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STAINLESS STEEL BARS

ROUND BARS

Type Number	UNS Number	Equivalent to			Chemical Composition, %								Other Elements
		JIS	BS	DIN	C	Mn	P	S	Si	Cr	Ni	Mo	
430F	(S43020)	SUS430F	--	4104	0.12	1.25	0.06	0.15	1.00	16.00 ~ 18.00	--	0.60*	
430FSe	(S43023)	--	--	--	0.12	1.25	0.06	0.06	1.00	16.00 ~ 18.00	--	--	Se 0.15 max.
431	(S43100)	SUS431	431S29	4037	0.20	1.00	0.04	0.03	1.00	15.00 ~ 17.00	1.25 ~ 2.50	--	
434	(S43400)	SUS434	434S19	4113	0.12	1.00	0.04	0.03	1.00	16.00 ~ 18.00		0.75 ~ 1.25	
436	(S43600)	--	--	--	0.12	1.00	0.04	0.03	1.00	16.00 ~ 18.00		0.75 ~ 1.25	Cb + Ta 5 x C min.
440A	(S44002)	SUS440A	--	--	0.60~ 0.75	1.00	0.04	0.03	1.00	16.00 ~ 18.00		0.75	
440B	(S44003)	SUS440B	--	--	0.75~ 0.95	1.00	0.04	0.03	1.00	16.00 ~ 18.00		0.75	
440C	(S44004)	SUS440C	--	--	0.95~ 1.20	1.00	0.04	0.03	1.00	16.00 ~ 18.00		0.75	
442	(S44200)	--	--	--	0.20	1.00	0.04	0.03	1.00	18.00 ~ 23.00			
446	(S44600)	SUH446	--	--	0.20	1.50	0.04	0.03	1.00	23.00 ~ 27.00			N0.25
501	(S50100)	--	--	--	0.10	1.00	0.04	0.03	1.00	4.00 ~ 6.00		0.40 ~ 0.65	
502	(S50200)	--	--	--	0.10	1.00	0.04	0.03	1.00	4.00 ~ 6.00		0.40 ~ 0.65	
	(S13800)	--	--	--	0.05	0.10	0.01	0.008	0.10	12.25 ~ 13.25	7.50 ~ 8.50	2.00 ~ 2.50	Al 0.09/1.35 N 0.01
	(S15500)	--	--	--	0.07	1.00	0.04	0.03	1.00	14.00 ~ 15.00	3.50 ~ 5.50	--	Cu 3.00/5.00 Cb + Ta 0.15/0.45
	(S17400)	SUS630	--	--	0.07	1.00	0.04	0.03	1.00	15.50 ~ 17.50	3.00 ~ 5.00	--	Cu 3.00/5.00 Cb + Ta 0.15/0.45
	(S17700)	SUS631	--	--	0.09	1.00	0.04	0.04	1.00	16.00 ~ 18.00	6.50 ~ 7.75	--	Al 0.75/1.50
	(K66286)	SUH660	--	--	0.08	2.00	0.04	0.03	1.00	13.50 ~ 16.00	24.00 ~ 17.00	1.00 ~ 1.50	Ti 1.90/2.35, Al 0.35 V 0.1/0.5, B 0.001/0.01

Note:

- * May be added at the manufacturer's discretion.
- Products excluded from the above table are also available upon request.

STAINLESS STEEL BARS

ROUND BARS (Sizes and Unit Weights)

Size	lb/ft	kg/ft	Size	lb/ft	kg/ft
3.00 mm	0.037	0.017	2"	10.803	4.901
1/8"	0.042	0.019	2 1/4"	13.673	6.202
4.50 mm	0.085	0.039	2 1/2"	16.88	7.657
3/16"	0.096	0.044	2 3/4"	20.425	9.265
6.00 mm	0.150	0.068	3"	24.307	11.026
1/4"	0.169	0.077	3 1/4"	28.526	12.94
7.50 mm	0.235	0.107	3 1/2"	33.084	15.007
5/16"	0.265	0.120	3 3/4"	37.978	17.227
9.00 mm	0.339	0.154	4"	43.211	19.601
3/8"	0.380	0.172	4 1/2"	54.689	24.807
12.00 mm	0.603	0.274	5"	67.516	30.626
1/2"	0.675	0.306	5 1/2"	81.695	37.057
5/8"	1.055	0.479	6"	97.224	44.101
3/4"	1.519	0.689	6 1/2"	114.10	51.757
7/8"	2.067	0.938	7"	132.33	60.026
1"	2.701	1.225	7 1/2"	151.91	68.907
1 1/8" 3.418	1.550	8"	172.84	78.401	
1 1/4" 4.220	1.914	9"	218.75	99.226	
1 3/8" 5.107	2.317	10"	270.07	122.50	
1 1/2" 6.076	2.756	11"	326.78	148.23	
1 5/8" 7.132	3.235	12"	388.89	176.40	
1 3/4" 8.271	3.752	13"	456.41	207.03	
1 7/8" 9.495	4.307	14"	529.33	240.10	

STAINLESS STEEL BARS

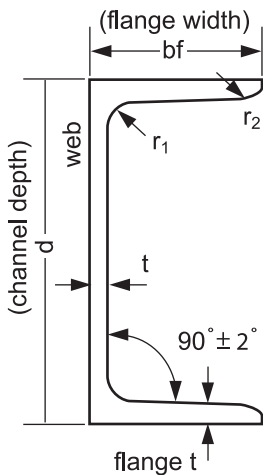
HEXAGON BARS (Sizes and Unit Weights)			SQUARE BARS (Sizes and Unit Weights)		
Size	lb/ft	kg/ft	Size	lb/ft	kg/ft
3.00 mm	0.041	0.019	3.00 mm	0.048	0.022
1/8"	0.046	0.021	1/8"	0.054	0.024
4.50 mm	0.093	0.042	4.50 mm	0.108	0.049
3/16"	0.105	0.047	3/16"	0.121	0.055
6.00 mm	0.166	0.075	6.00 mm	0.192	0.087
1/4"	0.186	0.084	1/4"	0.215	0.097
7.50 mm	0.259	0.118	7.50 mm	0.299	0.136
5/16"	0.290	0.132	5/16"	0.335	0.152
9.00 mm	0.373	0.169	9.00 mm	0.431	0.196
3/8"	0.418	0.190	3/8"	0.483	0.219
12.00 mm	0.664	0.301	12.00 mm	0.766	0.348
1/2"	0.744	0.337	1/2"	0.859	0.390
5/8"	1.162	0.527	5/8"	1.341	0.609
3/4"	1.673	0.759	3/4"	1.932	0.876
7/8"	2.277	1.033	7/8"	2.629	1.193
1"	2.974	1.350	1"	3.434	1.558
1 1/8" 3.765	1.708	1 1/8"	4.346	1.972	
1 1/4" 4.648	2.109	1 1/4"	5.366	2.434	
1 3/8" 5.624	2.552	1 3/8"	6.492	2.946	
1 1/2" 6.693	3.037	1 1/2"	7.727	3.506	
1 5/8" 7.854	3.564	1 5/8"	9.068	4.114	
1 3/4" 9.109	4.133	1 3/4"	10.517	4.772	
1 7/8" 10.457	4.745	1 7/8"	12.073	5.478	
2"	11.898	5.398	2"	13.736	6.232
2 1/4" 15.058	6.832	2 1/4"	17.385	7.888	
2 1/2" 18.590	8.435	2 1/2"	21.463	9.738	
2 3/4" 22.494	10.206	2 3/4"	25.970	11.783	
3"	26.770	12.146	3"	30.906	14.023
3 1/4" 31.418	14.255	3 1/4"	36.272	16.457	
3 1/2" 36.437	16.532	3 1/2"	42.067	19.086	
3 3/4" 41.828	41.828	18.978	3 3/4"	48.291	21.910
4"	47.591	21.593	4"	54.944	24.929

STAINLESS STEEL WELDED CHANNELS

AISI 304

Size (in.)	Tolerance		AISI 304. 304L	AISI 316. 316L
d x bf x t	d	bf	Unit weight (lb/ft)	Unit weight (lb/ft)
3 x 1 $\frac{1}{2}$ x $\frac{3}{16}$	$\pm\frac{1}{8}$	$\pm\frac{1}{8}$	3.70	3.70
4 x 2 x $\frac{3}{16}$	$\pm\frac{1}{8}$	$\pm\frac{1}{8}$	5.03	5.03
4 x 2 x $\frac{1}{4}$	$\pm\frac{1}{8}$	$\pm\frac{1}{8}$	6.58	6.58
5 x 2 $\frac{1}{2}$ x $\frac{1}{4}$	$\pm\frac{3}{16}$	$\pm\frac{1}{8}$	8.40	8.40
6 x 3 x $\frac{1}{4}$	$\pm\frac{3}{16}$	$\pm\frac{1}{8}$	10.14	10.14
6 x 3 x $\frac{3}{8}$	$\pm\frac{3}{16}$	$\pm\frac{1}{8}$	14.77	14.77
8 x 4 x $\frac{3}{8}$	$\pm\frac{3}{16}$	$+\frac{3}{16} - \frac{1}{8}$	20.02	20.02

Size (mm)	Tolerance		AISI 304 . 304L	AISI 316. 316L
d x bf x t	d	bf	Unit weight (kg/m)	Unit weight (kg/m)
80 x 40 x 5	± 2.00	± 2.00	6.08	6.08
100 x 50 x 4	± 2.00	± 2.00	6.28	6.28
100 x 50 x 6	± 2.00	± 2.00	9.16	9.16
130 x 65 x 6	± 3.00	± 3.00	12.20	12.20
150 x 75 x 6	± 3.00	± 3.00	14.13	14.13
150 x 75 x 9	± 3.00	± 3.00	20.53	20.53
200 x 100 x 10	± 4.00	± 4.00	30.80	30.80



MECHANICAL PROPERTIES OF AISI 304 BUTT WELDED JOINT

Joint Thickness (inch)	Tension test			Bend Test (180°)	
	Y.S. 0.2% P.S.I.	T.S. P.S.I.	E.L. %	Face	Root
$\frac{3}{16}$	50200	95800	57.7	Good	Good
$\frac{1}{4}$	48800	93100	57.2	Good	Good
$\frac{3}{8}$	41200	85200	43.2	Good	Good

Joint Thickness (mm)	Tension test			Bend Test (180°)	
	Y.S. 0.2% (kg/mm ²)	T.S. (kg/mm ²)	E.L. %	Face	Root
5	34	62	57.7	Good	Good
6	35	62	57.2	Good	Good
10	37	60	43.2	Good	Good

STAINLESS STEEL WELDED TUBINGS

WELDED STAINLESS STEEL TUBING

MECHANICAL PROPERTIES

Type	Round and square			Round			
	Tension test			Bending	Flattering		Flaring
	Yield strength kg/mm ²	Tensile strength kg/mm ²	Elongation (longitudinal) %		No.1	No.2	
AISI 430	> 25	> 42	> 20	180°4D	< ² / ₃ D	< ¹ / ₂ D	1.14D
AISI 304	> 21	> 53	> 35	180°3D	< ² / ₃ D	< ¹ / ₂ D	1.2D
AISI 316	> 21	> 53	> 35	180°3D	< ² / ₃ D	< ¹ / ₂ D	1.2D

Notes:

- (1) Yield strength test shall not be conducted unless requested. Other mechanical property values are available on request.
- (2) For the tension test of round tubing, test piece No. 11 or 12 specified in JIS Z 2201 (Tension Test Pieces for Materials) shall be used. Test piece No. 12 shall be taken from an unwelded portion.
- (3) For the tension test of square tubing, test tubing, test piece No. 5 specified in JIS Z 2201 shall be used. However, for a side length of less than 25 mm, the test shall be conducted on a test piece with a width equivalent to its side length. The test piece shall be taken from an unwelded portion.
- (4) Unless otherwise specified, the values of No. 1 shall be used.

TYPES OF MANUFACTURES

Surface Finish			
	Finish	Nisshin Code	Description
Round tubing	Unpolished	S2	As welded
	120 GRIT	S 240	Polished by #120 belt
	120 GRIT	SB 240	Buffed S 240
	180 GRIT	SB 400	Buffed after polishing by #180 belt
	180 GRIT	GL	Hair-line finished after polishing by #180 belt
	320 GRIT	SB 600	Buffed after polishing by #320 belt
	400 GRIT	BL 400	Buffed after polishing by #400 belt
	MIRROR	MR	Buffed after polishing by #600 belt
Square & rectangular tubing	Unpolished	S2	As welded
	180 GRIT	HL180	Hair-line finished after polishing by #180 belt

CHEMICAL COMPOSITION

Type	Chemical Composition (%)							
	C	Si	Mn	P	S	Ni	Cr	Mo
AISI 430	<0.12	<1.00	<1.00	<0.04	<0.03	--	16 ~18	--
AISI 304	<0.08	<1.00	<2.00	<0.04	<0.03	8~11	18~20	--
AISI 316	<0.08	<1.00	<2.00	<0.04	<0.03	11~14	16 ~18	200 ~300

STAINLESS STEEL WELDED TUBINGS

ROUND TUBING

TABLE OF UNIT WEIGHT

Outside diameter		Type	Wall Thickness (mm)																			
			0.5		0.6		0.7		0.8		1.0		1.2		1.5		2.0		2.5		3.0	
			kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'
1/2	12.7	AISI 304	0.152	0.927	0.181	1.10	0.209	1.27	0.237	1.44	0.291	1.77	0.344	2.10	0.418	2.55						
		AISI 430	0.148	0.902	0.176	1.07	0.203	1.24	0.203	1.40	0.283	1.73	0.334	2.04	0.406	2.47						
5/8	15.9	AISI 304	0.192	1.17	0.229	1.40	0.265	1.62	0.301	1.83	0.371	2.26	0.439	2.68	0.538	3.28						
		AISI 430	0.186	1.13	0.222	1.35	0.257	1.57	0.292	1.78	0.36	2.19	0.427	2.60	0.522	3.18						
	16	AISI 304	0.193	1.18	0.23	1.40	0.267	1.63	0.303	1.85	0.374	2.28	0.442	2.69	0.542	3.30						
		AISI 430	0.187	1.14	0.224	1.37	0.259	1.58	0.294	1.79	0.363	2.21	0.43	2.62	0.562	3.21						
3/4	19.1	AISI 304	0.232	1.41	0.277	1.69	0.321	1.96	0.365	2.23	0.451	2.75	0.533	3.26	0.658	4.01						
		AISI 430	0.225	1.37	0.269	1.64	0.321	1.90	0.354	2.16	0.438	2.67	0.52	3.17	0.639	3.90						
7/8	22.2	AISI 304			0.323	1.97	0.375	2.29	0.426	2.60	0.528	3.22	0.628	3.83	0.773	4.71	1.01	6.16				
		AISI 430			0.314	1.91	0.364	2.33	0.414	2.52	0.513	3.13	0.61	3.72	0.751	4.58	0.977	5.96				
	25	AISI 304			0.365	2.23	0.429	2.62	0.482	2.94	0.598	3.65	0.711	4.33	0.878	5.35	1.15	7.01				
		AISI 430			0.354	2.16	0.411	2.51	0.468	2.85	0.581	3.54	0.691	4.21	0.853	5.20	1.11	6.77				
1	25.4	AISI 304			0.371	2.26	0.431	2.63	0.49	2.99	0.608	3.71	0.723	4.41	0.893	5.44	1.17	7.13				
		AISI 430			0.36	2.19	0.417	2.54	0.476	2.90	0.59	3.60	0.702	4.28	0.867	5.29	1.13	6.89				
1 1/8	28.6	AISI 304			0.418	2.55	0.486	2.96	0.554	3.38	0.687	4.19	0.819	4.99	1.01	6.16	1.32	8.05				
		AISI 430			0.406	2.47	0.472	2.88	0.538	3.28	0.668	4.07	0.795	4.85	0.983	5.99	1.29	7.86				
1 1/4	31.8	AISI 304					0.542	3.30	0.618	3.77	0.767	4.68	0.915	5.58	1.13	6.89	1.48	9.02	1.82	11.1	2.15	13.1
		AISI 430					0.527	3.21	0.60	3.66	0.745	4.54	0.888	5.41	1.10	6.71	1.44	8.78	1.77	10.8	2.09	12.7
	32	AISI 304					0.546	3.33	0.622	3.79	0.772	4.71	0.921	5.61	1.14	6.95	1.49	9.08	1.84	11.2	2.17	13.2
		AISI 430					0.53	3.23	0.604	3.68	0.75	4.57	0.894	5.45	1.11	6.77	1.45	8.84	1.78	10.9	2.10	19.8
1 1/2	38.1	AISI 304					0.652	3.97	0.743	4.53	0.924	5.63	1.10	6.71	1.37	8.35	1.80	11.0	2.22	13.5	2.62	16.0
		AISI 430					0.633	3.86	0.722	4.40	0.897	5.47	1.07	6.52	1.33	8.11	1.75	10.7	2.15	13.1	2.55	15.3
1 5/8	41.3	AISI 304							0.807	4.92	1.00	6.10	1.20	7.32	1.49	9.08	1.96	11.9	2.42	14.8	2.86	17.4
		AISI 430							0.784	4.78	0.974	5.94	1.16	7.07	1.44	8.78	1.90	11.6	2.35	14.3	2.78	16.9
1 3/4	44.5	AISI 304							0.871	5.31	1.08	6.58	1.29	7.86	1.61	9.81	2.12	12.9	2.62	16.0	3.10	18.9
		AISI 430							0.846	5.16	1.05	6.40	1.26	7.68	1.56	9.51	2.06	12.6	2.54	15.5	3.01	18.3
	45	AISI 304							0.881	5.37	1.10	6.71	1.31	7.99	1.63	9.94	2.14	13.0	2.65	16.2	3.14	19.1
		AISI 430							0.855	5.21	1.06	6.46	1.27	7.74	1.58	9.63	2.08	12.7	2.57	15.7	3.05	18.6
1.9	48.3	AISI 304							0.947	5.77	1.18	7.19	1.41	8.60	1.75	10.7	2.31	14.1	2.85	17.4	3.39	20.7
		AISI 430							0.92	5.61	1.14	6.95	1.37	8.35	1.70	10.4	2.24	13.7	2.77	16.9	3.29	20.1
	50	AISI 304									1.22	7.44	1.46	8.90	1.81	11.0	2.39	14.6	2.96	18.0	3.51	21.4
		AISI 430									1.19	7.25	1.42	8.66	1.76	10.7	2.32	14.1	2.87	17.5	3.41	20.8
2	50.8	AISI 304									1.24	7.56	1.48	9.02	1.84	11.2	2.43	14.8	3.01	18.3	3.57	21.8
		AISI 430									1.20	7.32	1.44	8.78	1.79	10.9	2.36	14.4	2.92	17.8	3.47	21.2
	60.8	AISI 304											1.77	10.8	2.20	13.4	2.91	17.7	3.61	22.0	4.30	26.2
		AISI 430											1.72	10.5	2.14	13.0	2.83	17.3	3.51	21.4	4.17	25.4
2 1/2	63.5	AISI 304											1.86	11.3	2.32	14.1	3.06	18.7	3.80	23.2	4.52	27.6
		AISI 430											1.81	11.0	2.25	13.7	2.98	18.2	3.69	22.5	4.39	26.8
3	76.2	AISI 304											2.24	13.7	2.79	17.0	3.70	22.6	4.59	28.0	5.47	33.3
		AISI 430											2.18	13.3	2.71	16.5	3.59	21.9	4.46	27.2	5.31	32.4

STAINLESS STEEL WELDED TUBINGS

SQUARE TUBING

TABLE OF UNIT WEIGHT

Side length mm (inch)	Type	Wall thickness mm, BWG (inch)													
		1.0		1.2		1.5		2.0		3.0		BWG18(0.049)		BWG16(0.065)	
		kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/ft	kg/20'	kg/ft	kg/20'
12 x 12	AISI 304	0.347	2.12	0.409	2.49							0.128	2.57		
	AISI 430	0.337	2.05	0.397	2.42							0.125	2.49		
12.7 x 12.7 ($\frac{1}{2}$ x $\frac{1}{2}$)	AISI 304	0.369	2.25	0.436	2.66							0.137	2.74		
	AISI 430	0.359	2.19	0.423	2.58							0.133	2.66		
14 x 14	AISI 304	0.411	2.51	0.485	2.96							0.152	3.05		
	AISI 430	0.399	2.43	0.471	2.87							0.148	2.96		
15.9 x 15.9 ($\frac{5}{8}$ x $\frac{5}{8}$)	AISI 304	0.471	2.87	0.558	3.40							0.175	3.51		
	AISI 430	0.457	2.79	0.542	3.30							0.17	3.40		
16 x 16	AISI 304	0.474	2.89	0.562	3.43							0.176	3.53		
	AISI 430	0.46	2.8	0.545	3.32							0.171	3.43		
19 x 19	AISI 304	0.569	3.47	0.676	4.12	0.831	5.07					0.212	4.25	0.276	5.52
	AISI 430	0.553	3.37	0.656	4.00	0.807	4.92					0.206	4.13	0.268	5.36
19.1 x 19.1 ($\frac{3}{4}$ x $\frac{3}{4}$)	AISI 304	0.572	3.49	0.68	4.15	0.836	5.10					0.214	4.27	0.278	5.56
	AISI 430	0.556	3.39	0.66	4.02	0.811	4.94					0.207	4.15	0.27	5.39
20 x 20	AISI 304	0.601	3.66	0.714	4.35	0.878	5.35					0.224	4.49	0.292	5.85
	AISI 430	0.584	3.56	0.693	4.22	0.853	5.20					0.218	4.36	0.284	5.68
21 x 21	AISI 304	0.633	3.86	0.752	4.58	0.926	5.64					0.236	4.72	0.308	6.16
	AISI 430	0.614	3.74	0.73	4.45	0.899	5.48					0.23	4.59	0.299	5.99
22.2 x 22.2 ($\frac{7}{8}$ x $\frac{7}{8}$)	AISI 304	0.671	4.09	0.798	4.86	0.983	5.99					0.251	5.02	0.326	6.52
	AISI 430	0.651	3.97	0.774	4.72	0.955	5.82					0.244	4.88	0.317	6.34
25 x 25	AISI 304	0.76	4.63	0.904	5.51	1.12	6.83					0.284	5.69	0.372	7.44
	AISI 430	0.738	4.5	0.878	5.35	1.08	6.58					0.276	5.52	0.36	7.19
25.4 x 25.4 (1 x 1)	AISI 304	0.772	4.71	0.919	5.60	1.14	6.95					0.289	5.79	0.378	7.56
	AISI 430	0.75	4.57	0.893	5.44	1.10	6.71					0.281	5.61	0.369	7.38
28 x 28	AISI 304	0.855	5.21	1.02	6.22	1.26	7.68					0.32	6.40	0.421	8.41
	AISI 430	0.83	5.06	0.989	6.03	1.22	7.44					0.311	6.22	0.408	8.17
30 x 30	AISI 304	0.918	5.60	1.09	6.64	1.35	8.23	1.77	10.8			0.344	6.89	0.452	9.03
	AISI 430	0.892	5.44	1.06	6.46	1.32	8.05	1.72	10.5			0.335	6.71	0.439	8.78
31.8 x 31.8 ($1\frac{1}{4}$ x $1\frac{1}{4}$)	AISI 304	0.975	5.94	1.16	7.07	1.44	8.78	1.89	11.5			0.366	7.32	0.482	9.63
	AISI 430	0.947	5.77	1.13	6.89	1.40	8.53	1.83	11.2			0.357	7.13	0.466	9.33
32 x 32	AISI 304	0.982	5.99	1.17	7.13	1.45	8.84	1.90	11.6			0.369	7.38	0.485	9.69
	AISI 430	0.953	5.81	1.14	6.95	1.41	8.60	1.85	11.3			0.357	7.13	0.469	9.39
34 x 34	AISI 304	1.05	6.40	1.25	7.62	1.54	9.39	2.03	12.4			0.393	7.86	0.515	10.3
	AISI 430	1.01	6.16	1.21	7.38	1.50	9.14	1.97	12.0			0.381	7.62	0.50	10.0
35 x 35	AISI 304	1.08	6.58	1.28	7.80	1.59	9.69	2.09	12.7			0.405	8.11	0.531	10.6
	AISI 430	1.05	6.40	1.25	7.62	1.55	9.45	2.03	12.4			0.393	7.86	0.515	10.3
38 x 38	AISI 304	1.17	7.13	1.40	8.53	1.73	10.5	2.28	13.9			0.439	8.78	0.579	11.6
	AISI 430	1.14	6.95	1.36	8.29	1.68	10.2	2.22	13.5			0.427	8.53	0.564	11.3
38.1 x 38.1 ($1\frac{1}{2}$ x $1\frac{1}{2}$)	AISI 304	1.18	7.19	1.40	8.53	1.74	10.6	2.29	14.0			0.442	8.84	0.582	11.6
	AISI 430	1.14	6.95	1.36	8.29	1.69	10.3	2.22	13.5			0.43	8.60	0.564	11.3
40 x 40	AISI 304	1.24	7.56	1.48	9.02	1.83	11.2	2.41	14.7	3.52	21.5	0.463	9.27	0.613	12.3
	AISI 430	1.20	7.32	1.43	8.72	1.78	10.9	2.34	14.3	3.42	20.8	0.451	9.02	0.594	11.9
46 x 46	AISI 304	1.43	8.72	1.70	10.4	2.12	12.9	2.79	17.0	4.09	24.9	0.536	10.7	0.707	14.1
	AISI 430	1.38	8.41	1.65	10.1	2.05	12.5	2.71	16.5	3.97	24.2	0.521	10.4	0.686	13.7
50 x 50	AISI 304			1.86	11.3	2.31	14.1	3.04	18.5	4.47	27.2	0.585	11.7	0.771	15.4
	AISI 430			1.80	11.0	2.24	13.7	2.96	18.0	4.34	26.5	0.567	11.3	0.75	15.0
50.8 x 50.8 (2 x 2)	AISI 304			1.89	11.5	2.34	14.3	3.09	18.8	4.55	27.7	0.594	11.9	0.783	15.7
	AISI 430			1.83	11.2	2.28	13.9	3.00	18.3	4.42	26.9	0.576	11.5	0.762	15.2
60 x 60	AISI 304			2.24	13.7	2.78	16.9	3.68	22.4	5.42	33.0	0.704	14.1	0.93	18.6
	AISI 430			2.17	13.2	2.70	16.5	3.57	21.8	5.27	32.1	0.683	13.7	0.902	18.0

STAINLESS STEEL WELDED TUBINGS

RECTANGULAR TUBING

TABLE OF UNIT WEIGHT

Side length mm (inch)	Type	Wall thickness mm, BWG (inch)											
		1.0		1.2		1.5		2.0		BWG 18 (0.049)		BWG 16 (0.065)	
		kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/m	kg/20'	kg/ft	kg/20'	kg/ft	kg/20'
19 x 10	AISI 304	0.426	2.60	0.505	3.08					0.158	3.17		
	AISI 430	0.414	2.52	0.49	2.99					0.154	3.08		
20 x 10	AISI 304	0.442	2.69	0.524	3.19					0.164	3.29		
	AISI 430	0.43	2.62	0.508	3.10					0.16	3.19		
20 x 15	AISI 304	0.522	3.18	0.619	3.77					0.194	3.89		
	AISI 430	0.507	3.09	0.601	3.66					0.189	3.77		
22 x 10	AISI 304	0.474	2.89	0.562	3.43					0.176	3.53		
	AISI 430	0.46	2.80	0.545	3.32					0.171	3.43		
22 x 12	AISI 304	0.506	3.08	0.60	3.66					0.188	3.77		
	AISI 430	0.491	2.99	0.582	3.55					0.183	3.66		
24 x 14	AISI 304	0.569	3.47	0.676	4.12	0.831	5.07			0.212	4.25	0.276	5.52
	AISI 430	0.553	3.37	0.656	4.00	0.807	4.92			0.206	4.13	0.268	5.36
25 x 14	AISI 304	0.585	3.57	0.695	4.23	0.855	5.21			0.218	4.36	0.284	5.68
	AISI 430	0.568	3.46	0.675	4.11	0.83	5.06			0.212	4.24	0.276	5.52
25.4 x 12.7 (1 x 1/2)	AISI 304	0.571	3.48	0.678	4.13	0.833	5.08			0.213	4.26	0.277	5.54
	AISI 430	0.554	3.38	0.658	4.01	0.809	4.93			0.207	4.14	0.269	5.38
30 x 10	AISI 304	0.601	3.66	0.714	4.35	0.878	5.35			0.224	4.49	0.292	5.85
	AISI 430	0.584	3.56	0.693	4.22	0.853	5.20			0.218	4.36	0.284	5.68
30 x 20	AISI 304	0.76	4.63	0.904	5.51	1.12	6.83			0.284	5.69	0.372	7.44
	AISI 430	0.738	4.50	0.878	5.35	1.08	6.58			0.276	5.52	0.36	7.19
32 x 16	AISI 304	0.728	4.44	0.866	5.28	1.07	6.52			0.272	5.44	0.357	7.13
	AISI 430	0.707	4.31	0.841	5.13	1.04	6.34			0.265	5.29	0.344	6.89
32 x 17	AISI 304	0.744	4.54	0.885	5.39	1.09	6.64			0.278	5.57	0.363	7.25
	AISI 430	0.722	4.40	0.859	5.24	1.06	6.46			0.27	5.41	0.354	7.07
38.1 x 25.4 (1 1/2 x 1)	AISI 304	0.974	5.94	1.16	7.07	1.44	8.78			0.366	7.32	0.479	9.57
	AISI 430	0.945	5.76	1.13	6.89	1.40	8.53			0.354	7.07	0.466	9.33
40 x 10	AISI 304	0.76	4.63	0.904	5.51	1.12	6.83			0.284	5.69	0.372	7.44
	AISI 430	0.738	4.50	0.878	5.35	1.08	6.58			0.276	5.52	0.36	7.19
40 x 15	AISI 304	0.839	5.11	0.999	6.09	1.24	7.56			0.314	6.28	0.411	8.23
	AISI 430	0.815	4.97	0.97	5.91	1.20	7.32			0.305	6.10	0.399	7.99
40 x 18	AISI 304	0.886	5.40	1.06	6.46	1.31	7.99			0.332	6.64	0.436	8.72
	AISI 430	0.861	5.25	1.03	6.28	1.27	7.74			0.323	6.46	0.424	8.47
40 x 20	AISI 304	0.918	5.60	1.09	6.64	1.35	8.23	1.77	10.8	0.344	6.89	0.451	9.02
	AISI 430	0.892	5.44	1.06	6.46	1.31	7.99	1.72	10.5	0.335	6.71	0.439	8.78
40 x 25	AISI 304			1.19	7.25	1.47	8.96	1.93	11.8	0.375	7.50	0.491	9.81
	AISI 430			1.16	7.07	1.43	8.72	1.88	11.5	0.363	7.25	0.479	9.57
40 x 28	AISI 304			1.25	7.62	1.54	9.39	2.03	12.4	0.393	7.86	0.515	10.3
	AISI 430			1.21	7.38	1.50	9.14	1.97	12.0	0.381	7.62	0.50	10.0
50 x 25	AISI 304			1.38	8.41	1.71	10.4	2.25	13.7	0.433	8.66	0.57	11.4
	AISI 430			1.34	8.17	1.66	10.1	2.19	13.4	0.421	8.41	0.555	11.1
50.8 x 25.4 (2 x 1)	AISI 304			1.40	8.53	1.74	10.6	2.29	14.0	0.442	8.84	0.582	11.6
	AISI 430			1.36	8.29	1.69	10.3	2.22	13.5	0.43	8.60	0.564	11.3
51x 26	AISI 304			1.42	8.66	1.76	10.7	2.31	14.1	0.445	8.90	0.588	11.8
	AISI 430			1.38	8.41	1.71	10.4	2.25	13.7	0.433	8.66	0.57	11.4
60 x 20	AISI 304			1.48	9.02	1.83	11.2	2.41	14.7	0.463	9.27	0.613	12.3
	AISI 430			1.43	8.72	1.78	10.9	2.34	14.3	0.451	9.02	0.594	11.9
60 x 40	AISI 304			1.86	11.3	2.31	14.1	3.04	18.5	0.585	11.7	0.771	15.4
	AISI 430			1.80	11.0	2.24	13.7	2.96	18.0	0.567	11.3	0.75	15.0
75 x 45	AISI 304					2.78	16.9	3.68	22.4			0.93	18.6
	AISI 430					2.70	16.5	3.57	21.8			0.902	18.0
76.2 x 25.4 (3 x 1)	AISI 304					2.34	14.3	3.09	18.8			0.783	15.7
	AISI 430					2.28	13.9	3.00	18.3			0.762	15.2
80x40	AISI 304					2.78	16.9	3.68	22.4			0.93	18.6
	AISI 430					2.70	16.5	3.57	21.8			0.902	18.0

STAINLESS STEEL WELDED TUBINGS

*SWG = Standard Wire Gauge
BWG = Birmingham Wire Gauge
SWG can almost invariably be
replaced by BWG as the wall of
BWG is slightly thicker.

INSTRUMENTATION TUBE

Outside Diameter	Wall Thickness	Nominal* Size	Weight	3R12 SS 2352 AISI 304L	3R60 2353 316L	8R30 2337 321	8R70 2350 (316Ti)
mm	mm		kg/m				
6	1.0 1.5		0.125 0.169	•	•	•	•
6.35	0.89 0.91 1.22 1.24	1/4" X 20 BWG 20 SWG 18 SWG 18 BWG	0.122 0.124 0.157 0.159		• • • •		
8	1.0 1.5 2.0		0.175 0.244 0.300	• • •	• • •	• • •	• • •
9.53	0.89 0.91 1.22 1.24 1.63 1.65	3/8" x 20 BWG 20 SWG 18 SWG 18 BWG 16 SWG 16 BWG	0.193 0.196 0.254 0.257 0.322 0.326		• • • • • •		
10	1.0 1.5 2.0		0.225 0.319 0.400	• • •	• • •	• • •	• • •
10.2	2.0		0.411	•	•		
12	1.0 1.5 2.0		0.275 0.394 0.500	• • •	• • •	• • •	• • •
12.7	1.22 1.24 1.63 1.65	1/2" X 18 SWG 18 BWG 16 SWG 16 BWG	0.350 0.356 0.452 0.456		• • • •		
14	1.0 1.5 2.0		0.325 0.469 0.600	• • •	• • •	• • •	• • •
15	1.0 1.5 2.0		0.350 0.507 0.651	• • •	• • •	• • •	• • •
15.88	1.22 1.24	5/8" X 18 SWG 18 BWG	0.448 0.454		• •		
16	1.0 1.5 2.0 2.5 3.0		0.375 0.544 0.701 0.845 0.976	• • • • •	• • • • •	• • • • •	• • • • •
17.2	1.6 2.9		0.625 1.038	• •	• •	• •	• •
18	1.0 1.5 2.0 2.5 3.0		0.425 0.619 0.801 0.970 1.126	• • • • •	• • • • •	• • • • •	• • • • •
19.05	1.22 1.24	3/4" X 18 SWG 18 BWG	0.544 0.553		• •		
20	1.5 2.0 2.5 3.0 4.0		0.694 0.901 1.095 1.28 1.60	• • • • •	• • • • •	• • • • •	• • • • •

TOLERANCES

Metric sizes acc. to ISO 1127

Outside Diameter	Wall Thickness
±0.75%, but min. ±0.30 mm (D3)	±10%, but min. ±0.2 mm (T3)

Tube in imperial sizes acc. to ASTM A213/A450

Size	Outside Diameter	Wall Thickness average wall*	min wall
mm	mm	%	%
<25.4	+0.10/-0.10	+ 10/-10	+ 20/0

* Valid for stocked sizes.

SPECIFICATIONS

3R12 (AISI 304L) and 3R60 (AISI 316L)
SS 219711 (condition 22, i.e. 3R12=SS 2352-22)
DIN 17458, testing level 1
NFA 49-117
8R30 (AISI 321) and 8R70 (AISI 316Ti)
DIN 17458, testing level 1
Imperial size
3R60 (AISI 316L) ASTM A213, A269

STAINLESS STEEL PIPES

Symbol		Chemical Composition (%)								
JIS	ASTM	C	Si	Mn	P	S	Ni	Cr	Mo	Others
SUS 304TP SUS 304TB		≤0.08	≤1.00	≤2.00	≤0.04	≤0.03	8.00~ 11.00	18.00~ 20.00	--	
	TP 304	≤0.08	≤0.75	≤2.00	≤0.04	≤0.03	8.00~ 11.00	18.00~ 20.00	--	
SUS 304LTP SUS 304LTB		≤0.03	≤1.00	≤2.00	≤0.04	≤0.03	9.00~ 13.00	18.00~ 20.00	--	
	TP 304L	≤0.035*	≤0.75	≤2.00	≤0.04	≤0.03	8.00~ 13.00	18.00 ~ 20.00	--	
SUS 321TP SUS 321TB		≤0.08	≤1.00	≤2.00	≤0.04	≤0.03	9.00~ 13.00	17.00~ 19.00	--	Ti 5 x C% min.
	TP 321	≤0.08	≤0.75	≤2.00	≤0.04	≤0.03	9.00~ 13.00	17.00~ 19.00	--	Ti 5 x C% ~ 0.60
SUS 316TP SUS 316TB		≤0.08	≤1.00	≤2.00	≤0.04	≤0.03	10.00~ 14.00	16.00~ 18.00	2.00~ 3.00	
	TP 316	≤0.08	≤0.75	≤2.00	≤0.04	≤0.03	11.00~ 14.00	16.00~ 18.00	2.00~ 3.00	
SUS 316LTP SUS 316LTB		≤0.03	≤1.00	≤2.00	≤0.04	≤0.03	12.00~ 16.00	16.00~ 18.00	2.00~ 3.00	
	TP 316L	≤0.035*	≤0.75	≤2.00	≤0.04	≤0.03	10.00~ 15.00	16.00~ 18.00	2.00~ 3.00	
SUS 309STP SUS 309STB		≤0.15	≤1.00	≤2.00	≤0.04	≤0.03	12.00~ 15.00	22.00~ 24.00	--	
	TP 309	≤0.15	≤0.75	≤2.00	≤0.04	≤0.03	12.00~ 15.00	22.00~ 24.00	--	
SUS 310STP SUS 310STB		≤0.15	≤1.50	≤2.00	≤0.04	≤0.03	19.00~ 22.00	24.00~ 26.00	--	
	TP 310	≤0.15	≤0.75	≤2.00	≤0.04	≤0.03	19.00~ 22.00	24.00~ 26.00	--	
SUS 321HTP SUS 321HTB		0.04 ~ 0.10	≤0.75	≤2.00	≤0.03	≤0.03	9.00~ 13.00	17.00~ 20.00	--	Ti 4 x C% ~ 0.60
	TP 321H	0.04 ~ 0.10	≤0.75	≤2.00	≤0.04	≤0.03	9.00~ 13.00	17.00~ 20.00	--	Ti 4 x C% ~ 0.60

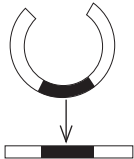
Note:

- The above chemical composition shall be applied even when product analysis is requested by the purchasers. However, the carbon content of SUS 340LTP/LTB and SUS 316 LTP/LTB shall be 0.035% or less.
- Those with an asterisk (*) having either a small diameter (i.e., outside diameter less than 0.5 in or 12.70 mm) or a thin wall (i.e., average wall thickness less than 0.049 in or 12.40 mm) or minimum wall thickness (less than 0.044 in or 1.12 mm) and requiring several drawings, shall have carbon content of 0.04% or less.

STAINLESS STEEL PIPES

All stainless steel pipes and tubes for piping, boilers and heat exchangers subjected to solution heat treatment as follows.

Symbol		Solution Heat Treatment	
For piping	For boilers and heat exchangers	Heat temperature (°C)	Cooling method
SUS 304 TP	SUS 304 TB	1010°C min.	Rapid cooling
SUS 304L TP	SUS 304L TB	1010°C min.	Rapid cooling
SUS 321 TP	SUS 321 TB	920°C min.	Rapid cooling
SUS 316 TP	SUS 316 TB	1010°C min.	Rapid cooling
SUS 316L TP	SUS 316L TB	1010°C min.	Rapid cooling
SUS 309S TP	SUS 309S TB	1030°C min.	Rapid cooling
SUS 310S TP	SUS 310S TB	1030°C min.	Rapid cooling
SUS 321H TP	SUS 321H TB	1095°C min.	Rapid cooling
TP 304		1040°C min.	Rapid cooling
TP 304L		1040°C min.	Rapid cooling
TP 321		1040°C min.	Rapid cooling
TP 316		1040°C min.	Rapid cooling
TP 316L		1040°C min.	Rapid cooling
TP 309		1040°C min.	Rapid cooling
TP 310		1040°C min.	Rapid cooling
TP 321H		1095°C min.	Rapid cooling

Symbol	Tension test				Flattening test	Flaring test	Reverse flattening test
	Tensile strength (kg/mm ²)	Yield strength (kg/mm ²)	Elongation (%)				
			Longitudinal	Transverse			
SUS 304TP/TB					$H = \frac{(1 + e) t}{e + \frac{t}{D}}$ where, H = Distance between flattening plates (mm) t = Wall thickness (mm) D = Outside diameter (mm) (mm) e = 0.09 (constant) Note: Welded section shall be affixed at an angle 90° to the direction of compression.	1.2 D (D = Outside diameter (mm))	 When flattened reversely, no cracks or breaks shall occur on its welded section.
TP 304	53 min.	21 min.	35 min.	25 min.			
SUS 304LTP/LTB							
TP 304L	49 min.	18 min.	35 min.	25 min.			
SUS 321TP/TB							
TP 321	53 min.	21 min.	35 min.	25 min.			
SUS 316TP/TB							
TP 316	53 min.	21 min.	35 min.	25 min.			
SUS 316LTP/LTB							
TP 316L	49 min.	28 min.	35 min.	25 min.			
SUS 309STP/STB							
TP 309	53 min.	21 min.	35 min.	25 min.			
SUS 310STP/STB							
TP 310S	53 min.	21 min.	35 min.	25 min.			
SUS 321HTP/HTB							
TP 321H	53 min.	21 min.	35 min.	25 min.			

Notes:

1. Tensile strength shall not be measured unless specifically requested.
2. Flaring test and reverse flattening test shall be conducted only on tubes for boilers and heat exchanger

STAINLESS STEEL PIPES

JIS	AISI	Characteristics	Uses
SUS 304	TP 304	Most widely used stainless steel. Because of its nickel content, it has excellent corrosion and heat resistance, strength under low temperature and improved mechanical properties. It has work hardening characteristic, and hardening by heat treatment is impossible. Has no magnetism.	Home appliances and kitchen utensils, architectural ornaments, vehicles, auto parts, medical equipment, and various applications in the food, chemical, and textile industries.
SUS 304L	TP 304L	Ni-Cr steel with extremely low carbon content. Although its degree of corrosion resistance under normal conditions is similar to that of SUS 304, it has greater resistance against intergranular corrosion after welding or stress-relief heat treatment. Has no magnetism.	Parts and structures in the chemical, petroleum, coal and pharmaceutical industries to which heat treatment is difficult to apply. Normally used at temperatures below 400°C.
SUS 321	TP 321	Excellent corrosion resistance. Other properties are similar to those of SUS 304. The resistance to intergranular corrosion is improved by addition of 18-8 type. It has no magnetism, and it is especially suited for use at temperatures between 430°C - 900°C. Though having no magnetism in annealed state, it becomes slightly magnetized by cold working.	Exhaust pipes of aircraft engines, boilers, jet engine parts, heating furnace parts, and other parts used in the chemical industry to which heat treatment after welding is difficult to apply.
SUS 316	TP 316	Substantial amounts of nickel and chromium content greatly improve heat and corrosion resistance. Work hardening characteristic but not magnetism.	Various uses in the chemical, food, photographic, textile, paper and pulp industries. Especially suited for exterior of structures located near coastal areas.
SUS 316L	TP 316L	Nickel-chromium steel containing molybdenum and an extremely small amount of carbon. Physical properties are similar to those of SUS 316. Excellent resistance against intergranular corrosion after welding or stress-relief heat treatment. Has no magnetism.	Typical applications similar to those of SUS 316. For machine parts and equipment which are not easily subjected to heat treatment after welding. Especially suited for use at temperatures below 420°C
SUS 309S SUS 310S	TP 309 TP 310	Substantial amount of nickel and chromium content greatly improve heat and corrosion resistance.	Jet engine parts, tanks for chemicals, combustion apparatus parts, boilers, and gasturbine parts.

STAINLESS STEEL PIPES

SEAMLESS/WELDED (Dimension and Weight)

ASTM A312TP 304/304L/316/316L/321/317/347/310, ETC. ANSI-B36.19

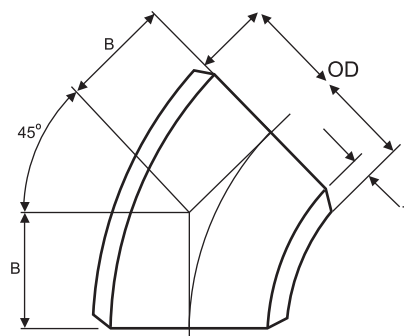
N.B.	Outside Diameter		SCH5S			SCH10S		
			Thick	Thick	Weight	Thick	Thick	Weight
	inch	mm	mm	mm	kg/m	inch	mm	kg/m
1/8"	0.405	10.29	--	--	--	0.049	1.24	0.277
1/4"	0.54	13.72	--	--	--	0.065	1.65	0.491
3/8"	0.675	17.15	--	--	--	0.065	1.65	0.631
1/2"	0.84	21.34	0.065	1.65	0.801	0.083	2.11	1.00
3/4"	1.05	26.67	0.065	1.65	1.02	0.083	2.11	1.28
1"	1.315	33.40	0.065	1.65	1.29	0.109	2.77	2.09
1 1/4"	1.66	42.16	0.065	1.65	1.65	0.109	2.77	2.69
1 1/2"	1.90	48.26	0.065	1.65	1.90	0.109	2.77	3.11
2"	2.375	60.33	0.065	1.65	2.39	0.109	2.77	3.93
2 1/2"	2.875	73.03	0.083	2.11	3.69	0.12	3.05	5.26
3"	3.50	88.90	0.083	2.11	4.52	0.12	3.05	6.46
3 1/2"	4.00	101.60	0.083	2.11	5.18	0.12	3.05	7.41
4"	4.50	114.30	0.083	2.11	5.84	0.12	3.05	8.37
5"	5.563	141.30	0.109	2.71	9.46	0.134	3.40	11.60
6"	6.625	168.28	0.109	2.77	11.30	0.134	3.40	13.80
8"	8.625	219.08	0.109	2.77	14.80	0.148	3.76	20.00
10"	10.75	273.05	0.134	3.40	22.60	0.165	4.19	27.80
12"	12.75	323.85	0.156	3.96	31.20	0.18	4.57	36.00
14"	14.00	355.60	0.156	3.96	34.30	0.188	4.78	41.40
16"	16.00	406.40	0.165	4.19	41.60	0.188	4.78	47.30
18"	18.00	457.20	0.165	4.19	46.80	0.188	4.78	53.30
20"	20.00	508.00	0.188	4.78	59.30	0.218	5.54	68.60
22"	22.00	558.80	0.188	4.78	65.30	0.218	5.54	75.60
24"	24.00	609.60	0.218	5.54	82.50	0.25	6.36	94.50
30"	30.00	762.00	0.25	6.36	118.30	0.312	7.92	147.30

ASTM A312 TP 304/304L/316/316L/321/317/347/310, ETC. ANSI-B36.19

N.B.	Outside Diameter		SCH40S			SCH80S		
			Thick	Thick	Weight	Thick	Thick	Weight
	inch	mm	mm	mm	kg/m	inch	mm	kg/m
1/8"	0.405	10.29	0.068	1.73	0.365	0.095	2.41	0.468
1/4"	0.54	13.72	0.088	2.24	0.634	0.119	3.02	0.797
3/8"	0.675	17.15	0.091	2.31	0.845	0.126	3.20	1.10
1/2"	0.84	21.34	0.109	2.77	1.27	0.147	3.73	1.62
3/4"	1.05	26.67	0.113	2.87	1.68	0.154	3.91	2.19
1"	1.315	33.40	0.133	3.38	2.50	0.179	4.55	3.42
1 1/4"	1.66	42.16	0.14	3.56	3.39	0.191	4.85	4.46
1 1/2"	1.90	48.26	0.146	3.68	4.05	0.20	5.08	5.41
2"	2.375	60.33	0.154	3.91	5.44	0.218	5.54	7.49
2 1/2"	2.875	73.03	0.203	5.61	8.64	0.276	7.01	11.40
3"	3.50	88.90	0.216	5.49	11.30	0.30	7.62	15.30
3 1/2"	4.00	101.60	0.226	5.74	13.60	0.318	8.08	18.60
4"	4.50	114.30	0.237	6.02	16.10	0.337	8.56	22.30
5"	5.563	141.30	0.258	6.55	21.80	0.375	9.25	30.90
6"	6.625	168.28	0.28	7.11	28.30	0.432	10.97	42.60
8"	8.625	219.08	0.322	8.18	42.50	0.50	12.70	64.60
10"	10.75	273.05	0.365	9.27	60.30	0.50	12.70	81.50
12"	12.75	323.85	0.375	9.52	73.80	0.50	12.70	97.40
14"	14.00	355.60	--	--	--	--	--	--
16"	16.00	406.40	--	--	--	--	--	--
18"	18.00	457.20	--	--	--	--	--	--
20"	20.00	508.00	--	--	--	--	--	--
22"	22.00	558.80	--	--	--	--	--	--
24"	24.00	609.60	--	--	--	--	--	--
30"	30.00	762.00	--	--	--	--	--	--

STAINLESS STEEL FITTINGS

FITTINGS

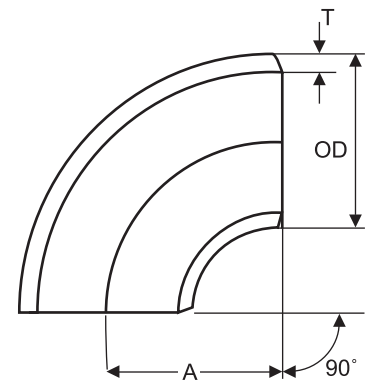


45° LONG RADIUS ELBOWS

Dimensions						Unit: inch
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Center to End B
		Sch.5S	Sch.10S	Sch.40S	Sch.80S	
1/2	0.840	0.065	0.083	0.109	0.147	0.625
3/4	1.050	0.065	0.083	0.113	0.154	0.438
1	1.315	0.065	0.109	0.133	0.179	0.875
1 1/4	1.660	0.065	0.109	0.140	0.191	1.00
1 1/2	1.900	0.065	0.109	0.145	0.200	1.12
2	2.375	0.065	0.109	0.154	0.218	1.38
2 1/2	2.875	0.083	0.120	0.203	0.276	1.75
3	3.500	0.083	0.120	0.216	0.300	2.00
3 1/2	4.000	0.083	0.120	0.226	0.318	2.25
4	4.500	0.083	0.120	0.237	0.337	2.50
5	5.563	0.109	0.134	0.258	0.375	3.12
6	6.625	0.109	0.134	0.280	0.432	3.75
8	8.625	0.109	0.148	0.322	0.500	5.00
10	10.750	0.134	0.165	0.365	0.500	6.25
12	12.750	0.156	0.180	0.375	0.500	7.50
14	14.000	0.250	0.312	0.375	0.500	8.75
16	16.000	0.250	0.312	0.375	0.500	10.00
18	18.000	0.250	0.312	0.375	0.500	11.25
20	20.000	0.250	0.375	0.375	0.500	12.50
22	22.000	0.250	0.375	0.375	0.500	13.50
24	24.000	0.250	0.375	0.375	0.500	15.00
26	26.000	0.312	0.500	0.375	0.500	16.00
28	28.000	0.312	0.500	0.375	0.500	17.25
30	30.000	0.312	0.500	0.375	0.500	18.50
32	32.000	0.312	0.500	0.375	0.500	19.75
34	34.000	0.312	0.500	0.375	0.500	21.00
36	36.000	0.312	0.500	0.375	0.500	22.25
38	38.000			0.375	0.500	23.62
40	40.000			0.375	0.500	24.88
42	42.000			0.375	0.500	26.00
44	44.000			0.375	0.500	27.38
46	47.000			0.375	0.500	28.62
48	48.000			0.375	0.500	29.88

Weights					Unit: kg
Nominal Pipe Size	Sch.5S	Sch.10S	Sch.40S	Sch.80S	
1/2	0.024	0.030	0.039	0.050	
3/4	0.031	0.039	0.052	0.078	
1	0.039	0.064	0.077	0.100	
1 1/4	0.063	0.103	0.130	0.177	
1 1/2	0.085	0.142	0.184	0.254	
2	0.143	0.239	0.326	0.471	
2 1/2	0.288	0.406	0.683	0.897	
3	0.405	0.571	1.02	1.44	
3 1/2	0.540	0.764	1.41	2.05	
4	0.696	0.985	1.92	2.81	
5	1.42	1.71	3.24	4.85	
6	2.01	2.43	4.97	8.02	
8	3.53	5.01	10.10	16.22	
10	6.62	7.77	17.70	25.37	
12	11.10	12.50	28.10	36.69	
14	23.33	28.63	32.93	42.40	
16	30.44	34.84	43.31	55.76	
18	38.10	44.21	54.99	70.94	
20	48.11	68.13	68.13	88.02	
22	57.26	81.13	81.13	104.87	
24	68.95	98.57	98.57	127.52	
26	91.78	147.63	114.11	147.68	
28	106.68	171.82	132.73	171.82	
30	122.72	197.78	152.75	197.78	
32	139.89	225.59	156.24	225.59	
34	158.14	255.22	196.96	255.13	
36	177.58	286.66	221.20	286.57	
38			248.15	321.58	
40			257.35	356.69	
42			301.34	391.85	
44			333.68	432.66	
46			365.07	473.25	
48			397.69	515.74	

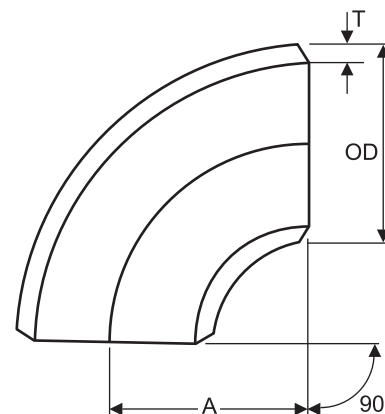
STAINLESS STEEL FITTINGS



90° LONG RADIUS ELBOWS

Dimensions							Unit: inch				
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Center to End A	Unit: kg				
		Sch.5S	Sch.10S	Sch.40S	Sch.80S		Nominal Pipe Size	Sch.5S	Sch.10S	Sch.40S	Sch.80S
1/2	0.840	0.065	0.083	0.109	0.147	1.50	1/2	0.049	0.061	0.078	0.100
3/4	1.050	0.065	0.083	0.113	0.154	1.12	3/4	0.062	0.078	0.103	0.104
1	1.315	0.065	0.109	0.133	0.179	1.50	1	0.078	0.128	0.154	0.204
1 1/4	1.660	0.065	0.109	0.140	0.191	1.88	1 1/4	0.125	0.206	0.260	0.349
1 1/2	1.900	0.065	0.109	0.145	0.200	2.25	1 1/2	0.170	0.283	0.368	0.507
2	2.375	0.065	0.109	0.154	0.218	3.00	2	0.286	0.478	0.652	0.938
2 1/2	2.875	0.083	0.120	0.203	0.276	3.75	2 1/2	0.576	0.812	1.37	1.79
3	3.500	0.083	0.120	0.216	0.300	4.50	3	0.810	1.14	2.04	2.87
3 1/2	4.000	0.083	0.120	0.226	0.318	5.25	3 1/2	1.08	1.53	2.82	4.10
4	4.500	0.083	0.120	0.237	0.337	6.00	4	1.39	1.97	3.84	5.62
5	5.563	0.109	0.134	0.258	0.375	7.50	5	2.83	3.42	6.48	9.69
6	6.625	0.109	0.134	0.280	0.432	9.00	6	4.03	4.87	9.94	15.99
8	8.625	0.109	0.148	0.322	0.500	12.00	8	7.06	10.00	20.10	32.16
10	10.750	0.134	0.165	0.365	0.500	15.00	10	13.20	15.50	35.40	49.83
12	12.750	0.156	0.180	0.375	0.500	18.00	12	22.30	25.90	56.20	72.48
14	14.000	0.250	0.312	0.375	0.500	21	14	46.98	57.58	66.36	85.53
16	16.000	0.250	0.312	0.375	0.500	24	16	61.56	70.03	87.11	112.34
18	18.000	0.250	0.312	0.375	0.500	27	18	78.14	89.01	110.67	142.79
20	20.000	0.250	0.375	0.375	0.500	30	20	96.67	136.99	136.99	176.90
22	22.000	0.250	0.375	0.375	0.500	33	22	114.93	163.03	163.03	210.83
24	24.000	0.250	0.375	0.375	0.500	36	24	139.57	198.05	198.05	256.13
26	26.000	0.312	0.500	0.375	0.500	39	26	184.10	296.76	229.17	296.53
28	28.000	0.312	0.500	0.375	0.500	42	28	214.04	345.05	266.50	344.96
30	30.000	0.312	0.500	0.375	0.500	45	30	246.16	397.15	306.59	396.74
32	32.000	0.312	0.500	0.375	0.500	48	32	280.59	453.00	349.53	452.68
34	34.000	0.312	0.500	0.375	0.500	51	34	317.19	512.16	395.24	511.98
36	36.000	0.312	0.500	0.375	0.500	54	36	356.01	575.13	443.71	574.95
38	38.000			0.375	0.500	57	38			497.71	644.75
40	40.000			0.375	0.500	60	40			551.98	715.51
42	42.000			0.375	0.500	63	42			606.16	785.77
44	44.000			0.375	0.500	66	44			669.04	867.40
46	46.000			0.375	0.500	69	46			731.69	948.81
48	48.000			0.375	0.500	72	48			797.28	1033.84

STAINLESS STEEL FITTINGS



90° SHORT RADIUS ELBOWS (24" and Smaller)

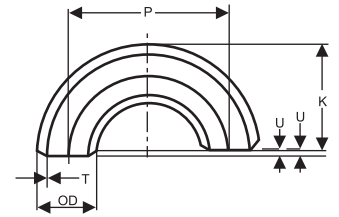
Dimensions						Unit: inch
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Center to End A
		Sch.5S	Sch.10S	Sch.40S	Sch.80S	
1	1.315	0.065	0.109	0.133	0.179	1.00
1 ¹ / ₄	1.660	0.065	0.109	0.140	0.191	1.25
1 ¹ / ₂	1.900	0.065	0.109	0.145	0.200	1.50
2	2.375	0.065	0.109	0.154	0.218	2.00
2 ¹ / ₂	2.875	0.083	0.120	0.203	0.276	2.50
3	3.500	0.083	0.120	0.216	0.300	3.00
3 ¹ / ₂	4.000	0.083	0.120	0.226	0.318	3.50
4	4.500	0.083	0.120	0.237	0.337	4.00
5	5.563	0.109	0.134	0.258	0.375	5.00
6	6.625	0.109	0.134	0.280	0.432	6.00
8	8.625	0.109	0.148	0.322	0.500	8.00
10	10.750	0.134	0.165	0.365	0.500	10.00
12	12.750	0.156	0.180	0.375	0.500	12.00
14	14.000	0.156	0.188	*	**	14.00
16	16.000	0.156	0.188	*	**	16.00
18	18.000	0.156	0.188	*	**	18.00
20	20.000	0.188	0.218	*	**	20.00
22	22.000	0.188	0.218	*	**	22.00
24	24.000	0.218	0.250	*	**	24.00

Weights					Unit: kg
Nominal Pipe Size	Sch.5S	Sch.10S	Sch.40S	Sch.80S	
1	0.077	0.094	0.118	0.122	
1 ¹ / ₄	0.101	0.152	0.204	0.217	
1 ¹ / ₂	0.114	0.189	0.245	0.317	
2	0.191	0.318	0.434	0.593	
2 ¹ / ₂	0.383	0.541	0.909	1.13	
3	0.540	0.761	1.36	1.82	
3 ¹ / ₂	0.719	1.02	1.88	2.61	
4	0.928	1.31	2.56	3.56	
5	1.89	2.28	4.32	6.20	
6	2.68	3.24	6.63	10.26	
8	4.71	6.68	13.40	20.84	
10	8.83	10.40	23.60	32.94	
12	14.80	16.70	37.50	47.30	
14	15.21	24.55	48.33*	59.25**	
16	25.60	32.00	63.37*	81.81**	
18	40.40	40.40	73.57*	103.96**	
20	50.42	60.25	99.66*	128.79**	
22	61.06	73.02	120.83*	156.29**	
24	86.98	101.56	144.06*	186.41**	

* std = 0.375

**XS = 0.500

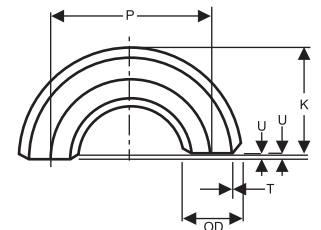
STAINLESS STEEL FITTINGS



180° LONG RADIUS RETURNS (12" and Smaller)

Dimensions		Unit: inch					
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Center to Center P	Back to Face K
		Sch.5S	Sch.10S	Sch.40S	Sch.80S		
1/2	0.840	0.065	0.083	0.109	0.147	3.00	1.88
3/4	1.050	0.065	0.083	0.113	0.154	2.25	1.69
1	1.315	0.065	0.109	0.133	0.179	3.00	2.19
1 1/4	1.660	0.065	0.109	0.140	0.191	3.75	2.75
1 1/2	1.900	0.065	0.109	0.145	0.200	4.50	3.25
2	2.375	0.065	0.109	0.154	0.218	6.00	4.19
2 1/2	2.875	0.083	0.120	0.203	0.276	7.50	5.19
3	3.500	0.083	0.120	0.216	0.300	9.00	6.25
3 1/2	4.000	0.083	0.120	0.226	0.318	10.50	7.25
4	4.500	0.083	0.120	0.237	0.337	12.00	8.25
5	5.563	0.109	0.134	0.258	0.375	15.00	10.31
6	6.625	0.109	0.134	0.280	0.432	18.00	12.31
8	8.625	0.109	0.148	0.322	0.500	24.00	16.31
10	10.750	0.134	0.165	0.365	0.500	30.00	20.38
12	12.750	0.156	0.180	0.375	0.500	36.00	24.38

Weights		Unit: kg			
Nominal Pipe Size		Sch.5S	Sch.10S	Sch.40S	Sch.80S
1/2		0.091	0.118	0.156	0.199
3/4		0.091	0.118	0.207	0.208
1		0.163	0.268	0.307	0.408
1 1/4		0.227	0.426	0.519	0.698
1 1/2		0.327	0.580	0.735	1.02
2		0.562	0.989	1.30	1.88
2 1/2		1.097	1.651	2.73	3.56
3		1.578	2.431	4.07	5.74
3 1/2		2.113	3.256	5.65	12.72
4		2.730	4.190	7.67	15.76
5		2.533	7.256	13.00	19.39
6		8.118	10.431	19.90	31.98
8		13.968	20.135	40.30	64.33
10		26.530	35.010	70.80	99.66
12		47.164	53.966	111.44	144.96

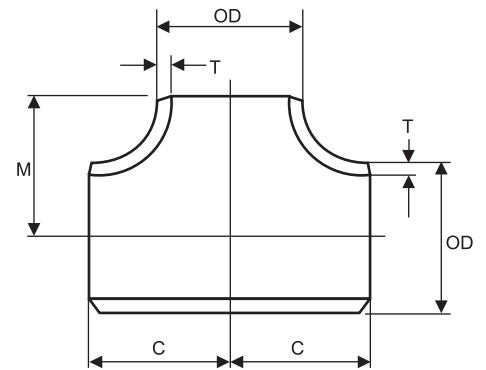


180° SHORT RADIUS RETURNS (12" and Smaller)

Dimensions		Unit: inch					
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Center to Center P	Back to Face K
		Sch.5S	Sch.10S	Sch.40S	Sch.80S		
1	1.315	0.065	0.109	0.133	0.179	2.00	1.63
1 1/4	1.660	0.065	0.109	0.140	0.191	2.50	2.06
1 1/2	1.900	0.065	0.109	0.145	0.200	3.00	2.44
2	2.375	0.065	0.109	0.154	0.218	4.00	3.19
2 1/2	2.875	0.083	0.120	0.203	0.276	5.00	3.94
3	3.500	0.083	0.120	0.216	0.300	6.00	4.75
3 1/2	4.000	0.083	0.120	0.226	0.318	7.00	5.50
4	4.500	0.083	0.120	0.237	0.337	8.00	6.25
5	5.563	0.109	0.134	0.258	0.375	10.00	7.75
6	6.625	0.109	0.134	0.280	0.432	12.00	9.31
8	8.625	0.109	0.148	0.322	0.500	16.00	12.31
10	10.750	0.134	0.165	0.365	0.500	20.00	15.38
12	12.750	0.156	0.180	0.375	0.500	24.00	18.38

Weights		Unit: kg			
Nominal Pipe Size		Sch.5S	Sch.10S	Sch.40S	Sch.80S
1		0.104	0.168	0.205	0.262
1 1/4		0.168	0.272	0.347	0.456
1 1/2		0.231	0.376	0.490	0.655
2		0.390	0.634	0.869	1.19
2 1/2		0.747	1.07	1.82	2.38
3		1.10	1.57	2.71	3.65
3 1/2		1.47	2.10	3.77	5.21
4		1.89	2.71	5.11	7.15
5		3.84	4.69	8.64	12.20
6		5.50	6.74	13.30	20.00
8		9.60	12.97	26.80	40.70
10		18.39	22.57	47.20	74.90
12		30.49	35.10	71.94	94.91

STAINLESS STEEL FITTINGS

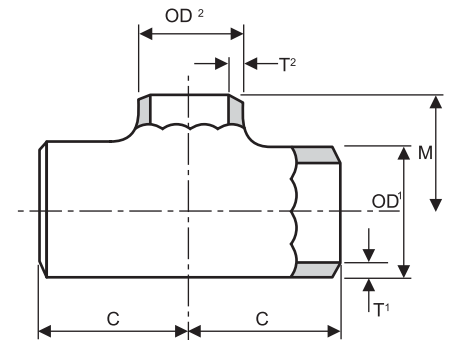


STRAIGHT TEES (24" and Smaller)

Dimensions		Unit: inch					
Nominal Pipe Size		Outside Dia. (OD)	Wall Thickness (T)				Center to End C, M
Run	Outlet		Sch.5S	Sch.10S	Sch.40S	Sch.80S	
1/2	1/2	0.840	0.065	0.083	0.109	0.147	1.00
3/4	3/4	1.050	0.065	0.083	0.113	0.154	1.13
1	1	1.315	0.065	0.109	0.133	0.179	1.50
1 1/4	1 1/4	1.660	0.065	0.109	0.140	0.191	1.88
1 1/2	1 1/2	1.900	0.065	0.109	0.145	0.200	2.25
2	2	2.375	0.065	0.109	0.154	0.218	2.50
2 1/2	2 1/2	2.875	0.083	0.120	0.203	0.276	3.00
3	3	3.500	0.083	0.120	0.216	0.300	3.38
4	4	4.500	0.083	0.120	0.237	0.337	4.13
5	5	5.563	0.109	0.134	0.258	0.375	4.88
6	6	6.625	0.109	0.134	0.280	0.432	5.63
8	8	8.625	0.109	0.148	0.322	0.500	7.00
10	10	10.750	0.134	0.165	0.365	0.500	8.50
12	12	12.750	0.156	0.180	0.375	0.500	10.00
14	14	14.000	0.156	0.188	*	**	11.00
16	16	16.000	0.165	0.188	*	**	12.00
18	18	18.000	0.165	0.188	*	**	13.50
20	20	20.000	0.188	0.218	*	**	15.00
22	22	22.000	0.188	0.218	*	**	16.50
24	24	25.000	0.218	0.250	*	**	17.00

Weights					Unit: kg	
Nominal Pipe Size	Sch.5S	Sch.10S	Sch.40S	Sch.80S		
1/2	0.090	0.090	0.181	0.204		
3/4	0.129	0.129	0.258	0.294		
1	0.223	0.223	0.256	0.399		
1 1/4	0.356	0.356	0.463	0.725		
1 1/2	0.497	0.497	0.648	1.02		
2	0.681	0.681	0.893	1.81		
2 1/2	1.20	1.20	1.98	3.17		
3	1.57	1.57	2.61	3.85		
4	2.46	2.46	4.76	7.11		
5	4.04	4.04	7.84	11.78		
6	5.50	5.50	9.54	18.12		
8	11.07	11.07	19.56	33.98		
10	19.87	19.87	32.61	47.57		
12	27.86	27.86	56.65	2.48		
14	28.74	35.98	71.21*	91.78**		
16	35.69	44.69	88.56*	114.23**		
18	45.21	56.62	112.33*	144.98**		
20	69.97	83.69	138.99*	179.66**		
22	84.72	101.35	170.07*	217.46**		
24	113.23	132.24	192.66*	243.13**		

STAINLESS STEEL FITTINGS



REDUCING OUTLET TEES (24" and Smaller)

Dimensions												Unit: inch	
Nominal Pipe Size		Out-side	Out-side	Wall Thickness (T 1, T 2)								Center to End	Center to End
Run	Out-let	Dia.	Dia.	Sch.5S		Sch.10S		Sch.40S		Sch.80S		C	M
		OD ₁	OD ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂		
3/4	1/2	1.050	0.840	0.065	0.065	0.083	0.083	0.113	0.109	0.154	0.147	1.13	1.13
1	1/2	1.310	0.840	0.065	0.065	0.109	0.083	0.133	0.109	0.179	0.147	1.50	1.50
1	3/4	1.315	1.050	0.065	0.065	0.109	0.083	0.133	0.113	0.179	0.154	1.50	1.50
1 1/4	3/4	1.660	1.050	0.065	0.065	0.109	0.083	0.140	0.113	0.191	0.154	1.88	1.88
1 1/4	1	1.660	1.315	0.065	0.065	0.109	0.109	0.140	0.133	0.191	0.179	1.88	1.88
1 1/2	1/2	1.900	0.840	0.065	0.065	0.109	0.083	0.145	0.109	0.200	0.147	2.25	2.25
1 1/2	3/4	1.900	1.050	0.065	0.065	0.109	0.083	0.145	0.113	0.200	0.154	2.25	2.25
1 1/2	1	1.900	1.315	0.065	0.065	0.109	0.109	0.145	0.133	0.200	0.179	2.25	2.25
1 1/2	1 1/4	1.900	1.660	0.065	0.065	0.109	0.109	0.145	0.140	0.200	0.191	2.25	2.25
2	3/4	2.375	1.050	0.065	0.065	0.109	0.033	0.154	0.113	0.218	0.154	2.50	1.75
2	1	2.375	1.315	0.065	0.065	0.109	0.109	0.154	0.133	0.218	0.179	2.50	2.00
2	1 1/4	2.375	1.660	0.065	0.065	0.109	0.109	0.154	0.140	0.218	0.191	2.50	2.25
2	1 1/2	2.375	1.900	0.065	0.065	0.109	0.109	0.154	0.145	0.218	0.200	2.50	2.38
2 1/2	1 1/2	2.875	1.900	0.083	0.065	0.120	0.109	0.203	0.145	0.276	0.200	3.00	2.63
2 1/2	2	2.875	2.375	0.083	0.065	0.120	0.109	0.203	0.154	0.276	0.218	3.00	2.75
3	1 1/2	3.500	1.900	0.083	0.065	0.120	0.109	0.216	0.145	0.300	0.200	3.13	2.88
3	2	3.500	2.375	0.083	0.065	0.120	0.109	0.216	0.154	0.300	0.218	3.38	3.00
3	2 1/2	3.500	2.875	0.083	0.083	0.120	0.120	0.216	0.203	0.300	0.276	3.38	3.25
4	1 1/2	4.500	1.900	0.083	0.065	0.120	0.109	0.237	0.145	0.337	0.200	4.13	3.38
4	2	4.500	2.375	0.083	0.065	0.120	0.109	0.237	0.154	0.337	0.218	4.13	3.50
4	2 1/2	4.500	2.875	0.083	0.083	0.120	0.120	0.237	0.203	0.337	0.276	4.13	3.75
4	3	4.500	3.500	0.083	0.083	0.120	0.120	0.237	0.216	0.337	0.300	4.13	3.88
5	3	5.563	3.500	0.109	0.083	0.134	0.120	0.258	0.216	0.375	0.300	4.88	4.38
5	4	5.563	4.500	0.109	0.083	0.134	0.120	0.258	0.237	0.375	0.337	4.88	4.63
6	2 1/2	6.625	2.875	0.109	0.083	0.134	0.210	0.280	0.203	0.432	0.276	5.63	4.75
6	3	6.625	3.500	0.109	0.083	0.134	0.120	0.280	0.216	0.432	0.300	5.63	4.88
6	4	6.625	4.500	0.109	0.083	0.134	0.120	0.280	0.237	0.432	0.337	5.63	5.13
6	5	6.625	5.563	0.109	0.109	0.134	0.134	0.280	0.258	0.432	0.375	5.63	5.38

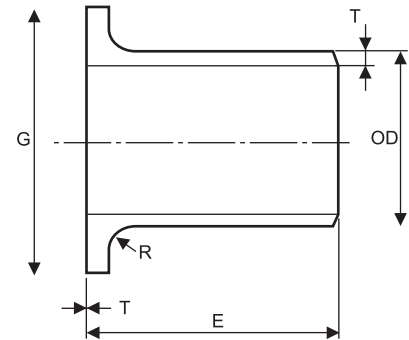
Weights						Unit: kg	
Nominal Pipe Size		Sch.5S	Sch.10S	Sch.40S	Sch.80S		
Run	Out-let						
3/4	1/2	0.123	0.123	0.123	0.186		
1	1/2	0.204	0.204	0.234	0.349		
1	3/4	0.212	0.212	0.244	0.353		
1 1/4	3/4	0.327	0.327	0.423	0.612		
1 1/4	1	0.339	0.339	0.441	0.643		
1 1/2	1/2	0.431	0.431	0.561	0.897		
1 1/2	3/4	0.446	0.446	0.578	0.901		
1 1/2	1	0.461	0.461	0.600	0.929		
1 1/2	1 1/4	0.482	0.482	0.629	0.978		
2	3/4	0.625	0.702	1.11	1.44		
2	1	0.643	0.725	1.14	1.46		
2	1 1/4	0.680	0.747	1.18	1.51		
2	1 1/2	0.734	0.766	1.23	1.57		
2 1/2	1 1/2	0.824	1.18	2.08	2.61		
2 1/2	2	0.838	1.21	2.12	2.66		
3	1 1/2	1.33	1.60	2.98	3.84		
3	2	1.42	1.63	3.01	3.86		
3	2 1/2	1.58	1.69	3.15	3.97		
4	1 1/2	2.45	2.51	5.04	6.63		
4	2	2.47	2.53	5.08	6.70		
4	2 1/2	2.52	2.59	5.22	6.79		
4	3	2.55	2.61	5.27	6.93		
5	3	3.38	4.17	8.13	10.95		
5	4	3.43	4.24	8.36	11.18		
6	2 1/2	4.62	5.69	11.87	17.05		
6	3	4.67	5.75	11.92	17.07		
6	4	4.75	5.85	12.10	17.19		
6	5	4.80	5.92	12.34	17.47		

STAINLESS STEEL FITTINGS

REDUCING OUTLET TEES (24" and Smaller)

Dimensions												Unit: inch		Weights						Unit: kg	
Nominal Pipe Size		Out-side	Out-side	Wall Thickness (T1, T2)								Center to End	Center to End	Nominal Pipe Size		Sch.5S	Sch.10S	Sch.40S	Sch.80S		
Run	Out-let	Dia.	Dia.	Sch.5S		Sch.10S		Sch.40S		Sch.80S		C	M	Run	Out-let						
		OD ₁	OD ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂										
8	4	8.625	4.500	0.109	0.083	0.148	0.120	0.322	0.237	0.500	0.337	7.00	6.13	8	4	7.35	9.80	21.26	30.76		
8	5	8.625	5.563	0.109	0.109	0.148	0.134	0.322	0.258	0.500	0.375	7.00	6.38	8	5	7.41	9.83	21.35	30.85		
8	6	8.625	6.625	0.109	0.109	0.148	0.134	0.322	0.280	0.500	0.432	7.00	6.63	8	6	7.44	9.90	21.62	31.25		
10	5	10.750	5.563	0.134	0.109	0.165	0.134	0.365	0.258	0.500	0.375	8.50	7.50	10	5	13.43	16.47	36.14	47.47		
10	6	10.750	6.625	0.134	0.109	0.165	0.134	0.365	0.280	0.500	0.432	8.50	7.63	10	6	13.49	16.58	36.27	47.95		
10	8	10.750	8.625	0.134	0.109	0.165	0.148	0.365	0.322	0.500	0.500	8.50	8.00	10	8	13.54	16.69	36.85	48.39		
12	6	12.750	6.625	0.156	0.109	0.180	0.134	0.375	0.280	0.500	0.432	10.00	8.63	12	6	22.93	25.10	51.96	67.48		
12	8	12.750	8.625	0.156	0.109	0.180	0.148	0.375	0.322	0.500	0.500	10.00	9.00	12	8	23.01	25.32	52.38	67.92		
12	10	12.750	10.750	0.156	0.104	0.180	0.165	0.375	0.365	0.500	0.500	10.00	9.50	12	10	23.13	25.48	53.72	68.81		
14	8	14.000	8.620	0.188	0.148	0.312	0.250	0.375	0.322	0.500	0.500	11.00	9.75	14	8	28.79	52.00	57.88	75.22		
14	10	14.000	10.750	0.188	0.165	0.312	0.250	0.375	0.365	0.500	0.500	11.00	10.13	14	10	28.96	52.37	58.34	75.89		
14	12	14.000	12.750	0.188	0.180	0.312	0.250	0.375	0.375	0.500	0.500	11.00	10.63	14	12	29.14	52.73	58.81	76.57		
16	10	16.000	10.750	0.188	0.165	0.312	0.250	0.375	0.365	0.500	0.500	12.00	11.13	16	10	34.26	61.89	68.88	89.51		
16	12	16.000	12.750	0.188	0.180	0.312	0.250	0.375	0.375	0.500	0.500	12.00	11.63	16	12	34.47	62.32	69.43	90.32		
16	14	16.000	14.000	0.188	0.188	0.312	0.312	0.375	0.375	0.500	0.500	12.00	12.00	16	14	34.67	62.76	69.98	91.13		
18	12	18.000	12.750	0.188	0.180	0.312	0.250	0.375	0.375	0.500	0.500	13.50	12.63	18	12	43.37	78.33	87.17	113.28		
18	14	18.000	14.000	0.188	0.188	0.312	0.312	0.375	0.375	0.500	0.500	13.50	13.00	18	14	43.62	78.88	87.87	114.30		
18	16	18.000	16.000	0.188	0.188	0.312	0.312	0.375	0.375	0.500	0.500	13.50	13.00	18	16	43.89	79.43	88.57	115.33		
20	14	20.000	14.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	15.00	14.00	20	14	64.47	107.63	107.63	139.86		
20	16	20.000	16.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	15.00	14.00	20	16	64.85	108.38	108.38	141.12		
20	18	20.000	18.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	15.00	14.50	20	18	65.24	109.14	109.14	142.39		
22	16	22.000	16.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	16.50	15.00	22	16	78.00	130.23	130.23	169.22		
22	18	22.000	18.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	16.50	15.50	22	18	78.47	131.14	131.14	170.74		
22	20	22.000	20.000	0.218	0.118	0.375	0.375	0.375	0.375	0.500	0.500	16.50	16.00	22	20	78.94	132.06	132.06	172.28		
24	16	24.000	16.000	0.250	0.188	0.375	0.312	0.375	0.375	0.500	0.500	17.00	16.00	24	16	96.84	138.24	138.24	179.64		
24	18	24.000	18.000	0.250	0.188	0.375	0.312	0.375	0.375	0.500	0.500	17.00	16.50	24	18	97.69	139.21	139.21	181.25		
24	20	24.000	20.000	0.250	0.218	0.375	0.375	0.375	0.375	0.500	0.500	17.00	17.00	24	20	98.00	140.18	140.18	182.89		
24	22	24.000	22.000	0.250	0.218	0.375	0.375	0.375	0.375	0.500	0.500	17.00	17.00	24	22	98.59	141.16	141.16	184.53		

STAINLESS STEEL FITTINGS



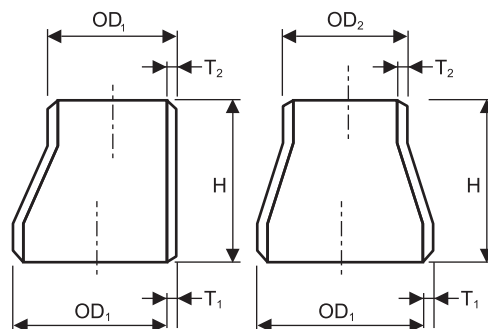
LAP-JOINT STUB ENDS (24" and Smaller)

Dimensions								Unit: inch		
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Dia of Lap (G)	Length F		R	
		Sch.5S	Sch.10S	Sch.40S	Sch.80S		ANSI	MSS	Type A (Max)	Type B (Max)
1/2	0.840	0.065	0.083	0.109	0.147	1.375	3.0	2.0	0.13	0.03
3/4	1.050	0.065	0.083	0.113	0.154	1.688	3.0	2.0	0.13	0.03
1	1.315	0.065	0.109	0.133	0.179	2.000	4.0	2.0	0.13	0.03
1 1/4	1.660	0.065	0.109	0.140	0.191	2.500	4.0	2.0	0.19	0.03
1 1/2	1.900	0.065	0.109	0.145	0.200	2.875	4.0	2.0	0.25	0.03
2	2.375	0.065	0.109	0.154	0.218	3.625	6.0	2.5	0.31	0.03
2 1/2	2.875	0.083	0.120	0.203	0.276	4.125	6.0	2.5	0.31	0.03
3	3.500	0.083	0.120	0.216	0.300	5.000	6.0	2.5	0.38	0.03
4	4.500	0.083	0.120	0.237	0.337	6.188	6.0	3.0	0.44	0.03
5	5.563	0.109	0.134	0.258	0.375	7.313	8.0	3.0	0.44	0.06
6	6.625	0.109	0.134	0.280	0.432	8.500	8.0	3.5	0.50	0.06
8	8.625	0.109	0.148	0.322	0.500	10.625	8.0	4.0	0.50	0.06
10	10.750	0.134	0.165	0.365	0.500	12.750	10.0	5.0	0.50	0.06
12	12.750	0.156	0.180	0.375	0.500	15.000	10.0	6.0	0.50	0.06
14	14.000	0.156	0.188	*	**	16.250	12.0	6.0	0.50	0.06
16	16.000	0.165	0.188	*	**	18.500	12.0	6.0	0.50	0.06
18	18.000	0.165	0.188	*	**	21.000	12.0	6.0	0.50	0.06
20	20.000	0.188	0.218	*	**	23.000	12.0	6.0	0.50	0.06
22	22.000	0.188	0.218	*	**	25.250	12.0	-	0.50	0.06
24	24.000	0.118	0.250	*	**	27.250	12.0	6.0	0.50	0.06

* Std 0.375 ** XS 0.500

Weights						Unit: kg	
Nominal Pipe Size	F	Sch.5S	Sch.10S	Sch.40S	Sch.80S		
1/2	ANSI	0.068	0.091	0.136	0.115		
	MSS	0.059	0.077	0.118	0.127		
3/4	ANSI	0.077	0.100	0.181	0.199		
	MSS	0.068	0.086	0.154	0.168		
1	ANSI	0.145	0.227	0.299	0.385		
	MSS	0.091	0.140	0.186	0.240		
1 1/4	ANSI	0.204	0.326	0.412	0.548		
	MSS	0.131	0.208	0.263	0.349		
1 1/2	ANSI	0.231	0.367	0.553	0.675		
	MSS	0.159	0.249	0.376	0.458		
2	ANSI	0.449	0.693	0.992	1.364		
	MSS	0.245	0.376	0.539	0.743		
2 1/2	ANSI	0.684	0.924	1.563	2.079		
	MSS	0.349	0.471	0.797	1.060		
3	ANSI	0.879	1.205	2.134	2.840		
	MSS	0.467	0.638	1.133	1.508		
4	ANSI	1.191	1.640	3.035	4.227		
	MSS	0.711	0.978	1.812	2.523		
5	ANSI	2.188	2.582	5.300	7.520		
	MSS	1.046	1.237	2.537	3.601		
6	ANSI	3.01	3.61	6.89	10.42		
	MSS	1.62	1.95	3.72	5.57		
8	ANSI	2.96	3.76	10.42	15.86		
	MSS	2.45	3.10	5.89	10.12		
10	ANSI	7.02	8.45	18.12	24.28		
	MSS	4.04	4.86	10.42	13.95		
12	ANSI	9.11	10.51	22.20	29.81		
	MSS	6.60	7.11	14.95	19.93		
14	ANSI	11.78	14.19	28.31*	37.80**		
	MSS	6.43	7.75	15.46*	20.62**		
16	ANSI	14.38	16.38	32.68*	43.57**		
	MSS	7.91	9.02	17.98*	23.98**		
18	ANSI	16.51	18.81	37.53*	50.03**		
	MSS	9.24	10.53	21.00*	28.00**		
20	ANSI	20.89	24.22	41.66*	55.55**		
	MSS	11.68	13.54	23.30*	31.07**		
22	ANSI	23.20	26.90	46.28*	61.07**		
	MSS			*	**		
24	ANSI	29.33	33.63	50.45*	67.27**		
	MSS	16.52	18.94	28.41*	37.88**		

STAINLESS STEEL FITTINGS

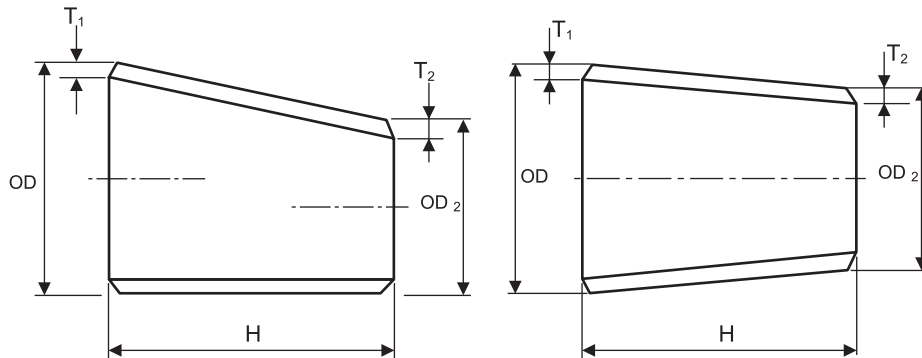


ECC., CON. REDUCERS (12" and Smaller)

Dimensions											Unit: inch
Nominal Pipe Size	Out- side Dia.	Out- side Dia.	Wall Thickness								Length
			Sch.5S		Sch.10S		Sch.40S		Sch.80S		
			OD ₁	OD ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	
3/4 X 1/2	1.050	0.840	0.065	0.065	0.083	0.083	0.113	0.109	0.154	0.147	1.50
1 X 1/2	1.315	0.840	0.065	0.065	0.109	0.083	0.133	0.109	0.179	0.147	2.00
1 X 3/4	1.315	1.050	0.065	0.065	0.109	0.083	0.133	0.113	0.179	0.154	2.00
1 1/4 X 1	1.660	1.315	0.065	0.065	0.109	0.109	0.140	0.133	0.191	0.179	2.00
1 1/2 X 1/2	1.900	0.840	0.065	0.065	0.109	0.083	0.145	0.109	0.200	0.147	2.50
1 1/2 X 3/4	1.900	1.050	0.065	0.065	0.109	0.083	0.145	0.113	0.200	0.154	2.50
1 1/2 X 1	1.905	1.315	0.065	0.065	0.109	0.109	0.145	0.133	0.200	0.179	2.50
1 1/2 X 1 1/4	1.900	1.660	0.065	0.065	0.109	0.109	0.145	0.140	0.200	0.191	2.50
2 X 3/4	2.375	1.050	0.065	0.065	0.109	0.083	0.154	0.113	0.218	0.154	3.00
2 X 1	2.375	1.315	0.065	0.065	0.109	0.109	0.154	0.133	0.218	0.179	3.00
2 X 1 1/4	2.375	1.660	0.065	0.065	0.109	0.109	0.154	0.140	0.218	0.191	3.00
2 X 1 1/2	2.375	1.900	0.065	0.065	0.109	0.109	0.154	0.145	0.218	0.200	3.00
2 1/2 X 1	2.875	1.315	0.083	0.065	0.120	0.109	0.203	0.133	0.276	0.179	3.50
2 1/2 X 1 1/4	2.875	1.660	0.083	0.065	0.120	0.109	0.203	0.140	0.276	0.191	3.50
2 1/2 X 1 1/2	2.875	1.900	0.083	0.065	0.120	0.109	0.203	0.145	0.276	0.200	3.50
2 1/2 X 2	2.875	2.375	0.083	0.065	0.120	0.109	0.203	0.154	0.276	0.218	3.50
3 X 1	3.500	1.315	0.083	0.065	0.120	0.109	0.216	0.133	0.300	0.179	3.50
3 X 1 1/4	3.500	1.660	0.083	0.065	0.120	0.109	0.216	0.140	0.300	0.191	3.50
3 X 1 1/2	3.500	1.900	0.083	0.065	0.120	0.109	0.216	0.145	0.300	0.200	3.50
3 X 2	3.500	2.375	0.083	0.065	0.120	0.109	0.216	0.154	0.300	0.218	3.50
3 X 2 1/2	3.500	2.875	0.083	0.083	0.120	0.120	0.216	0.203	0.300	0.276	3.50
4 X 1 1/2	4.500	1.900	0.083	0.065	0.120	0.109	0.237	0.145	0.337	0.200	4.00
4 X 2	4.500	2.375	0.083	0.065	0.120	0.109	0.237	0.154	0.337	0.218	4.00
4 X 2 1/2	4.500	2.875	0.083	0.083	0.120	0.120	0.237	0.203	0.337	0.276	4.00
4 X 3	4.500	3.500	0.083	0.083	0.120	0.120	0.237	0.216	0.337	0.300	4.00
5 X 3	5.563	3.500	0.109	0.083	0.134	0.120	0.258	0.216	0.375	0.300	5.00
5 X 4	5.563	4.500	0.109	0.083	0.134	0.120	0.258	0.237	0.375	0.337	5.00
6 X 2 1/2	6.625	2.875	0.109	0.083	0.134	0.120	0.280	0.203	0.432	0.276	5.50
6 X 3	6.625	3.500	0.109	0.083	0.134	0.120	0.280	0.216	0.432	0.300	5.50
6 X 4	6.625	4.500	0.109	0.083	0.134	0.130	0.280	0.237	0.432	0.337	5.50
6 X 5	6.625	5.563	0.109	0.109	0.134	0.134	0.280	0.258	0.432	0.375	5.50
8 X 4	8.625	4.500	0.109	0.083	0.148	0.120	0.322	0.237	0.500	0.337	6.00
8 X 5	8.625	5.563	0.109	0.109	0.148	0.134	0.322	0.258	0.500	0.375	6.00
8 X 6	8.625	6.625	0.109	0.109	0.148	0.134	0.322	0.280	0.500	0.432	6.00
10 X 5	10.750	5.563	0.134	0.109	0.165	0.134	0.365	0.258	0.500	0.375	7.00
10 X 6	10.750	6.625	0.134	0.109	0.165	0.134	0.365	0.280	0.500	0.432	7.00
10 X 8	10.750	8.625	0.134	0.109	0.165	0.148	0.365	0.322	0.500	0.500	7.00
12 X 6	12.750	6.625	0.156	0.109	0.180	0.134	0.375	0.280	0.500	0.432	8.00
12 X 8	12.750	8.625	0.156	0.109	0.180	0.148	0.375	0.322	0.500	0.500	8.00
12 X 10	12.750	10.750	0.156	0.134	0.180	0.165	0.375	0.365	0.500	0.500	8.00

Weights					Unit: kg
Nominal Pipe Size	Sch. 5S	Sch. 10S	Sch. 40S	Sch. 80S	
3/4 X 1/2	0.054	0.090	0.105	0.159	
1 X 1/2	0.060	0.089	0.105	0.163	
1 X 3/4	0.060	0.098	0.116	0.168	
1 1/4 X 1	0.066	0.125	0.157	0.245	
1 1/2 X 1/2	0.091	0.154	0.204	0.276	
1 1/2 X 3/4	0.095	0.156	0.201	0.294	
1 1/2 X 1	0.103	0.170	0.219	0.308	
1 1/2 X 1 1/4	0.118	0.195	0.263	0.331	
2 X 3/4	0.121	0.214	0.298	0.458	
2 X 1	0.143	0.237	0.322	0.458	
2 X 1 1/4	0.156	0.258	0.352	0.525	
2 X 1 1/2	0.164	0.273	0.372	0.544	
2 1/2 X 1	0.453	0.349	0.793	0.793	
2 1/2 X 1 1/4	0.276	0.394	0.670	0.825	
2 1/2 X 1 1/2	0.281	0.396	0.661	0.856	
2 1/2 X 2	0.307	0.432	0.724	0.938	
3 X 1	0.281	0.408	0.716	0.888	
3 X 1 1/4	0.299	0.430	0.775	1.037	
3 X 1 1/2	0.315	0.444	0.783	1.105	
3 X 2	0.339	0.478	0.846	1.178	
3 X 2 1/2	0.376	0.548	0.983	1.291	
4 X 1 1/2	0.435	0.630	1.241	1.730	
4 X 2	0.464	0.656	1.27	1.780	
4 X 2 1/2	0.499	0.706	1.37	2.007	
4 X 3	0.528	0.748	1.45	2.129	
5 X 3 1/2	0.991	1.20	2.27	3.547	
5 X 4	1.107	1.32	2.50	3.769	
6 X 2 1/2	1.12	1.35	2.74	4.51	
6 X 3	1.24	1.50	3.04	5.03	
6 X 4	1.34	1.62	3.30	5.44	
6 X 5	1.45	1.75	3.57	5.71	
8 X 4	1.80	2.56	5.10	8.43	
8 X 5	1.90	2.70	5.40	8.83	
8 X 6	2.01	2.85	5.71	9.24	
10 X 5	2.82	3.30	8.21	13.00	
10 X 6	3.30	3.87	8.78	13.50	
10 X 8	3.59	4.21	9.58	14.22	
12 X 6	4.97	5.73	12.40	18.39	
12 X 8	5.45	6.11	13.70	19.07	
12 X 10	5.83	6.55	14.70	19.48	

STAINLESS STEEL FITTINGS



ECC., CON. REDUCES (14" and Larger)

Dimensions												Unit: inch
Nominal Pipe Size		Out- side Dia.	Out- side Dia.	Wall Thickness								Length
				Sch.10S		Sch.20		Std.		XS		
		OD ₁	OD ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	H
14	12	14.000	12.750	0.188	0.180	0.312	0.250	0.375	0.375	0.500	0.500	13.000
14	10	14.000	10.750	0.188	0.165	0.312	0.250	0.375	0.365	0.500	0.500	13.000
14	8	14.000	8.625	0.188	0.148	0.312	0.250	0.375	0.322	0.500	0.500	13.000
14	6	14.000	6.625	0.188	0.134	0.312	-	0.375	0.280	0.500	0.432	13.000
16	14	16.000	14.000	0.188	0.188	0.312	0.312	0.375	0.375	0.500	0.500	14.000
16	12	16.000	12.750	0.188	0.180	0.312	0.250	0.375	0.375	0.500	0.500	14.000
16	10	16.000	10.750	0.188	0.165	0.312	0.250	0.375	0.365	0.500	0.500	14.000
16	8	16.000	8.625	0.188	0.148	0.312	0.250	0.375	0.322	0.500	0.500	14.000
18	16	18.000	16.000	0.188	0.188	0.312	0.312	0.375	0.375	0.500	0.500	15.000
18	14	18.000	14.000	0.188	0.188	0.312	0.312	0.375	0.375	0.500	0.500	15.000
18	12	18.000	12.750	0.188	0.180	0.312	0.250	0.375	0.375	0.500	0.500	15.000
18	10	18.000	10.750	0.188	0.165	0.312	0.250	0.375	0.365	0.500	0.500	15.000
20	18	20.000	18.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
20	16	20.000	16.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
20	14	20.000	14.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
20	12	20.000	12.750	0.218	0.180	0.375	0.250	0.375	0.375	0.500	0.500	20.000
22	20	22.000	20.000	0.218	0.218	0.375	0.375	0.375	0.375	0.500	0.500	20.000
22	18	22.000	18.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
22	16	22.000	16.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
22	14	22.000	14.000	0.218	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
24	22	24.000	22.000	0.250	0.218	0.375	0.375	0.375	0.375	0.500	0.500	20.000
24	20	24.000	20.000	0.250	0.218	0.375	0.375	0.375	0.350	0.500	0.500	20.000
24	18	24.000	18.000	0.250	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
24	16	24.000	16.000	0.250	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
24	14	24.000	14.000	0.250	0.188	0.375	0.312	0.375	0.375	0.500	0.500	20.000
26	24	26.000	24.000		0.250	0.500	0.375	0.375	0.375	0.500	0.500	24.000
26	20	26.000	20.000		0.218	0.500	0.375	0.375	0.375	0.500	0.500	24.000
26	18	26.000	18.000		0.188	0.500	0.312	0.375	0.375	0.500	0.500	24.000
26	16	26.000	16.000		0.188	0.500	0.312	0.375	0.375	0.500	0.500	24.000
28	26	28.000	26.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
28	24	28.000	24.000		0.250	0.500	0.375	0.375	0.375	0.500	0.500	24.000
28	20	28.000	20.000		0.218	0.500	0.375	0.375	0.375	0.500	0.500	24.000
28	18	28.000	18.000		0.188	0.500	0.312	0.375	0.375	0.500	0.500	24.000

Weights Unit: kg					
Nominal Pipe Size		Wall Thickness			
		Sch. 10S	Sch. 20	Std.	XS
14	12	14.27	25.65	28.55	37.18
14	10	13.21	23.73	26.41	34.40
14	8	12.07	21.70	24.15	31.44
14	6	11.01	19.78	22.01	28.66
16	14	17.24	30.98	34.48	44.90
16	12	16.52	29.69	33.05	43.03
16	10	15.37	27.62	30.75	40.04
16	8	14.15	25.43	28.30	36.86
18	16	20.94	37.62	41.87	54.52
18	14	19.71	35.41	39.41	51.32
18	12	18.94	34.02	37.87	49.32
18	10	17.70	31.81	35.41	46.11
20	18	38.01	62.40	62.40	81.25
20	16	36.01	59.11	59.11	76.97
20	14	33.98	55.83	55.83	72.70
20	12	32.76	53.78	53.78	70.02
22	20	42.01	68.96	68.96	89.80
22	18	40.01	65.68	65.68	85.53
22	16	38.01	62.40	62.40	81.25
22	14	36.01	59.08	59.08	76.97
24	22	52.91	75.53	75.53	98.36
24	20	50.61	72.25	72.25	94.08
24	18	48.31	68.96	68.96	89.80
24	16	46.01	65.68	65.68	85.53
24	14	43.71	62.40	62.40	81.25
26	24		128.28	98.53	128.03
26	20		118.02	90.65	117.79
26	18		112.88	86.70	112.66
26	16		107.76	82.76	107.54
28	26		138.54	106.41	138.27
28	24		133.41	102.47	133.15
28	20		123.15	94.59	122.91
28	18		118.02	90.65	117.78

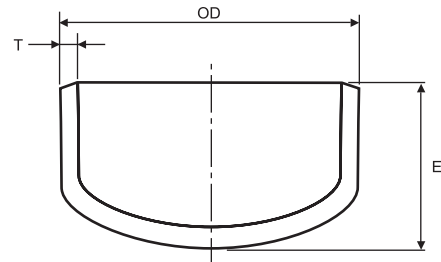
STAINLESS STEEL FITTINGS

ECC., CON. REDUCES (14" and Larger)

Dimensions												Unit: inch
Nominal Pipe Size		Out- side Dia.	Out- side Dia.	Wall Thickness								Length
				Sch.10S		Sch.20		Std.		XS		
				T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	T ₁	T ₂	
30	28	30.000	28.000	0.312		0.500	0.500	0.375	0.375	0.500	0.500	24.000
30	26	30.000	26.000	0.312		0.500	0.500	0.375	0.375	0.500	0.500	24.000
30	24	30.000	24.000	0.312	0.250	0.500	0.375	0.375	0.375	0.500	0.500	24.000
30	20	30.000	20.000	0.312	0.218	0.500	0.375	0.375	0.375	0.500	0.500	24.000
32	30	32.000	30.000		0.312	0.500	0.500	0.375	0.375	0.500	0.500	24.000
32	28	32.000	28.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
32	26	32.000	26.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
32	24	32.000	24.000		0.250	0.500	0.375	0.375	0.375	0.500	0.500	24.000
34	32	34.000	32.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
34	30	34.000	30.000		0.312	0.500	0.500	0.375	0.375	0.500	0.500	24.000
34	28	34.000	28.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
34	26	34.000	26.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
36	34	36.000	34.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
36	32	36.000	32.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
36	30	36.000	30.000		0.312	0.500	0.500	0.375	0.375	0.500	0.500	24.000
36	28	36.000	28.000			0.500	0.500	0.375	0.375	0.500	0.500	24.000
38	36	38.000	36.000					0.375	0.375	0.500	0.500	24.000
38	34	38.000	34.000					0.375	0.375	0.500	0.500	24.000
38	32	38.000	32.000					0.375	0.375	0.500	0.500	24.000
38	30	38.000	30.000					0.375	0.375	0.500	0.500	24.000
38	28	38.000	28.000					0.375	0.375	0.500	0.500	24.000
38	26	38.000	26.000					0.375	0.375	0.500	0.500	24.000
40	38	40.000	38.000					0.375	0.375	0.500	0.500	24.000
40	36	40.000	36.000					0.375	0.375	0.500	0.500	24.000
40	34	40.000	34.000					0.375	0.375	0.500	0.500	24.000
40	32	40.000	32.000					0.375	0.375	0.500	0.500	24.000
40	30	40.000	30.000					0.375	0.375	0.500	0.500	24.000
42	40	42.000	40.000					0.375	0.375	0.500	0.500	24.000
42	38	42.000	38.000					0.375	0.375	0.500	0.500	24.000
42	36	42.000	36.000					0.375	0.375	0.500	0.500	24.000
42	34	42.000	34.000					0.375	0.375	0.500	0.500	24.000
42	32	42.000	32.000					0.375	0.375	0.500	0.500	24.000
42	30	42.000	30.000					0.375	0.375	0.500	0.500	24.000
44	42	44.000	42.000					0.375	0.375	0.500	0.500	24.000
44	40	44.000	40.000					0.375	0.375	0.500	0.500	24.000
44	38	44.000	38.000					0.375	0.375	0.500	0.500	24.000
44	36	44.000	36.000					0.375	0.375	0.500	0.500	24.000
46	44	46.000	44.000					0.375	0.375	0.500	0.500	24.000
46	42	46.000	42.000					0.375	0.375	0.500	0.500	24.000
46	40	46.000	40.000					0.375	0.375	0.500	0.500	24.000
46	38	46.000	38.000					0.375	0.375	0.500	0.500	24.000
46	36	46.000	36.000					0.375	0.375	0.500	0.500	24.000
48	44	48.000	44.000					0.375	0.375	0.500	0.500	24.000
48	42	48.000	42.000					0.375	0.375	0.500	0.500	24.000
48	40	48.000	40.000					0.375	0.375	0.500	0.500	24.000

Weights						Unit: kg
Nominal Pipe Size		Wall Thickness				
		Sch. 10S	Sch. 20	Std.	XS	
30	28		148.80	114.29		148.51
30	26		143.67	110.35		143.39
30	24		138.54	106.41		138.27
30	20		128.28	98.53		128.03
32	30		159.07	122.17		158.75
32	28		153.93	118.23		153.63
32	26		148.80	114.29		148.51
32	24		143.67	110.40		143.39
34	32		169.33	130.06		168.99
34	30		164.20	126.12		163.88
34	28		159.07	122.17		158.75
34	26		153.93	118.51		153.63
36	34		179.59	137.94		179.24
36	32		174.46	133.99		174.12
36	30		169.33	130.06		168.99
36	28		164.20	126.12		163.88
38	36			145.82		189.48
38	34			141.88		184.36
38	32			137.94		179.24
38	30			133.99		174.12
38	28			130.06		168.99
38	26			126.12		163.88
40	38			153.70		199.72
40	36			149.76		194.60
40	34			145.82		189.46
40	32			141.88		184.36
40	30			137.94		179.24
42	40			161.59		209.97
42	38			157.64		204.85
42	36			153.70		199.72
42	34			149.76		194.60
42	32			145.82		189.48
42	30			141.88		184.36
44	42			169.47		220.21
44	40			165.53		215.09
44	38			161.59		209.97
44	36			157.64		204.85
46	44			177.35		230.45
46	42			173.41		225.33
46	40			169.47		220.21
46	38			165.53		215.09
46	36			165.53		215.09
48	44			181.29		235.57
48	42			177.35		230.45
48	40			173.41		225.33

STAINLESS STEEL FITTINGS



CAPS (24" and Smaller)

Dimensions						Unit: inch	Weights					Unit: kg
Nominal Pipe Size	Outside Dia. (OD)	Wall Thickness (T)				Length E	Nominal Pipe Size	Sch.5S	Sch.10S	Sch.40S	Sch.80S	
		Sch.5S	Sch.10S	Sch.40S	Sch.80S							
1/2	0.840	0.065	0.083	0.109	0.147	1.00	1/2	0.036	0.040	0.046	0.045	
3/4	1.050	0.065	0.083	0.113	0.154	1.00	3/4	0.049	0.052	0.059	0.086	
1	1.315	0.065	0.109	0.133	0.179	1.50	1	0.053	0.090	0.109	0.109	
1 1/4	1.660	0.065	0.109	0.140	0.191	1.50	1 1/4	0.066	0.113	0.145	0.159	
1 1/2	1.900	0.065	0.109	0.145	0.200	1.50	1 1/2	0.076	0.130	0.171	0.222	
2	2.375	0.065	0.109	0.154	0.218	1.50	2	0.099	0.168	0.234	0.344	
2 1/2	2.875	0.083	0.120	0.203	0.276	2.00	2 1/2	0.169	0.242	0.420	0.512	
3	3.500	0.083	0.120	0.216	0.300	2.50	3	0.253	0.362	0.664	0.888	
4	4.500	0.083	0.120	0.237	0.337	3.00	4	0.409	0.585	1.17	1.51	
5	5.563	0.109	0.134	0.258	0.375	3.50	5	0.805	0.978	1.90	2.89	
6	6.625	0.109	0.134	0.280	0.432	4.00	6	1.11	1.35	2.83	4.24	
8	8.625	0.109	0.148	0.322	0.500	5.00	8	1.74	2.49	5.11	7.76	
10	10.750	0.134	0.165	0.375	0.500	6.00	10	3.26	3.84	8.92	13.11	
12	12.750	0.156	0.180	0.375	0.500	7.00	12	5.47	6.15	14.10	17.94	
14	14.000	0.156	0.188	0.375*	0.500**	7.50	14	7.19	7.85	16.35	21.11	
16	16.000	0.165	0.188	0.375*	0.500**	8.00	16	10.99	9.56	18.60	25.73	
18	18.000	0.165	0.188	0.375*	0.500**	9.00	18	13.92	12.10	23.54	32.57	
20	20.000	0.188	0.218	0.375*	0.500**	10.00	20	17.19	17.90	29.06	40.21	
24	24.000	0.218	0.250	0.375*	0.500**	12.00	24	27.43	30.15	41.85	57.86	

*Std **XS

For fittings made from seamless and welded pipes ASTM A312 shall be applied as follows.

Grade	Carbon max, per cent	Mang- anese, max, percent	Phos- phorus per cent	Sulfur, max, per cent	Silicon, max, per cent	Chromium, percent	Nickel, percent	Other Elements	T.S Min ksi	Y.S Min ksi
TP 304	0.08	2.00	0.040	0.030	0.75	18.0~20.0	8.00~11.0		75	30
TP 304L	0.035	2.00	0.040	0.030	0.75	18.0~20.0	8.00~13.0		70	25
TP 316	0.08	2.00	0.040	0.030	0.75	16.0~18.0	11.00~14.0	Mo 2.0~3.0	75	30
TP 316L	0.035	2.00	0.040	0.030	0.75	16.0~18.0	10.00~15.0	Mo 2.0~3.0	70	25
TP 321	0.08	2.00	0.040	0.030	0.75	17.0~20.0	9.00~13.0	Ti min. CX5 max. 0.60	75	30
TP 347	0.08	2.00	0.040	0.030	0.75	17.0~20.0	9.00~13.0	Cb+Ta min. CX10 max. 1.00	75	30



MACHINERY STEEL PRODUCTS

HARDEN & TEMPERED STEEL - CARBON STEEL

248

PRE-HARDEN & TEMPERED ALLOY STEEL 249

COLD FINISHED STEEL BAR

251



YICK HOE

HARDEN & TEMPERED STEEL

CARBON STEEL KS D3752, JIS G4051

KS JIS	AISI	Chemical Composition %					Mechanical Properties					
		C	Si	Mn	P	S	YP	TS	EL	RA	Impact	Hardness
							kg/mm ²	kg/mm ²	%	%	charpy	HB
S10C	1010	.08/.13 .08/.13	.15/.35 -	.30/.60 .30/.60	≤0.03 ≤.035	≤.035 ≤.045	≥21	≥32	≥33			109/159
S12C	1012	.10/.15 .10/.15	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥24	≥38	≥30	-	-	111/167
S15C	1015	.13/.18 .13/.18	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥24	≥38	≥30	-	-	111/167
S17C	1017	.15/.20 .15/.20	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥25	≥41	≥28	-	-	116/174
S20C	1020	.18/.23 .18/.23	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥25	≥41	≥28	-	-	116/174
S22C	1023	.20/.25 .20/.25	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥27	≥45	≥27	-	-	123/183
S25C	1025	.22/.28 .22/.28	.15/.35 -	.30/.60 .30/.60	≤.030 ≤.035	≤.035 ≤.045	≥27	≥45	≥27	-	-	123/183
S28C	1029	.25/.31 .25/.31	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥29	≥48	≥25	-	-	137/197
S30C	1030	.27/.33 .28/.34	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥29	≥48	≥25	-	-	137/197
S33C		.30/.36	.15/.35	.60/.90	≤.030	≤.035	≥31	≥52	≥23	-	-	149/207
S35C	1035	.32/.36 .32/.38	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥31	≥52	≥23	-	-	149/207
S38C	1038	.35/.41 .35/.42	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.030	≤.035 ≤.045	≥33	≥55	≥22	-	-	156/217
S40C	1040	.37/.43 .37/.44	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.030	≤.035 ≤.045	≥33	≥55	≥22	-	-	156/217
S43C	1043	.40/.46 .40/.47	.15/.35 -	.60/.90 .70/1.00	≤.030 ≤.035	≤.035 ≤.045	≥35	≥58	≥20	-	-	167/229
S45C	1045	.42/.48 .43/.50	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥35	≥58	≥20	-	-	167/229
S48C		.45/.51	.15/.35	.60/.90	≤.030	≤.035	≥37	≥62	≥18	-	-	179/235
S50C	1049	.47/.53 .46/.53	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥37	≥62	≥18	-	-	179/235
S53C	1053	.50/.56 .48/.55	.15/.35 -	.60/.90 .70/1.00	≤.030 ≤.035	≤.035 ≤.045	≥40	≥66	≥15	-	-	183/255
S55C	1055	.52/.58 .50/.60	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥40	≥66	≥15	-	-	183/255
S58C	1060	.55/.61 .55/.65	.15/.35 -	.60/.90 .60/.90	≤.030 ≤.035	≤.035 ≤.045	≥40	≥66	≥15	-	-	183/255

PRE-HARDEN & TEMPERED ALLOY STEEL

ALLOY STEEL KS D3707 D3711, JIS G4104 G4105

KS JIS	AISI	Chemical Composition %							Mechanical Properties					
		C	Si	Mn	P	S	Cr	Mo	YP	TS	EL	RA	Impact	Hardness
									kg/mm ²	kg/mm ²	%	%	charpy	HB
SCR415		.13/.18	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	-	≥80	≥15	≥40	≥6	217/302
SCR420	5120	.18/.23	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	-	≥85	≥14	≥35	≥5	235/321
		.17/.22	.15/.30	.70/.90	≤.035	≤.040	.70/.90							
SCR430	5130	.28/.33	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥65	≥80	≥18	≥55	≥9	229/285
		.28/.33	.15/.30	.70/.90	≤.035	≤.040	.80/1.10							
SCR435	5135	.33/.38	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥75	≥90	≥15	≥50	≥7	255/311
		.33/.38	.15/.30	.60/.80	≤.035	≤.030	.80/1.05							
SCR440	5140	.38/.43	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥85	≥100	≥12	≥40	≥5	285/341
		.38/.43	.15/.30	.70/.90	≤.035	≤.040	.70/.90							
SCR445	5147	.43/.48	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	-	≥85	≥100	≥12	≥40	≥5	285/341
		.46/.51	.15/.30	.70/.95	≤.035	≤.030	.85/1.15							
SCM415		.13/.18	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	-	≥85	≥16	≥40	≥7	235/321
SCM418	4118	.16/.21	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	-	≥90	≥15	≥40	≥7	248/331
		.18/.23	.15/.30	.70/.90	≤.035	≤.040	.40/.60	.08/.15						
SCM420		.18/.23	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	-	≥95	≥14	≥35	≥6	285/363
SCM421		.17/.23	.15/.35	.70/1.00	≤.030	≤.030	.90/1.20	.15/.30	-	≥100	≥14	≥35	≥6	285/363
SCM822		.20/.25	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.35/.45	-	≥105	≥12	≥30	≥6	302/415
SCM430	4130	.28/.33	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥70	≥85	≥18	≥55	≥11	241/293
		.28/.33	.15/.30	.40/.60	≤.035	≤.040	.80/1.10	.15/.25						
SCM432	4135	.27/.37	.15/.35	.30/.60	≤.030	≤.030	1.00/1.50	.15/.30	≥75	≥90	≥16	≥50	≥9	225/321
		.33/.38	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥80	≥95	≥15	≥50	≥8	269/321
SCM435	4140	.33/.38	.20/.35	.70/.90	≤.035	≤.040	.80/1.10	.15/.25						
		.38/.43	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥85	≥100	≥12	≥45	≥6	285/341
SCM440	4145	.38/.43	.15/.30	.75/1.00	≤.035	≤.040	.80/1.10	.15/.25						
		.43/.48	.15/.35	.60/.85	≤.030	≤.030	.90/1.20	.15/.30	≥90	≥105	≥12	≥40	≥4	302/363
SCM445		.43/.48	.15/.30	.75/1.00	≤.035	≤.040	.80/1.10	.15/.25						

PRE-HARDEN & TEMPERED ALLOY STEEL

ALLOY STEEL KS D3708 D3709, JIS G4102 G4103

KS JIS	AISI	Chemical Composition %							Mechanical Properties						
		C	Si	Mn	P	S	Ni	Cr	Mo	YP kg/mm ²	TS kg/mm ²	EL %	RA %	Impact charpy	Hardness HB
SNC415		.12/.18	.15/.35	.35/.65	≤.030	≤.030	2.00/2.50	.20/.50	-	-	≤80	≥17	≥45	≥9	235/341
SNC815		.12/.18	.15/.35	.35/.65	≤.030	≤.030	3.00/3.50	.70/1.00	-	-	≤100	≥12	≥45	≥8	285/388
SNC631		.27/.35	.15/.35	.35/.65	≤.030	≤.030	2.50/3.00	.60/1.00	-	≥70	≥85	≥18	≥50	≥12	248/302
SNC236		.32/.40	.15/.35	.35/.65	≤.030	≤.030	2.50/3.00	.60/1.00	-	≥70	≥85	≥18	≥50	≥12	248/302
	3135	.33/.38	.20/.35	.60/.80	≤.040	≤.040	1.10/1.40	.55/.75	-						
SNC836		.32/.40	.15/.35	.35/.65	≤.030	≤.030	3.00/3.50	.60/1.00	-	≥80	≥95	≥19	≥49	≥8	269/321
SNCM415		.12/.18	.15/.35	.40/.70	≤.030	≤.030	1.60/2.00	.40/.65	.15/.30	-	≥90	≥16	≥45	≥7	255/341
SNCM815	.12/.18	.15/.35	.30/.60	≤.030	≤.030	4.00/4.52	.70/1.00	.15/.30	-	≥110	≥12	≥40	≥7	311/375	
SNCM616		.13/.20	.15/.35	.80/1.20	≤.030	≤.030	2.80/3.20	1.40/1.80	.40/.60	-	≥120	≥14	≥40	≥8	341/415
SNCM220		.17/.23	.15/.35	.60/.90	≤.030	≤.030	.40/.70	.40/.60	.15/.30	-	≥85	≥17	≥40	≥6	248/341
SNCM420		.17/.23	.13/.35	.40/.70	≤.030	≤.030	1.60/2.00	.40/.65	.15/.30	-	≥100	≥15	≥40	≥7	293/375
	4320	.17/.22	.15/.30	.45/.65	≤.035	≤.040	1.65/2.00	.40/.60	.20/.30						
SNCM625		.20/.30	.15/.35	.35/.60	≤.030	≤.030	3.00/3.50	1.00/1.50	.15/.30	≥85	≥95	≥18	≥50	≥8	269/321
SNCM630		.25/.35	.15/.35	.35/.60	≤.030	≤.030	2.50/3.50	2.50/3.50	.50/.70	≥90	≥110	≥15	≥45	≥8	302/352
SNCM431		.27/.35	.15/.35	.60/.90	≤.030	≤.030	1.60/2.00	.60/1.00	.15/.30	≥70	≥85	≥20	≥55	≥10	248/302
	4337	.35/.40	.20/.35	.60/.90	≤.040	≤.040	1.65/2.00	.70/.90	.20/.30						
SNCM439		.36/.43	.15/.35	.60/.90	≤.030	≤.030	1.60/2.00	.60/1.00	.15/.30	≥90	≥100	≥16	≥45	≥7	293/352
	4340	.38/.43	.15/.30	.60/.80	≤.035	≤.030	1.65/2.00	.70/.90	.20/.30						
SNCM240		.38/.43	.15/.35	.70/1.00	≤.030	≤.030	.40/.70	.40/.65	.15/.30	≥80	≥90	≥17	≥50	≥7	255/311

YICK HOE

COLD FINISHED STEEL BAR

FREE CUTTING STEEL KS D3567, JIS G4804

KS JIS	AISI	Chemical Composition %						Mechanical Properties					
		C	Si	Mn	P	S	Pb	YP	TS	EL	RA	Impact	Hardness
								kg/mm ²	kg/mm ²	%	%	charpy	HB
SUM11		.08/.13	-	.30/.60	≤.04	.08/.13	-						
	1109	.08/.13	-	.60/.90	≤.04	.08/.13	-						
SUM12		.08/.13	-	.60/.90	≤.04	.08/.13	-						
SUM32		.12/.20	-	.60/1.10	≤.04	.10/.20	-						
SUM31		.14/.20	-	1.00/1.30	≤.04	.08/.13	-						
	1117	.14/.20	-	1.00/1.30	≤.04	.08/.13	-	28.4	43.7	32.8	58	69	121
SUM41		.32/.39	-	1.35/1.65	≤.04	.08/.13	-						
	1137	.32/.39	-	1.35/1.65	≤.04	.08/.13	-	35.1	59.5	26.8	53.9	36.8	174
SUM42		.37/.45	-	1.35/1.65	≤.04	.08/.13	-						
	1141	.37/.45	-	1.35/1.65	≤.04	.08/.13	-						
SUM43		.37/.45	-	1.35/1.65	≤.04	.24/.33	-						
	1144	.40/.48	-	1.35/1.65	≤.04	.24/.33	-	35.2	59.5	24.8	47.3	48	167
SUM23		≤.09	-	.75/1.05	.04/.09	.26/.35	-						
SUM21		≤.13	-	.70/1.00	.07/.12	.16/.23	-						
	1213	≤.13	-	.70/1.00	.07/.12	.24/.33	-						
SUM22		≤.13	-	.70/1.00	.07/.12	.24/.33	-						

YICK HOE



NON-FERROUS METALS

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BRONZE

PRINCIPLE OF CONTINUOUS CASTING / SPECIFICATIONS /	
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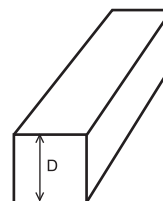
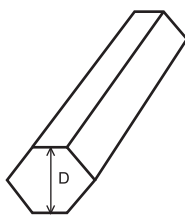
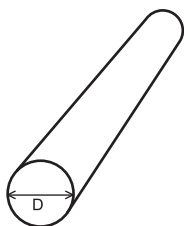
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YICK HOE

COPPER

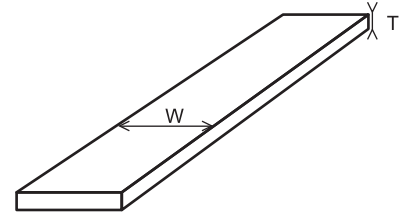
ROUND BARS, HEXAGON BARS & SQUARE BARS



WEIGHTS CHART (Imperial Units)

Dia. in	Round Bar			Hexagon Bar			Square Bar		
	kg/ft	lb/ft	kati/ft	kg/ft	lb/ft	kati/ft	kg/ft	lb/ft	kati/ft
1/8	0.022	0.048	0.036	0.024	0.052	0.040	0.027	0.060	0.045
3/16	0.049	0.107	0.081	0.054	0.118	0.089	0.062	0.136	0.103
1/4	0.086	0.189	0.142	0.095	0.209	0.157	0.109	0.241	0.180
5/16	0.134	0.296	0.222	0.148	0.326	0.245	0.171	0.377	0.283
3/8	0.193	0.426	0.319	0.213	0.470	0.352	0.246	0.543	0.407
7/16	0.263	0.580	0.435	0.290	0.640	0.480	0.335	0.739	0.554
1/2	0.344	0.758	0.569	0.379	0.835	0.627	0.438	0.965	0.724
9/16	0.435	0.959	0.719	0.479	1.057	0.792	0.554	1.221	0.916
5/8	0.537	1.184	0.888	0.592	1.305	0.979	0.684	1.507	1.131
11/16	0.650	1.432	1.075	0.716	1.579	1.184	0.827	1.824	1.368
3/4	0.773	1.705	1.279	0.853	1.880	1.411	0.984	2.170	1.628
13/16	0.907	2.000	1.500	1.002	2.210	1.657	1.157	2.550	1.914
7/8	1.052	2.320	1.740	1.160	2.558	1.919	1.340	2.954	2.216
15/16	1.211	2.670	2.003	1.334	2.940	2.206	1.542	3.400	2.550
1	1.374	3.030	2.273	1.515	3.341	2.506	1.750	3.858	2.895
1 1/8	1.740	3.835	2.878	1.918	4.229	3.172	2.215	4.883	3.664
1 1/4	2.148	4.735	3.553	2.368	5.221	3.917	2.735	6.029	4.524
1 3/8	2.599	5.729	4.299	2.865	6.317	4.739	3.309	7.295	5.473
1 1/2	3.093	6.818	5.116	3.410	7.518	5.640	3.938	8.681	6.513
1 5/8	3.630	8.002	6.004	4.003	8.824	6.621	4.622	10.19	7.645
1 3/4	4.210	9.281	6.963	4.636	10.22	7.668	5.362	11.82	8.869
1 7/8	4.831	10.65	7.990	5.330	11.750	8.816	6.151	13.56	10.174
2	5.498	12.12	9.094	6.065	13.37	10.032	6.999	15.43	11.576
2 1/8	6.214	13.70	10.278						
2 1/4	6.958	15.34	11.509	7.675	16.92	12.694	8.859	19.53	14.653
2 3/8	7.766	17.12	12.845						
2 1/2	8.605	18.97	14.233	9.448	20.83	15.627	10.941	24.12	18.096
2 5/8	9.485	20.91	15.688						
2 3/4	10.41	22.95	17.218	11.462	25.27	18.958	13.236	29.18	21.892
2 7/8	11.381	25.09	18.824						
3	12.388	27.31	20.490	13.662	30.12	22.597	15.753	34.73	26.055
3 1/4	14.542	32.06	24.052	16.007	35.29	26.476	18.484	40.75	30.573
3 1/2	16.865	37.18	27.895	18.566	40.93	30.708	21.442	47.27	35.465
3 3/4	19.360	42.68	32.021	21.315	46.99	35.255	24.612	54.26	40.708
4	22.027	48.56	36.433	24.249	53.46	40.108	28.001	61.73	46.314
4 1/4	24.866	54.82	41.128						
4 1/2	27.874	61.45	46.104				35.462	78.18	58.654

FLAT BARS



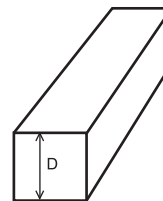
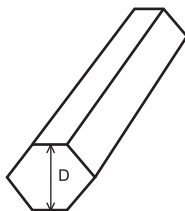
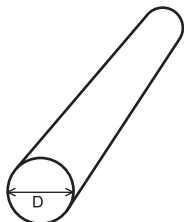
SIZES AND UNIT WEIGHTS (Approx. Wt. in lbs/ft)

Width	Thickness										
in	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{4}$	$\frac{5}{16}$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{5}{8}$	$\frac{3}{4}$	$\frac{7}{8}$	1
$\frac{1}{2}$	0.12	0.24	0.36	0.48	0.60	0.72					
$\frac{5}{8}$	0.15	0.30	0.45	0.60	0.75	0.91	1.21				
$\frac{3}{4}$	0.18	0.36	0.54	0.72	0.91	1.09	1.45	1.81			
$\frac{7}{8}$		0.42	0.63	0.85	1.06	1.27	1.69	2.11	2.54		
1	0.24	0.48	0.72	0.97	1.21	1.45	1.93	2.42	2.90	3.38	
$1\frac{1}{4}$		0.60	0.91	1.21	1.51	1.81	2.42	3.02	3.62	4.23	4.83
$1\frac{1}{2}$	0.36	0.72	1.09	1.45	1.81	2.17	2.90	3.62	4.35	5.07	5.80
$1\frac{3}{4}$		0.85	1.27	1.69	2.11	2.54	3.38	4.23	5.07	5.92	6.76
2		0.97	1.45	1.93	2.42	2.90	3.86	4.83	5.80	6.76	7.73
$2\frac{1}{4}$		1.09	1.63	2.17		3.26	4.35	5.43	6.52		8.69
$2\frac{1}{2}$		1.21	1.81	2.42	3.02	3.62	4.83	6.04	7.25	8.45	9.66
$2\frac{3}{4}$		1.33		2.66							
3		1.45	2.17	2.90	3.62	4.35	5.80	7.25	8.69		11.59
$3\frac{1}{4}$				3.14	3.92				9.42		
$3\frac{1}{2}$		1.69	2.54	3.38		5.07	6.76	8.45			
4		1.93	2.90	3.86	4.83	5.80	7.73	9.66	11.59		15.46
$4\frac{1}{2}$				4.35			8.69	10.87			
5			3.62	4.83			9.66				
6				5.80	7.25	8.69	11.59		17.39		23.18

BRASS

ROUND BARS, HEXAGON BARS & SQUARE BARS

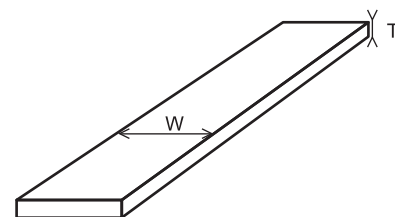
JIS H3250 C3604B/BS2874 CZ121/122



WEIGHTS CHART (Imperial Units)

Dia. in	Round Bar			Hexagon Bar			Square Bar		
	kg/ft	lb/ft	kati/ft	kg/ft	lb/ft	kati/ft	kg/ft	lb/ft	kati/ft
1/8	0.020	0.045	0.033	0.023	0.050	0.038	0.026	0.058	0.043
3/16	0.046	0.102	0.076	0.051	0.112	0.084	0.059	0.130	0.098
1/4	0.082	0.181	0.136	0.091	0.200	0.151	0.105	0.231	0.174
5/16	0.128	0.283	0.212	0.142	0.312	0.235	0.163	0.360	0.270
3/8	0.185	0.407	0.306	0.204	0.449	0.337	0.235	0.519	0.389
7/16	0.252	0.555	0.417	0.278	0.612	0.460	0.320	0.706	0.529
1/2	0.328	0.724	0.543	0.362	0.799	0.599	0.418	0.922	0.691
9/16	0.416	0.917	0.688	0.459	1.011	0.759	0.529	1.167	0.875
5/8	0.513	1.132	0.849	0.566	1.248	0.936	0.654	1.441	1.082
11/16	0.621	1.369	1.027	0.685	1.510	1.133	0.805	1.774	1.331
3/4	0.739	1.630	1.222	0.815	1.797	1.348	0.941	2.075	1.556
13/16	0.864	1.905	1.429	0.953	2.100	1.576	1.098	2.420	1.816
7/8	1.006	2.218	1.664	1.109	2.446	1.834	1.281	2.824	2.119
15/16	1.150	2.535	1.902	1.266	2.790	2.094	1.465	3.230	2.423
1	1.314	2.897	2.173	1.449	3.195	2.397	1.673	3.689	2.767
1 1/8	1.663	3.667	2.751	1.834	4.043	3.033	2.118	4.669	3.503
1 1/4	2.053	4.527	3.396	2.264	4.992	3.745	2.615	5.764	4.325
1 3/8	2.485	5.478	4.110	2.740	6.040	4.032	3.163	6.974	5.232
1 1/2	2.957	6.519	4.891	3.260	7.188	5.392	3.765	8.300	6.227
1 5/8	3.470	7.651	5.739	3.827	8.436	6.330	4.418	9.741	7.307
1 3/4	4.025	8.873	6.657	4.438	9.784	7.340	5.126	11.300	8.478
1 7/8	4.622	10.190	7.645	5.094	11.230	8.425	5.883	12.970	9.730
2	5.257	11.590	8.695	5.797	12.780	9.688	6.695	14.760	11.074
2 1/8	5.910	13.029	9.775						
2 1/4	6.654	14.670	11.006	7.335	16.170	12.132	8.473	18.680	14.014
2 3/8	7.380	16.270	12.207						
2 1/2	8.215	18.110	13.588	9.058	19.970	14.982	10.460	23.060	17.301
2 5/8	9.010	19.870	14.903						
2 3/4	9.938	21.910	16.437	10.959	24.160	18.126	12.655	27.900	20.931
2 7/8	10.810	23.840	17.880						
3	11.830	26.080	19.567	13.041	28.750	21.670	15.059	33.200	24.908
3 1/4	13.820	30.460	22.858				17.595	38.790	29.102
3 1/2	16.148	35.600	26.709				20.602	45.420	34.076
3 3/4	18.400	40.560	30.434				23.424	51.640	38.743
4	20.930	46.140	34.618				26.649	58.750	44.077

FLAT BARS



SIZES AND UNIT WEIGHTS (Approx. Wt. in lbs/ft)

Width	Thickness										
in	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{3}{16}$	$\frac{1}{4}$	$\frac{5}{16}$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{5}{8}$	$\frac{3}{4}$	$\frac{7}{8}$	1
$\frac{1}{2}$	0.11	0.23	0.34	0.46	0.57	0.69					
$\frac{5}{8}$	0.14	0.29	0.43	0.57	0.71	0.86	1.14				
$\frac{3}{4}$	0.17	0.34	0.51	0.69	0.86	1.03	1.37	1.72			
$\frac{7}{8}$		0.40	0.60	0.80	1.00	1.20	1.60	2.00	2.40		
1	0.23	0.46	0.69	0.92	1.14	1.37	1.83	2.29	2.75	3.20	
$1\frac{1}{4}$		0.57	0.86	1.14	1.43	1.72	2.29	2.86	3.43	4.00	4.58
$1\frac{1}{2}$	0.34	0.69	1.03	1.37	1.72	2.06	2.75	3.43	4.12	4.80	5.49
$1\frac{3}{4}$		0.80	1.20	1.60	2.00	2.40	3.20	4.00	4.80	5.60	6.41
2		0.92	1.37	1.83	2.29	2.75	3.66	4.58	5.49	6.41	7.32
$2\frac{1}{4}$		1.03	1.54	2.06		3.09	4.12	5.15	6.18		8.24
$2\frac{1}{2}$		1.14	1.72	2.29	2.86	3.43	4.58	5.72	6.86	8.01	9.15
$2\frac{3}{4}$		1.26		2.52							
3		1.37	2.06	2.75	3.43	4.12	5.49	6.86	8.24		10.98
$3\frac{1}{4}$					3.72		5.95		8.92		
$3\frac{1}{2}$		1.60	2.40	3.20		4.80	6.41	8.01			
4		1.83	2.75	3.66	4.58	5.49	7.32	9.15	10.98		14.64
$4\frac{1}{2}$				4.12			8.24	10.29			
5			3.42	4.58			9.15				
6				5.49	6.86	8.24	10.98		16.47		21.96

BRASS

SHEETS

JIS H3100 C2801P 1/2 HARD

IMPORTED BRASS PLATE FROM JAPAN - SAMBO

Gauge	Thickness	Width	Length	Weight
	mm			
6 OZ	0.22	14"	48"	0.83
8 OZ	0.30	14"	48"	1.13
10 OZ	0.35	14"	48"	1.31
12 OZ	0.54	14"	48"	1.54
14 OZ	0.50	14"	48"	1.88
16 OZ	0.55	14"	48"	2.06
16 G	1.50	2'	4'	9.64
18 G	1.20	2'	4'	7.72
20 G	0.90	2'	4'	5.79
22 G	0.70	2'	4'	4.50
24 G	0.50	2'	4'	3.21
	1.00	2'	4'	6.43
	2.00	2'	4'	12.86
	2.30	2'	4'	14.79
	3.00	2'	4'	19.29
	4.50	2'	4'	30.61
	6.00	2'	4'	40.83
	9.00	2'	4'	61.24
	12.00	2'	4'	81.65
	0.50	4'	8'	13.61
	0.70	4'	8'	18.00
	0.90	4'	8'	23.15
	1.20	4'	8'	30.86
	1.50	4'	8'	38.85
	2.00	4'	8'	51.44
	3.00	4'	8'	77.16
	4.50	4'	8'	122.43
	6.00	4'	8'	163.32
	9.00	4'	8'	244.98
	12.00	4'	8'	326.64

PRINCIPLE OF CONTINUOUS CASTING

Molten bronze is poured from a furnace into a crucible which is cemented to the neck of a graphite die. The lower half of the die is water cooled, the metal sets in this section and is withdrawn through two sets of rollers. A flying saw mounted below the roller cuts the bar to the required length. Continuous cast rod has greatly improved physical properties compared with normal sand-cast material. Because of the bottom pouring conditions and rapid cooling the metal receives in the cooling zone, inclusions are completely eliminated, grain structure is tight and even giving excellent pressure tightness qualities. Tensile strength and wearing qualities are increased by up to 35%.

SPECIFICATIONS

Our standard stock bronze is listed below. Other specifications can be supplied on request provided sufficient quantity is ordered.

Material Specification:

British Standard	-----BS.1400 LG2-C
Japanese Standard	-----JIS H5111 BC6C
American Standard	-----ASTM SAE 40
	ASTM B145-52.4A
	ASTM 584-85a C83600

Approximate Chemical Composition

-----Copper 85%

Tin 5%

Zinc 5%

Lead 5%

Typical Mechanical Properties : U.T.S.

-----18 Tons/in²

28 kg/mm²

17 Tons/in² BS.1400(min)

Yield Stress

-----10 Tons/in²

15 kg/mm²

Elongation

-----25%

15% BS.1400(min)

Hardness (Brinell No.)

-----80

ADVANTAGES

Constant research and improved production techniques ensure top quality material with many advantages :-

- ☐ Reduced machining allowance - less waste
- ☐ Reduced tool wear
- ☐ Higher machining speeds
- ☐ Excellent mechanical properties
- ☐ Freedom from gas and porosity
- ☐ Close grain structure provides good pressure resistance

APPLICATION

General purpose bearings, Automotive and Machine

- Hydraulic Components
- Gears, Worm Wheels
- Valve seats
- Slide valves
- Rock drill parts
- Crusher bearings
- Hose nozzles
- Irrigation equipment
- Rollers
- Washers
- Wear rings

BRONZE

STOCK SIZES & WEIGHTS

Outside Dia. mm(in)	Inside Dia. mm(in)	Solid kg/ft	Weight kg/ft
22(7/8)	12.5(1/2)	1.2	.93
25.5(1)	12.5(1/2)	1.54	1.28
	16(5/8)		1.10
	19(3/4)		0.89
28.5(1 1/8)	12.5(1/2)	1.93	1.66
	16(5/8)		1.49
	19(3/4)		1.27
	22(7/8)		1.01
31(1 1/4)	12.5(1/2)	2.34	2.09
	16(5/8)		1.91
	19(3/4)		1.74
	22(7/8)		1.44
	25.5(1)		1.14
35(1 3/8)	16(5/8)	2.82	2.39
	19(3/4)		2.17
	22(7/8)		1.71
	25.5(1)		1.61
38(1 1/2)	12.5(1/2)	3.34	3.07
	19(3/4)		2.68
	25.5(1)		2.13
	31.5(1 1/4)		1.40
41(1 5/8)	25.5(1)	3.89	2.68
	31.5(1 1/4)		1.95
44.5(1 3/4)	12.5(1/2)	4.49	4.23
	19(3/4)		3.84
	22(7/8)		3.60
	25.5(1)		3.29
	31.5(1 1/4)		2.55
	38(1 1/2)		1.65
51(2)	12.5(1/2)	5.81	5.55
	19(3/4)		5.17
	25.5(1)		4.61
	31.5(1 1/4)		3.88
	38(1 1/2)		2.98
57(2 1/4)	19(3/4)	7.31	6.67
	25.5(1)		6.11
	31.5(1 1/4)		5.38
	38(1 1/2)		4.48
	44.5(1 3/4)		3.40
	51(2)		2.16
63.5(2 1/2)	19(3/4)	8.98	8.34
	25.5(1)		7.78
	31.5(1 1/4)		7.05
	38(1 1/2)		6.14
	44.5(1 3/4)		5.07
	51(2)		3.82
69.5(2 3/4)	25.5(1)	10.83	9.62
	31.5(1 1/4)		8.89
	38(1 1/2)		7.90
	44.5(1 3/4)		6.91
	51(2)		5.67
	57(2 1/4)		4.25
76(3)	25.5(1)	12.84	11.64
	31.5(1 1/4)		10.90
	38(1 1/2)		10.00
	44.5(1 3/4)		8.93
	51(2)		7.68
	57(2 1/4)		6.27
	63.5(2 1/2)		4.68
82.5(3 1/4)	25.5(1)	15.02	13.82
	31.5(1 1/4)		13.09
	38(1 1/2)		12.19

Outside Dia. mm(in)	Inside Dia. mm(in)	Solid kg/ft	Weight kg/ft
82.5(3 1/4)	44.5(1 3/4)	15.02	11.11
	51(2)		9.87
	57(2 1/4)		8.45
	63.5(2 1/2)		6.87
	69.5(2 3/4)		5.11
89(3 1/2)	25.5(1)	17.38	16.18
	31.5(1 1/4)		15.44
	38(1 1/2)		14.54
	44.5(1 3/4)		13.47
	51(2)		12.22
	57(2 1/4)		10.81
	63.5(2 1/2)		9.22
	69.5(2 3/4)		7.46
	76(3)		5.53
96(3 3/4)	31.5(1 1/4)	19.91	17.97
	51(2)		14.76
	63.5(2 1/2)		11.75
	69.5(2 3/4)		9.99
	76(3)		8.06
101(4)	25.5(1)	22.6	21.40
	38(1 1/2)		19.76
	51(2)		17.45
	63.5(2 1/2)		14.45
	69.5(2 3/4)		12.69
	76(3)		10.76
	82.5(3 1/4)		9.23
	89(3 1/2)		6.99
108(4 1/4)	63.5(2 1/2)	25.47	17.32
	69.5(2 3/4)		15.56
	76(3)		13.63
	82.5(3 1/4)		12.09
115(4 1/2)	38(1 1/2)	28.51	26.40
	51(2)		23.36
	63.5(2 1/2)		20.36
	76(3)		16.67
	82.5(3 1/4)		15.13
	89(3 1/2)		12.90
	96(3 3/4)		10.50
	101(4)		7.93
121(4 3/4)	51(2)	31.73	26.58
	63.5(2 1/2)		23.57
	69.5(2 3/4)		21.81
	76(3)		19.88
	89(3 1/2)		16.12
	101(4)		11.14
127(5)	38(1 1/2)	35.11	33.50
	51(2)		29.96
	63.5(2 1/2)		26.96
	76(3)		23.27
	89(3 1/2)		19.50
	101(4)		14.53
	108(4 1/4)		11.78
	115(4 1/2)		8.89
139.5(5 1/2)	51(2)	42.40	37.24
	57(2 1/4)		35.83
	63.5(2 1/2)		34.24
	76(3)		30.55
	89(3 1/2)		26.78
	96(3 3/4)		24.38
	101(4)		21.81
	108(4 1/4)		19.07
	115(4 1/2)		16.15

Continuously Cast ☐
Centrifugally Cast ☐

Outside Dia. mm(in)	Inside Dia. mm(in)	Solid kg/ft	Weight kg/ft
152.5(6)	51(2)	50.36	45.21
	63.5(2½)		42.21
	76(3)		38.52
	89(3½)		34.75
	101(4)		29.78
	115(4½)		24.12
	127(5)		17.78
165(6½)	139.5(5½)	60.19	10.75
	76(3)		48.35
	89(3½)		44.57
	101(4)		39.61
	115(4½)		33.95
178(7)	127(5)	69.60	27.61
	139.5(5½)		20.58
	76(3)		57.78
	101(4)		49.03
	115(4½)		43.37
190(7½)	127(5)	79.70	37.03
	139.5(5½)		30.00
	152.5(6)		22.29
	76(3)		67.00
	101(4)		59.15
203(8)	127(5)	90.50	47.15
	139.5(5½)		40.12
	152.5(6)		32.41
	165(6½)		24.01
	76(3)		78.69
228(9)	101(4)	112.56	69.95
	127(5)		57.95
	152.5(6)		43.20
	165(6½)		34.80
	178(7)		25.72
254(10)	127(5)	138.82	93.60
	152.5(6)		81.60
	178(7)		66.86
	203(8)		49.38
	228(9)		29.14
279(11)	127(5)	167.78	108.00
	152.5(6)		93.26
	178(7)		75.78
	203(8)		55.55
	228(9)		32.58
304(12)	127(5)	200.00	138.00
	152.5(6)		123.00
	178(7)		105.00
	203(8)		85.00
	228(9)		60.67
330(13)	254(10)		68.00
	203(8)		170.00
	228(9)		155.00
	254(10)		137.00
	279(11)		117.00
355(14)	304(12)		94.00
	330(13)		81.00
	355(14)		68.00
	381(15)		55.00
	406(16)		43.00

Outside Dia. mm(in)	Inside Dia. mm(in)	Solid kg/ft	Weight kg/ft
355(14)	203(8)		189.00
	228(9)		166.00
	254(10)		140.00
	279(11)		112.00
	304(12)		81.00
381(15)	330(13)		47.00
	203(8)		229.00
	228(9)		206.00
	254(10)		181.00
	279(11)		152.00
406(16)	304(12)		121.00
	330(13)		87.00
	355(14)		50.00
	228(9)		249.00
	254(10)		223.00
432(17)	279(11)		195.00
	304(12)		164.00
	330(13)		130.00
	355(14)		93.00
	381(15)		54.00
457(18)	406(16)		264.00
	279(11)		235.00
	304(12)		203.00
	330(13)		167.00
	355(14)		130.00
483(19)	381(15)		90.00
	406(16)		56.58
	279(11)		283.00
	304(12)		251.00
	330(13)		216.00
508(20)	355(14)		179.00
	381(15)		138.00
	406(16)		95.00
	432(17)		49.00
	304(12)		313.00
533(21)	330(13)		268.00
	355(14)		230.00
	381(15)		190.00
	406(16)		146.00
	432(17)		101.00
559(22)	457(18)		51.00
	330(13)		329.00
	355(14)		292.00
	381(15)		253.00
	406(16)		210.00
584(23)	432(17)		165.00
	457(18)		118.00
	355(14)		348.00
	381(15)		309.00
	406(16)		267.00
609(24)	432(17)		222.00
	457(18)		174.00
	483(19)		124.00
	381(15)		369.00
	406(16)		326.00
635(25)	432(17)		281.00
	457(18)		234.00
	483(19)		183.00
	508(20)		130.00
	533(21)		93.00

SOLID
Machining Allowance O/D

<152.5mm	<6"	+ 1.5mm	+ .060"
>152.5mm	>6"	+ 3.0mm	+ .125"

CORED
Machining Allowance I/D

<76mm	<3"	-1.5mm	-.060"
>76mm	>3"	-3.0mm	-.125"





APPENDIX

GAUGE TABLE

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CONVERSION FACTORS

265



APPENDIX

GAUGE TABLE

S.W.G. B.G. U.S.G.	Standard Wire Gauge Birmingham Gauge US. government Standard Gauge				B.W.G. B.S.	Birmingham (or Stubs Iron) Wire Gauge American Brown and Sharp's Gauge								
No.	S	W	G	B	W	GB	G	B	S	U	S	G		
	inch		mm		inch		mm	inch	mm	inch		mm	inch	mm
0/7	.500		12.700	-	-			.666	16.932	-	-		.5000	12.70
0/6	.464		11.785	-	-			.625	15.875	.580		14.730	.4688	11.91
0/5	.432		10.973	.500		12.700		.588	14.943	.516		13.120	.4375	11.11
0/4	.400		10.160	.454		11.532		.542	13.757	.460		11.684	.4063	10.32
0/3	.372		9.449	.425		10.795		.500	12.700	.410		10.404	.3750	9.53
0/2	.348		8.839	.380		9.652		.445	11.308	.365		9.266	.3438	8.73
0	.324		8.23	.340		8.636		.396	10.068	.325		8.252	.3125	7.94
1	.300		7.620	.300		7.620		.353	8.971	.289		7.348	.2813	7.15
2	.276		7.010	.294		7.214		.315	7.993	.258		6.543	.2656	6.75
3	.252		6.401	.259		6.579		.280	7.122	.229		5.827	.2500	6.35
4	.232		5.893	.238		6.045		.250	6.850	.204		5.189	.2344	5.95
5	.212		5.385	.220		5.588		.223	5.651	.182		4.620	.2188	5.56
6	.192		4.877	.203		5.156		.198	5.032	.162		4.115	.2031	5.16
7	.176		4.470	.180		4.572		.176	4.480	.144		3.665	.1875	4.76
8	.160		4.064	.165		4.191		.157	3.988	.129		3.264	.1719	4.37
9	.144		3.658	.148		3.759		.140	3.551	.114		2.906	.1563	3.97
10	.128		3.251	.134		3.404		.125	3.157	.102		2.588	.1406	3.57
11	.116		2.946	.120		3.048		.111	2.827	.091		2.304	.1250	3.18
12	.104		2.642	.109		2.769		.099	2.517	.081		2.052	.1094	2.78
13	.092		2.337	.095		2.413		.088	2.240	.072		1.829	.0938	2.38
14	.080		2.032	.083		2.108		.079	1.994	.064		1.628	.0781	1.98
15	.072		1.829	.072		1.829		.070	1.775	.057		1.450	.0703	1.79
16	.064		1.626	.065		1.651		.063	1.587	.051		1.290	.0625	1.59
17	.056		1.422	.058		1.473		.056	1.412	.045		1.151	.0563	1.43
18	.048		1.219	.049		1.245		.050	1.257	.040		1.024	.0500	1.27
19	.0400		1.016	.042		1.067		.0440	1.118	.0359		.912	.0438	1.11
20	.0360		.914	.035		.889		.0392	.996	.0320		.813	.0375	.953
21	.0320		.813	.032		.813		.0349	.886	.0285		.724	.0344	.873
22	.0280		.711	.028		.711		.0313	.794	.0253		.643	.0313	.794
23	.0240		.610	.025		.635		.0278	.707	.0226		.574	.0281	.714
24	.0220		.559	.022		.559		.2248	.629	.0201		.511	.0250	.635
25	.0200		.508	.020		.508		.0220	.560	.0179		.455	.0219	.556
26	.0180		.457	.018		.457		.0196	.498	.0159		.404	.0188	.478
27	.0164		.417	.016		.406		.0175	.443	.0142		.361	.0172	.437
28	.0148		.376	.014		.356		.0156	.397	.0126		.320	.0156	.496
29	.0136		.345	.013		.330		.0139	.353	.0113		.287	.0141	.358
30	.0124		.315	.012		.305		.0123	.312	.0100		.254	.0125	.318
31	.0116		.295	.010		.254		.0110	.279	.0089		.227	.0109	.278
32	.0108		.274	.009		.228		.0098	.249	.0080		.202	.0102	.258
33	.0100		.254	.008		.203		.0087	.221	.0071		.180	.0094	.238
34	.0092		.234	.007		.178		.0077	.196	.0063		.160	.0086	.218
35	.0084		.213	.005		.127		.0069	.175	.0056		.143	.0078	.198
36	.0076		.193	.004		.101		.0061	.155	.0050		.127	.0070	.179
37	.0068		.173	-		-		.0054	.137	.0045		.113	.0066	.169
38	.0060		.152	-		-		.0048	.122	.0040		.101	.0063	.159
39	.0052		.132	-		-		.0043	.109	.0035		.090	.0059	.149
40	.0048		.122	-		-		.00386	.098	.0031		.080	.0055	.139

CONVERSION FACTORS

LINEAR MEASURE

	Millimeter mm	Centimeter cm	Meter m	Inch in	Foot ft	Yard yd	Mile mi
Millimeter (mm)	1	0.1	0.001	0.03937	0.0032808	0.0010936	0.0 ₆ 214
Centimeter (cm)	10	1	0.01	0.3937	0.032808	0.010936	0.0 ₅ 6214
Meter (m)	1,000	100	1	39.37	3.28083	1.0936	0.0 ₃ 6214
Inch (in)	25.4	2.54	0.0254	1	0.0833	0.02778	0.0 ₄ 1578
Foot (ft)	304.8	30.48	0.3048	12	1	0.3333	0.0 ₃ 1894
Yard (yd)	914.4	91.44	0.9144	36	3	1	0.0 ₃ 5682
Mile (mi)	1,609,347	160,934.7	1,609.35	63,360	5,280	1,760	1

SQUARE MEASURE

	Square Millimeter mm ²	Square Centimeter cm ²	Square Meter m ²	Square Inch in ²	Square Foot ft ²	Square Yard yd ²
Square millimeter (mm ²)	1	0.01	0.0 ₁	0.00155	0.0 ₄ 10764	0.0 ₅ 119599
Square centimeter (cm ²)	100	1	0.0001	0.154999	0.0010764	0.0 ₃ 119599
Square meter (m ²)	1,000,000	10,000	1	1,549.99	10.7639	1.19599
Square inch (in ²)	645.2	6.452	0.0 ₆ 452	1	0.006944	0.0 ₃ 7616
Square foot (ft ²)	92,900	929	0.0929	144	1	0.11111
Square yard (yd ²)	836,100	8,361	0.8361	1,296	9	1

CUBE MEASURE

	Cubic Centimeter cm ³	Cubic Meter m ³	Cubic Inch in ³	Cubic Foot ft ³	Cubic Yard yd ³
Cubic centimeter (cm ³)	1	0.0 ₁	0.06102	0.0 ₄ 3531	0.0 ₅ 1308
Cubic meter (m ³)	1,000,000	1	61,023	35.31	1.308
Cubic inch (in ³)	16.39	0.0 ₄ 1639	1	0.0 ₃ 5787	0.0 ₄ 2143
Cubic foot (ft ³)	28,317	0.028317	1,728	1	0.03704
Cubic yard (yd ³)	764,500	0.7645	46,660	27	1

WEIGHT

	Kilogram kg	Ounce oz	Pound lb	Net ton (2,000 lbs) nt	Gross ton (2,240 lbs) gt	Metric ton (1,000 kg) t
Kilogram (kg)	1	35.274	2.20462	0.001102	0.0 ₃ 9842	0.001
Ounce (oz)	0.02835	1	0.0625	0.0 ₄ 3125	0.0 ₄ 2790	0.0 ₄ 2835
Pound (lb)	0.45359	16	1	0.0005	0.0 ₃ 4464	0.0 ₃ 4536
Net ton (nt)	907.185	32,000	2,000	1	0.89286	0.90719
Gross ton (gt)	1,016.05	35,840	2,240	1.12	1	1.01605
Metric ton (t)	1,000	35,274	2,204.62	1.10231	0.98421	1

Note:

The small subnumeral following a zero indicates that the zero is to be repeated that number of time, thus 0.0₃4 = 0.0004

APPENDIX

WEIGHT PER LINEAR UNIT

	Gram per centimeter g/cm	Kilogram per meter kg/m	Pound per inch lb/in	Pound per foot lb/ft	Pound per yard lb/yd
Gram per centimeter (g/cm)	1 0.1	0.5056	0.0672	0.20159	
Kilogram per meter (kg/m)	10	1	0.056	0.67197	2.0159
Pound per inch (lb/in)	178.5	17.8579	1	12	36
Pound per foot (lb/ft)	148,816	1.48816	0.08333	1	3
Pound per yard (lb/yd)	4.96054	0.49605	0.02778	0.3333	1

WEIGHT PER UNIT AREA

	Kilogram per square centimeter kg/cm ²	Kilogram per square meter kg/m ²	Metric ton per square meter t/m ²
Kilogram per square centimeter (kg/cm ²)	1	10,000	10
Kilogram per square meter (kg/m ²)	0.0001	1	0.001
Metric ton per square meter (t/m ²)	0.1	1,000	1
Pound per square inch (lb/in ²)	0.0703067	703.06686	0.7031
Pound per square foot (lb/ft ²)	0.0004882	4.8824087	0.004882

WEIGHT PER UNIT VOLUME

	Kilogram per cubic centimeter kg/cm ³	Kilogram per cubic meter kg/m ³	Metric ton per cubic meter t/m ³
Kilogram per cubic centimeter (kg/cm ³)	1	1,000,000	1,000
Kilogram per cubic meter (kg/m ³)	0.000001	1	0.001
Metric ton per cubic meter (t/m ³)	0.001	1,000	1
Pound per cubic inch (lb/in ³)	0.02768	27,680.4	27.6804
Pound per cubic foot (lb/ft ³)	0.000016	16.0196	0.01602

ENERGY

	B.t.u.	Calories	Ft-lb	Kg-m	Hp-hr	Kw-hr	Joules
B.t.u. (mean)	1	0.252	778	107.563	0.0 ₃ 3929	0.0 ₃ 2931	1,054.8
Calories (mean)	3.968	1	3,091.36	426.84	0.001559	0.001163	4,185
Ft-lb	0.001285	0.0 ₃ 3239	1	0.1383	0.0 ₆ 505	0.0 ₆ 3767	1.355
Ft-ton	2.571	0.6478	2,000	276.511	0.00101	0.0 ₃ 7535	2,712.59
Kg-m	0.009297	0.002343	7.23301	--	0.0 ₅ 3653	0.0 ₅ 2725	9.806
Hp-hr	2,544.99	641.327	1,980,000	273,747	1	0.746	2,685,600
Kw-hr	3,411.57	859.702	2,654,200	366,959	1.34041	1	3,600,000
Joules (absolute)	0.0 ₃ 9477	0.0 ₃ 2389	0.73735	0.101937	0.0 ₆ 3725	0.0 ₆ 2778	1
Lb C	14,544	3,665	11,315,000	1,564,396	5.714	4.263	153,470,000
Lb H ₂ O	970.4	244.537	754,971	104,379	0.38127	0.284424	1,023,966

Note.
The small subnumeral following a zero indicates that the zero is to be repeated that number of time, thus 0.0₃4 = 0.0004.